



GE VERNOVA

ADMS

16.25.1

DNAF Advanced Applications Analyst Guide

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Change History

Date	Change Description
Jun 8, 2023	ADMS 3.16: • Updated screenshot in FISR Configuration. • Updated Problem Formulation Configuration. • Updated Enabling and Disabling Applications. • Updated Restoration Levels. • Updated Miscoordination. • Updated Loss of Voltage Triggered FISR. • Added Feeder Fault Triggered Loss of Voltage FISR. • Updated FISR Configuration. • Updated FISR Fault Event Triggering Details. • Updated FISR Loss of Voltage Event Triggering Details. • Editorial and formatting updates.
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1. Introduction

This document describes several configuration aspects of the advanced Distribution Network Analysis Functions (DNAF) applications.

The advanced DNAF applications include the following:

- **Fault Isolation and Service Restoration (FISR):** This function generates a switching sequence to isolate faulted sections and to restore service to the non-faulted sections.
- **Automated Feeder Reconfiguration (AFR):** This function provides plans to unload overloaded segments. AFR also generates switching plans to restore nominal network topology.
- **Planned Outage Study (PLOS):** This function provides plans to isolate a component, minimizing the number of interrupted customers. This function is available only in study mode.
- **Fault Isolation and Service Restoration Contingency Analysis (FISRCA):** This function executes a FISR analysis for every switchable section in the station. This function is available only in study mode.
- **Load and Volt/VAR Management (LVM):** This function optimizes and coordinates capacitors, voltage regulating transformers (regulators and LTCs), and generator and distributed energy resource VAR or power factor output to achieve an objective including demand minimization and reactive power support while maintaining operational voltage limits. LVM is used as a least-intrusive demand management tool.
- **Faulted Section Determination (FSD):** This function provides switching plans to identify a fault and isolate faulted sections. The FSD function applies a switching-based approach with step-by-step opening and closing switching cycles. At Stage 1, a faulted feeder downstream of a faulted power transformer is identified and isolated. At Stage 2, the faulted sections are identified and isolated.
- **Surgical Load Shed (SLS):** This function provides DMS data (critical loads, load types, transfer/throwover schemes, reserved capacity commitments, etc.) to the SCADA Load Shed function to accurately prioritize and shed circuits to obtain a desired target reduction in system load.

For a description of the standard DNAF applications and general concepts of DNAF, refer to the *Distribution Network Analysis Functions (DNAF) Standard Applications Analyst Guide*.

2. DNAF Configuration Editor

The DNAF Configuration Editor allows a user to easily modify the DNAF configurations. For more details about working with DNAF Configuration Editor, refer to the *Distribution Network Analysis Functions (DNAF) Standard Applications Analyst Guide*.

2.1. Surgical Load Shed Configuration

The Surgical Load Shed (SLS) functionality is part of a combined ADMS and Scada application for prioritizing and shedding circuits to obtain a desired target reduction in system load. [Surgical Load Shed](#) provides detailed information about SLS.

The following display contains the configuration parameters for SLS. For a detailed list of the configuration parameters, refer to [Surgical Load Shed Configuration](#).

The screenshot shows the DNAF Configuration Editor window. The title bar indicates the file path and revision: "DNAF Configuration Editor - D:\Eterra\Distribution\FantasyIsland\models\DNAFParams\DNAF_PARAMS.xml, Revision: 31". The menu bar includes "File" and "Help". The left sidebar contains a tree view with the following structure:

- Base Applications
 - Bus Load Allocation (BLA)
 - Distribution Power Flow (DPF)
 - Limit Monitor (LIM)
 - Fault Location (FL)
 - Protection Validation (PRV)
 - Short Circuit Calculation (SCC)
- Advanced Applications
 - Surgical Load Shed (SLS)**
- Switching Reconfiguration
 - Fault Isolation and Service Restoration (FISR)
 - Automatic Feeder Reconfiguration (AFR)
 - FISR Contingency Analysis (FISRCTA)
 - Planned Outage Study (PLOS)
 - Faulted Section Determination (FSD)
 - Problem Formulations
 - Operation Modes
- Load and Volt/VAR Management
 - Load and Volt/VAR Management (LVM)
 - Problem Formulations
 - Operation Modes
- Plugin Applications
 - Configuration

The main panel is divided into two sections:

Load Shed Classification

List of Classifications
(Lower Ids are more restrictive)

Id	Name
1	Do Not Shed
2	Critical
3	Medium
4	Low

Buttons: Add, Delete

Classification Info
Id: 1
Name: Do Not Shed

Rule Sets

List of Load Shed Rule Sets

Id	Name	Default Rule
1	Default	True
2	Restrictive	False

Buttons: Add, Copy, Delete

Rule Set Info
Name: Default ☒ Is Default
Default Class: Low

Figure 1. Surgical Load Shed Configuration

2.2. FISR Configuration

The Fault Isolation and Service Restoration function generates a switching sequence to isolate faulted sections and to restore service to the non-faulted sections. For detailed information about FISR, refer to [Faulted Section Determination](#).

This display contains the configuration parameters for FISR. For a detailed list of the configuration parameters, refer to [FISR Configuration](#).

The screenshot shows the 'DNAF Configuration Editor' window. The title bar indicates the file path 'D:\Eterra\Distribution\FantasyIsland\models\DNAFParams\DNAF_PARAMS.xml, Revision: 19'. The interface is divided into a left sidebar and a main configuration area.

Left Sidebar (Applications):

- Base Applications
 - Bus Load Allocation (BLA)
 - Distribution Power Flow (DPF)
 - Limit Monitor (LIM)
 - Fault Location (FL)
 - Protection Validation (PRV)
 - Short Circuit Calculation (SCC)
- Advanced Applications
 - Surgical Load Shed (SLS)
 - Switching Reconfiguration
 - Fault Isolation and Service Restoration (FISR)** (highlighted)
 - Automatic Feeder Reconfiguration (AFR)
 - FISR Contingency Analysis (FISRCTA)
 - Planned Outage Study (PLOS)
 - Faulted Section Determination (FSD)
 - Problem Formulations
 - Operation Modes
 - Load and Volt/VAR Management
 - Load and Volt/VAR Management (LVM)
 - Problem Formulations
 - Operation Modes
 - Plugin Applications
 - Configuration

Main Configuration Area (Solution Formulation):

Objectives

- Default Station Problem Formulation: Minimum Unserved kW (Auto Only)
- Default Transformer Problem Formulation: None: No Transformer Objective
- Default Feeder Problem Formulation: None: No Feeder Objective

Convergence Criteria

- Maximum Real Power Mismatch (kW): 2
- Maximum Reactive Power Mismatch (kVAR): 2
- Maximum Voltage Mismatch (p.u.): 0.0001

Switching Plan Generation Parameters

- Number of Selected Top Plans: 3
- Maximum Number of Internally Generated Plans: 3
- Maximum Number of Feeders: 40
- Plan Life - Real Time (hr): 4
- Look Ahead Duration (hr): 4
- Switch Transfer Flow Margin (%): 20

Triggering Logic Parameters

- ☒ Only Generate Plans that Restore Load
- ☐ Only Solve for the Triggering Fault
- ☐ Change Voltage Regulation Mode
- Time Tolerance for Old Events Not Considered (sec): 60
- ☒ Handle Miscoordinations
- Time Difference Tolerance (sec): 5
- ☐ Place Faults Conservatively when an Inconsistency is Detected Involving Directional Fault Indicators

Preprocessing Checks

- ☐ ...

Figure 2. Fault Isolation and Service Restoration Configuration

2.3. AFR Configuration

The Automated Feeder Reconfiguration function provides plans to unload overloaded segments. It also generates return to normal switching plans. Detailed information about AFR is provided in [Switching Reconfiguration Applications](#).

This display contains the configuration parameters for AFR. For a detailed list of the configuration parameters, refer to [AFR Configuration](#).

Figure 3. Automated Feeder Reconfiguration Configuration

The Fault Isolation and Service Restoration Contingency Analysis function executes a FISR analysis for every switchable section in the station. This function is available only in study mode. Detailed information about FISRC is provided in [Switching Reconfiguration Applications](#).

2. DNAF Configuration Editor
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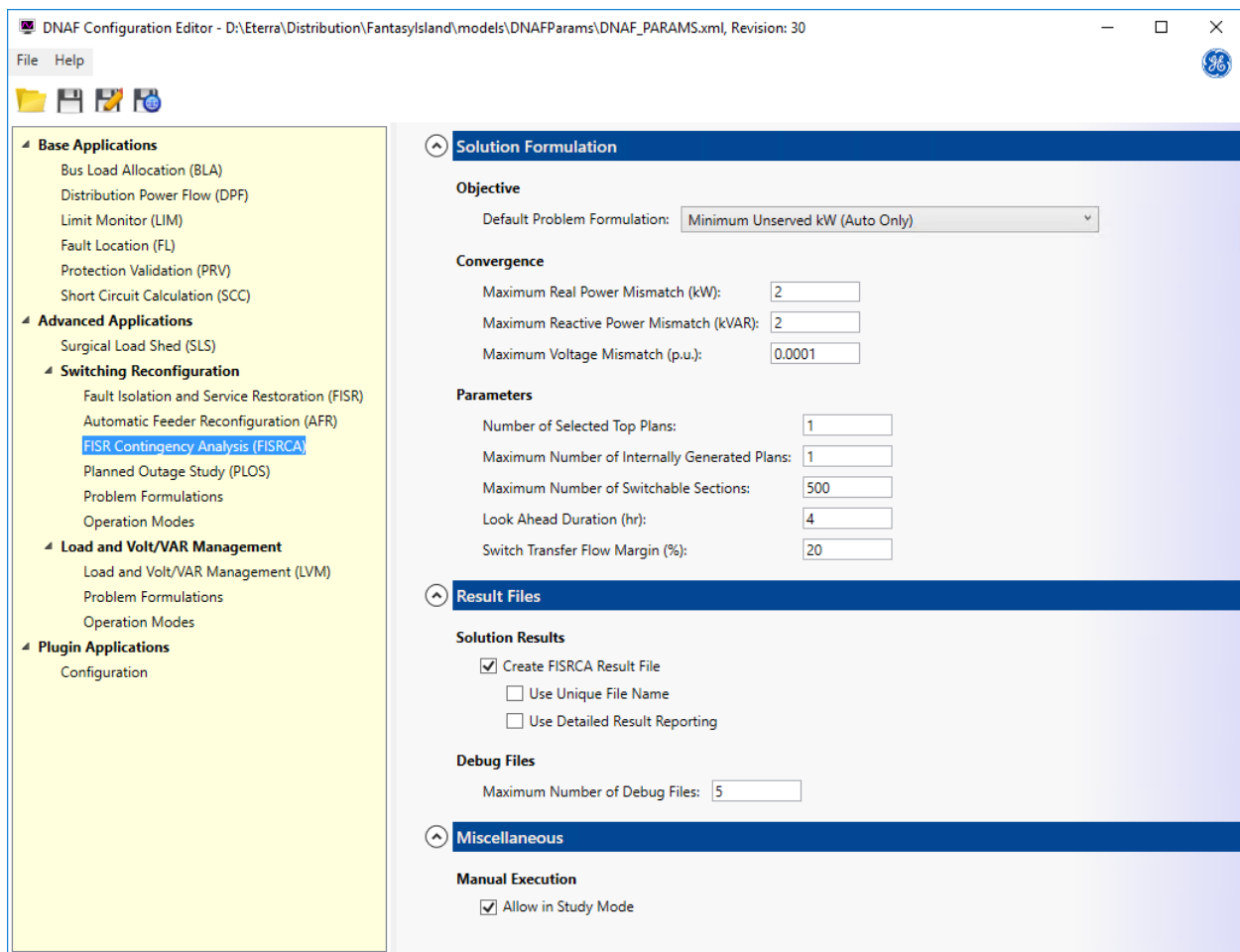


Figure 4. Fault Isolation and Service Restoration - Contingency Analysis Configuration

2.5. PLOS Configuration

The Planned Outage Study function provides plans to isolate a component, minimizing the number of interrupted customers. Detailed information about PLOS is provided in [Switching Reconfiguration Applications](#).

This display contains the configuration parameters for PLOS. For a detailed list of the configuration parameters, refer to [PLOS Configuration](#).

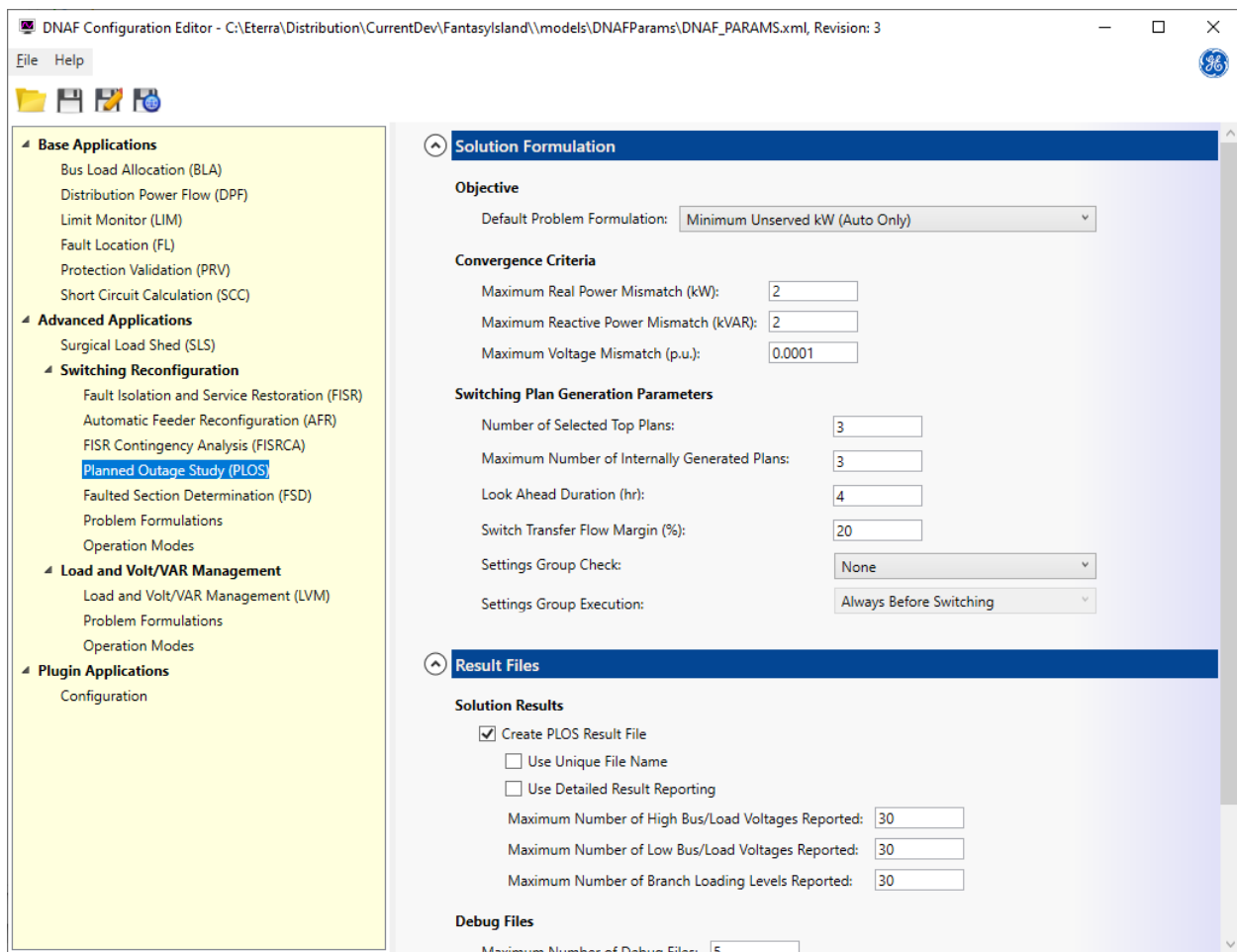


Figure 5. Planned Outage Study Configuration

2.6. FSD Configuration

The Faulted Section Determination (FSD) function provides switching plans to identify a fault and isolate faulted sections. Detailed information about FSD is provided in [Faulted Section Determination](#).

This display contains the configuration parameters for FSD. For a detailed list of the configuration parameters, refer to [FSD Configuration](#).

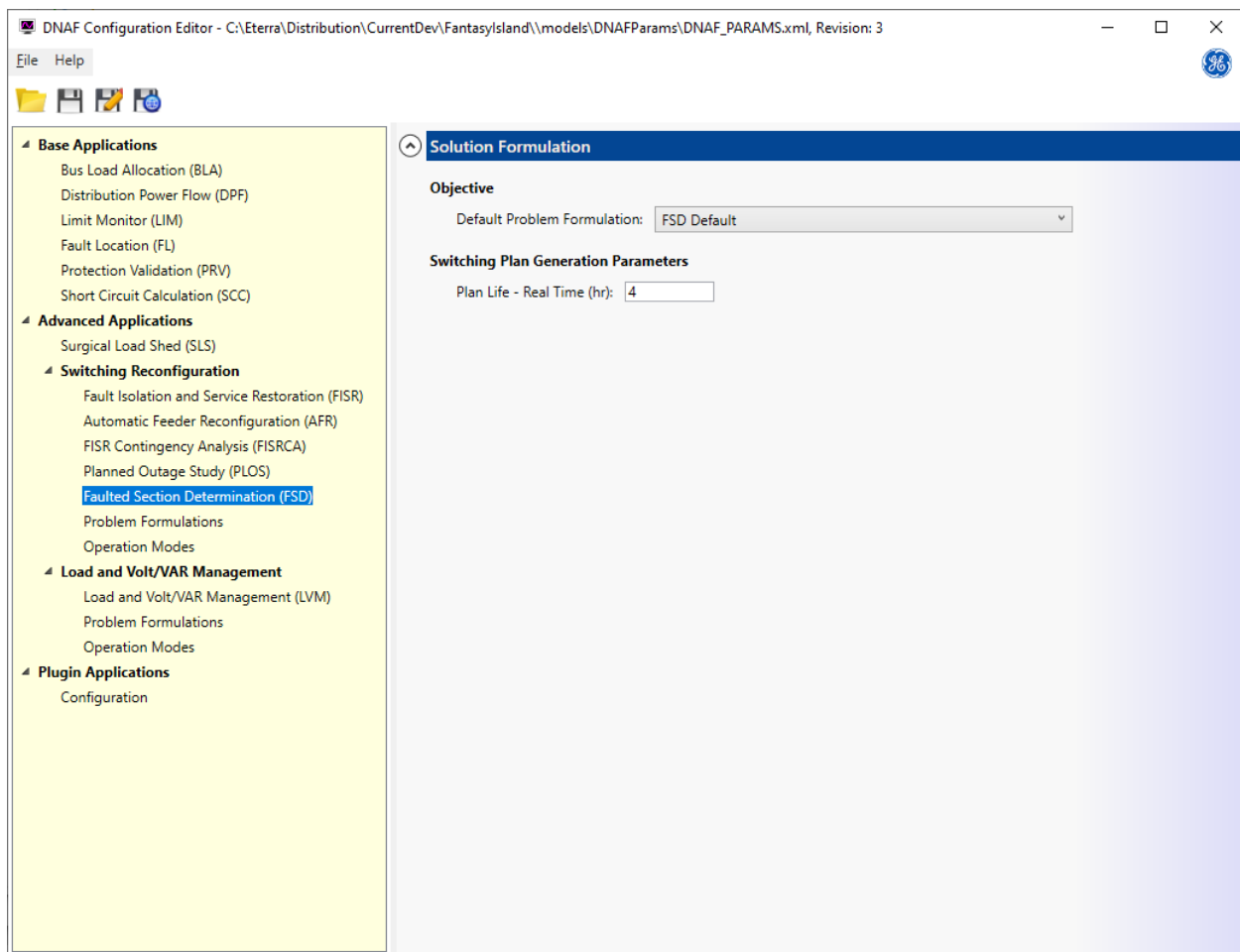


Figure 6. FSD Configuration

2.7. Problem Formulation - Reconfiguration Applications

For the reconfiguration applications (FISR, AFR, FISRCA, and PLOS), a problem formulation is required. The problem formulations are used to package the operational goals and constraints for the given problem. The problem formulation has an associated limit strategy and optimization control scheme. [Switching Reconfiguration Applications](#) provides detailed descriptions for creating specific problem formulations.

In the old DNAF Configurator, problem formulations were listed on a separate tab. In the new DNAF Configuration Editor, problem formulations are presented as an integral part of the "Switching Reconfiguration Applications".

This page displays the attributes used to define the reconfiguration goals. For a detailed list of the configuration parameters, refer to [Problem Formulation Configuration](#).

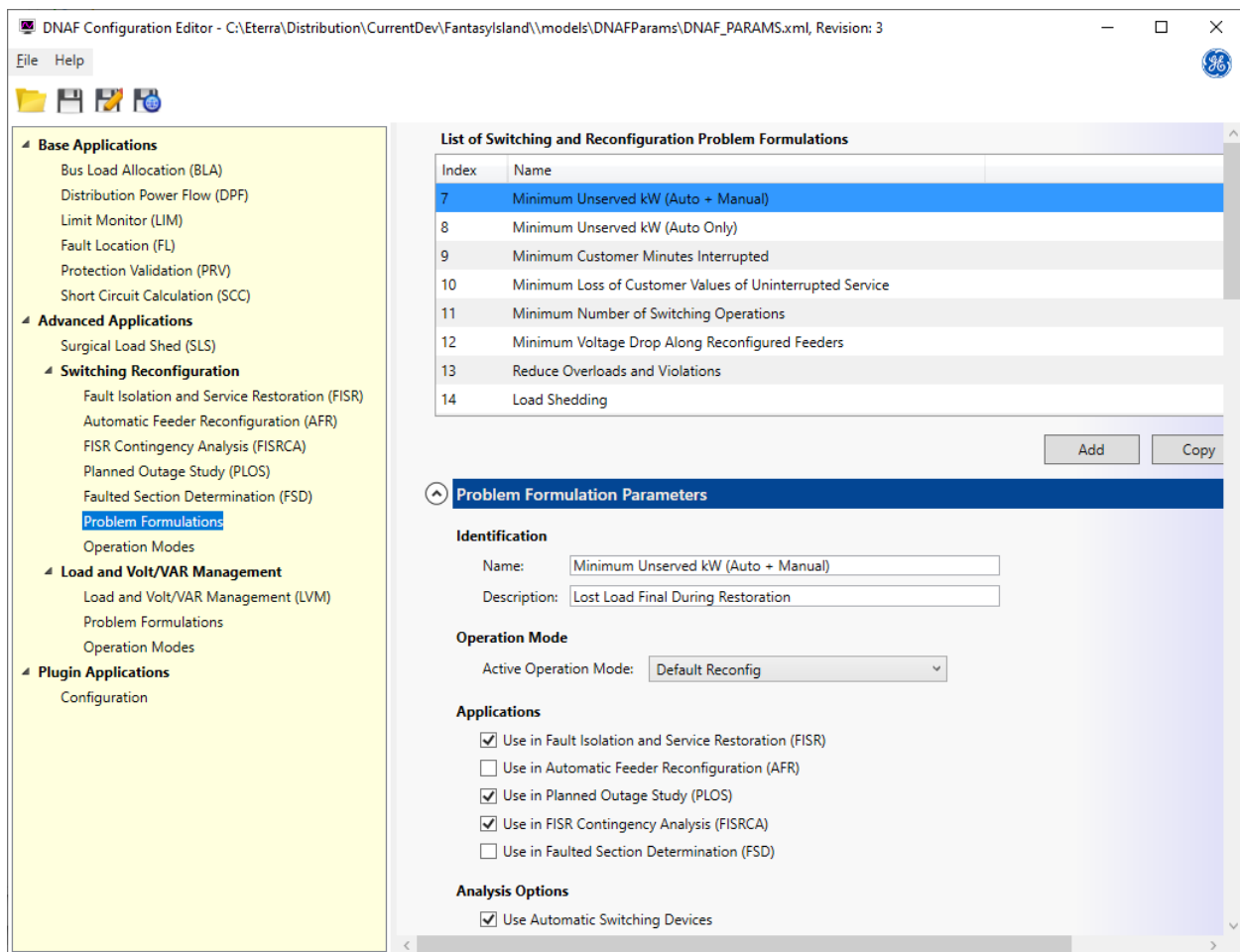


Figure 7. Problem Formulation - Reconfiguration Applications

2.8. Operation Modes - Reconfiguration Applications

This display allows users to configure the operation mode and plan execution settings for the real-time reconfiguration applications. For a detailed list of the configuration parameters, refer to [Operation Modes Configuration](#).

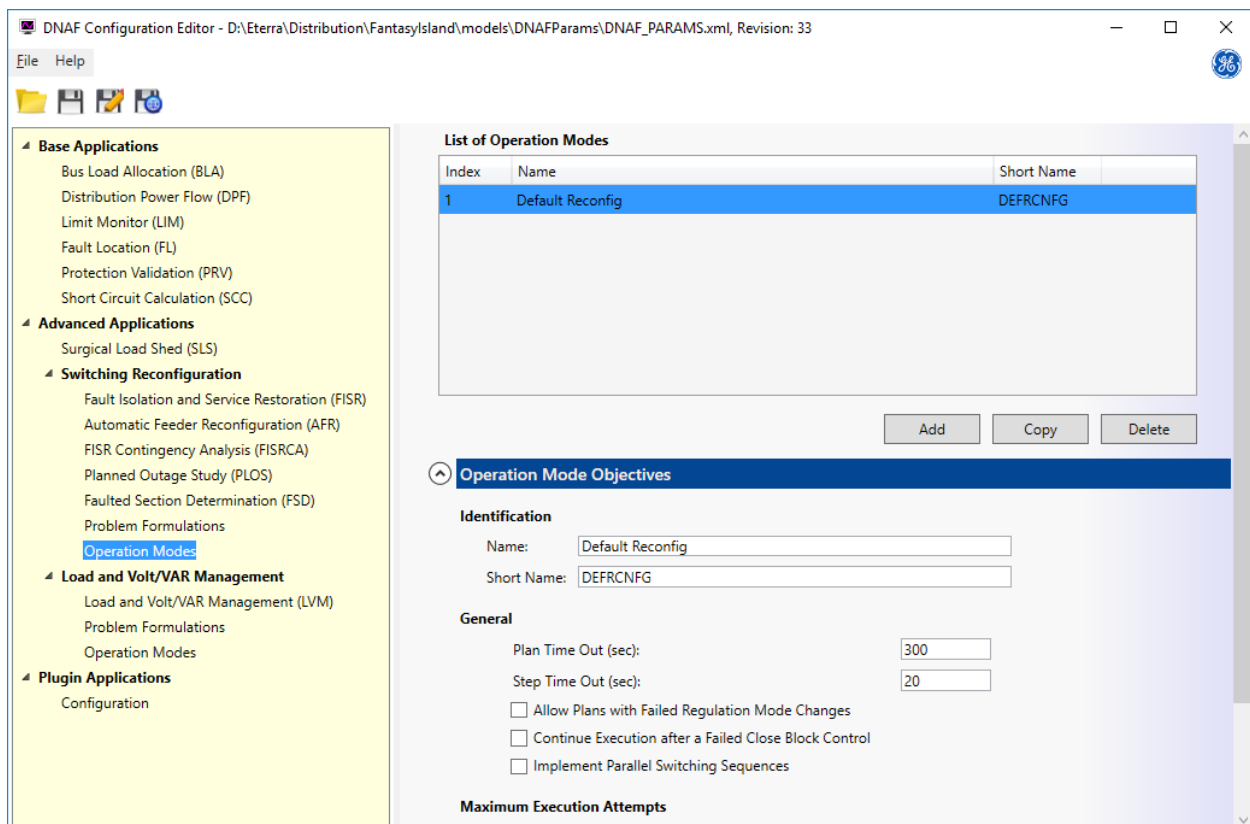


Figure 8. Operation Modes - Reconfiguration Applications

2.9. Load and Volt/VAR Management

The Load and Volt/VAR Management function optimizes and coordinates capacitors and voltage regulating transformers (regulators and LTCs) and DERS to achieve an objective including reactive power support or demand minimization while maintaining operational voltage limits. LVM is used as a least-intrusive demand management tool. Detailed information about LVM is provided in [Load and Volt/VAR Management](#).

This display contains the configuration parameters for LVM. For a detailed list of the configuration parameters, see [LVM Configuration](#).

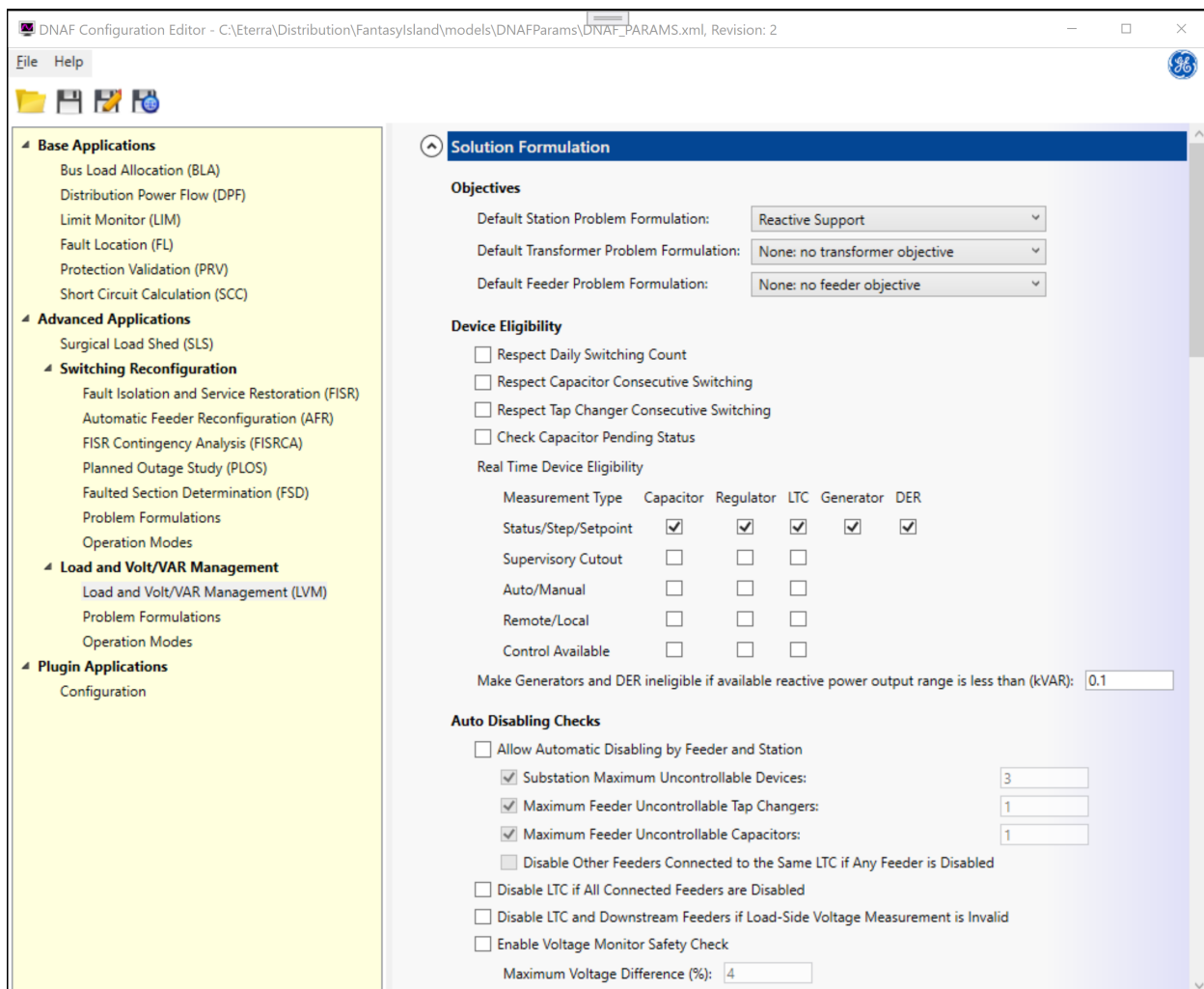


Figure 9. Load and Volt/VAR Management Configuration

2.10. Problem Formulation - LVM

For LVM, a problem formulation is required. The problem formulations are used to package the operational goals and constraints for the given problem. The problem formulation has an associated limit strategy and optimization control scheme. Detailed descriptions for creating specific problem formulations are provided in [Load and Volt/VAR Management](#).

In the old DNAF Configurator, problem formulations were listed on a separate tab. In the new DNAF Configuration Editor, problem formulations are presented as an integral part of the "Load and Volt/VAR Management" application.

This page displays the attributes used to define the optimization goal. For a detailed list of the configuration parameters, refer to [LVM Problem Formulation](#).

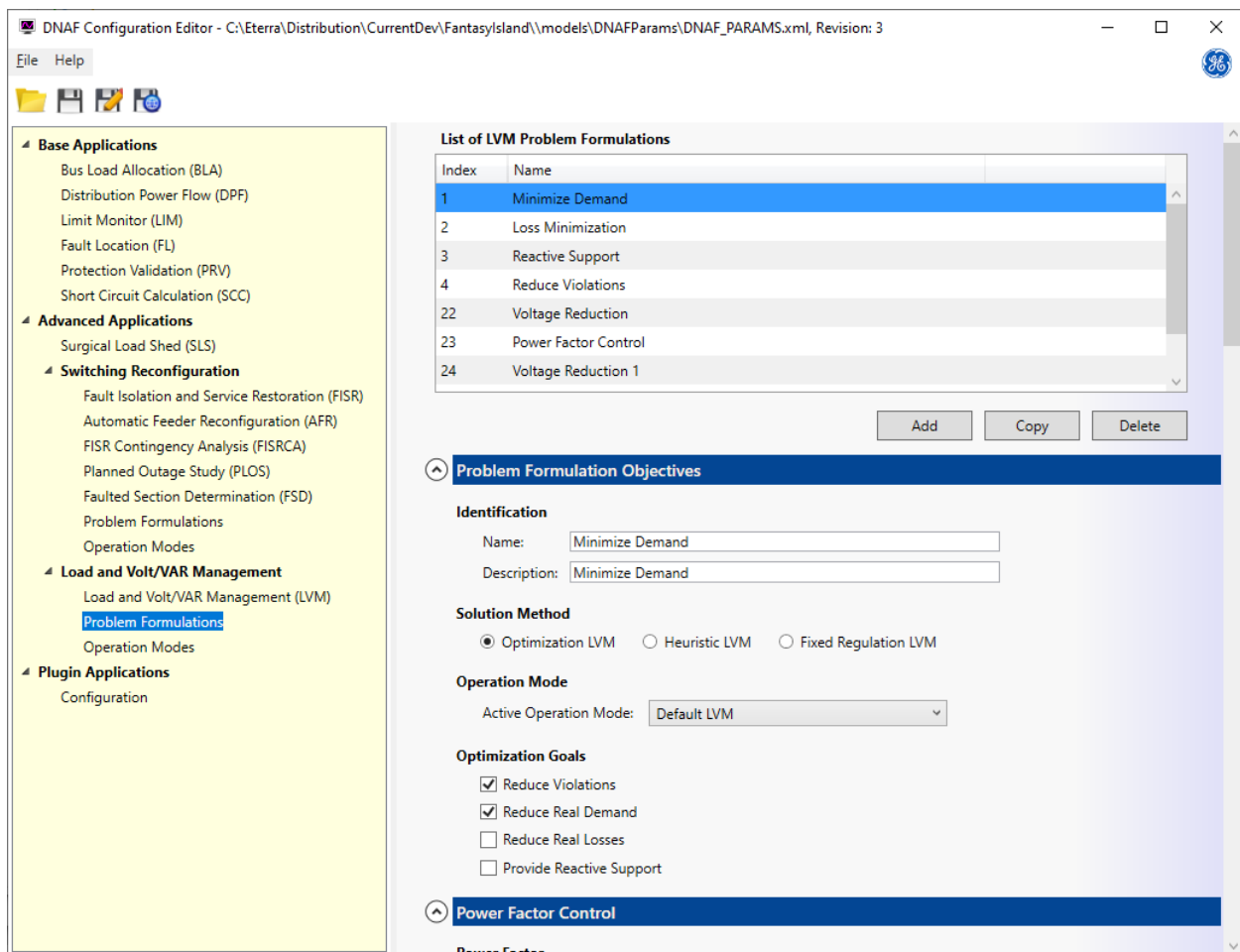


Figure 10. Problem Formulation - Load and Volt/VAR Management

2.11. Operation Modes - LVM

This display allows the user to configure the operation mode and plan execution settings for the real-time LVM application. For a detailed list of the configuration parameters, refer to [LVM Operation Modes](#).

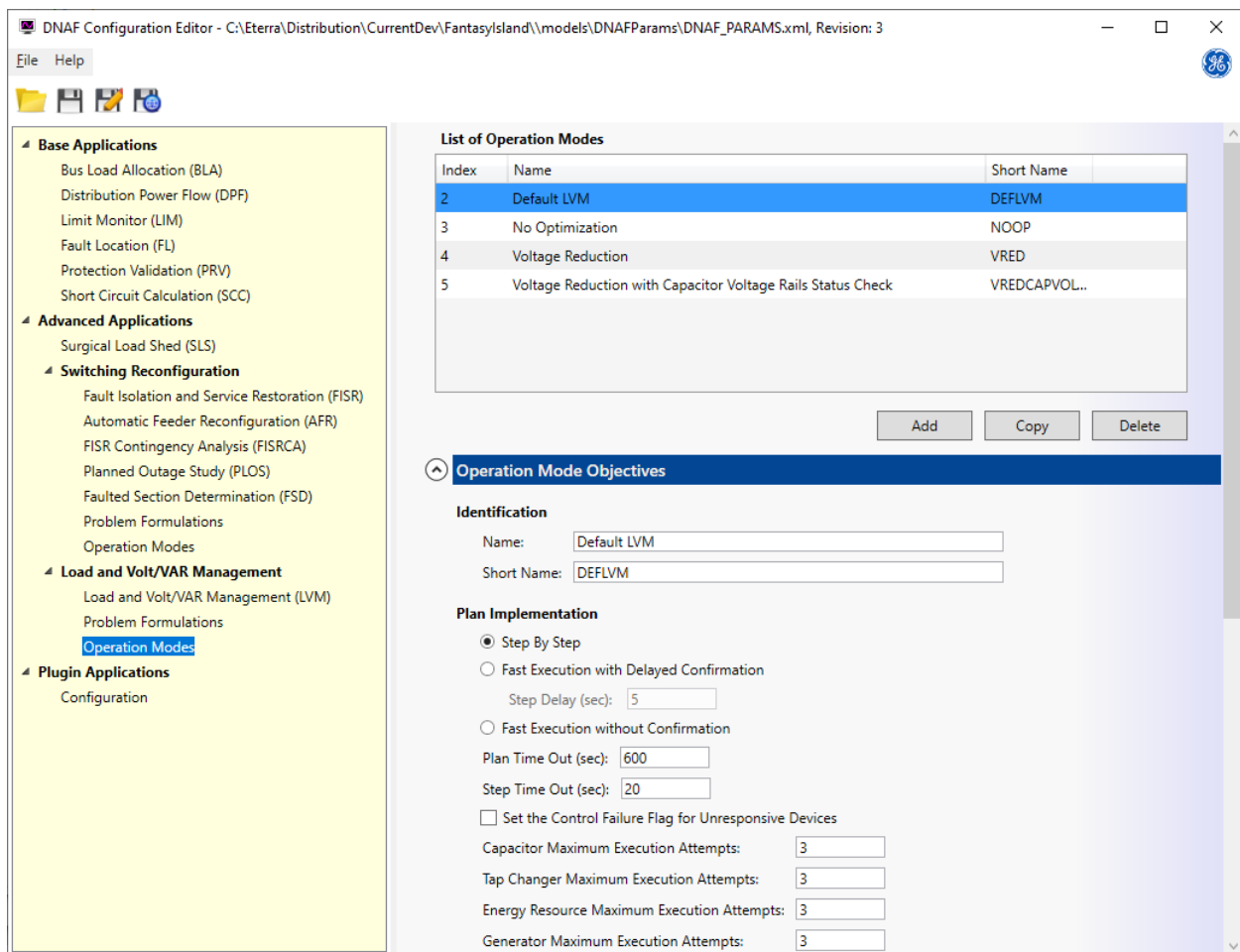


Figure 11. Operation Modes - Load and Volt/VAR Management

2.12. Plug-In Applications

This display allows the user to configure the LVM System Monitor settings. For a detailed list of the configuration parameters, refer to [LVM System Monitor Configuration](#).

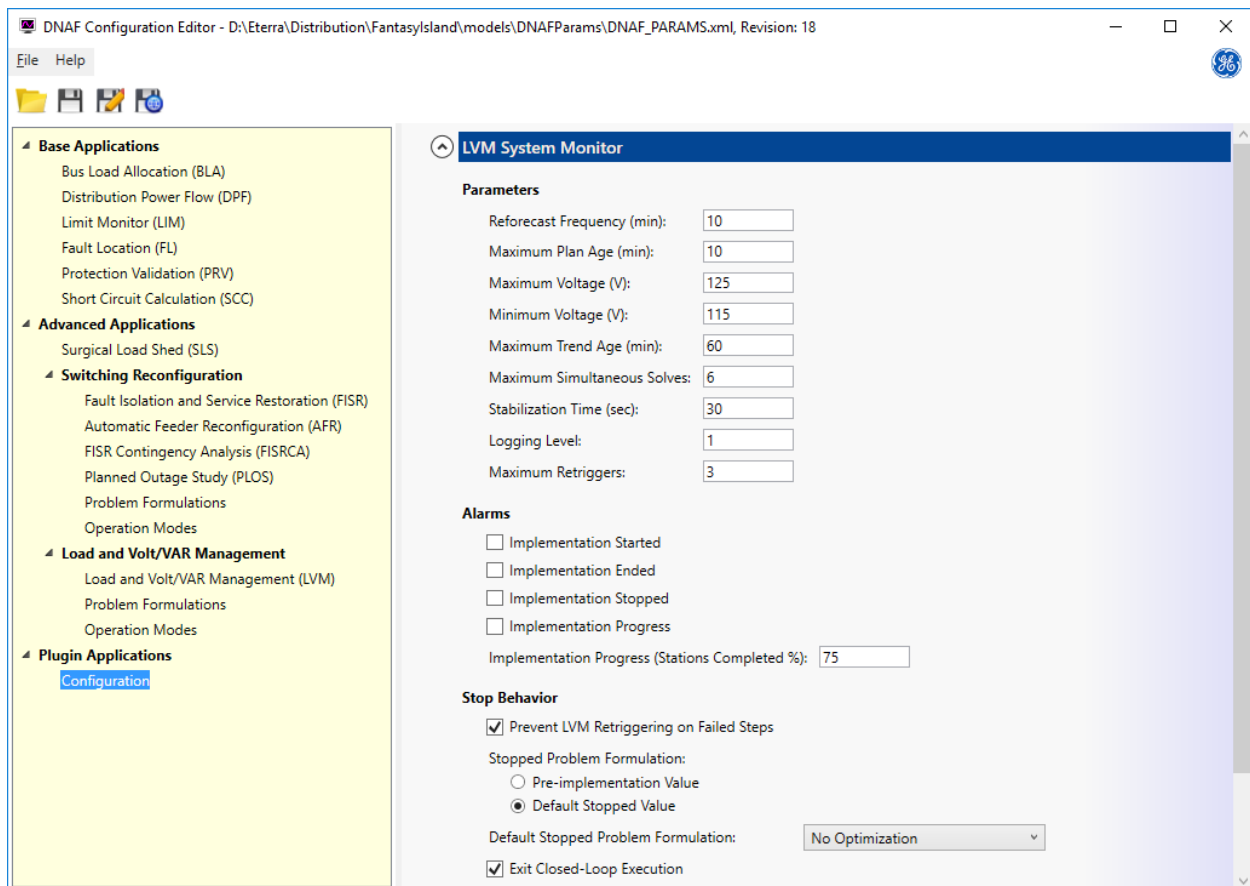


Figure 12. LVM System Monitor Configuration

3. Load and Volt/VAR Management

Load and Volt/VAR Management (LVM) is an important network optimization application. LVM generates a series of control actions to improve distribution system operation following a set of pre-defined objectives- for example, to keep the nodal voltages inside operational limits, to move the operating point to satisfy power factor requirements at specific locations in the network, to reduce total real power demand or real power losses in the network, or different combinations of these objectives.

3.1. Supported LVM Controls

LVM supports the following controls:

- Tap changing power transformers (LTCs) and regulators:
 - Tap position raise and lower controls,
 - Voltage reduction setpoint control, and
 - Voltage regulation bandcenter setpoint control. The following units are supported:
 - Per unit,
 - Actual kV (both L-G and L-L), and
 - User-configurable voltage base, such as 120 V.
- Shunt capacitors: open and close controls
- Generators and Distributed Energy Resources (DERs):
 - Reactive power output setpoint control, and
 - Power factor setpoint control.

3.2. Solution Methods

Three approaches to LVM are supported:

- **Optimization LVM** uses DPF results to determine the optimal series of control actions to meet a specified objective. An accurate network model and high-quality DPF results are required for accurate optimization LVM solutions. In the absence of high-quality input to Optimization LVM, Heuristic LVM or Fixed Regulation LVM can be used.
- **Heuristic LVM** uses real-time measurements, the dynamic network connectivity, and the impedance model instead of DPF results to determine violations and control sensitivities, such as voltage and power mismatches. It generates one control action at a time (a capacitor switching move, a tap position adjustment, or voltage setpoint adjustment) to achieve a specified objective.
- **Fixed Regulation LVM** is based on a specified voltage reduction percentage for all regulating

transformers (LTCs and regulators) in the network selection. It can be used to obtain the desired voltage reduction in an emergency voltage reduction situation or in cases where optimization LVM would not give good results due to a low-quality power flow solution. The voltage reduction LVM solution is a topology-based ordering of eligible LTCs or regulators; as a result of this, the Plan Manager subsystem issues controls to regulating transformers in a topology-based order.

Different LVM problem formulations are supported at the station, power transformer bank, and feeder levels. In order to solve LVM with potentially mixed problem formulation types, a station is divided into one or more LVM solution areas. LVM solution areas are contiguous subsets of energized LVM analysis network sections that are bounded by power transformer banks or feeder-heads with problem formulations belonging to the same LVM solution approach (optimization or fixed regulation). Network sections with different problem formulations of the fixed regulation LVM solution approach are treated as separate LVM solution areas. Currently, heuristic LVM problem formulations are not supported in mixed-problem formulations LVM solution.

An LVM solution for a station includes solutions from all the station's LVM solution areas. LVM device eligibility is calculated on the LVM solution area basis considering the problem formulation in effect for that solution area.

3.3. LVM Execution and Triggering

The LVM application can be executed using the following mechanisms:

- Manual Execution

This is done by requesting an LVM solution for a given station from the DNAF application pop-up.

The configuration option DNAF_MANUAL_TRIGGER must be set to **1** in the DNAF server configuration file for the manual trigger to work.

- Periodic Execution

LVM can be configured to be solved periodically in an automated manner, by setting the DNAF_LVM_PERIODIC_TRIGGER configuration option to **1** in the DNAF server configuration file. Also, the DNAF_LVM_MAX_PERIOD parameter must be set to a positive integer. This parameter specifies the periodicity, in minutes, for any station.

In addition to the "regular periodic" trigger, LVM can be configured to be executed with "fast periodic" triggers.

- To enable LVM fast triggers, set the DNAF_LVM_FAST_TRIGGER option to **1** in the DNAF server configuration file. If fast triggering is enabled, then the regular periodic trigger parameter DNAF_LVM_MAX_PERIOD has no effect. Separate periodicity parameters are used as described below.
- For optimization LVM, the DNAF_LVM_FAST_TRIGGER_PERIOD option should be set to a

reasonably small value, such as 3. This parameter specifies, in minutes, the interval between two periodic LVM executions for a given station when fast triggering is enabled.

- For heuristic LVM, several server configuration parameters must be set. For details, see [Heuristic LVM Methodology](#).

- Event Triggered Execution

LVM can be configured to be solved for network events such as topology changes or voltage quality changes.

- To trigger LVM in response to topology events such as opening/closing of breakers or switches or the addition/removal of temporary cuts and jumpers, set the `DNAF_LVM_EVENT_TRIGGER` configuration option to **1** in the DNAF server configuration file.
- For voltage quality triggering, set the `DNAF_LVM_EVENT_VOLTQUALITY_TRIGGER` option to **1** in the DNAF server configuration file. A voltage quality event is triggered when a voltage measurement exceeds its high voltage limit or falls below its low voltage limit.

When `DNAF_LVM_CHECK_MEAS_CHANGE` is also set to **1**, two additional validation checks are performed before the LVM voltage quality event trigger is issued:

1. **Value change check:** The difference between the current measurement value and the previous scan value must be greater than a threshold (0.1 V). If the measurement value has not changed significantly between scans, the LVM event trigger is skipped.
2. **Measurement age check:** The measurement timestamp must be recent, i.e., within the maximum age defined by the `DNAF_MAXTIME_CHECK_MEAS_CHANGE` parameter (default 300 seconds). If the measurement is outdated, the LVM event trigger is skipped.

These checks prevent redundant LVM evaluations caused by unchanged or stale SCADA voltage measurements.

- For LVM plan execution failure triggering, set the `DNAF_LVM_EVENT_PLANEXECFAILURE_TRIGGER` option to **1** in the DNAF server configuration file.

3.4. Optimization LVM Methodology

As a multi-dimensional non-linear problem, the distribution power system optimization is subject to different conditions, which can be described in the equation below:

$$\begin{aligned}
& \text{obj.} \quad \min \sum_{i=1}^N \lambda_i \cdot OG_i \\
& \text{st.} \quad f(x, u) = 0 \\
& \quad V_{min} \leq V \leq V_{max} \\
& \quad I_{min} \leq I \leq I_{max} \\
& \quad PF_{min} \leq PF \leq PF_{max} \\
& \quad \dots
\end{aligned}
\tag{Equation 3.1}$$

Where:

OG_i : The pre-defined optimization goal

λ_i : The weighting factor for different goals

x : The system variables, i.e., voltage magnitude and angle at each node

u : The control variable, i.e., LTC/regulator tap position or voltage setpoint, capacitor on/off status, or generator/DER VAR/PF output

$f()$: The power flow constraints

V, I, PF : The voltage/current/power factor operational limits

3.4.1. Cost Model

To solve the non-linear optimization problem described in Equation 3.1, a cost model-based optimization using Linear Programming (LP) is performed in the optimization LVM application. Most of the constraints are converted to corresponding costs in the final objective function. Equation 3.1 can then be converted to an unconstrained optimization problem, as described below:

$$\text{obj.} \quad \min \sum_{i=1}^N \lambda_i \cdot OG_i + \sum_{j=1}^M \mu_j \cdot CM_j
\tag{Equation 3.2}$$

Where:

CM : The cost model for the constraints

μ : The relative weighting factor (also called a scaling factor or scaling cost) for different cost models

Based on the three-phase unbalanced power flow model used in ADMS, different cost models are created for the different conditions.

The power flow constraints are hard constraints and they are strictly enforced. They are solved outside the LP engine. Based on the current value of the control variables, a DPF is performed to calculate the system state variables, and then the costs of other constraints can be calculated from these state variables.

The operational constraints are soft constraints; they can be violated if needed. When the constraints are met, the cost is zero; a linear penalty cost is incurred when the constraints are violated. The further the limit is violated, the larger the cost. For example, the graph in figure below indicates that when the nodal voltage is below the value on the low voltage limit constraint axis, the corresponding low voltage violation incurs a penalty cost. The lower the voltage below the voltage limit, the larger the cost. A similar cost curve can be created for high voltage violations.

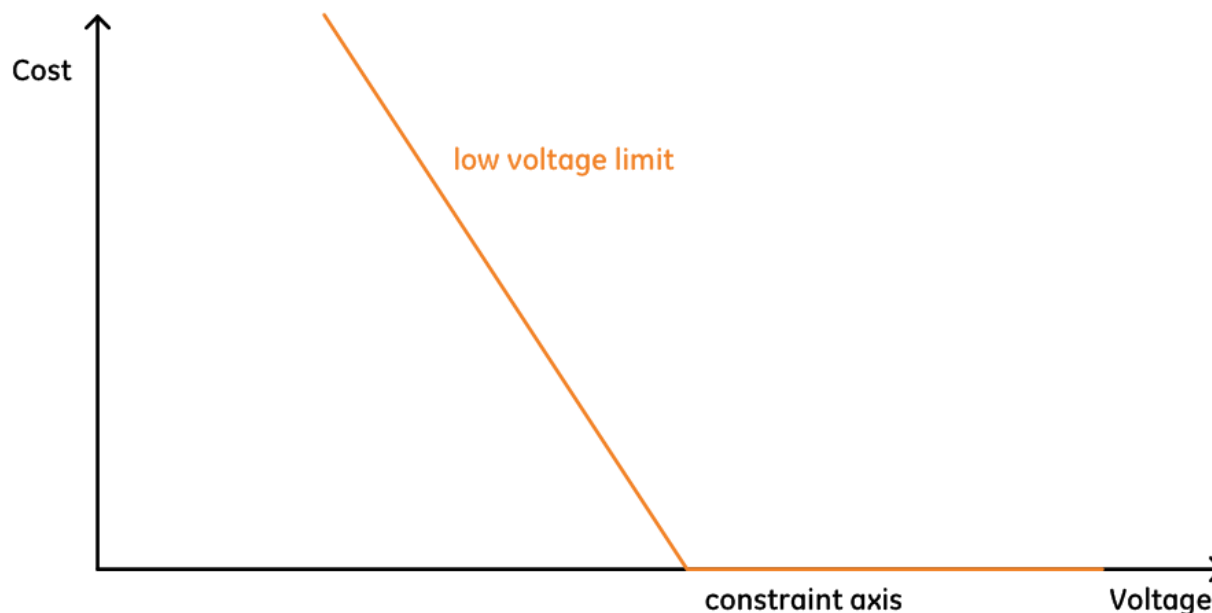


Figure 13. Linear Cost Model of Operational Constraints

A flow check point concept is introduced to handle power factor and demand control objectives. A flow check point is a general terminal flow summation of real or reactive power. It can be set at any point in the distribution network, or at any combination of points.

For example, to represent the power factor constraint on the high side of a station power transformer, a flow check point is created at that point. The active and reactive power can be calculated after DPF is solved, allowing the power factor to be determined.

Another example is loss minimization. To represent the loss in the system, a flow check point is created to include both ends of all lines, transformers, and series reactors. The sum of the flows on two ends of one line represents the loss on the line. Losses for transformers and series reactors are calculated similarly to line losses. After DPF is solved, the sum of flows of all terminals in the flow check point represents the loss in the system.

Each control move (tap move, capacitor switching, or generator/DER regulation) also incurs some predefined operation cost. Each control tries to stay in its original position, unless the contribution of the control move to the final objective is big enough to cover the move cost.

The operational cost of controls is represented by a "Vcurve". The cost is zero when the control stays at the original position, and then the cost appears when the control moves in either direction. The further the control moves, the larger the incurred cost. Figure below shows the Vcurve of one control.

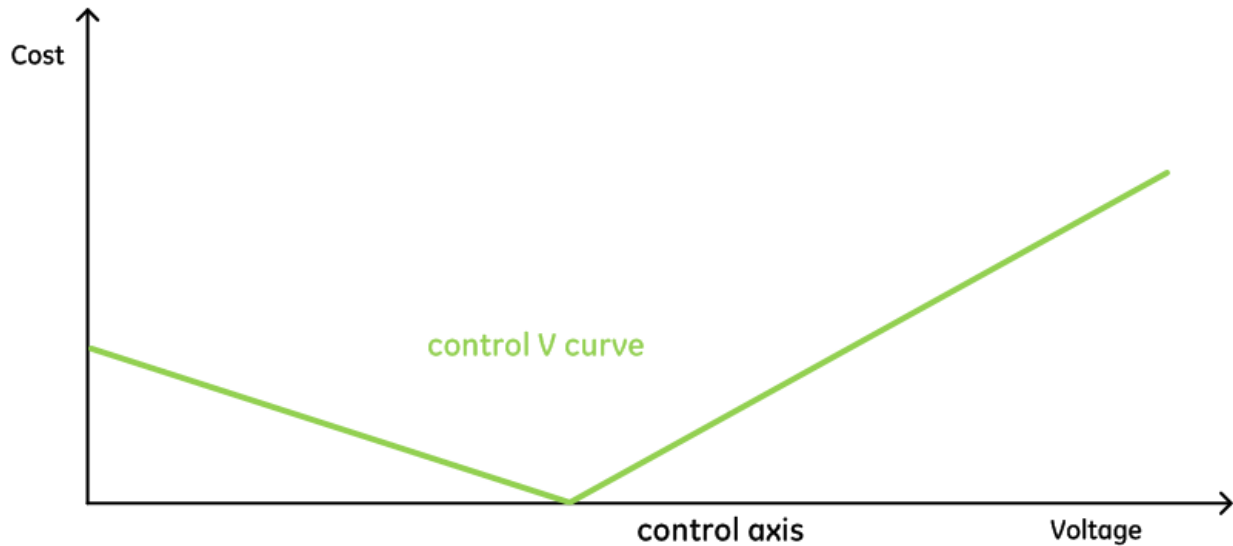


Figure 14. "V Curve" Cost Model of Control Movement

In cases where LTC/regulator, capacitor, and generator/DER control moves can achieve the same goal, the algorithm compares the integrated results of the benefit and the move cost of each control to determine the better move.

An objective function may contain different, competing goals. The competing goals are handled by the scaling factors for each objective. The scaling factors, λ and μ in Equation 3.2, can be adjusted using the DNAF Configuration Editor tool.

Different cost curves will be combined in the final model. For example, the combination of two cost curves, one for low voltage limit cost and the other for control movement cost, might look as shown in figure below:

- The green line represents control movement cost. The control cost is zero if the control does not move. Movement cost will be incurred by movement in either direction.
- The orange line represents the low voltage limit curve. If the voltage is above the low voltage limit, the cost is zero. A voltage lower than the low voltage limit will incur cost. The lower the voltage below the low voltage limit, the bigger the cost.
- The black line represents the combination of the two curves.

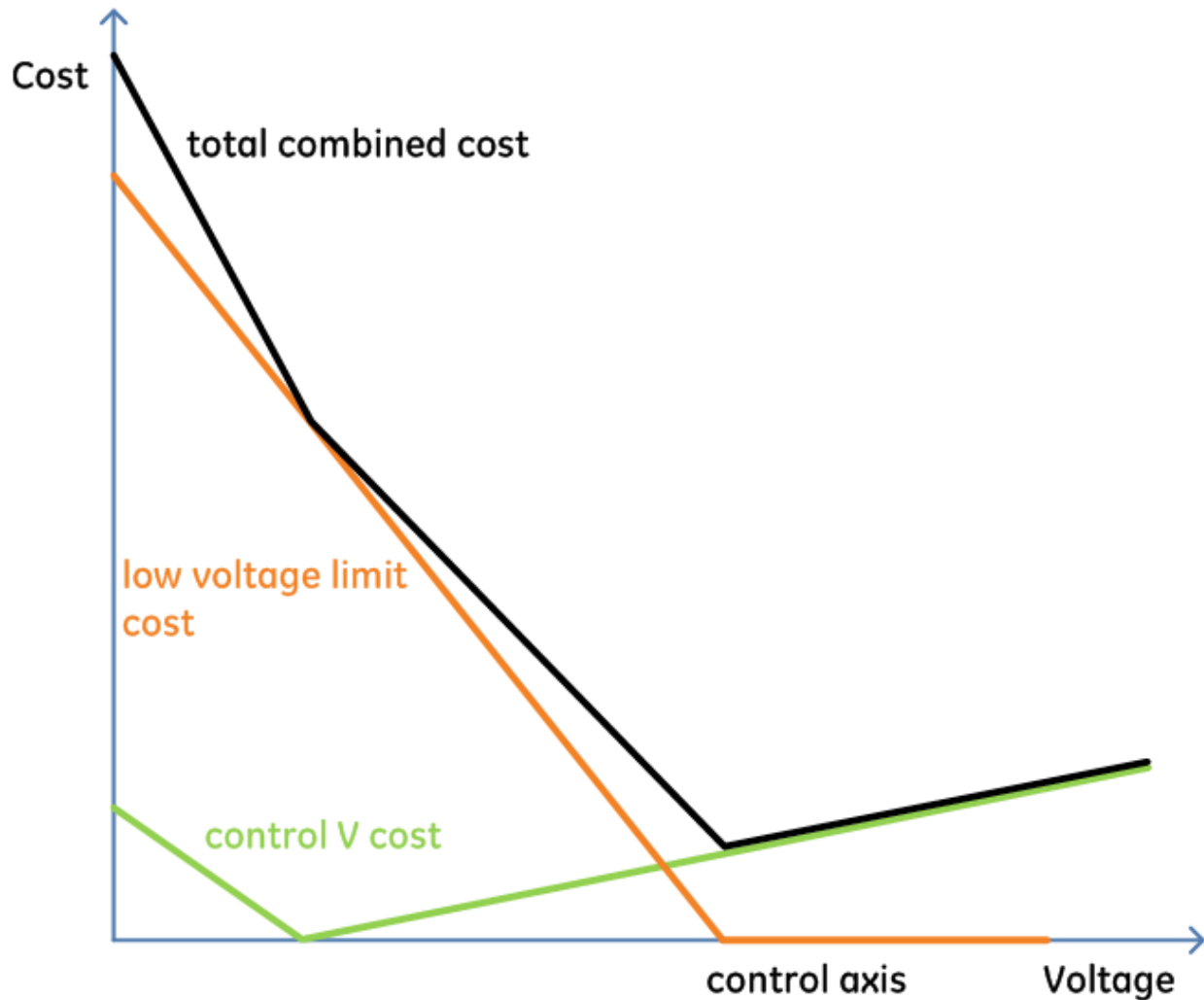


Figure 15. Combined Cost Model

3.4.2. Sensitivity Calculation

Creating the cost model changes the second part of Equation 3.2 to a linear combination of control and state variables. However, the first part of the equation, the optimization goal, is still a non-linear function of the control and state variables. To use the Linear Programming algorithm discussed in the next section, a sensitivity calculation is performed to convert the first part of the equation from a non-linear function to an approximate linear equivalent.

The sensitivities of control actions, with respect to the final objective, are calculated by chain rules. For example, the sensitivity of an active power change with respect to one regulator tap move can be calculated by the equation below:

$$\frac{\partial P}{\partial K} = \sum_{i=1}^N \frac{\partial P}{\partial U_i} \cdot \frac{\partial U_i}{\partial K} \quad (\text{Equation 3.3})$$

where:

P = active power

U_i = voltage at node i

K = tap move control

N = total number of nodes

The derivative dP/dU can be easily found in the corresponding element of the standard Jacobian matrix generated in the base case power flow calculation. To calculate the voltage change at node i with respect to the control move, the change in the admittance matrix when the tap is moved must be determined. The linear programming approximation is often only accurate enough around the continuous area of the base operation point; however, the tap position can only change discretely. To solve this problem, a fake small fraction of a tap move (such as a tap moved from position 3 to position 3.01) is introduced in the program to calculate the change in the admittance matrix; this ensures the accuracy of the sensitivity calculation. A similar assumption is used for capacitor controls and generator/DER controls.

3.4.3. Linear Programming

After the sensitivity calculation is finished, the sensitivity values, along with the generated cost model, are sent to the Linear Programming engine to solve the unconstrained linear functions described in Equation 3.2. Linear programming is a technique for optimization of linear objective functions with linear equality or inequality constraints. Because some of the unknown variables such as capacitor status and tap positions are required to be integers, the problem is also a Mixed Integer Linear Programming (MILP) problem. The MILP problem is generally NP-hard, which cannot be solved in polynomial time. However, with advanced algorithms such as the revised simplex method and the branch-and-bound method, the LP problem can be solved efficiently. After that, discretization efforts are made on the continuous LP solution to create a final discrete optimization solution.

3.4.4. Direct Discretization

The Linear Programming phase of LVM produces a plan with a set of continuous control values for the eligible control devices. For example, closing or opening a shunt capacitor will not be exactly 0 (open) or 1 (closed), but rather any possible continuous value around those integer values (for example, 0.4777). A plan with continuous control values must then be converted to a plan with discrete control values. For every control, the *direct discretization* process looks at the difference between the current control value and the new continuous value for the control produced by the LP solution. As long as this difference is greater than the threshold specified by the *ControlMoveTol* parameter in the *optalg.dat* file, the control will be executed. If the difference is less than the threshold, the control will remain unchanged. A different *ControlMoveTol* parameter exists for each type of control device except for generators/DERs that have continuous output. See *Optalg.dat* in the *Distribution Network Analysis Functions (DNAF) Standard Applications Analyst Guide* for more

information.

3.4.5. Limit Tolerance

One issue with this method is that rounding the continuous value to the appropriate discrete device position can cause monitored conditions to move just past the limits. For this reason, two user-configurable parameters named OverSatTol and FeasibleTol are defined in optalg.dat (see Optalg.dat in the *Distribution Network Analysis Functions (DNAF) Standard Applications Analyst Guide*) to adjust limit constraints internally before sending them to the LP solver.

- The OverSatTol is the buffer that is added to a limit, resulting in a revised range in the constraint region. For example, if the low voltage constraint is 0.95 p.u. and the OverSatTol is 0.005 p.u., then the internal target for the low voltage limit in the LP solver is 0.955 p.u. This gives the LP solver a small buffer to avoid an overshoot problem caused by discretization.
- The FeasibleTol is the amount a limit may be exceeded while still considering the constraint to be not violated. This parameter gives the LP solver more room if the user relaxes a constraint for other targets, such as demand change or reactive support. For example, if the low voltage constraint is 0.95 p.u. and the FeasibleTol is 0.005 p.u., then the plan is only considered to have a violation if the voltage is lower than 0.945 p.u.

In the above case, if the low voltage constraint is 0.95 p.u. and both OverSatTol and FeasibleTol are 0.005 p.u., the LP solver will try to create a solution with a 0.955 p.u. low voltage limit. After the solution is created, after discretization, some of voltage could be lower than 0.955 p.u. However, as long as the value is higher than 0.945 p.u., the solution is still marked as feasible without violation. This gives the solver much more flexibility to create a reasonable plan.

3.4.6. Voltage Limit Adjustment

In a real distribution system, there may be several voltage measurements along the feeder. In an ideal case, the voltages calculated from the distribution power flow should closely match the voltage measurements. However, due to an imperfect impedance model and an under-determined system, this rarely happens.

By setting the Limit Adjustment option for loads or buses to Regular or Enhanced on the LVM Problem Formulations page of the DNAF Configuration Editor, the voltage mismatch can be taken into consideration in LVM. This option adjusts the voltage limit rather than the voltage itself.

In figure below, the high/low voltage limit on the specified point can be adjusted as follows:

$$V_{newlimit} = V_{limit} + (V_{cal} - V_{meas}) \quad \text{(Equation 3.4)}$$

↘

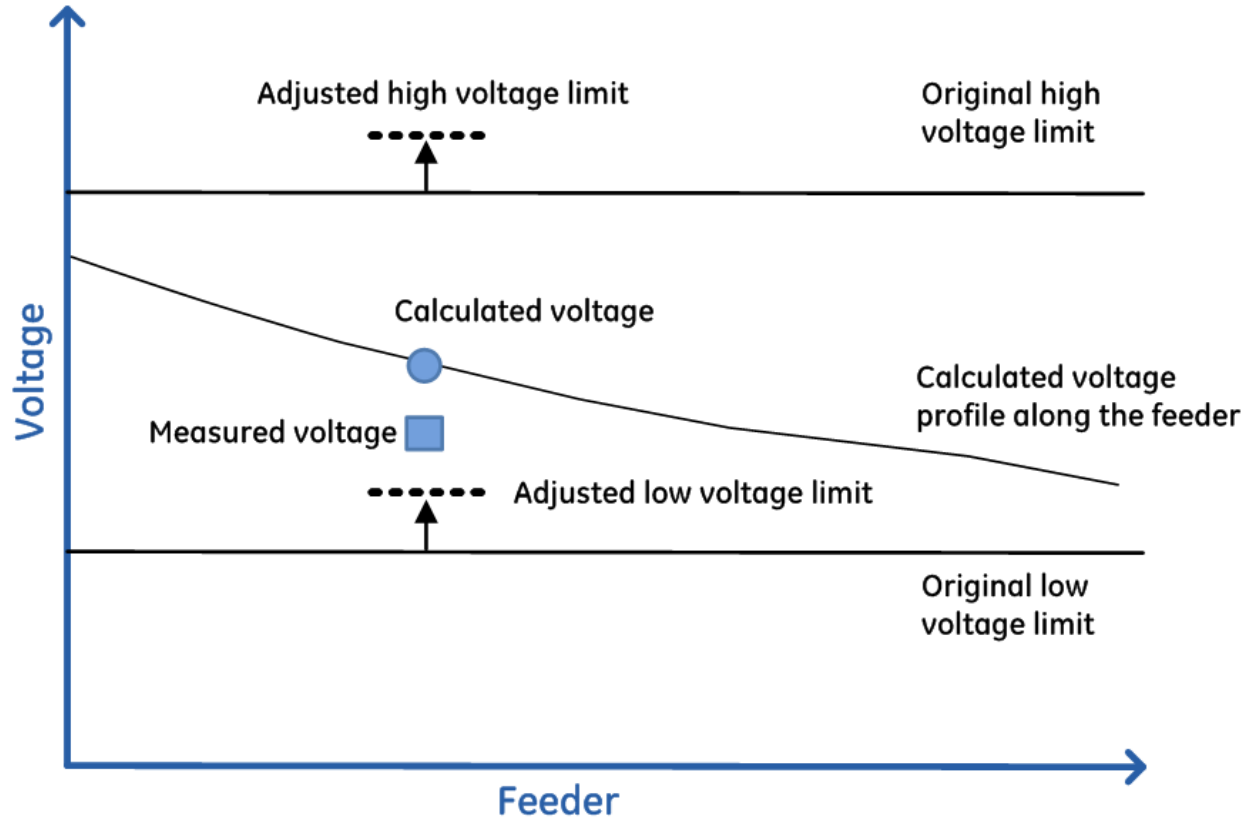


Figure 16. Voltage Limit Adjustment

An alternative voltage adjustment method is also available in the system. The enhanced voltage limit adjustment takes into account the highest and lowest voltage on the feeder.

The formula for this method is:

$$V_{lnew} = V_{lold} + (V_{cal} - V_{meas}) + (V_{cal} - V_l) = V_{lold} + 2 \cdot V_{cal} - V_{meas} - V_l \quad (\text{Equation 3.5})$$

$$V_{hnew} = V_{hoid} - (V_{meas} - V_{cal}) - (V_H - V_{cal}) = V_{hoid} + 2 \cdot V_{cal} - V_{meas} - V_H \quad (\text{Equation 3.6})$$

where V_{lnew} , V_{hnew} are the new low/high voltage limits, V_{lold} , V_{hoid} are the old low/high voltage limits, and V_l and V_H are the lowest/highest calculated voltages along the feeder.

In these equations, the first part is the adjustment due to the mismatch between the calculated and measured voltage, and the second part is the adjustment related to the calculated highest and lowest voltage on the feeder.

This method is represented in figure below.

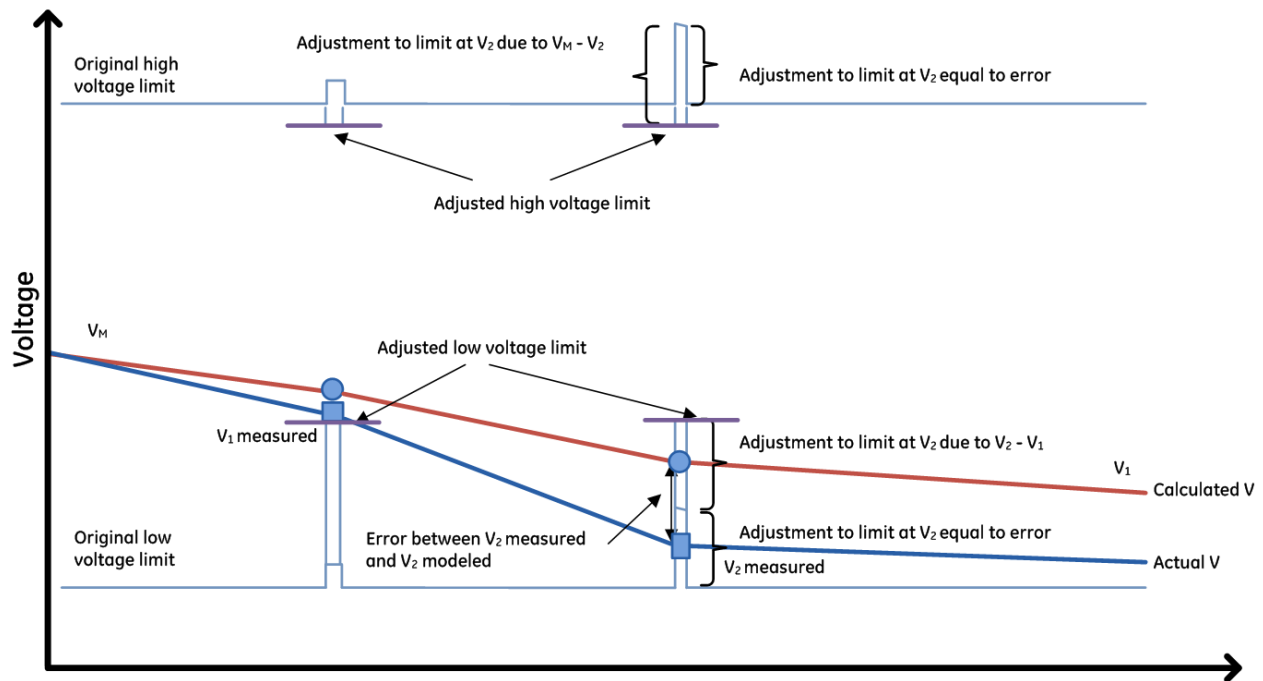


Figure 17. Enhanced Voltage Limit Adjustment

3.4.7. Regulator Group Voltage Imbalance

A "regulator group" is a pre-defined set of three single-phase voltage regulators connected in parallel. Regulator groups can be specified via the Substation Editor.

If regulator groups are not explicitly specified via the Substation Editor, the Converter attempts to detect paralleled single-phase regulators and then groups them into regulator groups. Each regulator group has pointers to the constituent single-phase regulators.

During operation, individual phase voltages on the regulating side of single-phase regulator groups may have significant imbalance. The optimization engine can take that imbalance into consideration and treat it as additional violation during LVM optimization. If the scaling penalty cost of voltage imbalance in the problem formulation defined in DCE is not zero, then the voltage imbalance constraint will be checked and a violation will be created if needed. If the scaling cost of voltage imbalance is zero in the problem formulation, then the voltage imbalance will not be considered in the LVM process and no corresponding violation will be created.

During optimization, for each regulator group, the voltage difference between phases on a regulated node is calculated. Then the calculated difference is compared to the maximum allowed mismatch to determine if violations exist. The maximum allowed mismatch is defined for each single-phase regulator in the MAXVOLTIMBALANCE field in the regulator object. For all regulators in a regulator group, the value specified in this field should be the same. If the values are different, the smallest value among the regulators in the regulator group is used.

3.4.8. Excluding Ineligible Components from LVM

When LVM runs, all control devices in the station are included in the analysis and all voltage/current limits on every component in the station are checked to determine solution feasibility.

However, in some cases ADMS cannot control devices on all of the feeders. It is better to ignore the control devices on some of the feeders where ADMS has no control. For this purpose, you can enable or disable LVM for a network element. For more information, see [Enable/Disable LVM](#).

3.4.9. Remote/Local Driver

The remote/local driver status measurement (the measurement type name is "RemoteLocalDriver") is an ADMS to Scada point for LVM capacitors and LTC/regulators. It is used for SCADA heartbeat functionality. ADMS sends Scada the calculated remote/local status of devices, which is determined based on the feeder LVM eligibility and the dynamic topology of the network.

The measurement status is determined as follows:

- For station-external capacitors and regulators, the point is calculated based on its dynamic energizing feeder LVM eligibility status. If the feeder is LVM-enabled and the execution mode is closed-loop, this status is set to Remote. Otherwise the status is set to Local.

The calculation is performed at server startup, topology processing, and feeder eligibility status changes, driven by a SCADA-measured Enabled/Disabled status or a user-toggled Enabled/Disabled status in the Real Time DNAF System Configuration Editor display.

- When the energizing feeder cannot be decided for station internal capacitors and regulators, DMS uses the LVM-Enabled status of its upstream power transformer to determine the value of the point. To effectively trace the upstream transformer, one internal device mapping relationship is built as part of the topology processing function. If the transformer is LVM enabled and the execution mode is closed-loop, the status is set to Remote; otherwise the status is set to Local.
- For parallel power transformers, DMS uses a conservative approach: if any transformer is LVM disabled, the device status is set to Local.
- If the remote local driver point is mapped to the station internal power transformer itself, the point is determined directly by the transformer LVM Enabled/Disabled status. When the status is Enabled and the execution mode is closed-loop, the point status is set to Remote.
- If LVM is disabled for a station (for example, due to auto-disabling checks or SCADA-Disabled measurement, or when manually disabled) the remote/local driver status measurement is set to Local.
- If the current problem formulation does not support the specified type of device (LTC/regulator or capacitor), the remote/local driver status measurement is set to Local.

3.4.10. Remote-Auto Capacitor Modeling

To inhibit LVM controls from interfering with a remotely controllable capacitor that has local control capability (remote auto device), optimization-based LVM uses the capacitor's local setpoints (On and Off levels) as constraints in the LP model. In contrast to other constraints in the model, the local setpoint constraints only constrain the control of the individual capacitor (or capacitor bank). Therefore, LVM still can issue controls that cause the capacitor's auto controller to act, but LVM does not issue a control to a remote-auto capacitor that is immediately reverted by the auto controller due to the move violating one of the local setpoints. For example, if a capacitor is disconnected, its "on level" is 0.95 p.u., its "off level" is 1.05 p.u., and its pre-plan voltage is 1.02 p.u., LVM will not issue a control to connect the capacitor if the post-plan voltage rises above 1.05 p.u., as that would cause the auto controller to re-disconnect the capacitor.

The local setpoints are displayed in the LVM result file as violations depending on the type of auto control. (Note that PF local setpoints are converted into VAR limits, similar to how station PF limits are handled.) To tighten the local setpoint constraints in the LVM engine, use the Auto Setpoint Slack parameter on the LVM Problem Formulations page in the DNAF Configuration Editor, which acts like other slack variables and tightens the range of the "On/Off levels" around the mean. For example, if the "on level" is 0.95 p.u. and the "off level" is 1.05 p.u., then the mean is 1.0 p.u. and the setpoint range is 0.1 p.u. Therefore, if the local setpoint slack variable is 5%, the new setpoint range would be $0.95 \times 0.1 \text{ p.u.} = 0.095 \text{ p.u.}$, giving an "on level" constraint of $1.0V - (0.095V/2) = 0.9525 \text{ p.u.}$ and an "off level" constraint of $1.0V + (0.095V/2) = 1.0475 \text{ p.u.}$ The greater this slack variable is, the more constrained the remote-auto capacitor's control will be and the lower the risk of an improper control of the local capacitor.

3.4.11. Voltage Constraint Management

The voltage high or low limit constraints at buses and loads incorporated into the Linear Programming model (see [Linear Programming](#)) are included on a regulation zone basis. The most violated voltage high or low network constraints ranked from each regulation zone are included, instead of including the most violated voltage high or low network constraints ranked in the entire network selection. This process is described below.

Definition: A regulation zone is defined as a contiguous portion of the network selection bounded by regulating transformers or by open tie switches with no other regulating transformers within the contiguous network area.

A tracing function is used to determine the regulation zones. Consider the following example:

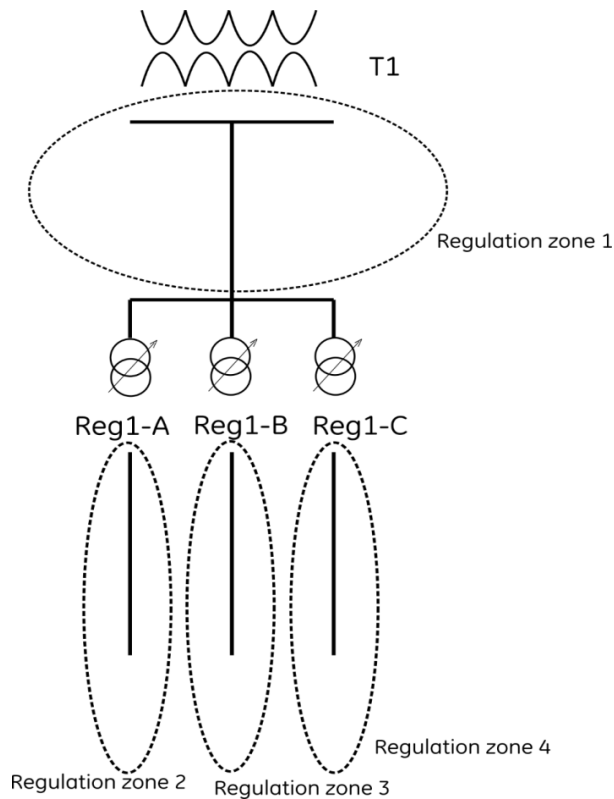


Figure 18. Regulation Zones Example 1

In this example, tracing starts from the load-side node of the station power transformer T1 and stops at the high-side nodes of the single-phase regulators Reg1-A, Reg1-B, and Reg1-C. The contiguous electrical network bounded by the load-side node of T1 and the high-side nodes of Reg1-A, Reg1-B, and Reg1-C is identified as one regulation zone. Then, tracing is resumed from the load-side nodes of each of the single-phase regulators until the tracing algorithm finds another regulator or an open tie switch. Each of the contiguous single-phase network sections bounded by the load-side nodes of Reg1-A, Reg1-B, or Reg1-C and a downstream regulator or an open tie switch is identified as a regulation zone. In this example, there are four regulation zones.

Consider another example:

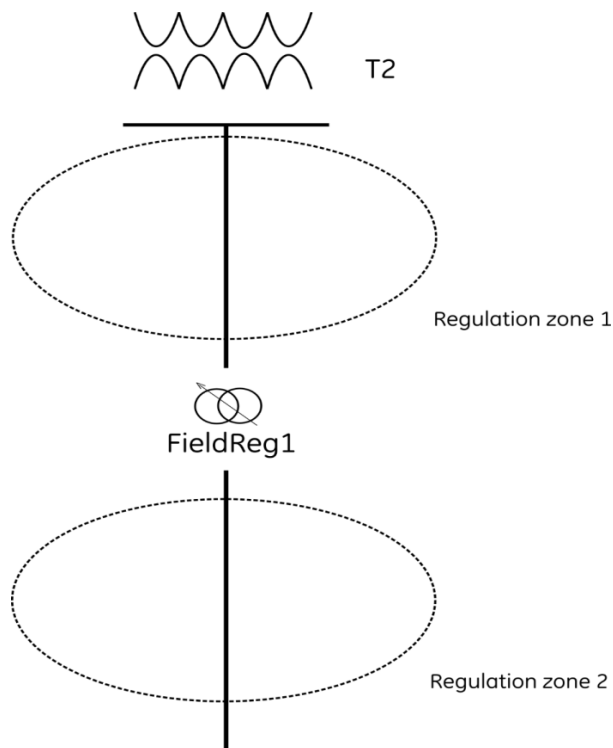


Figure 19. Regulation Zones Example 2

In this example, tracing starts from the load-side node of the station power transformer T2 and stops at the high-side node of the three-phase regulator FieldReg1. The contiguous electrical network bounded by the load-side node of T2 and the high-side node of FieldReg1 is identified as one regulation zone. Then, tracing is resumed from the load-side node of the three-phase regulator FieldReg1 until the tracing algorithm finds another regulator or an open tie switch. The contiguous network section bounded by the load-side node of the three-phase regulator FieldReg1 and a downstream regulator or an open tie switch is identified as another regulation zone. In this example, there are two regulation zones.

Optimization LVM works by iterating power flow with the LP engine. A base case power flow solution is obtained. Violations are evaluated and sensitivities of the violations with respect to optimization controls are updated. An LP model is formulated around this operating point. The LP model is solved to obtain the incremental change in the positions of the LVM controls such as tap-changers, capacitors, and generator reactive power and power factor setpoint output. Any incremental changes obtained from the solution to the LP model are applied to the internal three-phase unbalanced power system model. A power flow solution is obtained for the new operating point. Violations are evaluated and sensitivities are updated, and then the next LP model is formulated around the new operating point. This iterative procedure continues until the latest LP model has no more control moves to suggest, or the maximum number of power flow ~ LP model solution iterations has been reached.

For each power flow ~ LP model solution iteration, only the most violated constraints are included in the LP model. The number of additional most violated constraints to be included in the LP model formulated for that iteration is governed by the MaxExtraConstraints parameter in the optalg.dat file:

- For real-time LVM, this file is located in the %IDMS_ROOT%\models\DNAFParams folder.
- For study-mode LVM, the optalg.dat file from the %IDMS_ROOT%\models\StudyServer folder is used.

In this context, the formulated LP model constraints arising from the selected objectives in the LVM problem formulation are divided into two categories:

1. Network voltage high or low limit constraints: these arise from enforcing the voltage high or low limits at buses and at loads. For this category of constraints, the maximum additional top-ranking violations from each regulation zone that will be included in each LP model is given by $(1 + 5 * \text{MaxExtraConstraints} / \text{Number of Regulation Zones})$.
2. All other constraints: these arise from the other LVM objectives such as minimize demand, reactive power support, and minimize losses; and the enforced capacitor local set point control limits. For this category of constraints, the maximum additional top-ranking violations that will be included in each LP model is MaxExtraConstraints. Also, this category of constraints is not considered part of any regulation zone.

3.5. Optimization LVM Plan Enhancement and Selection

The sequence of Linear Programming (see [Linear Programming](#)) followed by the Direct Discretization process ([Direct Discretization](#)) can lead to voltage violations not anticipated in the linear programming solution. As a result, some generated LVM plans may lead to voltage violations if implemented.

There are two other options for finding the best LVM plan:

- Option 1: LVM Plan Enhancement
- Option 2: Discrete Move Improvement

These options can be selected independently. If both options are enabled, Option1 is executed first. However, in this case, if the LVM Plan Enhancement process can find a plan without violation, the Discrete Move Improvement (DMI) process will not be executed.

In addition, the Plan Selection function analyzes pre-plan and post-plan voltage and power factor violations and decides whether to accept or reject a plan. The plan selection process is independent of the plan enhancement processes mentioned earlier.

3.5.1. Plan Enhancement

The LVM Plan Enhancement option makes three attempts to find the best discrete solution that does not lead to voltage violations.

1. If at least one load or bus voltage (whether the load voltages or bus voltages are being monitored depends on the Limit Strategy settings in the DNAF configuration file) is expected to be outside the limits because of the LVM plan, pass 1 is implemented where all capacitor

statuses are fixed at the status taken from the Direct Discretization process, and LVM is executed again.

The logic will then verify whether all load and/or bus voltages are within their limits in the new plan. If all voltages are within their limits, LVM is ready for the next phase, LVM plan validation.

2. If all load and/or bus voltages are not within their limits, pass 2 is implemented where the LTC/regulator tap is moved up or down one position on the transformer connected to the feeder with the voltage violation. This tap position is fixed and LVM is executed one more time.

The logic will then verify whether all load and/or bus voltages are within their limits in the new plan. If all voltages are within their limits, LVM is ready for the next phase, LVM plan validation.

If regulators are present on the feeder that has a voltage problem, the regulator tap will be moved. If at least one feeder with voltage problems has no regulator, only the transformer LTC/regulator will be moved.

Before pass 2, if the tap cannot move further due to either a physical limit (such as a regulator/LTC at the lower or upper maximum tap position) or due to the limit specified by the "Maximum Number of Tap Moves Per Execution" parameter in the DNAF Configuration Editor, pass 2 is skipped.

3. If not all load and/or bus voltages are within their limits, pass 3 is implemented where the margin is adjusted by a user-specified value named "Voltage Margin Adjustment Amount" in the DNAF Configuration Editor. That value is added on top of the existing threshold, OverSatTol. (See figure below).

The purpose of pass 3 is to force LVM to look for solutions further away from the limits. The same additional margin is applied to both V_{low} (increased) and V_{high} (decreased). LVM is executed one more time.

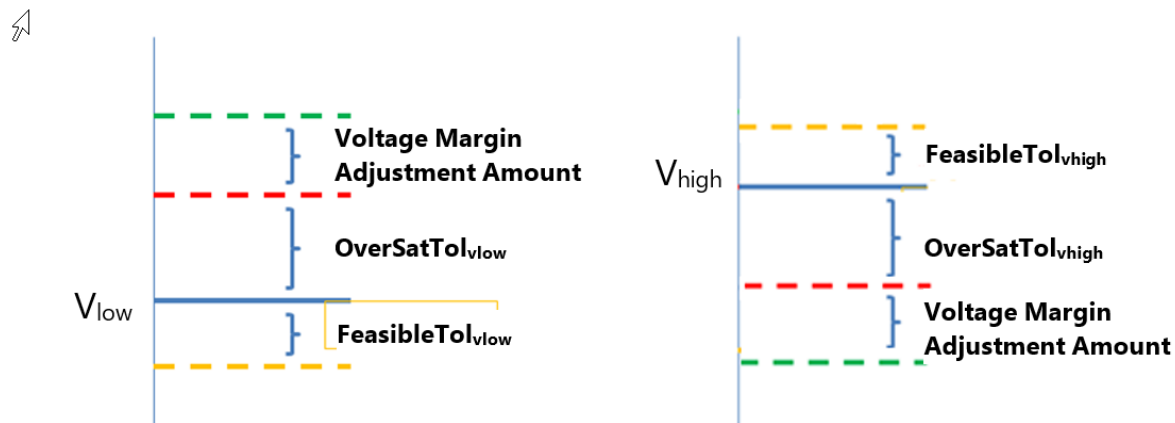


Figure 20. Discrete Margin Adjustment

Plan Enhancement Flowchart

The plan enhancement process is described in figure below. LVM generates an initial continuous

plan and then sends a single plan to the Direct Discretization. If the post-plan voltages are not inside limits, the three methods (fix capacitors, adjust tap positions, and increase voltage margin) are executed in sequence if needed. If the plan still has post-plan voltage violation, the DMI process is executed if the DMI option is enabled.

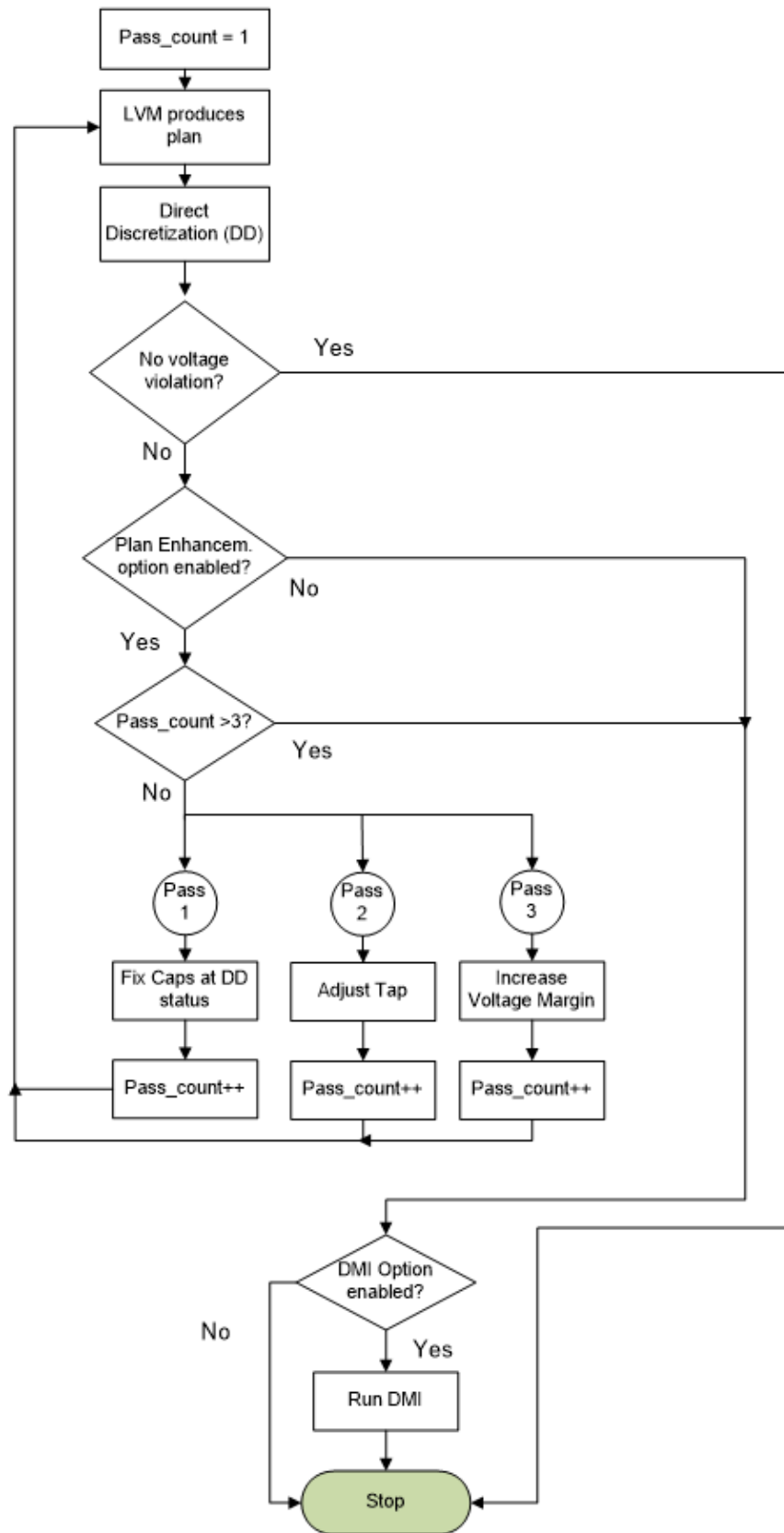


Figure 21. LVM Plan Enhancement Flowchart

3.5.2. Discrete Move Improvement

If the three passes in the Plan Enhancement option did not produce a plan with no expected voltage violations, or if the LVM Plan Enhancement option was not used, another option to improve the plan is the Discrete Move Improvement (DMI) method. To enable this option, set the `ImproveDiscreteMove` parameter in the `optalg.dat` file to 1.

Suppose there are N control devices. This procedure starts from the Direct Discretization solution if the LVM Plan Enhancement option is disabled or the three passes in the Plan Enhancement option did not produce a plan with no expected voltage violations. It performs N total Performance Index (PI) evaluations by reversing one control at a time. After comparing the total PIs, the control that delivers the worst PI is set to the original setting from the initial starting point in the final plan. Then the same process is executed on the other N-1 controls and one more control will be fixed for the final plan. This process continues until all controls are set to their final status.

3.5.3. Plan Selection

The Plan Selection option decides whether the plan is accepted (and therefore implemented if closed-loop mode is enabled) or rejected. The plan selection process is independent of the Plan Enhancement and Discrete Movement Improvement processes described earlier. It can be enabled/disabled using the Perform Plan Selection option on the LVM display in the DCE.

The plan selection is based on a truth table where it is decided, based on several pre-plan and post-plan binary performance indicators, whether the plan should be accepted. The selection process uses the concept of deadbands to prevent oscillation (one LVM plan is undoing the previously implemented LVM plan). It also prevents implementing a plan that results in minimal improvement.

Table 1. Binary Performance Indicators for LVM Plan Selection

Binary Indicator	Definition
Pre-Plan Voltage Violations	One or more voltage violations exist before the LVM plan is implemented.
Post-Plan Voltage Violations	One or more voltage violations exist after the LVM plan is implemented.
Pre-Plan Power Factor Violations	One or more power factor values are outside of limits before the LVM plan is implemented.
Post-Plan Power Factor Violations	One or more power factor values are outside of limits after the LVM plan is implemented.
Voltage Improved	The voltage performance index values have improved (by more than the required deadband) after implementing the LVM plan.
Demand Improved	Demand has been reduced after implementing the LVM plan.

The voltage performance indexes used to determine whether voltage has improved are determined by the following equations:

$$PI_{preplan} = \frac{\sum_{bus=1}^N (V_{bus}^{initial})}{N} \quad (\text{Equation 3.7})$$

$$PI_{postplan} = \frac{\sum_{bus=1}^N (V_{bus}^{final})}{N} \quad (\text{Equation 3.8})$$

where N is the number of voltage points. The points are selected according to the initial voltage conditions. If there are no voltage violations, all voltage points are selected and PI is the average of voltage magnitude on buses. Otherwise, the voltage points with violation (up to the number specified by the Number of Top Voltage Violations Used in Voltage PI Calculation option in the DCE) are selected. In this case, PI is the average of the voltage violation on the selected buses.

If there are no voltage violations before and after an LVM plan is generated, the voltage profile is considered to be improved when the post-plan average voltage is lower than the pre-plan average voltage. If there are voltage violations before and after an LVM plan is generated, the voltage profile is considered to be improved when the post-plan average violation is less than the pre-plan average violation.

The voltage profile is also considered to be improved when there is one or more voltage violation before an LVM plan is generated but no voltage violation after an LVM plan is generated. In this case, the PI calculation detail does not play a role for plan selection algorithm according to the truth tables referenced below.

If $PI_{preplan} - PI_{postplan} > AV_{deadband}$, the voltage profile in the system has improved enough for plan acceptance.

where $AV_{deadband}$ is the voltage change deadband (in per unit V) specified in the DNAF configuration file. The user can specify a different value for the system with or without initial voltage violation.

The flowchart of the LVM Plan Selection function shown in figure below is based on the truth table presented in [LVM Plan Selection Truth Tables](#).

The following are additional comments about the flowchart in figure below:

- If the initial system voltages are all within limits, but the system has at least one violation after the LVM plan execution, the plan will be discarded.
- If the initial system has at least one voltage outside the limits, and the system has all load and bus voltages within the limits after the LVM plan execution, the plan will be accepted, no matter how small the average voltage decrease. Restoring voltage within limits is valued over the deadband.
- Before the plan is finally accepted, the power factors at configured flow check points (power transformers and/or feeder heads, depending on the configuration) are verified against user-specified limits. This verification is only performed when both pre-plan and post-plan voltages are all within limits. If any power factors are outside the limits, the plan is rejected. If there are no power factor limits specified, this verification step is skipped.

- With the "Minimize Demand" problem formulation, the plan will be accepted only if the demand reduction is more than the specified demand deadband and all pre-plan and post-plan voltage levels are within the limits.
- All other problem formulation will be treated the same as the reduce violations problem formulation, as described in [LVM Plan Selection Truth Tables](#).

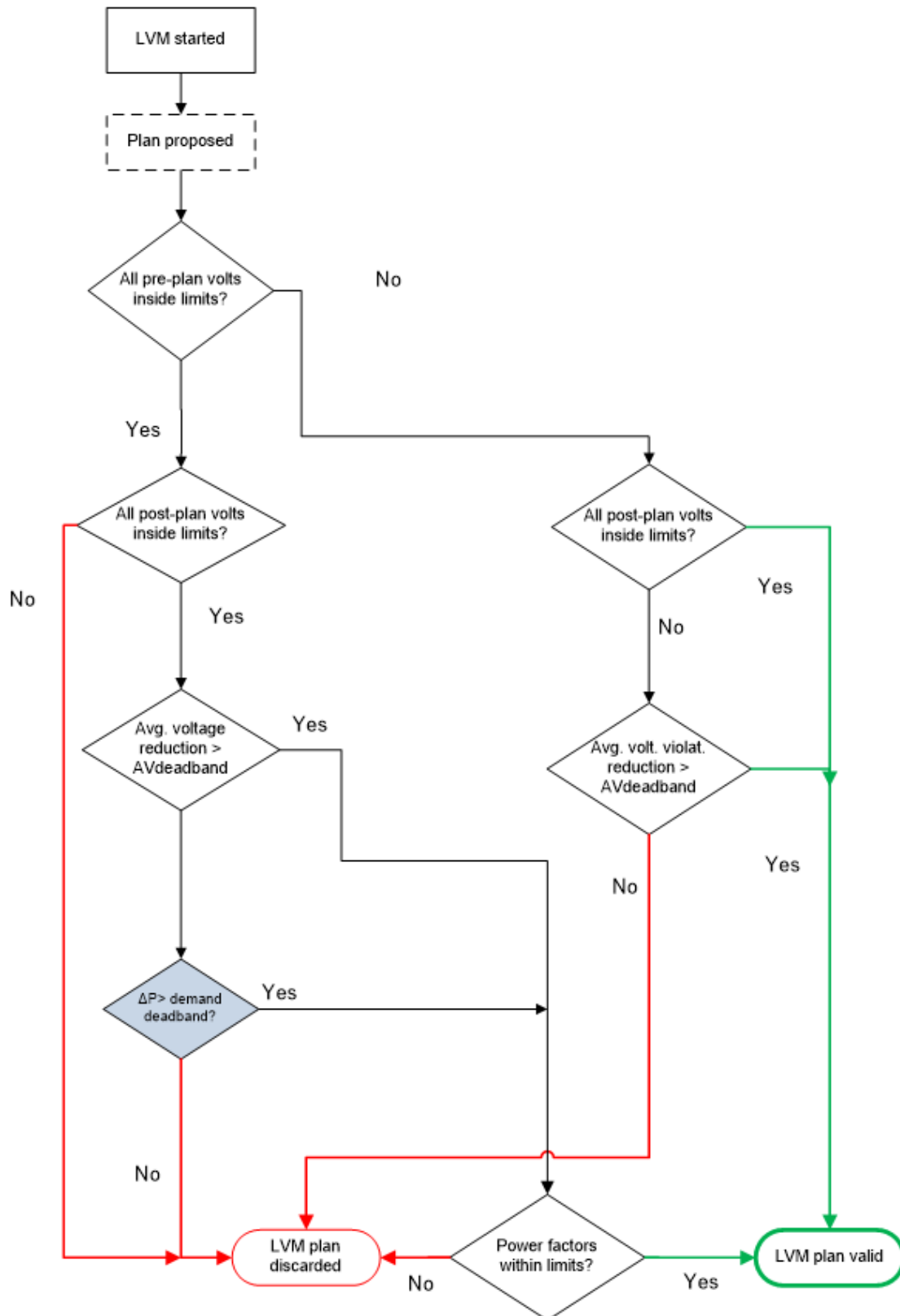


Figure 22. LVM Plan Selection Flowchart

The LVM application records key information for the LVM Plan Enhancement process in the LVM Status section of the LVM results file. The summary section of the results file lists the conditions that the Plan Enhancement logic uses to determine whether to accept the plan. These Yes/No conditions are as follows:

- Pre-Plan Voltage Violations
- Post-Plan Voltage Violations
- Pre-Plan Power Factor Violations
- Post-Plan Power Factor Violations
- Voltage Improved
- Demand Improved

An analyst can determine the plan rejection reason based on the value of the binary indicators.

3.5.4. Plan Enhancement and Selection Configuration

There are three options to enable/disable the plan enhancement and/or plan selection process:

- **Enable LVM Plan Enhancement (1):** This option on the LVM page in DCE is enabled by default.
- **ImproveDiscreteMove (2):** This parameter in the `optalg.dat` file enables/disables the Discrete Move Improvement (DMI) option. It is disabled by default.
- **Perform Plan Selection (3):** This option on the LVM page in DCE is enabled by default.

The following table shows plan selection stages based on enabling (=1) or disabling (=0) the three options:

Parameter Options			Plan Selection Stages	Comment
1	2	3		
0	0	0	• Direct Discretization • Final Plan	
0	0	1	• Direct Discretization • Plan Selection • Final Plan	
0	1	0	• Direct Discretization • Discrete Move Improvement (DMI) • Final Plan	

Parameter Options				
0	1	1	• Direct Discretization • Discrete Move Improvement (DMI) • Plan Selection • Final Plan	
1	0	0	• Direct Discretization • Plan Enhancement • Final Plan	
1	0	1	• Direct Discretization • Plan Enhancement • Plan Selection • Final Plan	This is the default configuration.
1	1	0	• Direct Discretization • Plan Enhancement • Discrete Move Improvement (DMI) • Final Plan	DMI executes only if plan enhancement fails to find a plan without voltage violation. If the plan enhancement can find a good plan, DMI is skipped even if it is enabled.
1	1	1	• Direct Discretization • Plan Enhancement • Discrete Move Improvement (DMI) • Plan Selection • Final Plan	DMI executes only if plan enhancement fails to find a plan without voltage violation. If the plan enhancement can find a good plan, DMI is skipped even if it is enabled.

3.6. Single-Step Mode

The output of the optimization LVM application is typically a multi-step plan that contains one or more control moves. However, due to field communications or control requirements, a plan with only one move is sometimes preferred. To this end, LVM can run in single-step mode to guarantee that at most one control per station appears in the final plan. This is done via post-processing of a multi-step plan.

Starting from the multiple-step plan generated by the LP optimization, each individual move is investigated. A set of single-step plans is generated; each plan contains only one individual move in the original multiple-step plan. The performance index is then computed for each single-step plan. The plan with the best (minimum) performance index is selected and presented as the final answer.

A single step will be selected among the individual steps that do not result in voltage violations if implemented. If all individual steps lead to one or more voltage violations, LVM will pick the step with the lowest PI value.

Figure below shows a simplified two-control space. In this figure, the original two-step plan contains two steps (control1 and control2) that reduce the Performance Index (PI) from 110 to 104. Two single-step plans are generated, one with control1, whose PI is 107, and another with control2, whose PI is 105. In this case, the latter plan with only control2 is selected as the final output in single-step mode.

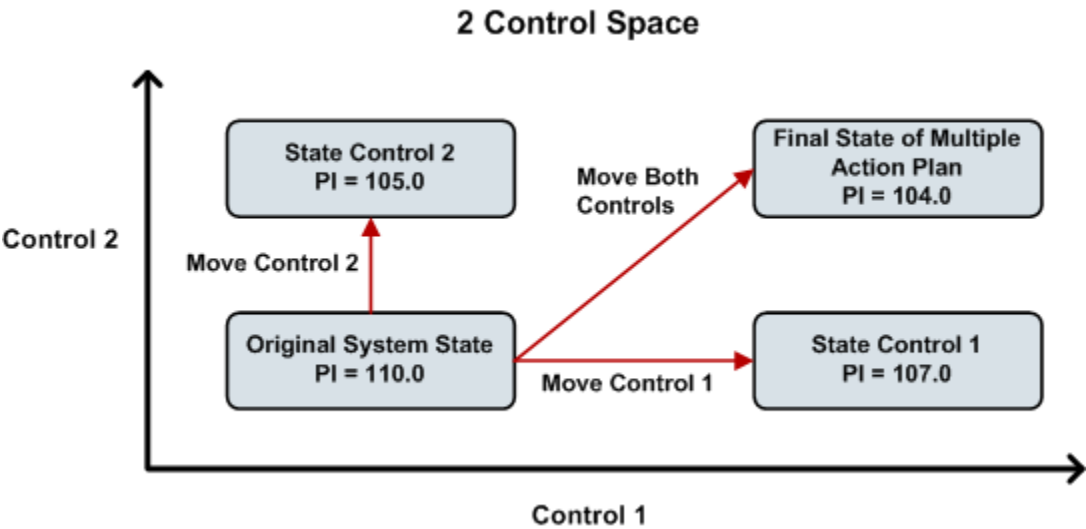


Figure 23. Single-Step Mode in LVM

The Plan Creation Method setting on the LVM page indicates the control modes, single step mode and multiple step mode. These are listed in table below.

Table 2. Plan Creation Methods

Plan Creation Method	Explanation
Multiple controls	The control plan may contain multiple moves
One control per station	The control plan contains at most one move in the whole station

3.7. Heuristic LVM Methodology

The optimization-based LVM (see [Optimization LVM Methodology](#) and [Optimization LVM Plan Enhancement and Selection](#)) solves a non-linear problem using linear programming, where many of the problem constraints are converted to a cost in an objective function. The power flow constraints are hard constraints that are strictly enforced. Based on the current value of the control

variables, a three-phase unbalanced power flow is performed to calculate the system state variables the costs of other constraints are then calculated using these state variables. The operational constraints are soft constraints. They can be violated if needed. When the constraints are met, the cost is zero. When the constraints are violated, a linear penalty cost is incurred. A higher degree of violation results in a larger penalty.

To obtain high quality results, a good power flow model is required. The model must have an accurate impedance model and a solid load model. In some cases, obtaining an accurate power flow model capable of generating high quality LVM results is not a simple task. Therefore, a heuristic-based LVM solution has been developed.

This approach depends less on the quality of the power flow model, although it still uses the primary network impedance model and the dynamic DMS network connectivity model. The impedance model, along with capacitor size, allows the calculation of the change in voltage (ΔV) and the change in Power Factor (ΔPF) in response to opening or closing capacitors. This approach also uses SCADA switch status and analog measurements such as voltages and real and reactive power flows in determining the appropriate control actions.

3.7.1. Mode of Operation

The heuristic LVM algorithm has two separate functions that work together:

- **Capacitor Control Function:** This function determines the capacitor to be switched based on the measured voltage and power factor values. The measured values can be control device locations or other monitoring points along the distribution circuits. The goal of this function is to keep the telemetered voltage profile relatively flat while also respecting power factor limits. Two weighting factors are used to determine the relative priority given to maintaining power factor targets vs. flattening the voltage.
- **LTC/Regulator Control Function:** This function determines the correct tap position based on the selected objective to lower the voltage level along a feeder without violating applicable voltage limits. If the LTC/regulator only supports voltage setpoint control, the tap position movement will be translated to the corresponding voltage setpoint value in the final plan.

The heuristic LVM algorithm first attempts to find and execute the one capacitor move that minimizes the objective. The expected objective is to obtain the lowest average voltage, lower voltages, and maintain the power factor target (depending on the selected objective function and the associated weights).

After all feasible capacitor moves are found and made in a series of LVM executions, the LTC/Regulator Control function is implemented. Similar to the Capacitor Control function, the LTC/Regulator Control function looks for only one device to operate.

Before executing the Capacitor Control function or the LTC/Regulator Control function, voltage violations are checked first. If bus voltage violations exist on LVM eligible feeders, then Heuristic LVM attempts to correct the voltage violation. If the voltage violation can be corrected, then an appropriate capacitor move or a tap-changer move is generated.

If no tap-changers or capacitors are available to correct a voltage violation, due to zero sensitivity, that voltage measurement is ignored. For example, a voltage violation could be downstream of a station power transformer that does not have any downstream LVM eligible tap-changers or capacitors, while all LVM eligible tap-changers or capacitors are downstream of other station power transformers. Clearly, the available tap-changers or capacitors cannot correct this voltage violation. Heuristic LVM ignores such voltage violations that cannot be addressed due to lack of controls with non-zero sensitivity.

Figure below shows the overall Heuristic LVM Algorithm control flow. The following sections describe the Voltage Monitoring, Capacitor Control, and LTC/Regulator Control functions.

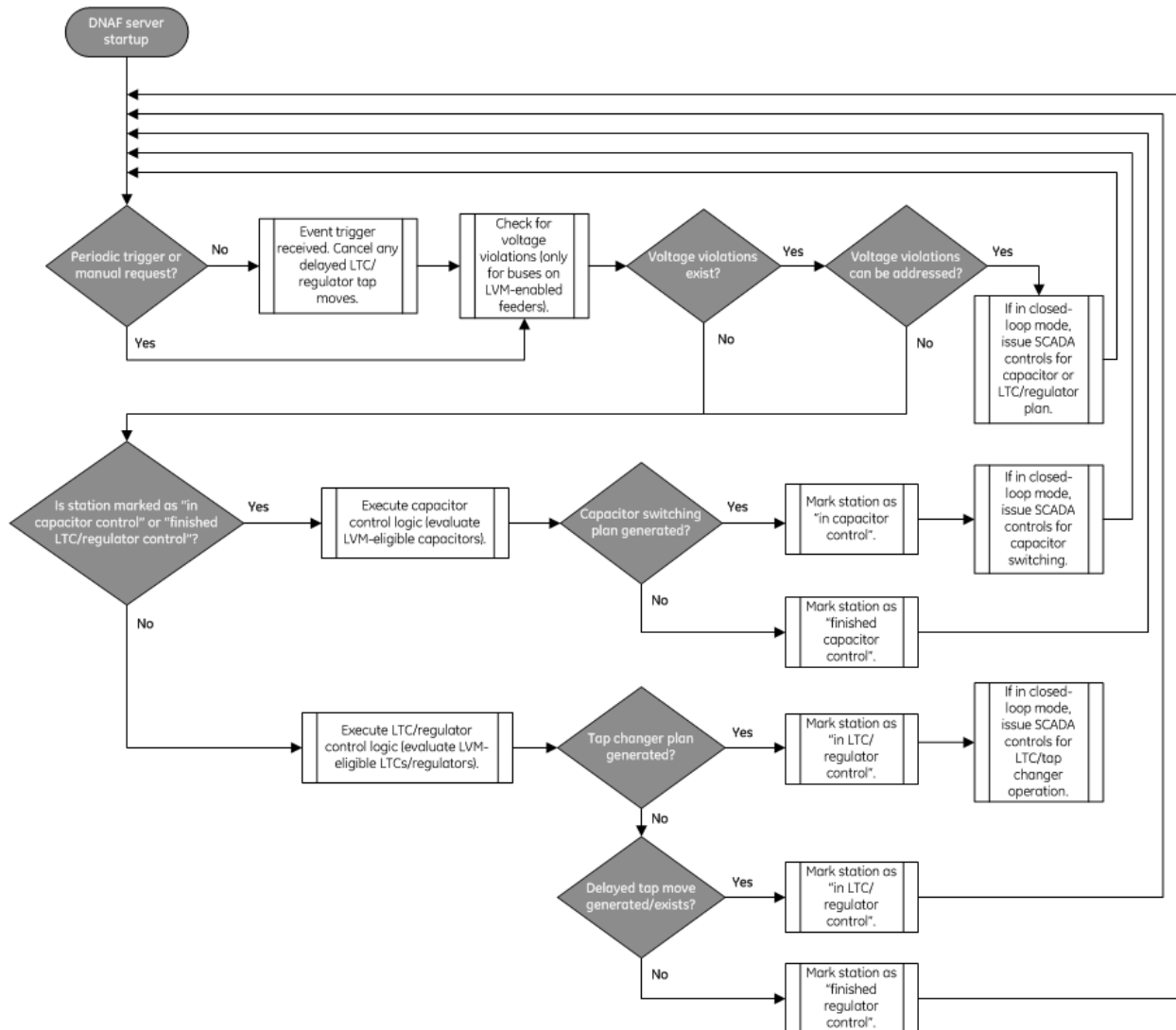


Figure 24. Overall Heuristic LVM Control Flow

3.7.2. Voltage Monitoring Function

This section describes the Voltage Monitoring Function and then discusses how the "best control

device" is identified to alleviate a high voltage or a low voltage condition.

3.7.2.1. General Principle

The Voltage Monitoring Function is performed at the beginning of each Heuristic LVM execution, regardless of whether the station in question is currently in capacitor control or in LTC/regulator control mode. This function checks if there are any high or low voltage violations on LVM-enabled feeders. If both high and low voltage violations exist, then high voltage violations are given higher priority. This is shown in the flowchart in figure below.

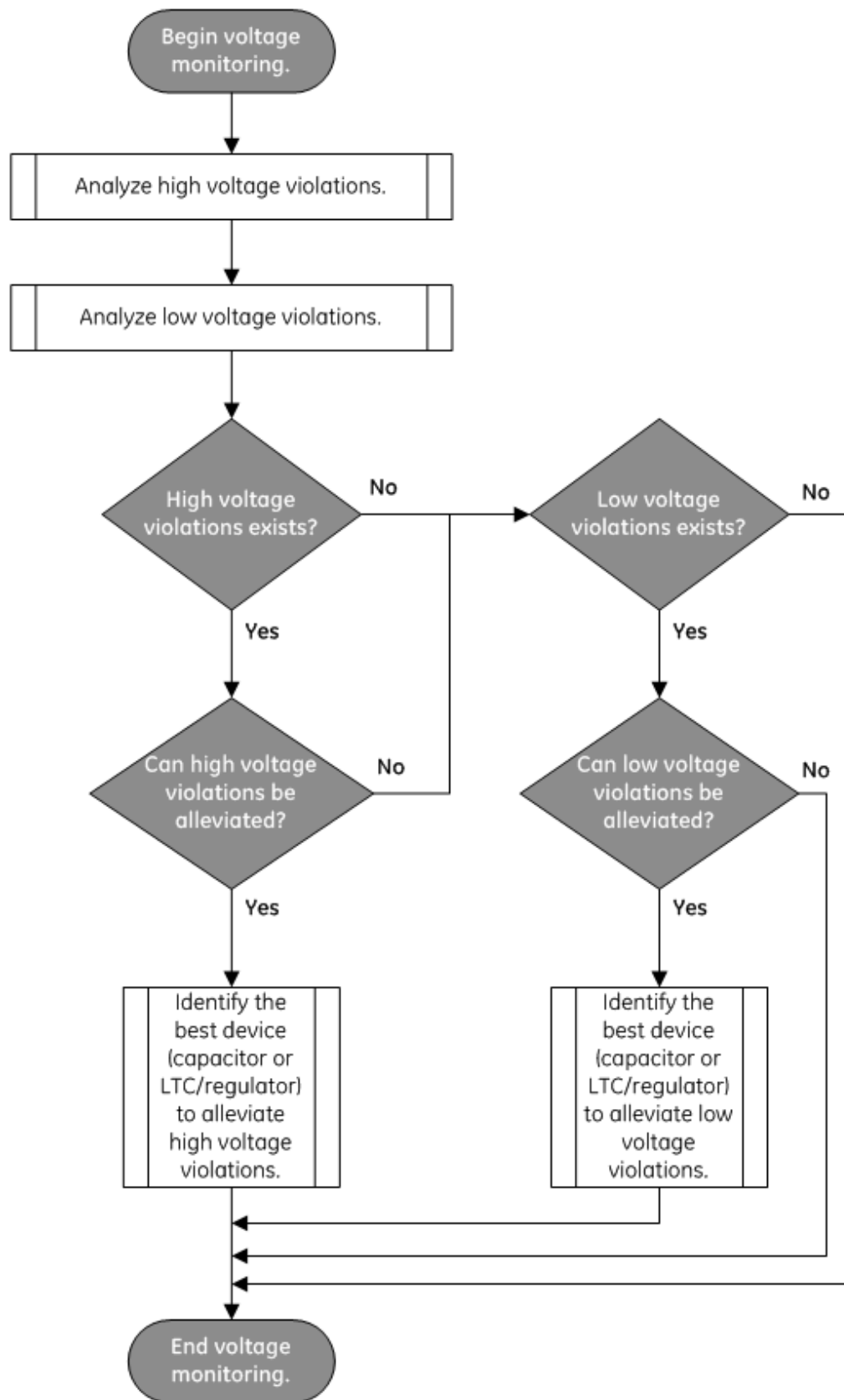


Figure 25. Voltage Monitoring Function

First, all telemetered bus voltages on LVM-enabled feeders are analyzed. In this analysis, the bus voltage measurements are checked against their corresponding limits. The excursions above the high voltage limit or below the low voltage limit are aggregated separately. In the end, the total relief required for high voltage violations and the total relief required for low voltage violations are determined. Note that during the aggregation process, the excursions are normalized over the available phases at any telemetered location. This is done to avoid undue weightage to locations that report measured voltages for all three phases versus locations that report measured voltages for only one phase.

Once the aggregation process is over, a search is performed for the "best control device" (a capacitor or an LTC/regulator) that can alleviate the violation.

High voltage violations are given higher priority. If a suitable control device can be found for alleviating a high voltage violation condition, then a control move (a capacitor switching plan or a lower tap plan) is generated. If a high voltage violation condition does not exist, or if it exists, but a suitable control device could not be found for addressing the high voltage violation condition, then the voltage monitoring function looks at alleviating a low voltage violation condition.

If a low voltage violation condition exists, a search is performed to identify the best control device that can alleviate the low voltage violation condition. If a suitable device is found, then a control move (a capacitor switching plan or a raise tap plan) is generated. If a suitable device cannot be found, Heuristic LVM resumes either the capacitor control or the LTC/regulator function.

3.7.2.2. Identifying the Best Control

To alleviate a high voltage condition or a low voltage condition, the "best control device" is identified. Generally, the best control is the capacitor or LTC/regulator that has the highest sensitivity in the direction of reducing the voltage violations. For a high voltage violation, the sensitivity should be negative (i.e., lower voltages), while for a low voltage violation, the sensitivity should be positive (i.e., raise voltages). However, in Heuristic LVM, depending on whether we need to relieve a high voltage or a low voltage violation condition, the best control device may not be the one with the highest sensitivity. This is described below.

- **High Voltage Violation:** For each available control device, the total relief provided by the control device towards high voltage violations is determined, excluding those voltage violations towards which the control device has negligible effect (less than 0.01V on a 120V base). For a capacitor control, it means switching the capacitor. For a tap-changing transformer, it means lowering the tap. From the total relief provided, the average relief provided by the control device is obtained. If the average relief is negative, then the control device can alleviate the high voltage violation condition. In this case, the device with the highest negative average relief, if it exists, is identified as the device for generating a plan.
- **Low Voltage Violation:** The goal of heuristic LVM is to maintain a flat voltage profile as close to the lower voltage limit as possible without violating the lower voltage limit. Therefore, identifying the best control device for alleviating a low voltage violation condition is somewhat different than the process for a high voltage violation condition. In this case, it is not simply a matter of identifying the device with the highest positive average relief, if it exists, and then

generating a plan with this device. The goal is to identify a device, with a positive average relief, that can correct the low voltage violation condition as close to the low voltage limit as possible. This is best explained with the help of a few examples.

Suppose that one voltage measurement is below its lower limit. The measured voltage is 113V, and the voltage low limit is 114V. The relief required in this case is 1V.

- Case 1: There are two control devices, A and B. Device A, providing a relief of 0.9V, can bring the voltage up to 113.9V (still below the low voltage limit). Device B, providing a relief of only 0.5V, can bring the voltage up to 113.5V. In this case, neither device can completely address the low voltage violation condition. The most helpful device in this case is device A, as it brings the node with the low voltage as close to the low voltage limit as possible.
- Case 2: Suppose, in addition to the two devices in case 1, there is another device C, providing a relief of 1.5V, which can bring the voltage up to 114.5V. In this case, device C is the best device as it can completely address the low voltage violation condition.
- Case 3: Suppose, in addition to devices A, B, and C described above, there is a fourth device D, which provides a relief of 2V. Clearly, this device has a higher sensitivity than device C. However, device D will not be selected for generating a corrective plan, because device D would take the voltage to 115V, thus overcorrecting the low voltage violation condition. Device C would be selected for plan generation.

To summarize, in a low voltage violation, the best control device is the one that provides the maximum relief without overcorrecting the low voltage violation condition. In a high voltage violation, the best control device is the one that provides the maximum relief.

3.7.3. Capacitor Control Function

The Capacitor Control function monitors the voltages at all available primary voltage monitoring points to determine the amount of voltage flattening. The lowest average voltage profile is the one where the difference between the average of all measured voltage locations and the lowest measured voltage is the smallest. If a location has three voltage measurements (one per phase), the average of these voltages is used to evaluate the average circuit voltage. If only a single phase voltage is known, that value is assumed for the other two phases. The flowchart for the Capacitor Control Function is shown in figure below.

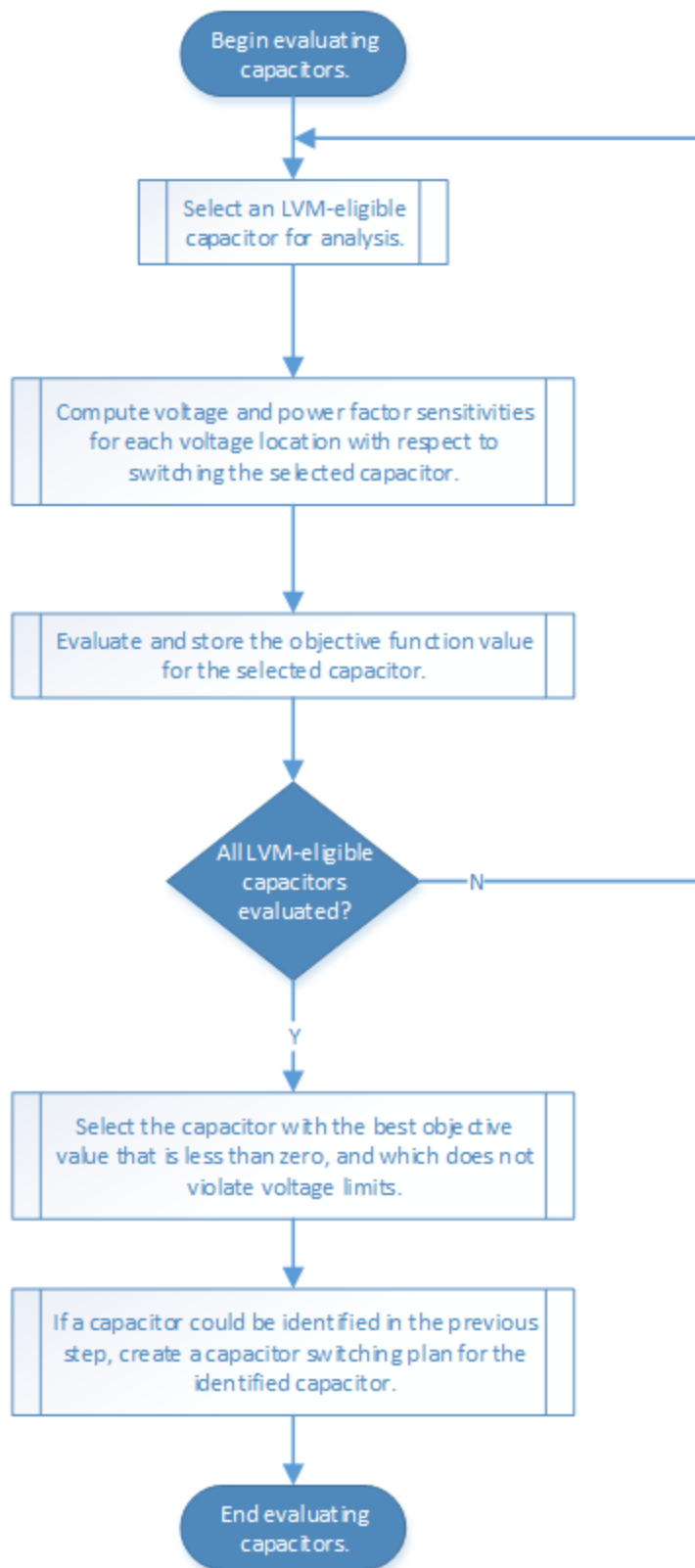


Figure 26. Capacitor Control Function

3.7.3.1. Voltage and Power Factor Change Calculation

For each capacitor on each circuit, the change in voltage (ΔV) is calculated for each monitored location, forming a table of values. Based on the size of the capacitor, the voltage at the capacitor, and the three-phase power factor calculated from flow measurements at the load side of the station transformer, the change in Power Factor (ΔPF) at the load side transformer bus can be determined for each capacitor.

The ΔV calculation uses the existing GE conductor series impedance matrix, in-line transformer impedance data (for step transformers and regulators), and the capacitor size (per phase VAR injection) to estimate the voltage change at each monitored voltage location. Because the VAR injection at a capacitor changes proportionally to the square of the bus voltage, the voltage change is adjusted based on the measured voltage.

3.7.3.2. Algorithm

The Capacitor Control function looks at flattening the voltage profile, reducing voltages, and maintaining the power factor at target levels. The power factor target is at either the source or the load side of the station transformer. The source side power factor is calculated based on transformer impedance values and flow measurements on the load side of the transformer. The objective for the capacitor control function is given below.

Minimize:

$$Obj_{Total} = Obj_{VoltFlattening} + Obj_{VoltReduction} + Obj_{PowerFactor} \quad (\text{Equation 3.9})$$

The components of the total objective are described below.

$Obj_{VoltFlattening}$ = Voltage Flattening Component, which is the sum of voltage flattening components of feeders in the substation where LVM can be performed.

$$Obj_{VoltFlattening} = \sum VF_{Feederj} \quad (\text{Equation 3.10})$$

For any feeder:

$$\sum VF_{Feederj} = u_1(x) * u_2(x) * W_{LV1} * \exp \{W_{LV2} * (|V_{avecalc} - V_{lowcalc}| - V_{flatthreshold})\} \quad (\text{Equation 3.11})$$

$$u_1(x) = \begin{cases} 1 & \text{if } abs(V_{avg} - V_{avecalc}) > 0.01 \\ 0 & \text{if } abs(V_{avg} - V_{avecalc}) \leq 0.01 \end{cases} \quad (\text{Equation 3.12})$$

$$u_2(x) = \begin{cases} -1 & \text{if } (V_{avg} - V_{low}) > (V_{avecalc} - V_{lowcalc}) \\ 1 & \text{if } (V_{avg} - V_{low}) \leq (V_{avecalc} - V_{lowcalc}) \end{cases} \quad (\text{Equation 3.13})$$

W_{LV1} and W_{LV2} are the exponential weighting factors for the lowest average voltage.

$V_{avecalc}$ is the average of the estimated voltages as a result of switching a particular capacitor. The average is calculated over all telemetered voltage locations on the feeder, in addition to the voltage measurements on the load side of the corresponding station power transformer.

$V_{lowcalc}$ is the lowest of the estimated voltages as a result of switching a particular capacitor. The lowest voltage is determined across all telemetered voltage (each phase) locations on the feeder, in addition to the voltage measurements on the load side of the corresponding station power transformer.

V_{avg} is the average measured voltage. The average is calculated over all telemetered voltage locations on the feeder, in addition to the voltage measurements on the load side of the corresponding station power transformer. For any telemetered location, if a measurement is available on only one phase, the same measurement value is assumed for the other phases.

V_{low} is the lowest measured voltage. It is determined across all telemetered voltage (each phase) locations on the feeder, in addition to the voltage measurements on the load side of the corresponding station power transformer.

$V_{flatthreshold}$ is the voltage flattening threshold, expressed in the same units as the other voltage terms in the formula.

$Obj_{VoltReduction}$ = Voltage Reduction Component, which is the sum of voltage reduction components of feeders in the substation where LVM can be performed.

$$Obj_{VoltReduction} = \sum VR_{Feederj} \quad \text{(Equation 3.14)}$$

For any feeder:

$$VR_{Feeder} = W_{VR} * ave(DeltaV) \quad \text{(Equation 3.15)}$$

W_{VR} is the Voltage Reduction weight.

$ave(DeltaV)$ is the average change in voltage as a result of switching a particular capacitor. The average is calculated over all telemetered voltage locations on the feeder, in addition to the voltage measurements on the load side of the corresponding station power transformer.

$Obj_{PowerFactor}$ = Power Factor Component, which is the sum of power factor components of all power factor control targets belonging to LVM-enabled LTCs and feeders in the substation.

$$Obj_{PowerFactor} = \sum PF_{Nodej} \quad \text{(Equation 3.16)}$$

For any power factor control target:

$$PF_{Nodej} = \{(1 - d(PF_{TargetMax})) * u(PF_{Actual} - PF_{TargetMax}) - (1 - d(PF_{TargetMin})) * u(PF_{TargetMin} - PF_{Actual})\} * (DeltaPF) * W_{PF1} * exp[abs(Delta PF) * W_{PF2}] \quad \text{(Equation 3.17)}$$

$PF_{TargetMax}$ = Maximum power factor target. Note that a leading power factor is shown as a negative

number.

$PF_{\text{TargetMin}}$ = Minimum power factor target. Note that a leading power factor is shown as a negative number.

$d(PF_{\text{TargetMax}})$ = 1 if maximum power factor target is specified as 0 (indicating power factor target maximum is not enforced); otherwise, it is 0.

$d(PF_{\text{TargetMin}})$ = 1 if minimum power factor target is specified as 0 (indicating power factor target minimum is not enforced); otherwise, it is 0.

$u(PF_{\text{Actual}} - PF_{\text{TargetMax}})$ = 1 if the measured power factor exceeds the maximum power factor target; otherwise, it is 0.

$u(PF_{\text{TargetMin}} - PF_{\text{Actual}})$ = 1 if the measured power factor is less than the minimum power factor target; otherwise, it is 0.

PF_{Actual} = Measured power factor at the power factor control target.

DeltaPF = Change in power factor as a result of capacitor switching.

W_{PF1} , W_{PF2} = Exponential weights for power factor control.

The capacitor with the lowest total objective, such that the total objective is negative, is selected for generating a capacitor switching plan. If all capacitors have non-negative total objectives, then no capacitor is selected for a switching plan and the capacitor control function ends.

3.7.4. LTC/Regulator Control Function

The LTC/regulator control function is invoked after the capacitor control function has finished determining the capacitor positions so that overall voltages along feeders are lowered without violating the low voltage limit. This function operates as follows.

1. If all measured voltages on feeders downstream of the LTC/regulator are below the voltage high limit and are higher than the lower voltage limit by the regulation margin parameter (0.75 Volts per step on a 120.0 Volt base + margin such as 0.50 Volts), lower the tap one step.
2. Repeat the above steps until no additional changes are required. This means the tap cannot be raised or lowered any further based on the voltage limits and the regulation margin.

For tap-changing devices where the taps are three-phase, lowering the tap one step would lower the tap across all phases simultaneously. For parallel single-phase regulators, each phase regulator is evaluated independently of the other phases. Lowering one step would lower the tap on the phase controlled by the regulator.

To avoid a large voltage imbalance, a voltage imbalance limit is enforced for parallel single-phase regulators. If the estimated voltage imbalance is higher than the voltage imbalance limit, a lower tap move for the regulator is not generated.

Tap-changing devices available for control are evaluated in the following order: first station LTCs or regulators are analyzed, then downstream or field regulators are analyzed. The flowchart for the

LTC/regulator function is shown in figure below.

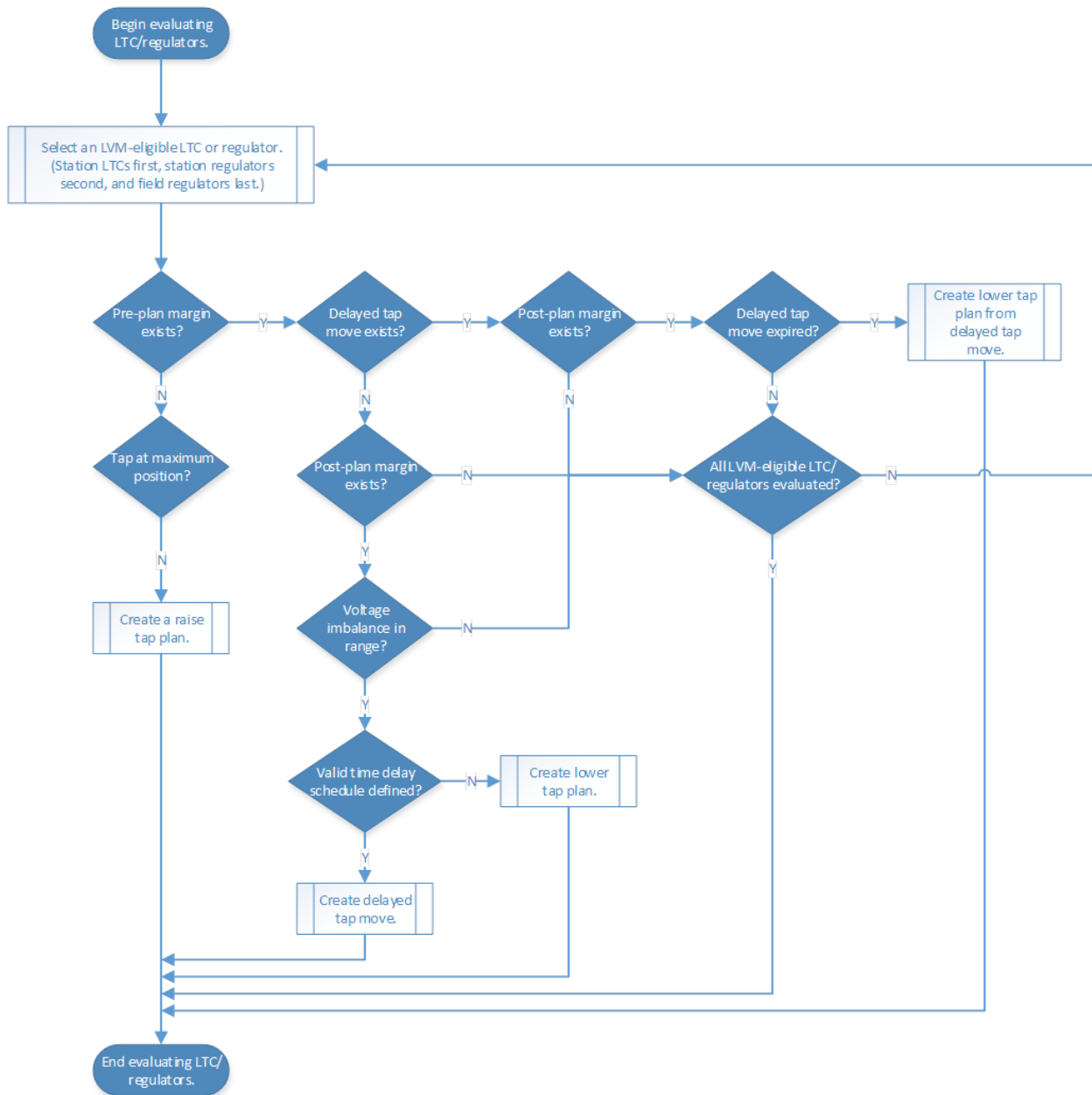


Figure 27. LTC/Regulator Control Function

3.7.4.1. Time Delay Schedule

To prevent excessive tap movements, an additional (optional) time delay is used. For example, if the measured voltages are just above the minimum voltage (with regulation margin and one tap step), the time delay is longer (multiple scan periods). If the measured voltages are well above the minimum voltage (with regulation margin and one tap step), the time delay is shorter. There is no time delay for the voltage correction steps in [LTC/Regulator Control Function](#).

The time delay is implemented using the existing DMS schedule functionality. To specify a time delay schedule for a regulator or an LTC, the time delay schedule name should be added to the applicable devices in the station CSV or XML file. The time delay schedule is optional: if it is not specified for a tap-changing device, then a tap move generated for that device is meant for immediate execution.

In the time delay schedule, the minimum voltage difference value (in percent) is specified on the x-axis, and the time delay (in seconds) is specified on the y-axis. The minimum voltage difference value is the smallest voltage difference between a measured voltage and the minimum voltage limit (incorporating the regulation margin, such as 0.50 Volts on a 120.0 Volt base and one tap step) for any telemetered voltage point downstream of the LTC/regulator.

The DMS schedule functionality allows any type of time delay curve (inverse or not) to be defined by simply adjusting the schedule point values for the time delay schedule. This approach is flexible and provides a configurable approach in determining when the LTC/regulator raise/lower tap steps should be executed.

3.7.4.2. Delayed Tap Moves

Depending on the time delay parameters described in the previous section, the LTC/Regulator Control Function may require several minutes before the LTC/regulator tap change move can be executed in the field. If an LTC/regulator move is generated with a time delay, it is initially marked as "Delayed". If the time delay is greater than the fast or regular periodic LVM interval, then in the subsequent LVM execution, measurements are checked again to determine if the delayed tap move is still valid.

The LVM application tracks the proposed tap move. If the topology has not changed and loading has not changed significantly (i.e., event trigger or voltage quality trigger has not been received), the tap move will be executed based on the time delay parameters. If an event trigger or a voltage quality trigger is received while there is a delayed tap move (tap move proposed in an earlier execution of LVM but not yet executed based on the time delay parameters), the delayed regulator move is cancelled and an alarm is issued. While a delayed tap move is valid, the Capacitor Control function is not used.

3.7.5. Triggering

The Heuristic LVM application has the following triggers:

- **Event trigger:** This includes any topology changes including the application of temporary modifications (cuts and jumpers). These changes can be enabled separately for the feeder main and the laterals.
- **Voltage quality trigger:** This includes changes in the measured voltages (inside the station or outside the station). This trigger also has a minimum time delay between consecutive voltage quality triggers. For example, if the delay is set to five minutes and a previous voltage quality triggered execution was completed four minutes ago, the new voltage quality trigger is

ignored.

- **Periodic trigger:** The following Heuristic LVM periodic fast trigger times (in minutes) are defined in the DNAF server configuration file:
 - **DNAF_HLVM_CAP_FAST_TRIGGER_PERIOD:** The waiting time between two consecutive capacitor control functions.
 - **DNAF_HLVM_CAP_TAP_TRIGGER_PERIOD:** The waiting time between the end of a capacitor control function and the start of the LTC/regulator control function.
 - **DNAF_HLVM_TAP_FAST_TRIGGER_PERIOD:** The waiting time between two consecutive LTC/regulator control functions.

In this context, the other influential DNAF server parameters are:

- **DNAF_LVM_FAST_TRIGGER:** If this parameter is set to 1, then the above-mentioned fast-trigger times are used. If this parameter is set to 0, then the regular LVM periodic timer (DNAF_LVM_MAX_PERIOD) is used.
- **DNAF_LVM_MAX_CONSECUTIVE_FAST_TRIGGERS:** If fast-triggering is enabled, and the number of consecutive fast triggers exceeds this value before reaching the end of a capacitor control and LTC/regulator control cycle, then fast-triggering is disabled. A server message is logged indicating the same. The subsequent triggers until the end of the cycle follow the regular LVM periodic timer (DNAF_LVM_MAX_PERIOD). At the end of the cycle, the number of consecutive fast triggers is reset to zero. A server message is logged indicating the same. The next cycle beginning with the capacitor control function will follow the fast trigger timers.

To avoid excessive LVM solutions, periodic triggering can be skipped for stations where LVM was recently solved as part of another station's network selection. The time period (in seconds) within which periodic application triggering for a station is skipped is configured by the DNAF_PERIODIC_DEADBAND parameter in the DNAF server configuration file. The time period counts after an application is solved for a station or after an application is triggered for an adjacent station in the network selection. This feature applies to manual or event-triggered solutions.

3.7.6. Plan Execution

The Heuristic LVM application first executes the capacitor control function followed by the LTC/regulator function. If no capacitor moves are required, taps are still considered. In other words, even if no capacitor moves were generated in the capacitor control function, potential tap-changer moves are evaluated in the ensuing LTC/regulator function.

3.8. Fixed Regulation LVM Methodology

The Fixed Regulation LVM approach can be used to perform emergency voltage reduction by changing the effective voltage regulation target of the LTCs/regulators. This LVM approach can also

be used as a failsafe approach if the power flow results are not good, likely resulting in a low-quality optimization-based LVM solution.

Most LTCs/regulators support one or more voltage reduction modes, which when active, lower the effective band-center by a pre-defined amount (such as 3%) and bypass the time-delay. The voltage reduction amount is typically a static setting, but it can also be measured via SCADA.

For a Fixed Regulation LVM problem formulation, a fixed voltage reduction value (in %) from the effective or nominal regulator/LTC voltage setting is specified in the DNAF Configuration Editor. The voltage reduction LVM solution is essentially the topology ordering: Plan Manager issues controls to field devices in a topology-based order.

A Fixed Regulation LVM problem formulation is implemented by issuing the appropriate Voltage Reduction setpoint to the most upstream set of regulators/LTCs (typically, inside the station) and then proceeding to issue setpoints to the next set of downstream (field) regulators based on the latest dynamic connectivity. The confirmation of the control actions is implemented based on the status and measurement validation options set in the LVM Operation Modes display in the DNAF Configuration Editor, where each problem formulation points to one operation mode.

When a Fixed Regulation LVM problem formulation is to be solved by LVM, estimates of the expected demand change (kW and kVAR) are approximated knowing the fixed voltage reduction amount and the load to voltage dependencies of the connected feeder loads. These estimates are obtained in a short time, which allows achieving the necessary execution time performance. The estimates are then updated each time the DPF is solved after the LVM solution is implemented in the field.

The Voltage Reduction (VR) status measurement and analog measurement (percentage) can be mapped on a controller. The VR status measurement type is MeasVReductRegTap. The VR percentage analog measurement type is AnalogVoltReductionValue. The voltage reduction value applies in power flow when it is mapped and the voltage reduction mode is active. When voltage reduction is enabled, the regulator's effective voltage target is calculated against the active band-center, which is the voltage setpoint measurement (type VoltSetpoint) or the nominal voltage target specified in the static model. For example, if the active band-center is 125 V, VR mode is active, and the VR analog value is 2%, the new "effective" band-center will be $125 - 125 \times 2\% = 122.5$ V.

Measurement points for voltage reduction control percentage can also be mapped to controller voltage reduction modes (VR1, VR2, or VR3 for the controllers that have 3 modes) when advanced regulation settings are modeled. Three status measurements (types StatusVoltReductionMode1, StatusVoltReductionMode2, and StatusVoltReductionMode3) are mapped to the VR modes. When the VR status (MeasVReductRegTap) is not available, the VR1, VR2, and VR3 status measurements are used to determine the voltage reduction mode, and the corresponding VR1, VR2, and VR3 percentages, which are defined in the static model, are used in the power flow "effective" band-center calculation.

Fixed voltage reduction problem formulations set all affected regulators to the same VR percent. The optimization problem formulations can set each regulator to a distinct optimized voltage reduction percent to achieve the LVM goals and mimic the effect of setting distinct band-center controls. The LVM solution engine converts the target tap step to a percent voltage reduction

setpoint control and sends it to Plan Manager.

3.9. LVM-Related Problem Formulations

A problem formulation is a pre-defined optimization or voltage reduction goal. It can be as simple as only requiring a good voltage profile, or it can be more complicated, possibly involving a combination of several different objectives. By changing the options and scaling factors of the problem formulation using the DNAF Configuration Editor tool, users can define different problem formulations according to their needs.

LVM supports three types of problem formulations: those related to the optimization approach, those related to the heuristic approach, and those related to fixed regulation approach. This is configured by the Solution Method options on the Load and Volt/VAR Management page in the DNAF Configuration Editor. For example, figure below shows that the "Minimize Demand" problem formulation is meant for use with the optimization approach.

Problem Formulation Objectives

Identification

Name:

Description:

Solution Method

☒ Optimization LVM ☐ Heuristic LVM ☐ Fixed Regulation LVM

Operation Mode

Active Operation Mode:

Optimization Goals

☒ Reduce Violations

☒ Reduce Demand

☐ Reduce Losses

☐ Provide Reactive Support

Figure 28. Selecting an LVM Solution Method

This section describes the various options related to LVM problem formulations and the steps needed to create a new problem formulation.

3.9.1. Problem Formulation Operation Mode

One of the pre-defined operation modes can be selected for a problem formulation as shown in figure below. The operation mode defines the plan execution options.



Figure 29. Selecting the Operation Mode in LVM

3.9.2. Problem Formulation Objectives

One or more optimization goals can also be selected. For example, if the goal is to have LVM reduce violations and reduce demand in the same problem formulation, select the appropriate check boxes as shown in figure below.



Figure 30. Selecting Optimization Objectives in LVM

The optimization goals include the following:

- Reduce Violations: This option instructs LVM to minimize voltage and flow violations if possible. This option applies to optimization LVM only.
- Reduce Demand: This option instructs LVM to minimize load demand without violating voltage limits. This option applies to optimization LVM only.
- Reduce Losses: This option instructs LVM to minimize losses without violating voltage limits. This option applies to optimization LVM only.
- Provide Reactive Support: This option instructs LVM to bring power factors monitored at station power transformers within specified power factor limits. This is described in more detail in [Power Factor Control](#). This option applies to both optimization and heuristic LVM.

3.9.3. Power Factor Control

This subsection of the LVM problem formulation is used to configure the options for maintaining power factor within specified limits. These options are effective if the Provide Reactive Support optimization goal is selected. The associated check boxes are used to set or clear the power factor targets for station power transformers. Minimum and maximum power factor targets are specified here, as shown in figure below.

Power Factor Control

Power Factor

Transformer Power Factor Monitoring Location: Source Side

☒ Default Minimum Power Factor:

☐ Default Maximum Power Factor:

Figure 31. Specifying Power Factor Control Options in LVM Problem Formulations

3.9.4. Limit Strategy

Each problem formulation allows the user to independently configure which limits are used when LVM is invoked with its respective problem formulation. In the example in figure below, bus voltage limits will not be used, while for loads normal operational limits will be used in the application. For branch flows, the most restrictive limit level, Level 1, will be used.

Limit Strategy

Voltage Limits

	Applicable Limit	Limit Adjustment
Buses with Voltage Measurements:	None	None
Buses without Voltage Measurements:	None	N/A
Loads with Voltage Measurements:	Normal Operational	None
Loads without Voltage Measurements:	Normal Operational	N/A

☐ Disable Capacitor Voltage Rails

☐ Discard Load Voltage Measurement with Large Deviations

☐ Voltage Bandwidth Reduction (%):

Branch Flow Limits

Respect Limit: None

Figure 32. Limit Strategy Options for LVM Problem Formulations

For voltage limits, the following options are available:

- **None:** If this option is selected, applicable limits are not enforced.
- **Normal Operational:** If this option is selected, normal operational limits are used.
- **Emergency Operational:** If this option is select, emergency operational limits are used.
- **Normal Optimization:** If this option is selected, normal (optional) optimization limits are used.
-

Emergency Optimization: If this option is selected, emergency (optional) optimization limits are used.

For branch flow limits, the following options are available:

- **None:** If this option is selected, branch flow limits are not enforced.
- **Level 1:** If this option is selected, the most restrictive branch flow limits are used.
- **Level 2:** If this option is selected, the least restrictive branch flow limits are used.

The "Discard Load Voltage Measurement with Large Deviation" option is used for load voltage constraint with AMI measurements. This option is combined with "Maximum Voltage Mismatch at Load Point (%)" on the DPF page in DCE. If the option is selected and the deviation between measured load AMI voltage and DPF calculated voltage is larger than the value of "Maximum Voltage Mismatch at Load Point (%)", this voltage measurement will not be used in LVM. The load will be treated as "Loads without Voltage Measurements".

3.9.5. Device Eligibility

This section specifies which types of controls are available for use by LVM. A user can independently allow devices located inside the station fence or outside the station fence to be considered in the optimization process. This is shown in figure below. They are excluded from Closed Loop operation.

If no device types are selected, the DNAF application terminates during a pre-processing check, which occurs before LVM is started. If LVM terminates due to no eligible control devices, no plan is generated, the solution status is not changed, and no LVM results file is generated.

Device Eligibility		
Eligible Controls	Substation Devices	Field Devices
Capacitors	☒	☒
Regulating Transformers	☒	☒
Generators	☐	☐
Distributed Energy Resources	☐	☐

Figure 33. Device Eligibility Options for LVM Problem Formulations

3.9.6. Optimization LVM Parameters

The parameters listed in this section are used only if the optimization method is selected for LVM. The options are shown below.



Optimization LVM Parameters

Performance Index Weights

	Final Step	Intermediate Steps
Flow Violations:	1	1
Bus Voltage Violations:	1	1
Load Voltage Violations:	1	1
Total Cost:	1	1
Real and Reactive Flow:	0	N/A
Control Moves:	1	N/A
Losses:	1	N/A
Bus Voltage Imbalance:	1	N/A

Violation Penalties and Costs

Bus Low Voltage Violation:	10000000
Bus High Voltage Violation:	20000000
Load Low Voltage Violation:	10000000

Figure 34. Options Specific to Optimization LVM

Performance Index Weights:

- **Flow Violations (Final Step):** The weight of the Flow Violations Performance Index (PI). This weight only applies for the final solutions. Valid value is any real numbers starting from 0.
- **Flow Violations (Intermediate Steps):** The weight of the Flow Violations Performance Index (PI). This weight is used for the PI only during the intermediate solutions. Valid value is any real numbers starting from 0.
- **Bus Voltage Violations (Final Step):** The weight of the Voltage Violations Performance Index (PI). This weight is applicable only for bus voltage violations in the final solution. Valid value is any real numbers starting from 0.
- **Bus Voltage Violations (Intermediate Steps):** The weight of the Voltage Violations Performance Index (PI). This weight is applicable only for bus voltage violations in the intermediate solutions. Valid value is any real numbers starting from 0.
- **Load Voltage Violations (Final Step):** The weight of the Voltage Violations Performance Index (PI). This weight is applicable only for load voltage violations in the final solution.
- **Load Voltage Violations (Intermediate Steps):** The weight of the Voltage Violations Performance Index (PI). This weight is applicable only for load voltage violations in the intermediate solutions.
- **Total Cost (Final Step):** The weight of the final step total cost performance index.

- **Total Cost (Intermediate Steps):** The weight of the Intermediate total cost Performance Index (PI).
- **Real and Reactive Flow:** The performance index weight for the real and reactive power violations at the flow monitoring points. Flow monitoring points are used in several LVM problem formulations including minimize demand and reactive area support.
- **Control Moves:** The weight of the final step number of control moves Performance Index (PI) (see [Problem Formulations](#) for more information).
- **Losses:** The weight of the final step losses Performance Index (PI) (see [Problem Formulations](#) for more information).
- **Bus Voltage Imbalance:** The performance index weight for the bus voltage imbalance. This PI is calculated using the standard maximum deviation from the average method. The PI value is included in the objective function value calculation, even if there are pre-plan voltage violations (similar to power factor).

Voltage Penalties and Costs:

- **Bus Low Voltage Violation:** The scaling factor for the bus low voltage violation cost curve. For more information about violation penalties and costs, see [Cost Model](#).
- **Bus High Voltage Violation:** The scaling factor for the bus high voltage violation cost curve.
- **Load Low Voltage Violation:** The scaling factor for the load low voltage violation cost curve.
- **Load High Voltage Violation:** The scaling factor for the load high voltage violation cost curve.
- **Low Voltage Violation Count:** The scaling factor for the number of customers associated with a load. This scaling factor is multiplied by the number of customers associated with a load, and then applied to the load low voltage violation cost curve.
- **High Voltage Violation Count:** The scaling factor for the number of customers associated with a load. This scaling factor is multiplied by the number of customers associated with a load, and then applied to the load high voltage violation cost curve.
- **Critical Customer Low Voltage Violation:** The scaling factor for critical customers associated with a load. This scaling factor is multiplied by the active power of a load with critical customers, and then applied to the load low voltage violation cost curve.
- **Critical Customer High Voltage Violation:** The scaling factor for critical customers associated with a load. This scaling factor is multiplied by the active power of a load with critical customers, and then applied to the load high voltage violation cost curve.
- **Flow Violation:** The scaling factor for the flow violation cost curve.
- **Voltage:** The scaling factor for voltage control setting (generators).
- **Real Flow:** The real power loss scaling cost (used in the minimize loss and minimize/maximize demand problem formulations).
- **Reactive Flow:** The scaling factor for the station's reactive power limit violation.
- **Tap Changer Move:** The scaling factor for LTC/regulator switching.

This scaling factor is multiplied by the LTC/regulator movement cost from the LTC/regulator

model and by a factor equal to $(1 + \text{CurrentDailyOperations}/\text{MaximumDailyOperations})$ to obtain the effective LTC/regulator movement cost that is used in formulating the Linear Programming model in Optimization LVM.

This parameter is used only if the current number of daily operations for the LTC/regulator is less than or equal to the maximum number of daily operations.

- **Tap Changer Movement (Exceed Daily Count):** The cost curve scaling factor for the LTC/regulator switching that exceeds the maximum number of daily switching operations.

This scaling factor is multiplied by the LTC/regulator movement cost from the LTC/regulator model and by a factor equal to $(1 + \text{CurrentDailyOperations}/\text{MaximumDailyOperations})$ to obtain the effective LTC/regulator movement cost that is used in formulating the Linear Programming model in Optimization LVM.

This parameter is used only if the current number of daily operations for the LTC/regulator exceeds the maximum number of daily operations.

- **Capacitor Switching:** The scaling factor for shunt impedance (capacitors) switching.

This scaling factor is multiplied by the capacitor switching cost from the capacitor model and by a factor equal to $(1 + \text{CurrentDailyOperations}/\text{MaximumDailyOperations})$ to obtain the effective capacitor switching cost that is used in formulating the Linear Programming model in Optimization LVM.

This parameter is used only if the current number of daily operations for the capacitor is less than or equal to the maximum number of daily operations.

- **Capacitor Switching (Exceed Daily Count):** The cost curve scaling factor for the capacitor switching that exceeds the maximum number of daily switching operations.

This scaling factor is multiplied by the capacitor switching cost from the capacitor model and by a factor equal to $(1 + \text{CurrentDailyOperations}/\text{MaximumDailyOperations})$ to obtain the effective capacitor switching cost that is used in formulating the Linear Programming model in Optimization LVM.

This parameter is used only if the current number of daily operations for the capacitor exceeds the maximum number of daily operations.

- **Generator VAR Change:** The scaling cost for generator's VAR output change.
- **Distributed Energy Resource VAR Change:** The scaling cost for distributed energy resource's VAR output change.
- **Large Loads:** The scaling cost for constraining the voltage limits of large loads.
- **Regulator Group Voltage Imbalance:** The scaling cost for voltage imbalance in regulation groups.
- **Local Control Capacitors (Amps):** The scaling cost for constraining the switching of locally regulating capacitors of the Amp control type.
- **Local Control Capacitors (Volt):** The scaling cost for constraining the switching of

locally regulating capacitors of the Volt control type.

- **Local Control Capacitors (VAR):** The scaling cost for constraining the switching of locally regulating capacitors of the VAR control type.
- **Local Control Capacitors (PF):** The scaling cost for constraining the switching of locally regulating capacitors of the PF control type.

Note

Capacitor and tap changer switching costs apply only in Real-Time LVM. In Study Mode, these switching actions are unmonitored and not penalized by the optimization.

Failsafe LVM:

- **Failsafe Problem Formulation:** This feature allows using a fixed voltage reduction problem formulation instead of the current power flow based LVM problem formulation to ensure failsafe operation if there are issues with the base case power flow solution. Please note that heuristic LVM problem formulations are not supported for failsafe operation.

The failsafe LVM feature is triggered when LVM is solved for a station, where the internal unbalanced power flow calculation is not converged or is of poor quality (the mismatch between the measured and calculated voltages at station internal and external buses is too large). Please be advised this unbalanced power flow is part of LVM and not the standalone BLA/DPF application, the failsafe is not determined by the standard BLA/DPF solution status. The problem formulations of the related system elements (station, power transformers, or feeders) are changed on the DNAF System Configuration display.

The failsafe LVM feature also applies to the Forecast server, which generates solutions for the LVM System Monitor. However, the problem formulations are not changed on the DNAF System Configuration display and the station-level failsafe flag is not updated by this form of the failsafe LVM feature.

The following options are available for the Failsafe Problem Formulation feature:

- **Manual Reversion:** With this option enabled, LVM does not re-apply the previous problem formulation for a system element (station, power transformer, or feeder) after the LVM Failsafe condition subsides (the station's internal unbalanced power flow solution status becomes converged or bus voltage mismatches are back within the tolerance).
- **Automatic Reversion:** With this option enabled, LVM monitors a station's DPF solution status and bus voltage mismatches before every run. If the station's LVM Failsafe condition subsides (the station's internal unbalanced power flow solution status becomes converged or bus voltage mismatches are back within the tolerance), the problem formulation for a system element is reset to the problem formulation used before the LVM Failsafe function was triggered.

Note

The failsafe status is de-activated when a user manually changes a failsafe problem formulation.

Note

The failsafe LVM feature depends on the LVM solution area. For example, if a station's unbalanced power flow solution is not converged, the failsafe LVM feature is triggered for the station and all of its sub-elements (power transformers and feeders).

If a station's unbalanced power flow solution is converged but the bus voltage mismatches are found, the failsafe LVM feature is triggered for the system element that the mismatch belongs to (for a station if the mismatch appeared upstream of a power transformer; for a power transformer if the mismatch appeared upstream of a feeder head but downstream of a power transformer; for a feeder if the mismatch appeared on a feeder).

Auto Capacitor Parameters:

- **Auto Setpoint Slack (%):** This entry specifies the amount of slack, in percentage, to be applied to the local setpoint values to tighten the control range of the individual capacitor. The slack is applied evenly to both the lower and upper setpoint values such that the median value between the two is maintained after the slack is applied.

3.9.7. Heuristic LVM Parameters

The parameters listed in the section are used only if the heuristic method is selected for LVM. These are described below.

Heuristic LVM Parameters

Parameters

Global Voltage Flattening Threshold (%):	1
Global Voltage Regulation Margin (%):	0.5
Voltage Reduction Weight:	0.5
Minimum Voltage for Power Factor Control (p.u.):	0.95
Maximum Delta V Deviation (%):	1

Exponential Components

Lowest Average Voltage Coefficient 1 (LV1):	1.5
Lowest Average Voltage Coefficient 2 (LV2):	1.7
Power Factor Coefficient 1 (PF1):	1.5
Power Factor Coefficient 2 (PF2):	15

Figure 35. Options Specific to Heuristic LVM

General parameters:

- **Global Voltage Flattening Threshold:** The default voltage flattening threshold that is used if a voltage flattening threshold is not defined for feeders in the station model file. It is expressed

as a percentage of the voltage base. For example, if the voltage flattening threshold is 1% and the voltage base is 120V, the voltage flattening threshold is 1.2V.

- **Global Voltage Regulation Margin:** The default voltage regulation margin that is used if a voltage regulation margin is not defined for regulators/LTCs in the station model file. It is expressed as a percentage of the voltage base.
- **Voltage Reduction Weight:** The weight for the voltage reduction component in the Heuristic LVM capacitor control algorithm.
- **Minimum Voltage for Power Factor Control:** The effective lower voltage limit for voltage flattening when the power factor control objective is used.
- **Maximum Delta V Deviation (future functionality):** The maximum allowed deviation between the capacitor voltage sensitivities calculated by Heuristic LVM and the ones received from SCADA. If any deviation is greater than this threshold, an alarm (Capacitor Delta V Mismatch) is issued.

Exponential components:

- **Lowest Average Voltage Coefficients 1 and 2 (LV1 and LV2):** These parameters are used in the lowest average voltage component in the Heuristic LVM capacitor control algorithm.
- **Power Factor Coefficients 1 and 2 (PF1 and PF2):** These parameters are used in the power factor component in the Heuristic LVM capacitor control algorithm.

The usage of these parameters is described in [Heuristic LVM Methodology](#).

3.9.8. Fixed Regulation LVM Parameters

The parameters listed in the section are used only if the fixed regulation method is selected for LVM. These are described below.

Figure 36. Options Specific to Fixed Regulation LVM

Fixed Regulation LVM Parameters:

- **Fixed Voltage Reduction:** This option specifies whether fixed band-center voltage reduction is enabled and the voltage reduction target specified in the Voltage Reduction field is applied.
- **Voltage Reduction:** This parameter specifies, in percentage, the voltage reduction target against the active band-center setpoint or the nominal voltage target value. For example, if the active band-center value is 125 V and the voltage reduction target is 2%, the effective reduction target is $125 - 125 \times 2\% = 122.5$ V.
- **Turn Off Voltage Reduction:** This option specifies whether the fixed band-center voltage reduction target is disabled and the voltage reduction target is reset to the value specified in

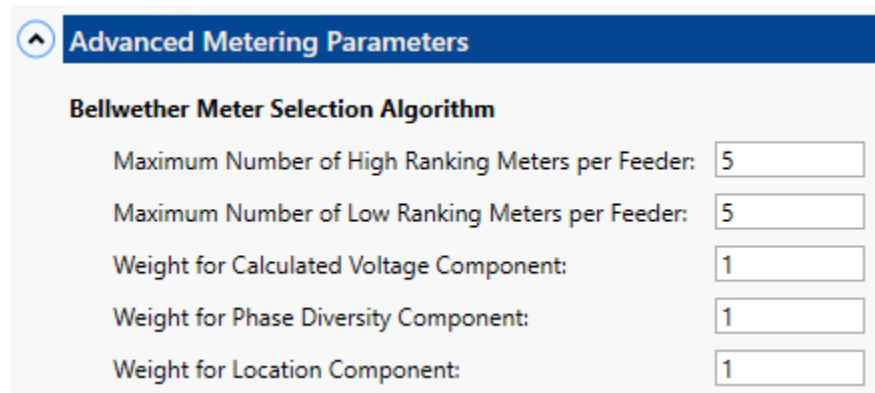
the Default Voltage Reduction field.

- **Default Voltage Reduction:** This parameter specifies, in percentage, the default voltage reduction target against the active band-center setpoint or the nominal voltage target value.

For more details about using the fixed regulation LVM approach, see [Fixed Regulation LVM Problem Formulations](#).

3.9.9. Advanced Metering Parameters

The parameters described in this section are used for the "Bellwether Meter Selection Algorithm" in connection with utilizing AMI load voltage measurements. For further details, see [Bellwether Meter Selection](#).



Advanced Metering Parameters	
Bellwether Meter Selection Algorithm	
Maximum Number of High Ranking Meters per Feeder:	5
Maximum Number of Low Ranking Meters per Feeder:	5
Weight for Calculated Voltage Component:	1
Weight for Phase Diversity Component:	1
Weight for Location Component:	1

Figure 37. Advanced Metering Parameters in LVM Problem Formulations

- **Maximum Number of High Ranking Meters per Feeder:** This parameter specifies the maximum number of high ranking AMI meters that can be selected for a request to pre-solve AMI readings.
- **Maximum Number of Low Ranking Meters per Feeder:** This parameter specifies the maximum number of low ranking AMI meters that can be selected for a request to pre-solve AMI readings.
- **Weight for Calculated Voltage Component:** This parameter specifies the weight for the calculated voltage component in ranking AMI loads for a pre-solve read request.
- **Weight for Phase Diversity Component:** This parameter specifies the weight for the phase diversity component in ranking AMI loads for pre-solve read request.
- **Weight for Location Component:** This parameter specifies the weight for the location component in ranking AMI loads for pre-solve read request.

3.9.10. Optimization LVM Problem Formulations

The following problem formulations are used with the optimization LVM solution approach.

3.9.10.1. Reduce Violations

This problem formulation checks for voltage violations in the base case power flow. The goal of this problem formulation is to remove or at least reduce violations if they exist. Figure below shows the typical configuration of this problem formulation in the DNAF Configuration Editor tool. In this figure, the Reduce Violations option is selected. If voltage or flow violations exist, a penalty cost is added in the final objective function.

The scaling factors for the cost curve on the left side indicate the slope of the cost curve, which represents μ in Equation 3.2. Note that the scaling factor of high voltage violation is larger than that of the low voltage violation, indicating that a high voltage violation is more severe than a low voltage violation and has higher priority to be removed if conflicts exist.

^
Problem Formulation Objectives

Identification

Name:

Description:

Solution Method

☒ Optimization LVM
☐ Heuristic LVM
☐ Fixed Regulation LVM

Operation Mode

Active Operation Mode:

Optimization Goals

☒ Reduce Violations

☐ Reduce Demand

☐ Reduce Losses

☐ Provide Reactive Support

v
Power Factor Control

v
Limit Strategy

v
Device Eligibility

^
Optimization LVM Parameters

Performance Index Weights

	Final Step	Intermediate Steps
Flow Violations:	1	1
Bus Voltage Violations:	1	1
Load Voltage Violations:	1	1
Total Cost:	1	1
Active and Reactive Flow:	0	N/A
Control Moves:	1	N/A
Losses:	1	N/A

Figure 38. Reduce Violations Problem Formulation

3.9.10.2. Reactive Area Support

This problem formulation attempts to maintain the power factor at the station power transformers within specified limits. For this problem formulation, the Provide Reactive Support option in the optimization goals should be selected. Specifying power factor targets and limits is described further in this section.

^
Problem Formulation Objectives

Identification

Name:

Description:

Solution Method

☒ Optimization LVM
☐ Heuristic LVM
☐ Fixed Regulation LVM

Operation Mode

Active Operation Mode:

Optimization Goals

☒ Reduce Violations

☐ Reduce Demand

☐ Reduce Losses

☒ Provide Reactive Support

^
Power Factor Control

Transformer Power Factor Monitoring Location:

☒ Default Minimum Power Factor:

☐ Default Maximum Power Factor:

Figure 39. Power Factor Target Monitoring

The typical settings for the Reactive Area Support problem formulation are shown in figure below.

Device Eligibility

Eligible Controls

	Substation Devices	Field Devices
Capacitors	☒	☒
Regulating Transformers	☒	☒
Generators	☐	☐
Distributed Energy Resources	☐	☐

Optimization LVM Parameters

Performance Index Weights

	Final Step	Intermediate Steps
Flow Violations:	1	1
Bus Voltage Violations:	1	1
Load Voltage Violations:	1	1
Total Cost:	1	1
Real and Reactive Flow:	0	N/A
Control Moves:	1	N/A
Losses:	1	N/A
Bus Voltage Imbalance:	1	N/A

Figure 40. Reactive Area Support Problem Formulation

Violation Penalties and Costs	
Bus Low Voltage Violation:	100000
Bus High Voltage Violation:	100000
Load Low Voltage Violation:	100000
Load High Voltage Violation:	100000
Low Voltage Violation Count:	1
High Voltage Violation Count:	1
Critical Customer Low Voltage Violation:	1
Critical Customer High Voltage Violation:	1
Flow Violation:	1000
Voltage:	1
Real Flow:	0
Reactive Flow:	5000

Figure 41. Reactive Area Support Violation Penalties and Costs

The scaling cost factor "Reactive Flow" is used as the weighting factor for reactive power. Because the weighting factor for the real power "Real Flow" is zero, this problem formulation targets controlling the reactive power.

When LVM is executed with this problem formulation, it searches for the applicable minimum and maximum power factor target limits in the following order:

For power transformers:

1. First, the application checks for manually entered power factor targets (entered on the Manual Input Parameter Entry display, accessed from the RT drop-down). If a manual entry exists, the specified value is used.

Manual Input Parameter Entry

System

CTL

BAKER

T978

T975

T2251

CANTERBURY

HEAVENLY

JUPITER

MERCURY

OLYMPIC

TFTEST

VAIL

WHISTLER

NEA

STH

Unknown

WST

CTL Division Parameters

Description	Value	Affected Sub-Elements	Sub-Element Values
General			
Temperature:		9 Stations	Auto
Load Scaling Factor:		65 Feeders	Auto
Violation Limits			
Trans. Flow Limit Scaling:		20 Power Transformers	Auto
Device Flow Limit Scaling:		65 Feeders	Auto
Bus High Voltage Limit (p.u.):		65 Feeders	Auto
Bus Low Voltage Limit (p.u.):		65 Feeders	Auto
Load High Voltage Limit (p.u.):		65 Feeders	Auto
Load Low Voltage Limit (p.u.):		65 Feeders	Auto
Optimization Limits			
Bus High Voltage Limit (p.u.):		65 Feeders	Auto
Bus Low Voltage Limit (p.u.):		65 Feeders	Auto
Load High Voltage Limit (p.u.):		65 Feeders	Auto
Load Low Voltage Limit (p.u.):		65 Feeders	Auto
Load and Volt/VAR Management (LVM)			
Feeder Max PF Target:		65 Feeders	Auto
Feeder Min PF Target:		65 Feeders	Auto
Station Max PF Target:		9 Stations	Auto
Station Min PF Target:		9 Stations	Auto

OK
Cancel
Apply

- Otherwise, the target power factor schedule for the station is checked. This is specified by the TARGETPFSCHED_AREAPF field in the global.csv file that was used to create the station model file.

In the schedule, each schedule point has two values:

- Value 1 stores the minimum target power factor.
 - Value 2 stores the maximum target power factor.
- If a power factor schedule is not available for power transformers, and the values specified in the Manual Input Parameter Entry display is "default", then the value of the Default Minimum/Maximum Power Factor parameter in the Power Factor Control subsection of the Load and Volt/VAR Management problem formulation page in DNAF Configuration Editor is used.

^
Power Factor Control

Transformer Power Factor Monitoring Location:
Source Side

☒ Default Minimum Power Factor:
0.99

☐ Default Maximum Power Factor:
0

If DER output and generation exceed load within a local area, the system could operate in backfeed condition, where power flows in the reverse direction through the station power transformers and feeder breakers. In this case, depending on the direction convention, the real power can be

negative. The power factor is calculated as shown in the equation below.

$$\begin{cases} PF = \frac{|P|}{S} = \frac{|P|}{\sqrt{P^2 + Q^2}} \\ \text{Leading if } P \cdot Q < 0 \\ \text{Lagging if } P \cdot Q > 0 \end{cases} \quad (\text{Equation 3.18})$$

Figure below shows the leading and lagging power factors in different active and reactive power combinations.

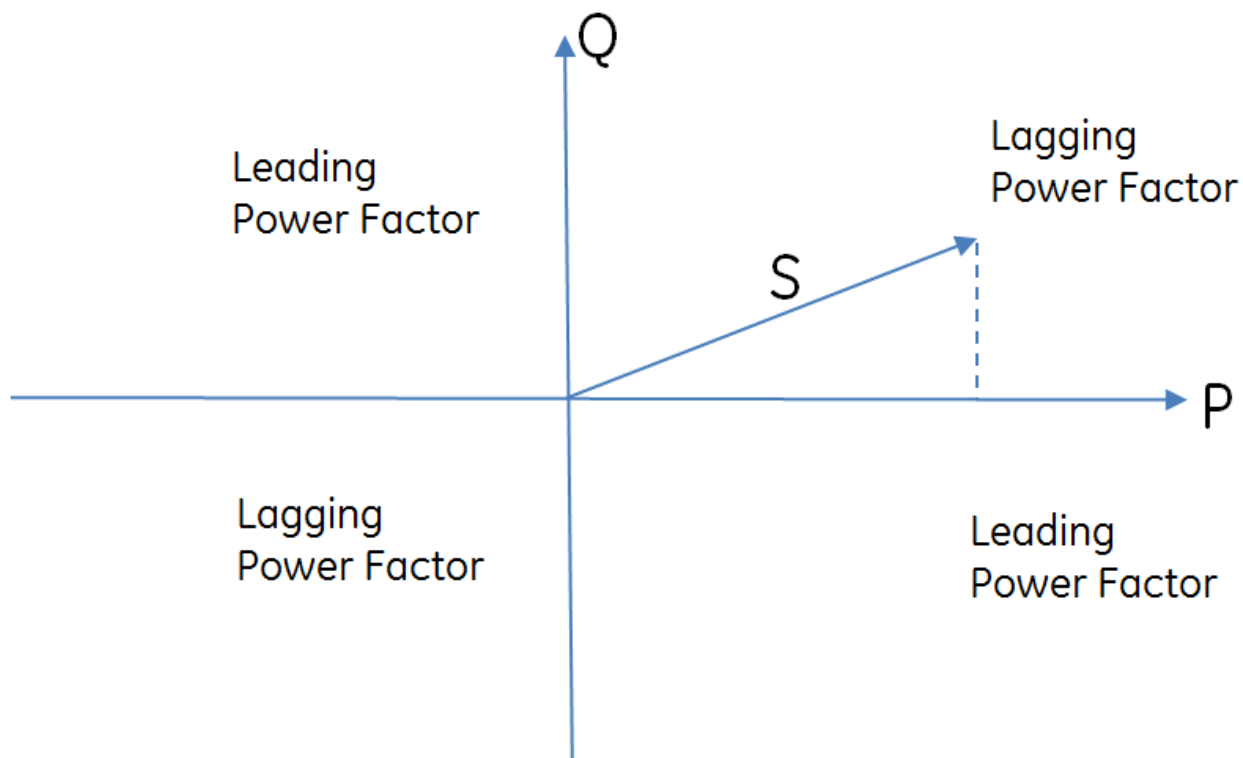


Figure 42. Power Factors for Different Real/Reactive Power Combinations

Table below lists some examples illustrating the allowed range of reactive power as a function of specified maximum and minimum power factors. The following sign convention is used:

- A positive value represents a lagging power factor.
- A negative value represents a leading power factor.

Table 3. Explanation of Sample Power Factor Target Range Values

Max PF	Min PF	Allowed Power Factor	Allowed Range of Q (suppose P = 1)
-0.92	0.92	Between leading 0.92 and lagging 0.92	$-\tan(\cos^{-1} 0.92) \sim \tan(\cos^{-1} 0.92)$
0.98	0.92	Between lagging 0.98 and lagging 0.92	$\tan(\cos^{-1} 0.98) \sim \tan(\cos^{-1} 0.92)$

Max PF	Min PF	Allowed Power Factor	Allowed Range of Q (suppose P = 1)
-0.92	-0.98	Between leading 0.92 and leading 0.98	$-\tan(\cos^{-1}0.92) \sim -\tan(\cos^{-1}0.98)$
1	0.98	No leading, lagging 0.98	$0 \sim \tan(\cos^{-1}0.98)$
-0.98	-1	No lagging, leading 0.98	$-\tan(\cos^{-1}0.98) \sim 0$
1	1	No leading and lagging	$0 \sim 0$
-1	-1	No leading and lagging	$0 \sim 0$
0	0.92	Maximum lagging 0.92	$-\infty \sim \tan(\cos^{-1}0.92)$
0.92	0	Minimum lagging 0.92	$\tan(\cos^{-1}0.92) \sim \infty$
0	-0.92	Maximum leading 0.92	$-\infty \sim -\tan(\cos^{-1}0.92)$
-0.92	0	Minimum leading 0.92	$-\tan(\cos^{-1}0.92) \sim \infty$
0.92	0.92	Sharp at lagging 0.92	$\tan(\cos^{-1}0.92)$
-0.92	-0.92	Sharp at leading 0.92	$-\tan(\cos^{-1}0.92)$
0.92	0.98	Not feasible, values are swapped in the code	
-0.98	-0.92	Not feasible, values are swapped in the code	

3.9.10.3. Loss Minimization

This problem formulation is used to reduce total losses in the distribution system. Typical settings are shown below.

^
Problem Formulation Objectives

Identification

Name:

Description:

Solution Method

☒ Optimization LVM
☐ Heuristic LVM
☐ Fixed Regulation LVM

Operation Mode

Active Operation Mode:

Optimization Goals

☒ Reduce Violations

☐ Reduce Demand

☒ Reduce Losses

☐ Provide Reactive Support

^
Power Factor Control

^
Limit Strategy

^
Device Eligibility

^
Optimization LVM Parameters

Performance Index Weights

	Final Step	Intermediate Steps
Flow Violations:	1	1
Bus Voltage Violations:	1	1
Load Voltage Violations:	1	1
Total Cost:	1	1
Active and Reactive Flow:	0	N/A
Control Moves:	1	N/A
Losses:	1	N/A
Voltage Imbalance:	0	N/A
Maximum Segment Margin:	0	N/A

Figure 43. Loss Minimization Problem Formulation

A sensitivity of losses with respect to the control variables is calculated inside the engine to achieve this goal. The scaling factor specified by "Real Flow" is used as the weight for loss reduction.

The loss in the distribution system is composed of several parts: line losses, power transformer losses, distribution transformer losses, regulator losses, and series reactor losses. Depending on the different load voltage dependencies, raising or lowering voltage may have different impacts on different parts of the losses. For example, for a constant power load, the lowering of voltages reduces transformer losses but increases the line losses. This problem formulation considers both effects and finds the optimal solution.

3.9.10.4. Minimize Demand

This problem formulation targets total real power demand reduction (loads plus loss) in the selected station. Because in most cases, the load to voltage curves of active loads have positive derivatives, this problem formulation tries to decrease system voltage to reduce demand until a constrained voltage hits its lower limit.

Problem Formulation Objectives

Identification
Name:
Description:

Solution Method
☒ Optimization LVM ☐ Heuristic LVM ☐ Fixed Regulation LVM

Operation Mode
Active Operation Mode:

Optimization Goals
☒ Reduce Violations
☒ Reduce Demand
☐ Reduce Losses
☐ Provide Reactive Support

Power Factor Control

Limit Strategy

Device Eligibility

Optimization LVM Parameters

Performance Index Weights

	Final Step	Intermediate Steps
Flow Violations:	1	1
Bus Voltage Violations:	1	1
Load Voltage Violations:	1	1
Total Cost:	1	1
Active and Reactive Flow:	0	N/A
Control Moves:	1	N/A
Losses:	1	N/A
Voltage Imbalance:	0	N/A
Maximum Segment Margin:	0	N/A

Figure 44. Minimize Demand Problem Formulation

3.9.11. Heuristic LVM Problem Formulations

LVM problem formulations intended for use with the heuristic approach are indicated by selecting the Heuristic LVM option in the Solution Method group on the LVM problem formulation page of the DNAF Configuration Editor.

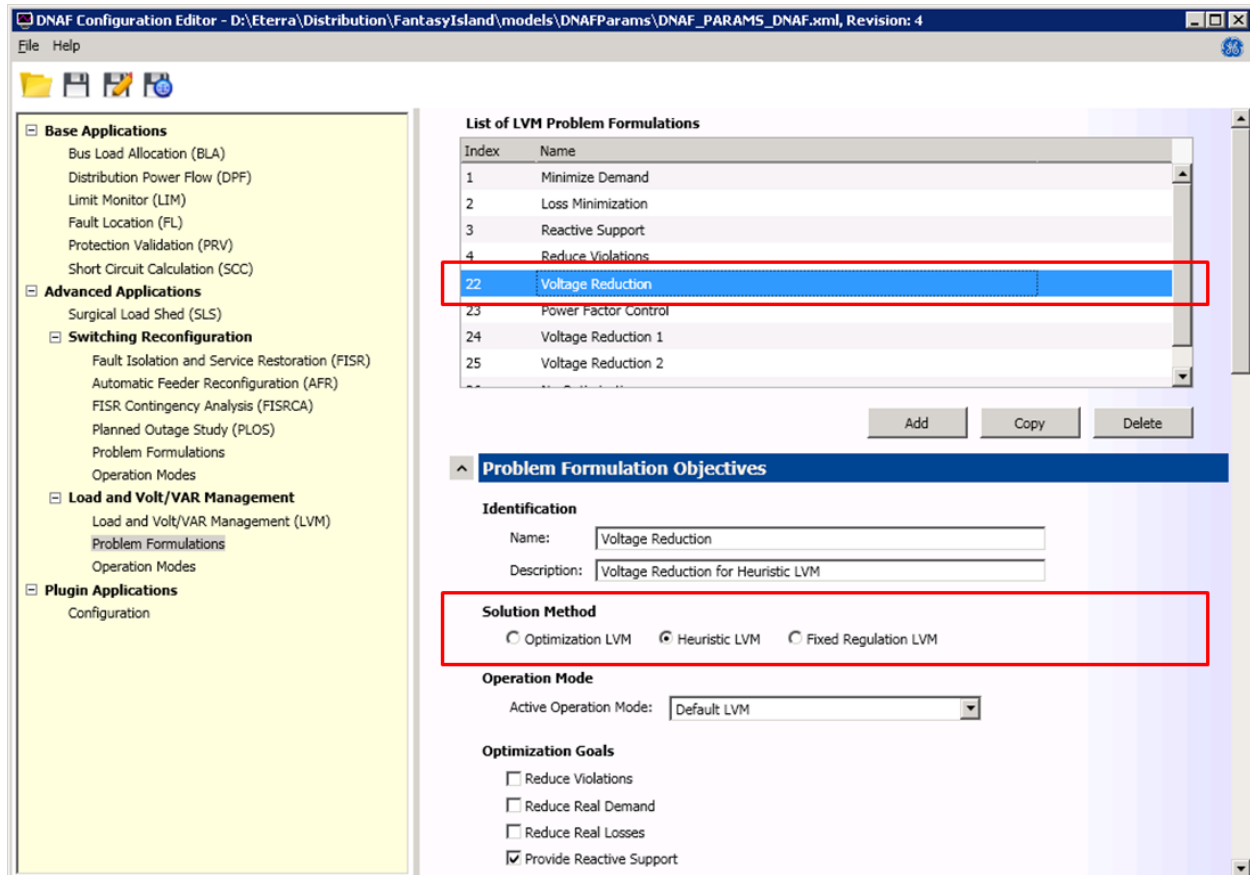


Figure 45. Specifying the Heuristic Approach

When the Heuristic LVM option is selected, the parameters specific to the heuristic approach are enabled while the ones specific to the optimization and fixed regulation approaches are grayed out.

3.9.11.1. Voltage Reduction

The purpose of this problem formulation is to perform voltage reduction and flattening using the Heuristic LVM approach. In this problem formulation, the exponential weights for lowest average voltage are non-zero, while the weights for power factor are zero. This problem formulation is similar to the Optimization LVM "Minimize Demand" problem formulation described in [Minimize Demand](#).

3.9.11.2. Power Factor Control

The purpose of this problem formulation is to perform power factor control using the Heuristic LVM approach. In this problem formulation, the exponential weights for power factor are non-zero, while the weights for lowest average voltage are zero. This problem formulation is similar to the Optimization LVM "Reactive Area Support" problem formulation described in [Reactive Area Support](#).

3.9.12. Fixed Regulation LVM Problem Formulations

To indicate LVM problem formulations intended for use with the fixed regulation approach, select the Fixed Regulation LVM option in the Solution Method group on the LVM problem formulation page of the DNAF Configuration Editor.

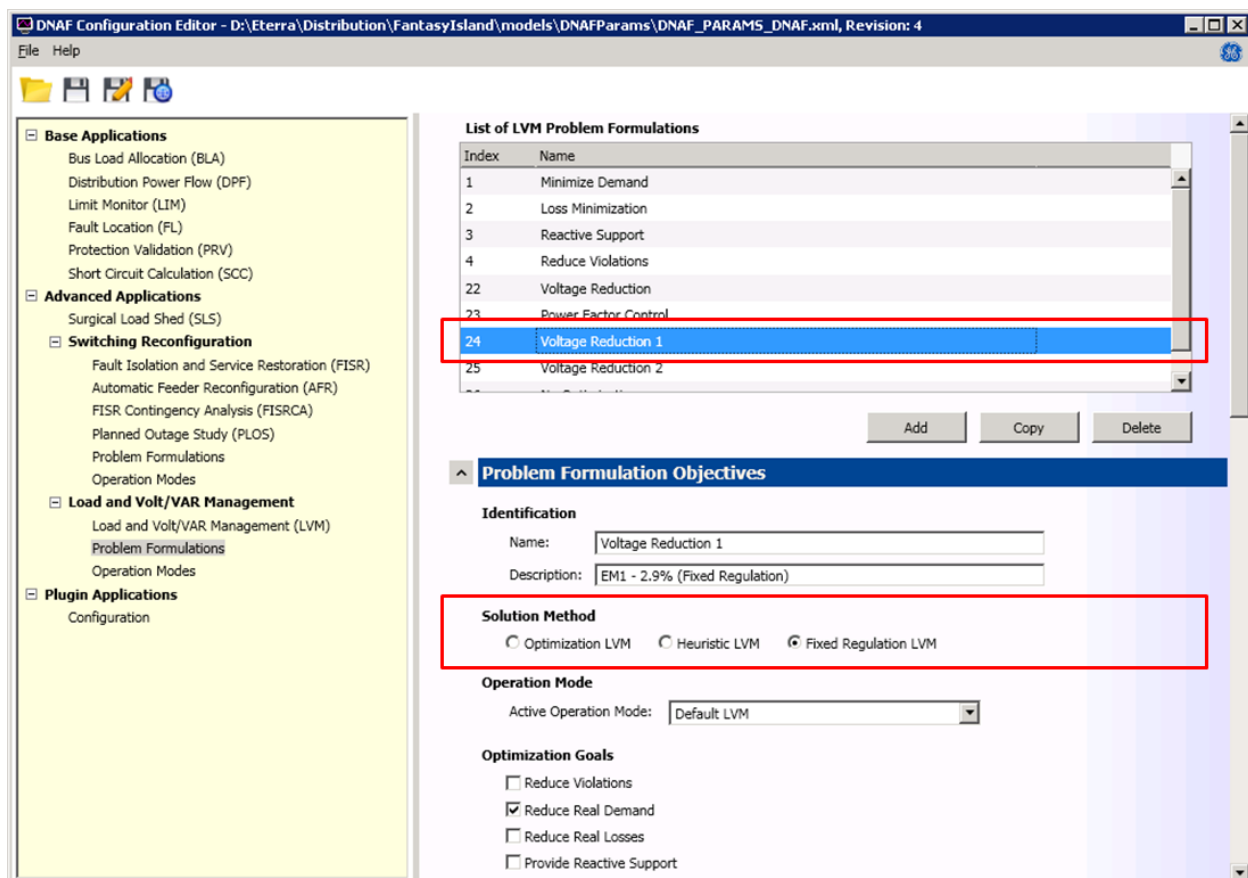


Figure 46. Specifying the Fixed Regulation Approach

When the Fixed Regulation LVM option is selected, the parameters specific to the fixed regulation approach are enabled while the ones specific to the optimization and heuristic approaches are grayed out.

3.9.12.1. Voltage Reduction 1

The purpose of this problem formulation is to perform the level 1 emergency voltage reduction mode to lower the effective voltage band-center by a common value, such as 2.9%.

When LVM solves in this mode, all eligible LTCs/regulators with a control measurement mapped for the associated voltage reduction mode are included in an LVM plan, where voltage reduction is set to a common value of 2.9%.

The screenshot displays the 'Fixed Regulation LVM Parameters' configuration window. It is divided into two main sections: 'Problem Formulation Objectives' and 'Fixed Regulation LVM Parameters'.

Problem Formulation Objectives:

- Identification:** Name: 'Voltage Reduction 1', Description: 'EM1 - 2.9% (Fixed Regulation)'.
- Solution Method:** Radio buttons for 'Optimization LVM', 'Heuristic LVM', and 'Fixed Regulation LVM' (selected).
- Operation Mode:** 'Active Operation Mode' dropdown menu set to 'Default LVM'.
- Optimization Goals:** Checkboxes for 'Reduce Violations', 'Reduce Real Demand' (checked), 'Reduce Real Losses', and 'Provide Reactive Support'.

Fixed Regulation LVM Parameters:

- Radio buttons for 'Fixed Voltage Reduction' (selected) and 'Turn Off Voltage Reduction'.
- For 'Fixed Voltage Reduction': 'Voltage Reduction (%)' input field set to '2.9'.
- For 'Turn Off Voltage Reduction': 'Default Voltage Reduction (%)' input field set to '0'.

Figure 47. Fixed Regulation Problem Formulation – Voltage Reduction 1

3.9.12.2. Voltage Reduction 2

The purpose of this problem formulation is to perform the level 2 emergency voltage reduction mode to lower the effective voltage band-center by a common value, such as 5%.

When LVM solves in this mode, all eligible LTCs/regulators with a control measurement mapped for the associated voltage reduction mode are included in an LVM plan, where voltage reduction is set to a common value of 5%. In addition, enabling this mode sends a control to all capacitor banks to turn off their high/low voltage regulation (disable capacitor voltage rails) so that low voltages are permitted.

^
Problem Formulation Objectives

Identification

Name: Voltage Reduction 2

Description: EM2 - 5.0% (Fixed Regulation, Disable Rails)

Solution Method

☐ Optimization LVM
☐ Heuristic LVM
☒ Fixed Regulation LVM

Operation Mode

Active Operation Mode: Voltage Reduction with Capacitor Voltage

Optimization Goals

☐ Reduce Violations
☒ Reduce Real Demand
☐ Reduce Real Losses
☐ Provide Reactive Support

^
Fixed Regulation LVM Parameters

☒ Fixed Voltage Reduction
Voltage Reduction (%): 5

☐ Turn Off Voltage Reduction
Default Voltage Reduction (%): 0

Figure 48. Fixed Regulation Problem Formulation – Voltage Reduction 2

3.9.12.3. No Optimization

The purpose of this problem formulation is to disable voltage reduction, effectively disabling LVM for all stations, power transformers, or feeders using this problem formulation. The voltage reduction value is also reset to the value given in the Default Voltage Reduction (%) field but voltage reduction is disabled. This problem formulation may be utilized when switching is being performed in the field.

When LVM solves in this mode, all eligible LTCs/regulators with a control measurement mapped for the associated voltage reduction mode are included in an LVM plan, where voltage reduction is set to a common default value, such as 2.5%.

^
Problem Formulation Objectives

Identification
Name:
Description:

Solution Method
☐ Optimization LVM ☐ Heuristic LVM ☒ Fixed Regulation LVM

Operation Mode
Active Operation Mode:

Optimization Goals
☐ Reduce Violations
☒ Reduce Real Demand
☐ Reduce Real Losses
☐ Provide Reactive Support

^
Fixed Regulation LVM Parameters

☒ Fixed Voltage Reduction ☐ Turn Off Voltage Reduction
Voltage Reduction (%): Default Voltage Reduction (%):

Figure 49. Fixed Regulation Problem Formulation – No Optimization

3.9.13. Creating a New Problem Formulation

A problem formulation can be fully user-customized. In addition, users can create a new problem formulation to suit their individual needs.

To create a new LVM problem formulation:

1. Using the DNAF Configuration Editor, open the appropriate DNAF_PARAMS_DNAF.xml file.
2. Select the Problem Formulation page.
3. Click Add. A new problem formulation is created, as shown below:

23 Power Factor Control (Heuristic)

24 New Formulation

Add Copy Delete

Problem Formulation Objectives

Identification

Name: New Formulation

Description: Problem Formulation Description

Solution Method

☒ Optimization LVM ☐ Heuristic LVM ☐ Fixed Regulation LVM

4. Edit the Name and Description as needed.
5. Select the solution method (Optimization LVM, Heuristic LVM, or Fixed Regulation LVM).
6. For Optimization LVM, select the goals for this problem formulation in the "Optimization Goals" sub-section.
7. Adjust the algorithm specific options such as scaling factors, violation penalties, and power factor options if necessary. For Heuristic LVM, adjust parameters in the Heuristic LVM parameters sub-section.
8. Click the "Save online" toolbar button to save the changes online.

If a variation of an existing problem formulation is required:

1. Using the DNAF Configuration Editor, open the appropriate DNAF_PARAMS_DNAF.xml file.
2. Select a problem formulation, and then click the Copy button.
3. Adjust the parameters of interest and then save the new configuration.

23 Power Factor Control (Heuristic)

24 New Formulation

25 Copy of Power Factor Control (Heuristic)

Add Copy Delete

Problem Formulation Objectives

Identification

Name: Copy of Power Factor Control (Heuristic)

Description: Power Factor Control for Heuristic LVM

Solution Method

☐ Optimization LVM ☒ Heuristic LVM ☐ Fixed Regulation LVM

3.9.14. Problem Formulation Parameter Tuning

The [Optimization LVM Parameters](#) and [Heuristic LVM Parameters](#) list some typical settings for LVM-related problem formulations.

 **Optimization LVM Parameters**

Performance Index Weights

	Final Step	Intermediate Steps
Flow Violations:	1	1
Bus Voltage Violations:	1	1
Load Voltage Violations:	1	1
Total Cost:	1	1
Real and Reactive Flow:	0	N/A
Control Moves:	1	N/A
Losses:	1	N/A
Bus Voltage Imbalance:	1	N/A

Violation Penalties and Costs

Bus Low Voltage Violation:	10000000
Bus High Voltage Violation:	20000000
Load Low Voltage Violation:	10000000
Load High Voltage Violation:	20000000
Low Voltage Violation Count:	1
High Voltage Violation Count:	1

Figure 50. Specifying LVM Problem Formulations for Feeders

For Optimization LVM, key problem formulation parameters are performance index weights and violation penalties and costs (scaling factors).

The performance indexes of the problem formulation are used to sort different candidate plans if multiple plans exist. Since, in regular multi-step mode, only one plan is generated by the LPengine, those values are not used. However, in single-step mode, because the LVM post-processing generates several single-step plans and there is a need to rank them, these values also must be filled in. The idea is to use bigger numbers for those performance indices that are more closely related to the optimization goal. The top plans are determined by the lowest total performance index numbers.

The values of the scaling factors can be adjusted as needed. A larger number means more emphasis on the corresponding item. For example, for the Minimize Demand problem formulation,

if the system can sacrifice part of the voltage profile to further reduce demand, the scaling factors of the active flow can be increased, or vice versa. The specific values can be determined by trial and error.

3.9.15. Feeder Specific Problem Formulation

As mentioned in [Enable/Disable LVM](#), LVM can be enabled or disabled by feeder. It is possible to specify a different problem formulation for different feeders to reach a different target. For example, the user can set a feeder to use Minimize Demand while setting another feeder to use Reactive Area Support. This can be specified via the DNAF System Configuration display from the RT drop-down as shown in figure below.

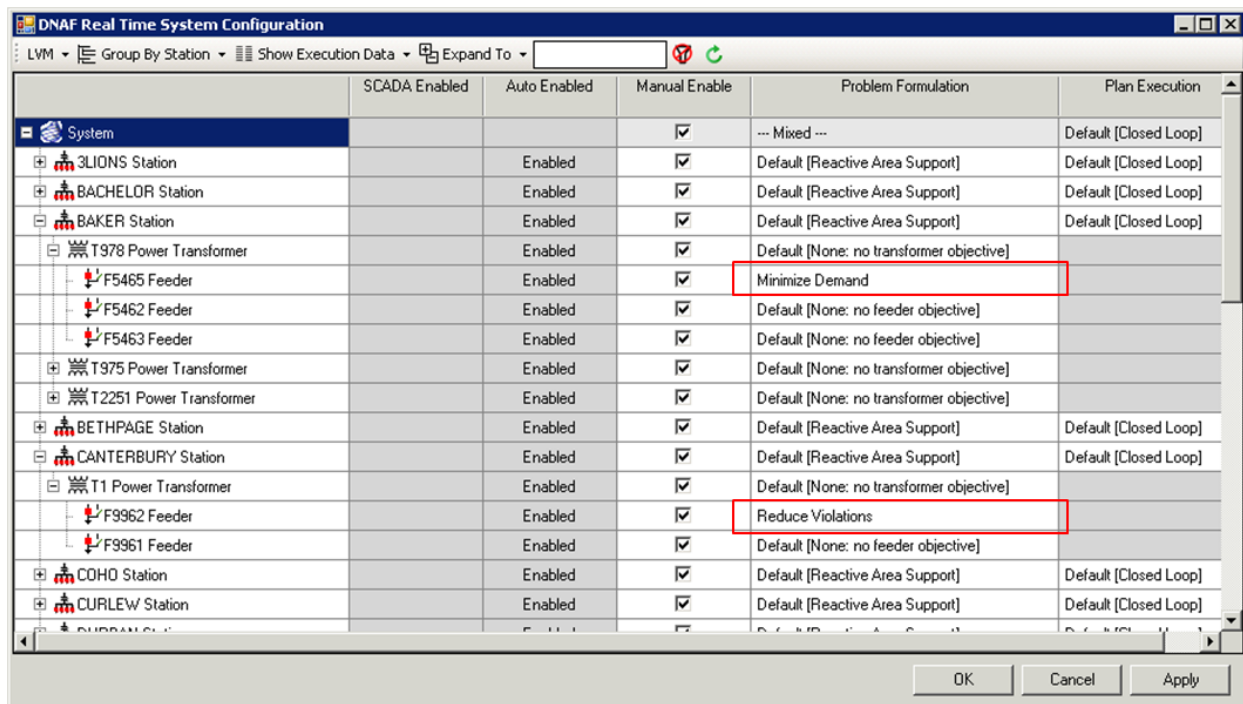


Figure 51. Specifying LVM Problem Formulations for Feeders

You can specify the default problem formulation for feeder using the feeder object's ProbFormuIndex field in the corresponding station XML file. The default value is -1, which means "no specific target". In this case, LVM is executed in regular station mode based on the target specified in the problem formulation of the station.

3.10. Key LVM Parameters

This section describes some key LVM-related parameters. These parameters can be edited with the DNAF Configuration Editor tool.

3.10.1. Optimization Device Eligibility

On the LVM page of the DNAF Configuration Editor, there are several parameters in the Device Eligibility group box. Those parameters impact the optimization eligibility of the capacitors, LTC/regulators, and DERs. If an optimization device is not eligible for control, the LVM application does not consider the device when generating an optimization plan. In addition, specify eligible devices for particular problem formulations (see [Device Eligibility](#).)

Device Eligibility
☐ Respect Daily Switching Count
☐ Respect Capacitor Consecutive Switching
☐ Respect Tap Changer Consecutive Switching
☐ Check Capacitor Pending Status
Real Time Device Eligibility

Measurement Type	Capacitor	Regulator	LTC	Generator	DER
Status/Step/Setpoint	☒	☒	☒	☒	☒
Supervisory Cutout	☐	☐	☐		
Auto/Manual	☐	☐	☐		
Remote/Local	☐	☐	☐		
Control Available	☐	☐	☐		

Make Generators and DER ineligible if available reactive power output range is less than (kVAR):

Figure 52. Device Eligibility

The easiest way to check the LVM eligibility of a capacitor is to use the LVM Device Eligibility tabular display. This display also provides the reason if a device is not eligible for use in LVM.

The screenshot shows a software window titled "BETHPAGE - LVM Device Eligibility". It contains a table with the following columns: Name, ID, Locate, Type, Energizing Station, Energizing Feeder, Analyst Attributes, Eligibility Status, Reason Device is not Eligible, and Energizing Feeder LVM Status. The table lists 14 capacitors, all of which are "Capacitor" type and located at "BETHPAGE". The "Energizing Feeder" column lists various feeder IDs (F4135, F4131, F4138, F4132, F4134). The "Eligibility Status" column indicates whether each capacitor is "Eligible" or "Not Eligible". The "Reason Device is not Eligible" column provides the reason for ineligibility, such as "The feeder is disabled for LVM.". The "Energizing Feeder LVM Status" column shows the status of the feeder, either "LVM Enabled" or "LVM Disabled".

Name	ID	Locate	Type	Energizing Station	Energizing Feeder	Analyst Attributes	Eligibility Status	Reason Device is not Eligible	Energizing Feeder LVM Status
1946	10216180		Capacitor	BETHPAGE	F4135		Eligible		LVM Enabled
3641	10216096		Capacitor	BETHPAGE	F4135		Eligible		LVM Enabled
1754	10216166		Capacitor	BETHPAGE	F4135		Eligible		LVM Enabled
1616	10217160		Capacitor	BETHPAGE	F4131		Not Eligible	The feeder is disabled for LVM.	LVM Disabled
1835	10217258		Capacitor	BETHPAGE	F4131		Not Eligible	The feeder is disabled for LVM.	LVM Disabled
1965	10217146		Capacitor	BETHPAGE	F4138		Eligible		LVM Enabled
1588	10217342		Capacitor	BETHPAGE	F4138		Eligible		LVM Enabled
1830	10216614		Capacitor	BETHPAGE	F4132		Not Eligible	The feeder is disabled for LVM.	LVM Disabled
5902	10216432		Capacitor	BETHPAGE	F4132		Not Eligible	The feeder is disabled for LVM.	LVM Disabled
5903	10216418		Capacitor	BETHPAGE	F4132		Not Eligible	The feeder is disabled for LVM.	LVM Disabled
1990	10216250		Capacitor	BETHPAGE	F4134		Eligible		LVM Enabled
4009	10216222		Capacitor	BETHPAGE	F4134		Eligible		LVM Enabled
1919	10217104		Capacitor	BETHPAGE	F4134		Eligible		LVM Enabled

Figure 53. LVM Device Eligibility Display

Each capacitor has a maximum daily switching count limitation. That value is specified when generating a model file. For example, if a capacitor can only switch 4 times per day, then after switching 4 times, the capacitor is not eligible for more controls and is not considered in the LVM application. In this case, the daily switching count is a "hard" constraint.

However, the maximum daily switching count is a "soft" constraint that can be violated. The maximum daily switching count limitation can be disabled by disabling the Respect Daily Switching Count option. However, the operation cost for the device that already reached the maximum daily switching count can be higher compared to a device that has not reached the daily switching limit. This is defined by the Capacitor Switching (Exceed Daily Count) parameter on the corresponding problem formulation page.

If Reset Daily Switching Count toZero at Midnight is enabled from server configuration i.e **DNAF_LVM_RESET_CAP_TAP_COUNT** = 1, the switching count is reset to zero at midnight every day.

Because capacitor operation circuits may need time to discharge before the next operation, many capacitors have a minimum consecutive operation time limit. For example, after an operation, some capacitors may need to wait for 30minutes before the next operation. So, the LVM application needs to check the last capacitor switching time to determine if it is eligible for control again. Those values can be found on the Operator Attribute display of the capacitor. However, if the Respect Capacitor Consecutive Switching option is cleared, this check is disabled.

The LVM application respects the on and off sequences of capacitor banks. Depending on the capacitor bank status, only one or two individual capacitors in a bank are eligible controls for LVM. For example, for a capacitor bank that contains three individual capacitors, if all capacitors are Off, only the first capacitor (capacitor 1) is eligible for LVM control. However, if the capacitor 1 is On, and the capacitors 2 and 3 are Off, the capacitors 1 and 2 are eligible for LVM control: LVM can turn off the capacitor 1 or turn on the capacitor 2, however LVM cannot operate both capacitors at the same time. The other capacitors will be ineligible with the following reason: "This capacitor is part of a

capacitor bank and cannot be operated according to the on/off sequence".

The device measurements can be selected to determine the eligibility in real-time mode. The types of measurements for capacitors, regulators, and LTCs include status/step, supervisory cutout status, auto/manual status, remote/local status, and control available status. If the check box is selected, the measurement status and quality will be evaluated in determining the device eligibility.

For a capacitor to be eligible for LVM application control, it must satisfy all of these conditions:

- The operation mode of the capacitor is **remote control**. A locally controlled or fixed controlled capacitor is not eligible for LVM consideration.
- The capacitor's Application Eligibility flag must be **"Optimization Enabled"**. This can be achieved by setting the Optimization Enabled flag to True. In addition, if the SCADA Decide flag is set to True, and there is no corresponding control available measurement, or the value of control available measurement is True, then the device is also Optimization Enabled.
- There are no control inhibit tags on the capacitor.
- There is no flow imbalance at the neutral connection of a capacitor. The corresponding Neutral Imbalance status measurement either does not exist or its value is set to False.
- In the DNAF_PARAMS.xml file, the capacitor is listed as an **eligible control** (the Eligible Controls value on the problem formulation page of the DNAF Configuration Editor).
- The time between the last switching time and now is **greater than** the required minimum consecutive switching time (unless the Respect Tap Changer Consecutive Switching flag is set to False).
- If the Respect Daily Switching Count flag is set to True, then the number of daily switchings is a "hard" limit. The capacitor can be eligible only if the number of daily switching operations **has not exceeded** its maximum daily limit.
- The status measurement of the capacitor position **exists and the quality of the measurement is good**.
- If a capacitor is part of a capacitor bank, the capacitor **can change the status according to the on/off sequence defined for the bank**.

i Note

Sometimes capacitors in the substation are controlled by the Energy Management System (EMS). Therefore, such capacitors should not be controlled by LVM.

For an LTC/regulator to be eligible for LVM application control, it must satisfy all of these conditions:

- The operation mode of the regulating transformer is **remote control**. A locally controlled or fixed regulating transformer is not eligible for LVM consideration. This rule applies only if the respect Remote/Local measurement check box is checked.
- The control mode of the regulating transformer is dependent on the control capability. If the transformer is expecting tap raise/lower control, then the transformer should be operated in **Manual mode**. If the transformer is expecting voltage setpoint or voltage reduction control, the

transformer should be operated in **Auto mode**. This rule applies only if the respect Auto/Manual measurement check box is checked.

- The LTC/regulator's Application Eligibility flag must be "**Optimization Enabled**". This can be achieved by setting the Optimization Enabled flag to True. In addition, if the SCADA Decide flag is set to True, and there is no corresponding control available measurement, or the value of control available measurement is True, then the device is also Optimization Enabled.
- There are no control inhibit tags on the regulating transformer.
- In the DNAF_PARAMS.xml file, the regulating transformer is listed as the **eligible control** (the Eligible Controls value on the problem formulation page of the DNAF Configuration Editor).
- If the tap is running in the tap raise/lower mode, the analog measurement of the regulating transformer tap position must exist and the quality of the measurement must be good. Otherwise, the tap estimation option must be on and the tap position must be successfully estimated by the DPF application.
- If the tap is running in the voltage setpoint mode, the voltage setpoint measurement of the regulating tap must exist and the quality of the measurement must be good.
- If the tap is running in the voltage reduction mode, the voltage reduction measurement of the regulating tap must exist and the quality of the measurement must be good.
- If the Respect Daily Switching Count flag is set to True, then the number of daily switching operations **has not exceeded** its maximum allowed value. If the flag is set to False, then the switching count is a "soft" constraint that can be violated.
- If the plan implementation method is set to Fast Execution with Delayed Confirmation and the tap is running in the raise/lower tap mode, the regulating transformer tap position analog measurement must exist and must be valid.

For a generator/DER to be eligible for LVM application control, it must satisfy all of these conditions:

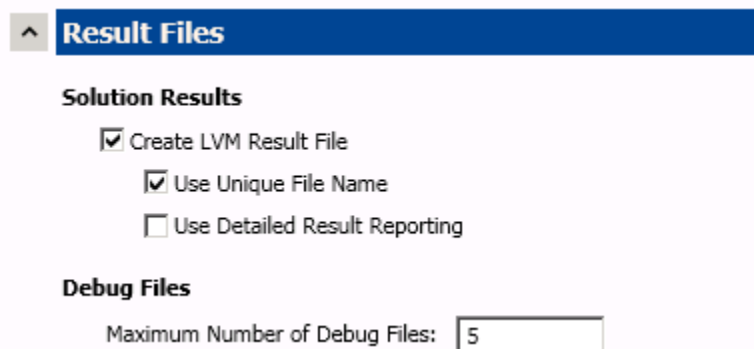
- The generator/DER's Application Eligibility flag must be "**Optimization Enabled**". This can be achieved by setting the Optimization Enabled flag to True. In addition, if the SCADA Decide flag is set to True, and there is no corresponding control available measurement, or the value of control available measurement is True, then the device is also Optimization Enabled.
- There are no control inhibit tags on the generator/DER.
- In the DNAF_PARAMS.xml file, the generator/DER is listed as the **eligible control** (the Eligible Controls value on the problem formulation page of the DNAF Configuration Editor).
- The kVAR/PF setpoint measurement of the generator/DER must exist and the quality of the measurement must be good.
- The minimum range of kVAR output of a generator or DER can be specified for the devices to be considered eligible. This kVAR output range is determined by the device capability curve and the current real power output. If this value is too low, this may cause not effectively controlled devices to be included in LVM analysis. This may cause the LVM solution to be Infeasible. If the value is too large, this may prevent smaller generators and DER from being used for VAR control. The kVAR output range value is computed as: $Q_{\max} - Q_{\min}$, where $Q_{\max}$ and $Q_{\min}$ are the limits from the device capability curve.

General LVM auto-disabling reasons include the following:

- Feeders and station transformers are not eligible for LVM if the station transformer has an invalid voltage measurement (including the load-side voltage measurement).
- A feeder is not eligible for LVM if the associated transformer has another feeder disabled on the same LTC.
- A transformer is not eligible for LVM if all of its feeders are not eligible for LVM.

3.10.2. LVM Solution Reporting

Figure below shows the options that determine how LVM generates solution files. If the Create LVM Result File option is selected, a text file containing the LVM report is generated when LVM has finished execution. The text file includes the current system conditions and solution results. Selecting the "Use Unique File Name" option forces the result file to be saved using a unique file name, which is constructed using the LVM execution time.



The screenshot shows a web-based configuration interface for LVM solution reporting. At the top, there is a blue header bar with a small upward-pointing arrow icon and the text "Result Files". Below this header, the interface is divided into two sections. The first section, titled "Solution Results", contains three checkboxes: "Create LVM Result File" (checked), "Use Unique File Name" (checked), and "Use Detailed Result Reporting" (unchecked). The second section, titled "Debug Files", contains a label "Maximum Number of Debug Files:" followed by a text input field containing the number "5".

Figure 54. Options for Reporting LVM Solution

3.11. Incorporating Voltage Measurements from Advanced Metering Infrastructure

LVM can use load voltage measurements from the Advanced Metering Infrastructure (AMI) system. To use these voltage measurements effectively, before solving LVM, recent voltage measurements from AMI meters are needed. Therefore, the ability to perform pre-solve AMI read requests has been added to LVM. Once the DNAF server is properly configured for LVM pre-solve AMI read, and a request to solve LVM comes in, whether by periodic trigger, event trigger, or user request, the Bellwether Meter Selection Algorithm is performed. The objective of this algorithm is to determine a small representative set of AMI loads for which a request for fresh voltage measurements will be sent out.

When the Bellwether Meter Selection Algorithm has recommended a set of AMI loads, the DNAF server sends out an analog scan request for these meters and waits for a configured amount of time. During this wait time, further requests for LVM for the station under consideration are

declined.

To enable the AMI voltage read function, the following parameters must be set in the DNAF server configuration file.

- Set **DNAF_AMI_LVM_PRESOLVE_READ_ENABLE** to 1 to identify bellwether meters and subsequently send out the pre-solve AMI read request.
- **DNAF_AMI_LVM_PRESOLVE_READ_WAIT_TIME** specifies the wait time in seconds after the pre-solve AMI read request has been sent before LVM execution can be resumed. Select a reasonable value depending on the communication latency in the AMI infrastructure.

Also, note that for AMI to work, additional customer-only parameters are needed in the DNAF server configuration file. Once the wait time elapses, the DNAF server resumes LVM execution for the station. In the LVM optimization process, load voltage measurements with measurement timestamps within the "AMI Voltage Measurement Time Tolerance" (from the Load and Volt/VAR Management page of the DNAF Configuration Editor) at the time of LVM execution are deemed recent and are used in the optimization process. Measurements with timestamps that do not meet the recent measurement criterion are not used.

The Bellwether Meter Selection algorithm and the usage of load voltage measurements for Optimization LVM and for Heuristic LVM are described in the following sections.

3.11.1. Bellwether Meter Selection

The bellwether meter selection is performed on each LVM-enabled feeder for the station under consideration. Voltage monitoring-enabled AMI loads are identified in the network selection and are ranked using the method described below.

$$Rank = W_1 * T_{Vcalc} + W_2 * T_{Phase} + W_3 * T_{Location} \quad (\text{Equation 3.18})$$

Where:

- W_1 = Weight for the Calculated Voltage Component
- W_2 = Weight for the Phase Diversity Component
- W_3 = Weight for the Location Component

These weights are specified in the "Bellwether Meter Selection Algorithm" section of LVM problem formulations in the DNAF Configuration Editor.

$T_{Vcalc} = 0$ for heuristic LVM. For optimization LVM, T_{Vcalc} is calculated as follows:

$$T_{Vcalc} = V_{Calc} - V_{limit,average} \quad (\text{Equation 3.19})$$

Where:

V_{Calc} : Calculated load voltage (average if polyphase).

$V_{limit,average}$: Average of high and low load voltage limits.

$$T_{Phase} = D_{high/low} * \frac{N_{ph}}{N_{loads}} \quad (\text{Equation 3.20})$$

Where:

$D_{high/low} = 1$ if $V_{Calc} > V_{Limit,average}$ (Optimization LVM); $D_{high/low} = -1$ if $V_{Calc} < V_{Limit,average}$ (Optimization LVM); $D_{high/low} = 1$ (Heuristic LVM).

N_{ph} : Number of AMI loads on the feeder with same phase.

N_{loads} : Number of AMI loads on the feeder.

$$T_{Location} = D_{high/low} * \frac{1}{N_{prgroup}} \quad (\text{Equation 3.21})$$

Where:

$N_{prgroup}$: Number of AMI loads on the feeder with same upstream protective device

Of these meters, at most X high-ranking and at most Y low-ranking meters are selected per LVM-enabled feeder. The DNAF server sends out a read request for the selected meters. The parameters X and Y are specified by the "Maximum Number of High Ranking Meters per Feeder" and "Maximum Number of Low Ranking Meters per Feeder" parameters, respectively, in the Bellwether Meter Selection Algorithm section of the LVM Problem Formulations in the DNAF Configuration Editor.

3.11.2. Load Voltage Measurement Usage in Optimization LVM

DPF calculated AMI load voltages, with voltage drop taken into account, are used in the objective functions for potential violations with respect to specified voltage limits. A separate set of violation penalty factors has been introduced for loads. This way, different weights can be assigned for bus voltage violations vs load voltage violations. For example, in figure below, bus voltage violations have higher priority than load voltage violations.

Violation Penalties and Costs	
Bus Low Voltage Violation:	10000000
Bus High Voltage Violation:	20000000
Load Low Voltage Violation:	100000
Load High Voltage Violation:	200000

Figure 55. Separate Voltage Violation Penalties for Buses and Loads

Voltage limit adjustment can be performed to take into account discrepancies between measured and calculated voltages, thereby changing the effective voltage of the AMI loads. The Limit Strategy section for LVM Problem Formulations has been enhanced to provide more granular controllability

over bus and loads. For example, in figure below, regular voltage limit adjustment is applied to loads, while no voltage limit adjustment is performed for buses.

Voltage Limits

	Applicable Limit	Limit Adjustment
Buses with Voltage Measurements:	None	None
Buses without Voltage Measurements:	None	N/A
Loads with Voltage Measurements:	Normal Operational	Regular
Loads without Voltage Measurements:	Normal Operational	N/A

Figure 56. Specifying Voltage Limit Adjustment Method for Loads

3.11.3. Load Voltage Measurement Usage in Heuristic LVM

In Heuristic LVM, AMI load voltage measurements that are considered recent are used in the same way as modelled bus voltage measurements coming in via SCADA. The only difference is that AMI voltage measurements are not used in the Voltage Monitoring Function described in [Voltage Monitoring Function](#). These voltage measurements are only used in the capacitor control and in the LTC/regulator control functions.

3.12. Real-Time Modes of Operation

Additional details about the real-time mode LVM application are discussed in this section.

3.12.1. Enable/Disable LVM

This section describes enabling and disabling components for LVM.

3.12.1.1. LVM

3.12.1.2. Auto Disabling Checks

LVM auto disabling pre-processing checks are configured on the LVM page on the LVM page of the DNAF Configuration Editor.

Auto Disabling Checks

☐ Allow Automatic Disabling by Feeder and Station	
☒ Substation Maximum Uncontrollable Devices:	3
☒ Maximum Feeder Uncontrollable Tap Changers:	1
☒ Maximum Feeder Uncontrollable Capacitors:	1
☐ Disable Other Feeders Connected to the Same LTC if Any Feeder is Disabled	
☐ Disable LTC if All Connected Feeders are Disabled	
☐ Disable LTC and Downstream Feeders if Load-Side Voltage Measurement is Invalid	
☐ Enable Voltage Monitor Safety Check	
Maximum Voltage Difference (%):	4

Figure 57. LVM Auto Disabling Checks

If the Allow Automatic Disabling by Feeder and Substation option is selected before the LVM application is run, the application tries to find the StatVVCReady SCADA LVM status measurement first. If the measurement does not exist, the LVM continues to run as usual. However, if the SCADA measurement value is False, which means the LVM application is disabled, then the LVM execution is skipped. With this method, the operator can manually enable/disable LVM runs by simply changing the value of the status measurement.

In some cases, if communication errors are observed among multiple control devices, it may be useful to disable LVM automatically. The Substation Maximum Uncontrollable Devices value is used to specify the maximum number of SCADA controllable devices within the substation that may be out of service before LVM is disabled. For example, if the value is zero, once the Control Available measurement is changed to False for the first controllable device, LVM is disabled automatically. Similarly, you can specify the maximum number of uncontrollable LTC/regulators and capacitors for feeders using the Maximum Feeder Uncontrollable Tap Changers and Maximum Feeder Uncontrollable Capacitors options.

If the Disable Other Feeders Connected to the Same LTC if Any Feeder is Disabled option is selected, and a feeder is found ineligible for LVM, all feeders downstream of the power transformer to which the given feeder is connected are considered ineligible for LVM.

The "Disable LTC and Downstream Feeders if Load-Side Voltage Measurement Is Invalid" option is used in conjunction with LTCs that do not have a tap position analog measurement. For such LTCs, the tap position can be estimated by enabling the "Perform Tap Estimation" option on the DPF page of the DNAF Configuration Editor. Depending on the economics of the case, if an LVM plan is generated involving such an LTC, the only way to confirm a tap change operation is to monitor the load-side voltage measurements. If this option is enabled, and load-side voltage measurements associated with an LTC without tap position measurement are invalid, then the LTC and its downstream feeders are made ineligible for LVM. The ineligible status is automatically removed as soon as the load-side voltage measurements are found to be of valid quality.

Setting the Enable Voltage Monitor Safety Check option allows the user to specify the maximum voltage difference (in %) for any node in a station that has a valid voltage measurement. If this relative value of the difference between the DPF calculated voltage for the node and its voltage

measurement is greater than the value specified in this field, then the voltage monitoring safety check for the station is said to have failed. In this case, LVM is not solved for that station.

3.12.1.3. Enable/Disable LVM for Feeder

Every feeder has an LVM Enable/Disable flag, which is defined by the EnableLVM parameter for a feeder in the station external XML file (or the ENABLELVM_FEEDER value in the station CSV file).

- If the flag is set to True, the feeder is considered during the LVM process.
- If the flag is set to False, both the control device on this feeder and the voltage/current limits are not considered in the LVM process. However, those feeders are still included in station-based targets, such as station demand reduction and area power factor support.

3.12.1.4. Enable/Disable LVM for a System Element

You can change the LVM Enable/Disable status for the entire system, optimization group, division, station, power transformer, or feeder at run time via the Manual Enable column on the LVM page in the DNAF System Configuration display. The LVM application can also be enabled/disabled by feeder from the Hazard Management System (HMS).

3.12.2. Plan Execution Mode

The effective plan execution mode for a generated LVM plan is determined by several factors. The following settings are considered in the decision process:

- The "Run in Closed Loop" option in the DNAF Configuration Editor, and
- Individual station and bank level plan execution mode settings on the DNAF System Configuration display.

Note

The execution mode settings on the DNAF System Configuration display override the default execution mode setting specified in the DNAF Configuration Editor.

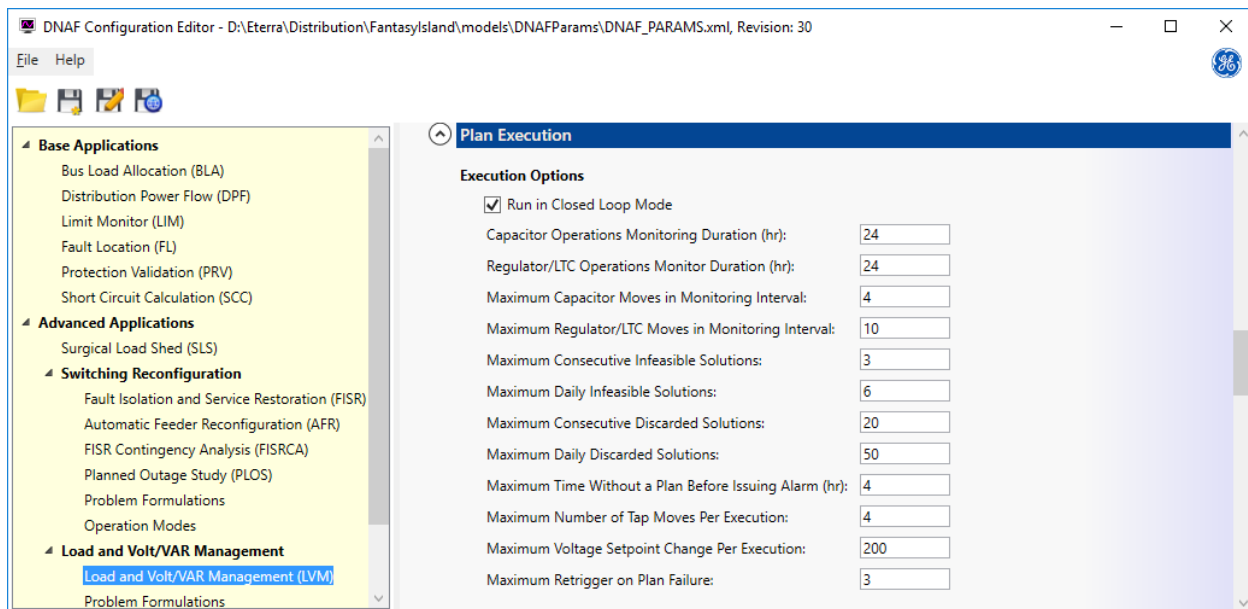


Figure 58. LVM Plan Execution

The following rules determine whether the plan should be executed in closed loop or if the plan is considered advisory only.

- If the station is in advisory mode (either from the DNAF Configuration Editor option or via override from the DNAF System Configuration display) the plan is considered advisory and will not be executed in closed loop.
- If the station is set to "closed loop", the individual plan steps are checked and the most restrictive mode from all device moves in the plan is selected. The effective execution mode for a device move is determined as follows:
 - If an upstream energizing power transformer can be found for a device move in the device's station, the upstream energizing power transformer's plan execution mode is used. Otherwise, the device move's station's plan execution mode is used.
 - If the device move is a station power transformer, the power transformer's own plan execution mode is checked.

3.12.3. Closed Loop Mode

If the effective execution mode for a generated LVM plan is closed loop, the generated plan is sent to the SCADA system directly, where the plan is executed automatically without the operator's involvement.

The capacitor control result can be monitored by a status measurement associated with it.

- If a capacitor is part of a capacitor bank and does not have an associated connection switching device, the control is sent to the capacitor and the control result is monitored by the capacitor status measurement.

- If the capacitor is part of a capacitor bank and has an associated connection switching device, the control is sent to the switching device and is monitored by the switching device status measurement instead of the capacitor status measurement.

In closed loop mode, the switching plan can be executed either all at once or one step at a time. The Step-by-Step option on the LVM > Operation Modes page is used to determine how the plan steps are executed. For the options to execute a plan with or without confirmation, the moves in the plan are sent out at the same time; otherwise, the moves are sent out sequentially, waiting for a response from the previous step before sending out the next move.

In Step-by-Step mode, the LTC/regulator movement is sent out as an accumulation of raise/lower controls. For example, tap movement from 3 to 5 requires two raise controls. LVM sends out one raise control first, waiting for the tap position measurement change from 3 to 4, and then sends out another raise control.

An LTC/regulator voltage setpoint is sent out as one setpoint control. The setpoint voltage analog measurement is monitored. If the voltage measurement value changes to the expected setpoint value within the voltage deadband, the setpoint control is considered successful.

A generator/DER kVAR or PF setpoint is sent out as one setpoint control. The setpoint kVAR or PF analog measurement is monitored for step confirmation. If the measurement value changes to the expected setpoint value, the setpoint control is considered successful.

The Plan Time Out parameter on the LVM Operation Modes page of the DNAF Configuration Editor determines the waiting period between each control before it reports a timeout. If LVM cannot receive the capacitor status measurement, tap position analog measurement change or voltage setpoint analog measurement change before the specified timeout period, LVM stops, marks the plan as failed, and does not continue to execute the plan. If the timeout parameter is zero, LVM continues waiting until the next LVM execution. At that time, the previous pending plan is deleted.

3.12.3.1. Indirectly Confirming a Capacitor Operation

In the case of indirect verification, the capacitor control result can be obtained by monitoring the reactive power measurement change upstream of the capacitor. This option can be enabled by the Monitor Upstream Reactive Power option on the relevant operation mode for the current problem formulation in the DNAF Configuration Editor.

When finding a reactive measurement used for confirmation, if there is a reactive measurement associated with the capacitor directly, that measurement is used. Otherwise, the system searches upstream for the first usable VAR measurements (three-phase and/or three single-phase VAR measurements) along the feeder up to the feeder head or the station power transformer, and associates those measurements with the capacitor.

The tracing logic takes SCADA quality and phasing of the VAR measurements into account. For example:

- If the upstream trace found a usable three-phase VAR measurement, but unusable single-phase measurements, then the tracing will continue to try and identify usable single-phase VAR

measurements from upstream locations along the feeder.

- LVM prioritizes single phase measurements over three-phase VAR measurements for indirect confirmation. In other words, if three single-phase VAR measurements and a three-phase VAR measurement were identified by the tracing logic, then the three single-phase measurements are used for indirect confirmation.
- If no usable single-phase VAR measurements could be found for the given capacitor, then the only available measurement for indirect confirmation is the single three-phase VAR measurement.

If no valid VAR measurement can be found for a capacitor, the indirect confirmation is not performed, and the step is marked as successful or failed depending only on the capacitor status value.

Indirect Measurement Confirmation

☒ Monitor Upstream Reactive Power

VAR Monitoring Criteria - High (%):

VAR Monitoring Criteria - Low (%):

☐ Monitor Load Side Voltage for Tap Position/Voltage Setpoint/Voltage Reduction Confirmation

For Tap Position Confirmation Only:

Maximum Voltage Change (%):

Minimum Voltage Change (%):

Figure 59. Indirectly Confirming a Capacitor Operation

For example, if the capacitor size is 100 kVAR and the VAR monitoring criteria in DCE is set to 70% to 120% as shown in figure above, then after the control command is sent, the plan manager continues to monitor the associated measurement value change and compare it to the recorded value before the control command is sent out. Before the capacitor step out timer expires, if the measurement change value is between 70 kVAR and 120 kVAR, then the plan step is marked as successful. Otherwise, the Plan Manager either resends the control or marks the step as failed if the maximum retry count for the capacitor is reached.

3.12.3.2. Load-Side Voltage Monitoring for Confirming Tap Changer Operation

For LTCs/regulators without a tap position measurement, voltage monitoring of the telemetered load-side voltage is the only indication that the tap position, voltage setpoint, or voltage reduction control was successfully executed for that LTC/regulator during LVM plan execution.

To enable voltage monitoring, the "Monitor Load Side Voltage for Tap Position/Voltage Setpoint/Voltage Reduction Confirmation" option must be selected on the LVM Operation Modes page of the DNAF Configuration editor, as shown in figure below.

-

For tap position validation, the associated parameters "Maximum Voltage Change (Tap Position Only) (%)" and "Minimum Voltage Change (Tap Position Only) (%)" must be set correctly.

- For voltage setpoint and voltage reduction controls, the confirmation is performed using the target voltage changes and actual voltage measurements after each control step.

If voltage monitoring cannot confirm the tap change operation, the plan step is marked as failed and the plan is aborted.

For example, if the LTC/regulator in question has a tap step change of 0.625% V, then the maximum and the minimum voltage change % could be set to 0.75% and 0.50% respectively. During LVM plan execution, the voltage monitoring logic will store the load-side voltage measurements before sending the LTC/regulator RAISE/LOWER control action. After the control action is sent, the load-side voltage measurements will be monitored. In this specific example, the voltage monitoring logic will expect the new voltage measurements to be within 0.0050 p.u. and 0.0075 p.u. of the previously stored values. If the load-side voltage measurements are not in this range after the maximum number of control attempts has been reached, the plan step is marked as failed.

Note that if the LTC/regulator is a three-phase device, but load-side voltage measurements are three single-phase measurements, then the measurements for all the phases must be within the expected range. If any phase voltage measurement is not within the expected range after the maximum number of control attempts has been reached, the plan step is marked as failed.

If the option Monitor Load Side Voltage for Tap Position/Voltage Setpoint/Voltage Reduction Confirmation is cleared, then the LTC/regulator control action is assumed to have been successful.

Operation Modes

- Load and Volt/VAR Management
 - Load and Volt/VAR Management (LVM)
 - Problem Formulations
 - Operation Modes
- Plugin Applications
 - Configuration

Indirect Measurement Confirmation

☐ Monitor Upstream Reactive Power

VAR Monitoring Criteria - High (%): 100

VAR Monitoring Criteria - Low (%): 30

☒ Monitor Load Side Voltage for Tap Position/Voltage Setpoint/Voltage Reduction Confirmation

For Tap Position Confirmation Only:

Maximum Voltage Change (%): 0.75

Minimum Voltage Change (%): 0.5

Wait for Analog Delay (sec): 6

Figure 60. Load-Side Voltage Monitoring for Confirming Tap Changer Operation

3.12.4. Advisory Mode

The LVM application can also be run in advisory mode, which means that the LVM application only generates plans. The operator can check the generated plan and decide whether to implement the plan. LVM runs in advisory mode if the Closed Loop Mode option is cleared on the LVM page of the DNAF Configuration Editor. The advisory mode can also be selected from the Plan Execution drop-down menu on the DNAF System Configuration display.

i Note

The execution mode settings on the DNAF System Configuration display override the default

execution mode setting specified in the DNAF Configuration Editor.

3.12.5. LVM Pre-Implementation Plan Check

When implementing an LVM plan for a station, Plan Manager checks that all devices in the plan are in the correct initial position, are eligible, and are still energized by the same feeder. If a check is failed for any device, the plan is failed.

3.12.6. LVM MW and MVAR Rate of Change Throttling

When implementing an LVM plan for a station, Plan Manager can manage the MW increase/decrease rate of change and the MVAR increase/decrease rate of change by transmission area. The monitoring can be enabled or disabled from DCE. The change rate limit is defined for the transmission area in the station model file.

Plan Manager uses the forecast effects on MW and MVAR change for each plan to proactively schedule plan implementation.

In the following example, LVM generated four plans and they arrived in Plan Manager at different times:

- Plan 1: 3MW, arrived in time = 0s
- Plan 2: 2MW, arrived in time = 4s
- Plan 3: 6MW, arrived in time = 5s
- Plan 4: -1MW, arrived in time = 7s

The MW throttling rate in the area for both increase and decrease is 60MW/min, which equals 1 MW/sec.

The plans would be scheduled for implementation as follows:

1. At time = 0s, plan 1 arrives and starts execution. The throttling timer starts and is set to 3 seconds.
2. At time = 3s, the throttling timer is up; the timer stops.
3. At time = 4s, plan 2 arrives and starts execution. The throttling timer starts and is set to 2 seconds.
4. At time = 5s, plan 3 arrives. Because the throttling timer is still running, the plan will be held in the queue.
5. At time = 6s, the throttling timer is up. Plan 3 execution starts; the throttling timer restarts and is set to 6 seconds.
6. At time = 7s, plan 4 arrives. Because the throttling timer is still running, the plan will be held in the queue.
7. At time = 12s, the throttling timer is up. Plan 4 execution starts; the throttling timer restarts and

is set to 1 second.

8. At time = 13s, the throttling timer is up; the timer stops.

The graph below shows the implementation of these plans. The red line is for plan implementation. The blue line is the throttling limit. Note that the blue line always gets reset after the throttling timer is up, and starts from zero after that. The plan implementation will cause the red line to be above the throttling limit but the additional wait after the implementation will ensure the rate of change is within limit in the longer term.

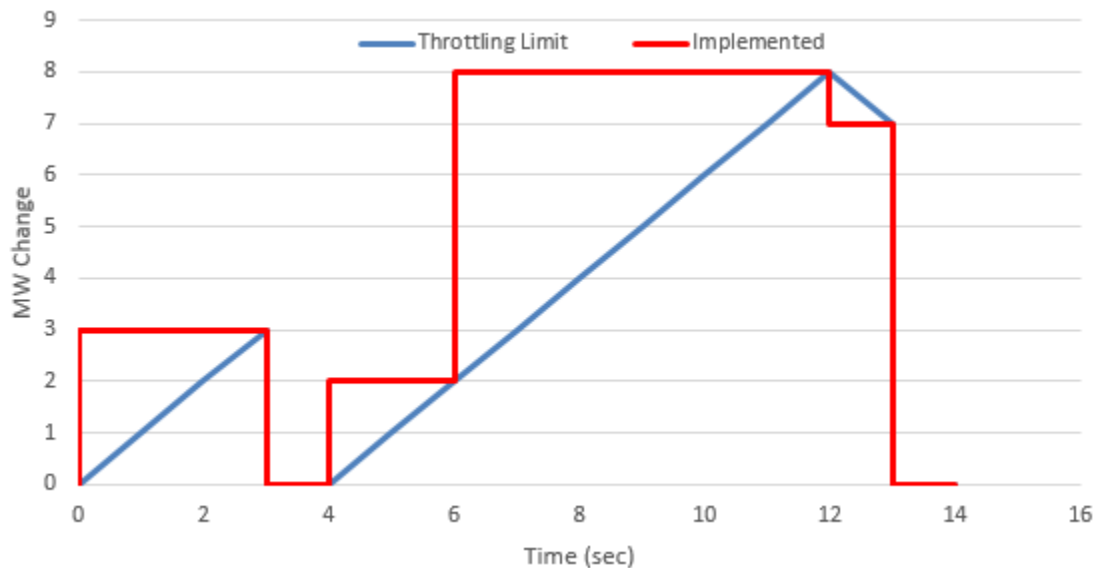


Figure 61. Plan Implementation Scheduling with Consideration of MW/MVAR Throttling

This schedule is conservative by assuming that power changes happen instantly at the initial time of plan implementation.

Both real and reactive power throttling limits are considered during implementation, with the most restrictive throttling limit determining the implementation schedule.

In addition to the scheduling described above, to ensure that throttling limits are respected, the LVM System Monitor will perform a secondary throttling check based on actual measured changes in load. Periodic checks will be used to compare actual measured power values to throttling limits. If limits are exceeded, the application will issue an alarm event indicating that throttling limits have been temporarily exceeded, and add a second delay during plan implementation. New plans will be suspended during the additional delay, to ensure that power changes drop below throttling rate limits. The delay time is configurable in DCE.

Note

The throttling function only schedules implementation of plans where implementation has not yet started. If a plan is sent for implementation and has already started at least one step before a throttling request was issued for a transmission area, the plan will be completed without throttling.

3.12.7. LVM Solution Sequence and Coordination

When LVM is solved for a station, LVM solution areas belonging to the fixed regulation LVM approach are solved first because a fixed voltage reduction setting on a tap changing transformer is a setpoint value, which is regulated, within the applicable bandwidth regardless of the movement of LVM controls in other parts of the station. However, the movement of LVM controls in other solution areas can be impacted by fixed regulation LVM implementations on neighboring feeders or transformer banks.

LVM solution areas belonging to the heuristic LVM approach are solved next. LVM solution areas belonging to the optimization LVM approach are solved last.

i Note

Although each LVM solution area is analyzed separately, the entire LVM network selection composed of individual solution areas is still an inter-connected network. When a plan is obtained from the solution of an LVM solution area, the plan steps are implemented in the internal unbalanced power system model and an internal power flow solution is obtained to provide the starting point for the next LVM solution area to be solved.

In case of looped conditions, the most restrictive approach is used to determine the effective problem formulation for an LVM solution area. The priority order follows the LVM solution area selection sequence: fixed regulation LVM problem formulations have the highest priority, followed by heuristic LVM problem formulations, which are followed by optimization LVM problem formulations.

The final plan generated by the LVM engine is a single plan. Plan steps coming from solutions of the individual LVM solution areas constitute sub-plans. The sub-plans are ordered in the final LVM plan in the order the corresponding LVM solution areas were solved. Each sub-plan indicates the corresponding power transformer bank name and the problem formulation.

If LVM is configured to be executed in closed-loop, the generated LVM sub-plans are sent to the plan manager in the same sequence as the solutions were obtained. Different problem formulations at different hierarchical levels can be associated with different operation modes. Each LVM sub-plan is presented to the plan manager as a separate plan. The plan manager executes the plans in the sequence received from the LVM application.

3.12.8. LVM Control Failure

In real time, if the "Set the Control Failure Flag for Unresponsive Devices" option in the LVM Operation Modes page of the DNAF Configuration Editor is enabled, Plan Manager flags devices that experience control failure during a plan implementation with the LVM Control Failure flag. Any device with this flag is ineligible for future LVM solutions.

The control failure conditions that may arise while Plan Manager is implementing a step of an LVM plan are as follows:

- **Invalid Measurement:** Control measurement is of invalid quality.
- **Missing Measurement:** Control measurement cannot be found by Plan Manager.
- **Status not Changed:** The control status measurement was not changed after implementation by Plan Manager.
- **Analog Check Failed:** VAR/Volt post-implementation monitoring failed.
- **ISD Failure:** Connection with ISD/SCADA failed.
- **Invalid Object:** Device cannot be found by Plan Manager.
- **Time Out:** Step timeout before a status or analog check is performed.

Note that the LVM Control Failure flag is placed on a device if its associated step in an LVM plan fails during implementation by Plan Manager. The LVM Control Failure flag is not placed on a device if the plan is failed as a whole, as opposed to an individual step failing.

The LVM Device Exclusions List tabular displays lists devices that have the LVM Control Failure flag set. On these displays, you can reset the LVM Control Failure flag for the devices.

The DNAF_LVM_EVENT_PLANEXECFAILURE_TRIGGER parameter in NetServerDnafConfig.txt specifies whether LVM automatically triggers to execute again after a device control failure. This ensures that the system is not left in a bad state (such as one phase in a regulator group tapped up and the other two phases are down, resulting in a large voltage imbalance) until the next periodic LVM solution. Server log messages also indicate that LVM is automatically triggered due to a control failure during an earlier plan execution.

3.12.9. LVM Device Exclusions List

The LVM Device Exclusions List display shows devices that meet criteria specified in the LVM Device Exclusions List section on the LVM page of the DNAF Configuration Editor. The display shows all devices in the system (or station if viewing the station-specific display) that have at least one exclusion reason.

Note

A device having an exclusion reason does not mean that the device is ineligible for LVM. Rather, it indicates which devices would be ineligible for LVM if eligibility criteria were to be changed.

The list also allows the operator to reset devices for control failure.

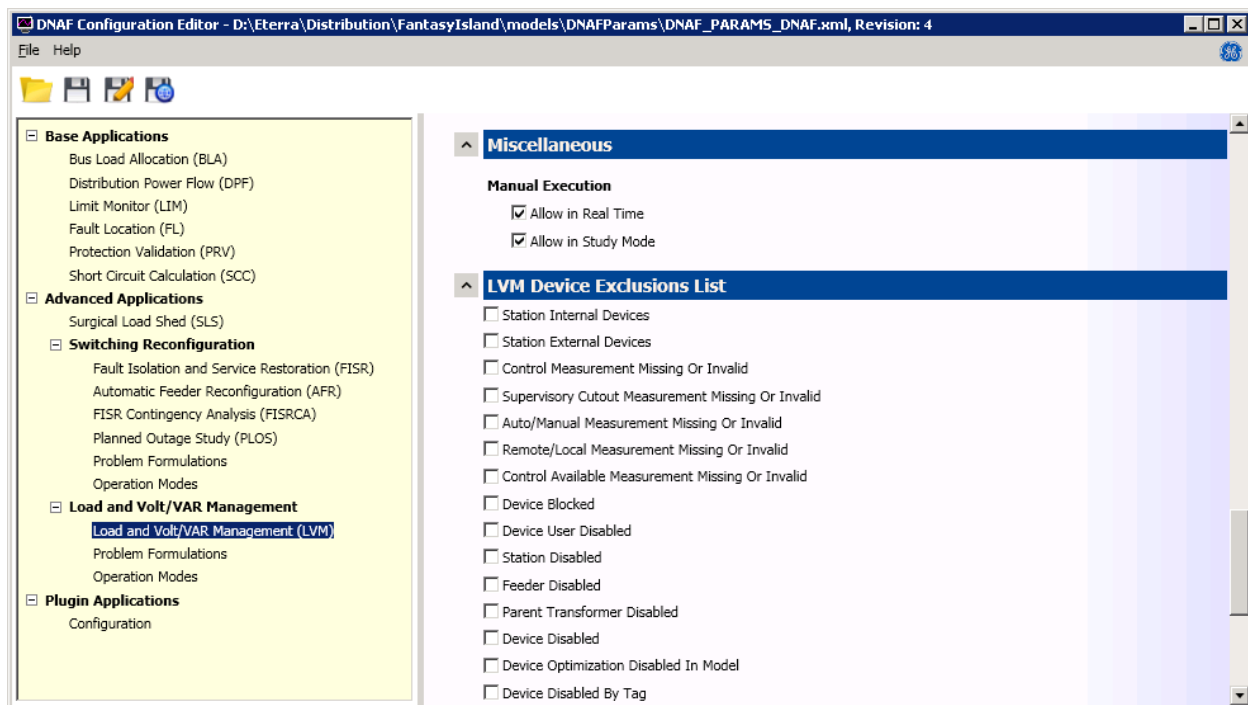


Figure 62. LVM Device Exclusions List Configuration

The list of currently supported exclusion reasons is as follows:

- Device is station internal
- Device is station external
- Control measurement is missing or invalid
- Supervisory cutout measurement is missing or invalid
- Auto/manual measurement is missing or invalid
- Remote/local measurement is missing or invalid
- Control available measurement is missing or invalid
- Device is blocked
- Device is user-disabled
- Device has manual exclusion
- Station is disabled for LVM
- Feeder is disabled for LVM
- Parent transformer is disabled for LVM
- Device is disabled for LVM
- Device is disabled for LVM in static model
- Device is disabled for LVM by tag
- Device has not met the consecutive operation time limit

- Device has met the maximum daily switching count
- Control failure occurred on the device
- Device is not in PQ mode
- Plan execution is not in advisory mode and device is only permitted for advisory mode plans
- Device has no control range
- Supervisory cutout measurement is on
- Control available measurement is off (unavailable)
- Neutral imbalance measurement is on for at least one phase
- Device is in auto mode
- Device is in manual mode
- Device is in local mode
- Optimization is not allowed on device
- Device does not have an energizing feeder

In addition, you can manually add or remove a device from the exclusions list using the appropriate device's shortcut menu options.

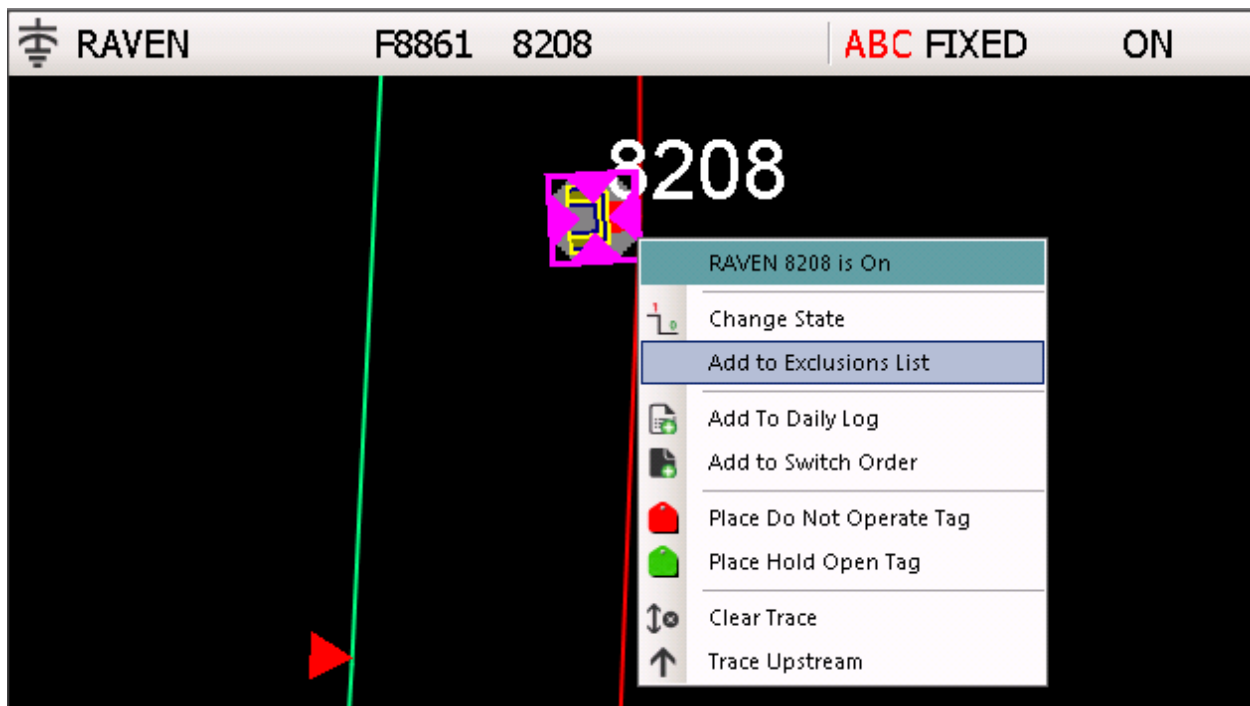


Figure 63. Adding a Device to the LVM Device Exclusions List

You can review the list of devices that have exclusion reasons on the LVM Device Exclusions List display, which you can access from the RT menu. To review all exclusion reasons, hover the mouse cursor over the Exclusion Reasons column. In addition, you can remove the LVM Control Failure flag and toggle the Control Processed flag for devices.

BAKER: LVM Device Exclusions List - Unprocessed						
Name	ID	Device Type	Energizing Station	Energizing Area	Energizing Optimization Group	Energizing Feeder
1875	10073884	Capacitor	BAKER	CTLB	SWTrans	F5462
2019	10073688	Capacitor	BAKER	CTLB	SWTrans	F5462
3692	10072358	Capacitor	BAKER	CTLB	SWTrans	F5467
3881	10072456	Capacitor	BAKER	CTLB	SWTrans	F5464
3884	10073604	Capacitor	BAKER	CTLB	SWTrans	F5462
3898	10072344	Capacitor	BAKER	CTLB	SWTrans	F5464
3948	10072484	Capacitor	BAKER	CTLB	SWTrans	F5464

BAKER: LVM Device Exclusions List - Processed						
Name	ID	Device Type	Energizing Station	Energizing Area	Energizing Optimization Group	Energizing Feeder
	ECD14953-C593-48...	Regulator	BAKER	CTLB	SWTrans	F5465
	BE54D4DA-72A0-4...	Regulator	BAKER	CTLB	SWTrans	F5465
	D295B806-C766-42...	Regulator	BAKER	CTLB	SWTrans	F5464
	74C77BD1-A8A7-4...	Regulator	BAKER	CTLB	SWTrans	F5461
	225F2495-0CFF-42...	Regulator	BAKER	CTLB	SWTrans	F5462

Figure 64. LVM Device Exclusions List Display

3.13. LVM System Monitor

The LVM System Monitor (LVMSM) allows users to easily monitor and control LVM execution across the entire system. It also provides the ability to evaluate potential changes to the system from implementing LVM plans, and then easily implement those changes.

3.13.1. Overview

A primary use case for the LVM System Monitor is to be able to apply voltage reduction schemes quickly, where regulating devices are used to reduce voltages on the system, thereby reducing the overall system demand. An operator would reduce demand by running the system normally and using the LVM System Monitor to continually analyze the effects of a voltage reduction scheme. Then when demand reduction is needed, the operator can easily implement the evaluated changes across the whole system, a desired optimization group (area) of stations, or a division.

The LVM System Monitor uses a separate LVM analysis server, which continually solves LVM for the current system conditions for a specified problem formulation profile. Problem formulation profiles are described in detail further below. A problem formulation profile has the same structure as the LVM problem formulation configuration provided in the DNAF System Configuration display. In a problem formulation profile, the operator can optionally specify problem formulations at the power transformer or the feeder levels, in addition to the problem formulation at the station level. Several problem formulation profiles can be created in the LVMSM. The operator can use any defined problem formulation profile to analyze post-LVM system conditions for the whole system, a desired optimization group, or a division. Solving LVM with a specified problem formulation profile using

the LVM System Monitor provides the operator with projected system conditions – such as demand, power factor, average voltage levels – if LVM were to be executed in closed-loop mode with that problem formulation profile. If a user chooses to implement a problem formulation change, the LVM analysis server will produce the necessary LVM plans, either from recent results, if available, or from a newly triggered analysis.

By default, LVM result files for plans that are generated by the LVM analysis server are stored in the %IDMS_ROOT%\log\Forecast folder. LVM System Monitor implementation result files are stored in the %IDMS_ROOT%\log\DNAF folder.

The LVM System Monitor can generate alarms to indicate the progress and state of an LVM mode implementation. Alarms can be generated when an implementation starts, ends, or is manually stopped, and when a given percentage of stations have completed. For alarm descriptions and configuration information, see [LVM System Monitor Configuration](#). Each alarm type is raised for each area included in the implementation.

Note

There is also a read-only version of the LVM System Monitor Overview display, where you can independently generate and study different evaluation scenarios without the ability to implement them.

3.13.2. Components

The LVM System Monitor is implemented as a plug-in that orchestrates execution and reporting among a number of components. Because it is implemented as a plug-in, it can be disabled if LVM System Monitor functionality is not needed. The Plugin Applications page of the DNAF Configuration Editor provides static configuration for the LVM System Monitor.

The DNAF Server is a client of the DMS Server and maintains the state of the system and runs DNAF analyses from various periodic, event, and manual triggers. The LVM System Monitor is used by the operator to override the current Real-Time LVM problem formulation configuration (viewable from the Real-Time DNAF System Configuration pop-up).

The LVM Analysis Server is a client of the DNAF Server and continually receives the current state of the system using the latest SCADA and real-time DMS dynamics. However, it does not run any analyses based on periodic or event triggers. Instead, it runs LVM analyses as requested by the LVM System Monitor plug-in to obtain LVM plans and post-plan system conditions for the specified problem formulation profile. The LVM Analysis Server runs with the ForecastSrv.exe executable using the NetServerForecastConfig.txt configuration file.

The LVM System Monitor plug-in runs within the DNAF Server as a module and does the core decision making for the LVM System Monitor. It requests LVM solutions as needed from the LVM Analysis Server. It triggers system-wide, area-wide, or optimization-group-wide closed-loop implementation of LVM plans and makes bulk changes to the current Real-Time LVM problem formulation configuration for the involved stations. It also provides data to the user interface, so users can monitor the current state of the system, and monitor LVM plan implementation progress

if triggered.

Note

You can also use a read-only version of the LVM System Monitor, which generates forecast plans, but cannot implement them.

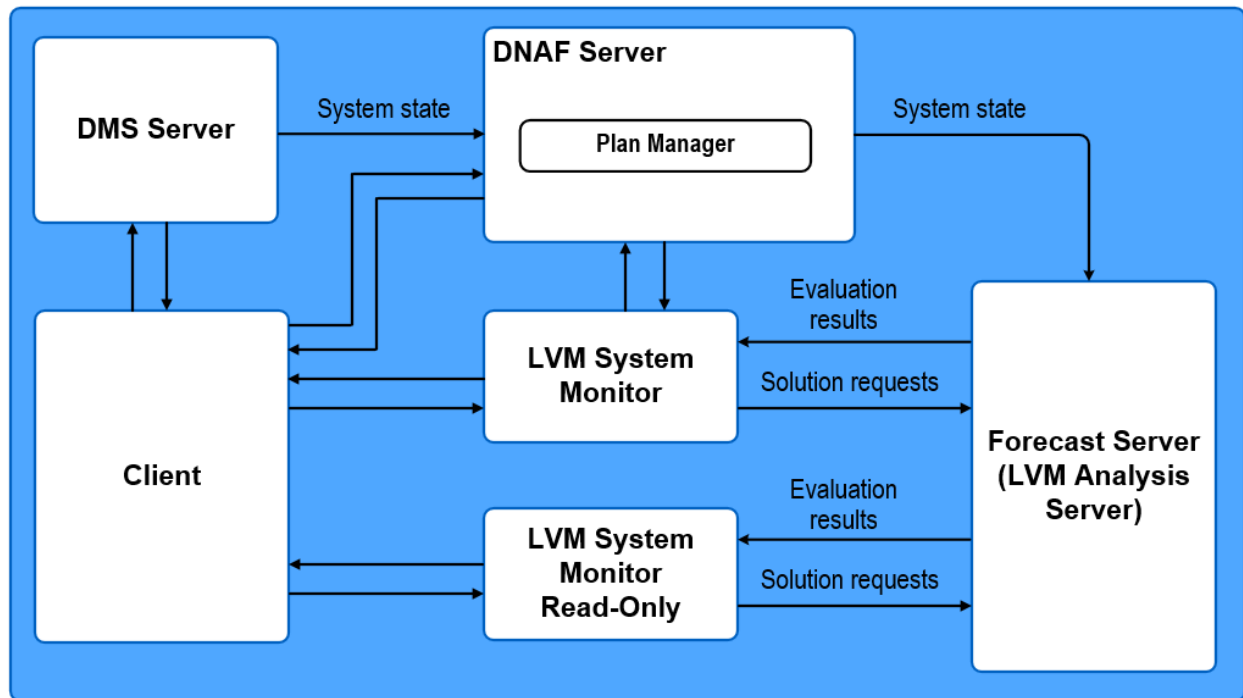


Figure 65. LVM System Monitor Flow

3.13.3. LVM System Monitor Display

To open the LVM System Monitor application, do one of the following:

- Click the DNAF Display Plugin option in the RT DMS menu.
- Use the following Netview command:

```
NETVIEW://PLUGIN.RT/DNAFDISPLAYPLUGIN
```

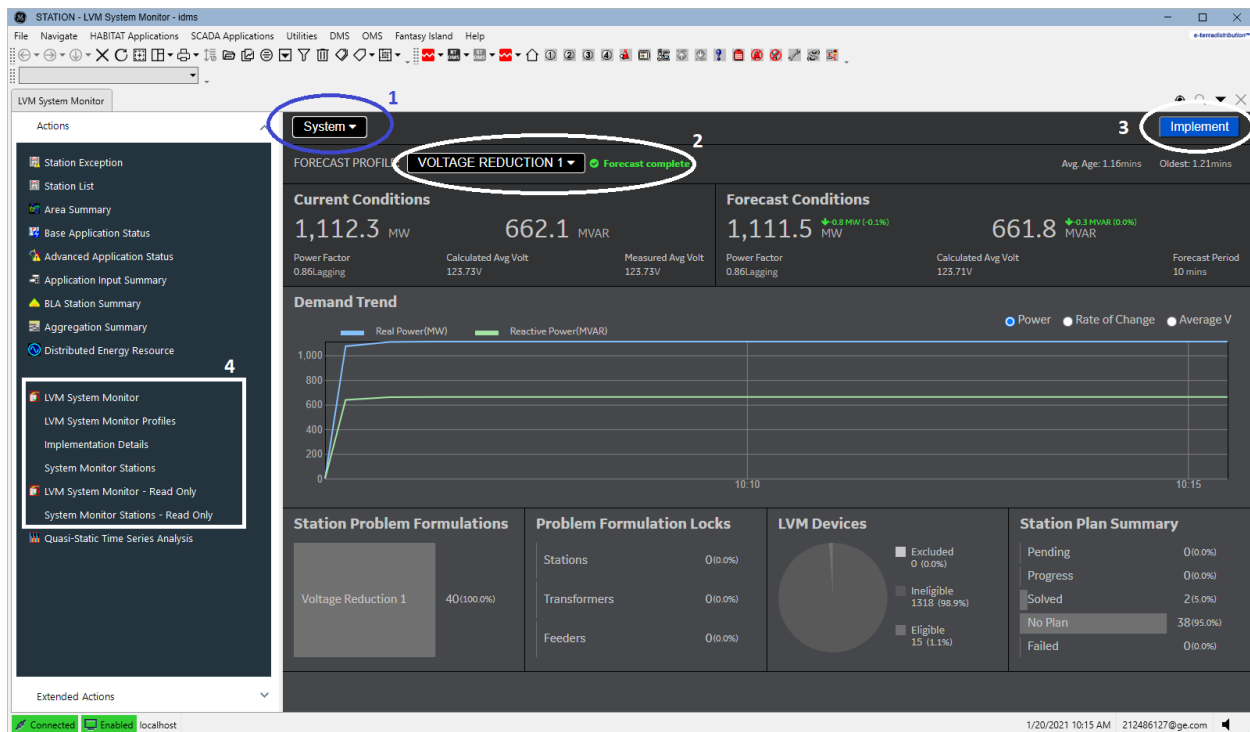


Figure 66. LVM System Monitor Display

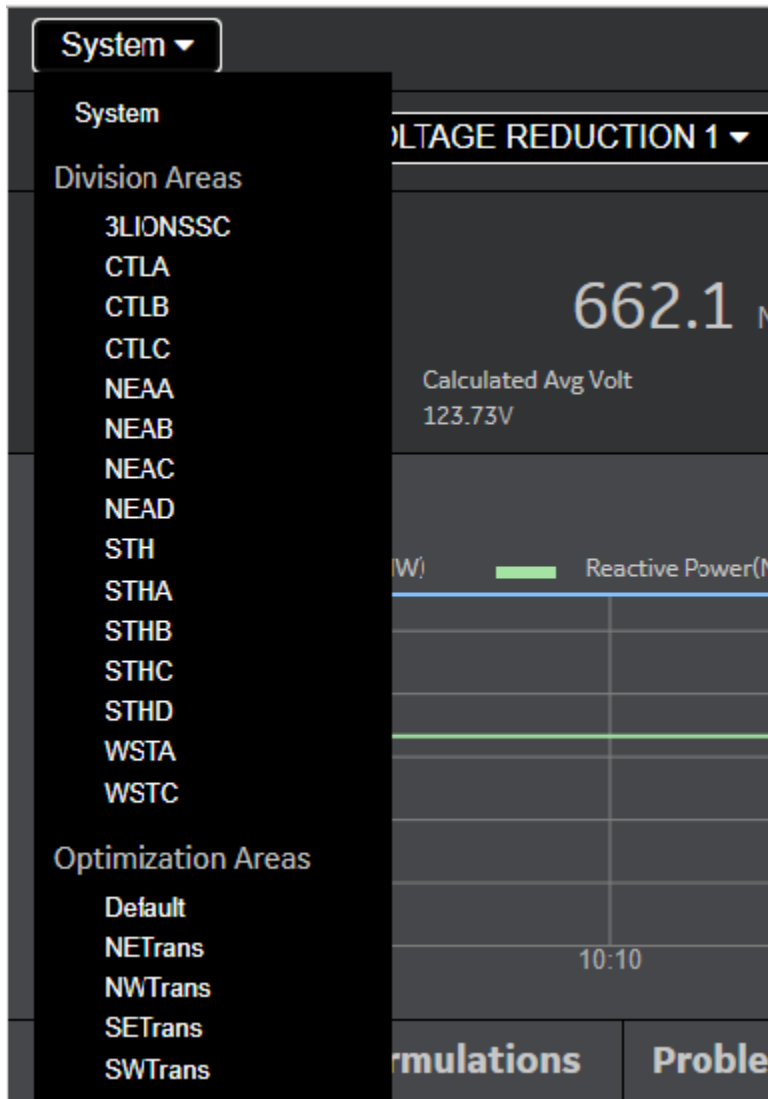
The LVM System Monitor main display includes the following components:

- The current conditions and the evaluated or post-plan conditions (labelled as "Forecast Conditions") for the entire system. The current and the evaluated or post-plan conditions are shown for the selected area.
- The "Demand Trend" section shows a continually updated chart for the selected area. The operator can choose to see real and reactive power or rate of change of real and reactive power or the average voltage for the selected area by using the option buttons in this section.
- The "Station Problem Formulations" section shows which problem formulations are being used in the LVM analysis for the selected area. The operator can specify different problems for different stations in a problem formulation profile.
- The "Problem Formulation Locks" section provides a summary of the real-time LVM problem formulations locks at station/bank/feeder levels for the selected area.
- The "LVM Devices" section shows the number of eligible/ineligible/excluded LVM devices for the selected area.
- The "Station Plan Summary" section shows the status of the current LVM analysis for the selected area.

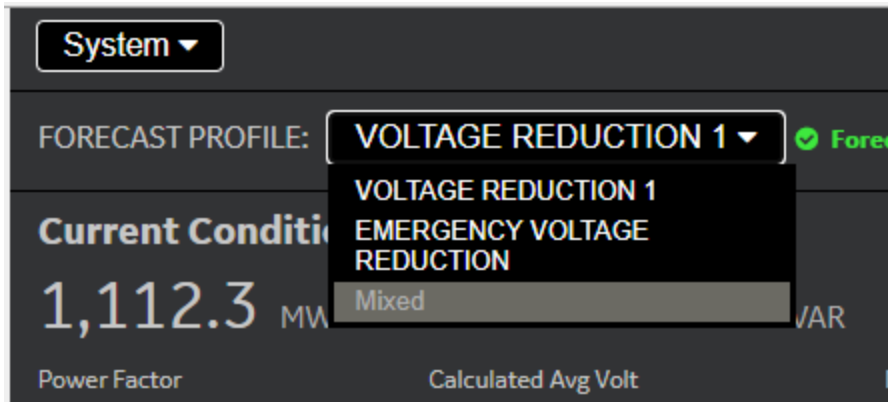
The following controls are available to the operator on this display:

- The area drop-down list in the upper-left corner of the display (label: 1) allows the operator to select Optimization Group or Division. The operator can specify whether to evaluate LVM for the entire system or a specified division or an optimization group. Changing the area using this

drop-down list also makes the other sections in this display show data pertinent to the selected area.



- The problem formulation profile selection drop-down list: using this drop-down list, the operator can specify the problem formulation profile to be used for performing an LVM analysis for the selected area.



- The "Implement" button allows the operator to directly implement a different problem formulation profile once the operator is satisfied with the results of the LVM analysis for the displayed LVM problem formulation profile or due to another reason, such as the need to implement voltage reduction.

The left pane provides navigation links to other displays used in the LVM System Monitor.

- "LVM System Monitor": Opens the main LVM System Monitor display.
- "LVM System Monitor Profiles": Opens the display that can be used to add, remove, or modify problem formulation profiles. A Problem Formulation Profile is described in more detail in a later section.
- "LVM Implementation Details": Opens the plan implementation details page. This is described in more detail in a later section.
- "LVM System Monitor Stations": Opens the station list.
- "LVM System Monitor – Read Only": Opens the read-only version of the main LVM System Monitor display. With this version of the display, the operator cannot make any changes to the real-time environment using the LVM System Monitor.
- "LVM System Monitor Stations – Read Only": Opens the read-only version of the station list.

The other links in the left side navigation pane are not related to the LVM System Monitor, but they provide other useful information regarding the real-time environment.

3.13.3.1. Problem Formulation Profiles Page

A problem formulation profile specifies the assignment of problem formulations at power transformer and feeder levels in addition to the station level problem formulation. Using a problem formulation profile defined on the LVM System Monitor, an operator can set up LVM evaluation scenarios for problem formulation configurations they are interested in. The operator can then look at LVM results obtained using the problem formulations in any given profile. When satisfied with the predicted results (such as resulting real/reactive power demand, power factor, and average voltage levels corresponding to the selected problem formulation profile), the operator can then enforce the problem formulation configuration in the real-time system by implementing the LVM plans obtained from this analysis. This action will change the real-time LVM problem formulations

for the stations in the selected area to the ones from the selected problem formulation profile.

Note that the LVM problem formulation configuration viewed on the real-time DNAF System Configuration display is the "active problem formulation profile". The LVM application in the DNAF server aims to maintain the parameters of the active problem formulation profile by generating LVM plans in response to changing system conditions.

Examples of LVM System Monitor profiles are shown below.

Example 1: Here, in the "Voltage Reduction 1" profile, a problem formulation is specified only at the station level. Obtaining LVM results using this profile is equivalent to solving the stations with a single station level specified problem formulation.

	Active Problem Formulation	Plan Execution	VOLTAGE REDUCTION 1
 System	--- Mixed ---	Default [Closed Loop]	Voltage Reduction 1
 3LIONS Station	Default [Reactive Support]	Default [Closed Loop]	Voltage Reduction 1
 BACHELOR Station	Default [Reactive Support]	Default [Closed Loop]	Voltage Reduction 1
 T1 Power Transformer	Default [None: no transformer...]	Default [Closed Loop]	None: no transformer objective
 F7931 Feeder	Default [None: no feeder obje...]		None: no feeder objective
 F7932 Feeder	Default [None: no feeder obje...]		None: no feeder objective
 F7933 Feeder	Default [None: no feeder obje...]		None: no feeder objective
 T2 Power Transformer	Default [None: no transformer...]	Default [Closed Loop]	None: no transformer objective
 F7934 Feeder	Default [None: no feeder obje...]		None: no feeder objective
 F7935 Feeder	Default [None: no feeder obje...]		None: no feeder objective
 F7936 Feeder	Default [None: no feeder obje...]		None: no feeder objective
 BAKER Station	Default [Reactive Support]	Default [Closed Loop]	Voltage Reduction 1
 BETHPAGE Station	Default [Reactive Support]	Default [Closed Loop]	Voltage Reduction 1
 CANTERBURY Station	Default [Reactive Support]	Default [Closed Loop]	Voltage Reduction 1
 COHO Station	Default [Reactive Support]	Default [Closed Loop]	Voltage Reduction 1

Example 2: Here, in the "Mixed Problem Formulations" profile, optimization LVM problem formulations are specified at some power transformers and feeders. Obtaining LVM results using this profile will result in enforcing the specified problem formulations at the indicated hierarchical levels. Specifically, problem formulation parameters and optimization goals from the respective problem formulations, such as minimize demand objective, power factor targets, reduce violation objective and bus/load voltage limits, as configured via the DNAF Configuration Editor are applied to that hierarchical level.

	Active Problem Formulation	Plan Execution	MIXED PROBLEM FORMULATIONS
 System	--- Mixed ---	Default [Closed Loop]	--- Mixed ---
 3LIONS Station	Default [Reactive Support]	Default [Closed Loop]	Minimize Demand
 BACHELOR Station	Default [Reactive Support]	Default [Closed Loop]	Minimize Demand
 T1 Power Transformer	Default [None: no transformer...	Default [Closed Loop]	None: no transformer objective
 F7931 Feeder	Default [None: no feeder obje...		None: no feeder objective
 F7932 Feeder	Default [None: no feeder obje...		Loss Minimization
 F7933 Feeder	Default [None: no feeder obje...		None: no feeder objective
 T2 Power Transformer	Default [None: no transformer...	Default [Closed Loop]	Reactive Support
 F7934 Feeder	Default [None: no feeder obje...		None: no feeder objective
 F7935 Feeder	Default [None: no feeder obje...		None: no feeder objective
 F7936 Feeder	Default [None: no feeder obje...		None: no feeder objective
 BAKER Station	Default [Reactive Support]	Default [Closed Loop]	Minimize Demand
 BETHPAGE Station	Default [Reactive Support]	Default [Closed Loop]	Minimize Demand
 CANTERBURY Station	Default [Reactive Support]	Default [Closed Loop]	Minimize Demand
 COHO Station	Default [Reactive Support]	Default [Closed Loop]	Minimize Demand

Example 3: Here, in the "Mixed Problem Formulations" profile, the configuration indicates an LVM solution involving mixed problem formulation types. "Voltage Reduction 1" and "Voltage Reduction 2" are fixed-voltage reduction type problem formulations. They do not involve obtaining a power flow solution, while "Minimize Demand" is an Optimization LVM type problem formulation requiring a power flow solution.

In this configuration, Feeder F7932 in the Bachelor station will be solved using the "Voltage Reduction 2" problem formulation, power transformer T2 in the Bachelor station and its downstream feeders will be solved using the "Voltage Reduction 1" problem formulation, and the rest of the stations will be solved using optimization LVM.

	Active Problem Formulation	Plan Execution	MIXED PROBLEM FORMULATIONS
 System	--- Mixed ---	Default [Closed Loop]	--- Mixed ---
 3LIONS Station	Default [Reactive Support]	Default [Closed Loop]	Minimize Demand
 BACHELOR Station	Default [Reactive Support]	Default [Closed Loop]	Minimize Demand
 T1 Power Transformer	Default [None: no transformer...	Default [Closed Loop]	None: no transformer objective
 F7931 Feeder	Default [None: no feeder obje...		None: no feeder objective
 F7932 Feeder	Default [None: no feeder obje...		Voltage Reduction 2
 F7933 Feeder	Default [None: no feeder obje...		None: no feeder objective
 T2 Power Transformer	Default [None: no transformer...	Default [Closed Loop]	Voltage Reduction 1
 F7934 Feeder	Default [None: no feeder obje...		None: no feeder objective
 F7935 Feeder	Default [None: no feeder obje...		None: no feeder objective
 F7936 Feeder	Default [None: no feeder obje...		None: no feeder objective
 BAKER Station	Default [Reactive Support]	Default [Closed Loop]	Minimize Demand
 BETHPAGE Station	Default [Reactive Support]	Default [Closed Loop]	Minimize Demand
 CANTERBURY Station	Default [Reactive Support]	Default [Closed Loop]	Minimize Demand
 COHO Station	Default [Reactive Support]	Default [Closed Loop]	Minimize Demand

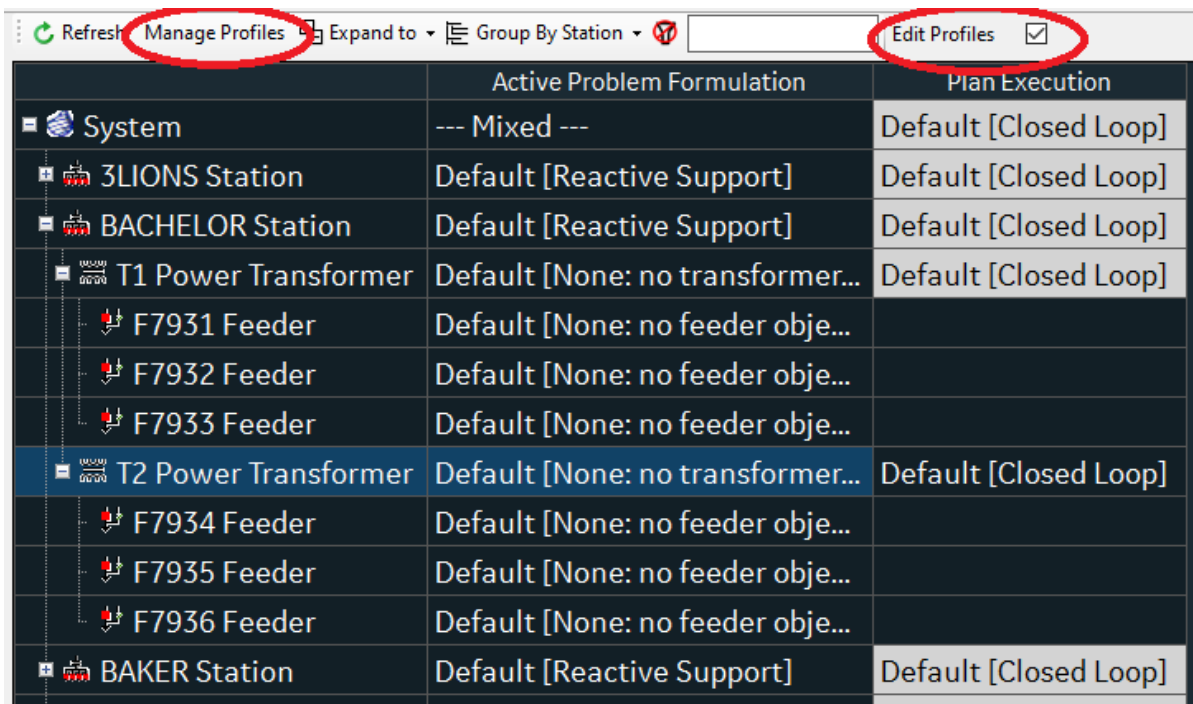
In addition to the LVM System Monitor problem formulation profiles, the LVM System Monitor

Profiles page also shows the current active real-time LVM problem formulation profile and the LVM plan execution mode (advisory or closed-loop) as viewed and configured on the real-time DNAF System Configuration display.

3.13.3.2. Managing Problem Formulation Profiles

The following steps can be used to add, remove, or modify problem formulation profiles in the LVM System Monitor.

1. Navigate to the LVM System Monitor Profiles display.



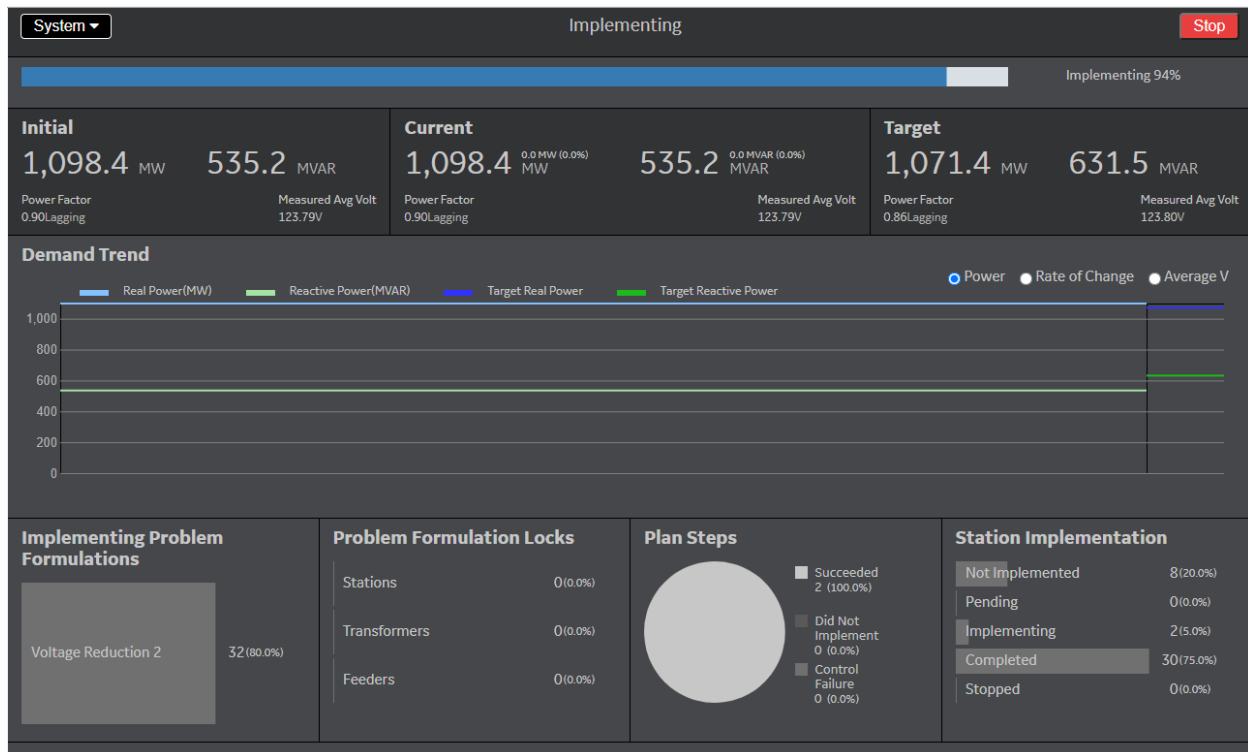
	Active Problem Formulation	Plan Execution
System	--- Mixed ---	Default [Closed Loop]
3LIONS Station	Default [Reactive Support]	Default [Closed Loop]
BACHELOR Station	Default [Reactive Support]	Default [Closed Loop]
T1 Power Transformer	Default [None: no transformer...]	Default [Closed Loop]
F7931 Feeder	Default [None: no feeder obje...]	
F7932 Feeder	Default [None: no feeder obje...]	
F7933 Feeder	Default [None: no feeder obje...]	
T2 Power Transformer	Default [None: no transformer...]	Default [Closed Loop]
F7934 Feeder	Default [None: no feeder obje...]	
F7935 Feeder	Default [None: no feeder obje...]	
F7936 Feeder	Default [None: no feeder obje...]	
BAKER Station	Default [Reactive Support]	Default [Closed Loop]

2. Select the Edit Profiles box. Unless this box is selected, problem formulation profiles cannot be changed.
3. Click Manage Profiles. The Manage Profiles dialog box appears.

7. Profiles page performs auto refresh to reload the data for all substations, power transformers and feeders. Each client can configure the desired refresh rate in seconds with parameter <LvmProfileRefreshRate> in file netconf_localhost.xml. Default value is 60.

3.13.3.3. LVM Implementation Details Page

This display shows the LVMSM plans implementation progress for the entire system or the selected division or optimization area.



- The area selection drop-down list in the upper-left corner of the display can be used to select the area of interest.
- The top portion of the display shows a continuously updated progress bar.
- The Stop button in the upper-right corner is used to cancel plan implementation.
- The "Initial" section shows the pre-plan real and reactive power demand, power factor, and average voltage values.
- The "Current" section shows the current real-time real/reactive power demand, power factor, and average voltage values as a result of the LVM plan implementation in progress.
- The "Target" section shows the projected or "post-plan" real/reactive power demand, power factor, and average voltage values.
- The "Implementing Problem Formulations" section shows the counts of various problem formulations used in the implementation.
- The "Problem Formulation Locks" section shows the number of real-time LVM problem

formulation locks at station/bank/feeder level.

- The "Plan Steps" section shows the number of plan steps that were completed successfully, cancelled, or encountered control failure during closed loop execution.
- The "Station Implementation" section shows the implementation status counts.

3.13.3.4. LVM System Monitor Stations Page

This display shows a grid-tabular view of the LVM evaluation state of the stations.

LVM System Monitor Stations																				
Station Name	Division	Forecast Problem Formulation	Forecast State	Plan Solved Time	Plan Steps	Eligible Devices	Ineligible Devices	Excluded Devices	Implementation State	Implemented Problem Formulation	Pre Implementation Problem Formulation	Measured Power Available	Measured Voltage Available							
3LIONS	3LIONSSC	Voltage Reduction 2	Solved - No Plan	2021/01/24 10:56...	0	0	27	0	Complete	Voltage Reducti...	Reactive Support	Yes	No							
BACHELOR	NEAD	Voltage Reduction 2	Solved - No Plan	2021/01/24 10:57...	0	2	14	0	Complete	Voltage Reducti...	Reactive Support	Yes	No							
BAKER	CTLB	Voltage Reduction 2	Solved - No Plan	2021/01/24 10:56...	0	0	51	0	Complete	Voltage Reducti...	Reactive Support	Yes	No							
BETHPAGE	NEAC	Voltage Reduction 2	Solved - No Plan	2021/01/24 10:55...	0	2	40	0	Complete	Voltage Reducti...	Reactive Support	Yes	No							
CANTERBURY	CTLB	Voltage Reduction 2	Solved - No Plan	2021/01/24 10:55...	0	0	15	0	Complete	Voltage Reducti...	Reactive Support	Yes	No							
COHO	WSTA	Voltage Reduction 2	Solved - No Plan	2021/01/24 10:55...	0	0	41	0	Complete	Voltage Reducti...	Reactive Support	Yes	No							
CORTLAND		Voltage Reduction 2	Solved - No Plan	2021/01/24 10:56...	0	0	9	0	No Implement...	--	--	Yes	No							
CURLWE	STHB	Voltage Reduction 2	Solved - No Plan	2021/01/24 10:56...	0	0	39	0	Complete	Voltage Reducti...	Reactive Support	Yes	No							
DURBAN	STHD	Voltage Reduction 2	Solved - No Plan	2021/01/24 10:56...	0	0	29	0	Complete	Voltage Reducti...	Reactive Support	Yes	No							
EQSI	STH	Voltage Reduction 2	Solved - No Plan	2021/01/24 10:54...	0	0	0	0	Complete	Voltage Reducti...	Reactive Support	Yes	No							
FISHER		Voltage Reduction 2	Solved - No Plan	2021/01/24 10:56...	0	0	14	0	No Implement...	--	--	Yes	No							
HEAVENLY	CTLA	Voltage Reduction 2	Solved - No Plan	2021/01/24 10:57...	0	0	23	0	Complete	Voltage Reducti...	Reactive Support	Yes	No							
JUPITER	CTLB	Voltage Reduction 2	Solved - No Plan	2021/01/24 10:55...	0	0	50	0	Complete	Voltage Reducti...	Reactive Support	Yes	No							
LABRADOR		Voltage Reduction 2	Solved - No Plan	2021/01/24 10:55...	0	0	0	0	No Implement...	--	--	Yes	No							
MARS	STHD	Voltage Reduction 2	Solved - No Plan	2021/01/24 10:55...	0	0	32	0	Complete	Voltage Reducti...	Reactive Support	Yes	No							
MERCURY	CTLA	Voltage Reduction 2	Solved - No Plan	2021/01/24 10:49...	0	0	32	0	Complete	Voltage Reducti...	Reactive Support	Yes	No							
MILLER		Voltage Reduction 2	Solved - No Plan	2021/01/24 10:50...	0	0	6	0	No Implement...	--	--	Yes	No							
MOBILE1		Voltage Reduction 2	Solved - No Plan	2021/01/24 10:48...	0	0	0	0	No Implement...	--	--	Yes	No							

3.13.3.5. LVM System Monitor - Read Only Page

This display connects to a read-only instance of the Forecast (LVM Analysis) server and shows the LVM System Monitor main display. The operator cannot make any changes to the real-time system using the read-only instance of the Forecast server.

3.13.3.6. LVM System Monitor Stations - Read Only Page

This display connects to a read-only instance of the Forecast (LVM Analysis) server and shows the System Monitor Stations display. The operator cannot make any changes to the real-time system using the read-only instance of the Forecast server.

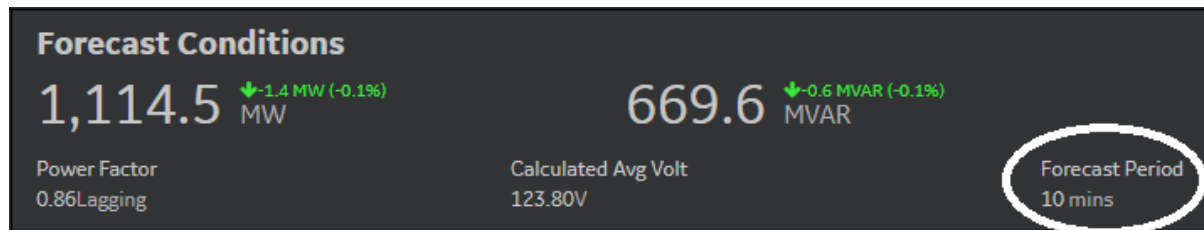
3.13.4. Configuring the LVM System Monitor

For more details about the LVM System Monitor configuration parameters, see [LVM System Monitor Configuration](#).

3.13.5. Obtaining LVM Evaluation Results

LVM results can be obtained via three approaches: periodic trigger, user request, or event triggers.

- **Periodic Triggering:** The LVM System Monitor periodically schedules stations to be solved for the currently selected problem formulation. This periodicity is displayed in the lower-right corner of the "Forecast Conditions" section. The periodicity can be configured on the plugins page of the DNAF Configuration Editor.



- **User Request:** Using the problem formulation profile selection drop-down list, the operator can change the LVM System Monitor problem formulation profile for analysis. This triggers new LVM solutions for the stations belonging to the area in the area selection drop down list.
- **Event Triggering:** Topology changes in the real-time environment make the LVM System Monitor automatically trigger new LVM solutions for the impacted stations.

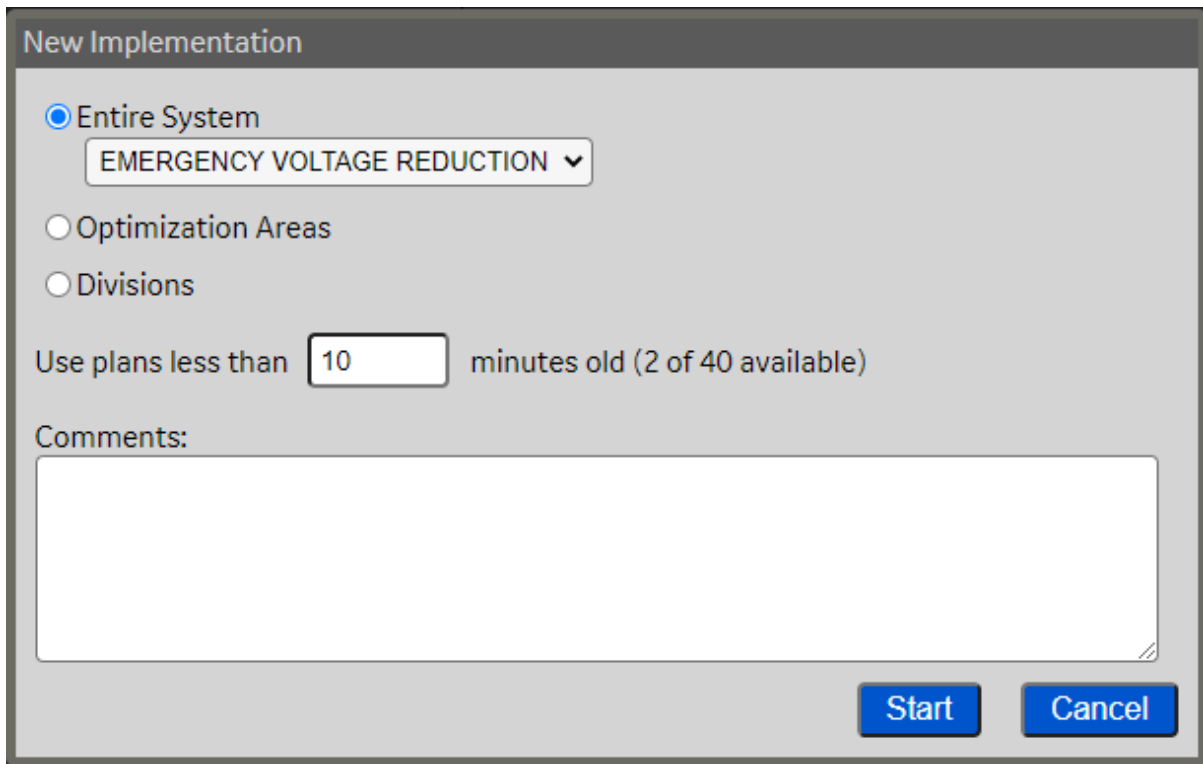
3.13.6. LVM Plan Implementation

Once the system operator is satisfied by the post-plan results obtained using the selected LVM problem profile for the selected area, the operator can proceed to perform a "problem formulation transition". A "problem formulation transition" is an operation that achieves the following:

- Changing the state of the real-time environment (capacitor open/close status, tap changing transformer positions/setpoints, or DER and generator VAR or power factor setpoints) to the post-plan states proposed by the LVM solutions for the stations in the selected area. LVM plans for these stations will be sent to the Plan Manager module in the DNAF server for closed-loop implementation.
- Additionally, the currently active real-time LVM problem formulation configurations for the impacted stations will be changed to the configuration as specified in the LVMSM problem formulation profile. This will enable the LVM application in the DNAF server to maintain the state of the impacted stations. For example, if the operator made a problem formulation transition to "Minimize Demand", the responsibility of the LVM application in the DNAF server is to generate LVM plans that will reduce demand – subject to voltage limits – in response to changing real-time conditions, such as load profile and topology.

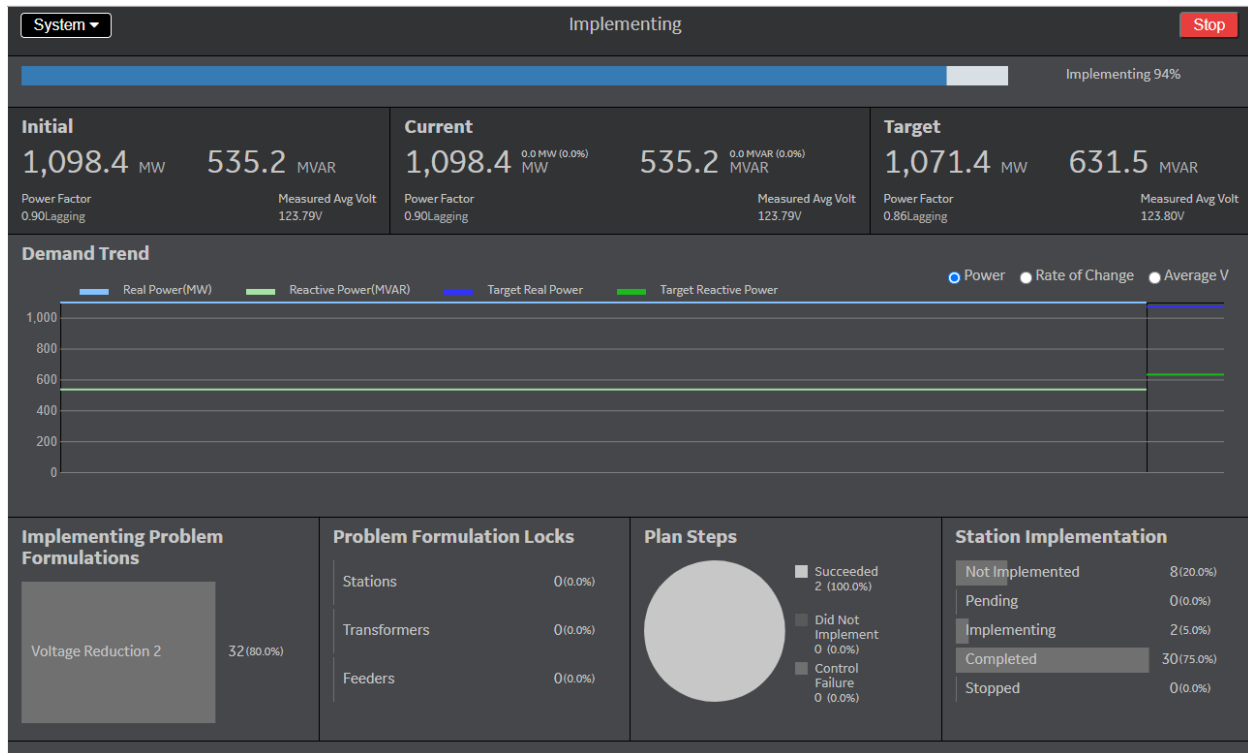
It is also possible that instead of analyzing LVM results from LVM System Monitor problem formulation profiles, the operator may directly want to execute a system-wide emergency voltage reduction. The "implement" button on the LVM System Monitor main display enables the operator to perform a problem formulation transition. Here,

- The operator can select between performing problem formulation transition for the entire system or an optimization area or a division.
- If the recent LVM plans corresponding to the selected problem formulation profile are available, then they will be implemented immediately. The age criterion for recent plans is specified in the dialog box. Otherwise, for the stations whose plans are older than the specified time limit, or if the plans were from a different problem formulation profile, new LVM solutions for those stations are obtained. The LVM plans are then sent to the Plan Manager module of the DNAF sever for closed-loop execution.



The image shows a software dialog box titled "New Implementation". It contains three radio buttons: "Entire System" (which is selected), "Optimization Areas", and "Divisions". Below the radio buttons is a dropdown menu currently showing "EMERGENCY VOLTAGE REDUCTION". Further down, there is a text field labeled "Use plans less than" followed by a numeric input box containing the value "10", and then the text "minutes old (2 of 40 available)". Below this is a section labeled "Comments:" with a large, empty text area. At the bottom right of the dialog are two buttons: "Start" and "Cancel".

The problem formulation transition progress can be viewed on the "Implementation Details" page of the LVM System Monitor.



3.14. Co-Ordination with the DNAF Server

The DNAF server and the LVMSM operate independently. The DNAF server could be solving LVM in response to periodic, event, or manual triggers and implementing LVM plans in closed-loop mode while the LVMSM is forecasting with a specified LVM problem formulation profile. If the operator choses to implement forecast LVM plans from the LVMSM user interface, there could be interference from the LVM operations currently being performed by the DNAF server. Therefore, it is necessary to implement measures to prevent the DNAF server from interfering with the closed-loop implementation initiated from the LVMSM UI.

A block mechanism is introduced to prevent interference between these two independent sub-systems during LVM problem formulation transitions. The block mechanism applies to the LVM application only. It does not apply to the reconfiguration applications, including Fault Isolation and Service Restoration (FISR) and Automatic Feeder Reconfiguration (AFR).

1. A "transition period" for a given station begins at the time when the operator, using the LVMSM UI, implements a forecast LVM solution for the system, area, or optimization group that the station belongs to. A transition period for a given station can also begin when the LVM plans from the DNAF server are sent to the Plan Manager module for closed-loop execution.
2. The transition period for a given station ends when LVM plan closed-loop execution for the station are completed successfully, or have failed, or are cancelled by the operator, including a user-configurable delay to allow for the system conditions to settle down. This is called the "stabilization stage" of the transition period. This stage allows sufficient time for station devices to respond adequately and for measurements to reflect steady-state conditions.

The DNAF_LVM_BLOCK_TIMER server configuration parameter in the DNAF server configuration file specifies, in minutes, the duration of the LVM solution stabilization period for any given station.

3. During the transition period, LVM solutions are coordinated as follows:

On the DNAF server, periodic and event-triggered LVM solutions are kept waiting until the end of the transition period, including the stabilization stage. As soon as the transition period ends, the waiting periodic or event-triggered LVM solution is executed. Manually triggered LVM solutions are rejected during the transition period, except in the stabilization stage.

4. The transition period is tracked individually for each station.
5. Timing issues between the DNAF LVM solutions and LVMSM initiated problem formulation transition are addressed as follows:

The DNAF LVM sub-system and the LVMSM sub-system are two independent subsystems. It is possible that both these subsystems generate LVM solutions from the same steady-state system conditions. If the operator chooses to implement LVMSM plans, the following timing-related race conditions can arise:

- The LVM plans generated by the DNAF LVM subsystem can be received by the Plan Manager Module just before the LVM plans from the LVMSM subsystem are received. This case is resolved as follows:
 - For the stations involved in the LVM plans generated by the DNAF LVM subsystem that **are not** part of the LVMSM-initiated problem formulation transition, nothing needs to be done.
 - For the stations involved in the LVM plans generated by the DNAF LVM subsystem that **are** part of the LVMSM-initiated problem formulation transition, the evaluation server LVM plans are rejected. The LVMSM is notified. The LVMSM monitors the LVM block period for these stations. As soon as the block period, including the stabilization period (only for problem formulation profiles involving optimization LVM problem formulations) has expired, the LVMSM will automatically reforecast and implement LVM solutions for the impacted stations.
 - The other possibility is that the LVM plans generated by the DNAF LVM subsystem are received by the Plan Manager Module just after the evaluation server LVM plans from the LVMSM subsystem are received. In this case, the LVM plans from the DNAF LVM subsystem are simply rejected and the forecast LVM plans from the LVMSM subsystem are allowed to be executed in closed-loop mode.
6. In the case of control failure retriggers (user-configurable), the LVM block period is respected. In other words, the LVMSM waits until the LVM block period has expired before automatically reforecasting another LVM solution due to control failures during closed-loop plan forecast LVM plan execution.

3.15. LVM Alarms

Table below lists the supported LVM alarms.

Table 4. LVM Alarms

Alarm	Description	Applicability
LVM Failed to Complete	When LVM fails to solve for a station (due to solution not converged, suspect solution, or other unexpected errors), this alarm is issued for the station.	Optimization LVM only
LVM Failed to Complete (Return to Normal)	When LVM is successfully solved for a station after an "LVM Failed to Complete" alarm was issued, the "Return to Normal" alarm is issued for the station.	Optimization LVM only
LVM No Plan Generated	When no LVM plan has been generated for a station within a specified number of hours, this alarm is issued. The "Maximum Time Without a Plan Before Issuing Alarm" parameter (on the Load and Volt/VAR Management page in the DNAF Configuration Editor) specifies the number of hours.	
LVM No Plan Generated (Return to Normal)	When an LVM plan is generated for a station after "LVM No Plan Generated" was issued for the station, the "Return to Normal" alarm is issued.	
LVM Maximum Consecutive Infeasible Plans	When the number of consecutive infeasible LVM plans for a station exceeds the value specified by "Maximum Consecutive Infeasible Solutions" (on the Load and Volt/VAR Management page in the DNAF Configuration Editor), this alarm is issued for that station.	Optimization LVM only
LVM Maximum Daily Infeasible Plans	When the number of infeasible LVM plans for a station on a given day exceeds the value specified by "Maximum Daily Infeasible Solutions" (on the Load and Volt/VAR Management page in the DNAF Configuration Editor), this alarm is issued for that station.	Optimization LVM only
LVM Maximum Consecutive Discarded Plans	When the number of consecutive discarded LVM plans for a station exceeds the value specified by "Maximum Consecutive Discarded Solutions" (on the Load and Volt/VAR Management page in the DNAF Configuration Editor), this alarm is issued for the station.	Optimization LVM only
LVM Maximum Daily Discarded Plans	When the number of discarded LVM plans for a station on a given day exceeds the value specified by "Maximum Daily Discarded Solutions" (on the Load and Volt/VAR Management page in the DNAF Configuration Editor), this alarm is issued for the station.	Optimization LVM only
LVM Too Many Capacitor Moves	When the number of capacitor moves for a station exceeds a limit ("Maximum Capacitor Moves in Monitoring Interval") within a specified duration ("Capacitor Operations Monitoring Duration"), this alarm is issued for the station. The limit and the monitoring duration are specified on the Load and Volt/VAR Management page in the DNAF Configuration Editor.	
LVM Too Many Tap Changer Moves	When the number of regulator/LTC moves for a station exceeds a limit ("Maximum regulator/LTC Moves in Monitoring Interval") within a specified duration ("Regulator/LTC Monitoring Duration"), this alarm is issued for the station. The limit and the monitoring duration are specified on the Load and Volt/VAR Management page in the DNAF Configuration Editor.	

Alarm	Description	Applicability
Feeder LVM Disabled	If a feeder is not eligible for LVM due to any of the reasons below, this alarm is issued for the feeder. • The feeder is disabled manually. • The connected power transformer is LVM disabled. • The number of uncontrollable regulator/LTCs or capacitors is greater than the allowed limit. The limit is specified in the DNAF Configuration Editor and optionally for each feeder in the station model file. If the limit for a feeder is not defined in the model file, then the default limit specified in the DNAF Configuration Editor is used for comparison for the given feeder. The comparison is performed only if the Allow Automatic Disabling by Feeder and Substation check box and the respective check boxes for regulators/LTCs and capacitors are enabled (on the Load and Volt/VAR Management page in the DNAF Configuration Editor).	
Feeder LVM Off due to Not Enough Controls	If a feeder is not eligible for LVM due to the following reason, this alarm is issued for the feeder. The number of uncontrollable regulator/LTCs or capacitors exceeds the allowed limit. The limit is specified in the DNAF Configuration Editor and optionally for each feeder in the station model file. If the limit for a feeder is not defined in the model file, then the default limit specified in the DNAF Configuration Editor is used for comparison for the given feeder. The comparison is performed only if the Allow Automatic Disabling by Feeder and Substation check box and the respective check boxes for regulators/LTCs and capacitors are enabled (on the Load and Volt/VAR Management page in the DNAF Configuration Editor).	
Tap Control Resend Failed	If a regulator/LTC control failed for the first two resend attempts, this alarm is issued for the regulator/LTC.	
Capacitor Control Resend Failed	If capacitor control failed for the first two resend attempts, this alarm is issued for the capacitor.	
Regulator Group Voltage Imbalance	For a group of three parallel single-phase regulators, if the voltage imbalance on the secondary side of the group exceeds a specified limit, this alarm is issued for that regulator group. The limit is specified in the station model file in the transformer object comprising the group. In Optimization LVM, DPF calculated voltages are used, while in Heuristic LVM, telemetered voltages are used.	
Regulator Group Voltage Imbalance (Return to Normal)	When an LVM plan is executed and voltage imbalance on the secondary side of the group is within a specified limit after "Regulator Group Voltage Imbalance" was issued for the station, the "Return to Normal" alarm is issued.	
Regulator Delayed Tap Move Cancelled	For a regulator, if a tap move that was generated based on a time delay schedule for that regulator is cancelled, this alarm is issued.	Heuristic LVM only
Controller Delta Volt Mismatch (FUTURE functionality)	For a capacitor, if the telemetered voltage sensitivity deviates significantly from the voltage sensitivity calculated by Heuristic LVM, this alarm is issued.	Heuristic LVM only
Energized Sensor Zero Voltage Event	If voltage measurement at an energized sensor shows zero voltage, this alarm is issued.	

Alarm	Description	Applicability
Energized Sensor Zero Voltage RTN Event	If voltage measurement at an energized sensor shows non-zero voltage after "Energized Sensor Zero Voltage Event" was issued, this alarm is issued.	
LVM Failsafe Started	If LVM failsafe conditions (DPF solution is suspect or not solved, or bus voltage mismatches) apply to a system element (station, transformer, or feeder), this alarm is issued.	Optimization LVM only
LVM Failsafe Ended	If an LVM failsafe condition subsides (DPF solution status becomes not-suspect or bus voltage mismatches are back within the tolerance) for a system element (station, transformer, or feeder), this alarm is issued. This alarm is also issued when manually changing the failsafe problem formulation or implementing an LVM plan with another problem formulation from the LVM System Monitor.	Optimization LVM only
Optimization Group Throttling Started	When an LVM plan is generated, if the LVM throttling function is enabled, the generated plan could be held in the queue before sending to Plan Manager if the real and/or reactive power demand change rate exceeds a pre-defined threshold for the LVM optimization group. In this case, this alarm is issued.	
Plan Execution Failure Retrigger	If LVM retriggers due to a plan execution failure, this alarm is issued.	
Plan State Check Failure	If LVM plan execution fails because a device is found ineligible during a pre-implementation check or device state changed, this alarm is issued.	

3.16. LVM Configuration

This section describes the Load and Volt/VAR Management configuration parameters.

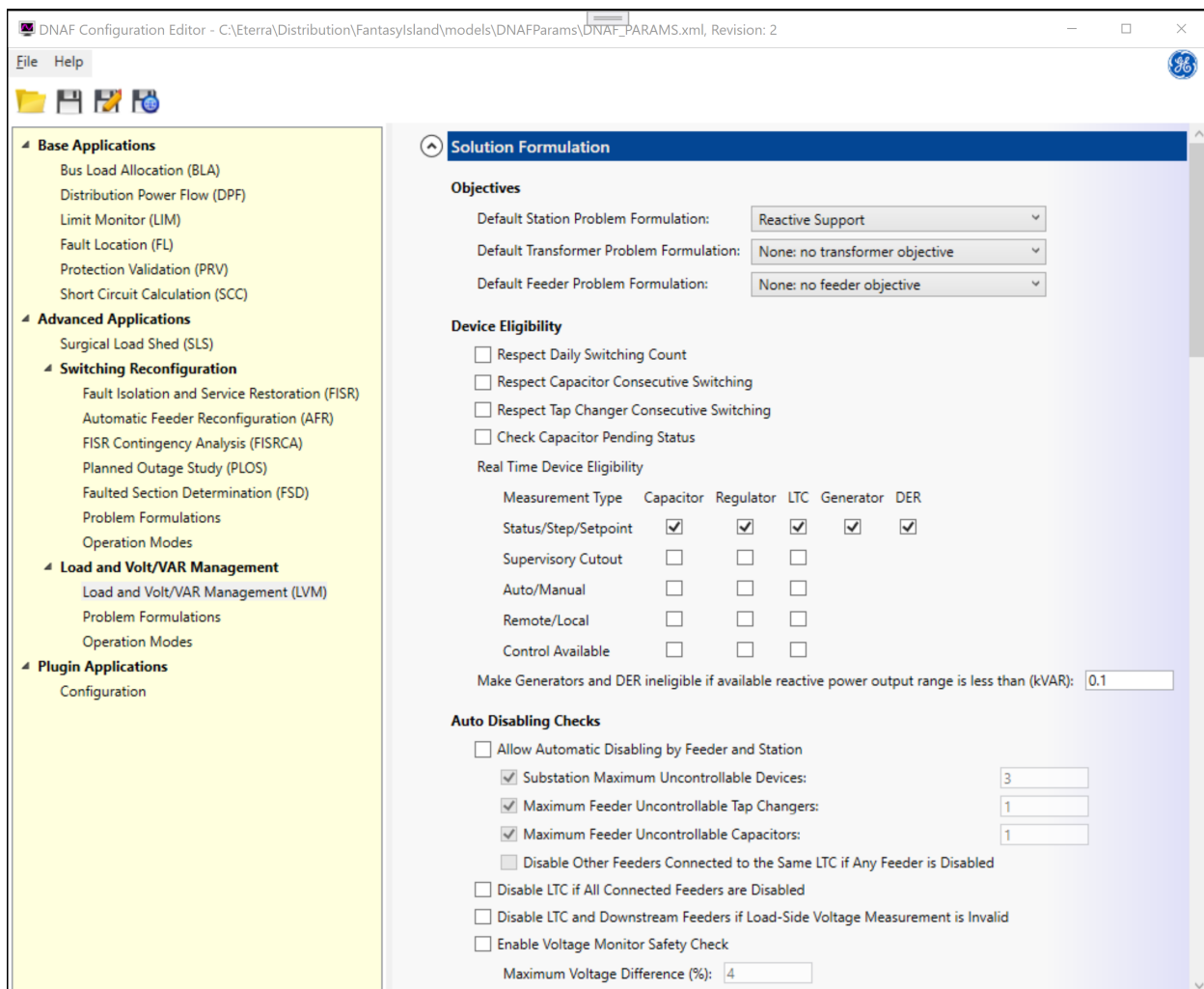


Figure 67. LVM Configuration

Table below lists the LVM configuration parameters. All listed parameters are contained by the <Params xsi:type = "VVCPParams"> element in the DNAF_PARAMS.xml configuration file.

Table 5. LVM Configuration Parameters

Configuration Editor UI Name	CSV File Field Name and XML File Attribute Name	Type	Description
Objectives			
Default Station Problem Formulation	ACTPROBFORM_VOLTMAN <DefaultProbIndex>	Drop-down list	This list shows the default LVM problem formulation for stations. All defined LVM problem formulations are shown in the Problem Formulations subsection on the Load and Volt/VAR Management page in the DNAF Configuration Editor. If LVM is executed for a station due to a periodic trigger or an event trigger, the setting specified for the station in the DNAF System Configuration (DSC) is checked. If the DSC indicates default problem formulation, the problem formulation specified by this field is used for the station.

Configuration Editor UI Name	CSV File Field Name and XML File Attribute Name	Type	Description
Default Transformer Problem Formulation	<DefaultTransProbIndex>	Drop-down list	This list shows the default LVM problem formulation for station transformers. To inherit the station's problem formulation, select "None". All defined LVM problem formulations are shown in the Problem Formulations subsection on the Load and Volt/VAR Management page in the DNAF Configuration Editor. If LVM is executed for a station due to a periodic trigger or an event trigger, the setting specified for the station transformers in the DNAF System Configuration (DSC) is checked. If the DSC indicates default problem formulation, the problem formulation specified by this field is used for the station transformers.
Default Feeder Problem Formulation	<DefaultFeederProbIndex>	Drop-down list	This list shows the default LVM problem formulation for feeders. To inherit the station's or transformer's problem formulation, select "None". All defined LVM problem formulations are shown in the Problem Formulations subsection on the Load and Volt/VAR Management page in the DNAF Configuration Editor. If LVM is executed for a station due to a periodic trigger or an event trigger, the setting specified for the station feeders in the DNAF System Configuration (DSC) is checked. If the DSC indicates default problem formulation, the problem formulation specified by this field is used for the station feeders.
Device Eligibility			
Respect Daily Switching Count	<RespectSwitchCount>	Check box	This check box indicates whether to respect the maximum daily operations count limit for capacitors and LTC/regulators. By default, it is cleared. If the box is cleared, the daily operations count for capacitors and LTC/regulators is compared to their respective maximum daily switching count limit. The limit is statically defined (by the parameters MaxSwitches and MaxOperations for capacitor and LTC/regulator objects, respectively). If the limit has been reached, the capacitor or LTC/regulator is made ineligible in the LVM process until the daily count has been reset. If the box is checked, the comparison is still performed. However, the capacitor or LTC/regulator can still be eligible even if the daily count has been reached. It will use a different operation cost defined in the corresponding problem formulation page.

Configuration Editor UI Name	CSV File Field Name and XML File Attribute Name	Type	Description
Respect Capacitor Consecutive Switching	<RespectConsecutiveCapSwitch>	Check box	This check box indicates whether to respect the capacitor minimum time interval limit between consecutive switching operations. If this box is checked, the time elapsed since the capacitor was last switched is compared with the capacitor minimum time interval limit between consecutive switching operations. The limit is statically defined (by the parameter ConsecSw, or CONSECSW_SHIMP for the CSV file) for capacitor objects in the station model. If the time elapsed is less than the limit, the capacitor is considered not eligible for LVM.
Respect Tap Changer Consecutive Switching	<RespectConsecutiveTapSwitch>	Check box	This check box indicates whether to respect the tap changer minimum time interval limit between consecutive switching operations. If this box is checked, the time elapsed since the tap changer was last switched is compared with the capacitor minimum time interval limit between consecutive switching operations. The limit is statically defined for tap changer objects in the station model. If the time elapsed is less than the limit, the tap changer is considered not eligible for LVM.
Check Capacitor Pending Status	CHECKPENDING_VOLTMAN <CheckPending>	Check box	This check box indicates whether LVM can be executed when one or more capacitors in the station have a Control Pending status measurement. This field applies only for some one-way communication systems that model capacitor control pending status measurements. If this box is checked, and one or more capacitors are pending, LVM is not executed. If this box is cleared, LVM execution bypasses the pending status check. The default state is cleared.
Real Time Device Eligibility – Current Status/Step/Setpoint (Capacitor)	<UseCapStatusMeas>	Check box	This check box indicates if the capacitor status SCADA measurement is used to determine device eligibility for LVM optimization. The device will be ineligible if the measurement is missing or invalid. Note The optimization eligibility status must be set to SCADA Decides for this setting to be considered.

Configuration Editor UI Name	CSV File Field Name and XML File Attribute Name	Type	Description
Real Time Device Eligibility – Current Status/Step/Setpoint (Regulator)	<UseRegulatorStepMeas>	Check box	This check box indicates if the regulator status/step/voltage setpoint SCADA measurements are used to determine device eligibility for LVM optimization. The device will be ineligible if the measurement is missing or invalid.  Note The optimization eligibility status must be set to SCADA Decides for this setting to be considered.
Real Time Device Eligibility – Current Status/Step/Setpoint (LTC)	<UseLTCStepMeas>	Check box	This check box indicates if the LTC status/step/voltage setpoint SCADA measurements are used to determine device eligibility for LVM optimization. The device will be ineligible if the measurement is missing or invalid.  Note The optimization eligibility status must be set to SCADA Decides for this setting to be considered.
Real Time Device Eligibility – Current Status/Step/Setpoint (Generator)	<UseDGSetpointMeas>	Check box	This check box indicates if the DG kVAR or PF setpoint SCADA measurements are used to determine device eligibility for LVM optimization. The device will be ineligible if the measurement is missing or invalid.  Note The optimization eligibility status must be set to SCADA Decides for this setting to be considered.
Real Time Device Eligibility – Current Status/Step/Setpoint (DER)	<UseDERSetpointMeas>	Check box	This check box indicates if the DER kVAR or PF setpoint SCADA measurements are used to determine device eligibility for LVM optimization. The device will be ineligible if the measurement is missing or invalid.  Note The optimization eligibility status must be set to SCADA Decides for this setting to be considered.
Real Time Device Eligibility – Supervisory Cutout (Capacitor)	<UseCapSuperCutoutMeas>	Check box	This check box indicates if the capacitor supervisory cutout SCADA measurement is used to determine device eligibility for LVM optimization. The device will be ineligible if the measurement is missing, invalid, or set to disabled.  Note The optimization eligibility status must be set to SCADA Decides for this setting to be considered.

Configuration Editor UI Name	CSV File Field Name and XML File Attribute Name	Type	Description
Real Time Device Eligibility – Supervisory Cutout (Regulator)	<UseRegulatorSuperCutoutMeas>	Check box	This check box indicates if the regulator supervisory cutout SCADA measurement is used to determine device eligibility for LVM optimization. The device will be ineligible if the measurement is missing, invalid, or set to disabled.  Note The optimization eligibility status must be set to SCADA Decides for this setting to be considered.
Real Time Device Eligibility – Supervisory Cutout (LTC)	<UseLTCSuperCutoutMeas>	Check box	This check box indicates if the LTC supervisory cutout SCADA measurement is used to determine device eligibility for LVM optimization. The device will be ineligible if the measurement is missing, invalid, or set to disabled.  Note The optimization eligibility status must be set to SCADA Decides for this setting to be considered.
Real Time Device Eligibility – Auto/Manual (Capacitor)	<UseCapAutoManualMeas>	Check box	This check box indicates if the capacitor auto/manual SCADA measurement is used to determine device eligibility for LVM optimization. The device will be ineligible if the measurement is missing or invalid, or the status is Auto.  Note The optimization eligibility status must be set to SCADA Decides for this setting to be considered.
Real Time Device Eligibility – Auto/Manual (Regulator)	<UseRegulatorAutoManualMeas>	Check box	This check box indicates if the regulator auto/manual SCADA measurement is used to determine device eligibility for LVM optimization. The device will be ineligible if the measurement is missing or invalid. It will also be ineligible if the status is Manual when using setpoint control, or Auto when using raise/lower controls.  Note The optimization eligibility status must be set to SCADA Decides for this setting to be considered.

Configuration Editor UI Name	CSV File Field Name and XML File Attribute Name	Type	Description
Real Time Device Eligibility – Auto/Manual (LTC)	<UseLTCAutoManualMeas>	Check box	This check box indicates if the LTC auto/manual SCADA measurement is used to determine device eligibility for LVM optimization. The device will be ineligible if the measurement is missing or invalid. It will also be ineligible if the status is Manual when using setpoint control, or Auto when using raise/lower controls.  Note The optimization eligibility status must be set to SCADA Decides for this setting to be considered.
Real Time Device Eligibility – Remote/Local (Capacitor)	<UseCapRemoteLocalMeas>	Check box	This check box indicates if the capacitor remote/local SCADA measurement is used to determine device eligibility for LVM optimization. The device will be ineligible if the measurement is missing, invalid, or set to local control.  Note The optimization eligibility status must be set to SCADA Decides for this setting to be considered.
Real Time Device Eligibility – Remote/Local (Regulator)	<UseRegulatorRemoteLocalMeas>	Check box	This check box indicates if the regulator remote/local SCADA measurement is used to determine device eligibility for LVM optimization. The device will be ineligible if the measurement is missing, invalid, or set to local control.  Note The optimization eligibility status must be set to SCADA Decides for this setting to be considered.
Real Time Device Eligibility – Remote/Local (LTC)	<UseLTCTRemoteLocalMeas>	Check box	This check box indicates if the LTC remote/local SCADA measurement is used to determine device eligibility for LVM optimization. The device will be ineligible if the measurement is missing, invalid, or set to local control.  Note The optimization eligibility status must be set to SCADA Decides for this setting to be considered.
Real Time Device Eligibility – Control Available (Capacitor)	<UseCapControlMeas>	Check box	This check box indicates if the capacitor control available SCADA measurement is used to determine device eligibility for LVM optimization. The device will be ineligible if the measurement is missing, invalid, or set to not available.  Note The optimization eligibility status must be set to SCADA Decides for this setting to be considered.

Configuration Editor UI Name	CSV File Field Name and XML File Attribute Name	Type	Description
Real Time Device Eligibility – Control Available (Regulator)	<UseRegulatorControlMeas>	Check box	This check box indicates if the regulator control available SCADA measurement is used to determine device eligibility for LVM optimization. The device will be ineligible if the measurement is missing, invalid, or set to not available.  Note The optimization eligibility status must be set to SCADA Decides for this setting to be considered.
Real Time Device Eligibility – Control Available (LTC)	<UseLTCControlMeas>	Check box	This check box indicates if the LTC control available SCADA measurement is used to determine device eligibility for LVM optimization. The device will be ineligible if the measurement is missing, invalid, or set to not available.  Note The optimization eligibility status must be set to SCADA Decides for this setting to be considered.
Make Generators and DER Ineligible if Output Below	<MinKVARRange>	Text box	This entry defines the minimum allowable range of kVAR output of a generator or DER for the device to be considered eligible. The kVAR output range is determined by the device capability curve and the current real power output. A value that is too low may cause devices not effectively controlled to be included in the LVM analysis. This may cause the LVM solution to be infeasible. A value that is too large may prevent smaller generators and DER from being used for VAR control.
Auto Disabling Checks			
Allow Automatic Disabling by Feeder	IVCSTATUSCHECK_VOLTMAN <CheckIVCFlag>	Check box	This check box indicates if the following additional conditions are checked before attempting to solve LVM for the indicated station: • The substation's number of uncontrollable devices (where the Control Available measurement status is False) is greater than the specified limit • The feeder's number of uncontrollable LTC/regulators is greater than the specified limit • The feeder's number of uncontrollable capacitors is greater than the specified limit If any of the above criteria are met, then LVM is skipped for that station. For more information about enabling and disabling LVM, see Enable/Disable LVM.

Configuration Editor UI Name	CSV File Field Name and XML File Attribute Name	Type	Description
Substation Maximum Uncontrollable Devices	MAXOUTDEVICE_VOLTMAN <CheckMaxOutDevice>	Check box	This check box is effective only if the Allow Automatic Disabling by Feeder option is checked. This check box enables or disables the Substation Maximum Uncontrollable Devices pre-processing check.
Substation Maximum Uncontrollable Devices	MAXOUTDEVICE_VOLTMAN <MaxOutDevice>	Text box	This entry is effective only if the Allow Automatic Disabling by Feeder option is checked. This entry specifies the total maximum number of controllable devices (capacitors and LTCs/regulators) that are allowed to be unavailable (i.e., "control available" status measurement is false or invalid) before LVM is disabled. Use the associated check box to enable or disable this pre-processing check. The allowable range is 0-20.
Maximum Feeder Uncontrollable Tap Changers	MAXUNCTRLTAPONFEEDER_VOLTMAN <CheckMaxUnCtrlTapChangersForFeeders>	Check box	This check box is effective only if the Allow Automatic Disabling by Feeder option is checked. This check box enables or disables the Maximum Feeder Uncontrollable Tap Changers pre-processing check.
Maximum Feeder Uncontrollable Tap Changers	MAXUNCTRLTAPONFEEDER_VOLTMAN <MaxUnCtrlTapChangersForFeeders>	Text box	This entry is effective only if the Allow Automatic Disabling by Feeder option is checked. This entry sets the maximum number of uncontrollable LTCs/regulators allowed on a feeder. If the number of uncontrollable LTCs/regulators on a feeder exceeds this limit, the feeder is disabled for the LVM execution. Note that a corresponding parameter (MaxUnCtrlTapChangers, or MAXUNCTRLTAPS_FEEDER for the CSV file) is statically defined for feeder objects in the station model. If the statically defined parameter for the feeder is set to -1 (default), the limit specified here is used. Otherwise, the limit specified in the feeder object is used. Use the associated check box to enable or disable this pre-processing check. The allowable range is 0-20.
Maximum Feeder Uncontrollable Capacitors	MAXUNCTRLCAPACITORONFEEDER_VOLTMAN <CheckMaxUnCtrlCapacitorsForFeeders>	Check box	This check box is effective only if the Allow Automatic Disabling by Feeder option is checked. This check box enables or disables the Maximum Feeder Uncontrollable Capacitors pre-processing check.

Configuration Editor UI Name	CSV File Field Name and XML File Attribute Name	Type	Description
Maximum Feeder Uncontrollable Capacitors	MAXUNCTRLCAPACITORON FEEDER_VOLTMAN <MaxUnCtrlCapacitors ForFeeders>	Text box	This entry is effective only if the Allow Automatic Disabling by Feeder option is checked. This field specifies the maximum number of uncontrollable capacitors allowed on a feeder. If the number of uncontrollable capacitors on a feeder exceeds this limit, then the feeder is disabled for LVM execution. Note that a corresponding parameter (MaxUnCtrlCapacitors) is statically defined for feeder objects in the station model. If the statically defined parameter for the feeder is set to -1 (default), the limit specified here is used. Otherwise, the limit specified in the feeder object is used. Use the associated check box to enable or disable this pre-processing check. The allowable range is 0-20.
Disable Other Feeders Connected to the Same LTC if Any Feeder is Disabled	<DisableOtherFeeders>	Check box	This check box is effective only if the Allow Automatic Disabling by Feeder option is checked. If this box is checked and a feeder is found ineligible for LVM, then all feeders downstream of the power transformer to which that feeder belongs are considered ineligible for LVM.
Disable LTC if All Connected Feeders are Disabled	DISABLECONNECTEDLTC_VOLTMAN <DisableConnectedLTC>	Check box	This check box is effective only if the Allow Automatic Disabling by Feeder option is checked. This check box indicates whether to disable an LTC/regulator if all feeders downstream of the station power transformer are determined to be ineligible for LVM due to maximum number of uncontrollable LTC/regulators or capacitors being exceeded. After LVM has finished executing, the LTC/regulators that were disabled due to this option are restored to their previous LVM enabled/disabled status.
Disable LTC and Downstream Feeders if Load-Side Voltage Measurement is Invalid	<DisableLTCandFeedersOnInvalidVoltageMeas>	Check box	This check box indicates whether an LTC and its downstream feeders should be disabled for LVM if the load-side voltage analog measurements associated with the LTC/regulator do not have a "valid" quality from SCADA.
Enable Voltage Monitor Safety Check	ENABLEVOLTAGESAFETYCHECK_VOLTMAN <EnableVoltageSafetyCheck>	Check box	This check box indicates if the voltage monitoring safety check is enabled. See the description for the parameter "Maximum Voltage Difference" for details on how this check is performed.

Configuration Editor UI Name	CSV File Field Name and XML File Attribute Name	Type	Description
Maximum Voltage Difference (%)	MAXVOLTPERCENTAGEDIFFERENCE FORSAFETYCHECK_VOLTMAN <MaxVoltPercentageDifference>	Text box	This entry specifies the maximum voltage difference (in %) for any node in a station that has a valid voltage measurement. This entry is used if Enable Voltage Monitor Safety Check is selected. If the relative value of the difference between the DPF calculated voltage for the node and its voltage measurement is greater than the value specified in this field, then the voltage monitoring safety check for the station is considered failed. In this case, LVM is not solved for the station. The measurement that caused the safety check to fail is logged in the LVM results file. The allowable range is 0.1-10.
Voltage Control			
AMI Voltage Measurement Time Tolerance	<AmiVoltMeasurementTimeTolerance>	Text box	This entry specifies the maximum difference between the evaluation time and AMI voltage measurement timestamp (in seconds). If the time difference is less than this number, then the AMI voltage measurement is used in LVM. Otherwise, the AMI voltage measurement is not considered in the LVM solution. The allowable range is 0-3600.
Plan Generation			
Support Mixed Problem Formulation Configuration	<LVMEEnableMixedProblemFormulations>	Check box	This check box indicates whether mixed LVM problem formulation types are supported. For example, this option enables solving an LVM solution using a power flow based problem formulation together with a fixed voltage reduction problem formulation when LVM failsafe conditions are triggered due to a bad-quality power flow (DPF) solution.
Create Station Power Transformer Level Plans	<LVMEEnableBankLevelPlanCreation>	Check box	This check box indicates whether station power transformer level plans can be created and executed in closed loop mode (if LVM is configured to run in closed loop mode).

Configuration Editor UI Name	CSV File Field Name and XML File Attribute Name	Type	Description
Voltage Setpoint/Reduction Control Conversion Method	<VoltSetpointConversionMethod>	Drop-down list	This selection specifies how the voltage setpoint control or voltage reduction control calculation is performed. There are two options (each option will respect the statically modeled voltage setpoint deadband and both the minimum and maximum voltage setpoint range values). Use DPF Results: The voltage setpoint control or the voltage reduction control is calculated based on the power flow results. A voltage setpoint control is determined as the internally solved DPF secondary/load side LTC or regulator voltage based on the recommended change in the tap position from the LVM plan. A voltage reduction (% Voltage change) control is determined based on the difference in the nominal voltage target and the internally solved DPF secondary/load side LTC or regulator voltage based on the recommended change in the tap position from the LVM plan. Use Tap Movement and Tap Size: The voltage setpoint control or the voltage reduction control is calculated based on the change in the number of tap steps in the LVM recommended plan and the modeled tap step size (fixed value). A voltage setpoint control is determined as the sum of the telemetered analog voltage setpoint value and the product of the tap step size and the number of tap moves recommended by the LVM plan. This product is positive if the tap position is raised, and the product is negative if the tap position is lowered. Similarly, a voltage reduction (% Voltage change) control is determined as the sum of initial voltage reduction value and product of the tap step size and the number of tap moves recommended by the LVM plan.
Plan Enhancement			
Filter Small Voltage Reduction Step	<FilterSmallIVRPercentage>	Text box	This entry specifies the minimum voltage reduction step allowed in LVM plans, in percentage. If a voltage reduction step is less than the Filter Small Voltage Reduction Step value multiplied by the tap step size, this plan step will be discarded. The allowable range is 0-100.
Plan Creation Method	SINGLECONTROLMODE_VOL TMAN <SingleControlMode>	Drop-down list	This selection indicates whether LVM issues one or more control moves per LVM execution per station. If the Single Step option is selected, LVM can issue only one control move at a time per station. If the Multiple Step option is selected, LVM can issue multiple control moves at a time per station.

Configuration Editor UI Name	CSV File Field Name and XML File Attribute Name	Type	Description
Enable LVM Plan Enhancement	ADDIPLANDISCLOGIC_VOLT MAN <LVMEEnableAdditionalPlan DiscretizationLogic>	Check box	This check box indicates if the plan discretization logic in optimization LVM is enabled (checked means it is enabled). See Optimization LVM Plan Enhancement and Selection for more details.
Voltage Margin Adjustment Amount	DISCADJUSTMARGIN_VOLT MAN <LVMDiscreteAdjustMargin>	Text box	This entry is the voltage margin adjustment that is used to relax the voltage limits during the plan discretization process optimization. This entry is expressed in p.u. See Optimization LVM Plan Enhancement and Selection for details. This entry is effective only if Enable LVM Plan Enhancement is checked. The allowable range is 0-0.4.
Perform Plan Selection	ADDIPLANSELELOGIC_VOLT MAN <LVMEEnableAdditionalPlan SelectionLogic>	Check box	This check box indicates if the LVM plan selection logic is executed. If this option is checked, then additional plan selection logic is performed in optimization LVM. See Optimization LVM Plan Enhancement and Selection for details.
Number of Top Voltage Violations Used in Voltage PI Calculation	NUMTOPAVVOLTVIO_VOLT MAN <NumofTopAverageVoltageV iolation>	Text box	This entry specifies the number of top voltage violations to be used to calculate the average voltage performance index. See Optimization LVM Plan Enhancement and Selection for details. This field is effective only if Perform Plan Selection is checked. The allowable range is 1-40.
Demand Change Deadband	DEMANDCHANGEDEADBAND _VOLT MAN <DemandChangeDeadb andAmount>	Text box	This entry specifies the demand change deadband (in kW). If the change in demand is less than this entry, the LVM plan is not generated. See Plan Selection for details. This entry is effective only if Perform Plan Selection is checked. The allowable range is 0-200.
Voltage Change Deadband (With Violation)	VOLTAGECHANGEDEADBAND _VOLT MAN <VoltageChangeDeadb andAmount>	Text box	This entry specifies the maximum voltage violation deadband (in per unit V). If the maximum voltage violation change is less than this value, the LVM plan is not generated. See Plan Selection for details. This entry is effective only if Perform Plan Selection is checked. The allowable range is 0-0.4.

Configuration Editor UI Name	CSV File Field Name and XML File Attribute Name	Type	Description
Voltage Change Deadband (Without Violation)	VOLTAGECHANGEDEADBAND WITHOUTVIOLATION_VOLTMAN <VoltageChangeDeadband AmountWithoutViolation>	Text box	This entry specifies a threshold that is used in the additional plan selection logic in optimization LVM. The plan selection logic considers a voltage improvement only if the voltage improvement performance index is greater than or equal to the value of this parameter. This entry is effective only if Perform Plan Selection is checked. The allowable range is 0-0.4.
Resource Aggregation			
Aggregate Small Energy Resources	<LVMGENDERAggregationLogic>	Check box	This check box indicates whether small energy resources can be aggregated internally by the LVM solution.
Threshold for Aggregating Small Energy Resources (kVA)	<LVMGENDERAggregationThreshold>	Text box	Threshold in kVA for aggregating small energy resources. The allowable range is 100-100000.
Execution Options			
Run in Closed Loop Mode	CLOSEDLOOP_VOLTMAN <ClosedLoop>	Check box	This check box indicates whether closed loop or advisory mode plan execution is utilized. This setting is used only if the LVM Plan Execution Mode in the DNAF System Configuration (from the RT DMS drop-down list on the RT geographic UI) is set to System Default. If this check box is selected, LVM plans (without violations) are sent to the plan manager and controls are sent to SCADA automatically. This is the closed loop mode of operation. If this check box is cleared, the operator must manually approve a plan for controls to be sent to SCADA.  Note The execution mode setting on the DNAF System Configuration display overrides the default operation mode setting specified in the DNAF Configuration Editor.
Capacitor Operations Monitoring Duration	CAPMONDURATION_VVC <CapacitorMonitorDuration>	Text box	This entry specifies the interval duration in hours for monitoring the number of capacitor switching operations. The monitoring intervals are counted from midnight. At the beginning of each day, the capacitor switching count is reset to zero. If the interval duration is not an integral divisor of 24 hours, then the last interval is truncated at the next midnight. The default value is 24 hours. The allowable range is 1-24.

Configuration Editor UI Name	CSV File Field Name and XML File Attribute Name	Type	Description
Regulator/LTC Operations Monitor Duration	REGMONDURATION_VVC <RegulatorMonitorDuration>	Text box	This entry specifies the interval duration in hours for monitoring the number of LTC/regulator moves. The monitoring intervals are counted from midnight. At the beginning of each day, the LTC/regulator moves count is reset to zero. If the interval duration is not an integral divisor of 24 hours, then the last interval is truncated at the next midnight. The allowable range is 1-24.
Maximum Capacitor Moves in Monitoring Interval	NUMCAPMOVESFORALARM_VVC <NumCapacitorMovesForMs g>	Text box	This entry specifies the maximum number of capacitor switching operations. If the number of operations for a capacitor in the current monitoring interval exceeds this entry, then an alarm is issued. The allowable range is 1-50.
Maximum Regulator/LTC Moves in Monitoring Interval	NUMREGMOVESFORALARM_VVC <NumRegulatorMovesForMs g>	Text box	This entry specifies the maximum number of LTC/regulator moves in the current monitoring interval. If the number of operations for an LTC/regulator in the current monitoring interval exceeds this entry, an alarm is issued. The allowable range is 1-200.
Maximum Consecutive Infeasible Solutions	MAXCONSINFEASIBLESOLU_VOLTMAN <MaxConsInfeasibleSolus>	Text box	This entry specifies the maximum number of consecutive infeasible solutions in optimization LVM in a day before an alarm is issued. The maximum number of consecutive infeasible solutions is counted from midnight. At the beginning of each day, the infeasible solutions count is reset to zero. If the interval duration is not an integral divisor of 24 hours, then the last interval is truncated at the next midnight. The allowable range is 1-20.
Maximum Daily Infeasible Solutions	MAXDAILYINFEASIBLESOLU_VOLTMAN <MaxDailyInfeasibleSolus>	Text box	This entry specifies the maximum number of infeasible solutions in a day before an alarm is issued. The maximum number of daily infeasible solutions is counted from midnight. At the beginning of each day, the infeasible solutions count is reset to zero. If the interval duration is not an integral divisor of 24 hours, then the last interval is truncated at the next midnight. The allowable range is 1-200.

Configuration Editor UI Name	CSV File Field Name and XML File Attribute Name	Type	Description
Maximum Consecutive Discarded Solutions	MAXCONSDISCARDSOLU_VO LTMAN <MaxConsDiscardSolus>	Text box	This entry specifies the maximum number of consecutive discarded solutions in optimization LVM in a day before an alarm is issued. The maximum number of consecutive discarded solutions is counted from midnight. At the beginning of each day, the discarded solutions count is reset to zero. If the interval duration is not an integral divisor of 24 hours, then the last interval is truncated at the next midnight. The allowable range is 1-20.
Maximum Daily Discarded Solutions	MAXDAILYDISCARDSOLU_VO LTMAN <MaxDailyDiscardSolus>	Text box	This entry specifies the number of discarded solutions in a day before an alarm is issued. The maximum number of daily discarded solutions is counted from midnight. At the beginning of each day, the discarded solutions count is reset to zero. If the interval duration is not an integral divisor of 24 hours, then the last interval is truncated at the next midnight. The allowable range is 1-200.
Maximum Time Without a Plan Before Issuing Alarm	<MaxDurationBeforeAlarm>	Text box	This entry specifies the maximum time without a valid LVM plan for a station before the LVM No Plan Generated alarm is issued for the station, in hours. The allowable range is 1-8.
Maximum Number of Tap Moves per Execution	MAXTAPPOSMOVE_VOLTMAN <MaxTapPosMoves>	Text box	This entry specifies the maximum allowable number of tap position moves for one LVM execution. The allowable range is 1-33.
Maximum Retrigger on Plan Failure	<MaximumRetriggerCount>	Text box	The maximum number of times to re-trigger LVM to create a new solution when a plan execution fails (for example, due to device control failure) before permanently failing to solve. (Retriggering must be enabled in the DNAF server configuration.) This parameter gives an upper bound to the number of times in a row the solution will retrigger if one or more devices in a station fail. To disable retriggering, set this parameter to zero. The allowable range is 0-100.
Power Rate of Change Throttling			
Real Power Throttling – Increase Value	<RealPowerThrottlingUp>	Text box	This entry specifies the rate at which real power increase is throttled during an LVM plan execution, in MW/min. The allowable range is 0-1000.

Configuration Editor UI Name	CSV File Field Name and XML File Attribute Name	Type	Description
Real Power Throttling – Decrease Value	<RealPowerThrottlingDown>	Text box	This entry specifies the rate at which real power decrease is throttled during an LVM plan execution, in MW/min. The allowable range is 0-1000.
Reactive Power Throttling – Increase Value	<ReactivePowerThrottlingUp>	Text box	This entry specifies the rate at which reactive power increase is throttled during an LVM plan execution, in MVAR/min. The allowable range is 0-1000.
Reactive Power Throttling – Decrease Value	<ReactivePowerThrottlingDown>	Text box	This entry specifies the rate at which reactive power increase is throttled during an LVM plan execution, in MVAR/min. The allowable range is 0-1000.
Throttling Started	<ThrottlingStartedAlarmEnabled>	Check Box	This check box indicates whether to issue an alarm to signify that throttling has been started for the specified optimization group.
Throttling Return To Normal	<ThrottlingRTNAlarmEnabled>	Check Box	This check box indicates whether to issue an alarm to signify that throttling has ended (returned to normal state) for the specified optimization group.
Throttling Paused	<ThrottlingPausedAlarmEnabled>	Check Box	This check box indicates whether to issue an alarm to signify that throttling has been paused for the specified optimization group.
Solution Results			
Create LVM Result File	PRINTOUTLOG_VOLTMAN <PrintVVCResult>	Check box	This check box indicates whether to print an LVM results file. If this box is checked, then a results file is generated. Otherwise, a results file is not generated. The results file is output in the folder specified by the parameter DNAF_OUTPUT_PATH in the appropriate server configuration file. The file name format depends on whether Create Unique Log option is enabled. See the description for Create Unique Log for details.
Use Unique File Name	UNIQUELOG_VOLTMAN <UniqueVVCLog>	Check box	This check box indicates whether to print a unique LVM solution file. If this check box is checked, then for each LVM execution, a unique solution file is generated. In this case, the filename has the format LVM Results_<stationname>_<executiontimestamp>.dat. If this check box is cleared, the filename has the format LVM Results_<stationname>.dat. The solution file is overwritten each time LVM is executed. The solution file is output in the folder specified by the parameter DNAF_OUTPUT_PATH in the appropriate server configuration file.

Configuration Editor UI Name	CSV File Field Name and XML File Attribute Name	Type	Description
Output LP Solver Debug Files	LPSOLVERREPORT_VOLTMAN <LPSolverReport>	Check box	If this check box is selected, LVM outputs an additional set of debug files that are generated by the LP solver in the DNAF output directory. The files contain optimization model information. The filenames have the format LVM_LP_<stationname><_optionaltimestamp>.
Output Base Case Power Flow Information	<OutputBaseCasePowerFlowInfo>	Check box	If this check box is selected and the base case power flow was solved for the LVM solution, the IMPROVED BLA LOG section with BLA information appears in the LVM results file.
Debug Files			
Maximum Number of Debug Files	MAXUPFOUTPUTFILE_VOLTMAN <MaxNumUpfOutFile>	Text box	This entry specifies the maximum number of accumulated debug output files created when LVM does not solve. These debug files are output in the folder specified by the DNAF_OUTPUT_PATH parameter in the appropriate server configuration file and can be identified by the prefixes upfin_export_ and alg_export_. Once the limit is reached, subsequent debug output files are not created. A server log message is generated indicating the limit has been reached. The allowable range is 0-50.
Manual Execution			
Allow in Real Time	REALTIME_VOLTMAN <RealTime>	Check box	This check box indicates if the LVM application can be invoked through the RT DNAF Solution pop-up display (checked means the application is shown). Note LVM can still be executed by periodic or event triggers in RT by setting the appropriate triggering parameters in the server configuration file.
Allow in Study Mode	STUDYMODE_VOLTMAN <Study>	Check box	This check box indicates if the LVM application can be invoked through the ST DNAF Solution pop-up display (checked means the application is shown).
LVM Device Exclusions List			
Station Internal Devices	<ExcludeStationInternalDevice>	Check box	This check box indicates if station internal devices are included in the LVM Device Exclusions list. Applies to capacitors, LTCs/regulators, DERs, and DGs.
Station External Devices	<ExcludeStationExternalDevice>	Check box	This check box indicates if station external devices are included in the LVM Device Exclusions list. Applies to capacitors, LTCs/regulators, DERs, and DGs.

Configuration Editor UI Name	CSV File Field Name and XML File Attribute Name	Type	Description
Control Measurement Missing or Invalid	<ExcludeControlMeas>	Check box	This check box indicates if: - capacitors with a missing or invalid control measurement, or - LTCs/regulators with missing or invalid tap position, voltage setpoint, or voltage reduction measurements are included in the LVM Device Exclusions list. Applies to capacitors and LTCs/regulators.
Supervisory Cutout Measurement Missing or Invalid	<ExcludeSupervisory CutoutMeas>	Check box	This check box indicates if devices with a missing or invalid supervisory cutout measurement are included in the LVM Device Exclusions list. Applies to capacitors and LTCs/regulators.
Auto/Manual Measurement Missing or Invalid	<ExcludeAutoManualMeas>	Check box	This check box indicates if devices with a missing or invalid auto/manual measurement are included in the LVM Device Exclusions list. Applies to capacitors and LTCs/regulators.
Remote/Local Measurement Missing or Invalid	<ExcludeRemoteLocalMeas>	Check box	This check box indicates if devices with a missing remote/local measurement are included in the LVM Device Exclusions list. Applies to capacitors and LTCs/regulators.
Control Available Measurement Missing or Invalid	<ExcludeControlAvailableMeas>	Check box	This check box indicates if devices with a missing or invalid control available measurement are included in the LVM Device Exclusions list. Applies to capacitors and LTCs/regulators.
Device Blocked	<ExcludeDeviceBlocked>	Check box	This check box indicates if blocked devices are included in the LVM Device Exclusions list. Applies to LTCs/regulators.
Device User Disabled	<ExcludeDeviceUserDisabled>	Check box	This check box indicates if user-disabled devices are included in the LVM Device Exclusions list. Applies to capacitors and LTCs/regulators.
Station Disabled	<ExcludeStationDisabled>	Check box	This check box indicates if devices in LVM-disabled stations are included in the LVM Device Exclusions list. Applies to capacitors, LTCs/regulators, DERs, and DGs.
Feeder Disabled	<ExcludeFeederDisabled>	Check box	This check box indicates if devices on LVM-disabled feeders are included in the LVM Device Exclusions list. Applies to capacitors, LTCs/regulators, DERs, and DGs.
Parent Transformer Disabled	<ExcludeParentTransformer Disabled>	Check box	This check box indicates if devices of under LVM-disabled transformers are included in the LVM Device Exclusions list. Applies to capacitors, LTCs/regulators, DERs, and DGs.
Device Disabled	<ExcludeDeviceDisabled>	Check box	This check box indicates if disabled devices are included in the LVM Device Exclusions list. Applies to LTCs/regulators.

Configuration Editor UI Name	CSV File Field Name and XML File Attribute Name	Type	Description
Device Optimization Disabled in Model	<ExcludeDeviceDSODisabledInModel>	Check box	This check box indicates if devices that are disabled for LVM in the static model are included in the LVM Device Exclusions list. Applies to capacitors and LTCs/regulators.
Device Disabled by Tag	<ExcludeDeviceDisabledByTag>	Check box	This check box indicates if devices that are disabled for LVM by tag are included in the LVM Device Exclusions list. Applies to capacitors and LTCs/regulators.
Device Consecutive Operations Time Limit not Met	<ExcludeDeviceConsecutiveOperationLimitNotMet>	Check box	This check box indicates if devices that have not met the consecutive operation time limit are included in the LVM Device Exclusions list. Applies to capacitors and LTCs/regulators.
Device Maximum Switching Count Met	<ExcludeMaxDailySwitchingCount>	Check box	This check box indicates if devices that have met the maximum daily switching count are included in the LVM Device Exclusions list. Applies to capacitors and LTCs/regulators.
Control Failure	<ExcludeControlFailureFlag>	Check box	This check box indicates if devices where control failure occurred are included in the LVM Device Exclusions list. Applies to capacitors, LTCs/regulators, DERs, and DGs.
Device not in PQ Control Mode	<ExcludeNotPQMode>	Check box	This check box indicates if devices that are not in PQ mode are included in the LVM Device Exclusions list. Applies to DERs and DGs.
Only Allowed in Advisory Execution	<ExcludeNotAdvisoryExecution>	Check box	This check box indicates if devices that are only permitted for advisory mode plans are included in the LVM Device Exclusions list if the plan execution is not in advisory mode. Applies to DERs and DGs.
Device has no Control Range or no Control Mode	<ExcludeNoControlRange>	Check box	This check box indicates if devices that have no control range are included in the LVM Device Exclusions list. Applies to LTCs/regulators, DERs, and DGs.
Supervisory Cutout On	<ExcludeSupervisoryCutoutOn>	Check box	This check box indicates if devices with the supervisory cutout measurement set to On are included in the LVM Device Exclusions list. Applies to capacitors and LTCs/regulators.
Control Available Off	<ExcludeControlAvailableOff>	Check box	This check box indicates if devices with the control available measurement set to Off (unavailable) are included in the LVM Device Exclusions list. Applies to capacitors and LTCs/regulators.
Neutral Imbalance On	<ExcludeNeutralImbalanceOn>	Check box	This check box indicates if devices with the neutral imbalance measurement set to On for at least one phase are included in the LVM Device Exclusions list. Applies to capacitors.
Device in Auto Mode	<ExcludeDeviceInAutoMode>	Check box	This check box indicates if devices that are in auto mode when they need to be in manual mode for proper control are included in the LVM Device Exclusions list. Applies to capacitors and LTCs/regulators.

Configuration Editor UI Name	CSV File Field Name and XML File Attribute Name	Type	Description
Device in Manual Mode	<ExcludeDeviceInManualMode>	Check box	This check box indicates if devices that are in manual mode when they need to be in auto mode for proper control are included in the LVM Device Exclusions list. Applies to LTCs/regulators.
Device in Local Mode	<ExcludeDeviceInLocalMode>	Check box	This check box indicates if devices that are in local mode when they need to be in remote mode for proper control are included in the LVM Device Exclusions list. Applies to capacitors and LTCs/regulators.
Optimization not Allowed	<ExcludeDSONotAllowed>	Check box	This check box indicates if devices where optimization is not allowed are included in the LVM Device Exclusions list. Applies to DERs and DGs.
No Feeder	<ExcludeNoFeeder>	Check box	This check box indicates if devices that do not have an energizing feeder are included in the LVM Device Exclusions list. Applies to DERs and DGs.
Old and Deprecated Parameters			
Function Name	NAME_VOLTMAN <Name>	Text box	This entry specifies the function name in the DNAF pop-up display. This field is exposed only in the old DNAF Configurator, not in the new DNAF Configuration Editor.
Function Enumeration	FUNCTENUM_VOLTMAN <FuncEnum>	Text box	This entry specifies the application enumeration: 0 = Other applications 1 = Distribution Power Flow 2 = Protection Validation 4 = Load and Volt/VAR Management 5 = Fault Isolation and Service Restoration 6 = Automatic Feeder Reconfiguration 7 = Planned Outage Study 8 = FISR Contingency Analysis 10 = Fault Location 11 = Short Circuit Calculation This field is exposed only in the old DNAF Configurator, not in the new DNAF Configuration Editor.
Function Index	INDEX_VOLTMAN	Text box	This entry represents the network analysis function index. This field is exposed only in the old DNAF Configurator, not in the new DNAF Configuration Editor.
Volt/Var Control Mode	MODE_VOLTMAN <ControlMode>	N/A	This feature has been deprecated.

Configuration Editor UI Name	CSV File Field Name and XML File Attribute Name	Type	Description
Associated Limit Strategy Index	LIMSTRAT_VOLTMAN	Drop-down list	This selection shows the associated limit strategy. All Limit Strategies are shown on the Limit Strategy page/tab of the DNAF Configuration Editor. A Limit Strategy is a method of specifying which type of limits (emergency or normal) and voltages (bus or load) are to be used by the optimization algorithm. This field is available in the old DNAF Configurator only. In the DNAF Configuration Editor, limit strategy attributes are part of problem formulations, and can be configured independently for each problem formulation.
Production Cost	PRODCOST_VOLTMAN <ProductCost>	Text box	This feature has been deprecated.
Customer Rate	CUSTOMERRATE_VOLTMAN <CustomerRate>	Text box	This feature has been deprecated.
Large Load Limit Size	LARGELOADSIZE_LIM_VOLTMAN <LargeLoadSize>	Text box	This feature has been deprecated.
Power Cost (\$/kW)	KWCOST_VOLTMAN <CostKW>	Text box	This feature has been deprecated.
Default Minimum Power Factor (for Station)	DEFAULTPOWERFACTOR_VOLTMAN	Text box	This entry specifies the default minimum target power factor for the station power transformers (should be close to unity). This default value is used when there is no target power factor schedule defined for the station. This field is available in the old DNAF Configurator only. In the DNAF Configuration Editor, power factor targets are configurable for each problem formulation defined for LVM.
Default Maximum Power Factor (for Station)	DEFAULTPOWERFACTOR_MAX_VOLTMAN	Text box	This entry specifies the default maximum target power factor for the station power transformers (should be close to unity). This default value is used when there is no target power factor schedule defined for the station. This field is available in the old DNAF Configurator only. In the DNAF Configuration Editor, power factor targets are configurable for each problem formulation defined for LVM.
Default Minimum Power Factor (for Feeder)	DEFAULTFEEDERPFMIN_VOLTMAN	Text box	This entry specifies the default minimum target power factor for the feeders (should be close to unity). This default value is used when there is no target power factor schedule. This field is available in the old DNAF Configurator only. In the DNAF Configuration Editor, power factor targets are configurable for each problem formulation defined for LVM.

Configuration Editor UI Name	CSV File Field Name and XML File Attribute Name	Type	Description
Default Maximum Power Factor (for Feeder)	DEFAULTFEEDERPFMAX_VOLT MAN	Text box	This entry specifies the default maximum target power factor for station feeder (should be close to unity). This default value is used when there is no target power factor schedule. This field is available in the old DNAF Configurator only. In the DNAF Configuration Editor, power factor targets are configurable for each problem formulation defined for LVM.
Feeder Power Factor Target	FEEDERPFSUPPORT_VOLTMA N	Check box	This check box indicates whether power factor targets are enforced for feeders. If this box is checked, then feeder-head power factor targets are enforced. Otherwise these power factor targets are not enforced in the optimization algorithm. The default state is cleared. This field is available in the old DNAF Configurator only. In the DNAF Configuration Editor, power factor targets are configurable for each problem formulation defined for LVM.
Source/Load Side Tf Power Factor	SUMFLOWPTFONHIGHSIDE_ VOLTMAN	Check box	This check box indicates the location of the power factor target for station power transformers. If this box is checked, then the power transformer source side node is used as the power factor target. If this box is cleared, then the load side node is used. The default state is checked, indicating the source side transformer node is used as the power factor target. This field is available in the old DNAF Configurator only. In the DNAF Configuration Editor, power factor targets are configurable for each problem formulation defined for LVM.
Ignore Daily Switching Count in RT	IGNORECAPSWTCNTRT_VVC	Check box	This check box indicates whether to ignore the capacitor maximum daily switching count limit in real time. The default state is checked. If the box is cleared, then the daily operation count for capacitors is compared to the capacitor maximum daily switching count limit. The limit is statically defined (by the parameter MaxSwitches) for capacitor objects in the station model. If the limit has been reached, then the capacitor is made optimization ineligible until the daily count has been reset. This field is available in the old DNAF Configurator only. In the DNAF Configuration Editor, this field has been deprecated in favor of IgnoreSwitchCount.

Configuration Editor UI Name	CSV File Field Name and XML File Attribute Name	Type	Description
Ignore Daily Switching Count in ST	IGNORECAPSWTCNTST_VVC	Check box	This check box indicates whether to ignore the capacitor maximum daily switching count limit in study mode. If the box is cleared, then the daily operation count for capacitors is compared to the capacitor maximum daily switching count limit. The limit is statically defined (by the parameter MaxSwitches) for capacitor objects in the station model. This field is available in the old DNAF Configurator only. In the DNAF Configuration Editor, this field has been deprecated in favor of IgnoreSwitchCount.
Ignore Consecutive Switching in RT	IGNORECAPSWTINTRT_VVC	Check box	This check box indicates whether to ignore the capacitor minimum time interval limit between consecutive switching operations in real time. The default state is checked. If this box is cleared, then the time elapsed since the capacitor was last switched is compared with the capacitor minimum time interval limit between consecutive switching operations. The limit is statically defined (by the parameter ConsecSw, or CONSECSW_SHIMP for the CSV file) for capacitor objects in the station model. If the time elapsed is less than the limit, then a switching plan is not generated for the capacitor. This field is available in the old DNAF Configurator only. In the DNAF Configuration Editor, this field has been deprecated in favor of IgnoreConsecSwitch.
Ignore Consecutive Switching in ST	IGNORECAPSWTINTST_VVC	Check box	This check box indicates whether to ignore the capacitor minimum time interval limit between consecutive switching operations in real time. The default state is checked. If this box is cleared, then the time elapsed since the capacitor was last switched is compared with the capacitor minimum time interval limit between consecutive switching operations. The limit is statically defined (by the parameter ConsecSw, or CONSECSW_SHIMP for the CSV file) for capacitor objects in the station model. If the time elapsed is less than the limit, then a switching plan is not generated for the capacitor. This field is available in the old DNAF Configurator only. In the DNAF Configuration Editor, this field has been deprecated in favor of IgnoreConsecSwitch.
Study Mode Same as Real Time	STCONTROLELIGIBLE_VVC <StudyAsRT>	Check box	This feature has been deprecated.

Configuration Editor UI Name	CSV File Field Name and XML File Attribute Name	Type	Description
Voltage Limit Adjustment	VOLTLIMITADJUST_VOLTMAN <VoltLimitAdjust>	Check box	This check box indicates how voltage limits are to be adjusted. If checked, then the voltage limits for the loads and/or buses are adjusted based on the differences in the power flow calculated voltages and the SCADA measurement voltages. If Enhanced Voltage Limit Adjustment is selected, then the enhanced voltage limit adjustment method is used to adjust voltage limit. For details, see Voltage Limit Adjustment. This field is available on the LVM tab in the old DNAF Configurator only.
Only Consider Adjusted Voltage Limits	ONLYADJVOLTLIMIT_VOLTMAN <OnlyConsiderAdjustedVoltageLimits>	Check box	This check box indicates if only load points (where voltage measurements exist) are to be used by LVM. This check box is effective only if Voltage Limit Adjustment or Enhanced Voltage Limit Adjustment is selected. This field is available on the LVM tab in the old DNAF Configurator only.
Enhanced Voltage Adjustment	ENHANCEDVOLTADJUST_VOLTMAN <EnhancedVoltageAdjustment>	Check box	This check box indicates how voltage limits are to be adjusted. If checked, then the enhanced voltage limit adjustment method is used to adjust voltage limit. For details, see Voltage Limit Adjustment. This field is available on the LVM tab in the old DNAF Configurator only.
Perform Tap Estimation	TAPESTIMATION_VOLTMAN	Check box	This check box indicates whether to perform tap estimation if the corresponding voltage measurement exists. There may be cases when no tap position measurement is available for some LTCs (and possibly regulators). In such a case, if this box is checked, and if both the source side and load side voltage measurements on the transformers are available, then the tap position is estimated using the measurements and the transformer model parameters. This field is available on the LVM tab in the old DNAF Configurator only. In the DNAF Configuration Editor, this field is available on the Distribution Power Flow page.
Issue Violation Alarm	ALARMOPT_VOLTMAN	Check box	This check box indicates if LVM is to generate alarms (sent to the Alarm Subsystem). If this box is checked, all appropriate LVM alarms documented in LVM Alarms will be issued. The default state is enabled. Deprecated: alarms issuance functionality is now available on the DNAF System Configuration display from the RT Geographic Display (RT DMS drop-down menu).

Configuration Editor UI Name	CSV File Field Name and XML File Attribute Name	Type	Description
Permission Area	ALARMPERMAREA_VOLTMAN	Text box	This entry specifies the default alarm permission area to be used for issuing LVM alarms. Deprecated: alarms issuance functionality is now available on the DNAF System Configuration display from the RT Geographic Display (RT DMS drop-down menu).
Control Method	SINGLEMODEPERFEEDER_VV C <SingleControlPerFeeder>	N/A	This field is no longer supported.
Ignore Controls in Substation	IGNORECOMPINSUB_VOLTMAN	Check box	This check box indicates whether capacitors and tap-changing transformers are ignored in LVM. If the box is checked, then capacitors and LTC/regulators are not considered in the LVM solution. This field is available on the LVM tab in the old DNAF Configurator only. In the DNAF Configuration Editor, this functionality is provided at a more granular level on a per LVM problem formulation basis, by the options "Substation Capacitors (Remote Controlled/Manually Controlled)" and "Substation Tap changers (Remote Controlled/Manually Controlled)" in the Device Eligibility section on the LVM Problem Formulations page.

Configuration Editor UI Name	CSV File Field Name and XML File Attribute Name	Type	Description
Use Additional Optimization Limit	USEADDITIONALLIMIT_VOLT MAN <UseAdditionalOptLimit>	Check box	If this box is checked, then a separate set of optimization specific limits are used by LVM. Otherwise, the operational limits are used. The optimization specific limits are optionally defined statically for bus objects and for load categories in the station model. If the heuristic approach is indicated as the solution method, and if the server is configured to use optimization limits, but optimization limits are not defined at a bus, then the voltage measurements at that bus cannot trigger an LVM Voltage Quality trigger. Further, the LVM application ignores those voltage measurements. If the optimization approach is indicated as the solution method, and if the server is configured to use optimization limits, but optimization limits are not defined for load categories, then voltage limits for the associated loads are not enforced by the LVM application. This field is available on the LVM tab in the old DNAF Configurator only. In the current DNAF Configuration Editor, this functionality is part of "Limit Strategy". To specify normal or optimization limits, use the "Respect bus voltage limits" and "Respect load voltage limits" drop-down lists.
Prevent Infeasible Solution	PREVENTINFEASIBLE_VOLTMAN <PreventInfeasiblePlan>	Check box	This field is no longer supported.
Maximum Retries for Switched Capacitors	MAXRETRYCAPACITOR_VOLTMAN <MaxRetryCapacitor>	Text box	This entry specifies the maximum number of attempts to send out a SCADA control for capacitors. If the total number of attempts for a capacitor in an LVM plan reaches this value, the plan is considered as failed, an alarm is optionally issued, and the plan is deleted. This field is available on the LVM tab in the old DNAF Configurator only.
Maximum Retries for LTCs and Regulators	MAXRETRYLTCREGULATOR_VOLTMAN <MaxRetryLTCRegulator>	Text box	This entry specifies the maximum number of attempts to send out a SCADA control for LTCs/regulators. If the total number of attempts for an LTC/regulator in an LVM plan reaches this value, the plan is considered as failed, an alarm is optionally issued, and the plan is deleted. This field is available on the LVM tab in the old DNAF Configurator only.

3.16.1. LVM Problem Formulation

This section describes problem formulation settings for the LVM application.

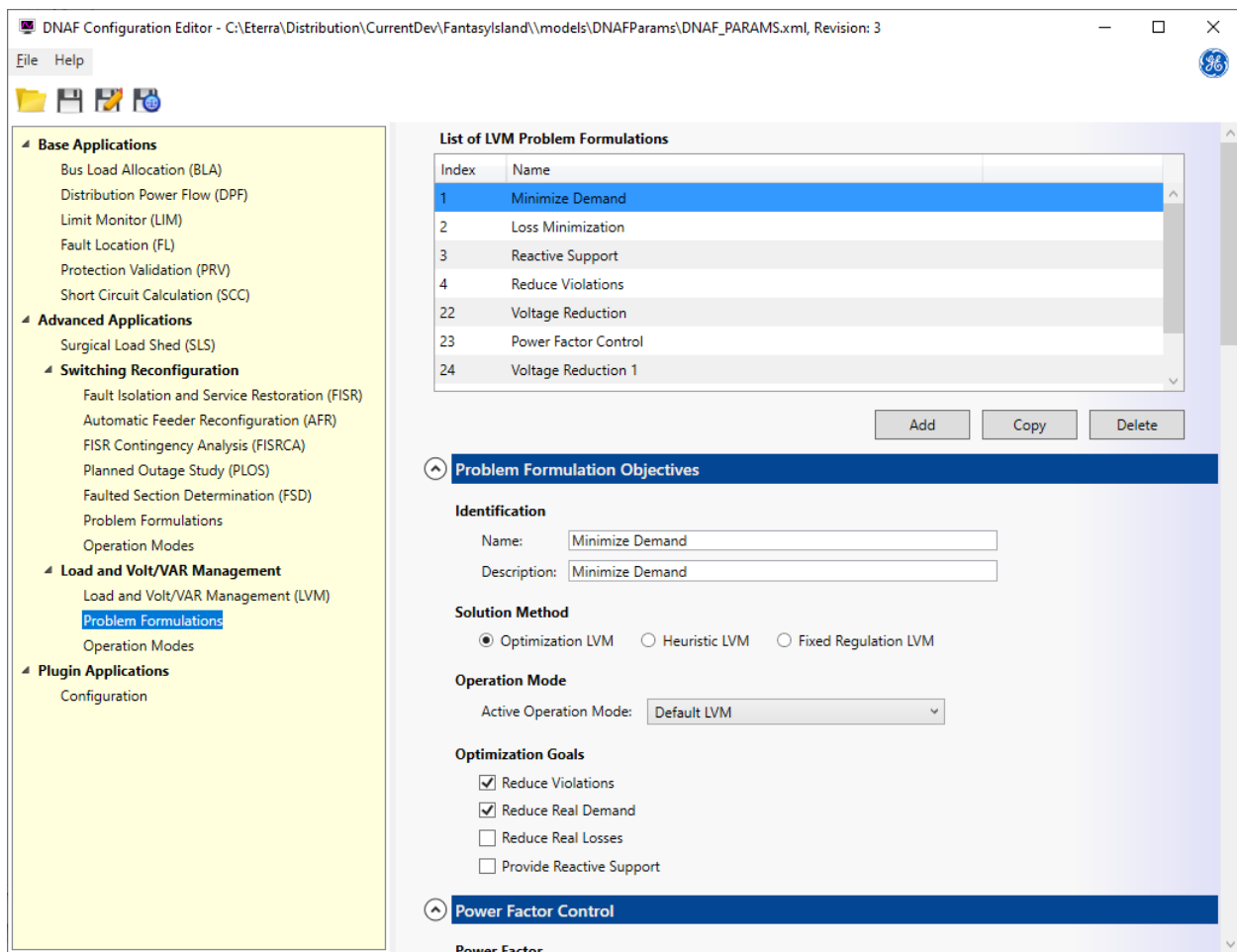


Figure 68. Problem Formulation LVM

Table below describes the parameters for the LVM application's problem formulations. All listed parameters are contained by the <Params xsi:type = "ProbFormulationParams"> element in the DNAF_PARAMS.xml configuration file.

Table 6. Problem Formulation - LVM

Configuration Editor Name	CSV File Field Name and XML File Attribute Name	Type	Description
Identification			
Index	INDEX_PROBFORM <NumberProb>	Int	This table entry is generated automatically and specifies the index of the problem formulation. This index is used to refer to a specific problem formulation in problem formulation schedules for stations.
Name	NAME_PROBFORM <Name>	Text box	This entry specifies the name of the problem formulation. The name specified here is used in various displays such as the DNAF Solution pop-up display and the DNAF System Configuration display.

Configuration Editor Name	CSV File Field Name and XML File Attribute Name	Type	Description
Description	DESC_PROBFORM <Desc>	Text box	This entry specifies a description for a problem formulation.
Operation Mode			
Active Operation Mode	<ActiveOperationModeID>	Drop-down list	This selection specifies which operation mode is used for the current problem formulation. This allows each problem formulation to have its own defined plan execution parameters.
Solution Method			
Optimization LVM	<LVMSolution> = 0	Option button	This setting specifies the LVM solution algorithm. When this option is selected, the optimization LVM Solution method is used. This approach uses a rigorous power flow based approach to determine the optimal series of control actions to meet a specified objective.
Heuristic LVM	<LVMSolution> = 1	Option button	This setting specifies the LVM solution algorithm. When this option is selected, the heuristic LVM Solution method is used. This approach does not use a power flow solution. It generates one control action at a time – either a capacitor switching move, or a tap position adjustment – to achieve a specified objective.
Fixed Regulation LVM	<LVMSolution> = 2	Option button	This setting specifies the LVM solution algorithm. When this option is selected, the fixed regulation LVM Solution method is used. The fixed regulation approach is based on the specific required voltage reduction percentage.
Optimization Goals			
Reduce Violations	<LVMTargetReduce Violation>	Check box	This setting indicates if LVM will attempt to reduce voltage and flow violations in the optimization process. This goal applies to Optimization LVM only.
Reduce Demand	<LVMTargetDemand Reduction>	Check box	This setting indicates if LVM will attempt to minimize load demand without violating voltage limits. This goal applies to Optimization LVM only.
Reduce Losses	USELOSSSENS_PROBFORM <LVMTargetLoss Reduction>	Check box	This setting indicates if LVM will attempt to minimize losses in the optimization process. This goal applies to Optimization LVM only.
Provide Reactive Support	<LVMTargetReactive Support>	Check box	This setting indicates if LVM will try to bring power factors at station power transformers within specified limits.
Power Factor Control			
Transformer Power Factor Monitoring Location	USESUMFLOWPTF_PROBFORM <PowerTransformerPFTargetOnSourceSide>	Drop-down list	This selection is used to indicate which side (source side or load side) of station power transformers should be used for power factor control.

Configuration Editor Name	CSV File Field Name and XML File Attribute Name	Type	Description
Default Minimum Power Factor	<EnforceMinimumPF>	Check box	This setting enables or disables the default minimum target power factor for system elements that use this problem formulation.
Default Minimum Power Factor	<MinimumPFTarget>	Text box	This entry specifies the default minimum target power factor for system elements that use this problem formulation. The values for stations, power transformers, and feeders should be close to unity. This default value is used when there is no manual target power factor defined for the system element using the DNAF System Configuration display. For more details about setting the default power factor target, see Table 3. The allowable range is -1~1.
Default Maximum Power Factor	<EnforceMaximumPF>	Check box	This setting enables or disables the default maximum target power factor for system elements that use this problem formulation.
Default Maximum Power Factor	<MaximumPFTarget>	Text box	This entry specifies the default maximum target power factor for system elements that use this problem formulation. The values for stations, power transformers, and feeders should be close to unity. This default value is used when there is no manual target power factor defined for the system element using the DNAF System Configuration display. The allowable range is -1~1.
Voltage Limits			
Buses with Voltage Measurements: Applicable Limit	<EnforceBusVoltage Limits>	Drop-down list	This selection specifies which voltage limit type is to be enforced for buses with recent voltage measurements. • If None is selected, voltage limits are not enforced. • If Normal Operational is selected, normal operational limits are enforced. • If Normal Optimization is selected, normal optimization limits are enforced. • If Emergency Operational is selected, emergency operational limits are enforced. • If Emergency Optimization is selected, emergency optimization limits are enforced.
Buses with Voltage Measurements: Limit Adjustment	<BusVoltageLimit AdjustmentMethod>	Drop-down list	This selection specifies which voltage limit adjustment will be applied for buses with recent voltage measurements. • If None is selected, limit adjustment is not applied. • If Regular is selected, regular limit adjustment is applied. • If Enhanced is selected, enhanced limit adjustment is applied.

Configuration Editor Name	CSV File Field Name and XML File Attribute Name	Type	Description
Buses without Voltage Measurements: Applicable Limit	<EnforceBusVoltageLimitsWithoutMeas>	Drop-down list	This selection specifies which voltage limit type is to be enforced for buses with no voltage measurements or without recent voltage measurements. • If None is selected, voltage limits are not enforced. • If Normal Operational is selected, normal operational limits are enforced. • If Normal Optimization is selected, normal optimization limits are enforced. • If Emergency Operational is selected, emergency operational limits are enforced. • If Emergency Optimization is selected, emergency optimization limits are enforced.
Loads with Voltage Measurements: Applicable Limit	<EnforceLoadVoltageLimits>	Drop-down list	This selection specifies which voltage limit type is to be enforced for loads with recent voltage measurements. • If None is selected, voltage limits are not enforced. • If Normal Operational is selected, normal operational limits are enforced. • If Normal Optimization is selected, normal optimization limits are enforced. • If Emergency Operational is selected, emergency operational limits are enforced. • If Emergency Optimization is selected, emergency optimization limits are enforced.
Loads with Voltage Measurements: Limit Adjustment	<LoadVoltageLimitAdjustmentMethod>	Drop-down list	This selection specifies which voltage limit adjustment will be applied for loads with recent voltage measurements. • If None is selected, limit adjustment is not applied. • If Regular is selected, regular limit adjustment is applied. • If Enhanced is selected, enhanced limit adjustment is applied.
Loads without Voltage Measurements: Applicable Limit	<EnforceLoadVoltageLimitsWithoutMeas>	Drop-down list	This selection specifies which voltage limit type is to be enforced for loads with no voltage measurements or without recent voltage measurements. • If None is selected, voltage limits are not enforced. • If Normal Operational is selected, normal operational limits are enforced. • If Normal Optimization is selected, normal optimization limits are enforced. • If Emergency Operational is selected, emergency operational limits are enforced. • If Emergency Optimization is selected, emergency optimization limits are enforced.

Configuration Editor Name	CSV File Field Name and XML File Attribute Name	Type	Description
Disable Capacitor Voltage Rails	<DisableVoltageRails>	Check box	This setting indicates if LVM will consider solutions where the voltage (high or low) exceeds or is below any capacitor voltage rails. If selected, the plan will include commands to disable the voltage rails on affected capacitors. These commands must be implemented within the time period defined by the Capacitor Voltage Rails Check Timeout (sec) option in the Operation Modes page.  Note Voltage reduction plan implementation should start after the voltage rails are disabled at all capacitors included in the plan.
Discard Load Voltage Measurement with Large Deviations	<DiscardVMeasLoadWithLargeDeviation>	Check box	This setting indicates whether the LVM solution discards a load voltage measurement from the voltage limit adjustment process if the deviation between the measurement and the DPF calculated value exceeds the tolerance set on the DPF page using the Maximum Voltage Mismatch at Load Point parameter.
Voltage Bandwidth Reduction	<EnableVoltBandwidthReduction>	Check box	This setting enables or disables the voltage bandwidth reduction feature. This feature only applies to optimization LVM approach problem formulations.
Voltage Bandwidth Reduction	<VoltBandwidthReduction>	Text box	This entry specifies the voltage bandwidth reduction percentage. The allowable range is 0~100.
Branch Flow Limits			
Respect Limit	<EnforceBranchFlowLimits>	Drop-down list	This selection specifies which branch flow limit level is to be enforced for branch objects. • None: Flow limits are not enforced. • Level 1: The most restrictive limit level. • Level 2: Less restrictive than level 1. • Level 3: Less restrictive than level 2. • Level 4: Less restrictive than level 3. • Level 5: The least restrictive limit level.
Eligible Controls			
Substation Capacitors	<UseStationCapacitors>	Check box	This setting specifies if LVM can use eligible capacitors from within the station fence in the optimization process.
Field Capacitors	<UseFieldCapacitors>	Check box	This setting specifies if LVM can use eligible field capacitors in the optimization process.
Substation Regulating Transformers	<UseStationTapChangers>	Check box	This setting specifies if LVM can use eligible LTC/regulators from within the station fence in the optimization process.
Field Regulating Transformers	<UseFieldTapChangers>	Check box	This setting specifies if LVM can use eligible field LTC/regulators in the optimization process.

Configuration Editor Name	CSV File Field Name and XML File Attribute Name	Type	Description
Substation Generators	<UseStationGenerators>	Check box	This setting specifies if LVM can use eligible generators from within the station fence in the optimization process.
Field Generators	<UseFieldGenerators>	Check box	This setting specifies if LVM can use eligible field generators in the optimization process.
Substation Distributed Energy Sources	<UseStationDERs>	Check box	This setting specifies if LVM can use eligible distributed energy sources from within the station fence in the optimization process.
Field Distributed Energy Sources	<UseFieldDERs>	Check box	This setting specifies if LVM can use eligible field distributed energy sources in the optimization process.
Performance Index Weights			
Flow Violations (Final Step)	PERFFLOWFINAL_PROBFORM <PIFlowViolationFinal>	Text box	This entry specifies the weight of the Flow Violations Performance Index (PI). This weight only applies for the final solutions. Valid value is any real numbers starting from 0. See Linear Programming, Problem Formulation Parameter Tuning, and Problem Formulations. The allowable range is 0-1000000000.
Flow Violations (Intermediate Steps)	PERFFLOWTEMP_PROBFORM <PIFlowViolationInt>	Text box	This entry specifies the weight of the Flow Violations Performance Index (PI). This weight is used for the PI only during the intermediate solutions. Valid value is any real numbers starting from 0. See Linear Programming, Problem Formulation Parameter Tuning, and Problem Formulations. The allowable range is 0-1000000000.
Bus Voltage Violations (Final Step)	PERFVOLTFINAL_PROBFORM <PIVoltageViolationFinal>	Text box	This entry specifies the weight of the Voltage Violations Performance Index (PI). This weight is applicable only for bus voltage violations in the final solution. Valid value is any real numbers starting from 0. See Linear Programming, Problem Formulation Parameter Tuning, and Problem Formulations. The allowable range is 0-1000000000.
Bus Voltage Violations (Intermediate Steps)	PERFVOLTTEMP_PROBFORM <PIVoltageViolationInt>	Text box	This entry specifies the weight of the Voltage Violations Performance Index (PI). This weight is applicable only for bus voltage violations in the intermediate solutions. Valid value is any real numbers starting from 0. See Linear Programming, Problem Formulation Parameter Tuning, and Problem Formulations. The allowable range is 0-1000000000.
Load Voltage Violations (Final Step)	<PILoadVoltageViolation Final>	Text box	This entry specifies the weight of the Voltage Violations Performance Index (PI). This weight is applicable only for load voltage violations in the final solution. The allowable range is 0-1000000000.
Load Voltage Violations (Intermediate Steps)	<PILoadVoltageViolation Int>	Text box	This entry specifies the weight of the Voltage Violations Performance Index (PI). This weight is applicable only for load voltage violations in the intermediate solutions. The allowable range is 0-1000000000.

Configuration Editor Name	CSV File Field Name and XML File Attribute Name	Type	Description
Total Cost (Final Step)	PERFCOSTFINAL_PROBFORM <PITotalCostFinal>	Text box	This entry specifies the weight of the final step total cost Performance Index (PI).
Total Cost (Intermediate Steps)	PERFCOSTTEMP_PROBFORM <PITotalCostInt>	Text box	This entry specifies the weight of the Intermediate total cost Performance Index (PI).
Real and Reactive Flow	<PISumflowFinal>	Text box	This entry specifies the performance index weight for the active and reactive power violations at the flow monitoring points. Flow monitoring points are used in several LVM problem formulations including minimize demand and reactive area support. The allowable range is 0-1000000000.
Control Moves	PERFCONTROLMOVES_FINAL_PROBFORM <PINoControlMove>	Text box	This entry specifies the weight of the final step number of control moves Performance Index (PI) (see Problem Formulations for more information). The allowable range is 0-1000000000.
Losses	PERFLOSSESFINAL_PROBFORM <Losses>	Text box	This entry specifies the weight of the final step losses Performance Index (PI) (see Problem Formulations for more information). The allowable range is 0-1000000000.
Bus Voltage Imbalance	<PIVoltImbalance>	Text box	This entry specifies the performance index weight for the bus voltage imbalance. This PI is calculated using the standard maximum deviation from the average method. The PI value is included in the objective function value calculation, even if there are pre-plan voltage violations (similar to power factor). The allowable range is 0-1000000000.
Violation Penalties and Costs			
Bus Low Voltage Violation	SCALECOSTVOLTLO_PROBFORM <SCLowVoltageViolation>	Text box	This entry specifies the scaling factor for the bus low voltage violation cost curve. For more information about violation penalties and costs, see Cost Model . The allowable range is 0-1000000000.
Bus High Voltage Violation	SCALECOSTVOLTHI_PROBFORM <SCHighVoltage Violation>	Text box	This entry specifies the scaling factor for the bus high voltage violation cost curve. The allowable range is 0-1000000000.
Load Low Voltage Violation	<LoadLowVoltViolation Penalty>	Text box	This entry specifies the scaling factor for load low voltage violation cost curve. The allowable range is 0-1000000000.

Configuration Editor Name	CSV File Field Name and XML File Attribute Name	Type	Description
Load High Voltage Violation	<LoadHighVoltViolationPenalty>	Text box	This entry specifies the scaling factor for load high voltage violation cost curve. The allowable range is 0-1000000000.
Low Voltage Violation Count	<SCLowVoltageViolationCount>	Text box	This entry specifies the scaling factor for the number of customers associated with a load. This scaling factor is multiplied by the number of customers associated with a load, and then applied to the load low voltage violation cost curve. The allowable range is 0-1000000000.
High Voltage Violation Count	<SCHighVoltageViolationCount>	Text box	This entry specifies the scaling factor for the number of customers associated with a load. This scaling factor is multiplied by the number of customers associated with a load, and then applied to the load high voltage violation cost curve. The allowable range is 0-1000000000.
Critical Customer Low Voltage Violation	<SCCriticalCustomerLowVoltageViolation>	Text box	This entry specifies the scaling factor for critical customers associated with a load. This scaling factor is multiplied by the active power of a load with critical customers, and then applied to the load low voltage violation cost curve. The allowable range is 0-1000000000.
Critical Customer High Voltage Violation	<SCCriticalCustomerHighVoltageViolation>	Text box	This entry specifies the scaling factor for the critical customers associated with a load. This scaling factor is multiplied by the active power of a load with critical customers, and then applied to the load high voltage violation cost curve. The allowable range is 0-1000000000.
Flow Violation	<SCFlowViolation>	Text box	This entry specifies the scaling factor for flow violation cost curve. The allowable range is 0-1000000000.
Voltage	SCALECOSTVOLT_PROBFORM <SCVoltage>	Text box	This entry specifies the scaling factor for voltage control setting (DER/generators). The allowable range is 0-1000000000.
Real Flow	SCALECOSTLOSSKW_PROBFORM <SCTotalActiveLoss>	Text box	This entry specifies the real power loss scaling cost (used in the minimize loss and minimize/maximize demand problem formulations). The allowable range is 0-1000000000.
Reactive Flow	SCALECOSTSUMFLOW_PROBFORM <SCSumFlow>	Text box	This entry specifies the scaling factor for the station's reactive power limit violation. The allowable range is 0-1000000000.

Configuration Editor Name	CSV File Field Name and XML File Attribute Name	Type	Description
Tap Changer	SCALECOSTTAP_PROBFORM <SCTapChanger>	Text box	This entry specifies the scaling factor for LTC/regulator switching that is less than or equal to the maximum number of daily switching operations. This scaling factor is multiplied by the LTC/regulator movement cost from the LTC/regulator model and by a factor equal to $(1 + \text{CurrentDailyOperations}/\text{MaximumDailyOperations})$ to obtain the effective LTC/regulator movement cost that is used in formulating the Linear Programming model in Optimization LVM. The allowable range is 0-1000000000.
Tap Changer Movement (Exceed Daily Count)	<SCTapChangerAfterDailyCount>	Text box	This entry specifies the cost curve scaling factor for the LTC/regulator switching that exceeds the maximum number of daily switching operations. This scaling factor is multiplied by the LTC/regulator movement cost from the LTC/regulator model and by a factor equal to $(1 + \text{CurrentDailyOperations}/\text{MaximumDailyOperations})$ to obtain the effective LTC/regulator movement cost that is used in formulating the Linear Programming model in Optimization LVM. The allowable range is 0-1000000000.
Capacitor Switching	SCALECOSTSWSH_PROBFORM <SCShimp>	Text box	This entry specifies the scaling factor for shunt impedance (capacitors) switching that is less than or equal to the maximum number of daily switching operations. This scaling factor is multiplied by the capacitor switching cost from the capacitor model and by a factor equal to $(1 + \text{CurrentDailyOperations}/\text{MaximumDailyOperations})$ to obtain the effective capacitor switching cost that is used in formulating the Linear Programming model in Optimization LVM. The allowable range is 0-1000000000.
Capacitor Switching (Exceed Daily Count)	SCALECOSTNUMCAPSWT_PROBFORM <SCNoCapSwitching>	Text box	This entry specifies the cost curve scaling factor for the capacitor switching that exceeds the maximum number of daily switching operations. This scaling factor is multiplied by the capacitor switching cost from the capacitor model and by a factor equal to $(1 + \text{CurrentDailyOperations}/\text{MaximumDailyOperations})$ to obtain the effective capacitor switching cost that is used in formulating the Linear Programming model in Optimization LVM. The allowable range is 0-1000000000.
Generator VAR Change	<SCGeneratorVAR>	Text box	This entry specifies the scaling cost for generator's VAR output change. The allowable range is 0-1000000000.

Configuration Editor Name	CSV File Field Name and XML File Attribute Name	Type	Description
DER VAR Change	<SCDERVAR>	Text box	This entry specifies the scaling cost for distributed energy resource's VAR output change. The allowable range is 0-1000000000.
Large Loads	SCALECOSTLGLDS_PROBFORM <SCLargeLoad>	Text box	This entry specifies the scaling cost for constraining the voltage limits of large loads. The allowable range is 0-1000000000.
Regulator Group Voltage Imbalance	<SCVoltImbalance>	Text box	This entry specifies the scaling cost for voltage imbalance in regulation groups. The allowable range is 0-1000000000.
Local Control Capacitors (Amps)	<SCLocalCapAmp>	Text box	This entry specifies the scaling cost for constraining the switching of locally regulating capacitors of the Amp control type. The allowable range is 0-1000000000.
Local Control Capacitors (Volt)	<SCLocalCapVolt>	Text box	This entry specifies the scaling cost for constraining the switching of locally regulating capacitors of the Volt control type. The allowable range is 0-1000000000.
Local Control Capacitors (VAR)	<SCLocalCapVAR>	Text box	This entry specifies the scaling cost for constraining the switching of locally regulating capacitors of the VAR control type. The allowable range is 0-1000000000.
Local Control Capacitors (PF)	<SCLocalCapPF>	Text box	This entry specifies the scaling cost for constraining the switching of locally regulating capacitors of the PF control type. The allowable range is 0-1000000000.
Failsafe LVM			
Failsafe LVM	<FailsafeEnabled>	Check box	This check box specifies whether a fixed voltage reduction LVM problem formulation is applied to a system element (station, power transformer, or feeder) when the station's DPF solution is suspect or not solved, or voltage mismatches at station internal and external buses are beyond the tolerance. The failsafe problem formulation overrides current problem formulations for the station and station sub-elements in the DNAF System Configuration display.
Failsafe Problem Formulation	<FailsafeID>	Drop-down selection	This entry specifies the name of the fixed voltage reduction problem formulation that is applied.
Manual Reversion	<FailsafeReversal Method> = 0	Option button	This option determines whether the LVM problem formulation automatically reverts to the one used before the failsafe problem formulation was enacted. If selected, LVM does not re-apply the previous problem formulation for a station after the LVM Failsafe condition subsides (the station's DPF solution status becomes not-suspect and bus voltage mismatches are back within the tolerance).

Configuration Editor Name	CSV File Field Name and XML File Attribute Name	Type	Description
Automatic Reversion	<FailsafeReversal Method> = 1	Option button	This option determines whether the LVM problem formulation automatically reverts to the one used before the failsafe problem formulation. If selected, LVM monitors a station's DPF solution status and bus voltage mismatches before every run. If the station's LVM Failsafe condition subsides (the station's DPF solution status becomes not-suspect and bus voltage mismatches are back within the tolerance), the problem formulation for the station is reset to the problem formulation used before the LVM Failsafe function was triggered.
Auto Capacitor Parameters			
Auto Setpoint Slack (%)	<LocalSetpointSlack>	Text box	This entry specifies the amount of slack, in percentage, to be applied to the local setpoint values to tighten the control range of the individual capacitor. The slack is applied evenly to both the lower and upper setpoint values such that the median value between the two is maintained after the slack is applied.
Fixed Regulation LVM Parameters			
Fixed Voltage Reduction	<FixedVRType> = 0	Option Button	This option specifies whether fixed voltage reduction is enabled and the voltage reduction target specified in the Voltage Reduction field is applied.
Turn Off Voltage Reduction	<FixedVRType> = 1	Option Button	This option specifies whether fixed voltage reduction is disabled and the voltage reduction target specified in the Default Voltage Reduction field is applied.
Voltage Reduction	<VRAmount>	Text box	This entry specifies the voltage reduction target against the active band-center setpoint or the nominal voltage target value, in percentage. For example, if the active band-center voltage is 125 V and the voltage reduction target is 2%, the effective reduction target is $125 - 125 * 2\% = 122.5$ V.
Default Voltage Reduction	<DefaultVRAmount>	Text box	This entry specifies the default voltage reduction target against the voltage base value, in percentage.
Heuristic LVM Parameters			
Global Voltage Flattening Threshold	<VoltFlatLimit>	Text box	This entry specifies the default voltage flattening threshold that is used if a voltage flattening threshold is not defined for feeders in the station model file. It is expressed as a percentage of the voltage base. For example, if the voltage flattening threshold is 1% and the voltage base is 120V, the voltage flattening threshold is 1.2V. For more information, see Heuristic LVM Parameters. The allowable range is 0-5.

Configuration Editor Name	CSV File Field Name and XML File Attribute Name	Type	Description
Global Voltage Regulation Margin	<VoltRegulationMargin>	Text box	This entry specifies the default voltage regulation margin that is used if a voltage regulation margin is not defined for regulators/LTCs in the station model file. It is expressed as a percentage of the voltage base. For more information, see Heuristic LVM Parameters. The allowable range is 0-5.
Voltage Reduction Weight	<VoltageReductionWeight>	Text box	This entry specifies the weight for the voltage reduction component in the Heuristic LVM capacitor control algorithm. For more information, see Heuristic LVM Parameters. The allowable range is 0-100.
Minimum Voltage for Power Factor Control	<MinVoltForPowerFactorControl>	Text box	This entry specifies the effective lower voltage limit for voltage flattening when the power factor control objective is used. For more information, see Heuristic LVM Parameters. The allowable range is 0-1.1.
Maximum Delta V Deviation	<MaxDeltaVForLVM>	Text box	This entry specifies the maximum allowed deviation between the capacitor voltage sensitivities calculated by the Heuristic LVM application and the ones received from SCADA. If any deviation is greater than this threshold, an alarm (Capacitor Delta V Mismatch) is issued. For more information, see Heuristic LVM Parameters. The allowable range is 0.1-10.
Heuristic LVM Exponential Components			
Lowest Average Voltage Coefficient 1 (LV1)	<HeuristicLVMExpVF1>	Text box	This entry specifies the first lowest average voltage coefficient that is used for exponential scaling of the lowest average voltage component in the objective function in the Heuristic LVM capacitor control algorithm. The lowest average voltage component is equal to $LV1 * \exp[\frac{\text{Average Calculated Voltage} - \text{Lowest Average Voltage}}{LV2}]$. The usage of this parameter is described in Heuristic LVM Methodology. The allowable range is 0-50.
Lowest Average Voltage Coefficient 2 (LV2)	<HeuristicLVMExpVF2>	Text box	This entry specifies the second lowest average voltage coefficient that is used for exponential scaling of the lowest average voltage component in the objective function in the Heuristic LVM capacitor control algorithm. The usage of this parameter is described in Heuristic LVM Methodology. The allowable range is 0-50.

Configuration Editor Name	CSV File Field Name and XML File Attribute Name	Type	Description
Power Factor Coefficient 1 (PF1)	<HeuristicLVMEExpPF1>	Text box	This entry specifies the value of the first coefficient (PF1) used for exponential scaling of power factor component in the objective function in the Heuristic LVM capacitor control algorithm. The power factor component is equal to $PF1 * \exp[\frac{u(PF_{Actual} - PF_{TargetMax}) * (PF_{Actual} - PF_{TargetMax}) + u(PF_{TargetMin} - PF_{Actual}) * (PF_{TargetMin} - PF_{Actual})}{PF2}]$. The usage of this parameter is described in Heuristic LVM Methodology. The allowable range is 0-50.
Power Factor Coefficient 2 (PF2)	<HeuristicLVMEExpPF2>	Text box	This entry specifies the value of the second coefficient (PF2) used for exponential scaling of power factor component in the objective function in the Heuristic LVM capacitor control algorithm. The usage of this parameter is described in Heuristic LVM Methodology. The allowable range is 0-50.
Bellwether Meter Selection Algorithm			
Maximum Number of High Ranking Meters per Feeder	<MaxHighBellWetherVolt MetersPerFeeder>	Text box	This entry specifies the maximum number of high-ranking voltage profile enabled AMI meters per LVM enabled feeder that can be selected for the pre-solve AMI read request by the Bellwether Meter Selection Algorithm. The allowable range is 1-100.
Maximum Number of Low Ranking Meters per Feeder	<MaxLowBellWetherVolt MetersPerFeeder>	Text box	This entry specifies the maximum number of low-ranking voltage profile enabled AMI meters per LVM enabled feeder that can be selected for the pre-solve AMI read request by the Bellwether Meter Selection Algorithm. The allowable range is 1-100.
Weight for Calculated Voltage Component	<WeightAmiCalculated Voltage>	Text box	This entry specifies the weight for the calculated load voltage component in ranking AMI meters by the Bellwether Meter Selection Algorithm. For more information, see Advanced Metering Parameters. The allowable range is 0-100.
Weight for Phase Diversity Component	<WeightAmiPhaseDiversity>	Text box	This entry specifies the weight for the load phase diversity component in ranking AMI meters by the Bellwether Meter Selection Algorithm. For more information, see Advanced Metering Parameters. The allowable range is 0-100.

Configuration Editor Name	CSV File Field Name and XML File Attribute Name	Type	Description
Weight for Location Component	<WeightAmiLocation>	Text box	This entry specifies the weight for the load location component in ranking AMI meters by the Bellwether Meter Selection Algorithm. For more information, see Advanced Metering Parameters. The allowable range is 0-100.

Old and Deprecated Parameters

TC Results	SCALECOSTTC_PROBFORM	Text box	This field is deprecated.
Flexible Large Load Limit	USEFLEXLARGELOAD_PROBFORM	Boolean	This field is deprecated.
Energize Lost Load First	ENERGIZELOSTLOAD_FIRST_PROBFORM	Boolean	This field is deprecated.
Use Loads in Loss	USELOADSINLOSS_PROBFORM	Boolean	This field is deprecated.
Maximize Load	USEMAXLOAD_PROBFORM	Boolean	This field is deprecated.
Economic	ECONOMIC_PROBFORM	Boolean	This field is deprecated.
Load Category	USESUMFLOWLOADCAT_PROBFORM	Boolean	This field is deprecated.
Loops / Final	ALLOWLOOPSFINAL_PROBFORM	Boolean	This field is deprecated.
Loops / Intermediate	ALLOWLOOPSTEMP_PROBFORM	Boolean	This field is deprecated.
Drop Load / Final	ALLOWDROPLoadFINAL_PROBFORM	Boolean	This field is deprecated.
Drop Load / Intermediate	ALLOWDROPLoadTEMP_PROBFORM	Boolean	This field is deprecated.
Shunt Impedance	SHUNTIMPCTRL_PROBFORM	Integer	This field is deprecated.
Generators	GENCTRL_PROBFORM	Integer	This field is deprecated.
Transformers	TFTAPCTRL_PROBFORM	Integer	This field is deprecated.
Switching Devices	SWITCHCTRL_PROBFORM	Integer	This field is deprecated.
No. of Infeasible Constraints	NUMINFEASCSTNTS_PROBFORM	Integer	This field is deprecated.
Switching Mode	SWITCHINGMODE_PROBFORM	Integer	This field is deprecated.
Application Type	APPLICATIONTYPE_PROBFORM	Integer	This field is deprecated.

3.16.2. Optimization Control Schemes

Each problem formulation also has an associated optimization control scheme. The control scheme indicates the type of switching devices that are to be considered (automated, manual, or both).

Note

The CSV file fields are deprecated in favor of more intuitive check-boxes "Use Automatic Switching Devices" and "Use Manual Switching Devices" in the "Analysis Options" subsection of "Switching Reconfiguration" problem formulations.

Table 7. Optimization Control Schemes

Configuration Editor UI Name	CSV File Field Name	Type	Description
Index	INDEX_OPTCTRL	Integer	This field is deprecated.
Name	NAME_OPTCTRL	String	This field is deprecated.
Device Type Considered	SWITCH_OPTCTRL	Integer	This field is deprecated.

3.16.3. LVM Operation Modes

This section describes plan execution configuration parameters for LVM.

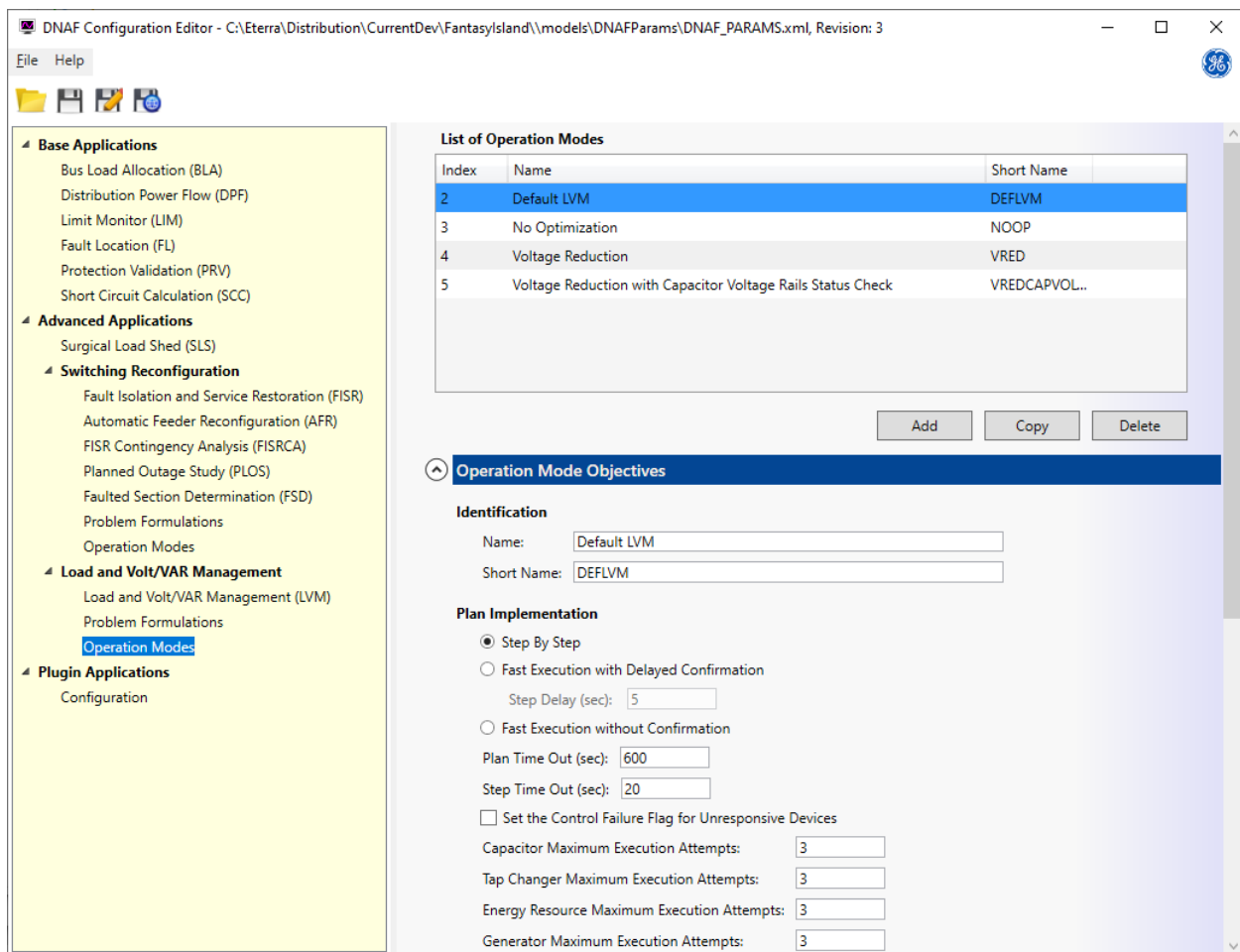


Figure 69. Operation Modes Settings - LVM

Table below lists the Operation Modes configuration parameters. All listed parameters are contained by the <Params xsi:type = "OperationModeParams"> element in the DNAF_PARAMS.xml configuration file.

Table 8. Operation Modes Configuration Parameters - LVM

Configuration Editor UI Name	CSV File Field Name and XML File Attribute Name	Type	Description
Identification			
Index	<ModeID>	Integer	This table entry is generated automatically and specifies the index of the operation mode. This index is used to refer to a specific operation mode for the problem formulation.
Short Name	<Acronym>	Text box	This entry specifies an optional description (acronym) for the operation mode.
Name	<Name>	Text box	This entry specifies the name of the operation mode. This name is used on the Problem Formulations tab to select the operation mode for a problem formulation.

Configuration Editor UI Name	CSV File Field Name and XML File Attribute Name	Type	Description
Plan Implementation			
Step by Step	<PlanImplementationMethod> = 0	Option button	This option indicates whether controls generated by an advanced application are sent out one step at a time. With this option enabled, the control steps are sent one by one. The plan manager waits for the response (confirmation) on the current control action before proceeding to the next action.
Fast Execution with Delayed Confirmation	<PlanImplementationMethod> = 1	Option button	This option indicates whether plan manager performs the fast plan execution with delayed confirmation. The plan moves are sent sequentially with the delay specified by the Step Delay parameter. If at least one step failed to execute properly, the plan is marked as failed.
Step Delay (sec)	<BurstModeStepDelay>	Text box	This entry specifies the delay used for the fast execution with delayed confirmation method. It is the maximum delay between each step during implementation. The allowable range is 0-120.
Fast Execution without Confirmation	<PlanImplementationMethod> = 2	Option button	This option indicates whether plan manager performs the fast plan execution without confirmation. The plan moves are sent sequentially without delay and without confirmation (i.e., burst mode).
Plan Time Out	CLOSELOOPTIMEOUTTOL_VOLTMAN <PlanTimeOut>	Text box	This entry specifies the maximum duration, in seconds, before a closed loop LVM plan timeout is reported. If LVM cannot receive and validate all control status confirmations including capacitor status measurement, capacitor VAR change (for indirect confirmation), tap position analog measurement change, voltage setpoint analog measurement change, or DER/DG VAR/power factor setpoint change, before the specified timeout period, LVM plan execution is stopped. The plan is marked as failed, and the plan is removed. The value 0 means this wait time is not checked and the Plan Manager subsystem waits indefinitely for a status confirmation to come back. The default value is 600 seconds to support a plan failure. The allowable range is 0-3,600.

Configuration Editor UI Name	CSV File Field Name and XML File Attribute Name	Type	Description
Step Time Out	CLOSELOOPSTEPTIME OUTTOL_VOLTMAN <StepTimeOut>	Text box	This entry specifies the default waiting time, in seconds, before the plan manager can consider a single control to have timed out. This value is used only when the timeout is not defined for the associated control measurement. If any step in a plan times out, the plan is considered to have failed and is removed. If the timeout is 0, the plan manager will wait indefinitely. The default value is 20 seconds. The allowable range is 0-300.
Set the Control Failure Flag for Unresponsive Devices	<ExcludeControlFailure>	Check box	This check box indicates whether LVM Control Failure flags should be set to devices where control during a plan implementation failed. The Control Failure flag makes devices ineligible for future LVM solutions.
Capacitor Maximum Execution Attempts	<MaximumAttemptsForCapacitors>	Text box	This entry specifies the maximum number of attempts to send out a SCADA control for capacitors. If the total number of attempts for a capacitor in an LVM plan reaches this value but the control action cannot be confirmed, the plan is considered as failed and an alarm is optionally issued. The control attempt interval is one minute. The allowable range is 1-10.
Tap Changer Maximum Execution Attempts	NUMREGMOVESFORALARM_VVC <MaximumAttemptsForTapChangers>	Text box	This entry specifies the maximum number of attempts to send out a SCADA control for LTCs/regulators. If the total number of attempts for an LTC/regulator in an LVM plan reaches this value but the control action cannot be confirmed, the plan is considered as failed and an alarm is optionally issued. The control attempt interval is one minute. The allowable range is 1-10.
Energy Resource Maximum Execution Attempts	<MaximumAttemptsForDERs>	Text box	This entry specifies the maximum number of attempts to send out a SCADA control for energy resources. If the total number of attempts for an energy resource in an LVM plan reaches this value but the control action cannot be confirmed, the plan is considered as failed and an alarm is optionally issued. The control attempt interval is one minute. The allowable range is 1-10.

Configuration Editor UI Name	CSV File Field Name and XML File Attribute Name	Type	Description
Generator Maximum Execution Attempts	<MaximumAttemptsForDGs>	Text box	This entry specifies the maximum number of attempts to send out a SCADA control for generators. If the total number of attempts for a generator in an LVM plan reaches this value but the control action cannot be confirmed, the plan is considered as failed and an alarm is optionally issued. The control attempt interval is one minute. The allowable range is 1-10.
Automatically Re Execute for a Control Failure	<EnableAutoReExecControlFailure>	Check box	This check box indicates whether to enable LVM retriggering due to a control failure for an operation mode NOTE: LVM retriggering for a control failure also depends on the DNAF_LVM_EVENT_PLANEXEC FAILURE_TRIGGER parameter in NetServerDnafConfig.txt.
Real/Reactive Power Throttling			
Enable Real Power Throttling - Increase	<EnableActivePowerThrottlingUp>	Check box	This check box indicates whether to enable the throttling of real power increases during LVM plan execution. If the real power sudden measurement change is larger than the pre-defined value for the station optimization group, the LVM plan execution will be delayed by the value specified by the Throttling Delay parameter.
Enable Real Power Throttling - Decrease	<EnableActivePowerThrottlingDown>	Check box	This check box indicates whether to enable the throttling of real power decreases during LVM plan execution. If the real power sudden measurement change is larger than the pre-defined value for the station optimization group, the LVM plan execution will be delayed by the value specified by the Throttling Delay parameter.
Enable Reactive Power Throttling - Increase	<EnableReactivePowerThrottlingUp>	Check box	This check box indicates whether to enable the throttling of reactive power increases during LVM plan execution. If the reactive power sudden measurement change is larger than the pre-defined value for the station optimization group, the LVM plan execution will be delayed by the value specified by the Throttling Delay parameter.

Configuration Editor UI Name	CSV File Field Name and XML File Attribute Name	Type	Description
Enable Reactive Power Throttling - Decrease	<EnableReactivePowerThrottlingDown>	Check box	This check box indicates whether to enable the throttling of reactive power decreases during LVM plan execution. If the reactive power sudden measurement change is larger than the pre-defined value for the station optimization group, the LVM plan execution will be delayed by the value specified by the Throttling Delay parameter.
Throttling Delay	<ThrottlingDelay>	Text box	This entry specifies the additional throttling delay time (in seconds) during real/reactive power throttling based on real-time measurement change. The allowable range is 0-600.
Status Confirmation			
Capacitor Voltage Rails Check Timeout (sec)	<CapVoltageRailsCheckTimeout>	Text box	This entry specifies the timeout value, in seconds, for confirming capacitors' voltage rails are disabled before implementing an LVM plan using a problem formulation that has the "Disable Capacitor Voltage Rails" option selected. The allowable range is 0-600.
Indirect Measurement Confirmation			
Monitor Upstream Reactive Power	<MonitorFeederHeadReactivePower>	Check box	This check box indicates whether to monitor the reactive power flow upstream of the capacitor. The monitoring is useful when the capacitor status is unavailable or suspect - change in the flow measurement shows whether the capacitor was operated. The parameters VAR Monitoring Criteria – High and VAR Monitoring Criteria – Low are used for determining the capacitor status (open/closed). If the upstream reactive power monitoring is enabled but the VAR measurement is not received for the device associated with the current step, the monitoring is skipped for the step.

Configuration Editor UI Name	CSV File Field Name and XML File Attribute Name	Type	Description
VAR Monitoring Criteria – High	<VARMonitorHigh>	Text box	This entry specifies, as a percentage value of the nominal reactive power rating of a capacitor, the upper limit for determining if a capacitor has switched. The Monitor Upstream Reactive Power check box must be selected for this entry to apply. If the change in the monitored reactive power at the nearest upstream reactive power measurement is within the specified high and low limits multiplied by the nominal reactive power rating of the capacitor, the capacitor has been successfully switched. The allowable range is 0.1-1.2.
VAR Monitoring Criteria – Low	<VARMonitorLow>	Text box	This entry specifies, as a percentage value of the nominal reactive power rating of a capacitor, the lower limit for determining if a capacitor has switched. The Monitor Upstream Reactive Power check box must be selected for this entry to apply. If the change in the monitored reactive power at the nearest upstream reactive power measurement is within the specified high and low limits multiplied by the reactive power rating of a capacitor, the capacitor has been successfully switched. The allowable range is 0.1-1.
Monitor Load Side Voltage for Tap Position/Voltage Setpoint/Voltage Reduction Confirmation	<MonitorLoadSideVoltChange>	Check box	This check box indicates whether the load side voltages for LTCs should be monitored for confirming LTC/regulator operation in closed-loop LVM execution, including tap RAISE/LOWER control, voltage setpoint control, and voltage reduction control. The parameters Maximum Voltage Change (Tap Position Only) and Minimum Voltage Change (Tap Position Only) are used to determine whether RAISE/LOWER tap control actions sent to SCADA were successful. The operation of these parameters is described in detail in Load-Side Voltage Monitoring for Confirming Tap Changer Operation. If the load side voltage change monitoring is enabled but the measurements are not available, the monitoring is skipped for the step.

Configuration Editor UI Name	CSV File Field Name and XML File Attribute Name	Type	Description
Maximum Voltage Change (Tap Position Only)	<VoltChangeMonitorHighPerc>	Text box	This entry specifies, in % voltage, the maximum change in the load-side voltage measurement for a regulator/LTC for which a tap position measurement is not available. The tap position is available when the change in load-side voltage measurement is within the minimum and maximum voltage change range. For example, if the maximum voltage change is 0.75% V, and the minimum voltage change is 0.50% V, then it is expected that the change in load-side voltage measurement following a tap RAISE/LOWER operation should be between 0.0075 p.u. and 0.0050 p.u. If the change in load-side voltage measurement is outside of this range, the LTC/regulator control action is considered failed. Otherwise, the control action is considered successful. The Monitor Load Side Voltage for Tap Position/Voltage Setpoint/Voltage Reduction Confirmation option must be selected for this entry to be applicable. The allowable range is 0.1-3.
Minimum Voltage Change (Tap Position Only)	<VoltChangeMonitorLowPerc>	Text box	This entry specifies, in % voltage, the minimum change in the load-side voltage measurement for a LTC/regulator for which a tap position measurement is not available. The tap position is available when the change in load-side voltage measurement is within the minimum and maximum voltage change range. For example, if the maximum voltage change is 0.75% V, and the minimum voltage change is 0.50% V, then it is expected that the change in load-side voltage measurement following a tap RAISE/LOWER operation should be between 0.0075 p.u. and 0.0050 p.u. If the change in load-side voltage measurement is outside of this range, the LTC/regulator control action is considered failed. Otherwise, the control action is considered successful. The Monitor Load Side Voltage for Tap Position/Voltage Setpoint/Voltage Reduction Confirmation option must be selected for this entry to be applicable. The allowable range is 0.1-3.

Configuration Editor UI Name	CSV File Field Name and XML File Attribute Name	Type	Description
Wait for Analog Delay	<WaitForAnalogDelay>	Text box	This entry specifies the waiting time, in seconds, after a command step is sent before evaluating the system state. This delay allows SCADA values to be updated after a control is sent, which may affect flows and voltages. This delay should be long enough to allow the SCADA measurements to be refreshed but shorter than the plan step timeout value. The analog delay is not applied for the voltage setpoint when the plan execution mode is set to fast execution with delayed confirmation and the control monitoring is disabled. The Monitor Upstream Reactive Power option or the Monitor Load Side Voltage for Tap Position/ Voltage Setpoint/ Voltage Reduction Confirmation option must be selected for this entry to be applicable. The allowable range is 1-120.

3.16.4. LVM System Monitor Configuration

This section describes LVM System Monitor configuration parameters.

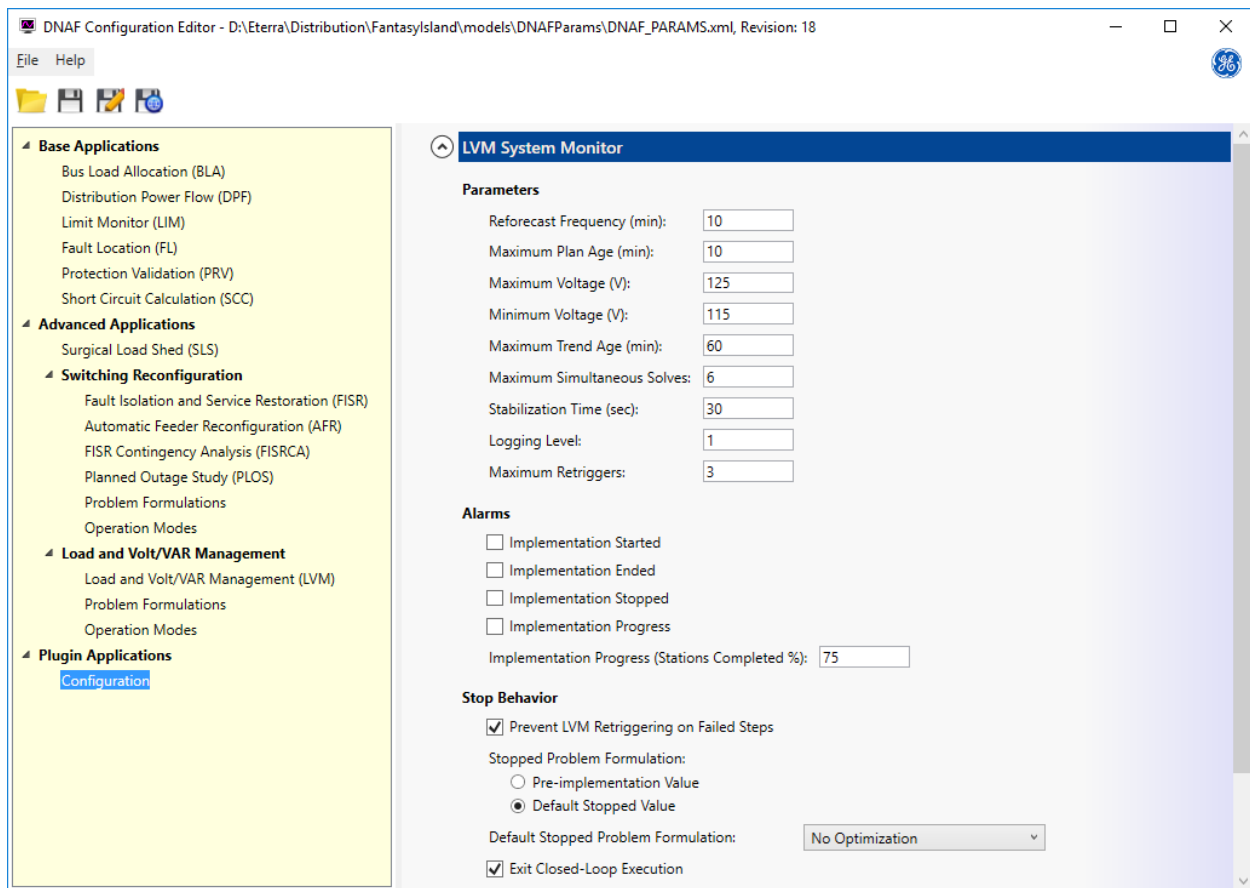


Figure 70. LVM System Monitor Configuration

Table below lists the LVM System Monitor configuration parameters. All listed parameters are contained by the <Params xsi:type = "PluginParams"> element in the DNAF_PARAMS.xml configuration file.

Table 9. LVM System Monitor Configuration Parameters

Configuration Editor UI Name	CSV File Field Name and XML File Attribute Name	Type	Description
Parameters			
Evaluation Frequency	<EvaluationFrequency>	Text box	The time period between LVM solutions, in minutes. All evaluated LVM results are updated at this frequency. The allowable range is 1-240.
Maximum Plan Age	<MaxPlanAge>	Text box	The maximum age for evaluated LVM plans, in minutes. Any plans older than the specified age are recalculated before usage when implementing an LVM change. This value can be overridden at run time in the LVM System Monitor New Implementation display. The allowable range is 0-240.

Configuration Editor UI Name	CSV File Field Name and XML File Attribute Name	Type	Description
Maximum Voltage	<MaxVoltLimit>	Text box	The upper limit to display on voltage trend graphs, in Volts. The allowable range is 0-1000.
Minimum Voltage	<MinVoltLimit>	Text box	The lower limit to display on voltage trend graphs, in Volts. The allowable range is 0-1000.
Maximum Trend Age	<MaxTrendAge>	Text box	The time range to show on voltage trend graphs, in minutes. Data older than the specified age is discarded. The allowable range is 5-240.
Maximum Simultaneous Solves	<MaxSolves>	Text box	The maximum number of simultaneous LVM solves to run on the LVM evaluation server. Once this limit is reached, new solution requests are queued until one or more in-progress solves are completed. The allowable range is 1-50.
Stabilization Time (sec)	<StabilizationTime>	Text box	Time period in seconds to wait after an implementation to allow the system to return to a steady-state condition. After completing an implementation, the system pauses for this time period before recording final implementation conditions and generating a report file. The allowable range is 0-300.
Logging Level	<LogLevel>	Text box	The amount of detail to include in the log. The options are: • 1: Basic information, such as system errors. • 2: More detailed information (errors and warnings). • 3: Very detailed information (errors, warnings, and information messages). The allowable range is 1-3.
Alarms			
Implementation Started	<Implementation StartedAlarmEnabled>	Check Box	This check box indicates whether to issue an alarm to signify the start of an implementation from the LVM System Monitor for the specified area.
Implementation Ended	<Implementation EndedAlarmEnabled>	Check Box	This check box indicates whether to issue an alarm to signify the end of an implementation from the LVM System Monitor for the specified area.

Configuration Editor UI Name	CSV File Field Name and XML File Attribute Name	Type	Description
Implementation Stopped	<ImplementationStoppedAlarmEnabled>	Check Box	This check box indicates whether to issue an alarm to signify the stop of an implementation from the LVM System Monitor for the specified area.
Implementation Progress	<ImplementationProgressAlarmEnabled>	Check Box	This check box indicates whether to issue an alarm to signify that implementation from the LVM System Monitor has been completed for the specified percentage of stations in the implementation area.
Implementation Progress (Stations Completed %)	<ImplementationPercentageCompletion>	Text box	This entry specifies the percentage of implementation progress (% of stations complete) for an area when the Implementation Progress alarm is to be issued. The allowable range is 1-100.
Stop Behavior			
Prevent LVM Retriggering on Failed Steps	<StopPreventRetriggering>	Check Box	This check box indicates whether to prevent any retriggering that may occur if a still-implementing plan fails due to a control failure when stopping an implementation.
Stopped Problem Formulation Value	<StopProblemFormulationTarget>	Option Button	This option defines whether the value of a stopped problem formulation is set to the value before the implementation was started or the value specified by the Default Stopped Problem Formulation Value parameter.
Default Stopped Problem Formulation Value	<StopDefaultProblemFormulation>	Drop-down list	The default "safe" problem formulation to use after an implementation stops if the Stopped Problem Formulation Value parameter is set to "Default Stopped Value".
Exit Closed-Loop Execution	<StopExitClosedLoop>	Check Box	This option defines whether to change LVM execution mode from closed-loop to advisory for all stations in the implementation area when stopping an implementation.
Forecast Engine			
File Folder	<ForecastLogFolder>	Text Box	Path to the file folder where Forecast server log files are created.

4. Switching Reconfiguration Applications

The switching reconfiguration applications take system conditions as input, and provide switching plans based on eligible switching devices. The applications use the concept of a switchable sections, which are subsets of the system, bounded by the following eligible feeder main switching devices: breaker, recloser, switch, sectionalizer, permanent jumper, temporary jumper, temporary cut, and fuse which is not the feeder main boundary. There are four switching reconfiguration applications in the DNAF suite:

- **Fault Isolation and Service Restoration (FISR):** This application generates switching plans to isolate faulted sections bounded by eligible switching devices and to restore service to the non-faulted sections.
- **Fault Isolation and Service Restoration – Contingency Analysis (FISRCA):** This application simulates one fault on one switchable section of the selected station. FISRCA generates a switching plan to isolate the faulted switchable section and restore service to the non-faulted sections. The process is repeated for all switchable sections in the station.
- **Automated Feeder Reconfiguration (AFR):** This application provides plans to optimize system operation. AFR can optimally unload overloaded segments, including power transformers and feeder main conductors. It can provide a plan to reduce real power losses in a station and its neighboring stations. It can also provide a plan to return a station to its nominal configuration (Return to Normal).
- **Planned Outage Study (PLOS):** This application provides plans to isolate a component in the network, such as a switch, a regulator, or a line segment.
- **Faulted Section Determination (FSD):** This function provides switching plans to identify a fault and isolate faulted sections. The FSD function applies a switching-based approach with step-by-step opening and closing switching cycles. At Stage 1, a faulted feeder downstream of a faulted power transformer is identified and isolated. At Stage 2, the faulted section is identified and isolated, the fault is placed in the faulted section.

These four applications share many similarities in functionality since they use the same engine to propose isolation and service restoration switching plans. They handle several objectives subject to network and operational constraints such as segment overloads, voltage violations and drops, number of switching operations, relay settings, and control-inhibit tags. These objectives and constraints are used to determine the best switching plans.

The switching reconfiguration applications can optionally incorporate adaptive relaying. The reconfigured network may have significantly different fault levels that call for changing settings on boundary reclosers. For information about adaptive relaying support, refer to the "Protection Settings" section in the *Distribution Network Analysis Functions (DNAF) Standard Applications Analyst's Guide*.

4.1. Network Loading Conditions

Generating the best switching plans requires an accurate estimate of the loads that must be

restored or transferred to adjacent feeders. This is achieved by analyzing the loading conditions over a look-ahead time window defined for each switching reconfiguration application. If the look-ahead time window duration (field named Look Ahead Duration (hr) in the DNAF Configuration Editor) is set to 0 hours for any switching reconfiguration application, the load values are taken from the last DPF solution. For the FISR application, this loading at the start of the look-ahead window is the pre-fault or pre-loss of voltage load values. Currently, the look-ahead duration has a range of 0 – 24 hours.

If the look-ahead duration is greater than 0 hours, the load values at the start of the look-ahead time window and the static load profiles defined in the load category are used to estimate the peak loading time and the future loading values. The sample load profile shown in figure below will be used to demonstrate this method.

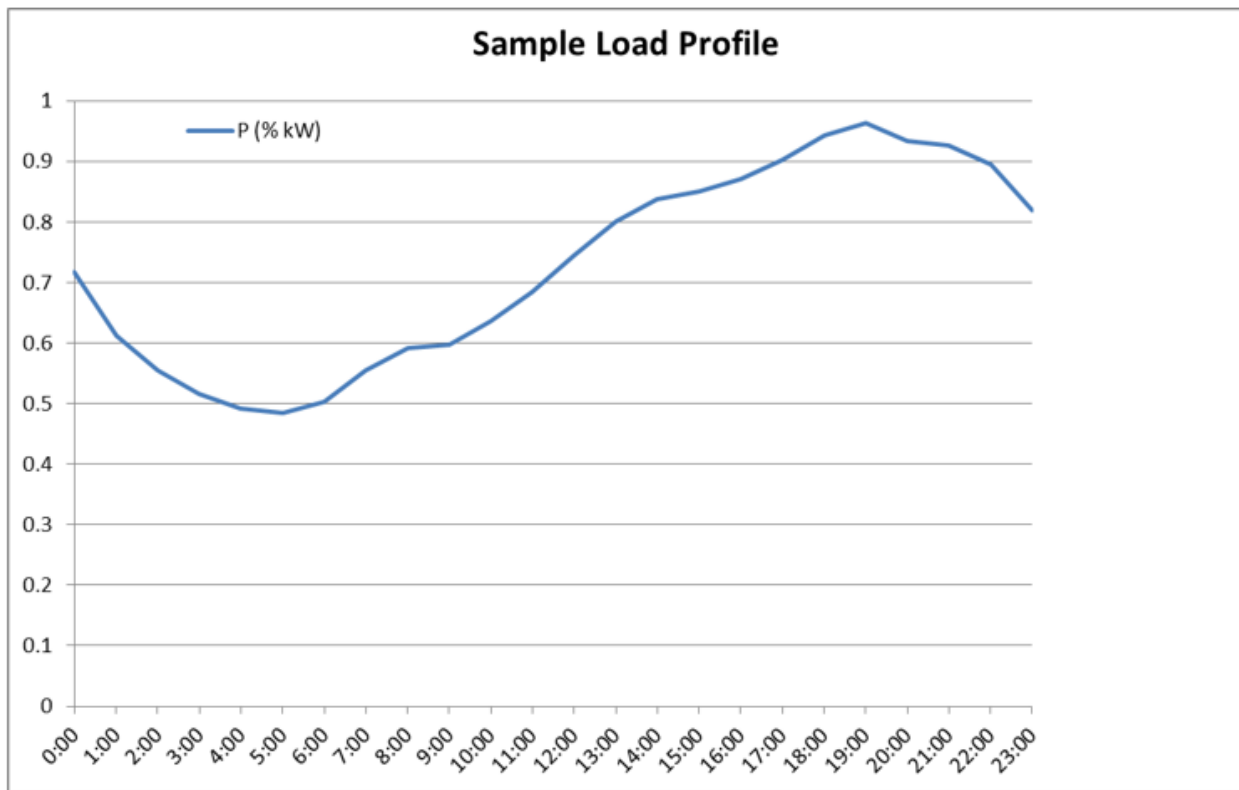


Figure 71. Sample Daily Load Profile

To demonstrate the peak load estimation method, consider a feeder with residential loads that all follow the sample load daily load profile shown in figure above. If a fault occurs at 3:00 PM and the FISR look-ahead duration is set to 6 hours, the real power (P) load multiplier for the next six hours is obtained from the static load profile. Table below shows these values.

Table 10. Load Profile Real Power Values

Time (hr)	Profile P Value (% kW)
15:00	0.850
16:00	0.870

Time (hr)	Profile P Value (% kW)
17:00	0.900
18:00	0.945
19:00	0.965
20:00	0.935
21:00	0.928

The first part of the peak load estimation process is determining the peak evaluation time. This is done by summing up all the loads at each hour for every station in the network selection over the look-ahead window. If a residential load, LD1, on the faulted feeder from the last solved DPF solution was 10.0 kW, the peak load estimation algorithm extrapolates the last known load value (from 3:00 PM) using the load profile values. The calculation of the estimated load value for the 10.0 kW load at 16:00 hr is performed utilizing the following equation:

$$Load_{16:00} = Load_{15:00} \times 0.87/0.85 = 10.0 \times 1.02 = 10.2 \text{ kW} \quad (\text{Equation 4.1})$$

This calculation is repeated for each load, and the total loading value for every station in the entire network selection is estimated for each hour in the look-ahead interval. The hour with the highest loading is used as the evaluation time in the switching reconfiguration analysis. All loads, both energized and de-energized, are determined using the equation above.

4.2. Integration of Cold Load Pickup (CLPU) Models in Reconfiguration Applications

CLPU models can provide better switching plans by considering cold load growth as a result of an outage. They are applied to real time as well as study mode operation.

The linear decay model, which is detailed in 'Cold Load Pickup (CLPU) Schedule' in the *Distribution Network Analysis Functions (DNAF) Standard Applications Analyst Guide*, is used to model the CLPU demand.

The "Consider Cold Load Pickup" option in the Switching Reconfiguration Problem Formulations tab in DCE specifies whether reconfiguration applications consider the cold load pickup effect.

☒ Consider Cold Load Pickup

If CLPU is considered, it will be used in the peak load estimation process. That is, the CLPU multipliers will be applied to every de-energized load at every hour over the look-ahead window. In the example of equation 4.1, if LD1's CLPU multiplier is 1.1 at hour one after outage, then its estimated load value at 16:00 hr is $10.2 \times 1.1 = 11.2 \text{ kW}$.

After the peak evaluation time is determined, the load values used by reconfiguration applications will also include the CLPU effect. In this example, if the 16:00 hr is the peak evaluation time, then

reconfiguration applications will use 11.2kW for this load to generate plans and calculate plan performance index and plan statistics.

Duration of Plan Validity will also be calculated if CLPU is considered. It is an estimate of the number of hours from when the outage occurred before there are overloads caused by CLPU.

4.3. Device Eligibility and Automatic Disabling Capabilities

There are some differences between real-time mode and study mode when determining the eligible switching devices to be considered in the reconfiguration applications. In addition, depending on the FISR/AFR configuration options, the tags and SCADA status measurements may also result in the FISR/AFR application automatically disabling itself by feeder.

4.3.1. Device Eligibility in Real-Time Mode

In real-time mode, the following switching device eligibility conditions are considered by the FISR and AFR applications:

- The device should not have a valid "DA Status" measurement with current status "Disabled". Additionally, the device should not be downstream of any device with "DA Status" disabled.
- The device must be a three-phase feeder main device or a three-phase station internal device.
- The two terminals (nodes) of the device must be at the same voltage level.
- If the device is a fuse, it must be a three-phase feeder main fuse (a fuse that is not a lateral or feeder boundary device).
- If the device is a permanent jumper, it must be enabled for the switching reconfiguration applications in the model by setting the DSOverride parameter to 1 for the permanent jumper in the station external XML file (or the DSOVER_JUMPER field in the station external CSV file).
- If the device is a composite switch (such as a throw-over device), it must be enabled for the switching reconfiguration applications in the model by setting the DSOverride parameter to 1 for the composite switch in the station external XML file (or the DSOVER_COMPSW field in the station external CSV file).
- If the device is a temporary jumper, the "Use Manual Switching Devices" option and the "Use Temporary Cuts and Jumpers as Manual Switching Devices" sub-option must be enabled in the selected problem formulation in the DCE.
- If the device is a temporary cut, the "Use Manual Switching Devices" option and the "Use Temporary Cuts and Jumpers as Manual Switching Devices" sub-option must be enabled in the selected problem formulation in the DCE.
- If the device is a composite switch, the "Use Composite Switches" option must be enabled in the selected problem formulation in the DCE.
-

The device is not a feeder main boundary device such as a lateral tap fuse.

- The device is not disabled for the switching reconfiguration applications in the model. To disable a device in the model, set the DSOOverride parameter to 2 for the device in the station external XML file (or the corresponding field in the station external CSV file).
- The device status is not Opt User-Disabled.
- For a device with the Opt SCADA Decides status, the following SCADA status measurements are considered:

- Communication Established status measurement

If the device has a communication established measurement with good quality, its status must be "Established". If the device has a communication established measurement with suspect quality, its status is not considered in deciding eligibility.

- Open/closed status measurement

If the device has single phase and/or three-phase open/closed measurements modeled, at least one of them must have good quality. If none has good quality and the option "Telemetered Switch Evaluated as a Manual Switch for Suspect Quality Status Measurement" is selected in DCE, this device is eligible as a manual device.

- Control Available status measurement

If the device has a control available measurement modeled, the quality must be good and its status must be "Available".

In the following two cases, this device is eligible as a manual device:

- The quality is suspect and the option "Telemetered Switch Evaluated as a Manual Switch for Suspect Quality Control Available Measurement" is selected in DCE.
- The quality is good but the status is Not Available and the option "Telemetered Switch Evaluated as a Manual Switch for Good Quality with Abnormal Value Control Available Measurement" is selected in DCE.

- Supervisory Cutout status measurement

If the device has a supervisory cutout measurement modeled, the quality must be good and its status must be "Off".

In the following case, this device is eligible as a manual device:

- The quality is suspect and the option "Telemetered Switch Evaluated as a Manual Switch for Suspect Quality Supervisory Cutout Measurement" is selected in DCE.

- Auto/Manual status measurement

If the device has an Auto/Manual measurement modeled, the quality must be good and its status must be "Auto".

In the following case, this device is eligible as a manual device:

- The quality is suspect and the option "Telemetered Switch Evaluated as a Manual Switch for

Suspect Quality Auto/Manual Measurement" is selected in DCE.

- If Hot Line status measurement is modeled for a switching device, its eligibility may be affected. Hot line status measurement indicates if a switching device can be closed. If the quality is good and the status is Not Applied, the device is eligible. If the quality is good and the status is Applied, the device is not eligible. In the following two cases, this device is eligible as a manual device:
 - The quality is suspect and the option "Telemetered Switch Evaluated as a Manual Switch for Suspect Quality Hot Line Measurement" is selected in DCE.
 - The quality is good but the status is Applied and the option "Telemetered Switch Evaluated as a Manual Switch for Good Quality with Abnormal Value Hot Line Measurement" is selected in DCE.
- The device status is not In Alarm, Remote Control Disabled, or Reclosing Disabled. If the In Alarm status measurement is set, or the device is operating in local control mode, or the reclosing has been disabled, the device is not eligible for FISR or AFR.
- The device respects tag types. In the real-time FISR and AFR applications, device eligibility from tag information is based on the tag type information defined in SCADA.
- The device is not part of a local automation switching scheme (based on SCADA measurements).
- For a device that is owned by a third-party company, the device eligibility is based on the setting in the Third-Party Equipment Options section in the Problem Formulations page.

The switching devices to be utilized in the application, both automatic and manual devices, are enabled or disabled in the selected problem formulation. If the problem formulation supports only automated switches, all manual switches are ineligible for all reconfiguration applications. Likewise, if the problem formulation supports only manual switches, all automated switches are ineligible for all reconfiguration applications.

For device protection settings group changes, the following eligibility conditions apply:

- The associated device is eligible for AFR and FISR (all eligibility conditions listed above are applied).
- The device has a protective settings group.
- The settings group status measurement is modeled and the measurement quality is good.

4.3.2. Device Eligibility in Study Mode

In study mode, the following switching device eligibility conditions are considered by the FISR, FISRCA, AFR, and PLOS applications:

- The device should not have a valid "DA Status" measurement with current status "Disabled". Additionally, the device should not be downstream of any device with "DA Status" disabled.
- The device must be a three-phase feeder main device or a three-phase station internal device.
-

The two terminals (nodes) of the device must be at the same voltage level.

- If the device is a fuse, it must be a three-phase feeder main fuse (a fuse that is not a lateral or feeder boundary device).
- If the device is a permanent jumper, it must be enabled for the switching reconfiguration applications in the model by setting the DSOOverride parameter to 1 for the permanent jumper in the station external XML file (or the DSOOVER_JUMPER field in the station external CSV file).
- If the device is a composite switch (like a throw-over device), it must be enabled for the switching reconfiguration applications in the model by setting the DSOOverride parameter to 1 for the composite switch in the station external XML file (or the DSOOVER_COMPSW field in the station external CSV file).
- If the device is a temporary jumper, the "Use Manual Switching Devices" option and the "Use Temporary Cuts and Jumpers as Manual Switching Devices" sub-option must be enabled in the selected problem formulation in the DCE.
- If the device is a temporary cut, the "Use Manual Switching Devices" option and the "Use Temporary Cuts and Jumpers as Manual Switching Devices" sub-option must be enabled in the selected problem formulation in the DCE.
- If the device is a composite switch, the "Use Composite Switches" option must be enabled in the selected problem formulation in the DCE.
- The device is not a feeder main boundary device such as a lateral tap fuse.
- The device is not disabled for the switching reconfiguration applications in the model. To disable a device in the model, set the DSOOverride parameter to 2 for the device in the station external XML file (or the corresponding field in the station external CSV file).
- The device is optimization user-enabled, or it is an optimization SCADA decides device.
- For a device that is owned by a third-party company, the device eligibility is based on the setting in the Third-Party Equipment Options section in the Problem Formulations page.
- The device respects tag types. In study mode, the tag type information is taken from the server configuration file. The following attributes in the netserverstudyconfig.txt file enable the tag information.

Set TAG_LOCAL to 1 to enable local (i.e., non-SCADA) tags:

```
TAG_LOCAL      = 1
The tag behavior is set using the following server configuration parameters:
TAGTYPE_COUNT = 13           // Number of tag types.
...
// Control inhibit tag type.
TAGTYPE_ID_4=CONTROL INHIBIT // Tag type identifier.
TAGTYPE_PRIORITY_4=4         // Tag type priority.
TAGTYPE_CONTROLRESTRICTIONS_4 = 1,8,32,64 // Control restrictions.
...
// Information tag type.
TAGTYPE_ID_10=INFO
TAGTYPE_PRIORITY_10=11
TAGTYPE_CONTROLRESTRICTIONS_10 = 0
```


...

The control restrictions field has the following values:

'1 = Open Restricted - used in study mode

'2 = Open Prevented - used in study mode

'4 = Close Restricted - used in study mode

'8 = Close Prevented - used in study mode

'16 = Setpoint Restricted - used in study mode

'32 = Setpoint Prevented - used in study mode

'64 = Block Optimization Operations - used in real-time mode and study mode

'128 = Tripped by Load Shed Application = used in real-time mode and study mode

For the tag types above, the control inhibit tag makes a switching device ineligible in study mode for the FISR, AFR, PLOS, and FISRCA applications since the open and close operations are prevented. If an information tag is placed on a switching device, the device is still eligible since there are no control restrictions. For FISR and AFR, users can select the "Make All Switching Devices Eligible" option on the study mode DNAF Solution display to consider all switching devices for switching plans.

The switching devices to be utilized in the application, both automatic and manual devices, are enabled or disabled in the selected problem formulation. If the problem formulation supports only automated switches, all manual switches are ineligible for all reconfiguration applications. Likewise, if the problem formulation supports only manual switches, all automated switches are ineligible for all reconfiguration applications.

4.3.3. Automatic Disabling Capabilities

Depending on the FISR/AFR configuration options, tags and SCADA status measurements may also result in the FISR/AFR application automatically disabling itself by feeder or feeder pair if the device is a currently open device between two different feeders.

This automatic disabling feature is configured in the DNAF Configuration Editor using the following parameters:

- **Allow Automatic Disabling by Feeder:** For enabling/disabling the feature.
- **Minimum Number of Unavailable Devices:** For setting the minimum number of devices that must be unavailable for the application to disable itself.
- **Respect Tags:** For disabling tags (that have the "Block Optimization Operations" bit in the Main server and DNAF server configuration files) as a criterion for automatically disabling FISR or AFR.
- **Hot Line:** For enabling the hot line status measurement for automatically disabling FISR or AFR.

- **In Alarm:** For enabling the in-alarm status measurement for automatically disabling FISR or AFR.
- **Remote Control Disabled:** For enabling the remote control status measurement for automatically disabling FISR or AFR (the device must be in Remote mode).
- **Reclosing Disabled:** For enabling the reclosing enabled status measurement for automatically disabling FISR or AFR (the device must have reclosing enabled).

4.3.4. Enabling and Disabling Applications

On the DNAF System Configuration display, the operator can enable or disable DNAF applications at run-time for system elements (system, optimization groups, divisions, stations, power transformers, or feeders).

The following fields on the DNAF System Configuration display relate to enabling and disabling DNAF applications:

- **Manual Enable:** Allows enabling or disabling applications for system elements. This field is also available from the Hazard Management System (HMS) at the feeder level for LVM, FISR, and AFR. Disabled applications cannot be run manually using the DNAF Solutions display or automatically due to periodic and event triggers. System elements (for example, feeders) where an application is disabled are not considered in application solutions.
- **SCADA Enabled:** For LVM, FISR, and AFR, this shows the SCADA Enabled status (Enabled, Disabled, or Invalid if status measurement has suspect quality). The field is blank if there is no SCADA status measurement for this system item. The SCADA Enabled status measurement can be mapped to the station, power transformer, and feeder for LVM. The SCADA Enabled status measurement can be mapped to the station and feeder for FISR and AFR.
- **Auto Enabled:** For LVM, FISR, and AFR, this shows the Auto Enabled status (Enabled or Disabled) based on the automatic disabling checks. See [Automatic Disabling Capabilities](#) for more details.

Note

In order for LVM, FISR and AFR applications to execute on system elements, all three statuses **SCADA Enabled**, **Auto Enabled** and **Manual Enable** must be enabled. For instance, if **Manual Enable** is checked, FISR will not operate if either **SCADA Enabled** or **Auto Enabled** is disabled.

Overriding the Manual Enable Status

The SCADA Enabled and Auto Enabled values can be configured to override the Manual Enable settings for LVM, FISR, and AFR.

The Manual Enable fields can be read-only for the user and automatically inherit the values of the SCADA Enabled or Auto Enabled fields. The override settings are defined by the LVM_USER_OVERRIDE, FISR_USER_OVERRIDE, and AFR_USER_OVERRIDE parameters in the DNAF server configuration files. Each parameter can include up to two of the following override options,

separated by a comma. If more than two options are defined, only the first two options are respected and the latter options are ignored.

- **SCADA_ON:** When SCADA Enabled is enabled, the Manual Enable is set to Enabled (selected) and cannot be edited by the user.
- **SCADA_OFF:** When SCADA Enabled is disabled, the Manual Enable is set to Disabled (cleared) and cannot be edited by the user.
- **AUTO_ON:** When Auto Enabled is enabled, the Manual Enable is set to Enabled (selected) and cannot be edited by the user.
- **AUTO_OFF:** When Auto Enabled is disabled, the Manual Enable is set to Disabled (cleared) and cannot be edited by the user.

	SCADA Enabled = True	SCADA Enabled = False	Auto Enabled = True	Auto Enabled = False
LVM_USER_OVERRIDE = SCADA_ON	Manual Enable = True	Manual Enable is user-specified	Manual Enable is user-specified	Manual Enable is user-specified
LVM_USER_OVERRIDE = SCADA_OFF	Manual Enable is user-specified	Manual Enable = False	Manual Enable is user-specified	Manual Enable is user-specified
LVM_USER_OVERRIDE = AUTO_ON	Manual Enable is user-specified	Manual Enable is user-specified	Manual Enable = True	Manual Enable is user-specified
LVM_USER_OVERRIDE = AUTO_OFF	Manual Enable is user-specified	Manual Enable is user-specified	Manual Enable is user-specified	Manual Enable = False

If a parameter value is not defined or the SCADA Enabled or Auto Enabled status is suspect, not available, or does not match the parameter setting, the Manual Enable setting is not overridden and remains user-specified. If conflicting SCADA measurement and automatic disabling check values are defined (for example, "SCADA_ON,AUTO_OFF" or "SCADA_OFF,AUTO_ON"), the latter option is ignored and the first option is respected.

For example, if "LVM_USER_OVERRIDE = SCADA_ON,SCADA_OFF", the Manual Enable fields are set as follows:

- For cases where SCADA Enabled fields are "Enabled", the Manual Enable fields are "Enabled" (selected) and cannot be edited by the user.
- For cases where SCADA Enabled fields are "Disabled", the Manual Enable fields are "Disabled" (cleared) and cannot be edited by the user.
- For cases where SCADA Enabled fields are not available or set to "Invalid", the Manual Enable fields remain user-specified and can be edited by the user (see figure below).

DNAF Real Time System Configuration

LVM ▾ Group By Station ▾ Show Execution Data ▾ Expand To ▾

	SCADA Enabled	Auto Enabled	Manual Enable	Problem Formulation	Plan Execution
PEBBLE BEACH Station	Enabled	Enabled	☒	Default [Reactive Support]	Default [Closed Loop]
T1 Power Transformer	Invalid	Enabled	☒	Default [None: no transformer objective]	
F6535 Feeder	Invalid	Enabled	☐	Default [None: no feeder objective]	
F6533 Feeder	Disabled	Enabled	☐	Default [None: no feeder objective]	
F6536 Feeder	Enabled	Enabled	☒	Default [None: no feeder objective]	
T2 Power Transformer	Enabled	Enabled	☒	Default [None: no transformer objective]	
F6531 Feeder	Disabled	Enabled	☐	Default [None: no feeder objective]	
F6534 Feeder	Enabled	Enabled	☒	Default [None: no feeder objective]	
F6532 Feeder	Enabled	Enabled	☒	Default [None: no feeder objective]	
PINE VALLEY Station		Enabled	☒	Default [Reactive Support]	Default [Closed Loop]
PINEHURST Station		Enabled	☒	Default [Reactive Support]	Default [Closed Loop]
PLUTO Station		Enabled	☒	Default [Reactive Support]	Default [Closed Loop]
RAVEN Station		Enabled	☒	Default [Reactive Support]	Default [Closed Loop]
ROYAL TROON Station		Enabled	☒	Default [Reactive Support]	Default [Closed Loop]
SATURN Station		Enabled	☒	Default [Reactive Support]	Default [Closed Loop]
SPYGLASS Station		Enabled	☒	Default [Reactive Support]	Default [Closed Loop]
STEVENS Station		Enabled	☒	Default [Reactive Support]	Default [Closed Loop]

Right-click to lock/unlock a cell

OK Cancel Apply

Figure 72. Overriding the Manual Enable Status for LVM

Note

When a SCADA measurement on an entire substation is set to OFF, the station has the "Disabled" SCADA Enabled measurement. In this case, all station components are shown with the "Disabled" SCADA Enabled status; however, the station components' Manual Enable fields are set to enabled (selected) or disabled (cleared) according to the actual components' SCADA Enabled measurements.

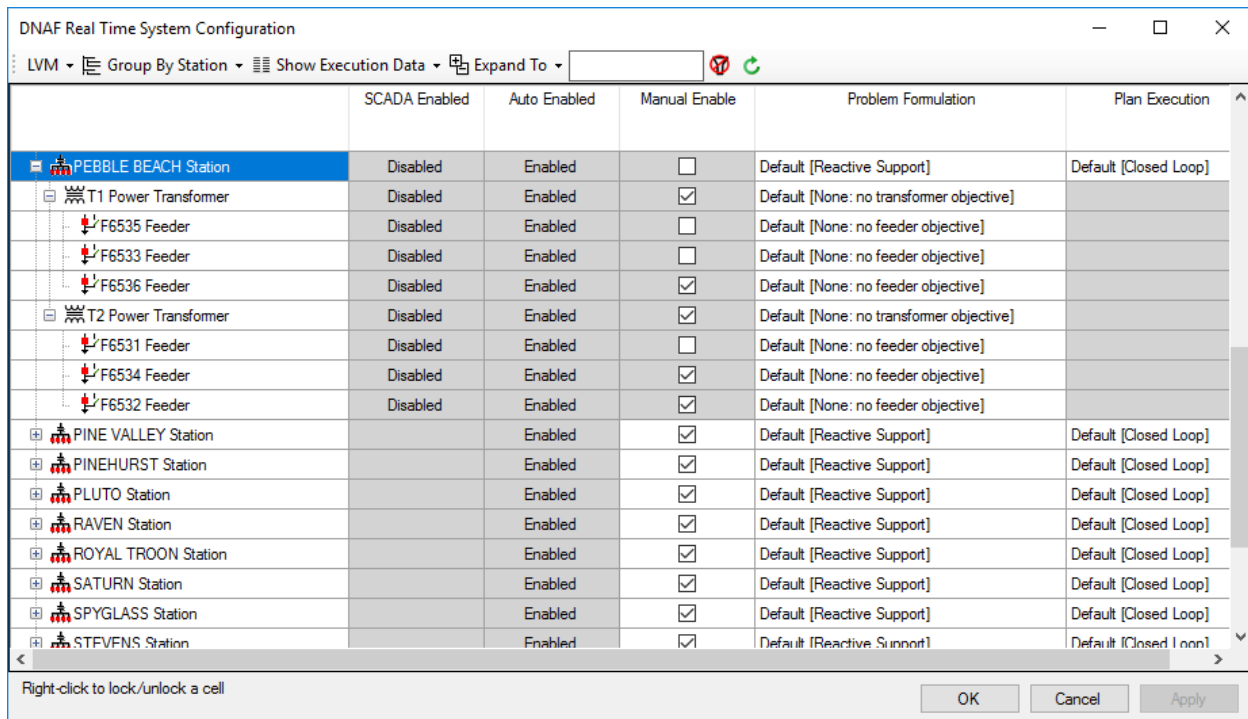


Figure 73. Station SCADA Measurement set to OFF

4.4. Solution Engine

The reconfiguration functionality is provided by a component referred to as the "Distribution System Plan Generation Engine" (DSPGE). This engine can solve both the Load and Volt/VAR Management and Reconfiguration functions based on the selection of objective functions, controls, and constraints. The DSPGE is used in different modes to solve the various distribution optimization functions needed for operational support (LVM, FISR, FISCA, AFR, and PLOS).

The reconfiguration algorithm is a heuristic switching technique using ranking indices. The algorithm can be used for a variety of purposes by activating selected constraints and controls and by applying different optimization objective functions. Within this framework, it is possible to experiment with different optimization approaches for the sub-problems while retaining the basic ranking and multi-step switching approach. The ranking is based on criteria such as the number of customers without service, losses, radial topology conditions, and flow and voltage violations.

Different types of constraints, such as the requirement for radial connections, minimum number of branch overloads, minimum number of high voltage or low voltage violations, or minimum number of switching actions, can be directly given costs used in the ranking. These constraints are given dollar or influencing dollar costs and are directly added to the continuous variable costs to determine a final ranking value. When violated constraints of infeasible plans enter the ranking, they are given a cost depending on the priority of the constraint and a scaling for the constraint type. This approach does not directly optimize these functions, but it allows the implementation of very complex objective functions. The end result is a useful compromise designed to consistently provide good answers — at the expense of no guarantee of providing the very best answer.

Objective functions can be specified "on the fly". These functions can be preset for different times of the day or for a given time interval.

The reconfiguration applications follow practical rules and objectives to isolate and restore the network. In every step of the isolation/restoration process, limit checking is available to guarantee normal operation. There are some switching rules to find the proper tie switches or sectionalizing switches for isolating and restoring the network.

The problem formulations for reconfiguration applications use a set of criteria, also called "Performance Indices" (PI), for determining the best reconfiguration solution. A weight is assigned to each performance index; by assigning different weights to the PIs, a wide range of problem formulations can be created for the various reconfiguration applications.

The solution always attempts to enforce voltage and flow limits. Relay protection settings and interruptive capacity are considered in each switching plan via a Protection Validation performance index.

When solving the reconfiguration applications, all stations that are n levels (configurable via the DNAF Configuration Editor) adjacent to the target station or feeder must be DNAF-capable and not diverging (Suspect or Solved DPF solution is acceptable).

4.4.1. Load Break Capability

Load break capability refers to the ability to open a switch when it is energized and current is flowing through it. Some switching devices are not designed to be opened when energized (not *load break capable*) and others have a limit (*load break rating*) on the maximum current in Amps that can be flowing for a switch to be openable.

Every type of switch, as defined in the Switch Model, has model parameters that determine whether the device is Load Break Capable and, if so, what the Load Break Rating is.

All reconfiguration applications may produce plans that require opening an energized switch. This is especially true for those applications which are used on fully energized systems, such as AFR and PLOS.

The reconfiguration analysis will check both the Load Break Capability and the Load Break Rating of a device before allowing the use of that switch as part of a reconfiguration plan when opening an energized switch.

4.4.2. Telemetered Regulators and LTCs

For telemetered regulators and LTCs, the switching reconfiguration applications either treat these devices as effectively having a fixed tap position (using the most recently telemetered or manually entered tap position value) or as devices that regulate voltage according to their internal controller settings; the internal controller settings include the voltage target, voltage bandwidth, Line Drop Compensation (LDC) resistance and reactance, tap step increment, voltage reduction, and advanced regulation modes with different regulated bus location and methods. This tap position logic is

determined by the "Enable Tap Emulation Logic for Telemetered Regulators and LTCs" option on the Switching Reconfiguration Problem Formulations tab in the DCE:

- If cleared, the SCADA-enabled regulators and LTCs with valid tap position measurements are treated as having a fixed tap position set to the latest analog tap position measurement value.
- If selected, the SCADA-enabled regulators and LTCs with valid tap position measurements internally regulate the voltage (and calculate a new tap position based on the proposed topology) during the switching reconfiguration plan generation process, respecting other measurements in the same manner as the base Distribution Power Flow (DPF) application. These measurements include the following:
 - Remote/Local measurement (status). If the tap changer is in Voltage Setpoint or Voltage Reduction mode, the tap changer is treated as a locally controlled device.
 - Auto/Manual measurement (status)
 - Blocked measurement (status)

The above three measurements determine whether the reconfiguration applications treat the device as having a fixed tap position (using the most recently set telemetered or manually entered tap position value) or whether the device self-regulates. For more details about how the DPF application determines if it can internally control the movable taps on a regulator or transformer LTC depending on certain SCADA measurements associated with the control status of the device, refer to section "Influence of Measurements on Effective Regulation Status" in the *Distribution Network Analysis Functions (DNAF) Standard Applications Analyst Guide*.

If the tap changing device self-regulates in the reconfiguration applications, its tap position is calculated in the internal power flow solution based on the new network topology after internal switching operations have been completed. These newly calculated tap positions are presented in the Plan Estimated Tap Position grid tabular display and logged in the Locally Controlled Tap Changers section under the PLAN INFORMATION major section of the reconfiguration application result files (FISR, AFR, and PLOS). These internally calculated tap positions will not affect the existing tap positions in the real time or study mode environments.

The tap regulation capability, either fixed or self-regulating, can impact the status of the reconfiguration application switching plans; this can also potentially impact closed-loop plan execution. This is illustrated in the following two examples.

The first example, shown in figure below, has two feeders A and B in their normal state with an open tie switch between them. There is a voltage regulator at the start of each feeder.

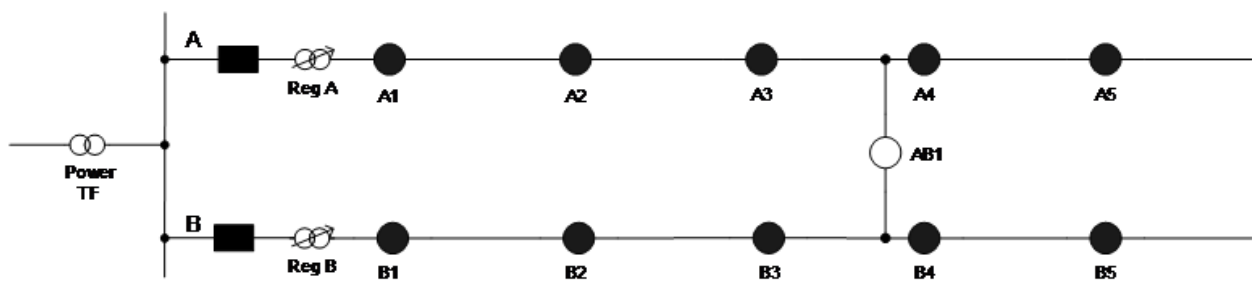


Figure 74. Normal Configuration for Regulators at the Feeder Head

When a fault occurs on feeder A upstream of switch A3, for example between A1 and A2, feeder B can restore the network downstream of the fault through the open tie switch AB1. This is shown in figure below.

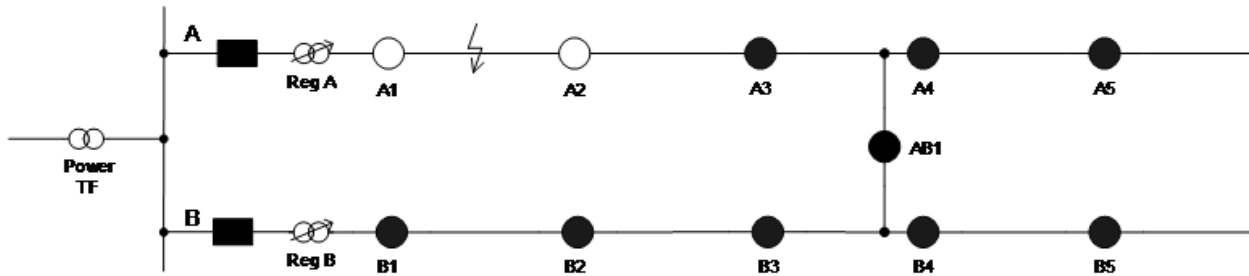


Figure 75. Configuration after FISR Plan is Implemented for Regulators at the Feeder Head

After restoration, there is more load downstream of the voltage regulator (or voltage regulator group) Reg B.

- If FISR treats the regulator as having a fixed tap position, there may be a low voltage violation at some downstream loads; the resulting FISR switching plan will have violations and it cannot be executed in closed-loop mode in real time.
- If FISR allows the regulator to self-regulate, emulating the behavior in the field, Reg B can change its tap position, eliminating the low voltage violations and allowing the top ranked FISR plan to be executed in closed-loop mode.

Similarly, after restoration, there is significantly less load downstream of the voltage regulator (or voltage regulator group) Reg A.

- If FISR treats the regulator as having a fixed tap position, there may be a high voltage violation at some downstream loads; the resulting FISR switching plan will have violations and it cannot be executed in closed-loop mode in real time.
- If FISR allows the regulator to self-regulate, emulating the behavior in the field, Reg A can change its tap position, eliminating the high voltage violations and allowing the top ranked FISR plan to be executed in closed-loop mode.

The second example, shown in figure below, has two feeders A and B in their normal state with an open tie switch between them. There is a bi-directional voltage regulator in the middle of feeder B.

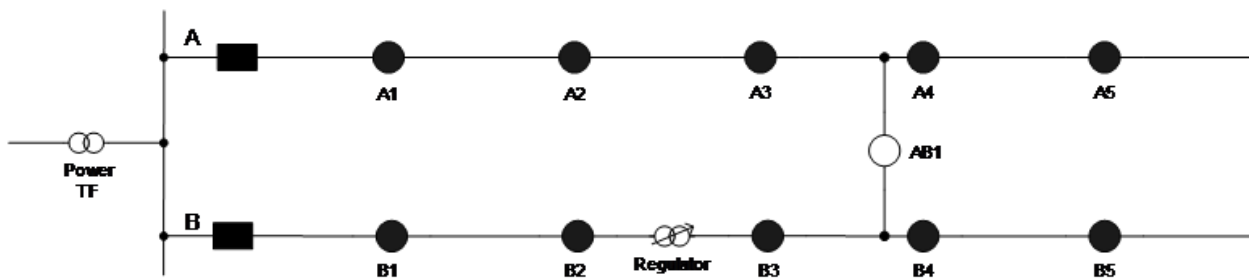


Figure 76. Normal Configuration for Bi-Directional Regulator in the Middle of the Feeder

When a fault occurs on feeder B, feeder A can restore the network downstream of the fault through

the open tie switch. This is shown in figure below.

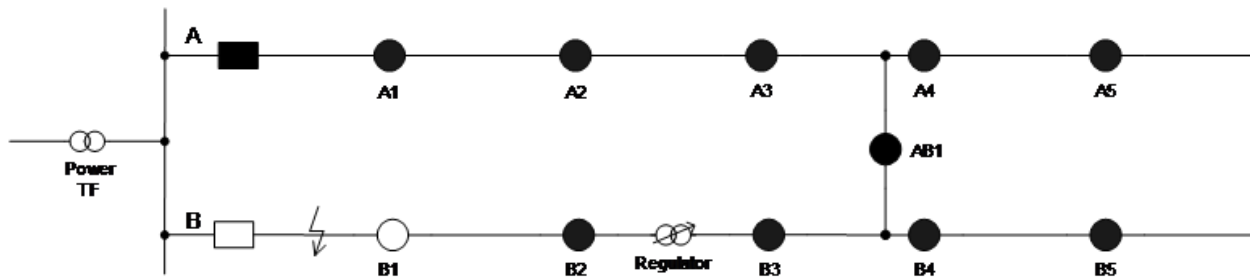


Figure 77. Configuration after FISR Plan is Implemented for Bi-Directional Regulator in the Middle of the Feeder

After restoration, the flow in the bi-directional voltage regulator is reversed.

- If FISR treats the bi-directional regulator as having a fixed tap position, there may be one or more high or low voltage violations at some downstream loads; the resulting FISR plan will have high or low voltage violations and the top ranked FISR plan cannot be executed in closed-loop mode in real time.
- If FISR allows the bi-directional regulator to self-regulate, emulating the behavior in the field, this bi-directional voltage regulator can change its regulation mode and tap position, eliminating the high or low voltage violations and allowing the top-ranked FISR plan to be executed in closed-loop mode.

In either example, if the internal self-regulation logic is used but the plan still has voltage or flow violations, it cannot be executed in closed-loop mode in real time.

4.4.3. Third-Party Owned Equipment

For third-party owned equipment, the switching reconfiguration applications can support the options below, which are configured in the "Third-Party Owned Equipment Options" sub-section on the Switching Reconfiguration Problem Formulations tab in the DCE:

- Treat third-party owned equipment as customer owned equipment
- Include third-party owned equipment in the switching plan but revert to semi-automatic/advisory plan execution mode
- Exclude third-party owned equipment in the switching analysis

Third-party owned equipment is always included in the network selection, no matter which option is selected. The difference is the switching device eligibility for the switching reconfiguration applications and the impact on the final plan execution mode.

The three options are illustrated in the FISR scenarios below. Figure below shows a simple system, which has two feeders A and B in their normal state with an open tie switch, AB1, between them. Switches B3, B4, and B5 are third-party owned equipment; all other switching devices are customer owned equipment. All the customer owned equipment is eligible for FISR.

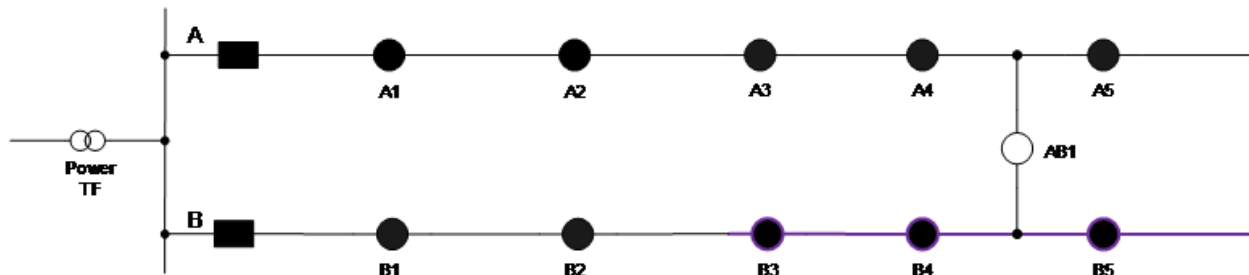


Figure 78. Normal Configuration

When a fault occurs on feeder B between switches B3 and B4, the feeder B breaker is tripped. This is shown in figure below.

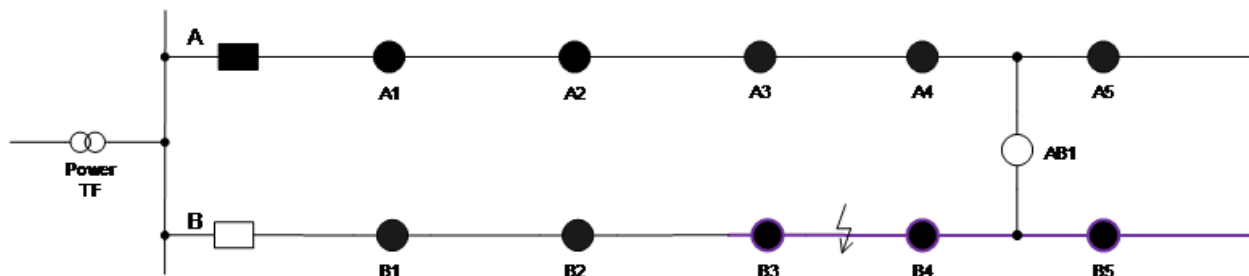


Figure 79. Configuration after Fault Occurs

If third-party owned equipment is treated as customer owned equipment, switches B3, B4, and B5 have the same eligibility as other switching devices. Feeder B can be partially restored through the tripped breaker and the open tie switch AB1. Because switches B3 and B4 are eligible devices, they are opened to isolate the fault and the feeder B breaker and the open tie switch AB1 are closed. This is shown in figure below. The plan execution mode when utilizing this option is not changed.

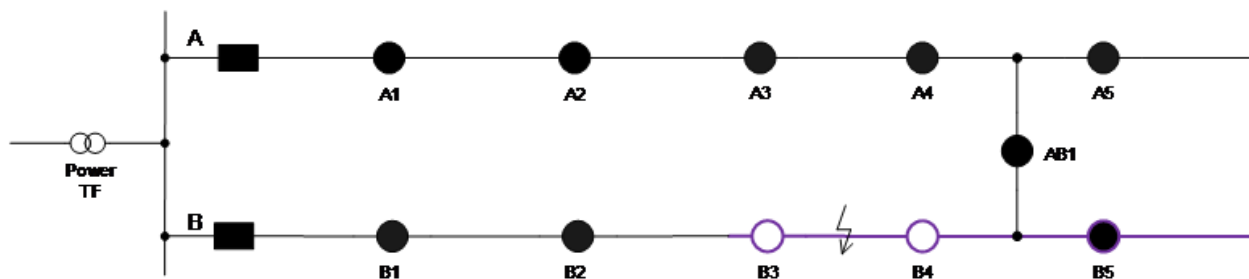


Figure 80. Configuration after FISR Plan is Implemented with B3 and B4 as Isolation Switches

If third-party owned equipment is included in the switching plan but the plan execution mode reverts to semi-automatic/advisory plan execution mode, switches B3, B4, and B5 have the same eligibility as other switches. Feeder B can be partially restored through the tripped breaker and the open tie switch AB1. Because switches B3 and B4 are eligible, they are opened to isolate the fault. This is also shown in figure above. However, the plan execution mode will automatically revert to a more restrictive one. If the current plan execution is "Closed Loop" or "Closed Loop – Calculated Voltage Limit Violation Not Worse", the plan execution mode for this plan will automatically revert to "Semi-Automatic". If the current plan execution mode is "Semi-Automatic", the plan execution

mode for this plan will automatically revert to "Advisory".

If third-party owned equipment is excluded in the switching analysis, switches B3, B4, and B5 are not eligible. Feeder B can be partially restored by isolating the fault with switch B2 and then closing the tripped breaker. This is shown in figure below. The plan execution mode will not be changed with this option.

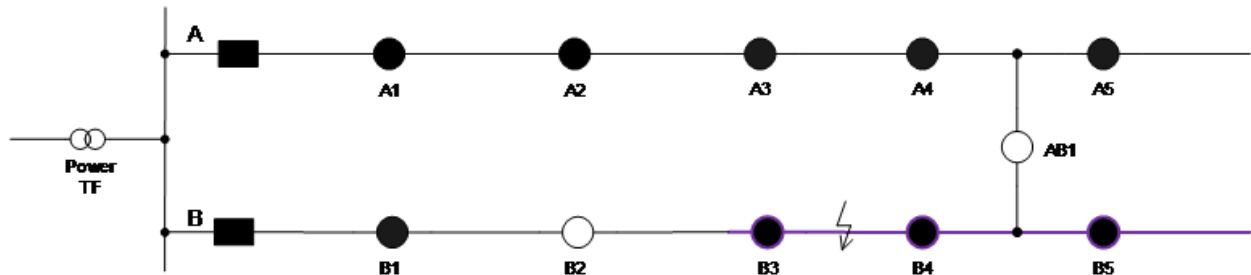


Figure 81. Configuration after FISR Plan is Implemented with B2 as an Isolation Switch

4.4.4. Composite Switching Devices

Multi-state switching devices, also called composite switches, can be considered eligible devices and included in the reconfiguration application switching plan generation logic. This option is configured by the Use Composite Switches option on the Switching Reconfiguration Problem Formulations tab in the DNAF Configuration Editor.

A composite switch can be either an automatic or manual device, depending on the configuration of the device. If the composite switch is an automatic device, the generated plan can be implemented in real-time closed loop or semi-automatic mode. If the composite switch is a manual device, the generated plan can only be implemented in study mode or implemented manually in the real-time advisory mode, similar to plans with regular manual two-state switching devices.

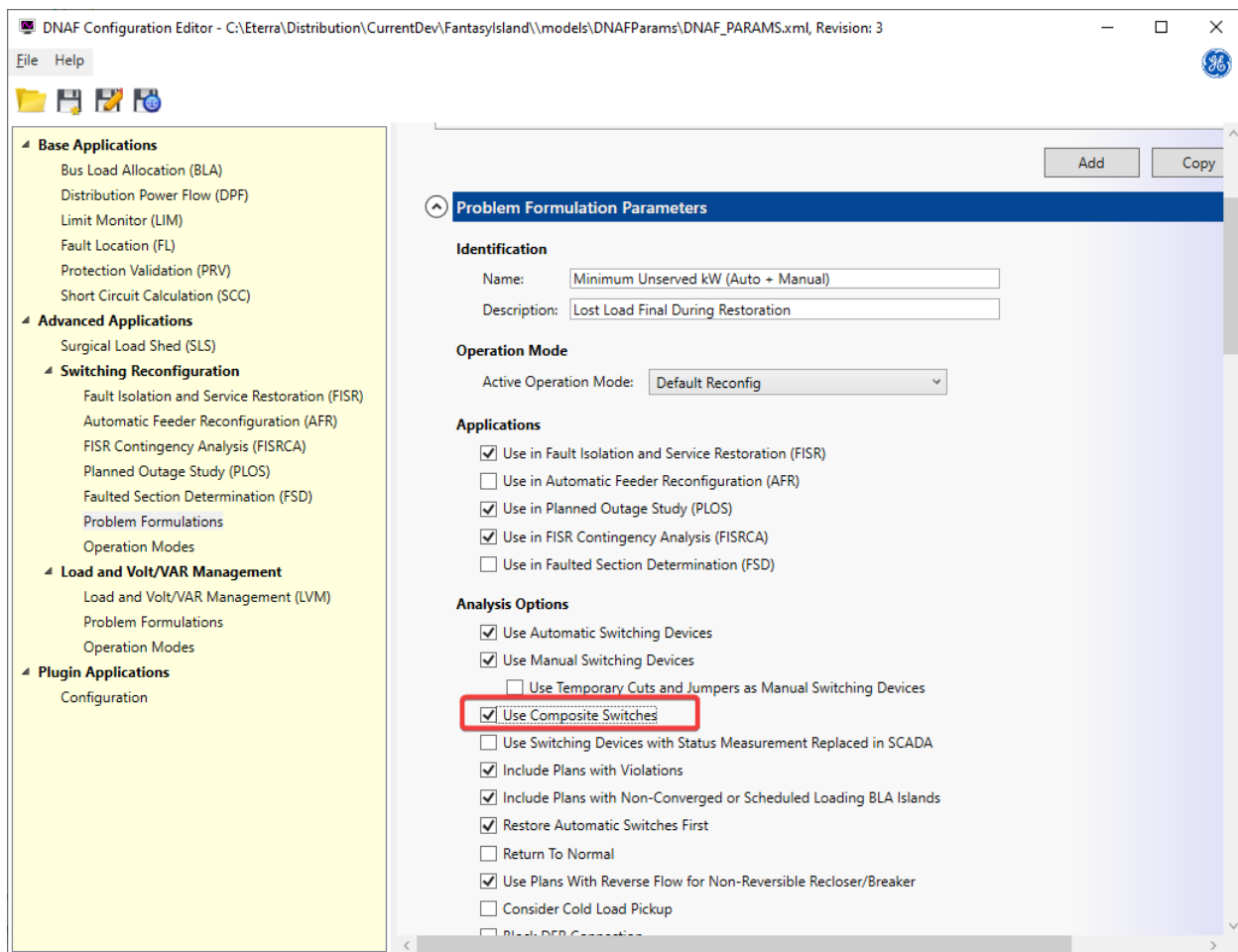
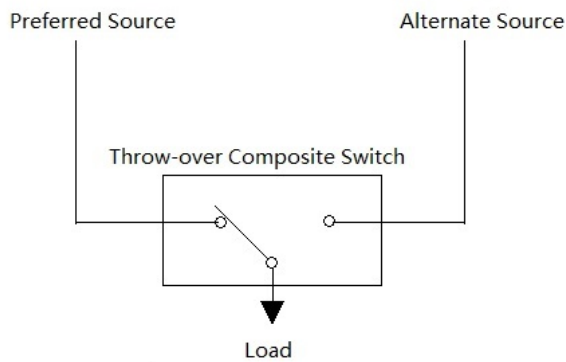


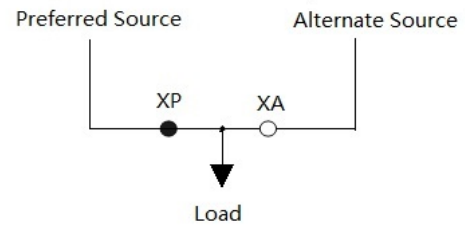
Figure 82. Using Composite Switches as Eligible Switching Devices

In the ADMS static model, a composite switch consists of a collection of regular two-state child switching devices. The status of a composite switch is determined based on the combination of the open/close statuses of the associated individual switches. A state table of valid composite switch statuses and their corresponding connection combinations is defined in the composite switch model in the global.csv file as part of the modeling phase. (Some switch status combinations do not represent a valid state for a given composite switch.)

For example, figure below shows a throw-over switch, which can switch load between two sources, the preferred source or the alternative source. This type of a transfer switch has two individual child switches and has three valid states (Preferred, Alternate, and Off). The Preferred child switch and the Alternate child switch cannot both be closed at the same time, therefore this is an invalid state for a throw-over composite switch. This invalid state is respected when creating a switching reconfiguration plan.



(a) Throw-over Composite Switch



(b) Internal Representation in DNOM

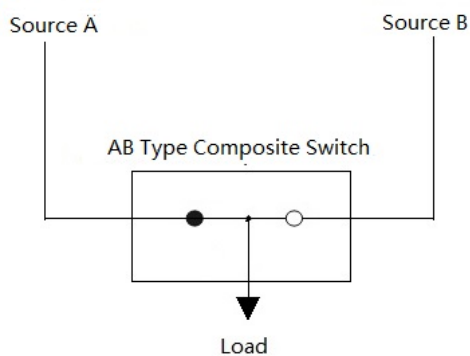
Figure 83. Throw-Over Type Composite Switch

In the static model, the throw-over composite switch is represented as two connected individual switches, Preferred switch XP and Alternative switch XA. The load is connected to the preferred source or alternative source according to the internal blade position. The DNOM model includes a table with valid states, as shown in table below.

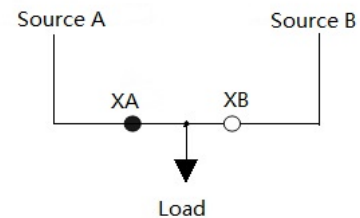
Table 11. Throw-Over Switch State Table

Composite switch state	XP switch status	XA switch status
Off	Open	Open
Preferred	Closed	Open
Alternate	Open	Closed

Figure below shows another type of a composite switch, AB switch.



(a) AB Switch



(b) Internal Representation in DNOM

Figure 84. AB Type Composite Switch

With the AB composite switch, the load can be connected to source A, source B, or both sources. In the DNOM model, both XA and XB internal switches can be closed at the same time to connect load

to both sources. Table below shows the valid state table for the AB switch.

Table 12. AB Switch State Table

Composite switch state	XA switch status	XB switch status
Off	Open	Open
A	Closed	Open
B	Open	Closed
AB	Closed	Closed

The RAC-type composite switch contains two individual switches: the main switch and the tie switch. An RAC-type composite switch has three valid states: it can operate in the Open, Close or Tie positions.

The RAC-type composite switch is often connected in a group, as shown below in figure. If RAC-1 and RAC-2 are both operated in the "Close" state, RAC-1 is connected to source A, and RAC-2 is connected to source B. If the RAC-1 state is changed to "Tie", RAC-1 and RAC-2 are tied together and connected to both Source A and Source B.

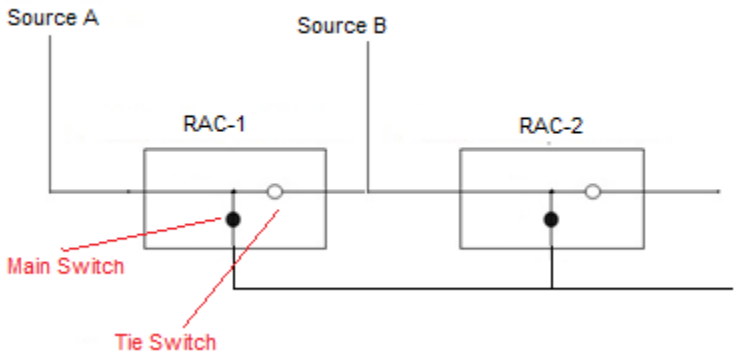


Figure 85. RAC Type Composite Switch Connections

Table below shows the valid state table for the RAC switch.

Table 13. RAC Switch State Table

Composite switch state	Main switch status	Tie switch status	STATE Value
Open	Open	Open	0 (00)
Tie	Open	Close	1 (01)
Close	Close	Open	2 (10)

Figure below shows a 4WRAC switch cabinet that contains one normal two-state switch (RAC-1) and three Composite Switches (RAC-2, RAC-3, and RAC-4). Each of the three composite switches has a different switching status, Tie, Close, or Open.

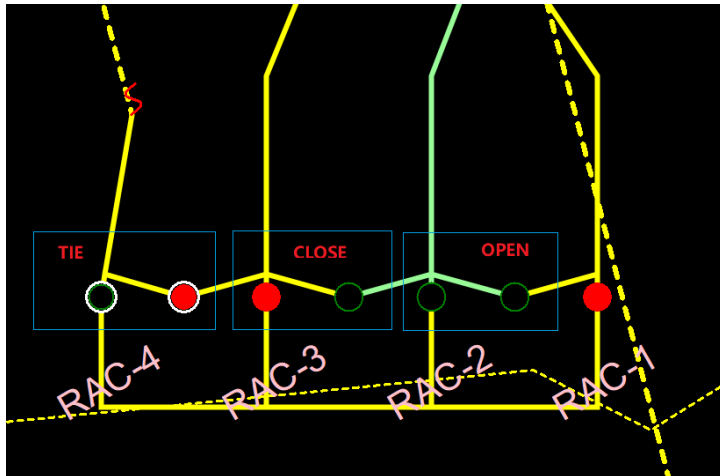


Figure 86. RAC Type Composite Switch Cabinet

An RAG-type composite switch has three valid states: it can operate in the Open, Close, or Ground positions. However, the Ground status is not telemetered and can only be manually set by the operator. You cannot switch directly between the Close and Ground states.

An RAG-type composite switch is shown figure below. It consists of a main switch and a grounding switch.

Figure 87. RAG Type Composite Switch Connections

Table below shows the valid state table for the RAG switch.

Table 14. RAG Switch State Table

Composite switch state	Main switch status	Ground switch status	STATE Value
Open	Open	Open	0 (00)
Close	Close	Open	2 (10)
Ground (not telemetered)	Open	Close	1 (01)

Figure below shows a 4WRAG switch cabinet that contains four composite switches (RAG-1, RAG-2, RAG-3, RAG-4).

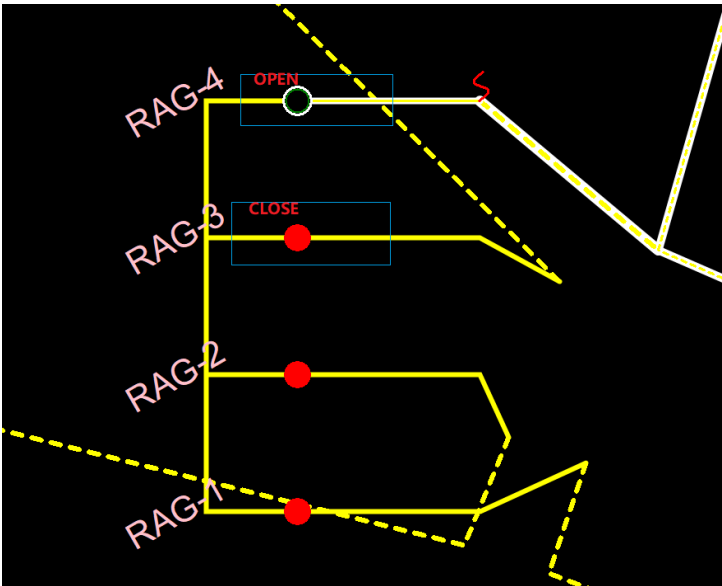


Figure 88. RAG Type Composite Switch Cabinet

Individual switches that make up a composite switch are considered when creating upstream and downstream restoration plans. The status of the composite switch is checked against the valid status truth table multiple times during the reconfiguration switching plan generation. If an individual switch operation results in an invalid composite switch status, additional individual switch operations are added to make sure the status of the composite switch is valid, or the operation is discarded if it is impossible to get a valid status.

After generating the original reconfiguration plan, the optimization engine performs additional processing to check if any plan steps include switches contained within a composite switch. If there are steps that include an individual switch inside a composite switch, those steps are combined, and the individual switch statuses are translated to the corresponding composite switch state in the final plan. The switching effect also shows the combined result of all related steps.

For example, for a throw-over switch, if the optimization engine generates an internal plan to open the XP (preferred) switch and close the XA (alternate) switch, the additional processing logic combines these two steps into a single composite switch status change from the "Preferred" state to the "Alternate" state in the final plan.

Special logic for regulator bypass composite switch devices and for moving a composite switch to a ground position (or other non-energizing status) are not considered in switching reconfiguration applications.

4.4.5. Close Bus Tie Switching Device

The reconfiguration applications can close an eligible tie switching device between two buses at the low voltage side of power transformers in the substation prior to closing an eligible tie switching device between two circuits fed from the same power transformers.

A Boolean flag in the static model indicates whether a switching device is an eligible Close Eligible Bus Tie Switching Device. The following rules are used to determine if a Close Eligible Bus Tie Switching Device is closed in a switching plan:

1. The function must be enabled using the Close Eligible Bus Tie Switching Device option on the Switching Reconfiguration Problem Formulations tab in the DNAF Configuration Editor.
2. The two power transformers must be in the same station. The reconfiguration applications will neither trace to the upstream power transformers in the sub-transmission network nor trace to the parent station of sub-fed stations.
3. The maximum voltage phase angle difference between the two sides of an energized open switching device must be within limits when checked by the reconfiguration application.
4. The two ends of the open bus tie to be closed must be at the same voltage level.

Two different scenarios are described below.

1. Station with Bus Tie Switch Devices between the power transformers:

The station is shown in figure below. Within this station, there are the following components:

- Two power transformers, T1 and T2.
Each power transformer serves one feeder.
- Two feeders, F1 and F2.
- Two feeder head breakers, B1 and B2.

The secondary side of power transformers T1 and T2 are at the same voltage level. BT12 is the normally open bus tie switching device at the low voltage side of the power transformers which can be closed to connect T1 and T2. R11, R12, R21, and R22 are normally closed reclosers alongside the feeders. T12 is a normally open tie recloser between the two feeders.

The Close the Bus Tie Switching Device feature is illustrated in the figure below using the AFR Reduce Overloads and Violations problem formulation. The FISR and PLOS applications uses the same rules.

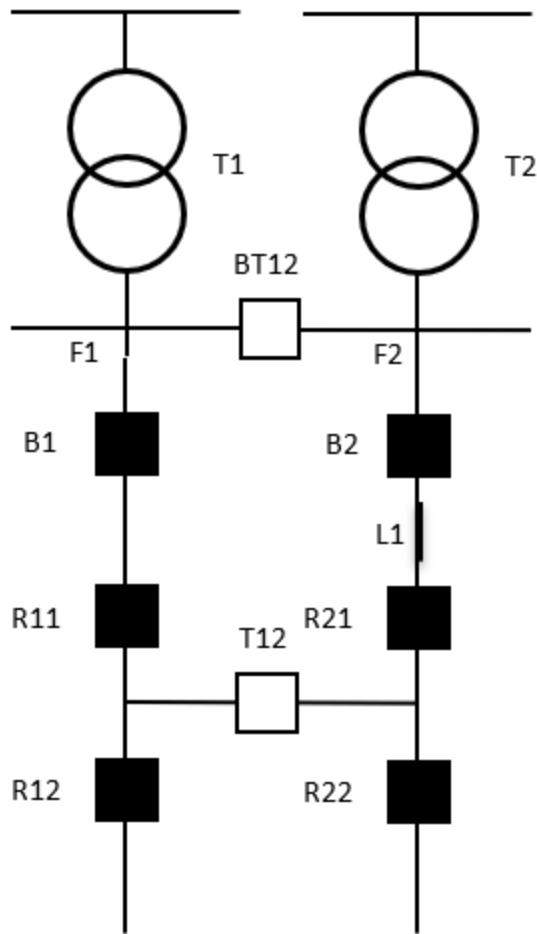


Figure 89. Station with Bus Tie between Power Transformers

For example, in the above AFR case, there is an overloaded line L1 between B2 and R21.

In this case, assume the flow violation in L1 can be relieved by transferring the load downstream of R21 to F1. AFR will generate a plan which first connects the two buses of T1 and T2 together, and then transfers load downstream of R21 from feeder F2 to F1. The AFR plan steps are as follows:

- Close BT12
- Close T12
- Open R21

In this plan, step 1 is the added step to connect the bus tie switching device, BT12, between the two buses of the power transformers inside the substation.

2. Station with no Bus Tie Switch devices between the power transformers:

The station is shown in figure below. The components and structure are similar as the station above with the exception that there is no eligible bus tie BT12 at the low voltage side of the power transformers. L1 is still the overloaded line.

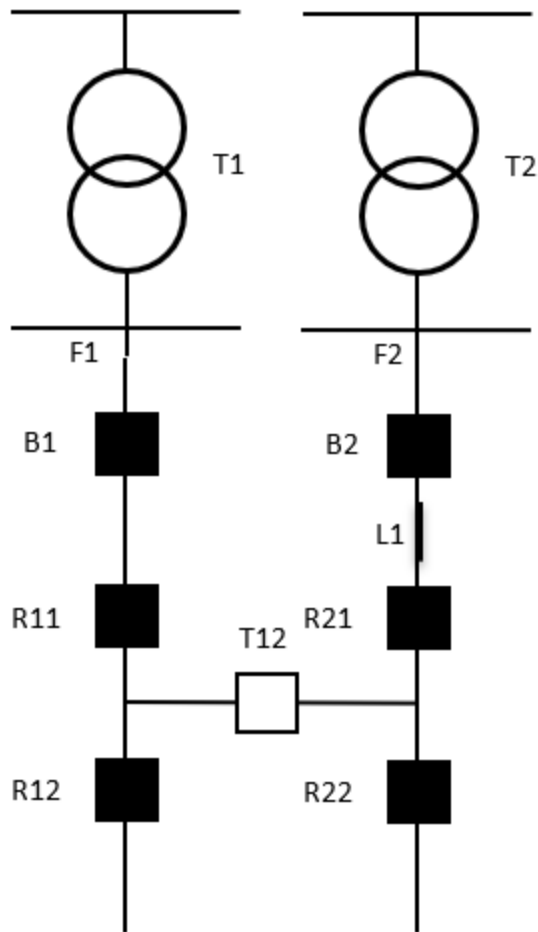


Figure 90. Station with no Bus Tie between Power Transformers

In this case, assume the flow violation for line L1 can be relieved by transferring load downstream of R21 to F1. AFR will generate a plan which transfers load downstream of R21 from feeder F2 to F1. The AFR plan steps are as follows:

- Close T12
- Open R21

4.5. Limit Check

The reconfiguration applications check the following limits:

- Flow limits for the feeder main branches in the reconfiguration path formed by the plan steps
- Voltage high and low limits for all feeder main and lateral loads and/or buses in the network selection affected by the plan

The limit check result impacts the plan state:

1. If there is no violation, the FISR plan is "Feasible".

2. If there is any flow or voltage violation in the network involved in the restoration, or if there are only voltage violations and at least one voltage violation is worse than the pre-plan voltage violations, the FISR plan is "Feasible - Has Violations".
3. If there is no flow violation and there are only voltage violations in the network involved in the restoration, and all the voltage violations are not worse than the pre-plan voltage violations, the FISR plan is "Feasible - Violations Execution Allowed".
4. If the violation is in the network not involved in the restoration, the FISR plan is "Feasible - Neighbor Has Violations".

The DCE parameter "Violation Not Worse Margin (%)" on the Problem Formulation tab is used in deciding if a post-plan voltage violation is worse than a pre-plan voltage violation:

Violation Not Worse Margin (%):

- For a high voltage violation, if the calculated post-plan voltage is not higher than the pre-plan calculated voltage plus this margin, the plan is considered not to make this calculated voltage limit violation worse.
- For a low voltage violation, if the calculated post-plan voltage is not lower than the pre-plan calculated voltage minus this margin, the plan is considered not to make this calculated voltage limit violation worse.

Plans that do not make the calculated voltage limit violation worse can be executed in closed loop mode if the Plan Execution parameter is set to Closed Loop - Calculated Voltage Limit Violation Not Worse.

4.6. Effects of DERs and Generators

When generators or Distributed Energy Resources (DERs) are present in a distribution system, they can have wide ranging effects on many aspects of the system behavior including increased fault levels and reverse power flows. The following sections describe how these devices are modeled, their effects on fault isolation and restoration, and how to prevent switching plans that result in reverse power flows.

4.6.1. Generator and DER Modeling

Generators are always modeled as individual devices. DER can also be modeled as individual devices connected directly to the system. DER that exists behind the meter can be implicitly modeled as an output profile or schedule within a load definition.

Load schedules are defined in the globals.csv as a Schedule object with a SCHEDTYPE_SCHEDULE field value of 1. The schedule points associated with this schedule represent the net load consumption over time.

The load schedule has an optional field LOADGEN_SCHEDULE which can reference another schedule

(with a SCHEDTYPE_SCHED field value of 20) which defines the power generation over time within the load.

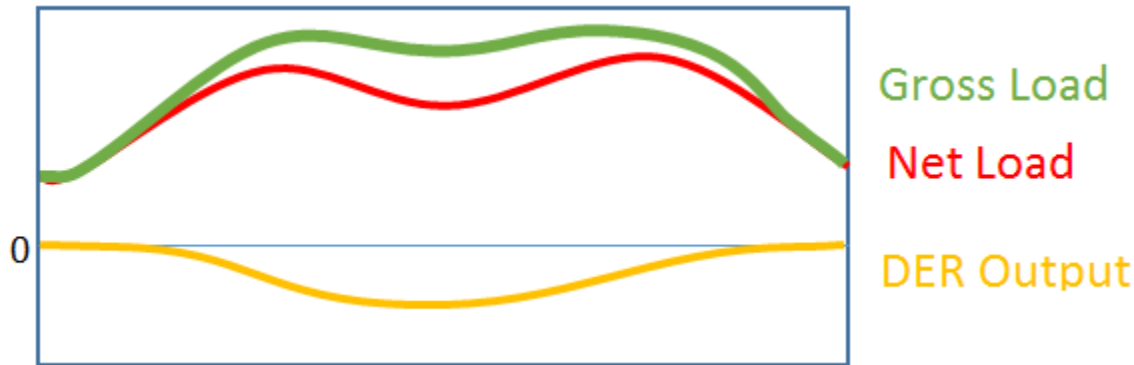


Figure 91. Load Generation Schedule

Thus, the gross load at a load point will equal the net load (normal load schedule) plus the DER generation (load generation schedule). The value for gross load becomes important during the FISR analysis.

4.6.2. Effect on Reconfiguration Applications

The standard behavior of DER and non-primary generation devices when distribution system supply is interrupted is to disconnect from the system and/or shut down. When the system supply is restored, the devices will reconnect after a finite time (typically in minutes). This behavior is defined in the IEEE1547 standard and is normal where the de-energized system is not intended to continue operation in an islanded or microgrid configuration.

This impacts FISR as any de-energized DER devices (either individually modeled or embedded within a de-energized load) are not initially available during restoration. Thus, the plan steps suggested by the FISR application must restore the gross load on the feeders and not just the net load. This will result in higher restoration currents and greater limitations on available switching operations.

After restoration is complete, re-energized DER will be restored and future analyses will include their effects.

During the AFR/PLOS process, a parallel circuit will be formed first before breaking the circuit. No load or DER/generator will be dropped during the process. In this case the system net load will remain the same. Therefore, no special handling is needed.

4.6.3. Inhibiting Plans with Reverse Power Flows

If DER output and generation exceed loads within a local area, it can cause reverse power flows through the station power transformers and feeder breakers. Unless the station is designed for this, the reverse flows can cause protection problems and voltage regulation problems as well as commercial issues.

The DNAF Configuration Editor contains an option called Enforce Station Transformer Backfeed Limit in the Limit Strategy section of the Switching Reconfiguration Problem Formulations page. When enabled, this limit will flag plans that contain reverse real power flow through a station power transformer as having a violation.

Plans with violations will not be implemented in closed-loop operation, and will appear in the user interface in red.

4.6.4. Cogeneration Voltage Regulators

On certain upstream voltage regulators, where DG is present on the load side, the regulator controller is programmed to operate in cogeneration mode. In this mode, the voltage regulator always operates in the forward direction, but allows for reverse settings (voltage setpoint, bandwidth, time delay, line drop compensation R/X settings) to be used if reverse power flow is detected.

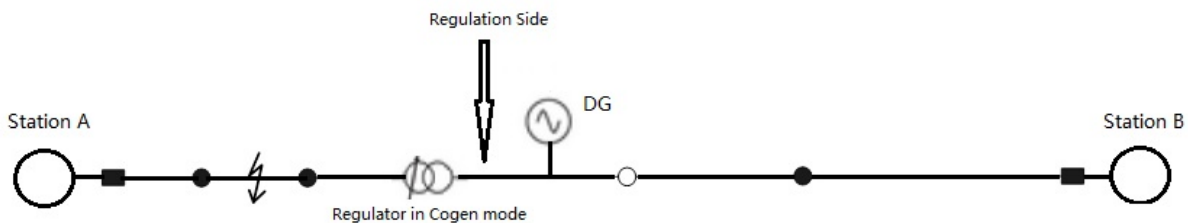


Figure 92. Normal Configuration for Cogen Voltage Regulator

Even though the controller can detect reverse power flow, it is difficult for the controller to determine if it is a true reverse power flow scenario or just reverse power flow due to the cogeneration source. In the case of a true reverse power flow, where the load side becomes the stiff source due to switching, the regulator would continue to operate in the forward regulation and the tap changer is likely to run to its maximum tap position. For this reason, the regulator should be changed to an alternate operating mode to better handle this reverse power flow condition.

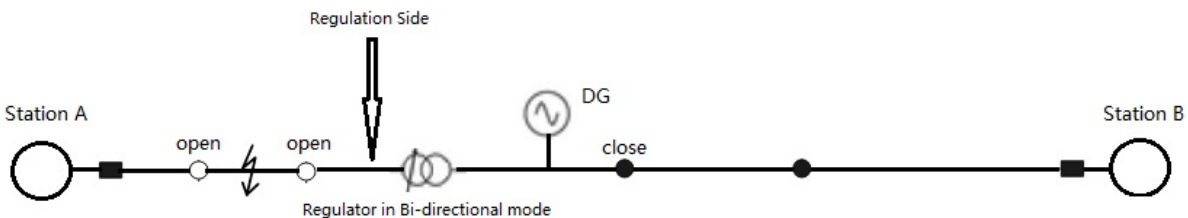


Figure 93. Alternate Topology for Cogen Voltage Regulator Changed to Bi-Directional Mode

The DNAF Configuration Editor contains an option called Change Voltage Regulation Mode in the Switching Plan Generation Parameters section of the Fault Isolation and Service Restoration page. When enabled, voltage regulators in the restoration area that have advanced regulation ability will

be checked to see if their voltage regulation mode needs to be changed. The voltage regulation mode change will be added in the final restoration plan as a separate plan step. To maintain voltage on the regulation side, the voltage regulation mode change step always appears before the switching steps in the plan.

Depending on the SCADA measurement configuration, Plan Manager will send either a setpoint value to change the active regulation mode analog measurement, or plan manager will send a control value to change the active regulation mode status measurement for closed-loop and semi-automatic plan execution modes. If both valid analog and status measurements are modeled for a regulator, the analog measurement will be used.

The DNAF Configuration Editor contains an option called Allow Plans with Failed Regulation Mode Changes in the Operation Mode page under the Switching Reconfiguration tab. If this option is enabled, during plan execution, even if the regulation mode change action failed, the plan will execute the next step in the plan instead of failing the current action and halting plan execution.

4.6.5. Block DER from Closing

For a large DER site, sometimes the DER site needs to be blocked from closing back in on circuit that cannot handle its load after reconfiguration. The DER site is connected to the network with a specific recloser. In the beginning of reconfiguration plan, the current status of a DER site whose capacity is larger than a certain configurable threshold will be checked. If the site is connected, then the recloser needs to be opened first to disconnect the DER site. Once the recloser is open and DER site is disconnected from the network, a command to prevent normal close control will be sent to the recloser to block the DER site from reconnecting back to network. Only the DER in the reconfiguration path will be considered. Plan steps for disconnecting and blocking DER sites always appear in the beginning of reconfiguration plan.

The DNAF Configuration Editor contains an option called Block DER Connection in the Analysis Options section of the Fault Isolation and Service Restoration Problem Formulation page. When enabled, FISR can generate plan steps to block DER connection for devices that are larger than the value specified in the Minimum DER (kVA) field.

The DNAF Configuration Editor also contains an option called Continue Execution after a Failed Close Block Control in the Operation Mode page under the Switching Reconfiguration tab. If this option is enabled, during plan execution, even if the prevent normal close (Block DER) action failed, the plan will execute the next step in the plan instead of failing the current action and halting plan execution.

4.7. Coordination with Automated Switching Schemes

Distribution utilities may have automated local peer-to-peer switching schemes on their distribution networks. The most commonly deployed system is the S&C Electric IntelliTeam® SG Automatic Restoration System composed of IntelliRupter® reclosers, Scada-Mate® switches, and automated PMH cabinet switchgear. These switches are generally controlled by field software applications to

automate the fault isolation and service restoration functions (decentralized FISR). The communication between the software and field devices happens over a fiber optic network or a radio network.

Figure below shows an example of a circuit with a local switching scheme composed of three switches, an IntelliRupter recloser (S1), a Scada-Mate switch (S2), and an automated switch inside a PMH cabinet (S3).

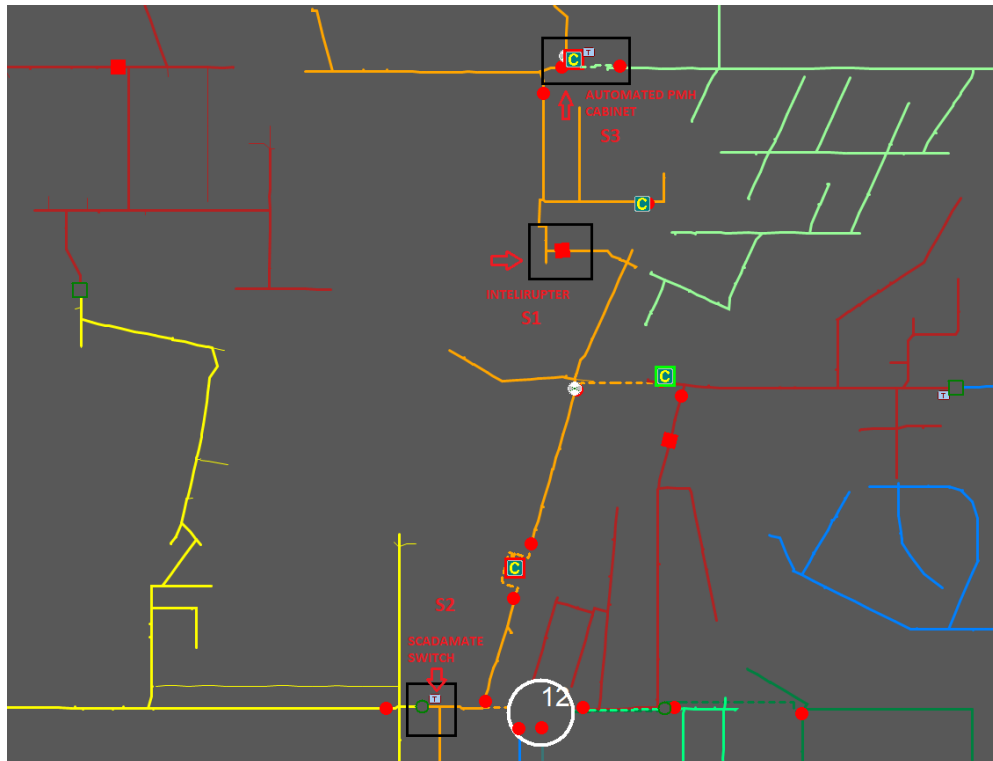


Figure 94. Local Automated Switching Scheme

When a fault occurs upstream of IntelliRupter recloser S1, the fault isolation and downstream restoration scheme is triggered, as shown in figure below. IntelliRupter recloser S1 opens and automated switch S3 closes. The feeder breaker will open after its reclosing cycle, but the scheme would trigger and restore much faster than the reclosing and lockout cycle.

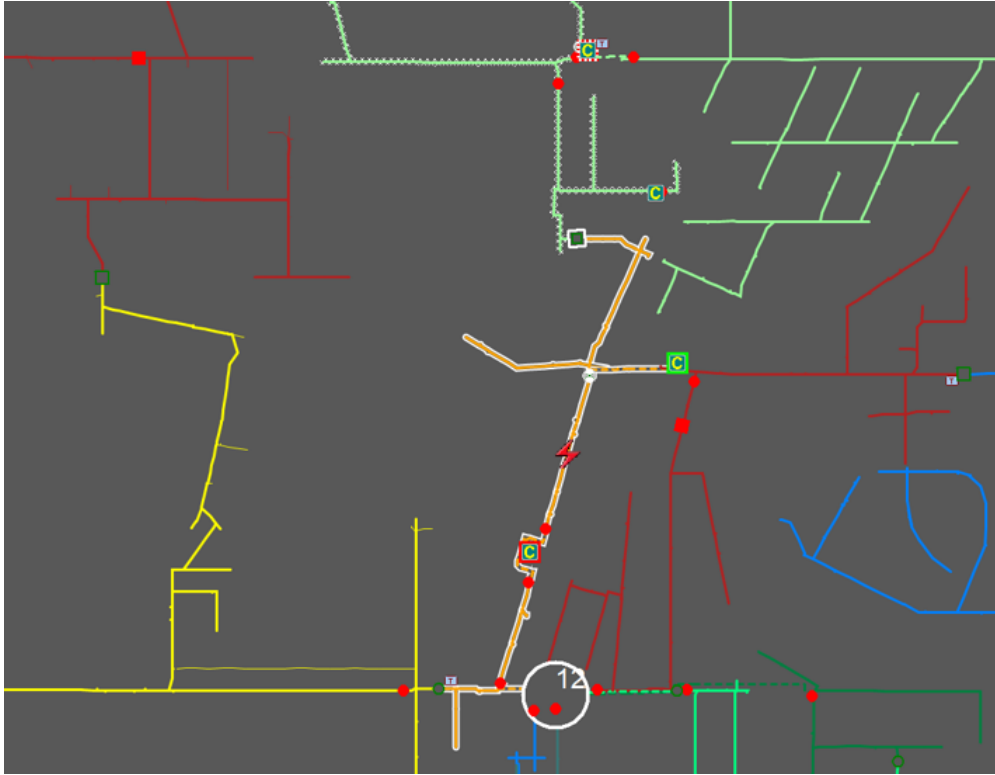


Figure 95. Restoration of a Fault Upstream of Recloser S1

When a fault occurs downstream of the IntelliRupter recloser S1, the restoration scheme is triggered to restore customers in the unfaulted sections of the feeder by closing the Scada-Mate switch S2 and opening IntelliRupter S1 to isolate the fault, as shown in figure below.

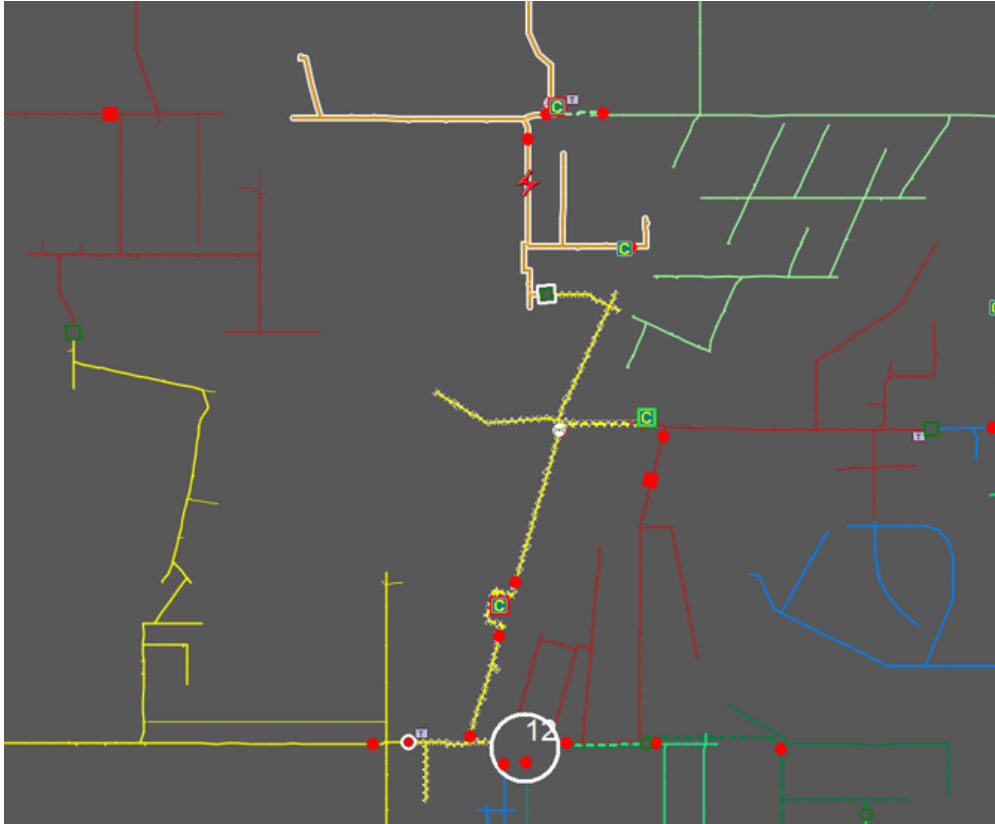


Figure 96. Restoration of a Fault Downstream of Recloser S1

Devices that are part of local automation switching schemes (based on SCADA measurements) are not eligible for FISR and AFR.

The ADMS system uses the following measurements to support local switching scheme functionality: SchemeAutoEnabled, MeasSchemeEnabled, MeasSwitchReady, MeasTeam1Ready, MeasTransferInProgress, and so on. In addition, the system can automatically issue controls to switching scheme devices based on the StatusDAEnabled, DAProhibit, DeviceReclosing, and GroundTripEnabled measurement statuses of feeder breakers.

The following options in the NetServerDnafConfig.txt file define how the centralized ADMS FISR and AFR applications integrate with local automation switching schemes.

- TAS_ENABLE_FISR_INTEGRATION_WITH_LOCAL_SCHEMES

This parameter defines whether integration between ADMS switching reconfiguration applications (AFR and FISR) and local peer to peer schemes is enabled.

- If enabled (=1), FISR and AFR recognize the presence of local schemes; the FISR and AFR operation is delayed until the local schemes complete operation in the field and the new reconfigured topology is updated within ADMS.
- If disabled (=0 or not defined), FISR and AFR operation does not consider the presence of switching schemes, and the applications have the default behavior.

- TAS_FISR_WAIT_FOR_LOCAL_SCHEMES_DELAY

Time delay in seconds for FISIR to wait until local schemes complete their operation. This parameter only comes into effect after the MeasTransferInProgress SCADA measurement that indicates that the scheme completion in the field is received.

- **TAS_FISR_CAN_USE_TEAM_MEMBERS**

Indicates whether FISIR can utilize and control switching devices participating in teams.

- If enabled (=1), switches that are part of teams are eligible for FISIR, after the switching schemes complete their operation.
- If disabled (=0 or not defined), switches that are part of teams are not eligible for FISIR.

The following options in the NetServerDnafConfig.txt file define automatic arming/disarming of local automation switching scheme devices.

- **RECLOSING_DISABLED_POINT_ENABLED**

A flag to enable (=1) or disable (=0) automatic "disarming" of all automation scheme team switching devices located on a feeder when a feeder breaker status is DA disabled, no restoration allowed, or reclosing disabled.

This option is used to automatically disable normal load restoration operations for devices on a feeder to be shed so that the load that is shed by the Load Shed function is not restored automatically by an automated switching scheme function or an ADMS automatic load restoration function. The "disarming" event is triggered by at least one of the following statuses on a breaker: StatusDAEnabled = 0, DAProhibit = 1, or DeviceReclosingEnabled = 0.

"Disarming" is issuing the following controls to the DeviceReclosingEnabled, ProhibitRestoration, and GroundTripEnabled measurement points:

- DeviceReclosingEnabled change from 1 to 0 (reclosing is disabled)
- ProhibitRestoration change from 0 to 1 (no restoration allowed)
- GroundTripEnabled change from 1 to 0 (ground trip protection turned off)

(Reclosers have the DeviceReclosingEnabled point modeled; switches have the ProhibitRestoration point modeled.)

In addition, this option specifies whether GroundTripEnabled is set to 0 for all automated switching scheme devices on a feeder if GroundTripEnabled = 0 on a feeder breaker.

- **ARM_DEACTIVATED_DEVICE_ENABLED**

A flag to enable (=1) or disable (=0) automatic "rearming" of all "disarmed" automation scheme team switching devices located on a feeder when a feeder breaker has the following statuses: closed, DA enabled, restoration allowed, and reclosing enabled.

This option is used to automatically enable normal load restoration operations for devices on a feeder after load shedding is finished. The "rearming" event is triggered when a breaker has all of the following statuses: Closed = 1, StatusDAEnabled = 1, DAProhibit = 0, and DeviceReclosingEnabled = 1.

"Rearming" is issuing the following controls to the DeviceReclosingEnabled, ProhibitRestoration, and GroundTripEnabled measurement points:

- DeviceReclosingEnabled change from 0 to 1 (reclosing is enabled)
- ProhibitRestoration change from 1 to 0 (restoration is allowed)
- GroundTripEnabled change from 0 to 1 (ground trip protection turned on)

In addition, this option specifies whether GroundTripEnabled is set to 1 for all automated switching scheme devices on a feeder if GroundTripEnabled = 1 on a feeder breaker.

4.8. Incorporation of BLA Status

When a measurement island involved in a reconfiguration application plan is not converged in the latest BLA/DPF solution, users cannot be completely confident in the generated restoration plans, particularly if FISR or AFR is running in closed-loop mode.

At the start of a reconfiguration application (Fault Isolation and Service Restoration (FISR), Automatic Feeder Reconfiguration (AFR), Planned Outage Study (PLOS), or Fault Isolation and Service Restoration Contingency Analysis (FISRCA)), the BLA status of all measurement islands in the network selection is optionally evaluated.

When a plan is being generated, the convergence of the measurement islands is checked to determine whether there are any non-converged or scheduled loading BLA measurement islands either inside or outside of the station fence in the reconfiguration path for a given plan.

The measurement island is considered BLA Non-Converged if the reconfiguration application's peak evaluation time is the current time, and BLA does not converge.

If there is at least one non-converged BLA measurement island in the reconfiguration path, this plan is set as non-converged BLA and the suffix "Non-Converged BLA Islands" is added to the plan status.

The reconfiguration application also checks for measurement islands where BLA was not performed and nominally scheduled values are used. These measurement islands are marked as scheduled loading for either of these two conditions:

- The reconfiguration application's peak evaluation time is the current time, and the scheduled load values are used.
- The reconfiguration application's peak evaluation time is not the current time, so the load values estimated as described in [Network Loading Conditions](#) are used.

If there is no non-converged BLA measurement island in the reconfiguration path for a given plan and there is at least one scheduled loading measurement island in the reconfiguration path, the plan is set as scheduled loading and the suffix "Scheduled Loading Islands" is added to the plan status.

The "Include Plans with Non-Converged or Scheduled Loading BLA Islands" option on the Switching Reconfiguration Problem Formulations tab in DCE specifies whether reconfiguration applications

generate plans with non-converged or scheduled loading measurement islands.

- If this option is not selected and there is at least one non-converged or scheduled loading BLA measurement island involved in a plan, the plan is not generated.
- If this option is selected, a plan is generated even if there is at least one non-converged or scheduled loading BLA measurement island involved in this plan. However, if the weight is not zero, the Non-Converged or Scheduled Loading BLA Islands performance index is added to the total PI and this plan is ranked lower than other plans.

☒ Include Plans with Non-Converged or Scheduled Loading BLA Islands

4.9. Incorporation of Bypass Switches in Reconfiguration Application Plans

Reconfiguration applications determine whether a recloser needs to be bypassed in these two scenarios:

- A plan will cause new reverse flows for a non-reversible protective recloser or breaker.
- Loading in a protective recloser or breaker is larger than the phase trip rating.

In these scenarios, reconfiguration applications include steps to close a bypass switch and open a recloser or breaker, if possible.

4.9.1. Reverse Flow for Non-Reversible Recloser/Breaker

The "Use Plans with Reverse Flow for Non-Reversible Recloser/Breaker" option in the Switching Reconfiguration Problem Formulations tab in DCE specifies whether reconfiguration applications generate plans with new reverse flow for non-reversible protective reclosers or breakers.

☒ Use Plans With Reverse Flow for Non-Reversible Recloser/Breaker

- If this option is not selected, any generated plan should not cause new reverse flow for non-reversible protective reclosers/breakers.
- If this option is selected, a plan is generated even if there is new reverse flow for at least one non-reversible protective recloser or breaker. However, if the weight of the Reverse Flow for Non-Reversible Recloser/Breaker performance index is not zero, it is added to the total PI and the plan may be ranked lower than other plans.

The performance index is calculated using the following equation:

$$PI_{ReverseFlow} = Count_{ReverseFlowRecloser} * Weight_{ReverseFlow} \quad (\text{Equation 4.2})$$

If there are non-reversible reclosers or breakers with reverse flow already in the system before a reconfiguration application starts to solve, these reclosers or breakers will not be considered by the

reconfiguration application and they will not impact the plan generation logic or PI calculation. Figure below shows an example two-feeder system in normal condition.

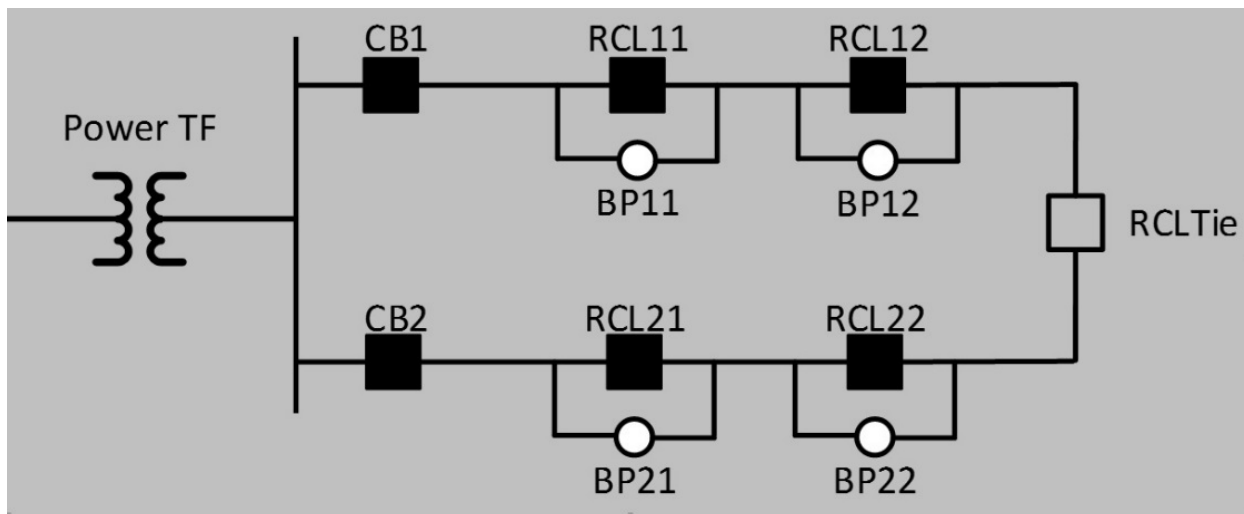


Figure 97. Example System for Reclosers with Bypass Switches

In this system, reclosers RCL11, RCL12, RCL21, and RCL22 are normally closed non-reversible reclosers with normal flow. BP11, BP12, BP21, and BP22 are their normally open tie switches. Tie recloser RCLTie is a normally open reversible recloser.

A fault happened between CB1 and RCL11; CB1 tripped and downstream sections were de-energized, as shown in figure below.

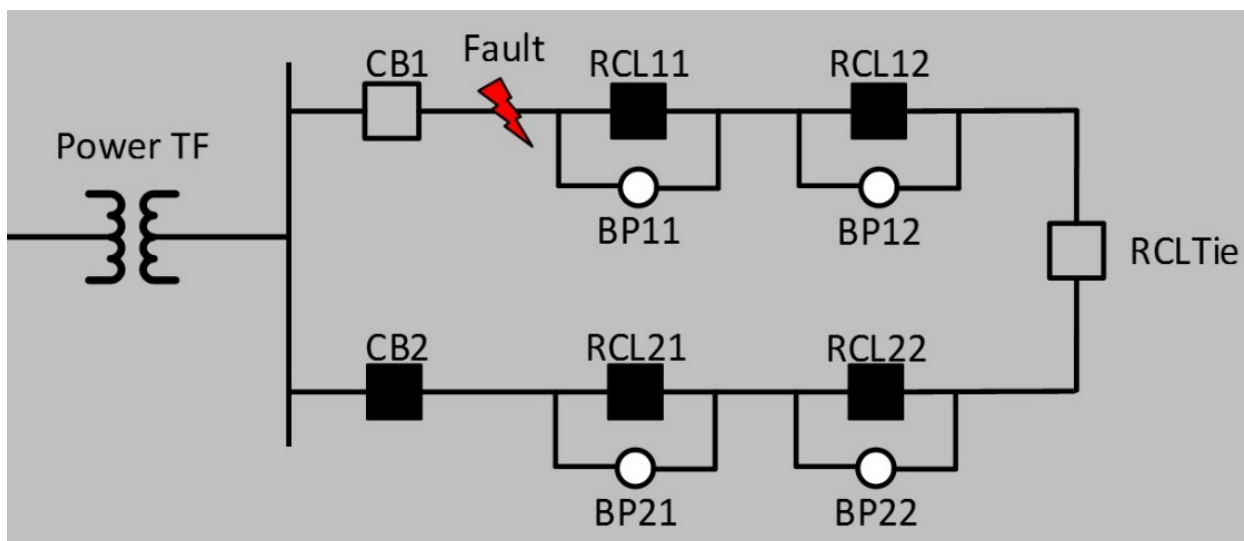


Figure 98. Fault on Example System for Reclosers with Bypass Switches

If FISR generates a plan of opening RCL11 and closing RCLTie, there will be reverse flow for RCL12. There are three cases:

- Case 1: If reverse flow is allowed, this plan is generated.

- Case 2: If reverse flow is not allowed and the bypass is eligible, two more steps of closing the bypass and opening the non-reversible recloser are added. A four-step plan of opening RCL11, closing BP12, opening RCL12, and closing RCLTie is generated.
- Case 3: If reverse flow is not allowed and the bypass is not eligible, FISR will restore only part of the sections not causing reverse flow. In this example, if BP12 is not eligible, a two-step plan of opening RCL12 and closing RCLTie is generated, and the load between RCL11 and RCL12 is not restored.

4.9.2. Phase Trip Rating Violation

Reconfiguration applications consider the recloser's or breaker's phase trip rating.

When the flow in a breaker or recloser is larger than its phase trip rating during the implementation of the FISR plan, the system tries to include steps for closing the bypass switch and opening a recloser or breaker in the plan. If a non-reversible breaker or recloser with reverse flow cannot be bypassed, the plan is generated with the phase trip violation status and cannot be implemented in closed loop mode.

4.10. Switching Validation Checks in Reconfiguration Applications

Additional switching validation checks can be optionally incorporated into the reconfiguration applications solution process, including the following: Ferroresonance, 3-Wire/4-Wire, and Paralleling Circuits with Reclosing-Capable Protective Devices.

Each switching validation check can be independently enabled or disabled for any given reconfiguration application problem formulation in the DNAF Configuration Editor. For example:

- ☐ Prevent Closed Loop Execution if a Switching Validation Violation Exists
- ☐ Enable Ferroresonance Switching Validation Check
- ☐ Enable 3-Wire/4-Wire Switching Validation Check
- ☐ Enable Paralleling Circuits with Reclosing Capable Protective Devices Switching Validation Check

If a specific switching validation check fails for a plan, the plan is marked as switching validation failed for that specific check. For plan ranking purposes, a performance index is calculated for each failed check based on its weight specified in the Performance Index Weights subsection in the reconfiguration problem formulations page in the DNAF Configuration Editor. The formulae are described in [Problem Formulations](#).

Switching Validation Check - Ferroresonance:	10	N/A
Switching Validation Check - 3-Wire/4-Wire:	10	N/A
Switching Validation Check - Parallel Recloser:	10	N/A

Ferroresonance Switching Validation Check

If this check is enabled (using the "Enable Ferroresonance Switching Validation Check" option for a Switching Reconfiguration Application Problem Formulation in the DNAF Configuration Editor), a ferroresonance check will be performed for each switching operation in the plan.

Ferroresonance is a nonlinear behavior that occurs in both transmission and distribution systems. In a typical distribution system, there are inductive components (transformers), capacitive components (overhead line or cable charging), and resistive components (loads). When one or two phases of a circuit are disconnected, unbalanced current will flow through the series connection of the inductor-capacitor pair in the remaining connected phases. If the resistive component is small, ferroresonance may occur because the current is not limited by the resistance in the circuit. This will lead to overvoltage and/or overcurrent in the system and possible equipment insulation breakdown.

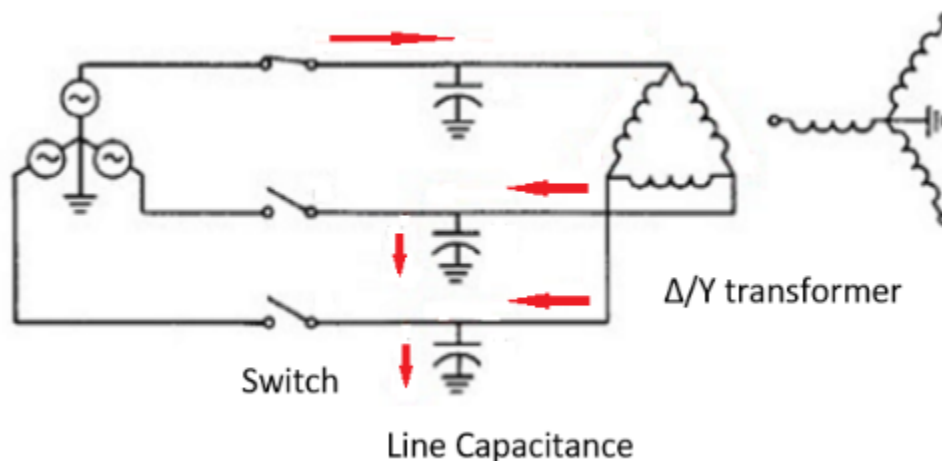


Figure 99. Ferroresonance Effect when Closing a Phase Switch

As shown in figure above, the transformer is operating in a no-load condition; when the phase A switch is closed before the other phases, there are current flows from the voltage source on phase A to ground through the line capacitance on phases B and C. If the resistance in the circuit is small, the current magnitude could be much larger than in normal operation after all phases are closed.

A ferroresonance violation tends to happen on lightly loaded transformers that have a small transformer resistance. Research indicates that the ratio of the capacitive reactance to the transformer magnetizing reactance of the ferroresonance circuit will determine whether the circuit will experience over voltages. One way to reduce the ferroresonance effect is to connect a small resistive load on the transformer secondaries to limit the current in a circuit. Another way is use three-phase interrupting devices such as ganged-switching devices so all three phases are connected or disconnected at the exact same time.

The following logic is used for the ferroresonance switch validation check:

- First, the switching device switching mechanism is retrieved. If the switching device operates in ganged mode (as defined in the static model), meaning all three phases are operated at the

same time, the switching step has no ferroresonance concern and no further validation is performed.

- If the switching device is not operated in ganged mode (phase-independent switching operation is supported), a downstream trace is performed to retrieve the list of all downstream three-phase transformers.
- For each transformer in the list, the criteria listed below are checked. If all three conditions listed below are satisfied for any transformer in the list, this switch has a ferroresonance concern. The plan is marked as failed due to a ferroresonance switching validation, and the associated performance index is calculated (the index weight is configurable in the DCE) and is added to the overall total plan performance index.
 - Transformer resistance is less than a configurable threshold that is provided in the Limit Monitor page in the DCE;
 - Transformer loading is less than a configurable percentage of full load that is provided in the Limit Monitor page in the DCE;
 - The ratio of the total line capacitance between the switch and the transformer to the transformer magnetizing reactance is calculated first. The difference between this ratio and 1.0 is then compared to the user-configured threshold provided in the Limit Monitor page in the DCE. The logic for this criterion is shown below.

$$\left\{ \begin{array}{l} \text{Check failed if } \left| \frac{X_c}{X_L} - 1.0 \right| \leq \frac{\text{threshold}}{100.0} \\ \text{Check passed if } \left| \frac{X_c}{X_L} - 1.0 \right| > \frac{\text{threshold}}{100.0} \end{array} \right.$$

3-Wire/4-Wire Switching Validation Check

If this check is enabled in the problem formulation for a switching reconfiguration application, each open to close switching operation in the plan steps performs an upstream topology trace to determine if closing the switch involves energized circuits with grounded loads served through conductors that do not have a neutral wire.

The following logic is used:

- If both sides of the switch are energized:
 - Both sides are traced. If no grounded loads are found, the check is considered passed.
 - If grounded loads are found and both sides do not have non-neutral lines, the check is considered passed.
 - If grounded loads are found and either side has a non-neutral line, the check is considered failed and a contributing performance index is calculated (the index weight is configurable) and is added to the overall plan performance index.
- If only one side of a switch is energized:
 -

The de-energized side is traced. If no grounded loads are found, the check is considered passed.

- If grounded loads are found on the de-energized side, the energized side is traced upstream to check if a non-neutral line is found on the energized side. If a non-neutral line exists on the energized side of the switch, the check is considered failed and a contributing performance index is calculated (the index weight is configurable) and is added to the overall plan performance index. Otherwise, the check is considered passed.

Parallel Circuits with Reclosing-Capable Protective Devices Switching Validation Check

If this check is enabled in the problem formulation for a switching reconfiguration application, each open to close switching operation in the plan steps performs an upstream topology trace from both sides of the switch to determine if closing the switch results in a parallel circuit configuration with reclosing capable protective devices on either of the circuits.

The following devices are considered "reclosing-capable protective devices":

- Reclosers
- Three-phase fuses
- Non-feeder head breakers

4.11. Changing Protection Settings by Protection Validation

When the Protection Validation (PRV) application is executed in the scope of the reconfiguration applications, PRV needs to evaluate the protection reach for each PRV zone bounded by breakers and reclosers in recloser mode. The PRV zone formation in the reconfiguration applications is similar to the base PRV application including the handling of transformer self-protection.

When there is a protection reach violation or protection margin violation, and protection settings are enabled to be transitioned from non-recloser operation mode (switch or sectionalizer mode) to recloser mode, the recloser that is currently set to the switch or sectionalizer mode can also bound the PRV zone in the intermediate process of finding the feasible protection settings group change. This will allow the appropriate settings groups transitions to be determined based on the user-configured reach and neutral safety factor and peak load safety factor.

Note

Only the reclosers in recloser mode and breakers are considered as protection devices in the forming PRV protection zones (not the reclosers in switch mode or sectionalizer mode).

Like the standard PRV application, the PRV application that is run in the scope a reconfiguration application can calculate fault currents using a bolted fault and/or a user-configured fault impedance. PRV violations for the reconfiguration applications are determined as follows:

- When checking for a maximum interruption violation which is based on the calculated maximum capacity ratio, the largest fault current value calculated from either the bolted fault or the user-configured fault impedance in the PRV zone is used.
- When checking for a minimum phase pick-up violation and reach violation which are based on the calculated minimum phase pick-up ratio (the phase fault sensitivity is the inverse of this ratio), the smallest phase fault current value calculated from either the bolted fault or the user-configured fault impedance in the PRV zone is used. For the standard PRV application, the calculated phase-to-phase fault current is utilized as the phase fault current. For the reconfiguration application PRV, either the phase-to-phase fault current or the minimum of the phase-to-phase and the phase-to-ground fault current is used depending on the Adaptive Relaying option in the Switching Reconfiguration Problem Formulations page in the DCE. For either method used in this violation check, the Reach and Neutral Safety Factor from the Switching Reconfiguration Problem Formulations page in the DCE is used.
- When checking for a minimum ground pick-up violation which is based on the calculated minimum ground pick-up ratio (ground fault sensitivity is the inverse of this ratio), the smallest phase-to-ground fault current value calculated from either the bolted fault or the user-configured fault impedance in the PRV zone is used.

To improve the performance of calculating fault currents using both a bolted fault and a user-configured fault impedance, the maximum fault current can be calculated with a bolted fault, and the minimum phase fault current, protection reach, and ground fault current values can be calculated with the user-configured fault impedance. This can be achieved by enabling the Perform the Minimum Number of Fault Evaluations option in Switching Reconfiguration Problem Formulations page in the DCE.

Some rules are applicable when generating settings group steps in a FISR/AFR plan to resolve a protection reach violation and/or protection margin violation. The simple system shown in figure below is used to illustrate some of these rules.



Figure 100. System in Nominal Configuration

There are two feeders in the system, Feeder 1 and Feeder 2, and two feeder breakers, B1 and B2, that do not have protection settings groups. R1, R3, and R4 are closed reclosers, and R2 is a normally opened tie recloser between the two feeders. The breaker and reclosers in red are closed and the ones in green are opened. All reclosers are eligible for settings group control. A fault occurs between R4 and B2 and B2 is tripped to clear the fault as shown in figure below.



Figure 101. Faulted System with Settings Groups Enabled

In figure below, FISR tries to open R4 and close R2 to restore the downstream load for Feeder 2.

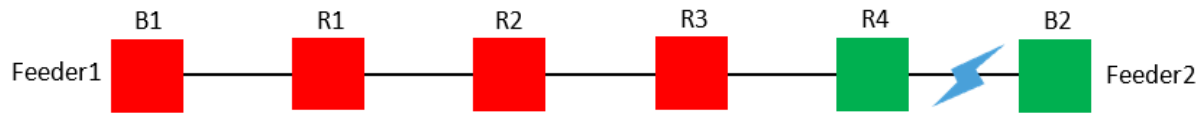
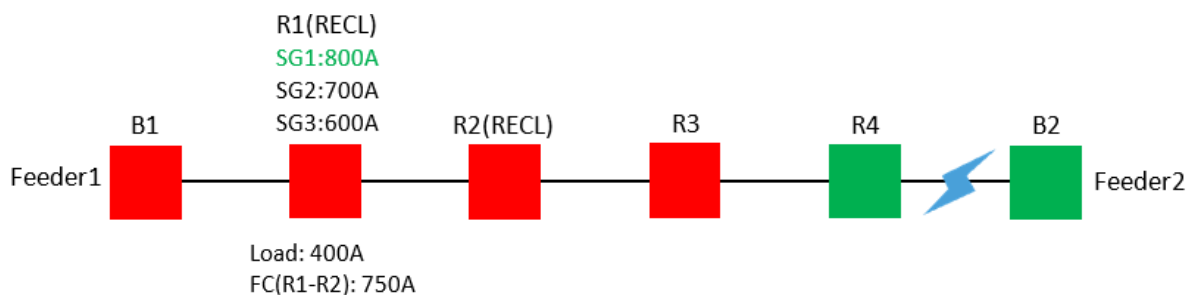


Figure 102. Faulted System Restored with Settings Groups Enabled

In the examples below, the user-configured reach and neutral safety factor and peak load safety factor are 1.0 and only the phase-to-phase fault current is used to check for reach violations.

The rules for generating settings group steps are as follows:

- Only eligible protection devices will form a PRV zone for the Protection Validation (PRV) application in the scope of the reconfiguration applications.
- The settings group of a device is eligible if the associated device is eligible and there is a valid settings group status or analog measurement indicating the current settings group.
- The settings group of a device in a protective operation mode (recloser mode) can be changed to a different protective operation mode if the associated transitions are supported in the settings group template. However, a device in recloser mode cannot be changed to a non-protective operation mode (switch mode or sectionalizer mode).
- If there is a protection reach violation and/or protection margin violation, the settings group will be changed to the one with the highest phase trip value to allow for a potential downstream load increase.



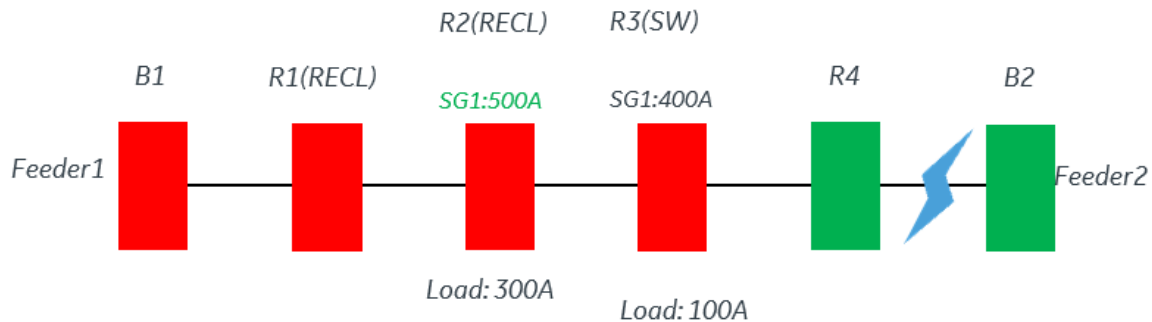
For example, in the system in the above figure, R1 and R2 are the reclosers operating in recloser mode. After restoration, the load for R1 is 400 Amps, the minimum phase Fault Current (FC) for R1 for the zone from R1 to R2 is calculated by PRV to be 750 Amps, and the current phase trip setting is 800 Amps, meaning there is a protection reach violation for R1. There are two other R1 settings groups with phase trip setting values of 700 Amps and 600 Amps. Both the 700 Amps and 600 Amps settings groups have a phase trip setting that can resolve the protection reach violation because they are both larger than the load (700.0 is greater than 400.0, and 600.0 is greater than 400.0), but smaller than the minimum phase fault current (700.0 is less than 750.0, and 600.0 is less than 750.0). Since 700 is greater than 600, the settings group with the 700 Amps phase trip setting is selected.

Table below shows the results before and after the settings group change.

Table 15. R1 Results Before and After Picking Highest Phase Trip Setting

	Before	After
Phase Trip Setting (A)	800	700
Fault Current (A)	750	750
Violation	Yes	No

- A device in a non-protective operation mode (switch mode or sectionalizer mode) can be changed to a protective mode (recloser mode) if the associated transitions are supported in the settings group template.



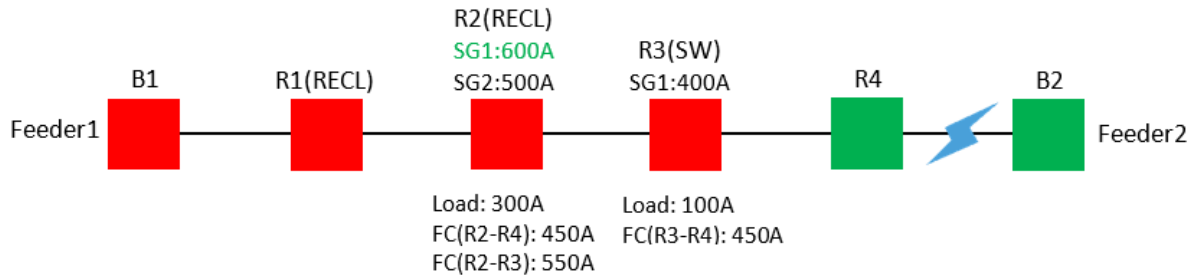
For example, in the system in the above figure, R2 is a recloser operating in recloser mode and R3 is a recloser operating in switch mode. After restoration, the load for R2 is 300 Amps, the PRV calculated minimum phase fault current in R2 is 450 Amps, and the current phase trip setting is 500 Amps, meaning there is a protection reach violation for R2. The load for R3 is 100 Amps. R3 has a settings group in recloser mode with a phase trip setting of 400 Amps, and after R3 is changed to recloser mode, the minimum phase fault current in R3 is 450 Amps and the minimum phase fault current in R2 is 550 Amps. The settings group with a 400 Amps phase trip setting for R3 will be selected.

Table below shows the Results Before and After the Settings Group Change.

Table 16. Results Before and After Changing Non-Protective Operation Mode to Recloser Mode

	Before			After		
	Phase Trip Setting (A)	Fault Current (A)	Violation	Phase Trip Setting (A)	Fault Current (A)	Violation
R2	500	450	Yes	500	550	No
R3	—	—	—	400	450	No

- Settings groups will be changed for multiple devices only if there is a protection reach violation and/or protection margin violation that has to be resolved by changing modes for more than one device.



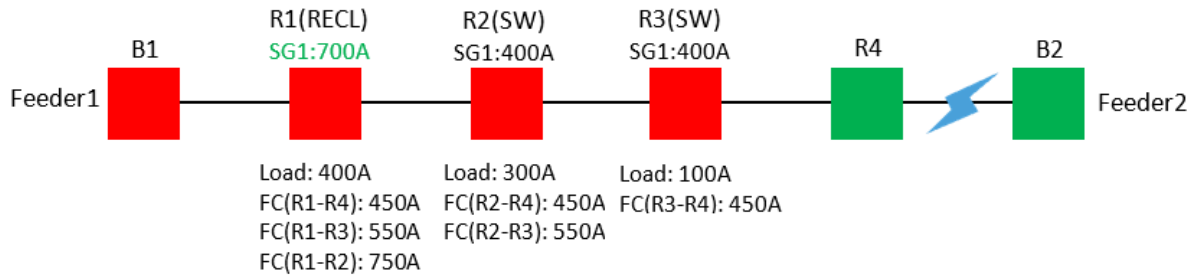
For example, in the system in the above figure, R2 is a recloser operating in recloser mode and R3 is a recloser operating in switch mode. After restoration, the load for R2 is 300 Amps, the PRV calculated minimum phase fault current in R2 is 450 Amps, and the current phase trip setting is 600 Amps, meaning there is a protection reach violation for R2. The load for R3 is 100 Amps. If R3 has a settings group in recloser mode with a phase trip setting of 400 Amps, and after R3 is changed to recloser mode, the minimum phase fault current in R3 is 450 Amps and the minimum phase fault current in R2 is 550 Amps. There is still a protection reach violation for R2. If there is another settings group with a phase trip setting of 500 Amps for R2, then changing R2 to this settings group will solve this violation. There are two devices involved to resolve the violation because changing the settings group of one device cannot resolve the violation.

Table below shows the results before and after the settings group change.

Table 17. Results Before and After Changing Two Devices

Before				After		
	Phase Trip Setting (A)	Fault Current (A)	Violation	Phase Trip Setting (A)	Fault Current (A)	Violation
R2	600	450	Yes	500	550	No
R3	—	—	—	400	450	No

- If the settings group for multiple devices can be changed, the settings group check starts from the most downstream device. If the most downstream device cannot resolve the violation, then the next upstream device is checked until one device that can resolve the violations has been found or all feasible devices have been checked. If all feasible devices have been checked, and one device cannot resolve the violations, two devices will be checked in the same order from downstream to upstream. If two devices cannot resolve the violation, three devices will be checked. This process goes on until multiple devices that can resolve the violations have been found or all combinations have been checked.



For example, in the system in the above figure, R1 is the only recloser in recloser mode. R2 and R3 are reclosers in switch mode. After restoration, the load for R1, R2, and R3 are 400 Amps, 300 Amps, and 100 Amps, respectively.

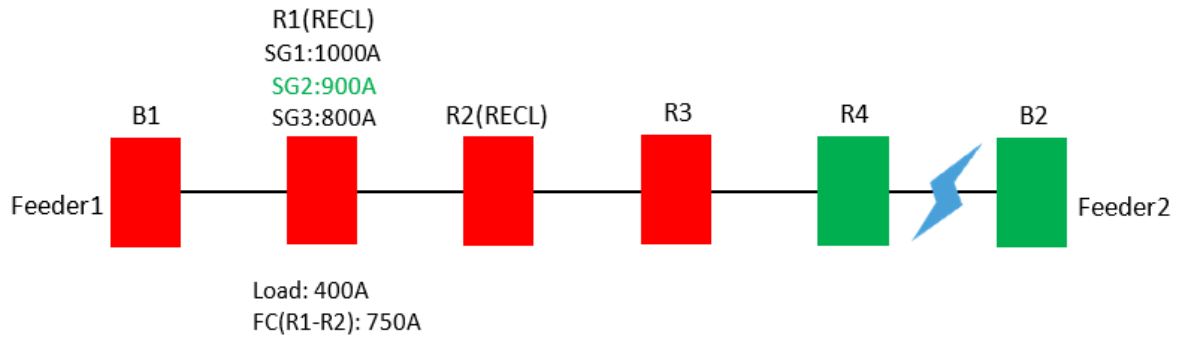
The PRV calculated minimum phase fault current in R1 is 450 Amps, and the current phase trip setting is 700 Amps, meaning there is a protection reach violation for R1. There are two reclosers R2 and R3 that can be changed to recloser mode and the settings group in recloser mode has a phase trip setting of 400 Amps. Since R3 is the most downstream device, R3 is checked first. If R3 cannot resolve the violation, R2 is then checked. After R3 is changed to recloser mode, the minimum phase fault current in R3 is 450 Amps and the minimum phase fault current in R1 is 550 Amps. There is still a protection reach violation for R1. So, changing R3 cannot resolve the violation, and the next upstream device, R2, must be checked. After R2 is changed to recloser mode, the minimum phase fault current in R2 is 450 Amps and the minimum phase fault current in R1 is 750 Amps. There is no violation for R1 or R2, so changing R2 to recloser mode will resolve this violation.

Table below shows the results before and after the settings group change.

Table 18. Results Before and After Changing Two Devices from Most Downstream Device

Before				After		
	Phase Trip Setting (A)	Fault Current (A)	Violation	Phase Trip Setting (A)	Fault Current (A)	Violation
R1	700	450	Yes	700	750	No
R2	—	—	—	400	550	No
R3	—	—	—	400	450	No

If a protection reach violation and/or protection margin violation cannot be resolved within one PRV zone, no settings group will be changed in the devices in that zone.



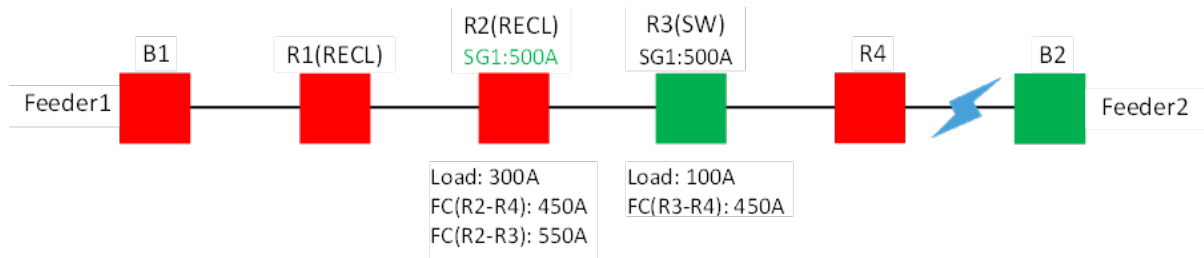
For example, in the system in the above figure, R1 and R2 are reclosers operating in recloser mode. After restoration, the load for R1 is 400 Amps, the PRV calculated minimum phase fault current for R1 is 750 Amps, and the current phase trip setting is 900 Amps, meaning there is a protection reach violation for R1. There are two other R1 settings groups with phase trip setting values of 800 Amps and 1000 Amps. Neither the 800 Amps nor the 1000 Amps settings groups have a phase trip setting that can resolve the protection reach violation because they are both larger than the minimum phase fault current (800.0 is larger than 750.0, and 1000.0 is larger than 750.0). In this case, a new settings group will not be selected, resulting in a FISR plan that cannot be executed in closed-loop mode if the "Allow Closed Loop Execution if Protection Validation Violations Exist" option in the DCE is not selected.

Table below shows the results before and after the settings group check.

Table 19. R1 Results Before and After when Settings Group Violation Not Resolved

	Before	After
Phase Trip Setting (A)	900	900
Fault Current (A)	750	750
Violation	Yes	Yes

- If a protection reach violation and/or protection margin violation cannot be resolved within one PRV zone, partial restoration will be attempted in addition to generating a plan with one or more violations.



For example, in the system in the figure above, R2 is a recloser operating in recloser mode and R3 is a recloser operating in switch mode. After restoration, the load for R2 is 300 Amps, the PRV calculated minimum phase fault current for R2 is 450 Amps, and the current phase trip setting is 500 Amps, meaning there is a protection reach violation for R2. If R3 has a settings

group in recloser mode with phase trip setting 500 Amps, and after R3 is changed to recloser mode, the minimum phase fault current in R3 is 450 Amps and the minimum phase fault current in R2 is 550 Amps. The reach violation for R2 is resolved but there is still protection reach violations for R3. This means R3 has to be opened to resolve the violation. The system after partial restoration is shown in the following figure.

Table below shows the results before and after the settings group change.

Table 20. Results Before and After Partial Restoration

Before				After		
	Phase Trip Setting (A)	Fault Current (A)	Violation	Phase Trip Setting (A)	Fault Current (A)	Violation
R2	500	450	Yes	500	550	No
R3	—	—	—	—	—	—

4.12. Switching Reconfiguration Application Parameters

This section provides additional information about several reconfiguration application specific parameters.

4.12.1. Switching Costs for All Reconfiguration Applications

The switching Cost group can be accessed from the Problem Formulations page. This group involves four parameters. The default value for FISR is shown in figure below.

Switching Cost

Number of Adjacency Levels:

Feeder Adjacency Levels	1	
Manual Switch Cost	4	
Automatic Switch Cost	1	

Figure 103. Default Value for FISR Switching Costs

The Number of Adjacency Levels parameter sets the number of feeder adjacent switching levels that are considered in the analysis. A cost for manual (Manual Switch Cost) and automatic (Automatic Switch Cost) switching devices is assigned for each switching level. Based on the values set for these switching costs, the solutions can be biased to use switching devices closer to the devices that are disconnected due to a fault or outage, resulting in less driving time and improved reliability by using the closest switching devices.

A sample system is shown in figure below.

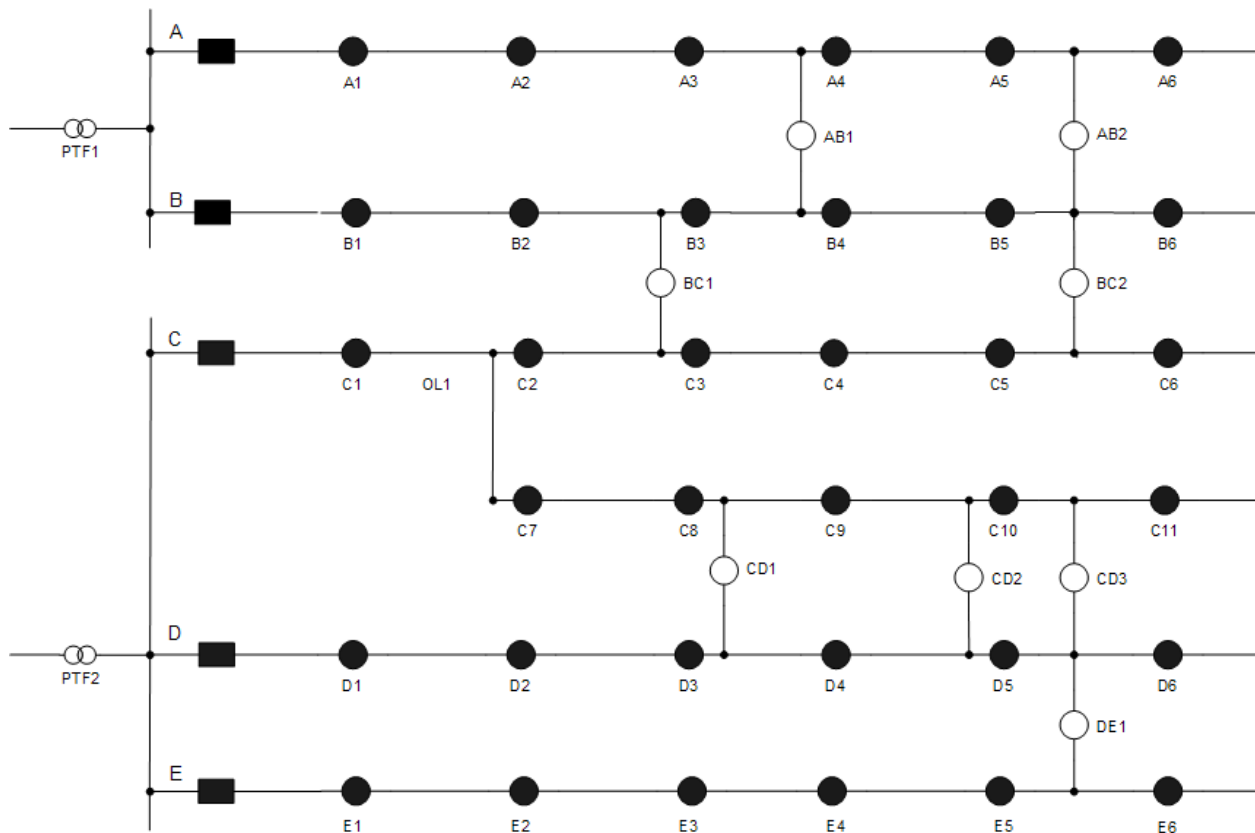


Figure 104. Sample Power System

If the fault is in Feeder B, Feeders A and C are the Level 1 feeders and Feeder D is a Level 2 feeder. These four feeders are included in the network selection.

Since Feeder E is three feeders away from Feeder B (more than the two levels specified by FISIR), Feeder E is not included. The cost of automatic switches in Feeders A and C is 1, and the cost of automatic switches in Feeder D is 2. The cost of manual switches in Feeders A and C is 4, and the cost of manual switches in Feeder D is 8.

4.12.2. Problem Formulation Parameters

There are several problem formulations for each switching reconfiguration application. Every problem formulation has a different emphasis or goal within the reconfiguration application.

FISIR, FISIRCA, and PLOS share the same problem formulations. AFR and FSD have their own set of problem formulations.

The goal of every problem formulation is determined by the Performance Indices. [Problem Formulations](#) summarizes the problem formulation parameters.

The weight of a Performance Index (PI) can be any real number starting from 0. If the weight is 0, it means that the Distribution System Plan Generation Engine (DSPGE) does not take this PI into consideration. More importance is placed on a PI with a larger weight.

All the PI weights can be modified using the DNAF Configuration Editor tool. When modifying an existing problem formulation, setting the new weight from 0.1 to 10 times the range of the default weight should be sufficient in most cases. If a new problem formulation is created, setting the new weight to 0.01 to 100 times the range of the default weight listed in following tables are recommended.

For the FSD application specifically, weighting factors are used for ranking and ordering feeders in FSD Stage 1.

4.12.2.1. Performance Indices for FISR, FISRCA, and PLOS

The default problem formulations used by FISR, FISRCA, and PLOS are listed in table below. However, other problem formulations can be easily created as needed.

Table 21. FISR, FISRCA, and PLOS Problem Formulations

Problem Formulation	Description
Minimum unserved kW	The function finds a restoration plan that restores as much load (kW) as possible. A large weight is applied to the intermediate and final lost load PI.
Minimum customer minutes interrupted	The function finds a restoration plan that minimizes the total number of customer interruption minutes. A large weight is assigned to the intermediate and final dropped customer count PI. See Problem Formulations for more information.
Minimize loss of customer values of uninterrupted service	The function finds a restoration plan that minimizes the loss of customer value of uninterrupted service. A large weight is assigned to the intermediate and final dropped customer priority PI. The customer priorities are assigned based on the costs of uninterrupted service that are given for each load category.
Minimum number of switching operations	The function finds a restoration plan that minimizes the total number of switching operations. A large weight is assigned to the number of control moves PI and intermediate and final total cost PI.
Minimum voltage drop along the reconfigured feeders	The function finds a restoration plan that minimizes the voltage drop between the highest voltage and the lowest voltage along the feeder. A large weight is assigned to the final feeder voltage drop PI.

Table below lists the default weights of the PI for FISR, FISRCA and PLOS.

Table 22. Default PI Weights for FISR, FISRCA, and PLOS Problem Formulations

Performance Index	Minimum Unserved kW	Minimum Customer Minutes Interrupted	Minimum Loss of Customer Values of Uninterrupted Service	Minimum Number of Switching Operations	Minimum Voltage Drop Along the Reconfigured Feeders
Flow Violations - Final	200000	1000	1000	1000	1000
Flow Violations - Intermediate	100	100	100	100	100
Bus Voltage Violations - Final	100	100	100	100	100

Performance Index	Minimum Unserved kW	Minimum Customer Minutes Interrupted	Minimum Loss of Customer Values of Uninterrupted Service	Minimum Number of Switching Operations	Minimum Voltage Drop Along the Reconfigured Feeders
Bus Voltage Violations - Intermediate	10	10	10	10	10
Load Voltage Violations - Final	100	100	100	100	100
Load Voltage Violations - Intermediate	100	100	100	100	100
Total Cost - Final	0.5	0.5	0.5	0.5	0.5
Total Cost - Intermediate	0.05	0.05	0.05	0.05	0.05
Lost Load - Final	1	0.1	0.1	1	1
Lost Load - Intermediate	0.1	0	0	0	0
Dropped Customer Count - Final	1	10	1	1	1
Dropped Customer Count - Intermediate	0.1	1	0	0	0
Dropped Customer Priority - Final	0	0	10	0	0
Dropped Customer Priority - Intermediate	0	0	1	0	0
Radial Condition - Final	10	10	10	10	10
Radial Condition - Intermediate	0	0	0	0	0
Maximum Margin	0	0	0	0	0
Protection Margin	0	0	0	0	0
Control Moves	10	10	10	1000	10
Third-Party Owned Equipment Control Moves	100	100	100	100	100
Losses	0	0	0	0	0
Protection Validation	10	0	10	10	10
Protection Reach	0	0	0	0	0
Voltage Equalization	0	0	0	0	0
Feeder Voltage Drop	0	0	0	0	10000
Load Minutes Interrupted	0	0	0	0	0
Priority Customer Minutes Interrupted	0	0	0	0	0
Customer Minutes Interrupted	0	0	0	0	0
Momentary Frequency - MAIFI	0	0	0	0	0

Performance Index	Minimum Unserved kW	Minimum Customer Minutes Interrupted	Minimum Loss of Customer Values of Uninterrupted Service	Minimum Number of Switching Operations	Minimum Voltage Drop Along the Reconfigured Feeders
Interruption Frequency - SAIFI	0	0	0	0	0
Interruption Duration - SAIDI	0	0	0	0	0
Maximum Number of Operations	1	1	1	1	1
Maximum Segment Margin	0	0	0	0	0
Non-Converged or Scheduled Loading BLA Islands	10	10	10	10	10
Reverse Flow for Non-Reversible Recloser/Breaker	10	10	10	10	10
Switching Validation Check - Ferroresonance	10	10	10	10	10
Switching Validation Check - Grounded Load	10	10	10	10	10
Switching Validation Check - Parallel Recloser	10	10	10	10	10

4.12.2.2. Performance Indices for AFR

Table below lists the default problem formulations used by AFR, but other problem formulations can be easily created as needed.

Table 23. AFR Problem Formulations

Problem Formulation	Description
Reduce Overloads and Violations	The nominal problem formulation that is used to relieve violations, taking into account the peak load within the time window of interest. No overloads or voltage violations are allowed in the given distribution network, including power transformers, distribution lines, or buses.
Load Shedding	This problem formulation performs the action of shedding non-critical load to reduce overloads in transformers, lines, and breakers within the time window of interest.
Return to Normal	This problem formulation restores the switching devices in energized circuits back to their nominal state.

The default weights of the PI for the AFR problem formulations are listed in table below.

Table 24. Default PI Weights for AFR Problem Formulations

Performance Index	Reduce Overloads and Violations	Load Shedding	Return to Normal
Flow Violations - Final	1000	200000	1000
Flow Violations - Intermediate	100	0	100
Bus Voltage Violations - Final	100	100	100
Bus Voltage Violations - Intermediate	10	0	10
Load Voltage Violations - Final	100	100	100
Load Voltage Violations - Intermediate	100	0	10
Total Cost - Final	0.5	0.5	0.5
Total Cost - Intermediate	0.05	0	0.05
Lost Load - Final	1	0.1	1
Lost Load - Intermediate	0	0	0
Dropped Customer Count - Final	20	0.1	20
Dropped Customer Count - Intermediate	2	0	2
Dropped Customer Priority - Final	1	1	1
Dropped Customer Priority - Intermediate	1	0	1
Radial Condition - Final	10	1	1
Radial Condition - Intermediate	0	0	0
Maximum Margin	1000	0	0
Protection Margin	0	0	0
Control Moves	10	0.1	10
Third-Party Equipment Control Moves	100	100	100
Losses	0.001	0	0.001
Protection Validation	10	10	0
Protection Reach	0	0	0
Voltage Equalization	0	0	0
Feeder Voltage Drop	0	0	0
Load Minutes Interrupted	0	0	0
Priority Customer Minutes Interrupted	0	0	0
Customer Minutes Interrupted	0	0	0
Momentary Frequency - MAIFI	0.01	0	0
Interruption Frequency - SAIFI	0.01	0	0
Interruption Duration - SAIDI	0.01	0	0
Maximum Number of Operations	1	1	1
Maximum Segment Margin	0	0	0
Non-Converged BLA Islands	10	10	10
Reverse Flow for Non-Reversible Recloser/Breaker	10	10	10

Performance Index	Reduce Overloads and Violations	Load Shedding	Return to Normal
Switching Validation Check - Ferroresonance	10	10	10
Switching Validation Check - Grounded Load	10	10	10
Switching Validation Check - Parallel Recloser	10	10	10

4.12.2.3. Weighting Factors for FSD

For the FSD application, weighting factors are used for ranking and ordering feeders in FSD Stage 1.

Table below lists the default problem formulations used by FSD, but other problem formulations can be easily created as needed.

Table 25. FSD Problem Formulations

Problem Formulation	Description
FSD Default	This problem formulation specifies that feeders with set fault indicators have a higher ranking so that they are processed first during FSD switching cycles.

The default weighting factors for the FSD problem formulations are listed in table below.

Table 26. Default Weighting Factors for FSD Problem Formulations

Weighting Factor	FSD Default
Fault Detector	100
Customer Priority	10
Number of Customers	10
Total kW load	10
Feeder Overhead Length	10

4.13. Problem Formulation Configuration

This section describes the configuration parameters for the problem formulations that set goals for the switching reconfiguration applications.

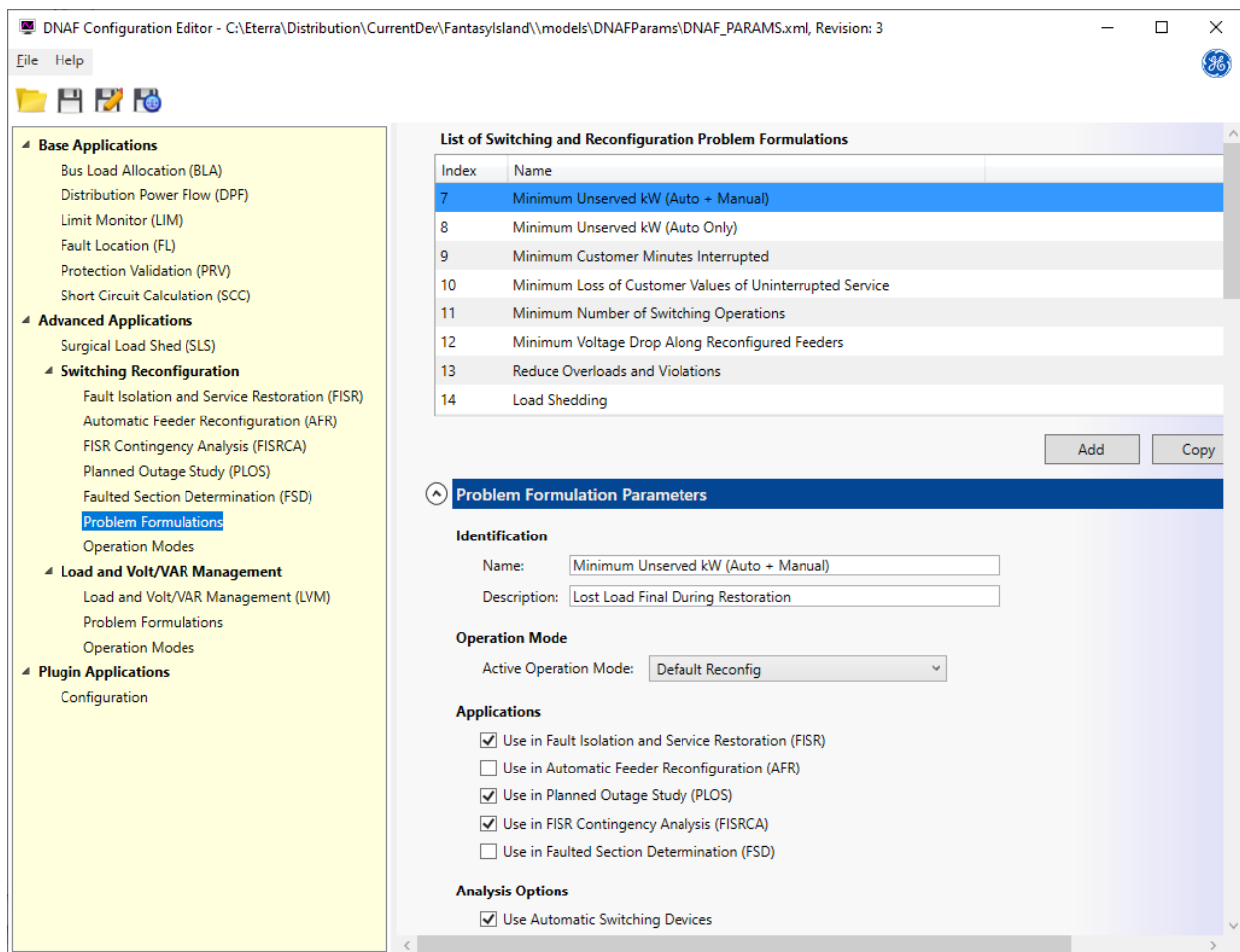


Figure 105. Problem Formulation - Reconfiguration Applications

Table below describes the parameters for the switching reconfiguration applications' problem formulations. The <Params xsi:type = "ProbFormulationParams"> element in the DNAF_PARAMS.xml configuration file contains all the listed parameters.

Table 27. Problem Formulation - Reconfiguration Applications

Configuration Editor UI Name	CSV File Field Name and XML File Attribute Name	Type	Description
Identification			
Index	INDEX_PROBFORM <NumberProb>	Integer	This table entry is generated automatically and specifies the index of the problem formulation. This index is used to refer to a specific problem in problem formulation schedules for stations.
Name	NAME_PROBFORM <Name>	Text box	This entry specifies the name of the problem formulation. The name is used in various displays such as the DNAF popup display and the DNAF System Configuration dialog box.
Description	DESC_PROBFORM <Desc>	Text box	This entry specifies an optional description for the problem formulation.

Configuration Editor UI Name	CSV File Field Name and XML File Attribute Name	Type	Description
Operation Mode			
Active Operation Mode	<ActiveOperationModeID>	Drop-down list	This selection specifies which operation mode is used for the current problem formulation. This allows each problem formulation to have its own defined plan execution parameters.
Applications			
Use in Fault Isolation and Service Restoration (FISR)	<UseInFISR>	Check box	This check box indicates if the problem formulation can be used by the FISR application.
Use in Automatic Feeder Reconfiguration (AFR)	<UseInAFR>	Check box	This check box indicates if the problem formulation can be used by the AFR application.
Use in Planned Outage Study (PLOS)	<UseInPLOS>	Check box	This check box indicates if the problem formulation can be used by the PLOS application.
Use in FISR Contingency Analysis (FISRCA)	<UseInFISRCA>	Check box	This check box indicates if the problem formulation can be used by the FISRCA application.
Use in Faulted Section Determination (FSD)	<UseInFSD>	Check box	This check box indicates if the problem formulation can be used by the FSD application.
Analysis Options			
Use Automatic Switching Devices	<UseAutomaticSwitches>	Check box	This check box indicates if automatic (remotely controllable) switching devices can be used by the reconfiguration applications.
Use Manual Switching Devices	<UseManualSwitches>	Check box	This check box indicates if manual switching devices can be used by the reconfiguration applications.
Use Temporary Cuts and Jumpers as Manual Switching Devices	<UseTemporaryCutAndJumperAsManualSwitch>	Check box	This check box specifies whether a switching reconfiguration application can consider temporary cuts and temporary jumpers as eligible manual switching devices. If the option is unchecked, temporary cuts and jumpers are not considered eligible manual switching devices.
Use Composite Switches	<UseCompositeSwitches>	Check box	This check box specifies whether a switching reconfiguration application can consider composite switches as eligible switching devices. If the option is unchecked, composite switches are not considered eligible switching devices.

Configuration Editor UI Name	CSV File Field Name and XML File Attribute Name	Type	Description
Use Switching Devices with Status Measurement Replaced in SCADA	<UseStatusMeasurementReplacedSwitches>	Check box	This check box indicates whether switching devices whose SCADA measurement values have been manually replaced can be used by the reconfiguration applications.
Include Plans with Violations	<IncludeSwitchingPlanWithViolations>	Check box	This check box specifies whether the reconfiguration application will generate plans with flow or voltage violations. If this option is checked, the reconfiguration application can generate plans that have flow or voltage violations.
Include Plans with Non-Converged or Scheduled Loading BLA Islands	<IncludeSwitchingPlanWithNonConvergedBLAIslands>	Check box	This check box specifies whether a switching reconfiguration application generates plans with non-converged or scheduled loading measurement islands involved in a switching plan. If this option is cleared and at least one measurement island is involved in a switching plan for which BLA is not converged or scheduled loading is used, this plan will not be generated.
Restore Automatic Switches First	<RestoreAutomaticSwitchesFirst>	Check box	This check box specifies whether FISR will first restore de-energized sections with SCADA controlled switching devices, and then restore de-energized sections only with manually operated switching devices. If this option is checked, FISR will first restore de-energized sections with SCADA controlled switching devices.
Return to Normal	<ReturnToNormal>	Check box	This check box specifies whether AFR is to return the system to its normal state. This option should only be set when using the AFR to generate a Return to Normal (RTN) switching plan.
Reduce Real Power Losses	<AFRTargetLossReduction>	Check box	This check box specifies whether AFR is to reduce real power losses utilizing a heuristic method. This option should only be selected when using AFR to generate a reduced real power loss switching plan.
Use Plans with Reverse Flow on Non-Reversible Recloser/Breaker	<PlansWithRevFlowNonRevDevice>	Check box	This check box specifies whether to allow steps with reverse flow on non-reversible devices.
Consider Cold Load Pickup	<ConsiderColdLoadPickup>	Check box	This check box specifies whether cold load pickup effect is considered in the reconfiguration applications.
Block DER Connection	<BlockDERConnection>	Check box	This check box specifies whether FISR generates plans that block DER connections. If this option is checked, FISR can generate plans to block DER connection for devices with size greater than the value specified in the Minimum DER (kVA) field.
Minimum DER Size (kVA)	<MinimumDERSize>	Check box	This entry specifies the minimum DER size in kVA to block DER connection. The allowable range is 10-10000.

Configuration Editor UI Name	CSV File Field Name and XML File Attribute Name	Type	Description
Use Network Reduction Algorithm	<UseNetworkReductionAlgorithm>	Check box	This check box specifies whether the FISIR application will internally use a reduced network model in the plan generation logic. If this option is checked, the FISIR network reduction algorithm will be executed.
Bound Protection Area by Fuses	<BoundProtectionAreaFuses>	Check box	This check box specifies whether the FISIR Protection Validation (PRV) zone is bounded by fuses. If this option is checked, the FISIR PRV analysis calculates the lowest fault current utilizing a network location within the breaker or recloser PRV zone; it does not evaluate locations downstream of a fuse. If this option is not checked, the lowest fault current network location could be downstream of a fuse.
Perform the Minimum Number of Fault Evaluations	<PerformMinFaultEvaluation>	Check box	This check box specifies whether the PRV analysis evaluates multiple PRV fault impedances (as in the base PRV solution) or simulates fewer faults. • If this option is selected, the maximum fault current is calculated with a bolted fault (zero impedance). The minimum phase/ground fault currents are calculated with the user-configured fault impedance (set on the Protection Validation page). • If this option is not selected, all fault currents (maximum and minimum phase/ground currents) are calculated twice using a bolted fault (zero impedance) and the user-configured fault impedance (set on the Protection Validation page). This check box is only enabled when the Fault Impedance Evaluation option on the PRV page is set to "Evaluate a Bolted Fault and the User Configured Fault Impedance".
Only Generate First Valid Plan	<OnlyGenerateFirstValidPlan>	Text box	This checkbox specifies whether FISIR generates only one valid plan and then stops plan generation. A valid plan is a "green" plan (no overloads or voltage violations) that restores the maximum number of customers (all customers in the healthy network sections).

Configuration Editor UI Name	CSV File Field Name and XML File Attribute Name	Type	Description
Enable Tap Emulation Logic for Telemetered Regulators and LTCs	<EnableTapEmulation Logic>	Check box	This check box determines whether reconfiguration applications treat SCADA-enabled LTCs/regulators that have valid tap position measurements as devices that have a fixed tap position (using the most recently set telemetered or manually entered value) or as devices that are regulated according to their internal controller settings, including voltage target, voltage bandwidth, Line Drop Compensation (LDC) resistance and reactance, tap step increment, and voltage reduction. • If unchecked, the SCADA-enabled regulators and LTCs with valid tap position measurements are treated as having a fixed tap position and are not regulated during the internal power flow analysis executed within the switching reconfiguration plan generation process unless they are blocked. • If checked, the SCADA-enabled regulators and LTCs with valid tap position measurements are regulated during the internal power flow analysis executed within the switching reconfiguration plan generation process unless they are blocked.
Prevent Closed Loop Execution if an Additional Switching Violation Exists	<PreventClosedLoopExecution OnSwitchingValidationViolations>	Check box	This check box specifies whether the plans generated will be implemented in Closed Loop mode if there are switching reconfiguration advanced application switching validation check violations. The evaluated checks are enabled with the following Problem Formulation options: Enable Ferroresonance Check, Enable 3-Wire/4-Wire Check, and Enable Paralleling Circuits with Reclosing Capable Protective Devices Check. If this option is checked, valid plans (no overloads and no high/low voltage violations) will be implemented in closed loop if the enabled checks are not violated.
Enable Ferroresonance Switching Check	<FerroresonanceSwitching ValidationCheckEnabled>	Check box	This check box specifies whether the ferroresonance switching check is enabled. If checked, this option will impact both the Prevent Closed Loop Execution if an Additional Switching Violation Exists option, and the calculation of the weighted Switching Validation Check - Ferroresonance Performance Index.
Enable 3-Wire/4-Wire Check	<GroundedLoadNeutral SwitchingValidationCheck Enabled>	Check box	This check box specifies whether the 3-Wire/4-Wire check is enabled. If checked, this option will impact both the Prevent Closed Loop Execution if an Additional Switching Violation Exists option, and also the calculation of the weighted Switching Validation Check - 3-Wire/4-Wire Performance Index.
Enable Paralleling Circuits with Reclosing Capable Protective Devices Check	<ParallelRecloserSwitching ValidationCheckEnabled>	Check box	This check box specifies whether the paralleling circuits with reclosing capable protective devices check is enabled. If checked, this option will impact both the Prevent Closed Loop Execution if an Additional Switching Violation Exists option, and the calculation of the weighted Switching Validation Check - Parallel Recloser Performance Index.

Configuration Editor UI Name	CSV File Field Name and XML File Attribute Name	Type	Description
Close Eligible Bus Tie Switching Device	<CloseEligibleBusTieSwDeviceCheckEnabled>	Check box	This check box indicates if an eligible open tie switching device between two buses at the low voltage side of two power transformers in the substation needs to be closed prior to closing an eligible tie switching device between two feeders that is downstream of the two power transformers.
Maximum Number of Switching Steps	MAXSWITCHOPERATIONS_PROBFORM <MaxNoSwitching>	Integer	This value specifies the maximum number of switching steps allowed in any plan. This parameter applies to all AFR problem formulations except the Reduce Overloads and Violations problem formulation. The allowable range is 1-40.
Adaptive Relaying			
Adaptive Relaying	<AdaptiveRelayingSelectionMethod>	Option Button	This setting specifies how the trip setting evaluation is performed. If Compare Phase Trip Setting with the Minimum Phase-to-Phase Fault Current is selected, the trip setting evaluation is performed through comparing phase trip setting with the phase-to-phase fault current. If Compare Phase Trip Setting with the Minimum of the Minimum Phase-to-Phase and the Minimum Phase-to-Ground Fault Current Values is selected, the trip setting evaluation is performed through compare phase trip setting with the minimum of the phase-to-phase and the phase-to-ground fault current values.
Third-Party Owned Equipment Options			
Include as Customer-Owned, and Respect Plan Execution Mode	<UseThirdPartyEquipment>	Option Button	If this option is selected, third-party owned equipment is treated as customer owned equipment and can be included in reconfiguration plans. The plan execution for plans with third-party equipment is not changed. (The execution mode is determined by settings in the DNAF Configuration Editor or DNAF System Configuration display.)
Include as Customer-Owned, but Revert Plan Execution Mode to Semi-Automatic/Advisory	<UseThirdPartyEquipment>	Option Button	If this option is selected, third-party owned equipment is treated as customer owned equipment and can be included in reconfiguration plans. The plan execution mode for plans with third-party equipment is changed as follows: • If the plan execution mode was "Closed Loop" or "Closed Loop – Calculated Voltage Limit Violation Not Worse", the plan execution mode automatically changes to "Semi-Automatic". • If the plan execution mode was "Semi-Automatic", the plan execution mode automatically changes to "Advisory".
Do Not Include Third-Party Owned Equipment	<UseThirdPartyEquipment>	Option Button	If this option is selected, third-party owned equipment cannot be included in reconfiguration plans.

Configuration Editor UI Name	CSV File Field Name and XML File Attribute Name	Type	Description
Voltage Limits			
Respect Bus Voltage Limits	<EnforceBusVoltage Limits>	Drop-down list	This selection specifies which limit type is to be enforced for buses: • None: The voltage limits are not enforced. • Normal Operational: The normal operational limits are enforced. • Normal Optimization: The normal optimization limits are enforced. • Emergency Operational: The emergency operational limits are enforced. • Emergency Optimization: The emergency optimization limits are enforced.
Respect Load Voltage Limits	<EnforceLoadVoltage Limits>	Drop-down list	This selection specifies which limit type is to be enforced for loads: • None: The voltage limits are not enforced. • Normal Operational: The normal operational limits are enforced. • Normal Optimization: The normal optimization limits are enforced. • Emergency Operational: The emergency operational limits are enforced. • Emergency Optimization: The emergency optimization limits are enforced.

Configuration Editor UI Name	CSV File Field Name and XML File Attribute Name	Type	Description
Violation Not Worse Margin (%)	<VoltageViolationNotWorseMarginPercent>	Text box	This entry specifies the voltage margin in percentage to decide if a plan does not make any calculated voltage limit violations worse. For a high voltage violation, if the calculated post-plan voltage is lower than the pre-plan calculated voltage plus this margin in percent voltage, the plan is considered not to make this calculated voltage limit violation worse. For example, if this entry is set as 3%, that is 0.03 per unit. The high voltage limit is 1.05 per unit, the post-plan voltage is 1.09 per unit, and the pre-plan voltage is 1.07 per unit. The plan is considered not to make this calculated voltage limit violation worse since $1.09 < 1.07 + 0.03$. For a low voltage violation, if the calculated post-plan voltage is higher than the pre-plan calculated voltage minus this margin in percent voltage, the plan is considered not to make this calculated voltage limit violation worse. For example, if this entry is set as 3%, that is 0.03 per unit. The low voltage limit is 0.95 per unit, the post-plan voltage is 0.92 per unit, and the pre-plan voltage is 0.93 per unit. The plan is considered not to make this calculated voltage limit violation worse since $0.92 > 0.93 - 0.03$. The allowable range is 0-10.
Branch Flow Limits			
Respect Limit	<EnforceBranchFlowLimits>	Drop-down list	This selection specifies which branch flow limit level is to be enforced for branch objects. • None: Flow limits are not enforced. • Level 1: The most restrictive limit level. • Level 2: Less restrictive than level 1.
Branch Flow Limit for DER Delay	<BranchFlowLimitforDERDelay>	Drop-down list	This selection specifies which flow limit type is to be enforced for DERs/DGs for which the re-energization delay specified by the DG/DER Delay Time parameter expired: • None: Flow limits are not enforced. • Level 1: The most restrictive limit level. • Level 2: Less restrictive than level 1.
Allow Post-Plan Violations if not Worse than Pre-Plan Value	<AllowFlowNotWorse>	Check box	This check box specifies whether to allow the plan state to be Feasible if post-plan branch flow violations are not worse compared to the pre-plan violations (taking into account the percentage defined using the Violation not Worse Margin parameter). If selected, the branch limit violation which is not worse than the pre-plan violation is not considered, and the plan state is set to Feasible. If not selected, the branch limit violation which is not worse than the pre-plan violation is considered, and the plan state is set to Feasible - Has Violation.

Configuration Editor UI Name	CSV File Field Name and XML File Attribute Name	Type	Description
Violation not Worse Margin	<FlowViolationNot WorseMarginPercent>	Text box	This entry specifies the flow margin in percentage to decide if a plan does not make any flow limit violations worse. For a flow violation, if the post-plan branch flow is lower than the pre-plan branch flow plus this margin, the plan is considered not to make this branch flow limit violation worse. The allowable range is 0-10.
Backfeed			
Enforce Station Transformer Backfeed Limit	<EnforceTransformer Backfeed>	Check box	This check box indicates that reconfiguration plans that result in reverse real power flow through a station power transformer should be flagged as containing violations. This will prevent the plan being implemented in closed-loop mode and the plan will be flagged as containing violations in the plan tabular output.
Injection Scaling			
DG/DER Injection Scaling Factor	<DERInjectionScaling Factor>	Check box	This entry specifies the "confidence" parameter for scaling up or down the DER component of the load for the de-energized section that is to be restored. The allowable range is 0.1-5.
Reserved Capacity			
Adjust Limits Based on Reserved Capacity	<AdjustLimForReserved Capacity>	Check box	This check box specifies whether reserved capacity commitments are used to adjust limits in the switching reconfiguration applications using the selected problem formulation.
Only Consider Firm Commitments	<ReservedCapacityConsider OnlyFirmCommitments>	Check box	This check box specifies whether only firm reserved capacity commitments are considered for adjusted limits in the switching reconfiguration applications using the selected problem formulation. This check box is effective only if the option "Adjust Limits Based on Reserved Capacity" is checked.
Limits			
Branch Flow	<FlowAppLimitEnforce>	Text box	This entry specifies the percentage of the original branch flow limit that is considered in the switching reconfiguration applications. The default value of 100% means that the original model specified limit is used. The flow limits are set in the Limit Strategy section using the Branch Flow Limits Respect Limit control. The allowable range is 0-200.

Configuration Editor UI Name	CSV File Field Name and XML File Attribute Name	Type	Description
Delta High Voltage	<DeltaVoltageHighAppLimitEnforce>	Text box	This entry specifies the increase of the original high voltage limit for a load or a bus in percent of per unit voltage for the switching reconfiguration application. The original high voltage limit is set using Respect Bus Voltage Limits or Respect Load Voltage Limits fields in the Limit Strategy section. The adjusted high voltage limit is the original high voltage plus Delta High Voltage/100 per unit. The allowable range is 0-10.
Delta Low Voltage	<DeltaVoltageLowAppLimitEnforce>	Text box	This entry specifies the decrease of the original low voltage limit for a load or a bus in percent of per unit voltage for the switching reconfiguration application. The original low voltage limit is set using the Respect Bus Voltage Limits or Respect Load Voltage Limits fields in the Limit Strategy section. The adjusted low voltage limit is the original low voltage minus Delta Low Voltage/100 per unit. The allowable range is 0-10.
Protection			
Peak Load Safety Factor	<PeakLoadBuffer>	Text box	This entry specifies the peak load safety factor (scalar multiplier). It is a multiplier of the peak load, which provides the minimum load to be considered when selecting the appropriate protection settings group. The allowable range is 0.1-10.
Reach and Neutral Safety Factor	<ReachBuffer>	Text box	This entry specifies the reach and neutral safety factor. This safety factor is used to determine the minimum phase pick-up limit, which is used to find the appropriate protection settings group. This safety factor is also used to validate the neutral relay current protection status. The allowable range is 0.1-10.
Ground Safety Factor	<GroundBuffer>	Text box	This entry specifies the Ground Safety Factor. This safety factor is used to determine the minimum phase ground pick-up limit which is used to find the appropriate protection settings group. The allowable range is 0.1-10.
Performance Index Weights			
Flow Violations (Final Step)	PERFFLOWFINAL_PROBFORM <PIFlowViolationFinal>	Text box	This entry specifies the weight of the Flow Violations Performance Index (PI). This weight only applies for the final solutions. Valid values are real numbers starting from 0. For more information see Linear Programming, Problem Formulation Parameter Tuning, and Problem Formulations. The allowable range is 0-1000000000.

Configuration Editor UI Name	CSV File Field Name and XML File Attribute Name	Type	Description
Flow Violations (Intermediate Steps)	PERFFLOWTEMP_PROBFORM <PIFlowViolationInt>	Text box	This entry specifies that weight of the Flow Violations Performance Index (PI). This weight is used for the PI only during the intermediate switching state/solution. Valid values are real numbers starting from 0. For more information see Linear Programming, Problem Formulation Parameter Tuning, and Problem Formulations. The allowable range is 0-1000000000.
Bus Voltage Violations (Final Step)	PERFVOLTFINAL_PROBFORM <PIVoltageViolationFinal>	Text box	This entry specifies the weight of the Voltage Violations Performance Index (PI). This weight is applicable only for bus voltage violations in the final solution. Valid values are real numbers starting from 0. For more information see Linear Programming, Problem Formulation Parameter Tuning, and Problem Formulations. The allowable range is 0-1000000000.
Bus Voltage Violations (Intermediate Steps)	PERFVOLTTEMP_PROBFORM <PIVoltageViolationInt>	Text box	This entry specifies the weight of the Voltage Violations Performance Index (PI). This weight is applicable only for bus voltage violations in the intermediate switching state/solution. For more information see Linear Programming, Problem Formulation Parameter Tuning, and Problem Formulations. The allowable range is 0-1000000000.
Load Voltage Violations (Final Step)	<PILoadVoltageViolationFinal>	Text box	This entry specifies the weight of the Voltage Violations Performance Index (PI). This weight is applicable only for load voltage violations in the final solution. The allowable range is 0-1000000000.
Load Voltage Violations (Intermediate Steps)	<PILoadVoltageViolationInt>	Text box	This entry specifies the weight of the Voltage Violations Performance Index (PI). This weight is applicable only for load voltage violations in the intermediate switching state/solution. The allowable range is 0-1000000000.
Total Cost (Final Step)	PERFCOSTFINAL_PROBFORM <PITotalCostFinal>	Text box	This entry specifies the weight of the final step total cost performance index. The allowable range is 0-1000000000.
Total Cost (Intermediate Steps)	PERFCOSTTEMP_PROBFORM <PITotalCostInt>	Text box	This entry specifies the weight of the Intermediate total cost Performance Index (PI). The allowable range is 0-1000000000.
Lost Load (Final Step)	PERFLOSTLOADFINAL_PROBFORM <PILostLoadFinal>	Text box	This entry specifies the weight of the final step lost load Performance Index (PI). The allowable range is 0-1000000000.

Configuration Editor UI Name	CSV File Field Name and XML File Attribute Name	Type	Description
Lost Load (Intermediate Steps)	PERFLOSTLOADTEMP_PROBFORM <PILostLoadInt>	Text box	This entry specifies the weight of the Intermediate lost load Performance Index (PI). The allowable range is 0-1000000000.
Dropped Customer Count (Final Step)	PERFDROPCUSTOMERFINAL_PROBFORM <PIDroppedCustomerCount>	Text box	This entry specifies the weight of the final number of the dropped customers count Performance Index (PI). The allowable range is 0-1000000000.
Dropped Customer Count (Intermediate Steps)	PERFDROPCUSTOMERTEMP_PROBFORM <PIDroppedCustomerCountInt>	Text box	This entry specifies the weight of the intermediate step number of the dropped customers count Performance Index (PI). The allowable range is 0-1000000000.
Dropped Customer Priority (Final Step)	PERFDROPPRIORITYFINAL_PROBFORM <PIDroppedCustomerPriorityFinal>	Text box	This entry specifies the weight of the final step dropped customer priority Performance Index (PI). Priority is specified as Value of interrupted service in \$/kW. The allowable range is 0-1000000000.
Dropped Customer Priority (Intermediate Steps)	PERFDROPPRIORITYTEMP_PROBFORM <PIDroppedCustomerPriorityInt>	Text box	This entry specifies the weight of the intermediate step dropped customer priority Performance Index (PI). Priority is specified as value of interrupted service in \$/kW. The allowable range is 0-1000000000.
Radial Condition (Final Step)	PERFRADIALFINAL_PROBFORM <PIRadialControl>	Text box	This entry specifies the weight for the performance index for the number of loops in the network selection after the plan is executed. For more information see Problem Formulations for more information. The allowable range is 0-1000000000.
Radial Condition (Intermediate Steps)	PERFRADIALTEMP_PROBFORM <PIRadialInt>	Text box	This entry specifies the weight for the performance index for the number of energization loops in the network selection after the intermediate steps of the plan are executed. For more information see Problem Formulations for more information. The allowable range is 0-1000000000.
Maximum Margin	PERFMAXMARGINFINAL_PROBFORM <PIMaxMargin>	Text box	This entry specifies the weight for the performance index for the final step maximum margin. For more information see Problem Formulations for more information. The allowable range is 0-1000000000.

Configuration Editor UI Name	CSV File Field Name and XML File Attribute Name	Type	Description
Protection Margin	<PIProtectionMargin>	Text box	This entry specifies the weight for the protection margin performance index, which considers the distance from the current flow to the flow limit. For more information see Problem Formulations for more information. The allowable range is 0-10000000000.
Control Moves	PERFCONTROLMOVESFINAL_PROBFORM <PINoControlMove>	Text box	This entry specifies the weight of the final step number of control moves Performance Index (PI). For more information see Problem Formulations for more information. The allowable range is 0-10000000000.
Third-Party Owned Equipment Control Moves	<PIThirdPartyControl Move>	Text box	This entry specifies the weight of the final step number of control moves involving third-party equipment Performance Index (PI). For more information see Problem Formulations for more information. The allowable range is 0-10000000000.
Losses	PERFLOSSESFINAL_PROBFORM <Losses>	Text box	This entry specifies the weight of the final step losses Performance Index (PI). For more information see Problem Formulations for more information. The allowable range is 0-10000000000.
Protection Validation	PERFPROTECT_PROBFORM <PIFaultProtection>	Text box	This entry specifies the weight of the performance index for the final step execution of the Protection Validation function (for the top-ranked plans). For more information see Problem Formulations for more information. The allowable range is 0-10000000000.
Protection Reach	<PIProtectionReach>	Text box	This entry specifies the weight for the protection reach performance index, which checks whether a protection settings group can satisfy the minimum fault current. For more information see Problem Formulations for more information. The allowable range is 0-10000000000.
Voltage Equalization	PERFCRITVOLT_PROBFORM <PIVoltageEqualization>	Text box	This entry specifies the weight of the performance index for the final step equalization of critical voltages (maximize tap range). For more information see Problem Formulations for more information. The allowable range is 0-10000000000.

Configuration Editor UI Name	CSV File Field Name and XML File Attribute Name	Type	Description
Feeder Voltage Drop	PERFFVOLTDROP_PROBFORM <PIFeederVoltageDrop>	Text box	This entry specifies the weight of the performance index, defined by the largest voltage drop among all feeders. For more information see Problem Formulations for more information. The allowable range is 0-1000000000.
Load Minutes Interrupted	PERFLMI_PROBFORM <PILMI>	Text box	This entry specifies the weight of the LMI performance index. For more information see Problem Formulations for more information. The allowable range is 0-1000000000.
Priority Customer Minutes Interrupted	PERFPRIORITYCMI_PROBFORM <PIPRIORITYCMI>	Text box	This entry specifies the weight of the priority Customer Minutes Interruption (CMI) performance index. For more information see Problem Formulations for more information. The allowable range is 0-1000000000.
Customer Minutes Interrupted	PERFCMI_PROBFORM <PICMI>	Text box	This entry specifies the weight of the CMI performance index. For more information see Problem Formulations for more information. The allowable range is 0-1000000000.
Momentary Frequency - MAIFI	PERFMAIFIFINAL_PROBFORM <PIMAIFI>	Text box	This entry specifies the weight of the performance index for the final step system Momentary Average Interruption Frequency Index (MAIFI). See Reliability Index Calculations for more information. The allowable range is 0-1000000000.
Interruption Frequency - SAIFI	PERFSAIFIFINAL_PROBFORM <PISAIFI>	Text box	This entry specifies the weight of the performance index for the final System Average Interruption Frequency Index (SAIFI) step. For more information see Reliability Index Calculations for more information. The allowable range is 0-1000000000.
Interruption Duration - SAIDI	PERFSAIDIFINAL_PROBFORM <PISAIDI>	Text box	This entry specifies the weight of the performance index for the final System Average Interruption Duration Index (SAIDI) step. For more information see Reliability Index Calculations for more information. The allowable range is 0-1000000000.

Configuration Editor UI Name	CSV File Field Name and XML File Attribute Name	Type	Description
Maximum Number of Operations	MAXSWITCHCONTROL_PROBFORM <MaxNoOperation>	Text box	This entry specifies the weight of the performance index for the maximum number of switching operations for any switching device in a plan. This parameter applies to all AFR problem formulations except the Reduce Overloads and Violations problem formulation. The allowable range is 0-1000000000.
Maximum Segment Margin	<PIMaxSegmentLoad>	Text box	This entry specifies the weight of the performance index for the maximum segment loading in a plan. This PI is useful for ranking plans that do not have flow violations: the least loaded plan is ranked higher. The PI is zero if the flow limit is not enforced. The allowable range is 0-1000000000.
Non-Converged or Scheduled Loading BLA Islands	<PINonConvergedBLA Islands>	Text box	This entry specifies the weight of the non-converged or scheduled loading BLA islands Performance Index (PI) term. For more information see Problem Formulations. The allowable range is 0-1000000000.
Reverse Flow for Non-Reversible Recloser/Breaker	<RevFlowNonRev DevicePI>	Text box	This entry specifies the weight of the non-reversible devices with reverse flow Performance Index (PI) term. For more information see Problem Formulations. The allowable range is 0-1000000000.
Switching Validation Check - Ferroresonance	<Ferroresonance SwitchingValidationPI>	Text box	This entry specifies the weight of a switching step with the "Ferroresonance switching conditions" switching violation. This performance index is only applied if the Enable Ferroresonance Switching Check option is selected and the switching validation check Severity is set to Warning or Prohibit in the Check Before Operate tab in the Viewer Configuration Editor. For more information, see Problem Formulations. The allowable range is 0-1000000000.
Switching Validation Check - 3-Wire/4-Wire	<GroundedLoad SwitchingValidationPI>	Text box	This entry specifies the weight of a switching step with the "Feeding single phase-ground loads from 3 or 4 wire feeder sections" switching violation. This performance index is only applied if the Enable 3-Wire/4-Wire Check option is selected and the switching validation check Severity is set to Warning or Prohibit in the Check Before Operate tab in the Viewer Configuration Editor. For more information, see Problem Formulations. The allowable range is 0-1000000000.

Configuration Editor UI Name	CSV File Field Name and XML File Attribute Name	Type	Description
Switching Validation Check - Parallel Recloser	<ParallelRecloser SwitchingValidationPI>	Text box	This entry specifies the weight of a switching step with the "Paralleling circuits with reclosing capable protective devices" switching violation. This performance index is only applied if the Enable Paralleling Circuits with Reclosing Capable Protective Devices Check option is selected and the switching validation check Severity is set to Warning or Prohibit in the Check Before Operate tab in the Viewer Configuration Editor. For more information, see Problem Formulations. The allowable range is 0-100000000.
Plan Parameters			
Maximum Plan Duration	<MaximumSwitching PlanDuration>	Text box	This entry specifies the maximum time (in minutes) needed to execute a reconfiguration plan. It is used for the loads that the reconfiguration application cannot restore to calculate several performance indices including CMI, LMI, and Priority CMI. The allowable range is 1-480.
Manual Switch Operation Time	<ManualSwitchOperation Time>	Text box	This entry specifies the time (in minutes) to operate a manually operated switching device. It is used to estimate the time to restore the load for a switching plan, which in turn is used to calculate several performance indices including CMI, LMI, and Priority CMI. The allowable range is 1-240.
Automatic Switch Operation Time	<AutomaticSwitch OperationTime>	Text box	This entry specifies the time (in minutes) to operate a SCADA-controllable switching device. It is used to estimate the time to restore the load for a switching plan, which in turn is used to calculate several performance indices including CMI, LMI, and Priority CMI. The allowable range is 0-60.
Switching Cost			
Number of Adjacency Levels	<NumberOfAdjacency Levels>	Text box	This entry specifies the number of feeder or station adjacency levels used to determine the network selection for the switching reconfiguration applications. The number of levels determines the number of columns that are enabled in the Feeder Adjacency Levels grid box. The maximum adjacency level of 2 is allowed. The allowable range is 1-2.
Feeder Adjacency Levels	<FeederAdjacencyLevels>	Text box	This entry is the feeder or station adjacency level, depending on the network selection method. This value is shown based on the Number of Adjacency Levels field. The allowable range is 1-2.

Configuration Editor UI Name	CSV File Field Name and XML File Attribute Name	Type	Description
Manual Switch Cost	<ManualSwitchCost>	Text box	This entry is the relative switching cost for manual switching devices at a given adjacency level. The allowable range is 0-200.
Automatic Switch Cost	<AutomaticSwitchCost>	Text box	This entry is the relative switching cost for automated switching devices at a given adjacency level. The allowable range is 0-200.
Faulted Section Determination Weighting Factors			
Fault Detector	FAULTDETECTOR_PROBFORM <FaultDetector>	Text box	This entry is the weighting factor for set fault indicators when estimating the feeder island ranking. It is recommended to set this weighting factor higher than the other weighting factors because feeders that have set fault indicators are the most probable faulted feeders, and therefore they should be first in the ranked list. The allowable range is 0-1000.
Customer Priority	CUSTOMERPRIORITY_PROBFORM <CustomerPriority>	Text box	This entry is the weighting factor for priority customers when estimating the feeder island ranking. It is recommended to set this weighting factor to a reasonably low value so that the feeders with highest priority customers are tested last in the list because the test involves disconnecting those customers to check for fault presence. The allowable range is 0-100.
Number of Customers	NUMBEROFCUSTOMERS_PROBFORM <NumberOfCustomers>	Text box	This entry is the weighting factor for the total number of customers when estimating the feeder island ranking. The allowable range is 0-100.
Total kW load	TOTALKWLOAD_PROBFORM <TotalKWLoad>	Text box	This entry is the weighting factor for the total load in kW when estimating the feeder island ranking. The allowable range is 0-100.
Feeder Overhead Length	FEEDEROVERHEAD_LENGTH_PROBFORM <FeederOverhead Length>	Text box	This entry is the weighting factor for the overhead line length when estimating the feeder island ranking. It is recommended to set this weighting factor to a reasonably high value because the longer overhead feeders are most subject to fault and should be ranked higher. The allowable range is 0-100.
Old and Deprecated Parameters			
Index	INDEX_LIMSTRAT	Integer	This field is deprecated.
Name	NAME_LIMSTRAT	String	This field is deprecated.
Use Normal Voltage Limit Level	USENORMVLTLIM_LIMSTRAT	Boolean	This field is deprecated.

Configuration Editor UI Name	CSV File Field Name and XML File Attribute Name	Type	Description
Use Emergency Voltage Limit in Intermediate Steps	USEEMRGLIMINTSTEP_LI MSTRAT	Boolean	This field is deprecated.
Flow Limit Level	FLOWLIMLEVEL_LIMSTRA T	Integer	This field is deprecated.
Sumflow: Final	PERFSUMFLOWFINAL_PR OBFORM	Text box	This field is deprecated.
Sumflow: Intermediate	PERFSUMFLOWTEMP_PR OBFORM	Text box	This field is deprecated.

4.14. Operation Modes Configuration

This section describes plan execution configuration parameters for the reconfiguration applications.

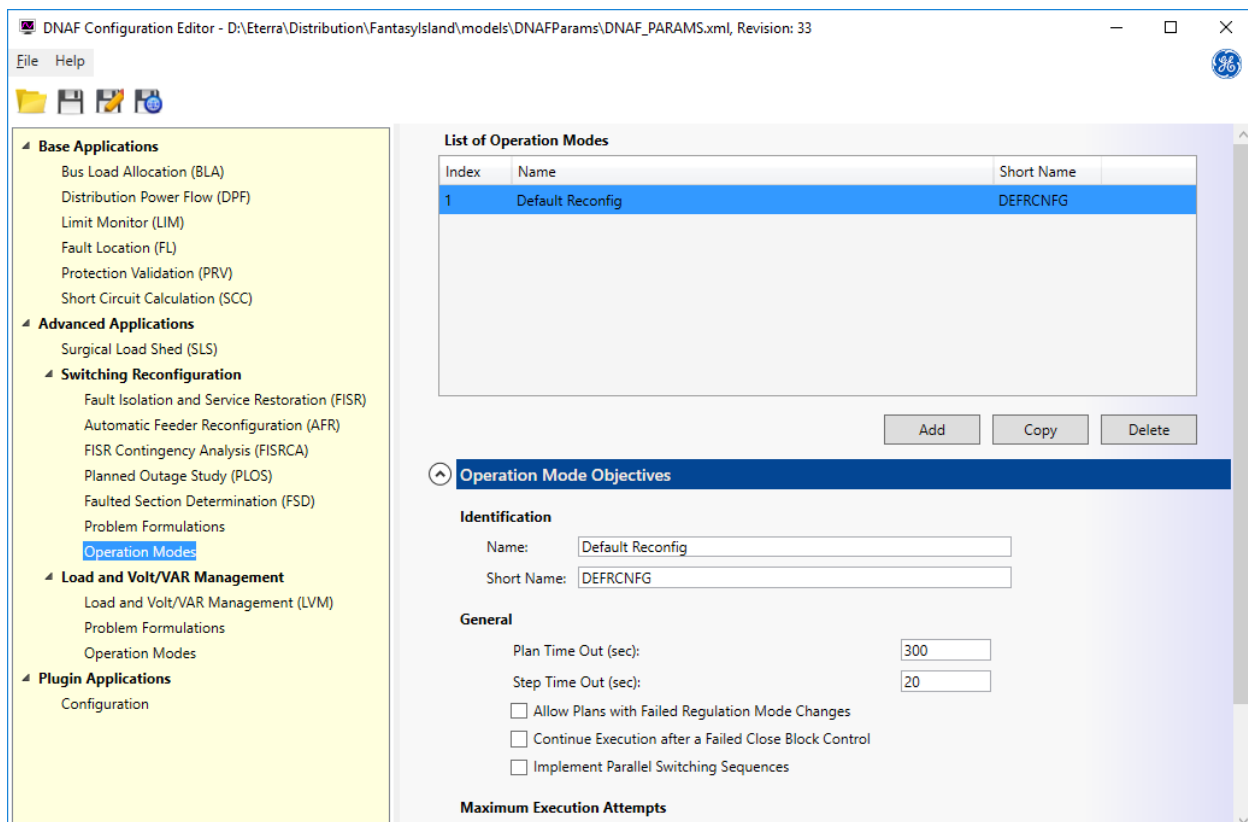


Figure 106. Operation Modes Settings - Reconfiguration Applications

Table below lists the Operation Modes configuration parameters. All listed parameters are contained by the <Params xsi:type = "OperationModeParams"> element in the DNAS_PARAMS.xml

configuration file.

Table 28. Operation Modes Configuration Parameters

Configuration Editor UI Name	CSV File Field Name & XML File Attribute Name	Type	Description
Identification			
Index	<ModeID>	Integer	This table entry is generated automatically and specifies the index of the operation mode. This index is used to refer to a specific operation mode for the problem formulation.
Short Name	<Acronym>	Text box	This entry specifies an optional description for the operation mode.
Name	<Name>	Text box	This entry specifies the name of the operation mode. This name is used on the Problem Formulations tab to select the operation mode for a problem formulation.
General			
Plan Time Out	CLOSELOOPTIME OUTTOL_VOLTMAN <PlanTimeOut>	Text box	This entry specifies the timeout tolerance, in seconds, for closed loop controls for an advanced application. If a status confirmation for all controls in a plan has not been received by the time specified in this text box, then the plan is considered to have failed and is removed. The value 0 means the wait time is not checked and the plan waits indefinitely for a status confirmation to come back. The default value is 200 seconds. The allowable range is 0-600.
Step Time Out	CLOSELOOPSTEPTIME OUTTOL_VOLTMAN <StepTimeOut>	Text box	This entry specifies the default waiting time, in seconds, before the plan manager can consider a single control to have timed out. This value is used only when the timeout is not defined for the associated control measurement. If any step in a plan times out, the plan is considered to have failed and is removed. If the timeout is 0, the plan manager will wait indefinitely. The default value is 20 seconds. The allowable range is 0-300.

Configuration Editor UI Name	CSV File Field Name & XML File Attribute Name	Type	Description
Allow Plans with Failed Regulation Mode Changes	<AllowFailedRegulatorModeChange>	Check box	This check box indicates whether the FISR plan is executed even if regulator mode cannot be changed and/or correctly confirmed. The regulator mode change logic is configured using the Change Voltage Regulation Modes option in the FISR page.
Continue Execution after a Failed Close Block Control	<ExecuteFailedBlockCloseControl>	Check box	This check box indicates if the FISR plan execution will continue if a block DER connection control cannot be applied or correctly confirmed
Implement Parallel Switching Sequences	<ImplementParallelSwitchingSequences>	Check box	This checkbox specifies whether FISR internally applies the parallel switching sequence functionality in the plan generation and plan execution logic. (Plans are split into sub-plans and executed in parallel.) When a plan is generated, parallel switching sequences with independent switching steps are determined by the network topology and stored as individual sub-plans. When the plan is sent to the plan manager, the plan is separated into sub-plans, which are executed independently in parallel. Any failed step in a sub-plan has no impact on execution of other sub-plans.
Maximum Execution Attempts			
Closing Switching Devices	<MaximumAttemptsForSwitchingDevicesClose>	Text box	This entry specifies the maximum number of attempts to send out a SCADA control to close switching devices. If the total number of attempts for a switching device in a plan step reaches this value but the control action cannot be confirmed, the plan is considered as failed and an alarm is optionally issued. The allowable range is 1-10.
Opening Switching Devices	<MaximumAttemptsForSwitchingDevicesOpen>	Text box	This entry specifies the maximum number of attempts to send out a SCADA control to open switching devices. If the total number of attempts for a switching device in a plan step reaches this value but the control action cannot be confirmed, the plan is considered as failed and an alarm is optionally issued. The allowable range is 1-10.

Configuration Editor UI Name	CSV File Field Name & XML File Attribute Name	Type	Description
Settings Groups	<MaximumAttemptsForSettingsGroup>	Text box	This entry specifies the maximum number of attempts to send out a SCADA control for protection settings group. If the total number of attempts for a given protective device in a plan reaches this value but the control action cannot be confirmed, the plan is considered as failed and an alarm is optionally issued. The allowable range is 1-10.
Control Monitoring			
Wait for Analog Delay	<WaitForAnalogDelay>	Text box	This entry specifies the waiting time, in seconds, after a command step is sent before evaluating the system state. This delay allows SCADA values to be updated after a control is sent, which may affect flows and voltages. This delay should be long enough to allow the SCADA measurements to be refreshed but shorter than the plan step timeout value. The allowable range is 0-120.

4.15. Plan Execution Mode

The effective plan execution mode for a switching reconfiguration application is determined by the several factors. The following settings are considered in the decision process:

- The default plan execution mode setting in the DNAF Configuration Editor.
- Individual feeder-level plan execution mode settings on the DNAF System Configuration display. The user can specify plan execution modes individually at the feeder level.
- The Use Triggering Feeder Plan Execution Mode option in the DNAF Configuration Editor, which indicates whether only the triggering feeder's plan execution mode should be used.

The settings on the DNAF System Configuration display override the default plan execution mode setting from the DNAF Configuration Editor. The plan execution mode displayed at the system, the station, or the power transformer level reflects the downstream feeder level settings.

For example, if all feeders downstream of a bank are set to "Advisory", the bank level setting is shown as "Advisory". If one feeder is set to "Advisory" and another feeder from the same transformer bank is set to "Closed Loop", the bank level setting is shown as "mixed".

In the DNAF System Configuration display, you can change the plan execution mode at the feeder level only. The settings at the transformer and upper levels depend on feeders' settings and show Mixed if feeders contain different settings.

After a switching reconfiguration plan is generated, the Plan Manager subsystem determines the most restrictive plan execution mode based on the solved network, the Use Triggering Feeder Plan Execution Mode option, and the specified plan execution mode fields on the DSC. The most

restrictive plan execution mode order is Advisory, followed by Semi-Automatic (FISR and AFR only), followed by Closed Loop.

The Use Triggering Feeder Plan Execution Mode options in the FISR and AFR Plan Execution subsections the DNAF Configuration Editor specify how the FISR and AFR plan execution mode is selected when there is a conflict. If this option is set, the plan execution mode is based only on the triggering dynamic (energizing) feeder plan execution mode. For the small system shown in figure below, the following logic is used if this option is set:

- When a fault happens on Feeder B, all generated switching plans will not switch automatically (Semi-Automatic or Advisory).
- When a fault happens on Feeder A, all generated switching plans will switch automatically (Closed Loop), including any plans that use the tie switch (N.O. Tie), even though Feeder B is set to Open Loop.

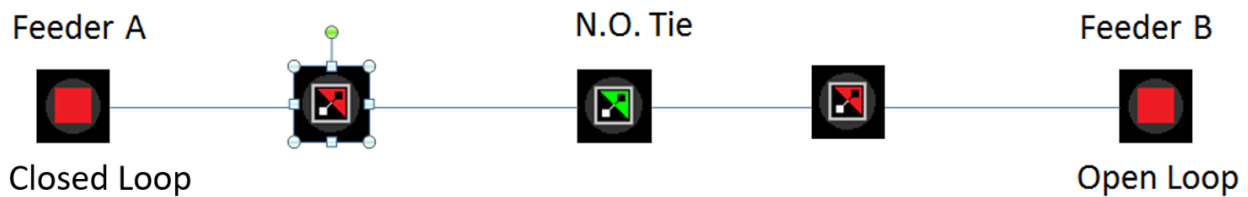


Figure 107. FISR Triggering Feeder Plan Execution

For the FISR application, if this DCE option is set but a triggering feeder cannot be found (such as with a station internal fault), the most restrictive mode from all the station's feeders is used.

For the AFR application, if this DCE option is set but the triggering device is a station power transformer, the most restrictive mode from all the power transformer bank's downstream feeders is used.

If this DCE option is not set, the effective plan execution mode is determined by the most restrictive mode from the involved energizing feeders' settings for the switching devices in the plan steps.

Note

The plan execution in the closed-loop mode also depends on the "Allow Closed Loop Execution if Protection Validation Violations Exist" and "Prevent Closed Loop Execution if an Additional Switching Violation Exists" options, which specify whether plans with protection validation and switching validation check violations can be executed in closed-loop mode.

5. Fault Isolation and Service Restoration

5.1. FISR Plan Generation

The FISR application proposes one or more switching plans to isolate faulted segments and to assist the operator in the restoration of the distribution network. If there are more than one un-faulted de-energized sections to be restored, the order of these sections to be restored are decided by the following rules with descending priority:

1. FISR always first isolates the faulted section from the un-faulted de-energized section, and then restores the un-faulted de-energized section. This is to avoid closing in on the fault.
2. If one un-faulted de-energized section can only be restored from another un-faulted de-energized section after the latter section is restored from an energized section, FISR will restore the latter de-energized section first, and then restore the former section from the latter section that was just restored. This condition is illustrated in figure below.

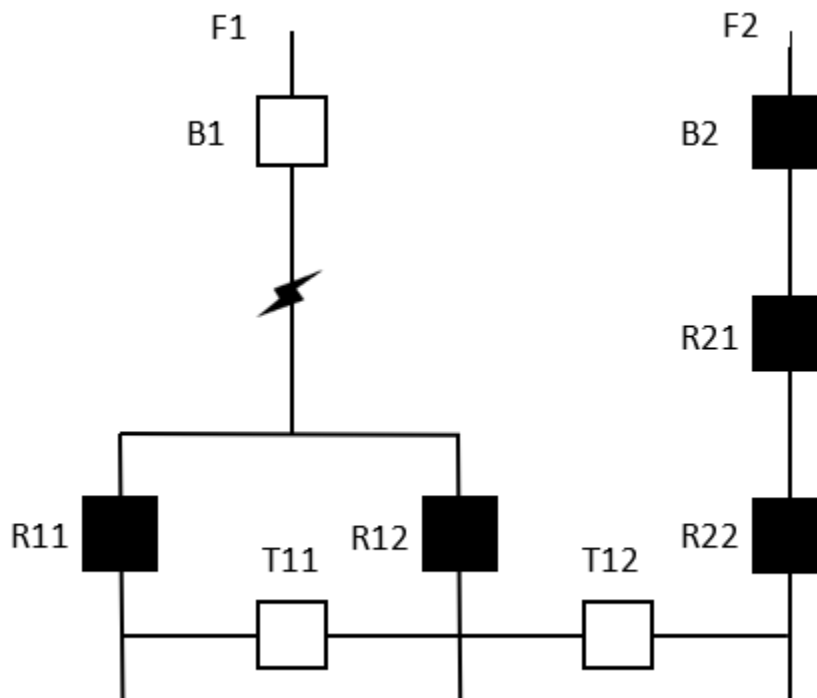


Figure 108. One De-Energized Section Can Only Be Restored from Another De-energized Section

In the system above, there are two feeders, F1 and F2, and B1 and B2 are the feeder head breakers. R11, R12, R21, and R22 are closed reclosers along the feeders, and T11 and T12 are open tie reclosers. A fault occurs downstream of B1, and B1 is tripped to clear the fault. The de-energized section downstream of R11 can only be restored from the de-energized section downstream of R12 after it is first restored from T12. FISR will generate a plan that has the following switching steps:

○

- Open R12
 - Close T12
 - Open R11
 - Close T11
3. If the Restore Automatic Switches First option in the DCE Switching Reconfiguration Problem Formulations tab in the Analysis Options section is selected, FISR will restore the de-energized sections that can be restored with SCADA controlled switching devices before restoring the de-energized sections that can only be restored with manually operated switching devices.
 4. For every un-faulted de-energized section, FISR will calculate the sum of three performance indices: Lost Load, Dropped Customer Count, and Dropped Customer Priority. Then, FISR will calculate an ordering index by dividing this sum by the total time in minutes to operate the switching devices to restore this section. The switching device operation time is set by the Manual Switch Operation Time and Automatic Switch Operation Time parameters in the DCE Switching Reconfiguration Problem Formulations tab in the Plan Parameters section. FISR will restore de-energized sections with the larger ordering index before restoring the de-energized sections that have a smaller ordering index.

5.1.1. Restoration Levels

If there is no load in the upstream area or downstream area, FISR will not generate plan steps to restore that upstream or downstream network section. To restore the loads upstream of the fault, FISR closes the upstream protective device (for example, recloser, breaker, or sectionalizer) that was tripped. For loads downstream of the fault, there are four levels of restoration:

Level 0: Restored from own feeder

If the load in the downstream area is connected to the upstream part of the faulted feeder through a normally open switch, closing that switch is the best choice. This is illustrated in figure below.

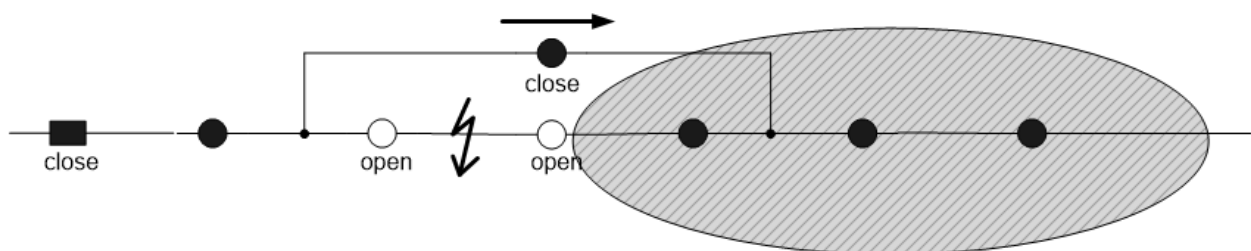


Figure 109. Level 0 Restoration

Level 1: Restored from one adjacent feeder

If the load in the downstream area is connected to an adjacent feeder through an open tie switch, it is possible that the adjacent feeder can pick up the dropped load. This is illustrated in figure below.

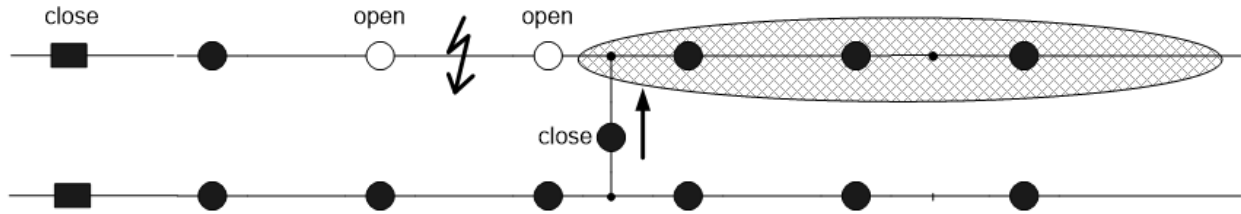


Figure 110. Level 1 Restoration

Level 2: Restored from two adjacent feeders or with load transfer

There are two cases for this level:

Case 1 involves two or more adjacent feeders. It opens one or more normally closed switches in the downstream area, and then transfers the loads to two or more adjacent feeders by using neighboring tie switches. Figure below illustrates the case of transferring loads to two adjacent feeders.

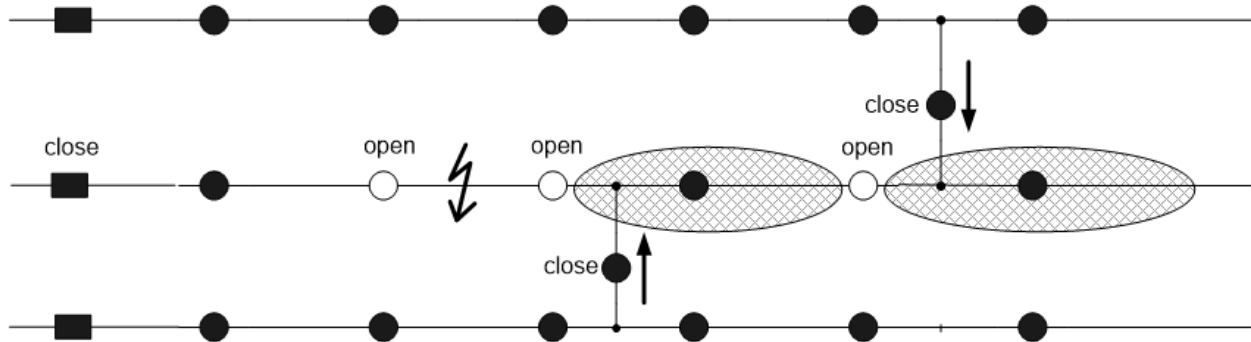


Figure 111. Level 2 Restoration Case 1

Case 2 involves adjacent feeders and other feeders that are two levels away from the faulted feeder. In this case, the adjacent feeders do not have enough transfer capacity (Amps) to pick up all dropped loads.

The solution is to transfer some live loads in an adjacent feeder which will restore the load to its adjacent feeders by performing the following:

1. Closing a tie switch between the two adjacent feeders to form a loop.
2. Opening the switch in the adjacent feeder which will break the loop, providing more transfer capacity to restore the dropped loads on the faulted feeder.
3. Closing the tie switch between the adjacent feeder and the loads in the downstream area to restore the dropped loads. This is illustrated in figure below.

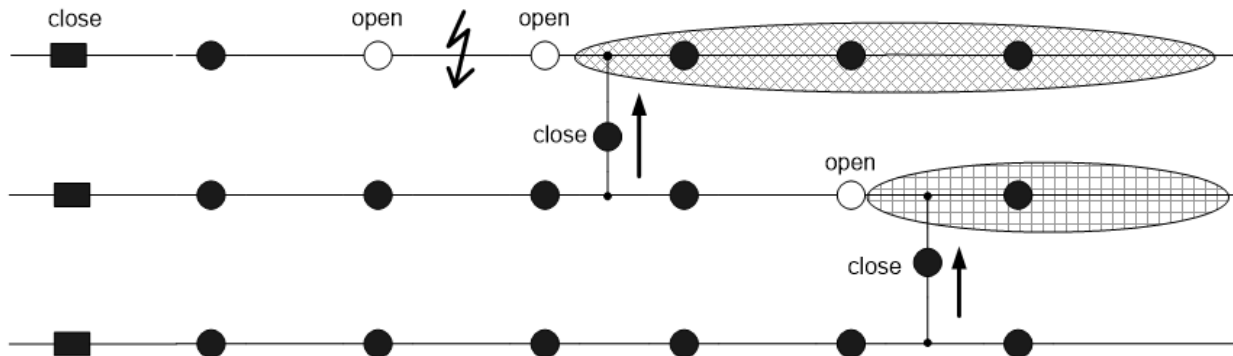


Figure 112. Level 2 Restoration Case 2

Level 3: Partial restoration

FISR only restores part of the dropped loads because it cannot restore all the loads without causing violations. This is illustrated in figure below.

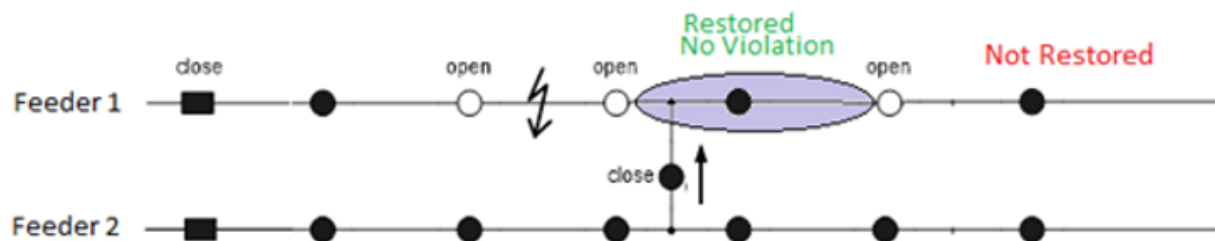


Figure 113. Level 3 Restoration

The above four restoration levels are the major approaches that FISR uses to solve a single fault case. In case of multiple faults in the same network selection, FISR internally solves the faults one by one, and then combines the steps to generate plans in one solution. The plans isolate all the faults and restore as many de-energized loads as possible.

To reduce the total FISR solution time, FISR can be configured to use a reduced network model in the plan generation logic. When network reduction is enabled, loads on the laterals will be aggregated to a lump sum load which is connected directly to the primary network. A lateral will not be reduced under several specific circumstances, which include the presence of DER/DG, regulators, capacitors, or primary meter load on the lateral, multiple branch flow directions on the lateral, or nodes on the lateral that are specified as non-reducible in the network (these are the maximum fault node, minimum phase fault node, and minimum ground fault node determined in the Protection Validation (PRV) application). After the plans are generated utilizing the reduced network, the final plan statistics for the selected top plans are calculated using the original full network to ensure the results are accurate.

5.1.2. Switch Transfer Flow Margin

The transfer margin of an open switch is the maximum flow in Amps that can be transferred

without causing any flow violations on that switch and its branch components upstream at the source side once it is closed. The Switch Transfer Flow Margin parameter specifies the additional transfer margin in percentage allowed for tie switches and load transfer switches when determining whether a FISR plan should be generated. This parameter improves the FISR performance by not generating some plans which are estimated to cause significant flow violations, which is applied under the following two conditions:

1. FISR generates a Level 2 Case 1 plan as described in [Restoration Levels](#). In this case, FISR restores the loads from two or more adjacent feeders by using neighboring tie switches. The transfer margin of all used neighboring tie switches will be calculated using the Switch Transfer Flow Margin parameter. All the calculated transfer margins must be larger than the load to be restored. This condition is illustrated in the figure below.

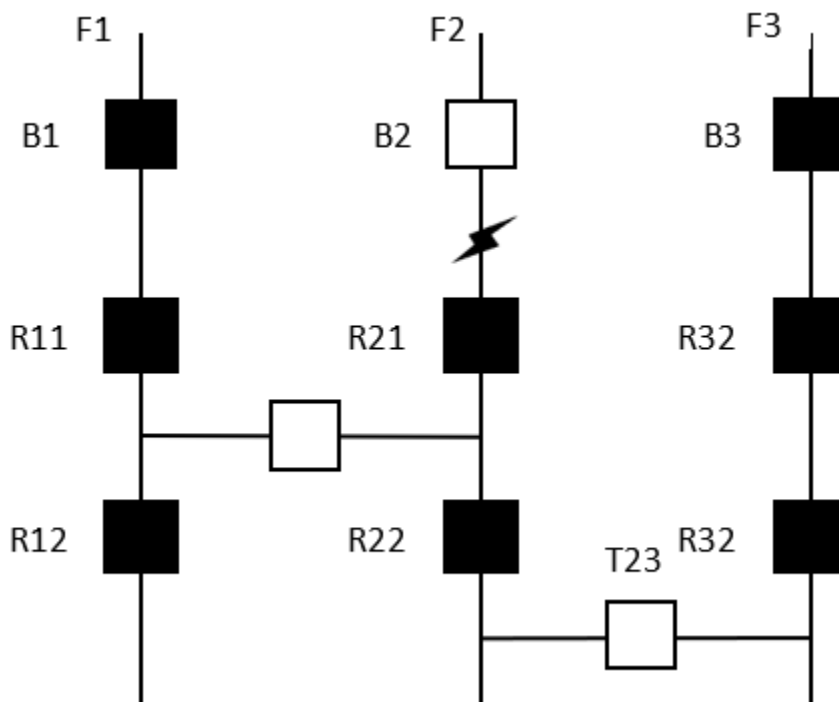


Figure 114. Switch Transfer Flow Margin for Level 2 Restoration Case 1

In the system above, there are three feeders in the station, F1, F2, and F3. B1, B2, and B3 are the feeder head breakers. R11, R12, R21, R22, R31, and R32 are closed reclosers along the feeders, and T12 and T23 are open tie reclosers between the feeders. A fault occurs downstream of B2, and B2 is tripped to clear the fault.

FISR may generate a Level 2 Case 1 plan that has the following switching steps:

- 1. Open R22
- 2. Close T23
- 3. Open R21
- 4. Close T12

FISR will check the transfer margin of the two ties switches, that is T23 and T12, against the load to be restored. For example, for this plan, the source side of T12 is feeder F1. To calculate the transfer margin of T12, FISR traces upstream from T12 to the reference bus interface and calculate flow margin of every branch component which is the internal DPF calculated flow minus the flow limit in Amps. The flow limit level used is set by the Branch Flow Limits Respect Limit parameter in the DCE Switching Reconfiguration Problem Formulations tab in the Limit Strategy section. The minimum of these calculated flow margins is the transfer margin for T12.

The load to be transferred is estimated in Amps by converting from the load's real and reactive power using the load's nominal voltage. For the scenarios below, assume the transfer margin for T12 is 80 Amps and for T23 it is 90 Amps. The load to be restored between reclosers R21 and R22 is 85 Amps and the load to be restored downstream of R22 is 120 Amps.

- **Scenario 1:** The Switch Transfer Flow Margin is set to 20%. Since $80 \times (1 + 0.2) = 96 > 85$, the transfer margin requirement is met for T12. However, since $90 \times (1 + 0.2) = 108 < 120$, the transfer margin requirement is not met for T23 and so this switching plan will not be generated.
 - **Scenario 2:** The Switch Transfer Flow Margin is set to 50%. Since $80 \times (1 + 0.5) = 120 > 85$ and $90 \times (1 + 0.5) = 135 > 120$, the transfer margin requirement is met for both T12 and T23 and this switching plan will be generated.
2. FISR generates a Level 2 Case 2 plan as described in [Restoration Levels](#). In this case, FISR first transfers part of the load from an adjacent feeder to an additional adjacent feeder by closing the load transfer switch and opening the isolation switch, then FISR restores the loads from this adjacent feeder by using the neighboring tie switch. The combined transfer margin that is increased by this parameter for both the load transfer switch and the tie switch must be larger than the total load to be restored. This condition is illustrated in the figure below.

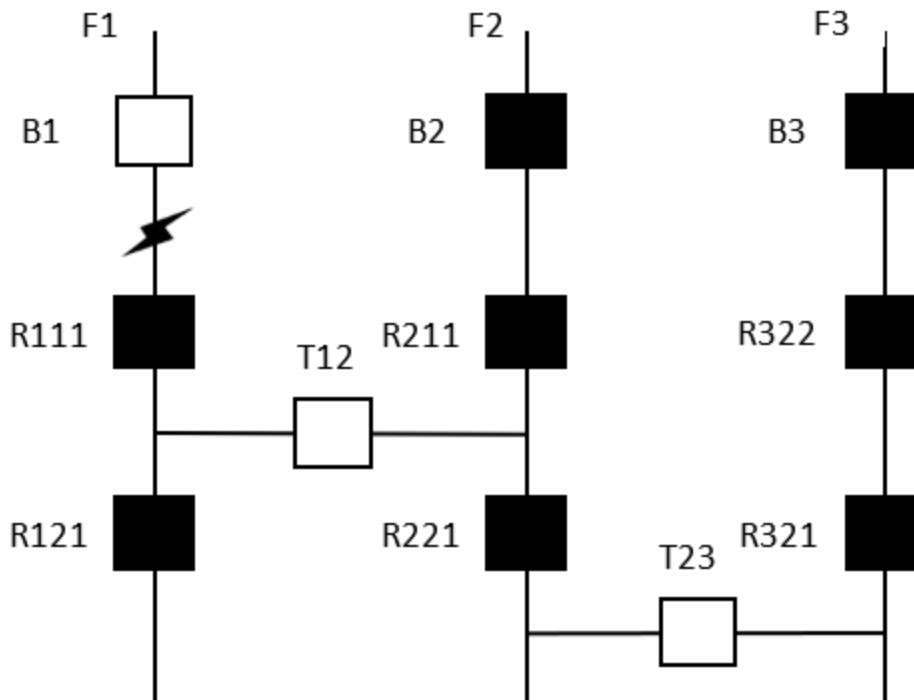


Figure 115. Switch Transfer Flow Margin for Level 2 Restoration Case 2

This system above is the same as the one shown in the earlier figure. A fault occurs downstream of B1, and B1 is tripped to clear the fault.

FISR may generate a Level 2 Case 2 plan that has the following switching steps:

- Close T23
- Open R22
- Open R11
- Close T12

FISR will check the total transfer margin of the load transfer switch and the neighboring tie switch against the total load to be restored. In the scenarios below, assume the transfer margin for T12 is 60 Amps and for T23 it is 90 Amps. The load to be restored downstream of R11 is 200 Amps.

- **Scenario 1:** The Switch Transfer Flow Margin is set to 20%. Since $(60+90) \times (1+0.2) = 180 < 200$, the transfer margin requirement is not met and so this switching plan will not be generated.
- **Scenario 2:** The Switch Transfer Flow Margin is set to 50%. Since $(60+90) \times (1+0.5) = 225 > 200$, the transfer margin requirement is met, and this plan will be generated.

When FISR performs this load transfer margin check to see if certain plans need to be generated, FISR does not run a power flow. FISR performs a flow estimation for the switching devices from the affected load values but does not consider aspects such as load to voltage dependency and phase angle. To ensure no potentially good plans are missed, it is recommended to make this parameter 20% or larger and to take any estimation inaccuracy into consideration.

5.1.3. Maximum Number of Feeders

When generating the network selection, FISR checks if the number of feeders in the network selection is larger than the "Maximum Number of Feeders" defined in the Fault Isolation and Service Restoration (FISR) tab in the DNAF Configuration Editor (DCE). If the number of feeders is larger than this limit, FISR forms the largest possible network selection (in terms of number of feeders) and continues with its solution, generating accurate isolation and restoration switching plans. The network selection tries to first include level 1 adjacent feeders and then the level 2 adjacent feeders (if configured in the Switching Reconfiguration Problem Formulations tab via the Number of Adjacency Levels parameter).

The solution is illustrated in the system below in figure below. There are two stations S1 and S2, and six feeders in each station. A fault is on feeder F21, and the adjacent feeders are as follows:

- Faulted feeder: F21
- Level 1 feeders: F13, F16, F22, and F23
- Level 2 feeders: F11, F12, F14, F15, F24, F25, and F26

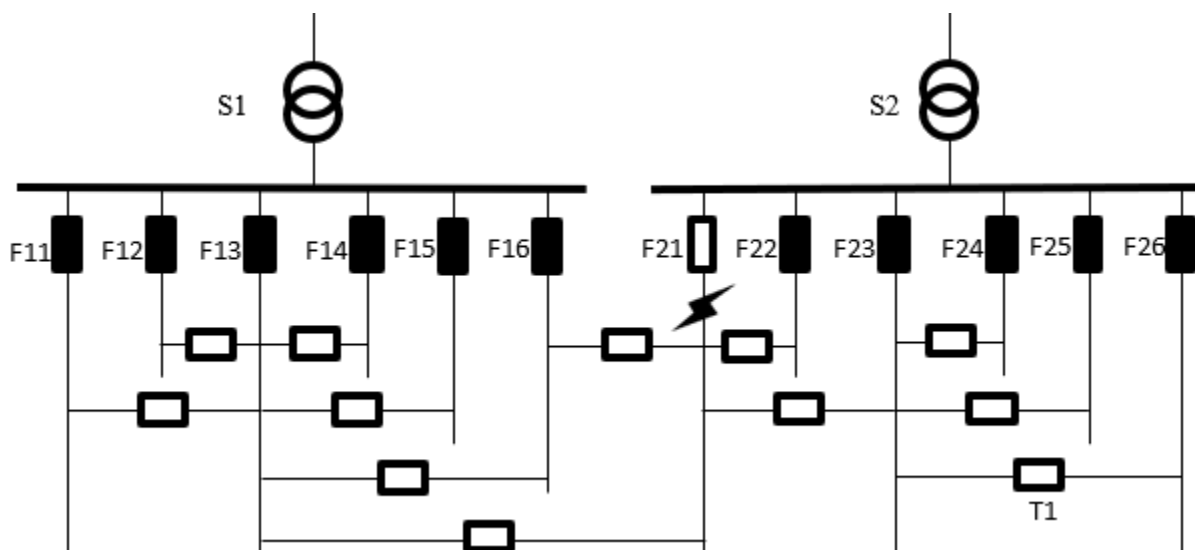


Figure 116. Illustrating System for FISR Network Selection Reaching Maximum Number of Feeders

Assume the "Number of Adjacency Levels" in the Switching Reconfiguration Problem Formulations tab in the DCE is set to 2.

1. If the "Maximum Number of Feeders" in the FISR tab in the DCE is set to 4, and since there are four level 1 feeders in the network selection, FISR will include the faulted feeder F21 and three of the four Level 1 feeders in the network selection; FISR will be solved and generate plans. The three chosen Level 1 feeders will be selected based on having the highest flow margin upstream of the eligible tie switching device that restores the faulted area.
2. If the "Maximum Number of Feeders" in the FISR tab in the DCE is set to 10, and since there are twelve feeders in the network selection, FISR will include the faulted feeder F21 and all four Level 1 feeders along with five out of seven Level 2 feeders in the network selection; FISR will be

solved and generate plans. The five Level 2 feeders will be selected based on their flow margin upstream of the eligible tie switching device that can transfer loads from the Level 1 feeders to the Level 2 feeders. For example, T1 is an eligible tie switch between level 1 feeder F23 and level 2 feeder F26, and the flow margin in feeder F26 upstream of T1 is used to evaluate and decide the level 2 feeders.

3. If the "Maximum Number of Feeders" in the FISIR tab in the DCE is set to 12 or higher, and since there is a total of twelve feeders in the network selection, FISIR will include all feeders in the network selection; FISIR will be solved and generate plans.

5.2. Modes of Operations

This section describes the modes of operation for FISIR. In both real-time and study mode, DPF must be solved before executing FISIR.

The FISIR application can be executed in real-time mode or study mode.

- **Real-Time Mode:** In real time, FISIR is triggered manually or automatically. If the dispatcher places a fault object on the ADMS display at the fault location, FISIR can be executed manually by using the RT DNAF Solution pop-up display.

The FISIR application can be triggered by an event, either by a fault indicator status change from Not Set to Set (with or without an event-triggered Fault Location application), or by a loss of voltage measurement value. The event trigger can be enabled and disabled in the appropriate DNAF server configuration file.

- **Study Mode:** In study mode, FISIR can be executed from the ST DNAF solution pop-up display. It proposes switching plans in response to user requests.

5.2.1. Fault Triggered FISIR

In real time, the FISIR application can be triggered by a fault event. Currently, these fault event types are supported:

- A fault event triggered by both fault indicator status change(s) from Not Set to Set and an event-triggered FL solution.
- A fault event triggered by only a fault indicator status change from Not Set to Set.

For either fault trigger, FISIR tries to place a fault after the tripped switching device is locked out.

- If there is a valid lockout status measurement on the tripped switching device, FISIR checks whether the device is set to lockout. The FISIR triggering logic is automatically terminated and no fault is placed if the device is not set to lockout within the maximum wait time specified in the appropriate DNAF server configuration file by the DNAF_FISR_NO_LOCKOUT_WAITTIME parameter (the default value is 150 seconds).
- If there is no valid lockout status measurement or the measurement is not valid on the tripped

switching device that clears the fault, FISR waits for the maximum lockout wait time specified by the DNAF_FISR_MAX_LOCKOUT_WAITTIME parameter (the default value is 65 seconds). If the fault placement does not require a valid lockout measurement (the DNAF_FISR_LOCKOUT_FAULTPLACEMENT parameter is set to 0), FISR places the fault. If the parameter is set to 1, FISR does not place the fault.

Note

The lockout status is set for a breaker or recloser when the associated measurement is received from SCADA or when the device is manually tripped by the operator.

The fault indicator and fault location event triggers are controlled by the following three DNAF server parameters:

- **DNAF_FISR_FAULTIND_EVENT_TRIGGER (1):** Enables (=1) or disables (=0) the fault indicator status change event trigger for FISR.
- **DNAF_FISR_FL_EVENT_TRIGGER (2):** Enables (=1) or disables (=0) the fault location event trigger for FISR.
- **DNAF_FL_EVENT_TRIGGER (3):** Enables (=1) or disables (=0) event triggering for the fault location application.

The table below describes the FISR behavior for different combinations of the above parameters, where 1 is enabled and 0 is disabled.

Parameter 1	Parameter 2	Parameter 3	FISR Behavior
1	1	1	Place a fault at the fault location based on fault indicator statuses and results from the fault location application Execute FISR
1	1	0	Place a fault at the fault location based on fault indicator statuses Do not execute FISR
1	0	1	Place a fault at the fault location based on fault indicator statuses Execute FISR
1	0	0	Place a fault at the fault location based on fault indicator statuses Execute FISR
0	Any	Any	Fault indication is needed to identify the faulted section. Do not place a fault Do not execute FISR

When a fault event is triggered by both a fault indicator status and an event-triggered Fault

Location application, FISIR finds the faulted area after waiting for the time set by `DNAF_FISR_MAX_LOCKOUT_WAITTIME`, or after waiting for tripping a switching device that is locked out in that station. FISIR searches through the faulted area between the switching device having a Set fault indicator status and the switching device having Unset fault indicator status in the fault location application results. FISIR then places a fault at the location with fault currents that are closest to the measured fault current. If the fault location results cannot be found in the faulted area decided by the fault indicators, the fault will be placed on a feeder main line around 200 feet downstream of the most downstream set fault indicator, but FISIR will not be calculated in this case.

When the FISIR application is only triggered by a fault indicator status, after a fault indicator is changed from Unset to Set, FISIR finds the faulted area after waiting for the time set by `DNAF_FISR_MAX_LOCKOUT_WAITTIME`, or after waiting for tripping a switching device that is locked out in that station. FISIR then places a fault in the de-energized area between the switching device having a Set fault indicator status and the switching device having an Unset fault indicator status on a feeder main line around 200 feet downstream of the most downstream switching device having a Set fault indicator status.

When a fault indicator phase mismatch is observed between two fault indicator islands, FISIR inserts a fault in each island. However, for phase-agnostic fault indicators, FISIR checks whether the `PhaseAgnosticFaultIndication` flag is set to True for a fault indicator in an island (faulted phase is disregarded). If this flag is True, FISIR merges the two islands into one, applies the faulted phase from the island with a regular fault indicator, and inserts only one fault.

5.2.1.1. Miscoordination

Miscoordination between reclosing capable protective devices (for example, a recloser in recloser mode or a breaker) occurs when the protective device immediately upstream of a fault fails to operate before another protective device further upstream operates. In the simple one-feeder system below, B1 is the feeder head breaker for feeder F1 and R11 and R12 are downstream reclosers. A fault occurs between R11 and R12, and R11 does not trip but B1 does trip. The fault indicator associated with B1 and R11 are both set, and the fault indicator associated with R12 remains unset.



A check box named “Handle Miscoordinations” on the FISR tab in the DNAF Configuration Editor enables FISR miscoordination processing. If the check box is checked, and there is a miscoordination such that a protective device further upstream than the closest upstream protective device clears the fault and cycles to lockout, FISR will predict the fault is downstream of the closest upstream protective device and generate a correct upstream isolation and restoration plan. In this simple system, the fault location is predicted between R11 and R12. If the check box is not checked, no miscoordination is considered, and FISR will predict one fault downstream of the one tripped protective device B1. The predicted fault also needs to be downstream of the closed devices whose associated fault indicator is set. In this simple system, R11 is closed and its associated fault indicator is set, so the fault should be downstream of R11. R12 is also closed but its associated fault indicator is unset. So, the fault is predicted between R11 and R12.

☒ **Handle Miscoordinations**

Time Difference Tolerance (sec):

Figure 117. Miscoordination Options

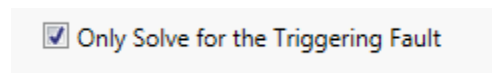
The associated parameter “Time Difference Tolerance (sec)” specifies the time limit for determining if a miscoordination has occurred. That is, if the time difference between two fault indicator status measurements on the sequential protective devices is less than the predefined limit, FISR detects a miscoordination and the fault is downstream of the downstream fault indicator. In this simple system, the fault is predicted between R11 and R12. Otherwise, FISR will predict there is no miscoordination and determine there are two faults in the area; one fault is between the two fault indicators, and the other fault is downstream of the furthest downstream fault indicator. In this simple system, two faults are predicted, one is between R11 and R12, and the other is between B1

and R11.

5.2.1.2. Multiple Faults

If multiple new faults occur in the network selection, FISR solves them together and finds the plan that is trying to restore the de-energized loads caused by the faults.

If faults occurred previously in the network selection (which was solved by an earlier FISR solution), whether the new FISR solution solves for these solved faults depends on the parameter shown in figure below. This parameter is defined on the FISR tab in DCE.



☒ Only Solve for the Triggering Fault

Figure 118. FISR Triggering Option for Multiple Fault

The "Only Solve for the Triggering Fault" option specifies whether FISR only generates a plan that restores load for the triggering faults. If it is selected, FISR solves only the triggering faults, but other previously solved faults in the network selection are respected. If this option is not selected, FISR solves the triggering fault, and all faults in the network selection are also solved.

5.2.2. Loss of Voltage Triggered FISR

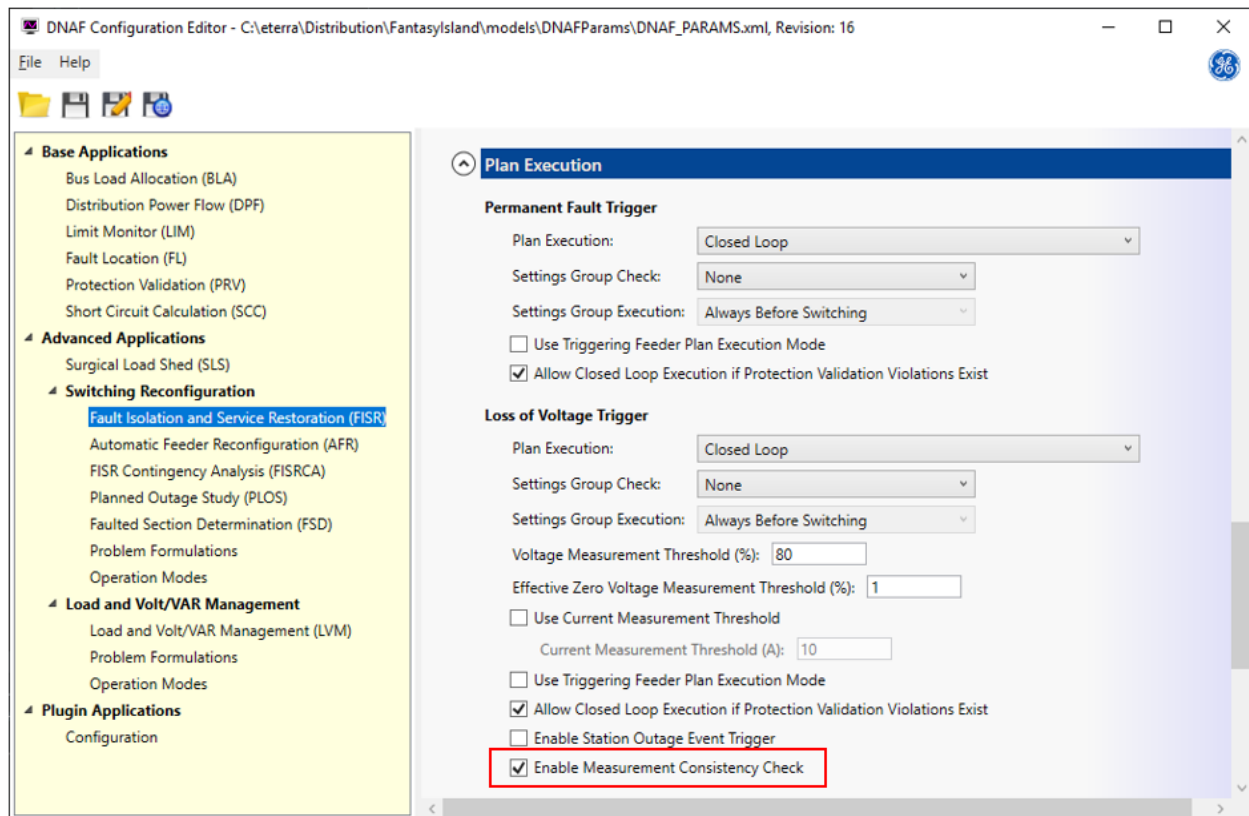
The FISR application can also be triggered by detecting the change of voltage measurements in the system, which is also called the Loss of Voltage (LoV) trigger.

When a circuit opens unexpectedly (not due to switching), the downstream voltage measurement changes to zero. After a sustained period of time, FISR tries to place a cut. This time delay is defined as DNAF_LOV_DELAY in the DNAF server configuration file and is used to prevent false triggering that may occur because of reclosing operations due to temporary loss of voltage. Note that cuts are placed only in the case of station loss of voltage events.

If a loss of voltage condition is detected inside the substation, there is another delay in addition to the original delay before any cuts are placed. This time delay is defined as DNAF_LOV_2ND_DELAY in the DNAF server configuration file. This timer allows additional time for substation restoration schemes to operate.

After the time delay expires (or both delays for a station internal loss of voltage), a "downstream measurement consistency check", if enabled, is performed. The purpose of this check is to ensure that voltage measurements, and optionally current measurements, in the network section experiencing loss of voltage are effectively zero.

The "Enable Measurement Consistency Check" option is used to enable or disable this check.



If the "Use Current Measurement Threshold" option is also enabled, then current measurements are also tested if they are effectively zero.

The following rules are used:

1. Only voltage and current measurements with valid SCADA status are considered. Additionally, these measurements should also be on feeder main or in station internal and not associated with any open switches.
2. For station interface loss of voltage, all voltage and optionally current measurements downstream of the interface that lost voltage, but excluding any on laterals, are considered.
3. For station power transformer loss of voltage, all voltage and optionally current measurements downstream of the power transformer that lost voltage, but excluding any on laterals, are considered.
4. For feeder loss of voltage, all voltage and optionally current measurements downstream of the most upstream breaker/recloser that lost voltage, but excluding any on laterals, are considered.
5. Voltage measurements are compared to the parameter "Voltage Measurement Threshold". Voltage measurements with analog values above this parameter are considered not consistent with loss of voltage. Such voltage measurements are marked as "inconsistent", and the measurement consistency check is considered failed. In this case the FISR application is not triggered.
6. Current measurements are compared to the parameter "Current Measurement Threshold". Current measurements with analog values above this parameter are considered not consistent

with loss of voltage. Such current measurements are marked as "inconsistent", and the measurement consistency check is considered failed. In this case the FISR application is not triggered.

If the "downstream measurement consistency check" passed, or if it was disabled and if the loss of voltage measurements are still present, the FISR application locates the LoV event and places a cut in the appropriate location. In most cases, this location is upstream of the first device on the circuit that lost voltage. FISR will not be LoV-triggered for a commanded opening of a telemetered switch, recloser, or breaker. Once the cut is placed, FISR triggers after a short delay to prevent a timing issue in the real-time mode. This delay is defined as DNAF_EVENT_DELAY in the DNAF server configuration file.

Consider a substation one line diagram as shown in figure below.

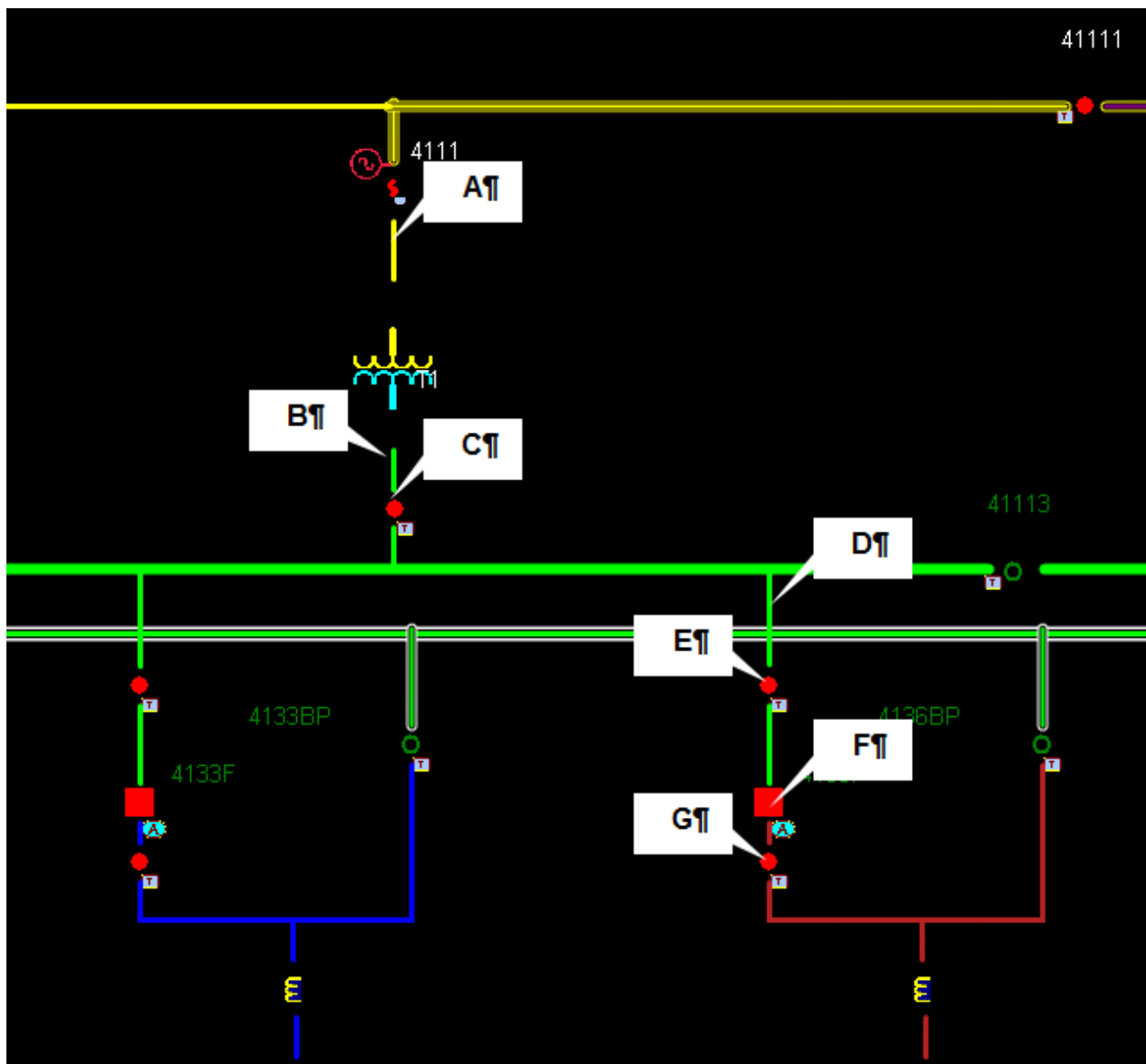


Figure 119. One Line Diagram of an LOV Station

Location A is at a reference bus interface location (which has a voltage magnitude measurement); B is a jumper on the load side of the power transformer; C is a switch on the load side of the power transformer; D is a jumper; E is a disconnect switch at the beginning of a feeder; F is a feeder head breaker; and G is a disconnect switch downstream of the feeder breaker.

Among the switches and breakers, the open/closed status of devices C, E, F, and G are telemetered. There are voltage measurements at location C (measurement V_C) and location F (measurement V_F).

Table below describes the different loss of voltage scenarios and FISR responses for the cases where the loss of voltage trigger is enabled.

Table 29. Station and Feeder LOV Scenarios

Scenario	Initial Condition (in field or in Simulator)	System Response and Analysis	Application Response
1	Voltage magnitude at interface A changed to zero.	Power transformer and all connected downstream feeders in Real Time will become de-energized. Voltages V_C and V_F change to zero.	Station loss of voltage on all feeders is triggered.
2	Jumper B is opened.	Power transformer and all connected downstream feeders in Real Time will remain energized. Voltages V_C and V_F change to zero.	Station loss of voltage on all feeders is triggered.
3	Switch C is opened.	Power transformer and all connected downstream feeders in Real Time will become de-energized. Voltages V_C and V_F change to zero.	Station wide loss of voltage on all feeders is triggered.
4	Jumper D is opened.	The feeders not energized by breaker F in Real Time will remain energized. Voltage V_C stays normal but voltage V_F changes to zero.	A feeder loss of voltage is triggered for the feeder energized by breaker F. The plan should open breaker F and try to restore the whole feeder using adjacent energized feeders.
5	Switch E is opened (commanded operation).	The feeder energized by breaker F in Real Time will become de-energized. Voltage V_C stays normal but voltage V_F changes to zero.	No loss of voltage is triggered because the voltage change is caused by a commanded opening of a SCADA monitored switch. However, if the switch is not telemetered, the feeder will stay energized in Real Time. A feeder loss of voltage will be triggered. Similar to jumper D being opened, the plan should open breaker F and try to restore the whole feeder using adjacent energized feeders.
6	Breaker F is opened.	The feeder energized by breaker F in Real Time will become de-energized. Voltage V_C stays normal but voltage V_F changes to zero.	No loss of voltage is triggered because the voltage change is caused by commanded opening of a SCADA monitored breaker.

Scenario	Initial Condition (in field or in Simulator)	System Response and Analysis	Application Response
7	Switch G is opened.	The feeder energized by breaker F in Real Time will become de-energized. Both voltages V_C and V_F stay normal, and the voltage measurements on the downstream devices change to zero.	No loss of voltage is triggered because the voltage change is caused by the commanded opening of a SCADA monitored switch. However, if the switch is not telemetered, the feeder will stay energized in Real Time, and a feeder loss of voltage will be triggered. The plan should open the first recloser or breaker on the feeder and restore the remaining downstream loads.

A Loss of Voltage symbol is shown near a substation circle and a Loss of Voltage cut is placed on a line near a faulted component (for station external, feeder LoV events) after change of voltage measurements is detected and predicted cut is placed. There are two parameters on the FISR page in DCE specifies the time in seconds on the behavior how to remove the predicted cut and station loss of voltage symbol.

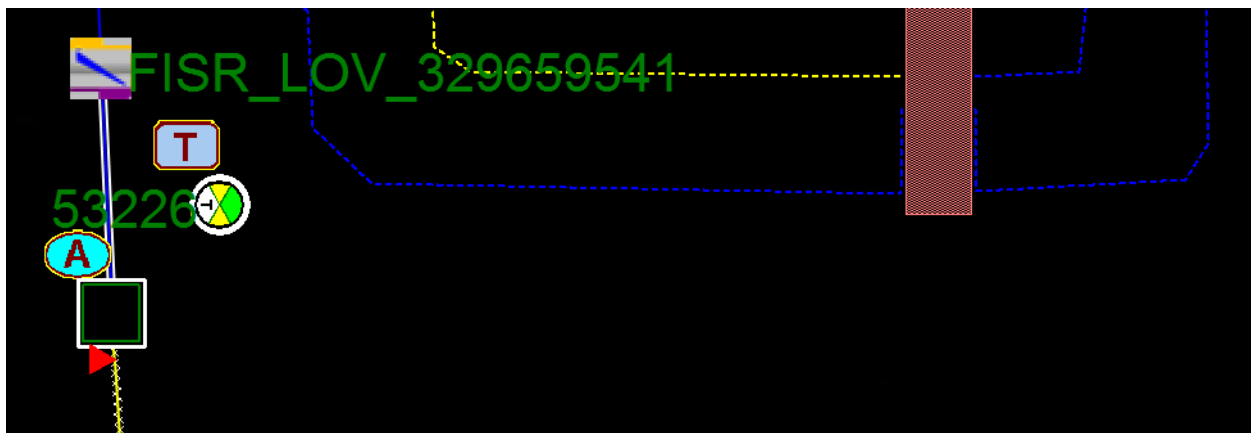


Figure 120. Example of a Feeder Loss of Voltage Event

The feeder and station loss of voltage statuses are checked when performing network selection for a fault triggered or loss of voltage triggered FISR solution. If a station is in the loss of voltage status, all the feeders in the station are considered ineligible to restore load for the neighboring station.

The station or feeder loss of voltage status is set when zero (or low) voltage measurement is detected and a Loss of Voltage cut is placed.

For feeders, the status is reset when the cut is removed. The cut can be removed manually or automatically. If the related voltage measurement value change back to normal, and then after the time delay specified as "Time to Remove Predicted Cut" parameter in DCE, the cut will be removed automatically.

The station loss of voltage symbol can also be removed manually or automatically. To manually remove the symbol, user can use "Clear Loss of Voltage Flag" button on the station toolbar. It will also disappear automatically after the time delay specified as "Time to Remove Station Loss of Voltage Symbol" parameter in DCE.

Feeder and station loss of voltage status will be reset after cut or symbol is removed.

5.2.2.1. Station Outage Triggered FISR

Station internal faults may lead to power transformer or bus differential protection scheme operation, which could open one or multiple breakers on the source or load side of a power transformer, bus, or feeder head. Examples of differential protection scheme operation scenarios are provided below.

Scenario 1: Transformer Relay

When a fault occurs on a power transformer, a transformer relay scheme may open the transformer source side breaker. This de-energizes all connected downstream feeders.

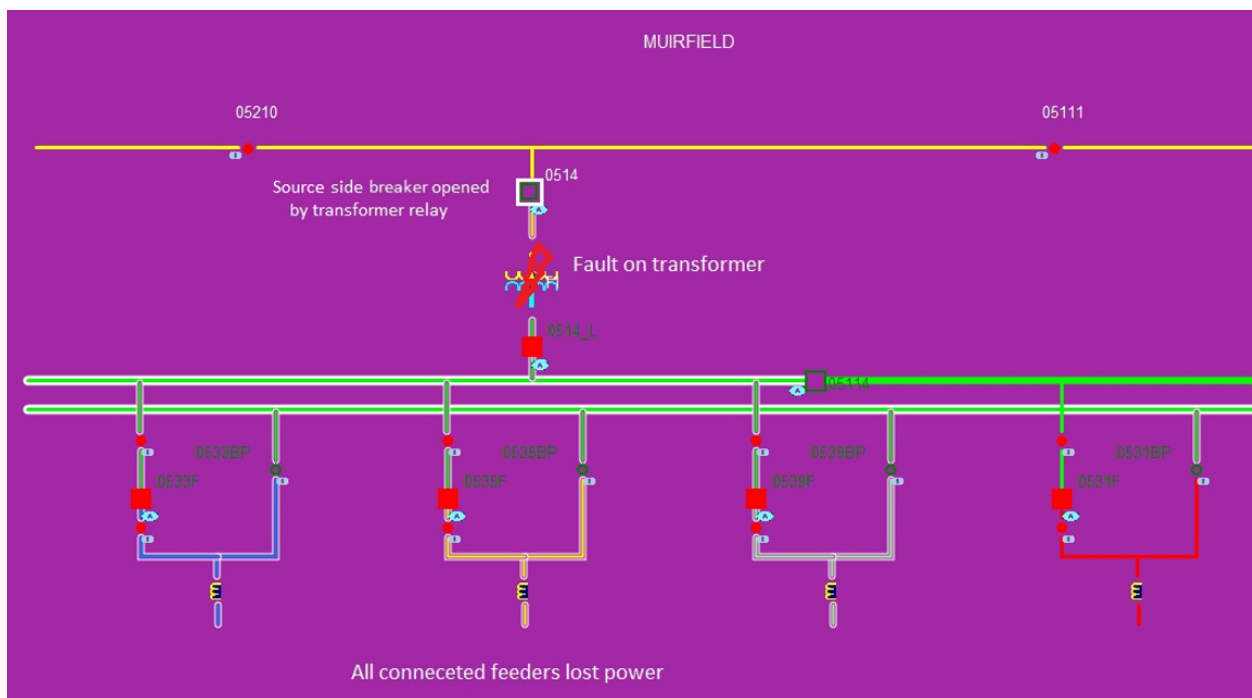


Figure 121. Station Internal Fault - Transformer Relay Operation

Scenario 2: Transformer Differential Relay

When a fault occurs on a power transformer, a transformer differential relay scheme may open the transformer source side and load side breakers. This de-energizes all connected downstream feeders.

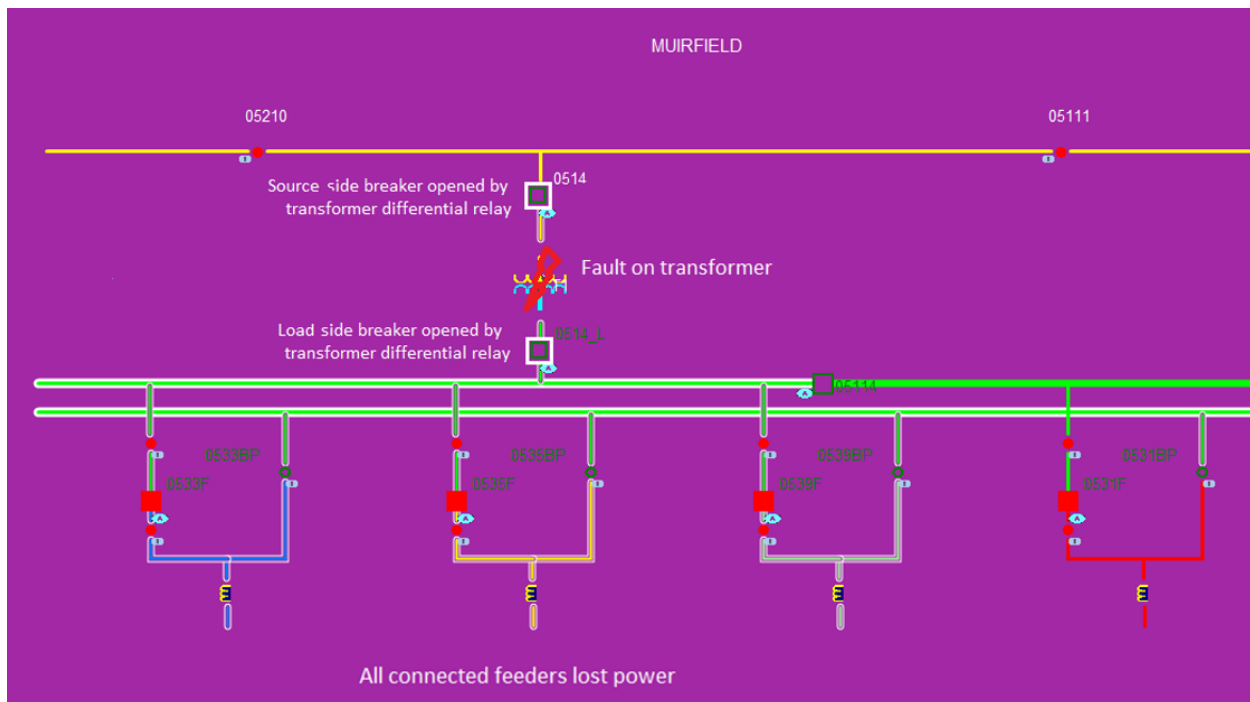


Figure 122. Station Internal Fault - Transformer Differential Relay Operation

Scenario 3: Bus Differential Relay

After a fault occurs on a station bus, a bus differential relay scheme may open the transformer load side breaker and all connected feeder head breakers. This de-energizes all feeders connected to the bus.

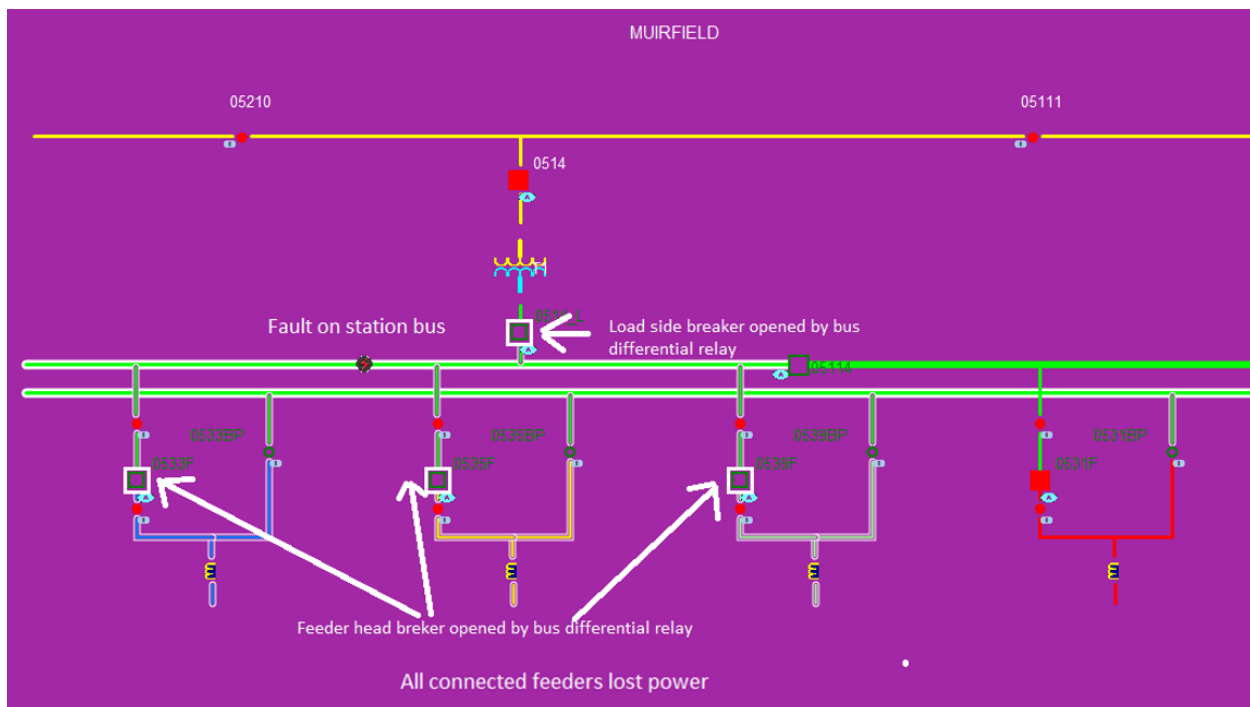


Figure 123. Station Internal Fault - Bus Differential Relay Operation

Scenario 4: Transformer Differential Relay with Automatic Throw-Over Scheme

After a fault occurs on a power transformer, a transformer differential relay scheme may open both the transformer source and load side breakers. This momentarily de-energizes all feeders connected to the bus. Next, the automatic bus transfer scheme may close the bus tie breaker. This restores power to the connected feeders from another power transformer.

In this case, the station outage event triggers first when the voltage is lost momentarily. After a time delay, the related measurements are checked again. At that time, the power is already restored and the voltage measurements change back to values above the user-configured threshold. The loss of voltage condition is no longer detected, FISR triggering stops, and FISR is not executed.

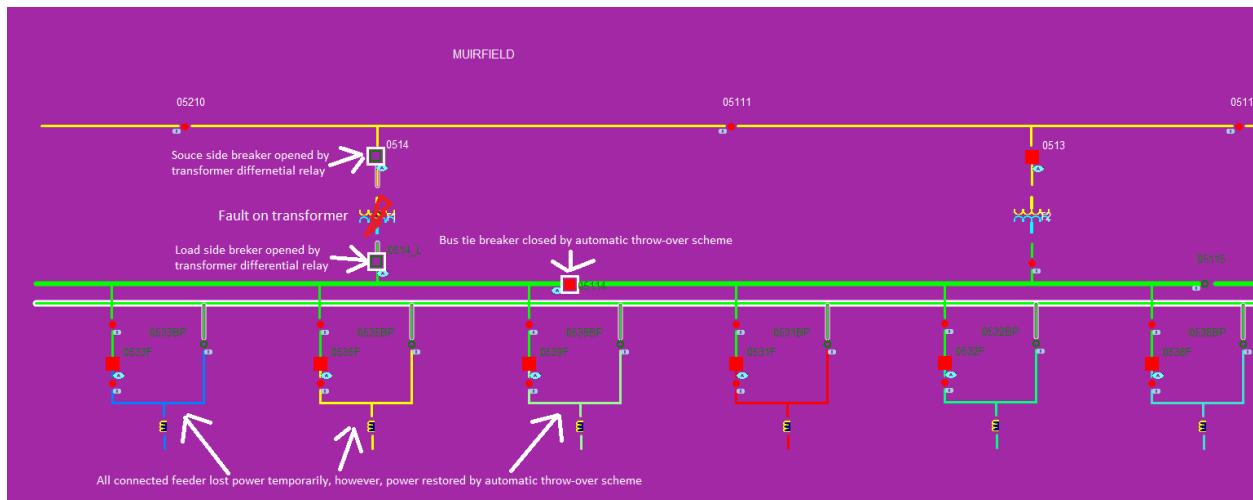


Figure 124. Station Internal Fault - Bus Differential Relay Operation

Station Outage Event Triggering

When a station internal fault occurs, the voltage on the load side of the power transformer, the bus, or at the feeder heads may change to zero or a very low value, and the connected downstream feeder head current and power measurements also change to zero or a very low value.

- As shown in the examples above, depending on the protection scheme, the transformer and feeder breakers can be either open or closed. If any breaker is opened by protection relays, FISR may be triggered by the station outage event trigger. (However, if any breaker opened via an operator or load shed application command, a station outage event is not triggered because the loss of voltage is expected behavior caused by a known (commanded) operation.)
- Alternatively, if a voltage measurement inside the station changed to zero or a very low value, but all related breakers remain closed, FISR may be triggered by the station loss of voltage event trigger.

FISR triggering for station internal faults is configured by the Enable Station Outage Event Trigger parameter in the Loss of Voltage group on the FISR page in the DNAF Configuration Editor.

FISR triggering for station internal faults works as follows:

1. When a power transformer load side voltage, a bus voltage, or a feeder head voltage that has no impedance between the measurement node and the power transformer load side node changes to zero or a value below a threshold defined using the "Voltage Measurement Threshold" option, a timer starts. The voltage measurement here represents the voltage at the load side of the power transformer, station bus, or feeder head.
2. After the time delay defined as the sum of the DNAF_LOV_DELAY and DNAF_LOV_2ND_DELAY parameters in the DNAF server configuration file expires, the voltage measurement is checked again.
 - If the voltage returns to a value above the threshold, FISR is not triggered, and the process is stopped.
 - If the bus voltage is still below the threshold, all connected feeder head current measurements that are connected to the common bus are checked to make sure that all current measurements are below the threshold defined using the "Use Current Measurement Threshold" and "Current Measurement Threshold" parameters. If any of the feeder head current measurements has a value higher than the defined threshold, FISR is not triggered, and the process is stopped.
3. The FISR triggering function checks transformer and feeder breakers that are associated with the affected bus.
 - If all breakers are closed, the station loss of voltage FISR is triggered.
 - If any of the breakers is open, the FISR triggering function checks the last operation time of the breaker. If the breaker was opened before the measurement change (with a safety margin), the breaker is not used for the triggering consideration because the breaker status change is not related to the station outage event. Otherwise, the Commanded Open flag for the open breakers is checked. If any of the related breakers are opened by a command, FISR is not triggered, and the triggering process is stopped. If all open breakers are un-commanded open, the station outage FISR is triggered.

The Station Outage decoration  appears near a substation circle when FISR is triggered for a station outage event. The decoration is processed similar to regular station or feeder loss of voltage events, where the decoration can be removed either manually or automatically. The user can manually remove the decoration using the "Clear Loss of Voltage Flag" button on the station toolbar. If the related FISR solution is completed and the plan is successfully implemented, the decoration will also be removed automatically.

5.2.2.2. Feeder Fault Triggered Loss of Voltage FISR

For feeder head breakers that only have a telemetered open/close status measurement modeled in the ADMS, and if a fault occurs on the feeder downstream of this breaker and upstream of another fully telemetered device with all the necessary SCADA measurements including fault indicator status measurements, loss of voltage FISR will be triggered if all the following conditions are true:

- A zero voltage is detected on a feeder switching device.
- The breaker has a valid open/close status, and the status is open.

- The breaker has no fault indicator status measurement(s).
- The breaker opening was un-commanded.
- There is a device with valid SCADA measurements including fault indicator status measurements downstream of the breaker.
- The breaker last operation time is close enough to the zero voltage measurement change time (within a safety margin).

The predicted fault location will be immediately upstream of the first device which has both effective zero voltage measurement(s) and "unset" fault indication measurement(s). This is illustrated in the simple system in figure below.

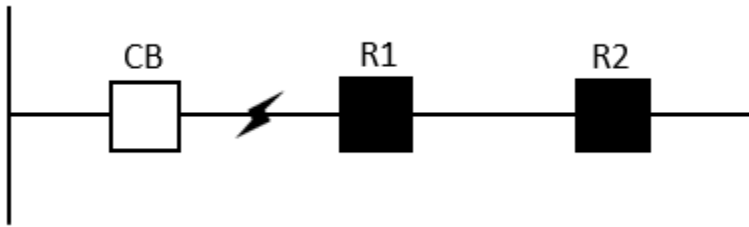


Figure 125. Feeder Fault with Breaker that has only a Status Measurement

In this one feeder system, the feeder head breaker CB only has a valid status measurement and reclosers R1 and R2 have valid voltage and fault indicator measurements. A fault occurs between CB and R1 and the CB is tripped to clear the fault. Reclosers R1 and R2 have an effective zero voltage measurement and unset fault indicator measurement, FISR is triggered, and the predicted location is immediately upstream of R1. A Loss of Voltage decoration with text "FAULTED FEEDER" is shown near the substation circle and a Loss of Voltage cut is placed on the line immediately upstream of R1.

In the same system, if R1 has valid voltage and fault indicator measurements, but R2 does not, the predicted location is the same as the case above, that is, immediately upstream of R1.

In the same system, if R2 has valid voltage and fault indicator measurements, but R1 does not, the predicted location is immediately upstream of R2.

In the same system, if R1 and R2 do not have valid fault indicator measurements, the location of the fault cannot be determined.

5.2.2.3. Running Loss of Voltage Triggered FISR in Study Mode

After a FISR loss of voltage event is triggered in the real-time environment and predicted cuts are placed on the feeders outside the substation, you can initialize the study mode to real-time environment (using the "Initialize Study to Real Time" option) to reproduce the same loss of voltage scenario in study mode, to analyze the scenario in detail. You can then create a savecase to permanently save the case.

Note

For feeder loss of voltage events, the predicted cuts are included in study mode. Since no predicted cuts are placed for the station loss of voltage events in the real-time environment, for the station loss of voltage events, the predicted cut has to be manually placed.

In the real-time environment, the predicted cut may be removed automatically after a time delay when the related voltage measurement is restored to normal value (specified by the Time to Remove Predicted Cut and Station Loss of Voltage Symbol option) after the related voltage measurements are restored from zero. In the study environment, the predicted cut is not removed automatically.

In study mode, you can manually place a predicted loss of voltage cut inside or outside the station internals by selecting the Loss of Voltage type when creating a cut, as shown in figure below. Both station and feeder loss of voltage scenarios can be created.

The image shows a 'Create Cut Dialog' window with the following fields and options:

- Name:** PredictedCUT
- Station:** BETHPAGE
- Line Id:** 339947135
- Comment:** (empty text box)
- Type:** A dropdown menu is open, showing the following options: Cut, Temporary Switch, Elbow Switch, Recloser, Sectionalizer, Gang Switch, Disconnect, Fuse, and Loss of Voltage (which is highlighted at the bottom).
- Label Placement:** A grid of 9 checkboxes, with the top-left checkbox checked.
- Current State:** A section with two radio buttons labeled 'Open' and 'Close'. The 'Close' radio button is selected.
- Buttons:** Evaluate, OK, and Cancel.

Figure 126. Manually Placing a Predicted Cut for Executing a FISR Loss of Voltage Case

The predicted cut is an indicator to decide whether a FISR Loss of Voltage condition needs to be solved. The cut state always must be Closed. If you manually change the cut status to Open after it has been inserted, FISR will not try to restore the downstream de-energized section because it is caused by a known open point in the network.

For station-based FISR loss of voltage cases, a predicted cut can be placed on a station internal device upstream of the feeder head breakers, including power transformer or jumper. For example,

Figure 127 shows a predicted cut named "CUT4" manually placed on the power transformer T1 inside the substation. In this case, all downstream feeders supplied by transformer T1 lose power and FISR tries to restore power to all downstream deenergized feeders using adjacent energized donor feeders.

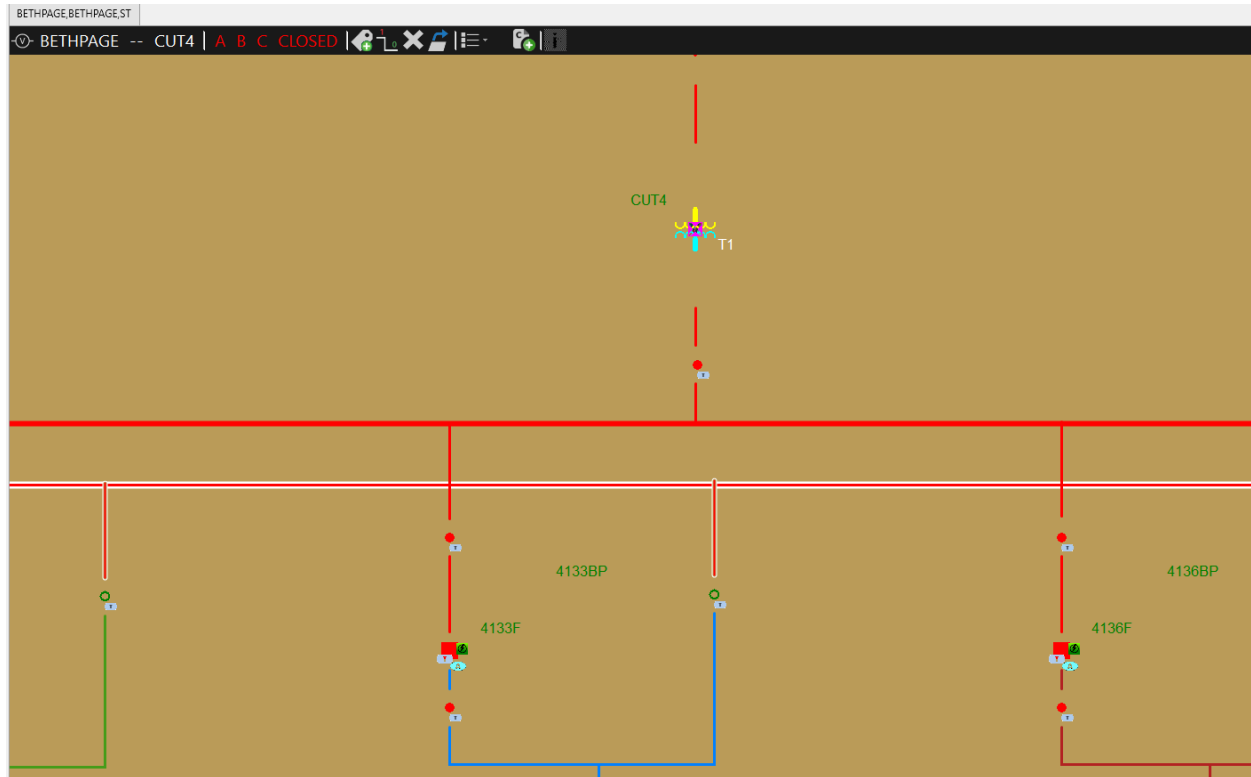


Figure 127. Predicted Cut Placed Inside Substation

For feeder-based FISR loss of voltage cases, a predicted cut needs to be placed on a device, typically a jumper or line, in the station external geographic diagram. The cut needs to be placed upstream of a switch where voltage measurements changed to zero. Multiple cuts can be placed. In Figure 128, two predicted cuts "CUT1" and "CUT2" are placed on different feeders. FISR tries to restore load on both affected feeders.

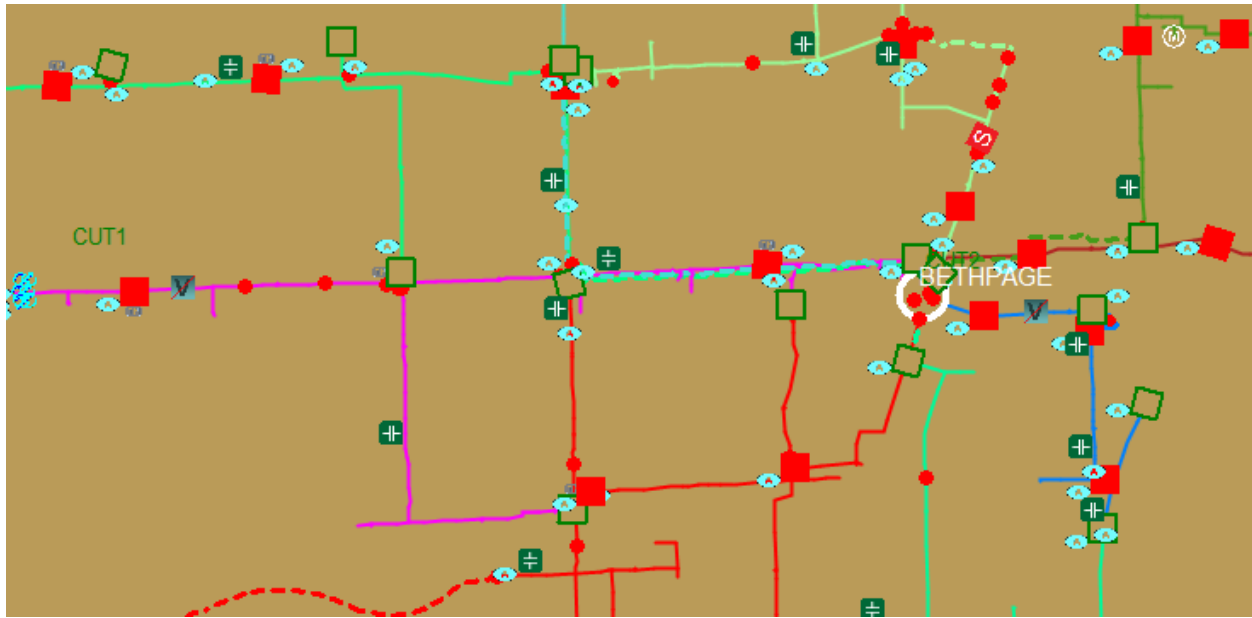


Figure 128. Predicted Cut Placed Outside the Substation

5.2.3. Manually Executed FISR

In the study mode environment, you can manually execute loss of voltage triggered FISR and fault triggered FISR.

To run FISR:

1. Open the DNAF Solution pop-up display.
2. In the Application list, select Fault Isolation and Service Restoration.
3. Select the problem formulation.
4. Click Run.

You can also manually execute the FISR application by selecting Network Analysis > Fault Isolation and Service Restoration from the line context menu from the geographic view.

The following sequence occurs when manually executing FISR:

1. If a fault is present in the selected station, the fault-triggered FISR is run to isolate the faulted section and restoration loads in the healthy network sections. The faults to be solved by FISR are impacted by the "Only Solve for the Triggering Fault" parameter (for more details, see [Multiple Faults](#)).
 - If "Only Solve for the Triggering Fault" is selected, FISR solves only the unsolved faults in the network selection. However, the other faults in the network selection are respected.
 - If the "Only Solve for the Triggering Fault" option is not selected, FISR solves all faults in the network selection.
2. If there is no fault in the system, but there are predicted cuts placed within the substation,

station-based loss of voltage FISR is run to try to restore loads on all affected feeders. In this case, any predicted cuts placed outside the substation are ignored.

3. If there is no fault in the system and no predicted cuts inside the substation, but there are predicted cuts on the feeders outside the substation, feeder-based loss of voltage is run to try to restore loads on the impacted feeders.
4. If there is no fault or predicted cut in the system, FISR is rejected with the reason "Unsolved - Fault or Loss of Voltage Indication Not Found".

5.2.4. Fault and Loss of Voltage Triggered FISR on Sub-Fed Stations

For sub-fed stations, FISR will only be triggered once in the whole process, either on the parent station or the child station when a fault or loss of voltage occurs in one of the stations.

When a fault or loss of voltage occurs in the child station, the parent station is not impacted. Only the child station will see the fault or loss of voltage condition and trigger the FISR application.

When a fault or loss of voltage occurs in the parent station, both the parent and the child station(s) are impacted.

Fault Scenario

When a fault occurs in the parent station, the parent station will see the fault, and the child station will see the downstream voltage measurements change from their near nominal values to an approximately zero value.

The triggering flow is as follows:

1. If FISR is enabled in the parent station, the FISR fault trigger will be executed in the parent station. The resulting zero voltage measurement changes in the child station(s) will not trigger a loss of voltage in the child station(s) because the switch that has a voltage measurement change from a normal operation voltage to an effective zero value in the child station is de-energized. The de-energization is caused by a known fault or opened switch in the upstream parent station. Fault triggered FISR will isolate the fault and restore both the parent and the child station(s).
2. If the fault triggered FISR is disabled in the parent station, FISR will not be triggered in the parent station. In this case, the loss of voltage FISR trigger is executed in the child station(s) to restore power in the child station(s).

Loss of Voltage Scenario

When a loss of voltage occurs in the parent station, the parent and child station(s) may or may not see the voltage measurement changes depending on the loss of voltage location. Two distinct locations are possible:

- Inside the child station zone – This is the first parent station switchable section upstream of the

child station bounded by a telemetered switching device that has a voltage measurement.

- Outside the child station zone - This is the portion of the parent station that is upstream of the furthest downstream switching device that has a voltage measurement.

These locations are demonstrated in the figure below.

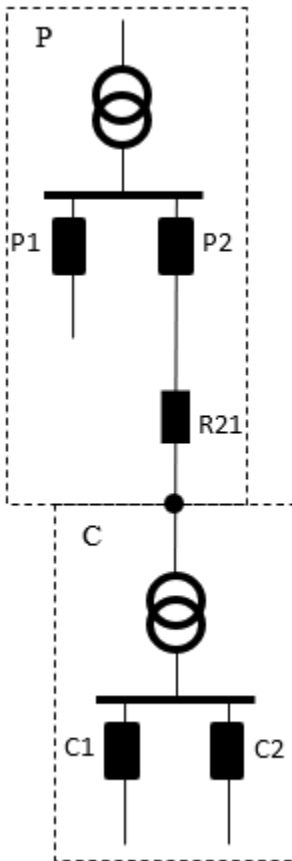


Figure 129. Schematic Diagram of Parent and Child Station

In this figure, feeder P2 in station P energizes station C. In the parent station P, the network upstream of R21 is the "outside the child station zone" location, and the network downstream of R21 is the "inside the child station zone" location. This is a radial subfed configuration where the parent station stop energize other feeders at child station boundary.

There is also other different configuration, for example, a feedthrough configuration where parent station continues provide energy to other feeders after child station. In those case, a fault happend "inside child zone" can still has impact on the rest of parent station, then the triggering flow described below is not fully applicable.

The triggering flow for radial subfed configuration is as follows:

1. If a loss of voltage occurs inside the child station zone (downstream of R21 in the figure above), the parent station cannot detect the zero-voltage measurement change because all voltage measurements associated with the switching devices in the parent station will not change. In this case, the loss of voltage can only be triggered in the child station.

2. If a loss of voltage occurs outside the child station zone (upstream of R21 in the figure above), both the parent and the child station(s) can detect the zero-voltage measurement change. In this case, the loss of voltage will be triggered in the parent station if FISR is enabled. However, if FISR is disabled in the parent station, the loss of voltage will be triggered in the child station if FISR is enabled in the child station.

Table below describes the expected behavior for the different parent and child station configurations and the FISR enable/disable status.

Table 30. Behavior for Different Parent and Child Station Configurations

FISR Ability		Scenario	Expected Behavior
Parent	Child		
Enabled	Enabled	Fault on parent station, outside child station zone	The parent station isolates the fault and restores itself and possibly the child's station/feeder(s).
Enabled	Enabled	Fault on parent station, inside child station zone	The parent station isolates the fault and restores itself and possibly the child's station/feeder(s).
Enabled	Enabled	Loss of voltage on parent station, outside child station zone	The parent station restores the de-energized feeders on its own network and (possibly) the child's station/feeders.
Enabled	Enabled	Loss of voltage on parent station, inside child station zone	The parent station cannot detect the loss of voltage. Loss of voltage triggered FISR is executed in the child station and restore its own network.
Enabled	Disabled	Fault on parent station, outside child station zone	The parent station solves FISR for its own network and (possibly) restores the child station/feeders.
Enabled	Disabled	Fault on parent station, inside child station zone	The parent station solves FISR for its own network and (possibly) restores the child station/feeders.
Enabled	Disabled	Loss of voltage on parent station, outside child station zone	The parent station solves loss of voltage FISR for its own network and (possibly) restores the child station/feeders.
Enabled	Disabled	Loss of voltage on parent station, inside child station zone	The parent station cannot detect the loss of voltage. Loss of voltage FISR is not triggered in the child station because FISR is disabled in the child station. No triggering files for parent and child station are created. There is message added to DNAF server log indicate the measurement is on disabled feeder or station if the trace level is set to "info".
Disabled	Enabled	Fault on parent station, outside child station zone	Fault triggered FISR in the parent station is executed, but then rejected for solving because FISR is disabled. The child station solves loss of voltage triggered FISR for itself.
Disabled	Enabled	Fault on parent station, inside child station zone	Fault triggered FISR in the parent station is executed, but then rejected for solving because FISR is disabled. The child station solves loss of voltage triggered FISR for itself.

FISR Ability		Scenario	Expected Behavior
Disabled	Enabled	Loss of voltage on parent station, outside child station zone	Loss of voltage FISR is not trigger in parent station because FISR is disabled. No triggering file for parent station is created. There is message added to DNAF server log indicate the measurement is on disabled feeder or station if the trace level is set to "info". The child station solves loss of voltage triggered FISR for itself.
Disabled	Enabled	Loss of voltage on parent station, inside child station zone	The parent station cannot detect the loss of voltage. The child station solves loss of voltage triggered FISR for itself.
Disabled	Disabled	Fault on parent station, outside child station zone	Fault triggered FISR on the parent station is executed, but then rejected for solving because FISR is disabled. The child station does not execute loss of voltage triggered FISR for itself because FISR is disabled.
Disabled	Disabled	Fault on parent station, inside child station zone	Fault triggered FISR in the parent station is executed, but then rejected for solving because FISR is disabled. The child station does not execute loss of voltage triggered FISR for itself because FISR is disabled.
Disabled	Disabled	Loss of voltage on parent station, outside child station zone	Loss of voltage FISR is not triggered in either parent or child station because FISR is disabled for both stations. No triggering file for either station is created. There is message added to DNAF server log indicate the measurement is on disabled feeder or station if the trace level is set to "info".
Disabled	Disabled	Loss of voltage on parent station, inside child station zone	The parent station cannot detect the loss of voltage. Loss of voltage FISR is not triggered in child station because FISR is disabled. No triggering file for either station is created. There is message added to DNAF server log indicate the measurement is on disabled feeder or station if the trace level is set to "info".
Enabled	Enabled	Fault on child station	Fault triggered FISR only is executed in the child station and restore its own network.
Enabled	Enabled	Loss of voltage on child station	Loss of voltage triggered FISR only is executed in the child station and restore its own network.
Enabled	Disabled	Fault on child station	Fault triggered FISR in the child station is executed, but then rejected for solving because the feeder is disabled.
Enabled	Disabled	Loss of voltage on child station	Loss of voltage FISR is not triggered in child station because FISR is disabled. No triggering file is created. There is message added to DNAF server log indicate the measurement is on disabled feeder or station if the trace level is set to "info".
Disabled	Enabled	Fault on child station	Fault triggered FISR only is executed in the child station and restore its own network.
Disabled	Enabled	Loss of voltage on child station	Loss of voltage triggered FISR only is executed in the child station and restore its own network.

FISR Ability		Scenario	Expected Behavior
Disabled	Disabled	Fault on child station	Fault triggered FISR in the child station is executed, but then rejected for solving because FISR is disabled.
Disabled	Disabled	Loss of voltage on child station	Loss of voltage FISR is not triggered in child station because FISR is disabled. No triggering file is created. There is message added to DNAF server log indicate the measurement is on disabled feeder or station if the trace level is set to "info".

5.2.5. FISR Consistency Checks

FISR requires a certain amount of time to solve. However, crucial system changes may occur during the solution. In such cases, the generated switching plans may not be compatible with the new system conditions. To handle this situation and ensure that the plans are valid, FISR performs two different checks:

- **Switching Consistency**

FISR compares the state of all switches in the network selection before and after the FISR solution. If the state of any of these switches changes, each effected plan in the FISR solution is marked to indicate that topology changed, and the plan is not implemented in closed-loop mode.

For cases where topology changes occurred in the network selection during a FISR solution, the FISR Completed alarm (plan is invalid because of topology changes) will only be issued if all plans in the solution are marked as having a topology change.

Note

Depending on the configuration, the FISR network selection may include feeders that are a number of adjacency steps away from the fault or loss of voltage. This means that any switching changes in the same part of the system, even if not associated with the FISR solution, may cause this check to trigger.

- **Fault Indicator Consistency**

For permanent fault triggering, FISR also compares any changes in the state of relevant fault indicators. The initial state is noted at the time the fault is placed. The final state is noted after the FISR solution is computed. The fault indicators checked are those in the path between the fault and supply sources. Fault Indicators on neighboring feeders or other nearby parts of the system are not considered. For information on multiple faults related to this case, see [Multiple Faults](#).

If any of the evaluated fault indicators changes from Unset to Set during the comparison period, each plan in the FISR solution is marked to indicate that the fault indicator state changed. Such plans are not implemented in closed-loop mode.

5.2.6. Stop Plan Implementation on New Fault

If a new fault event occurs during FISR plan execution, the remaining steps in the plan will be cancelled and the plan stopped. A new fault event is defined as a new fault target set in the impacted network for each switching step.

At the end of the FISR solution, all the valid set phases (phases A, B, C) and set reverse phases (phases A, B, C) of all the fault indicators for every switching step are found and saved. These fault indicators have the following characteristics:

- The same dynamic feeder as the device of the switching step, or
- Located upstream of the switching step, which may be in the station internals or an adjacent station.

Before implementing each switching step in the Plan Manager subsystem, the found fault indicators are checked. If there is any new phase (phases A, B, C) set, indicating there is a new fault, the plan will be canceled and a SCADA alarm will be issued (configuration option).

5.3. Feeder Fault Performance

The Series 3 ADMS Fault Isolation and Service Restoration (FISR) application supports the following event triggers:

- Permanent fault on a feeder (outside the station fence)
- Permanent fault on station internal equipment such as the station power transformer
- Loss of supply (voltage) along a feeder
- Loss of supply (voltage) for a power transformer or a station (a station may have multiple power transformers)

The most common usage of the FISR application is for permanent faults on a feeder. For a feeder fault, FISR must generate switching plans that isolate the faulted section of the feeder network and restore service to as many customers in the healthy section(s) of the network utilizing adjacent "donor" feeders. If the closed-loop plan execution mode is enabled, the Plan Manager subsystem is responsible for executing the switching (and protection related) steps via the Supervisory Control and Data Acquisition (SCADA) system in sequential order, confirming correct operation for each step before issuing the next control.

One metric for measuring the FISR application feeder fault solution performance is the total elapsed time between a protective device lockout (breaker or recloser) to the time when the FISR solution has been completed and the top-ranked set of plans have been generated. This time is highly dependent on a number of different factors including model size and quality, power flow solution quality, hardware capability, number of available control devices, and many FISR application configuration parameters.

Many customers prefer that the feeder fault solution performance time also includes the time to fully implement the top-ranked plan in the field (all device controls have been issued, executed

successfully, and confirmed). However, this time cannot be guaranteed by GE since there is no way to account for field device control performance including the associated communications network and any associated network/communication delays.

5.3.1. Performance Factors

There are two major groups of factors that impact the overall FISR performance: FISR Solution Algorithm, and Plan Implementation. GE has some control over the FISR Algorithm group, but GE has very little to no control over the Plan Implementation group which is tightly coupled with the customer's SCADA communication network.

5.3.1.1. Solution Algorithm Factors

Several key FISR Solution Algorithm performance factors are summarized in the table below. This table highlights several of the most critical performance factors that impact the time needed by FISR to generate the fault isolation and restoration switching plans. There may be other algorithm related factors that impact performance, but the major ones are listed in this table.

Table 31. Solution Algorithm Performance Factors

Performance Factor	Description/Comment
Solution Quality	The FISR algorithm utilizes the Distribution Power Flow (DPF) with Bus Load Allocation (BLA) to verify the quality of a proposed switching plan, specifically looking for branch overloads and high/low bus/load voltages. If the DPF/BLA solution quality is poor, more BLA and/or DPF iterations are required for each solution, increasing the overall solution time.
Model Complexity	The complexity of the network model, including the number and location of Distributed Energy Resources (DER), and the number of locally controlled devices including tap changing transformers, capacitors, and DERs (with/without autonomous modes), will directly impact the power flow solution time. Since power flow is called multiple times within a FISR solution, model complexity and solution quality will have a large impact on the overall FISR performance.
Network Size	Using the DNAF configuration Editor (DCE), the user can configure the number of tie-point adjacency levels. The adjacency levels determine the FISR network selection which is also known as the study area. Higher adjacency levels (level 2) will increase the size of the network that is evaluated, increasing the internal power flow solution times and the overall FISR solution time. Sub-fed configurations (a higher voltage distribution station feeding one or more lower voltage distribution stations) can also dramatically increase the size of the network that is solved.
Look-Ahead Interval	The FISR pre-processing code finds the single time interval where the loading level is the highest. A longer look-ahead window has more time intervals which requires more loading evaluations.
Number of Plans Generated	The number of plans that are internally generated and fully evaluated will impact the overall solution time. This may require more Protection Validation (PRV) evaluations (hence more power flow solutions).

Performance Factor	Description/Comment
Automatic Switch Count (Eligible Control Devices)	For closed loop FISR, the plan generation logic only utilizes automated switching devices. With more automated switching devices in the network, there are more potential plans that can be generated, increasing the number of internal power flow solutions and the FISR solution time.
Protective Device Count/ Protection Checks	Many customers utilize the PRV application internal checks within the FISR solution (enabled using a DCE problem formulation Performance Index, specifically Protection Validation). The PRV application simulates multiple faults for each application protection area ("zone") which is formed based on the location of breakers and reclosers. Each fault simulation requires a power flow solution. The PRV analysis can utilize one or two fault impedance values. Evaluating two fault impedances doubles the number of power flow solutions. There is a PRV performance improvement option that will calculate the maximum fault current using only a bolted fault, and the phase/ground fault simulations are performed using only the user-configured fault impedance.
Adaptive Protection	The adaptive protection (settings group check) that utilizes a pre-configured look-up table takes the results from the internal network application topology processor to determine the appropriate settings group based on the energizing station and feeder for each affected recloser. This method does not require much additional processing and so the performance is reasonably fast. The minimum fault current adaptive protection requires multiple PRV executions, including solutions done within the plan generation logic itself so that the correct continuous current limit is checked, and the appropriate settings group is applied based on the pre-defined settings group phase trip settings. Additional PRV executions require additional power flows which increases the overall FISR solution time.
Block DER Connection	For a large DER site, sometimes the DER site needs to be blocked from closing back in on a circuit that cannot handle its load after reconfiguration. After the initial creation of a reconfiguration plan, the current status of a DER site whose capacity is larger than a certain configurable threshold will be checked. If the site is connected, then the connecting recloser needs to be opened first to disconnect the DER site. Once the recloser is open and the DER site is disconnected from the network, a command to prevent normal close control will be sent to the recloser to block the DER site from reconnecting back to the network. This logic adds steps to the final plan and increases the overall solution time.
Change Voltage Regulation Mode	When enabled, voltage regulators in the restoration area that have advanced regulation ability will be checked to see if their voltage regulation mode needs to be changed. The voltage regulation mode change will be added in the final restoration plan as a separate plan step. To maintain voltage on the regulation side, the voltage regulation mode change step always appears before the switching steps in the plan. This logic requires additional checks of the current and power directions, increasing the overall solution time.
Additional Network Checks	Check Before Operate (CBO, aka Pre-Switching Validation) checks including Ferroresonance, 3-Wire/4-Wire, and Paralleling Circuits with Reclosing Capable Protective Devices, also require more processing time in the performance index calculation code.
Integration with LVM	The FISR application can be configured to internally solve the Load and Volt/VAR Management (LVM) application to solve voltage violations that may occur in the new topology (after a switching plan has been implemented). This LVM solution requires more evaluation time (the optimization based LVM also internally solves multiple power flows).

Performance Factor	Description/Comment
Performance Improvement Options	The network reduction algorithm is able to reduce "normal" laterals to a lumped load equivalent that still accounts for the voltage drop from the lateral tap fuse to the lowest downstream voltage location. Enabling this option effectively reduces the size of the network that must be evaluated in each power flow solution, including power flows that are part of the PRV and LVM applications.
Result File	The FISR result file can be configured to show a lot of detailed (analyst) information. This requires more time to write out a larger, more detailed file.

5.3.1.2. Plan Implementation Factors

The Plan Implementation related performance factors are listed in the table below. Note that GE has no control over the speed of the communications network (second item in the table), and thus **GE cannot provide any guarantee on the plan execution time.**

Table 32. Plan Implementation Performance Factors

Performance Factor	Description/Comment
Parallel Switching Sequences	In the DCE, an option is available that allows a multi-step switching plan to be divided up into independent sub-plans and executed in parallel. This helps with restoring the maximum number of customers when there is a plan step failure (which prevents any further switching plan operations). The execution of parallel plans may reduce the overall plan implementation time.
Communication Network	The performance of the customer's communication network will impact the overall plan implementation speed. Many customers have a cellular/radio network to monitor and control the feeder devices. If a plan with many steps is to be implemented, a significant amount of time may be required to issue the control and confirm the device response. GE has no control over the communication network and its performance characteristics/capabilities, and thus FISR performance times (including plan execution duration) cannot be guaranteed.
Maximum Execution Attempts	There are options that allow for a failed control, either switching device or setting group controls, to be executed multiple times. However, this will add to the overall time required to implement the FISR switching plan. The re-execution is based on a global timeout parameter (in seconds), or the timeout specified on the individual SCADA control measurement.

5.3.2. Performance Analysis

There are numerous configuration options that directly impact the FISR solution performance. The key assumption made for this analysis is that the "maximum FISR DPF required functionality" is enabled as applicable when obtaining the overall FISR solution times. Due to the Fantasy Island model's limited Adaptive Protection modeling and sub-fed network selection configurations, some of the tables and graphs provided in this section were obtained using customer models.

Unless otherwise noted, the following parameters were specified to determine the performance

numbers that are provided later in this section.

1. The size of the distribution network that is being analyzed in a single FISR analysis, also known as the network selection, is one of the largest factors impacting the overall solution performance of the FISR application. The open switch (tie point) adjacency is set in the DCE Switching Reconfiguration Problem Formulations page in the Switching Algorithm Parameters group in the Switching Cost section:
 - Number of Adjacency Levels (2, default is 1)
 - Automatic Switch Cost (2 for level 2, 1 for level 1)
2. Plan generation is limited to the three top selected plans, with a maximum of three internally generated plans. The FISR solution will require more time as more plans need to be generated and evaluated. This is covered with the following DCE FISR page parameters in the Switching Plan Generation Parameters group in the Switching Plan Generation Parameters section:
 - Number of Selected Top Plans (3)
 - Maximum Number of Internally Generated Plans (3)
3. Both intermediate and post-plan PRV Analyses are enabled, including adaptive protection using settings group templates defined in the static model. This is covered with the following DCE FISR page parameter in the Plan Execution group in the Permanent Fault Trigger section:
 - Settings Group Check (By Minimum Fault Current)
4. The post-plan PRV analysis is enabled with the following DCE parameters in the Switching Reconfiguration Problem Formulations in the Switching Algorithm Parameters group in the Protection and Performance Index Weights sections:
 - Protection Validation (default is 10, value is greater than 0)
 - Protection Reach (default is 10, value is greater than 0)
 - Peak Load Safety Factor (default = 1.5, value is greater than 0)
 - Reach and Neutral Safety Factor (default = 1.5, value is greater than 0)
5. Other PRV evaluation options that will impact the overall PRV solution performance are configured with the DCE Switching Reconfiguration Problem Formulations page in the Problem Formulation Parameters group in the Analysis Options and Adaptive Relaying sections:
 - Perform the Minimum Number of Fault Evaluations (checked, the base PRV Fault Impedance Evaluation option must be set to Evaluate a Bolted Fault and the User Configured Fault Impedance)
 - Compare Phase Trip Setting with the Minimum of the Minimum Phase-to-Phase and Minimum Phase to Ground Fault Current Values (option selected)
6. For this analysis, one key FISR network size reduction performance option is enabled, which is configured in the DCE Switching Reconfiguration Problem Formulations in the Problem Formulation Parameters group in the Analysis Options section:
 - Use Network Reduction Algorithm (checked)
7. There are many additional configuration DCE parameters that may impact the overall FISR solution time including the following items listed below:

- Maximum Real Power Mismatch (kW) (2.000, FISR page, Solution Formulation group under the Convergence Criteria section)
- Maximum Reactive Power Mismatch (kVAR 2.000, FISR page, Solution Formulation group under the Convergence Criteria section)
- Maximum Voltage Mismatch (p.u.) (0.0001, FISR page, Solution Formulation group under the Convergence Criteria section)
- Change Voltage Regulation Mode (unchecked, FISR page, Solution Formulation group under the Switching Plan Generation Parameters section)
- Integrate with LVM (unchecked, FISR page, Solution Formulations group under the Other section)
- Use Detailed Solution Reporting (unchecked, FISR page, Result Files group under the Solution Results section) – the number of high voltage, low voltage, and branch loading values configured in the associated sub-options may also impact performance
- Consider Cold Load Pickup (unchecked, Switching Reconfiguration Problem Formulations page, Problem Formulation Parameters group in the Analysis Options section)
- Block DER Connection (unchecked Switching Reconfiguration Problem Formulations page, Problem Formulation Parameters group in the Analysis Options section)
- Enable Ferroresonance Switching Validation Check (unchecked, Switching Reconfiguration Problem Formulations page, Problem Formulation Parameters group in the Analysis Options section)
- Enable 3-Wire/4-Wire Switching Validation Check (unchecked, Switching Reconfiguration Problem Formulations page, Problem Formulation Parameters group in the Analysis Options section)
- Enable Paralleling Circuits with Reclosing Capable Protective Devices Switching Validation Check (unchecked, Switching Reconfiguration Problem Formulations page, Problem Formulation Parameters group in the Analysis Options section)
- Branch Flow Limit for DER Delay (Level 2, Switching Reconfiguration Problem Formulations page, Limit Strategy group in the Branch Flow Limits section)
- Enforce Station Transformer Backfeed Limit (unchecked, Switching Reconfiguration Problem Formulations page, Limit Strategy group in the Backfeed section)
- Adjust Limits Based on Reserved Capacity (unchecked, Switching Reconfiguration Problem Formulations page, Limit Strategy group in the Reserved Capacity section)

The Simulator, aka Distribution Operator Training Simulator (DOTS), does not have a control time delay. As soon as a control is issued via the ADMS Plan Manager subsystem, SCADA will implement the control and the status confirmation will be received almost instantaneously. For this reason, **the times recorded in this section are only for the FISR plan generation time.**

The FISR generated plan quality, either invalid or valid (due to power flow or protection results), is not important in this analysis. The important item is the total plan generation time.

The FISR Triggering method is also not important. Several larger, more advanced distribution

models were evaluated utilizing customer save cases and FISR was manually executed in study mode.

5.3.2.1. Hardware

The FISR solution times recorded in this section were compiled using a laptop with the following specifications:

- **Windows:** Windows 10 Enterprise
- **CPU:** Intel® Core™ i7-9750H CPU @ 2.60GHz, 6 cores, 12 logical processors
- **RAM:** 32 GB

5.3.2.2. Fantasy Island Model Node and Device Collection Sizes

The size of several Fantasy Island stations that were used in determining the FISR solution times are provided in the table below. The key node and device collection counts are listed.

Station	ND + BUS	FDR	LN	LD	GEN + ER	TF + REG	CAP	SWT	REFB
Bethpage	9289	9	4040	2491	5	2523	32	4119	2
Canterbury	3961	2	1671	1009	2	1133	8	1617	1
Coho	12753	6	5598	3298	1	3314	22	5381	2
Muirfield	5712	9	2542	1628	15	1568	16	2373	3
Pebble Beach	6503	6	2739	1776	1	1701	13	2793	2
Raven	11460	5	4143	2447	1	2436	19	5537	2
Royal Troon	19485	8	7685	4495	0	4594	42	9637	3
Winged Foot	12809	6	5158	3113	0	3115	25	5759	2

Legend:

ND – Node

BUS – Bus (voltage limit monitoring location)

FDR – Feeder

LN – Line

LD – Load

GEN – Generator

ER – Energy Resource

TF – Transformer (all types including inline transformers and distribution transformers)

REG - Regulator

CAP – Shunt Capacitor

SWT – Switching Device

REFB – Reference Bus Interface

5.3.3. Fantasy Island Performance Results

The Fantasy Island model has a limited number of stations that have automated switching devices. For this reason, there is a limited number of test cases provided in the following table (all test cases in this section were configured to generate three plans).

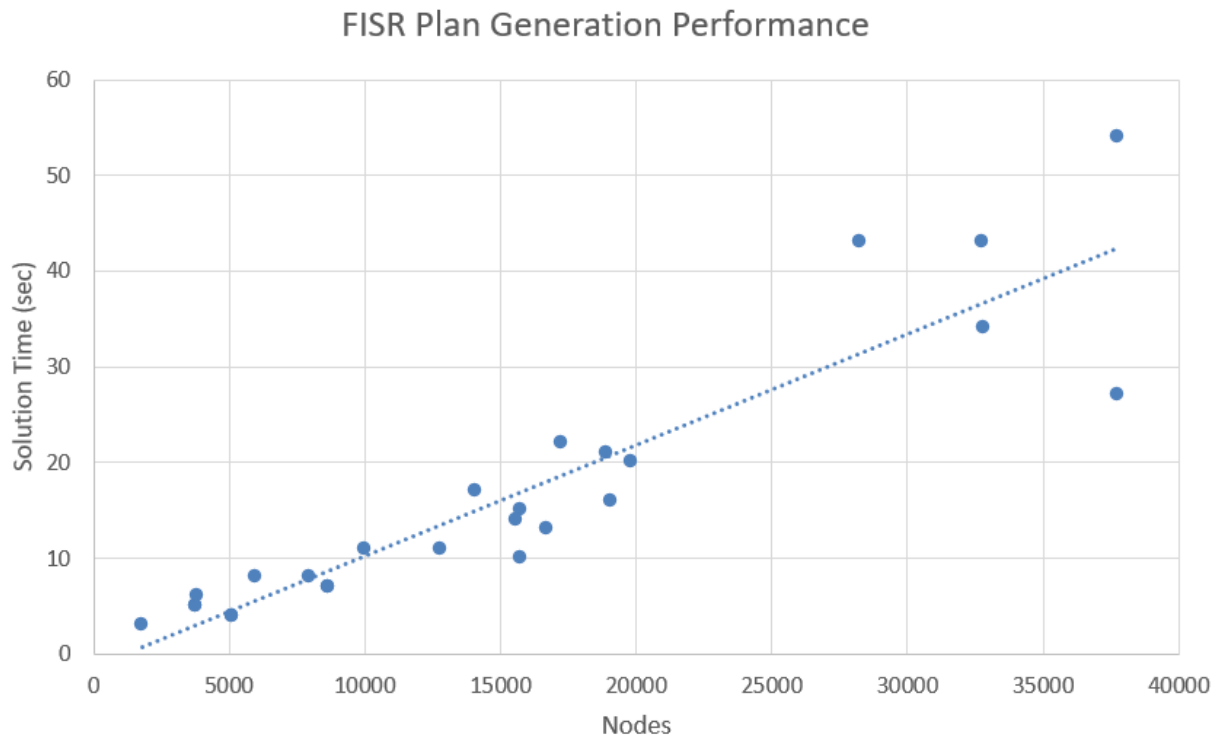
Case Number	Initiating Station	Faulted Line Section Name	Adjacency Level	Nodes	Dist. Tf	Loads	Switching Devices	Solution Time (Sec)
1	Bethpage	22278713	1	5126	1582	1598	33	4
2	Bethpage	22278713	2	8688	2535	2565	63	7
3	Bethpage	22271212	1	1788	523	565	18	3
4	Bethpage	22271212	2	3806	1163	1134	30	5
5	Bethpage	222758985	2	3805	1163	1134	30	5
6	Bethpage	22271062	2	3809	1163	1134	30	6
7	Bethpage	22341424	2	10010	2472	2499	29	11
8	Bethpage	22273973	2	19079	4918	4952	72	16
9	Bethpage	22338792	2	18891	4663	4664	89	21
10	Bethpage	104294856	2	19810	5191	5239	99	20
11	Muirfield	22354331	2	15580	3793	3827	71	14
12	Muirfield	22349388	2	5995	1864	1882	44	8
13	Raven	22385224	2	32793	7866	7695	42	34
14	Raven	22385224	2	14094	3265	3367	33	17
15	Raven	22382416	1	15744	3754	3858	25	10
16	Raven	22382416	2	37779	9101	8973	49	27
17	Raven	22455220	1	15744	3754	3858	25	15
18	Raven	22455220	2	37779	9101	8973	49	54
19	Raven	22464845	2	12776	3037	3062	22	11
20	Raven	22460434	2	28277	6856	6665	35	43
21	Raven	22460434	2	32789	7866	7695	41	43
22	Raven	22460434	2	17264	4025	4179	36	22
23	Royal Troon	126569516	2	16721	4026	4088	66	13
24	Vail	9617872	2	7949	2253	2215	26	8

The solution times in Fantasy Island ranged from 3 to 54 seconds with a corresponding node model size ranging from 1788 to 37779 nodes.

With Fantasy Island being a test model, the overall model quality is very good. This results in reasonably fast FISR solution times.

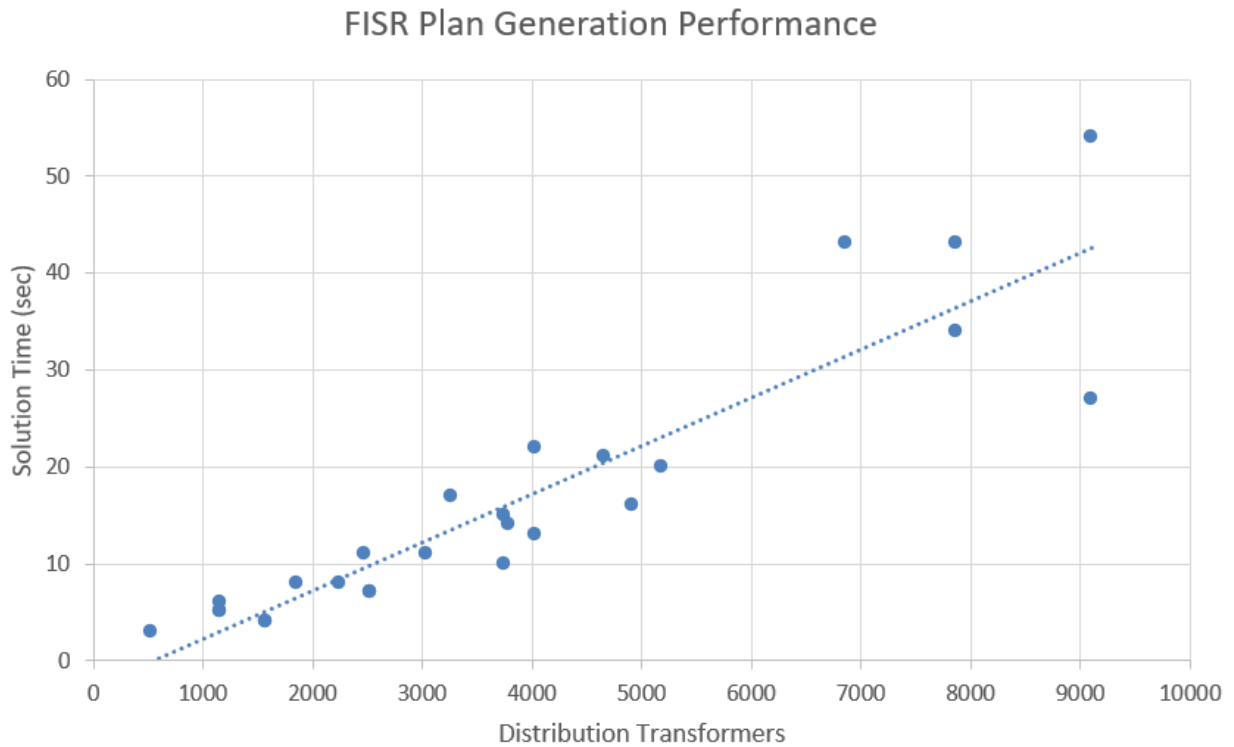
5.3.3.1. Nodes vs. Solution Time

The plot of node count (horizontal axis) vs FISR solution time in seconds (vertical axis) is displayed in the following scatterplot with a linear trendline. The solution performance is reasonably linear with the number of nodes in the analyzed network (network selection).



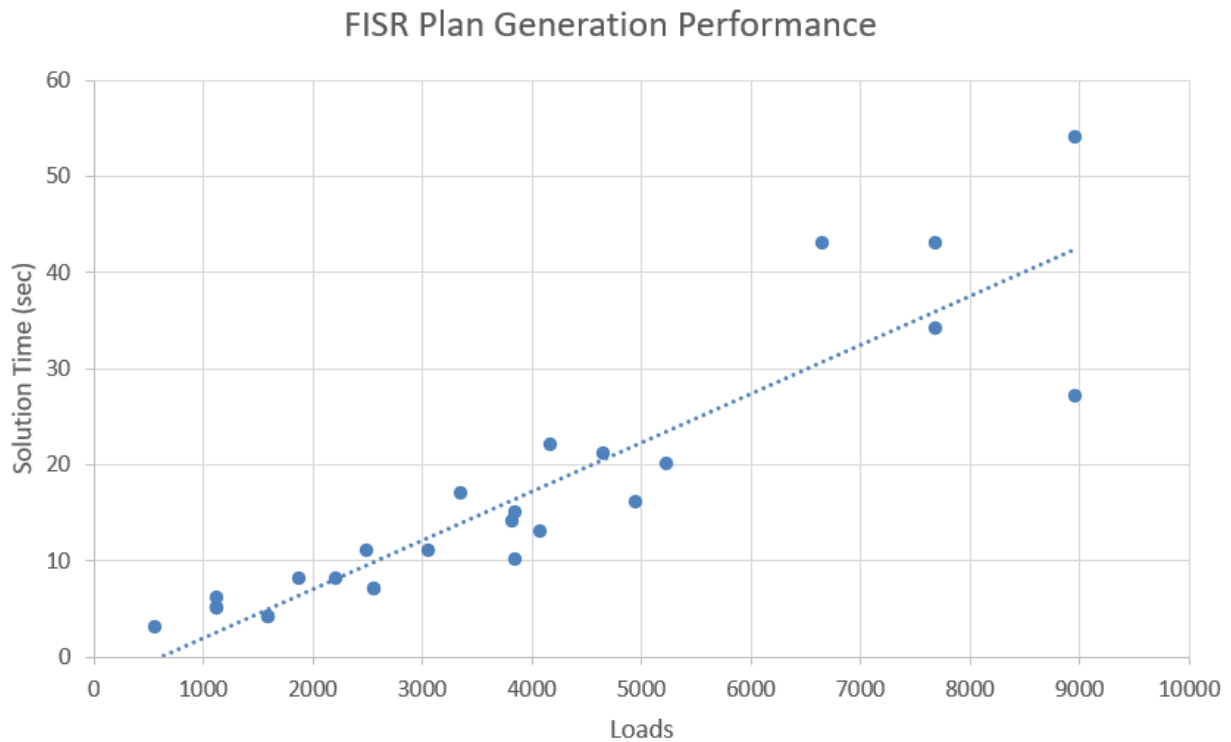
5.3.3.2. Distribution Transformers vs. Solution Time

The plot of distribution transformer count (horizontal axis) vs FISR solution time in seconds (vertical axis) is displayed in the following scatterplot with a linear trendline. The solution performance is reasonably linear with the number of distribution transformers in the analyzed network (network selection).



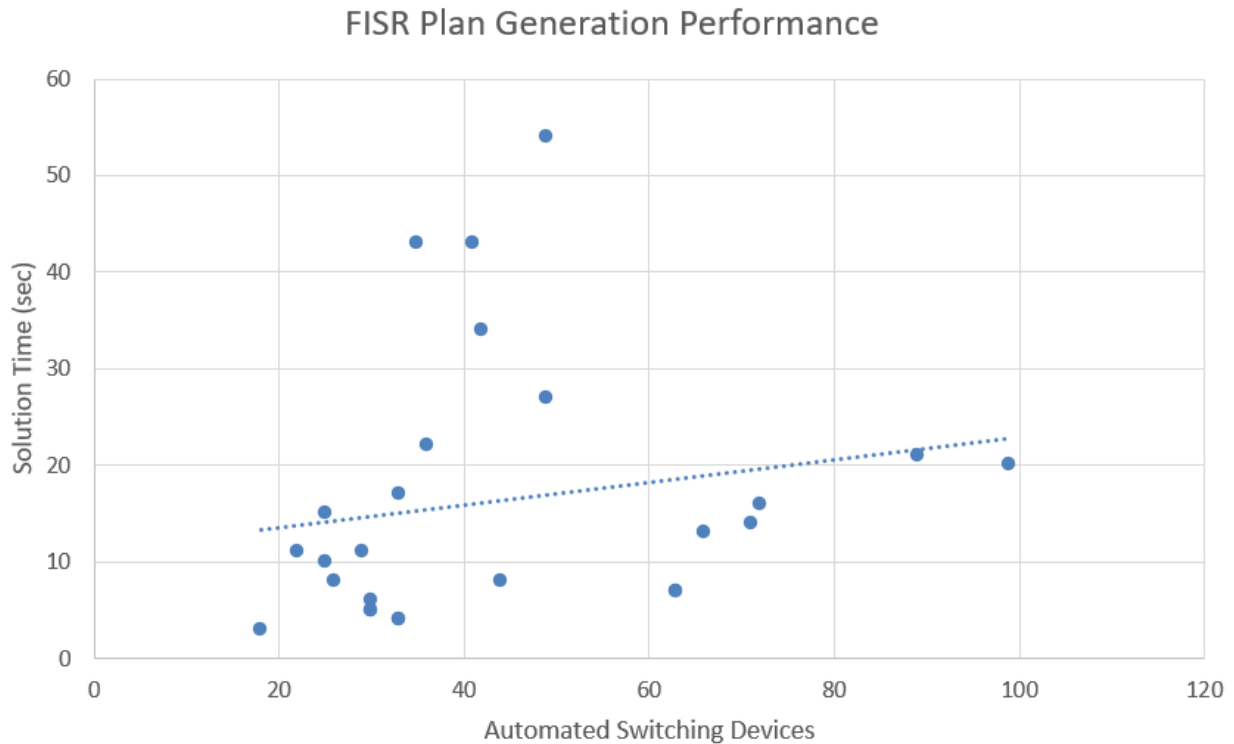
5.3.3.3. Loads vs. Solution Time

The plot of load count (horizontal axis) vs FISR solution time in seconds (vertical axis) is displayed in the following scatterplot with a linear trendline. The solution performance is reasonably linear with the number of loads in the analyzed network (network selection).



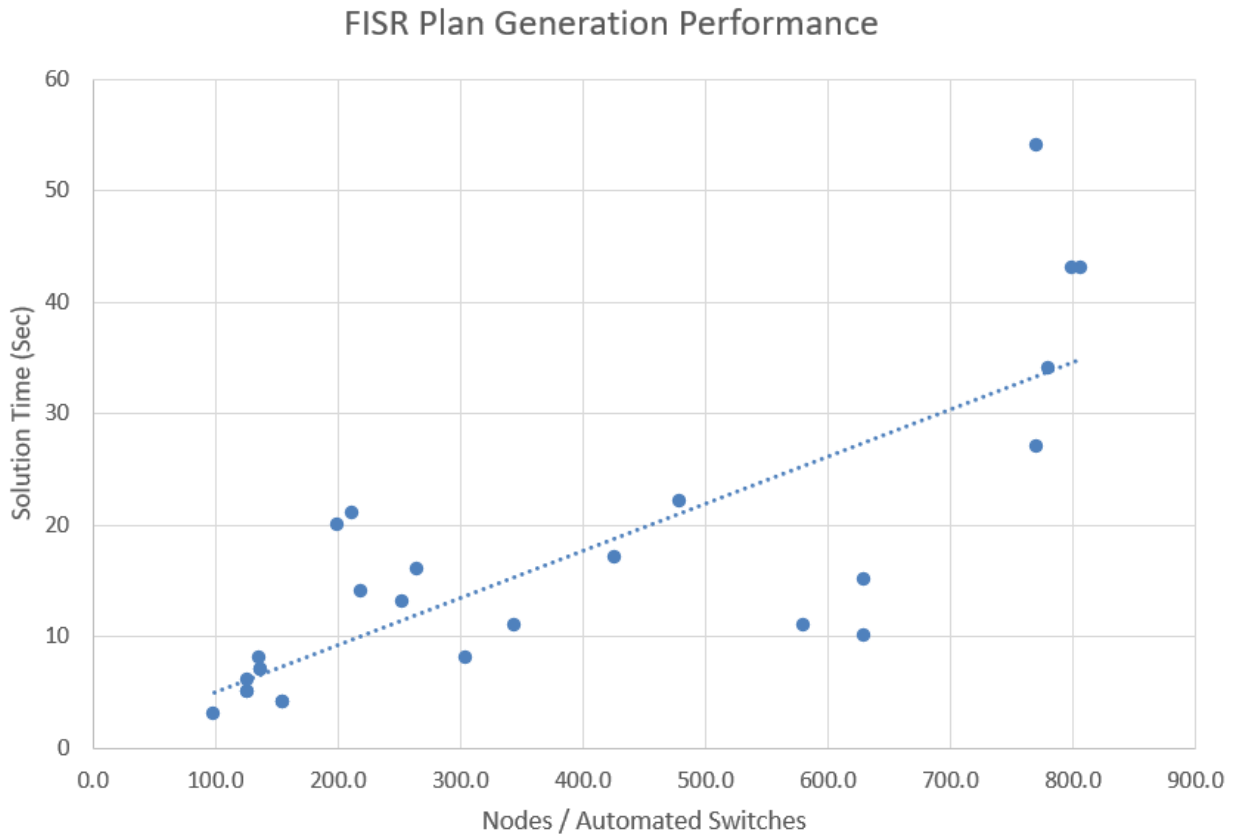
5.3.3.4. Switching Devices vs. Solution Time

The plot of switching device count (horizontal axis) vs FISR solution time in seconds (vertical axis) is displayed in the following scatterplot with a linear trendline. Unlike the node, distribution transformer, and load counts, the relationship between the number of automated switching devices and FISR solution time is not linear (other factors such as the node size also impact the solution time).



5.3.3.5. Section Size vs. Solution Time

The plot of nodes divided by automated switching devices (horizontal axis) vs FISR solution time in seconds (vertical axis) is displayed in the following scatterplot with a linear trendline. This is an attempt to incorporate the average size of the switchable section bounded by the automated switching devices which here is defined as the node count divided by the number of automated switching devices.



5.3.4. Customer Model Performance Results

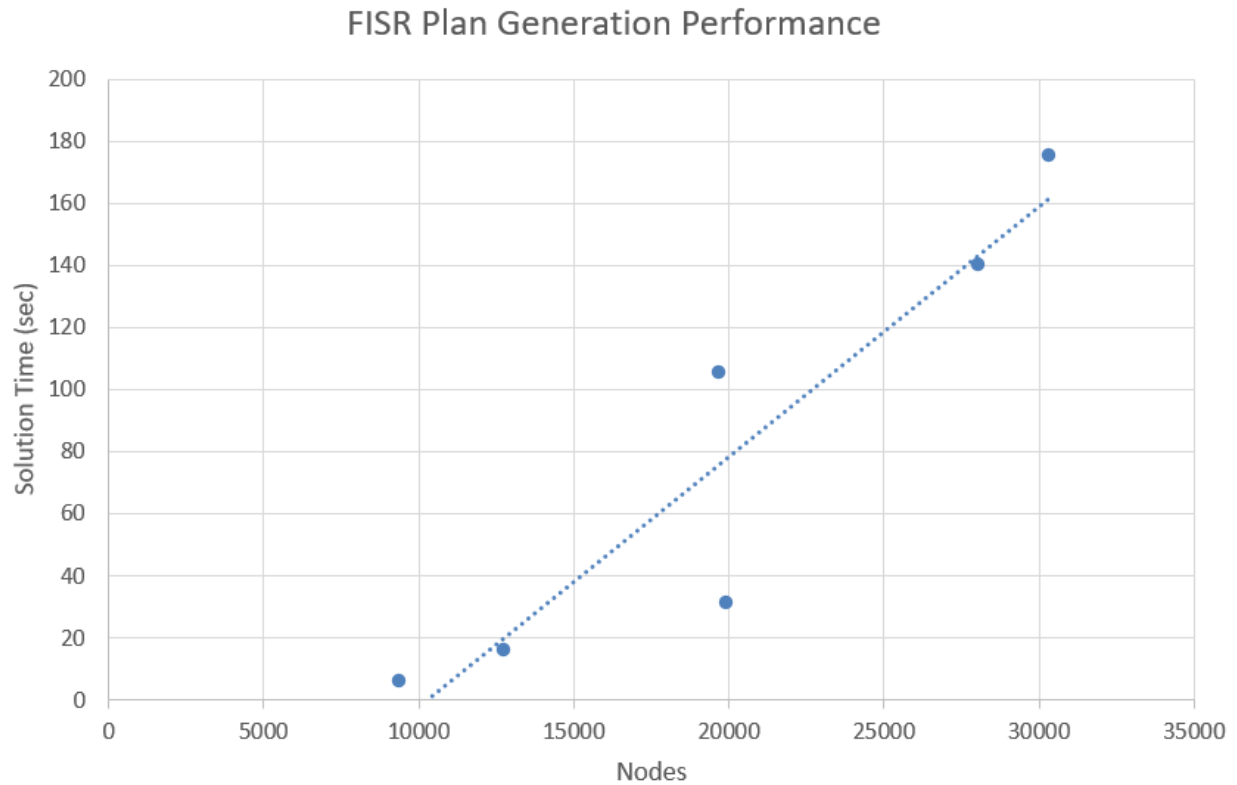
The results from a customer model are summarized in this section. The network size is slightly larger than the Fantasy Island model, but the solution times are longer due to adaptive protection modeling and more switching plans being generated.

The number of test cases is much smaller than what was presented in the Fantasy Island Performance Results section. The following table summarizes the key model data and solution times (all cases generated 10 plans and settings group/adaptive protection logic was enabled).

Case Number	Adjacency Level	Nodes	Dist. Tf	Loads	Switching Devices	Solution Time (Sec)
1	1	28089	7333	7229	149	140
2	2	30351	7977	7863	150	175
3	1	9408	2312	2306	12	6
4	2	19942	5863	5743	109	31
5	2	19694	5014	4949	74	105
6	2	12790	3734	4061	62	16

5.3.4.1. Nodes vs. Solution Time

The plot of node count (horizontal axis) vs FISR solution time in seconds (vertical axis) is displayed in the following scatterplot with a linear trendline. Similar to the Fantasy Island model, solution performance is reasonably linear with the number of nodes in the analyzed network (network selection).



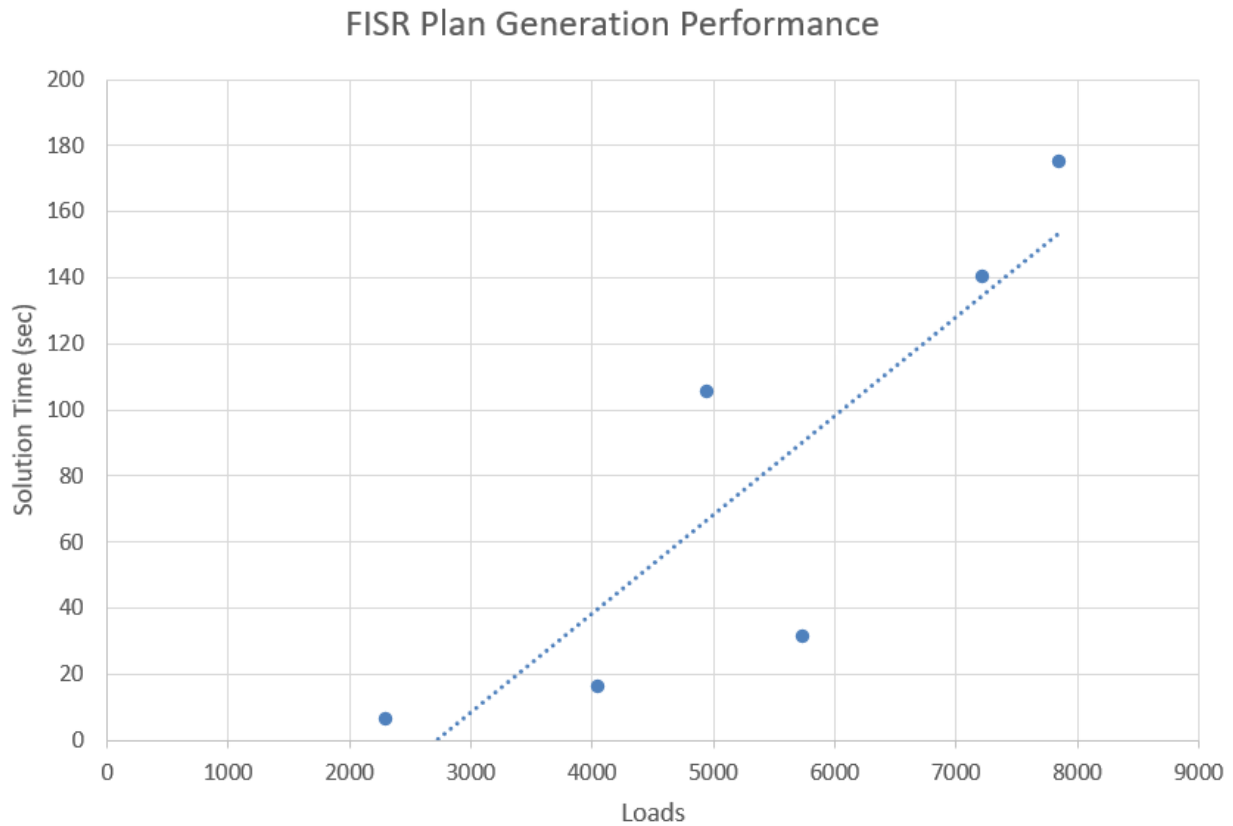
5.3.4.2. Distribution Transformers vs. Solution Time

The plot of distribution transformer count (horizontal axis) vs FISR solution time in seconds (vertical axis) is displayed in the following scatterplot with a linear trendline. Similar to the Fantasy Island model, solution performance is reasonably linear with the number of distribution transformers in the analyzed network (network selection).



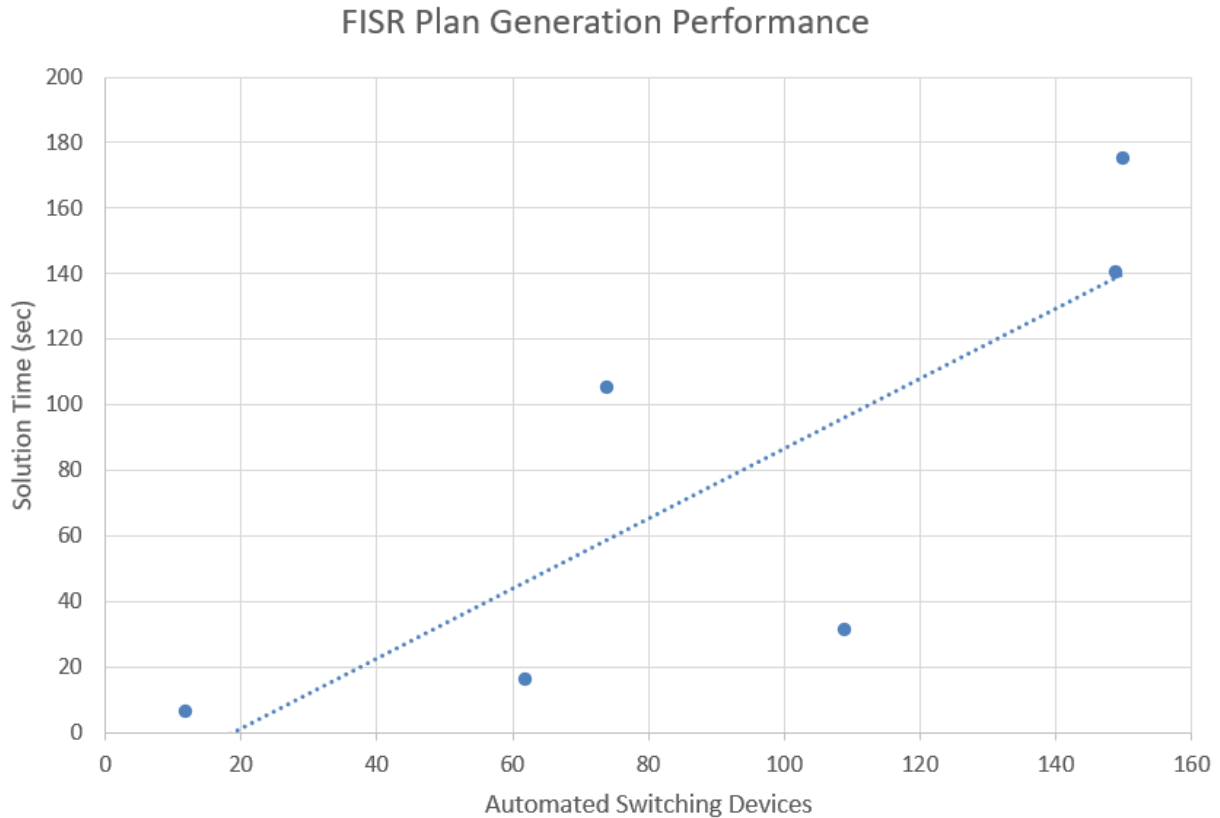
5.3.4.3. Loads vs. Solution Time

The plot of load count (horizontal axis) vs FISR solution time in seconds (vertical axis) is displayed in the following scatterplot with a linear trendline. Similar to the Fantasy Island model, solution performance is reasonably linear with the number of loads in the analyzed network (network selection).



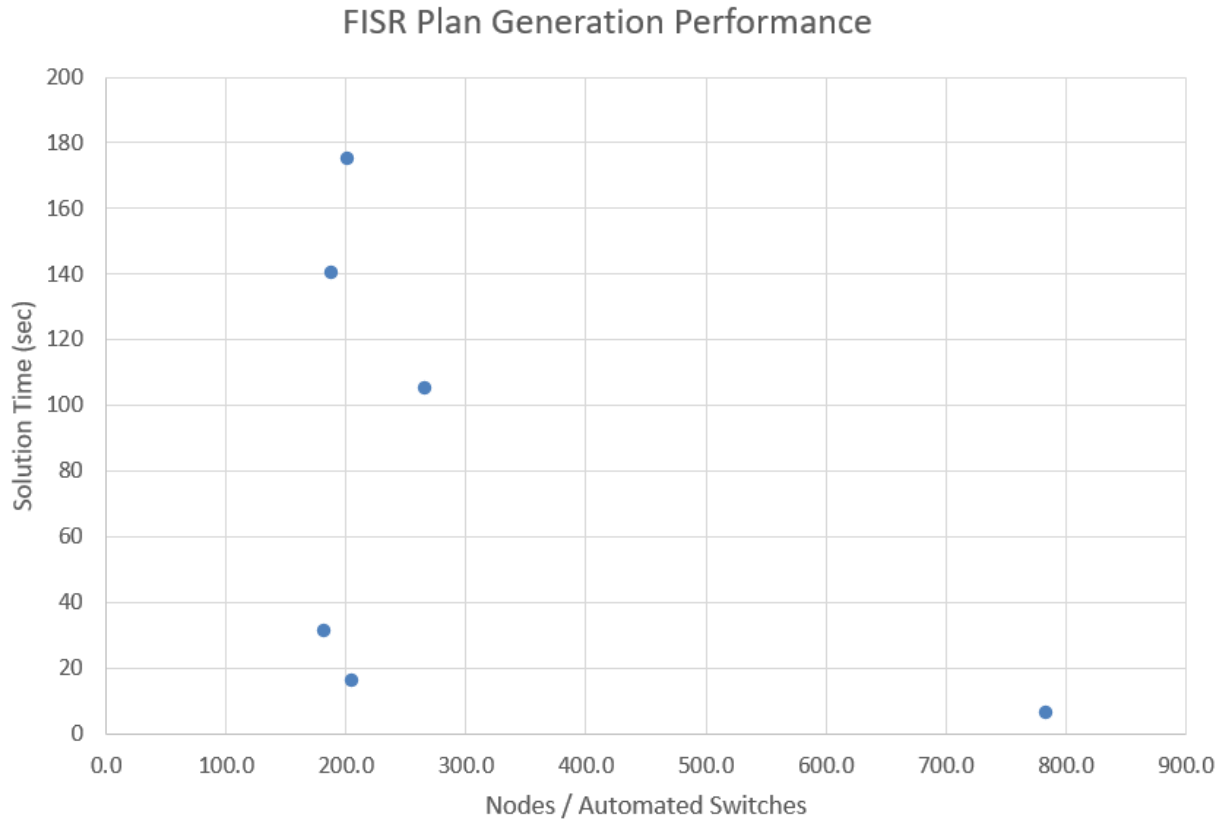
5.3.4.4. Switching Devices vs. Solution Time

The plot of switching device count (horizontal axis) vs FISR solution time in seconds (vertical axis) is displayed in the following scatterplot with a linear trendline. Unlike the node, distribution transformer, and load counts, the relationship between the number of automated switching devices and FISR solution time is not linear (other factors such as the node size also impact the solution time).



5.3.4.5. Section Size vs. Solution Time

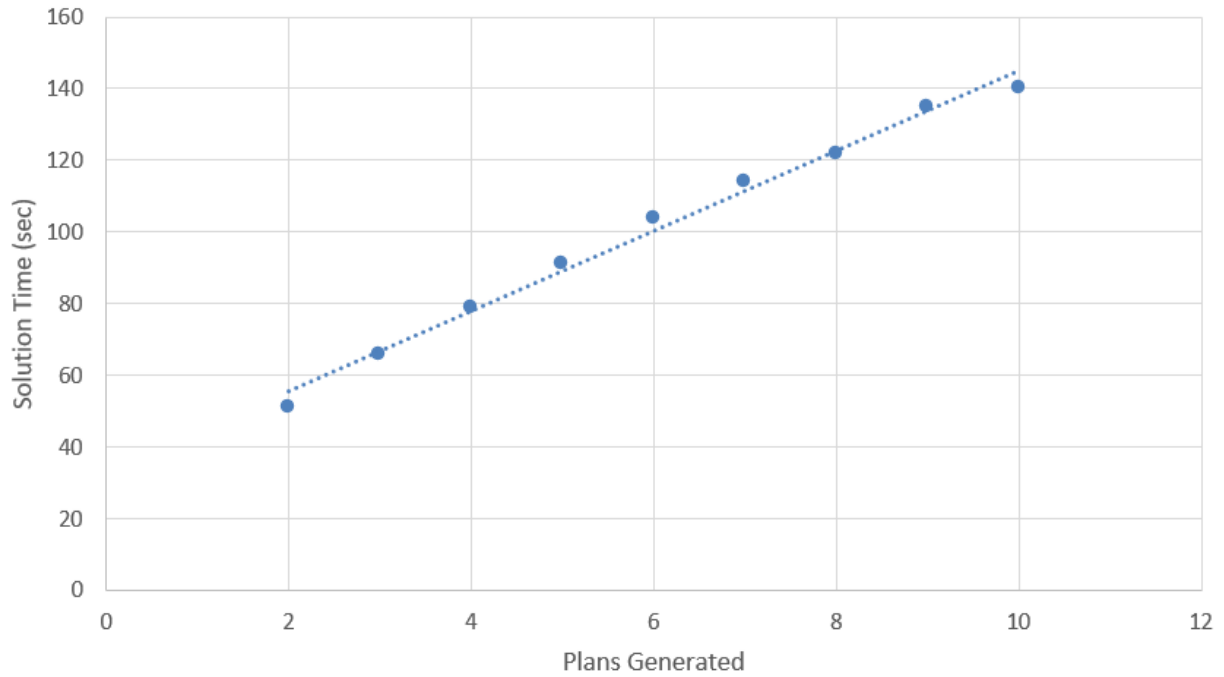
The plot of nodes divided by automated switching devices (horizontal axis) vs FISR solution time in seconds (vertical axis) is displayed in the following scatterplot without a trendline. This is an attempt to incorporate the average size of the switchable section bounded by the automated switching devices which here is defined as the node count divided by the number of automated switching devices.



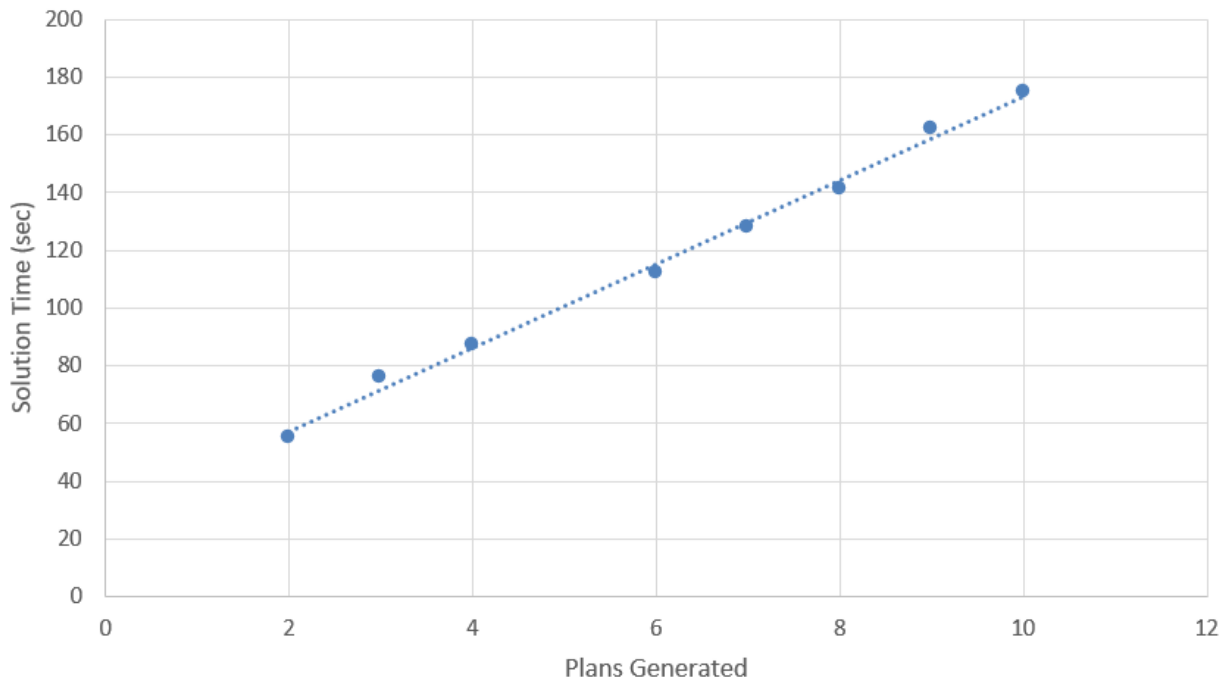
5.3.4.6. Plans vs. Solution Time

The plot of plans vs FISR solution time in seconds (vertical axis) is displayed in the following two scatterplots with a linear trendline. This was done for the largest two test cases, Cases 1 and 2. The FISR solution time linearly increase as the number of internal power flow executions is increased.

FISR Plan Generation Performance



FISR Plan Generation Performance



5.4. FISR Alarms

In real time, the FISR application can generate several alarms and send them to the alarm subsystem.

Table below lists the supported FISR alarms. Note that the alarms can be enabled or disabled at the station level using the DNAF System Configuration display. More information on the Alarm naming format is detailed in DNAF Alarm Format in the *Distribution Network Analysis Functions (DNAF) Standard Applications Analyst Guide*.

Table 33. FISR Alarms

Alarm	Description
Enabled by SCADA	When FISR is enabled for a feeder by the StatusDAEnabled status measurement, this alarm is issued for the feeder.
Disabled by SCADA	When FISR is disabled for a feeder by the StatusDAEnabled status measurement, this alarm is issued for the feeder.
Disabled by DMS	When FISR is disabled for a feeder by the DMS server due to auto-disabling checks, this alarm is issued for the feeder.
Enabled Status is not Updated	When FISR is disabled for a feeder by the StatusDAEnabled status measurement, but this feeder is already disabled by the DMS server due to auto-disabling checks, this alarm is issued for the feeder.
FISR Not Triggered	When FISR is not triggered by a detected fault, this alarm is issued for the station. There are two reasons for this alarm to be issued: • The fault should be detected by both fault indicator status change(s) from Not Set to Set and an event-triggered FL solution according to the DNAF server parameters, but the fault is not confirmed by the FL solution. • FISR should not be triggered by a fault event according to the DNAF server parameter.
FISR Started	When FISR starts to solve for a station, this alarm is issued for the station. There are three reasons for this alarm to be issued: • FISR is triggered because a fault is detected. • FISR is triggered because a loss of voltage is detected. • FISR is triggered because a user manually executes FISR from the DNAF Solution pop-up display.
FISR Completed	When FISR solves for a station, this alarm is issued for the station. There are several potential outcomes: • FISR solves but no plan is generated (for example, due to an absence of tie switches or existence of non-converged BLA islands). • FISR solves and plans are generated, but all plans have voltage and/or flow violations. • FISR solves and plans are generated, and at least one plan does not have any voltage or flow violations. • FISR solves and plans are generated, but there were topology changes in the network selection while FISR was solving (Switching Consistency functionality).

Alarm	Description
FISR Failed	When FISR fails to solve for a station, this alarm is issued for the station. There are several possible reasons: • There is no valid base DPF solution for the station. • The FISR internal power flow does not converge. • The faulted section becomes energized while FISR is solving. • The fault is on a FISR disabled feeder. • FISR does not find a faulted location in the network. • Fault is removed before FISR completed. • Incorrect problem formulation is used for FISR.
Plan Approved	If in closed loop mode, or in semi-automatic mode and the FISR plan is approved to be implemented by the user, this alarm is issued. The approval comment is also included in the alarm message.
Plan Rejected	If in semi-automatic mode, the FISR plan is rejected by user, this alarm is issued. The rejection comment is also included in the alarm message.
Implementation Complete	When FISR plan implementation is completed successfully and the status is confirmed for all switching steps and settings group changes in the plan, this alarm is issued.
Implementation Failed	When FISR plan implementation fails to complete or the status is not confirmed for any switching step or settings group change in a plan step, this alarm is issued.
Implementation Canceled	When FISR plan implementation is canceled due to a new fault event during plan execution, or a restrictive execution mode is applied because of third-party owned equipment, this alarm is issued.
Make Before Break Low Flow	After a make-before-break switching step in a FISR plan is implemented, the measured flow through that switching device is checked against the flow threshold specified by the Flow Threshold field in the DNAF Configuration Editor. If the measured flow is smaller than the threshold, this alarm is issued.
LVM Support Mode	FISR can be configured to internally solve the Load and Volt/VAR Management (LVM) application to correct voltage violations that may occur in the new topology (after a switching plan has been implemented). If FISR cannot generate a plan without flow or voltage violations, but the internal LVM solution is able to generate plans to correct the violations in the new system configuration (due to switching), this alarm is issued. The name of the internal LVM problem formulation that is used (user-configurable in the DCE) is also displayed.
LVM Support Mode Return To Normal	FISR can be configured to internally solve the Load and Volt/VAR Management (LVM) application to correct voltage violations that may occur in the new topology (after a switching plan has been implemented). If FISR cannot generate a plan without flow or voltage violations, but LVM is able to generate plans to correct the violations in the new system configuration (due to switching), the correctable FISR plan is implemented. Later, when the return to normal plan is generated and implemented, and all the switching devices are back in their normal state, this alarm is issued along with the name of the original LVM problem formulation (the one that was configured before the LVM Support Mode was enabled).

5.5. FISR Configuration

This section describes the Fault Isolation and Service Restoration configuration parameters.

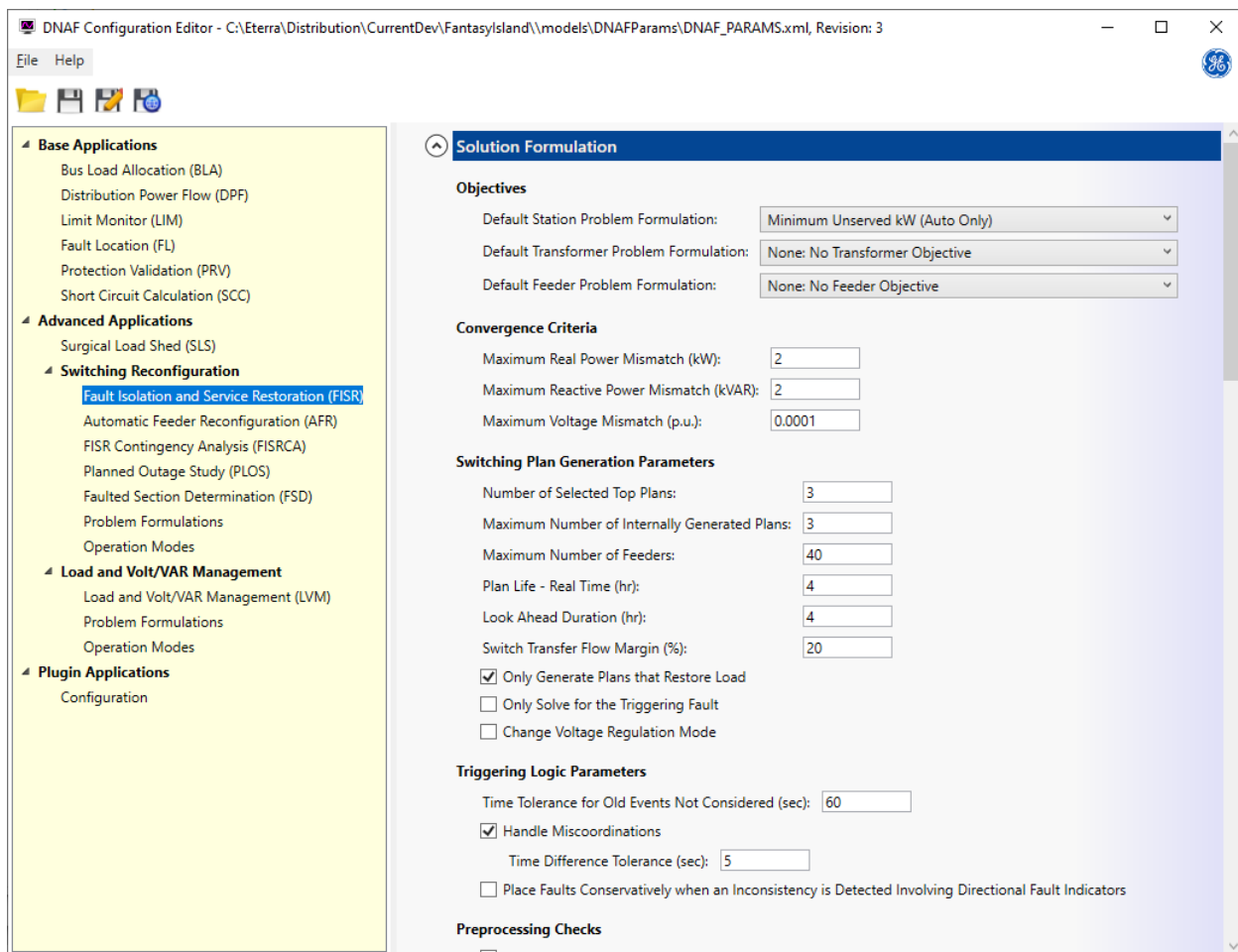


Figure 130. Fault Isolation and Service Restoration Configuration

Table below lists the FISR configuration parameters. All listed parameters are contained by the <Params xsi:type = "FISRPparams"> element in the configuration file DNAF_PARAMS.xml.

Note

All FISR problem formulations are shown on the Switching and Reconfiguration Problem Formulations page of the DNAF Configuration Editor.

Table 34. FISR Configuration Parameters

Configuration Editor UI Name	CSV File Field Name and XML File Attribute Name	Type	Description
Objectives			
Default Station Problem Formulation	DEFPROBFORM_SERVREST <DefaultProbIndex>	Drop-down list	This list shows the default FISR problem formulation for stations. If FISR runs due to an event trigger, the setting specified for the station in the DNAF System Configuration (DSC) is checked. If the DSC indicates default problem formulation, the problem formulation specified by this field is used for the station.

Configuration Editor UI Name	CSV File Field Name and XML File Attribute Name	Type	Description
Default Transformer Problem Formulation	<DefaultTransProbIndex>	Drop-down list	This list shows the default FISR problem formulation for power transformers. To inherit the station's problem formulation, select "None". If FISR runs due to an event trigger, the setting specified for the station transformers in the DSC is checked. If the DSC indicates default problem formulation, the problem formulation specified by this field is used for the station transformers.
Default Feeder Problem Formulation	<DefaultFeederProbIndex>	Drop-down list	This list shows the default FISR problem formulation for feeders. To inherit the station's or transformer's problem formulation, select "None". If FISR runs due to an event trigger, the setting specified for the station transformers in the DSC is checked. If the DSC indicates default problem formulation, the problem formulation specified by this field is used for the station feeders.
Convergence Criteria			
Maximum Real Power Mismatch	PMISMATCH_SERVREST <PMismatch>	Text box	This entry specifies the internal power flow's real power convergence criterion in kW for the FISR application. The power flow checks the calculated real power difference for every load between the current iteration and the previous iteration and the mismatch between the real power target and calculated output of every PQ and PV generator. The FISR solution status is set to Diverged if any load's real power difference between internal power flow iterations or any generator's real power mismatch between its target and output is larger than this specified value. The allowable range is 0-10.
Maximum Reactive Power Mismatch	QMISMATCH_SERVREST <QMismatch>	Text box	This entry specifies the internal power flow's reactive power convergence criterion in kVAR for the FISR application. The power flow checks the calculated reactive power difference for every load between the current iteration and the previous iteration and the mismatch between the reactive power target and calculated output of every PQ generator. The FISR solution status is set to Diverged if any load's reactive power difference between internal power flow iterations or any generator's reactive power mismatch between its target and output is larger than this specified value. The allowable range is 0-10.

Configuration Editor UI Name	CSV File Field Name and XML File Attribute Name	Type	Description
Maximum Voltage Mismatch	VMISMATCH_SERVREST <VMismatch>	Text box	This entry specifies the internal power flow's voltage convergence criterion in per unit for the FISR application. The power flow checks the differences between the voltage target and the calculated voltage at the regulated node for every PV generator and reference bus interface. If the voltage mismatch is larger than this specified value, the FISR solution status is set to Diverged. The allowable range is 0-0.1.
Switching Plan Generation Parameters			
Number of Selected Top Plans	TOPPLANEVAL_SERVREST <NoTopPlan>	Text box	This entry specifies the number of top-ranked fully evaluated FISR plans to be listed in the tabular output displays. Plans with smaller performance indexes are ranked higher. If there are both feasible and infeasible plans, the plan with the smallest performance index among the plans that are qualified to be executed in closed loop is ranked as the first. The other plans are ranked next based on their performance indexes. The allowable range is 1-50.
Maximum Number of Internally Generated Plans	<MaxNoGeneratedPlan>	Text box	This entry specifies the number of plans to be generated and fully evaluated by FISR and listed in the FISR results file. The allowable range is 1-50.
Maximum Number of Feeders	MAXNUMFEEDER_SERVREST <MaxNumFeeder>	Text box	This entry specifies the maximum number of feeders that are considered in the network selection. If the number of feeders in the network selection is larger than this entry, FISR will form the largest possible network selection (in terms of number of feeders) and continue with its solution, generating accurate isolation and restoration switching plans. The allowable range is 1-200.
Plan Life - Real Time	PLANLIFE_SERVREST <PlanLife>	Text box	This entry specifies the number of hours a FISR plan is saved in the real time DMS. After this period expires, the plan is deleted. This parameter is applicable only in real time. In study mode, a FISR plan is never deleted. The allowable range is 0.5-168.

Configuration Editor UI Name	CSV File Field Name and XML File Attribute Name	Type	Description
Look Ahead Duration	LOOKAHEAD_SERVREST <LookAheadDuration>	Text box	This entry specifies the look-ahead duration in hours to find the peak loading time used in the FISR analysis. FISR is solved at the peak loading time to make sure the plan is sufficient from a loading perspective once the plan is implemented in the field, especially if the plan requires manual switching. If this entry is set to 0, FISR will use the current system loading values. An entry of 0 typically means that the devices are all automated or loading is not a concern. The allowable range is 0-24.
Switch Transfer Flow Margin (%)	<SwitchTransferMargin>	Text box	This entry specifies the additional transfer margin in percentage allowed for tie switches and load transfer switches when determining whether a FISR plan should be generated. This parameter will affect FISR plan generation logic under the following two conditions: - FISR generates a Level 2 Case 1 plan as described in Restoration Levels. In this case, FISR restores the loads from two or more adjacent feeders by using neighboring tie switches. The transfer margin of all used neighboring tie switches will be calculated using this parameter. All the calculated transfer margins must be larger than the load to be restored. - FISR generates a Level 2 Case 2 plan as described in Restoration Levels. In this case, FISR first transfers part of the load from an adjacent feeder to an additional adjacent feeder by closing the load transfer switch and opening the isolation switch, then FISR restores the loads from this adjacent feeder by using the neighboring tie switch. The combined transfer margin that is increased by this parameter for both the load transfer switch and the tie switch must be larger than the total load to be restored. The allowable range is 0-200.
Only Generate Plans that Restore Load	<OnlyGenerateRestoreLoadPlans>	Check box	This check box specifies whether FISR only generates plans that restore load. In some scenarios, like single phase fault tripping of a three-phase protective device, the candidate plan may drop more load than it can restore. If this option is checked, this plan will be filtered out and it will not be presented.

Configuration Editor UI Name	CSV File Field Name and XML File Attribute Name	Type	Description
Only Solve for the Triggering Fault	<OnlySolveTriggeringFault>	Check box	This check box specifies whether FISR only generates plan that restores load for the triggering faults and ignores the other faults in the system. If this option is selected: • If FISR is triggered by a fault event, FISR solves only the triggering faults. However, the other faults in the network selection are respected. • If FISR is manually executed from the DNAF Solution pop-up display, FISR solves only the unsolved faults in the fault zone the user selects when opening the DNAF Solution display. However, the other faults in the network selection are respected. If this option is not selected: • If FISR is triggered by a fault event, FISR solves the triggering faults as well as all faults in the network selection. • If FISR is manually executed from the DNAF Solution pop-up display, FISR solves the faults in the selected station as well as all faults in the network selection.
Change Voltage Regulation Mode	<ChangeVoltageRegulationModes>	Check box	This check box specifies whether FISR determines if a switching plan results in a new source for voltage regulators and adds plan steps to change regulator operation mode.
Triggering Logic Parameters			
Handle Mis coordinations	ENABLEHANDLEMISCOORDINATION_SERVREST <EnableHandleMisCoordination>	Check box	This check box specifies whether FISR needs to handle miscoordination of protective devices. For more information, see Miscoordination .
Time Difference Tolerance (sec)	TIMEDIFFTOL_SERVREST <TimeDiffTolerance>	Text box	This entry specifies the time difference tolerance in seconds to differentiate between a miscoordination case or simultaneous overlapping faults. The allowable range is 1-60.

Configuration Editor UI Name	CSV File Field Name and XML File Attribute Name	Type	Description
Time Tolerance for Old Events Not Considered (sec)	<OldEventsTimeTolerance>	Text box	This entry specifies the maximum time (in seconds) FISR is allowed to incorporate old measurements/events in its triggering logic, including the following conditions: (1) This margin is used to set the time window that FISR will use to trace back from the set time of fault indicators that trigger FISR. Any device that has a set time value prior to the current time minus this time tolerance will not be considered in the FISR triggering logic, including any fault indicator, tripped device, or locked out device. (2) This margin is used to process FISR fault events triggered by a feeder head breaker that does not have all of the necessary SCADA measurements. When the feeder head breaker opens due to a downstream fault (an un-commanded operation) and zero voltage is detected on the switching devices along the feeder, the last switching operation time of the breaker must be close enough to the zero voltage measurement change time of the downstream switching devices (within this margin) to meet the triggering requirement. (3) This margin is used to process FISR triggering on sub-fed parent/child stations. When a fault occurs in a parent station which has FISR disabled, the parent station will see the fault, and the child station which has FISR enabled will see the voltage measurement change(s). The fault clearing device's (breaker or recloser) last operation time must be close enough to the zero voltage measurement change time (within this margin) to meet the station-wide loss of voltage triggering requirement for the child station. The allowable time range is 5 – 300 seconds.
Place Faults Conservatively when an Inconsistency is Detected Involving Directional Fault Indicators	<ConservativeFaultPlacementForInconsistentDirectionalFaultIndicator>	Check box	This check box specifies whether FISR places faults conservatively when an inconsistency is detected involving directional fault indicators. If this option is selected, FISR places a fault in every section upstream of the furthest downstream set fault indicator if any inconsistency is detected in faulted phases or direction involving at least one set directional fault indicator. This effectively forces FISR to only perform downstream isolation and restoration switching. If this option is not selected, FISR places faults only in the sections where an inconsistency is detected.

Preprocessing Checks

Configuration Editor UI Name	CSV File Field Name and XML File Attribute Name	Type	Description
Allow Automatic Disabling by Feeder	AUTODISABLEBYFEEDER_SERVREST <AutoDisableByFeeder>	Check box	This check box specifies whether to allow automatic disabling of FISR by feeder. If this option is checked and the number of optimization ineligible devices for one feeder is greater than or equal to the specified Minimum Number of Unavailable Devices parameter, DMS will disable FISR automatically for this feeder.
Minimum Number of Unavailable Devices	MAXNUMUNAVAILABLEDEVICE_SERVREST <MaxUnavailableDevice>	Text box	This entry specifies the minimum number of optimization ineligible devices. If automatically disabling FISR by feeder is allowed and the number of optimization ineligible devices in one feeder is larger than or equal to this parameter, DMS will disable FISR automatically for this feeder. The allowable range is 1-20.
Respect Tags	DISABLESWITCHTAGGED_SERVREST <DisableSwitchTagged>	Check box	This check box specifies whether switching devices with advanced application eligibility blocking tags are counted as optimization ineligible devices. The types of tags are specified by the TAGTYPE_CONTROLRESTRICTIONS_n parameter in the Main server and DNAF server configuration files. For a tag to block advanced application operations, the seventh bit of this parameter must be set. If this option is checked, a switching device with a block advanced application eligibility tag is counted as an optimization ineligible device. If the total number of optimization ineligible devices for one feeder is greater than or equal to the specified Minimum Number of Unavailable Devices, FISR will be automatically disabled for this feeder.
Hot Line	DISABLESWITCHHOTLINETAGGED_SERVREST <DisableSwitchHotLineTagged>	Check box	This check box specifies whether switching devices with an applied hot line are counted as optimization ineligible devices. If this option is checked, a switching device with an applied Hot Line is counted as an optimization ineligible device. If the total number of optimization ineligible devices for one feeder is greater than or equal to the specified Minimum Number of Unavailable Devices, FISR will be automatically disabled for this feeder.

Configuration Editor UI Name	CSV File Field Name and XML File Attribute Name	Type	Description
In Alarm	DISABLESWITCHINALARM_SERVREST <DisableSwitchInAlarm>	Check box	This check box specifies whether switching devices with an enabled alarm status measurement are counted as optimization ineligible devices. If this option is checked, a switching device with an enabled alarm status measurement is counted as an optimization ineligible device. If the total number of optimization ineligible devices for one feeder is greater than or equal to the specified Minimum Number of Unavailable Devices, FISR will be automatically disabled for this feeder.
Remote Control Disabled	DISABLESWITCHLOCAL_CTRL_SERVREST <DisableSwitchLocalControl>	Check box	This check box specifies whether switching devices with a remote/local status measurement set to local are counted as optimization ineligible devices. If this option is checked, a switching device with a remote/local status measurement set to local is counted as an optimization ineligible device. If the total number of optimization ineligible devices for one feeder is greater than or equal to the specified Minimum Number of Unavailable Devices, FISR will be automatically disabled for this feeder.
Reclosing Disabled	<DisableSwitchReclosing>	Check box	This check box specifies whether switching devices with a reclosing status measurement set to disabled are counted as optimization ineligible devices. If this option is checked, a switching device with a disabled reclosing status measurement is counted as an optimization ineligible device. If the total number of optimization ineligible devices in one feeder is greater than or equal to the specified Minimum Number of Unavailable Devices, FISR will be automatically disabled for this feeder.
Telemetered Switch Evaluated as a Manual Switch			
Suspect Quality - Status	<EligibleForManualDeviceWithSuspectStatusMeas>	Check box	This check box specifies whether telemetered switching devices with a suspect quality switch status measurement are treated as manual switching devices.
Suspect Quality - Hot Line	<EligibleForManualDeviceWithSuspectHotLineTagMeas>	Check box	This check box specifies whether telemetered switching devices with a suspect quality Hot Line status measurement are treated as manual switching devices.
Suspect Quality - In Alarm	<EligibleForManualDeviceWithSuspectInAlarmMeas>	Check box	This check box specifies whether telemetered switching devices with a suspect quality enabled alarm status measurement are treated as manual switching devices.

Configuration Editor UI Name	CSV File Field Name and XML File Attribute Name	Type	Description
Suspect Quality - Remove/Local	<EligibleForManualDeviceWithSuspectRemoteLocalMeas>	Check box	This check box specifies whether telemetered switching devices with a suspect quality remote/local status measurement are treated as manual switching devices.
Suspect Quality - Control Available	<EligibleForManualDeviceWithSuspectCtrlAvailMeas>	Check box	This check box specifies whether telemetered switching devices with a suspect quality reclosing disabled status measurement are treated as manual switching devices.
Suspect Quality - Supervisory Cutout	<EligibleForManualDeviceWithSuspectSupervisoryCutoutMeas>	Check box	This check box specifies whether telemetered switching devices with a suspect quality supervisory cutout status measurement are treated as manual switching devices.
Suspect Quality - Auto/Manual	<EligibleForManualDeviceWithSuspectAutoManualMeas>	Check box	This check box specifies whether telemetered switching devices with a suspect quality auto/manual status measurement are treated as manual switching devices.
Good Quality with Abnormal Value - Hot Line	<EligibleForManualDeviceHotLineTagged>	Check box	This check box specifies whether telemetered switching devices with an abnormal Hot Line status measurement with good quality are treated as manual switching devices.
Good Quality with Abnormal Value - In Alarm	<EligibleForManualDeviceInAlarm>	Check box	This check box specifies whether telemetered switching devices with an abnormal alarm status measurement with good quality are treated as manual switching devices.
Good Quality with Abnormal Value - Control Available	<EligibleForManualDeviceCtrlNotAvail>	Check box	This check box specifies whether telemetered switching devices with an abnormal reclosing disabled status measurement with good quality are treated as manual switching devices.
Other			
Integrate With LVM	INTEGRATELVM_SERVREST <IntegrateLVM>	Check box	This check box specifies whether to run LVM inside the FISR application when FISR cannot generate a plan that restores all possible de-energized loads without flow or voltage violations. If this option is checked and FISR cannot generate a plan that can restore all de-energized loads without violation, LVM is automatically executed within FISR to check whether LVM can correct the voltage violations after implementing the best FISR plan. Note that any perceived violations are determined based on the limit strategy configured in the FISR Problem Formulation. However, the selected LVM Problem Formulation (explained below), may employ a different limit strategy, and the resulting FISR solution may have voltage violations that are not considered as violations in the internal LVM solution. Users have the flexibility to opt for the same limit strategy across both applications."

Configuration Editor UI Name	CSV File Field Name and XML File Attribute Name	Type	Description
LVM Problem Formulation	<IntegrateLVMProbForm>	Drop-down list	This selection specifies the default LVM problem formulation when the Integrate with LVM option is enabled for FISR. This problem formulation automatically becomes the default LVM problem formulation until the station topology is returned to its nominal state.
Check Flow After Make Before Break Switching	<MakeBeforeBreakCheck>	Check box	This check box specifies whether a load check is enabled as part of make-before-break switching operations. When a distribution dispatcher performs make-before-break switching (such as closing an open switch or recloser) the dispatcher will check the loading of the closed device. This is a good check that a jumper did not burn off because of the sudden loading change. In some cases, the recloser may have either a light loading situation or it may be caused by a larger customer who is slower in energizing their equipment. If the MakeBeforeBreakAmps threshold is not exceeded, a message is sent to the dispatcher.
Flow Threshold	<MakeBeforeBreakAmps>	Text box	This entry specifies the minimum single-phase Amp flow that should be exceeded when closing a switching device as part of a make before break switching sequence. The allowable range is 0.5-200.
Display System Generated Fault Symbol	<DisplayEstimatedFaultSymbol>	Check box	This check box specifies whether to display the fault symbol in the User Interface if the location of the fault is predicted by FISR (using fault indicators and/or the results from the Fault Location application).
Display System Generated Loss of Voltage Symbol	<DisplayEstimatedLOVSymbol>	Check box	This check box specifies whether to show the loss of voltage symbol in the User Interface if the location of the loss of voltage fault is predicted by FISR.
Time to Remove Predicted Cut	<TimerToRemoveCut>	Text box	This entry specifies the time in seconds before removing a predicted cut that is generated from a loss of voltage event after the related voltage measurement returns to normal. The predicted cut is removed from the system, regardless of whether it was shown. The allowable range is 1-600.
Time to Remove Station Loss of Voltage Symbol	<TimerToRemoveStationLovSymbol>	Text box	This entry specifies the time in seconds that eclipsed to remove the station loss of voltage symbol. The allowable range is 1-600.

Configuration Editor UI Name	CSV File Field Name and XML File Attribute Name	Type	Description
DG/DER Delay Time	<DERDelayTime>	Text box	This entry specifies the threshold for planned restoration time, in seconds. DERs/DGs with planned restoration time outside of the threshold are not considered for FISR plans. The allowable range is 0-18000.
Default Planned Restoration Time	<DefPlannedRestoreTime>	Text box	This entry specifies the default planned restoration time for DERs/DGs after re-energization of a feeder, in seconds. Used for FISR plan generation purposes. This value is used if the planned restoration time for a DER/DG is not provided in the station model or the static model. The allowable range is 1-18000.
Permanent Fault Trigger			
Plan Execution	PLANEXEFAULT_SERVREST <PermFaultPlanExecution>	Drop-down list	This selection specifies how a fault triggered FISR plan is executed in real time. There are four options: • Advisory: The plan cannot be executed automatically; it is displayed in the FISR plans tabular display. The plan can be executed manually by the operator (plan can include automatic and/or manual devices). • Semi-Automatic: The plan can be executed automatically after operator approval. It will be displayed in the FISR plans tabular display and the plan can be implemented by clicking the "Implement the Plan" button if all devices in the plan are remotely controlled. • Closed Loop: The best plan without voltage and flow violations is executed automatically if all devices in the plan are remotely controlled. For this option, if there is no plan without any violations, the plans can be executed automatically after operator approval, similar to the Semi-Automatic option. • Closed Loop - Calculated Voltage Limit Violation Not Worse: Works similarly to the Closed Loop option, but also considers the plans with voltage violations. A plan can be executed automatically if the calculated voltage violations are not made worse by the plan. A voltage margin specified by the Violation Not Worse Margin (%) parameter is used to determine if a plan resulting in voltage violations is acceptable. A plan can be turned into a switch order for all four plan execution modes.

Configuration Editor UI Name	CSV File Field Name and XML File Attribute Name	Type	Description
Settings Group Check	INCLUDESETTINGGROUPFAULT_SERVREST <PermFaultIncorpProtectCoord>	Drop-down list	This selection specifies how the protection settings group steps are added to a plan for a fault triggered FISR execution. The options are: • None: No settings group steps are added to the FISR plan. • By Feeder Source: Settings group steps are added to the FISR plan. • By Minimum Fault Current: If this option is set, Protective Validation (PRV) is forced to run after the FISR plan is generated. The active protection settings group is selected by finding the settings group with the highest phase trip current, which is below the minimum phase fault current on the protected protection area (multiplied by the appropriate safety factor) and also above the estimated load current at the peak load time and the continuous current limit of the protective device.
Settings Group Execution	<PermFaultSettingsGroupExecution>	Drop-down list	This selection specifies how the protection settings group steps are executed for a fault-triggered FISR. The selection options depend on the Settings Group Selection option. The options are: • None: Execution options are disabled when no settings group steps are added to the FISR plan. • Always Before Switching: Settings group steps are executed before the switching step that results in the settings group change. • Always After Switching: Settings group steps are executed after the switching step that results in the settings group change. • Switch Before for Trip Raise, Switch After for Trip Lower: (This option is only valid when setting groups are checked by minimum fault current.) Settings group steps are executed before the switching step that results in the settings group change if the phase trip settings are being raised. Settings group steps are executed after the switching step that results in the settings group change if the phase trip settings are being lowered. Note: Raise is the phase trip setting change to a higher value (for example, from 300 A to 500 A. Lower is the phase trip setting change to a lower value (for example, from 500 A to 300 A).

Configuration Editor UI Name	CSV File Field Name and XML File Attribute Name	Type	Description
Use Triggering Feeder Plan Execution Mode	<PermFaultUseFeederPlan ExecutionMode>	Check box	This check box specifies the effective closed loop plan execution mode determination logic for the FISR permanent fault trigger. If this box is checked and a permanent fault occurs on a feeder, then the triggering feeder's plan execution mode will be used. If unchecked, the execution mode of the feeder associated with each plan step will be examined, and the most conservative mode will be applied. For more information, see Plan Execution Mode.
Allow Closed Loop Execution if New Protection Validation Violations Occur	<AllowClosedLoopPermanent FaultFISRIfPRVViolation>	Check box	This check box specifies whether the closed loop execution is allowed for fault-triggered FISR plans with any type of protection validation violation in the post-plan Protection Validation (PRV) function (maximum interruption violation, minimum phase or ground pickup violation, protection peak load margin violation, or protection reach violation). If this option is selected and the plan execution mode is set to closed-loop, FISR automatically implements valid plans (without overloads or high/low voltage violations) regardless of PRV violations.

Loss of Voltage Trigger

Configuration Editor UI Name	CSV File Field Name and XML File Attribute Name	Type	Description
Plan Execution	PLANEXELOV_SERVREST <LossVoltPlanExecution>	Drop-down list	This selection specifies how the loss of voltage triggered FISR plan is executed in real time. There are four options: • Advisory: The plan cannot be executed automatically; it is displayed in the FISR plans tabular display. The plan can be executed manually by the operator (automatic or manual devices) or possibly turned into a switch order. • Semi-Automatic: The plan cannot be executed automatically. It is displayed in the FISR plans tabular display and the plan can be implemented by clicking the "Implement the Plan" button if all devices in the plan can be remotely controlled. • Closed Loop: The best plan without flow and voltage violations is executed automatically if all devices in the plan can be remotely controlled. • Closed Loop - Calculated Voltage Limit Violation Not Worse: Works similarly to the Closed Loop option, but also considers the plans with voltage violations. A plan can be executed automatically if the calculated voltage violations are not made worse by the plan. A voltage margin specified by the Violation Not Worse Margin (%) parameter is used to determine if a plan resulting in voltage violations is acceptable. A plan can be turned into a switch order for all four plan execution modes.
Settings Group Check	INCLUDESETTINGGROUPLOV_SERVREST <LossVoltIncorpProtectCoord>	Drop-down list	This selection specifies how the protection settings group steps are added to a loss of voltage triggered FISR execution. The options are: • None: No settings group steps are added to the FISR plan. • By Feeder Source: Settings group steps are added to the FISR plan. • By Minimum Fault Current: If this option is set, Protective Validation (PRV) is forced to run after the FISR plan is generated. The active protection settings group is selected by finding the settings group with the highest phase trip current, which is below the minimum phase fault current on the protected protection area (multiplied by the appropriate safety factor) and also above the estimated load current at the peak load time and the continuous current limit of the protective device.

Configuration Editor UI Name	CSV File Field Name and XML File Attribute Name	Type	Description
Settings Group Execution	<LossVoltSettingsGroupExecution>	Drop-down list	This selection specifies how the protection settings group steps are executed for a loss of voltage triggered FISR. The selection options depend on the Settings Group Selection option. The options are: • None: Execution options are disabled when no settings group steps are added to the FISR plan. • Always Before Switching: Settings group steps are executed before the switching step that results in the settings group change. • Always After Switching: Settings group steps are executed after the switching step that results in the settings group change. • Switch Before for Trip Raise, Switch After for Trip Lower: (This option is only valid when setting groups are checked by minimum fault current.) Settings group steps are executed before the switching step that results in the settings group change if the phase trip settings are being raised. Settings group steps are executed after the switching step that results in the settings group change if the phase trip settings are being lowered.  Note Raise is the phase trip setting change to a higher value; for example, from 300 A to 500 A. Lower is the phase trip setting change to a lower value; for example, from 500 A to 300 A.
Voltage Measurement Threshold	<LossOfVoltageMeasThreshold>	Text box	This entry specifies the voltage threshold in percentage to trigger the loss of voltage for FISR. If there is no fault in the affected area, the loss of voltage FISR may be triggered under two conditions: • Amp measurement is not used. If the voltage measurement value is below this voltage threshold percentage, FISR is triggered due a loss of voltage. • Amp measurement is used. If the voltage measurement value is below this voltage threshold percentage and the current measurement value is less than the current measurement threshold in Amps, specified by the Current Measurement Threshold field, FISR is triggered due to a loss of voltage. The allowable range is 0-100.

Configuration Editor UI Name	CSV File Field Name and XML File Attribute Name	Type	Description
Effective Zero Voltage Measurement Threshold	<LossOfVoltageEffectiveZeroVoltageMeasThreshold>	Text box	This entry specifies the voltage threshold in percentage to determine whether a reference bus interface SCADA voltage measurement is treated as an effective zero voltage for the FISR loss of voltage trigger. If the absolute value of a valid voltage measurement is smaller than this threshold (in percent), it will be treated as a zero voltage.
Use Current Measurement Threshold	<UseLossOfVoltageCurrMeas>	Check box	This check box specifies whether to evaluate current (Amp) measurements for the FISR loss of voltage trigger. If the option is checked, and if the voltage measurement value is below this voltage threshold percentage and the current measurement value is less than the current measurement threshold (in Amps), FISR is triggered due to a loss of voltage.
Current Measurement Threshold	<LossOfVoltageCurrMeasThreshold>	Text box	This entry specifies the current threshold in Amps for the FISR loss of voltage trigger. This entry takes effect only when the Use Current Measurement Threshold option is checked. In this case, if the voltage measurement value is below the Voltage Measurement Threshold (%) percentage and the current measurement value is less than this parameter's value, FISR is triggered due to a loss of voltage. The allowable range is 0.5-200.
Use Triggering Feeder Plan Execution Mode	<LoVTriggerUseFeederPlanExecutionMode>	Check box	This check box specifies the effective closed loop plan execution mode determination logic for the FISR loss of voltage trigger. If this box is checked and a loss of voltage is detected on a feeder, then the triggering feeder's plan execution mode will be used. If the loss of voltage occurs due to a fault on a power transformer or is a station-wide loss of voltage event, then the most conservative mode of the feeder(s) supplied by the transformer/reference bus interface will be applied. If unchecked, the execution mode of the feeder associated with each plan step will be examined, and the most conservative mode will be applied. For more information, see Plan Execution Mode.

Configuration Editor UI Name	CSV File Field Name and XML File Attribute Name	Type	Description
Allow Closed Loop Execution if New Protection Validation Violations Occur	<AllowClosedLoopLOVFISRIfPRVViolation>	Check box	This check box specifies whether the closed loop execution is allowed for loss-of-voltage-triggered FISR plans with any type of protection validation violation in the post-plan Protection Validation (PRV) function (maximum interruption violation, minimum phase or ground pickup violation, protection peak load margin violation, or protection reach violation). If this option is selected and the plan execution mode is set to closed-loop, FISR automatically implements valid plans (without overloads or high/low voltage violations) regardless of PRV violations.
Enable Station Outage Event Trigger	<EnableStationOutageEventTriggerFISR>	Check box	If this option is selected, FISR is triggered when a station outage event occurs. For example, when a bank differential, bus differential, or power transformer breaker trips.
Enable Feeder Fault Triggered Loss of Voltage Trigger	<EnableFeederFaultLOVTriggerFISR>	Check box	If this option is selected, FISR is triggered when a feeder fault triggered loss of voltage event occurs. Note: More details about feeder fault triggered loss of voltage functionality can be found in the section Feeder Fault Triggered Loss of Voltage FISR .
Enable Measurement Consistency Check	<EnableLovMeasurementConsistencyCheck>	Check box	When this box is checked, voltage measurements downstream from the device experiencing a loss of voltage are evaluated against the Voltage Measurement Threshold. If all applicable downstream voltage measurements are found to be below the threshold, the FISR loss of voltage solution is triggered. Otherwise, the FISR loss of voltage solution is not triggered.
Solution Results			
Create FISR Results File	PRINTOUT_SERVREST <PrintFISRResult>	Check box	This check box specifies whether to create a FISR results file. If this box is checked, a results file is generated. The results file is created in the folder specified by the DNAF_OUTPUT_PATH parameter in the appropriate server configuration file. The file name format depends on whether the "Use Unique File Names" option is enabled. Refer to the description for "Use Unique File Names" for more details.

Configuration Editor UI Name	CSV File Field Name and XML File Attribute Name	Type	Description
Use Unique File Names	UNIQUELOG_SERVREST <UniqueLog>	Check box	This check box specifies whether to create a unique FISR results file. If this check box is checked, a unique results file is created for every FISR execution. In this case, the file name is in the format "FISR Results_<stationname>_<executiontimestamp>.dat". If this check box is cleared, the file name is in the format "FISR Results_<stationname>.dat". The results file is overwritten each time FISR is executed for that station.
Use Detailed Solution Reporting	DETAILLOG_SERVREST <DetailLog>	Check box	This check box specifies whether to create a detailed FISR results file. If it is checked, additional information is included in the FISR results file, including the de-energized island switching devices and a list of components with voltage and flow violations.
Top High Bus/Load Voltages Reported	<MaxNumHighVoltReported>	Text box	This entry specifies the maximum number of top-ranking high bus or load voltages to be reported in the FISR results file.
Top Low Bus/Load Voltages Reported	<MaxNumLowVoltReported>	Text box	This entry specifies the maximum number of top-ranking low bus or load voltages to be reported in the FISR results file.
Top Branch Loading Levels Reported	<MaxNumBranchLoading LevelsReported>	Text box	This entry specifies the maximum number of top-ranking branch segment loading levels to be reported in the FISR results file.
Debug Files			
Maximum Number of Debug File	MAXUPFOUTPUTFILE_SERVREST <MaxNumUpfOutFile>	Text box	This entry specifies the maximum number of accumulated debug output files created when FISR does not solve. These debug files are output in the folder specified by the parameter DNAF_OUTPUT_PATH in the appropriate server configuration file and can be identified by the prefixes "upfin_export_", "optalg_export_" and "alg_export_" with station name and application name in the file name. Once the limit is reached, no subsequent debug output files are created. The allowable range is 0-50.
Manual Execution			
Allow in Real Time	REALTIME_SERVREST <RealTime>	Check box	This check box indicates if the FISR application can be invoked through the RT DNAF Solution pop-up display. If checked the application is shown. Note FISR can still be executed by event triggers in RT by setting the appropriate triggering parameters in the server configuration file.

Configuration Editor UI Name	CSV File Field Name and XML File Attribute Name	Type	Description
Allow in Study Mode	STUDYMODE_SERVREST <Study>	Check box	This check box indicates if the FISR application can be invoked through the ST DNAF Solution pop-up display. If checked the application is shown.
Old and Deprecated Parameters			
Function Name	NAME_SERVREST <Name>	Text box	This entry specifies the application name in the DNAF pop-up display. <i>i</i> Note This field is exposed only in the old DNAF Configurator, not in the new DNAF Configuration Editor.
Function Enumeration	FUNCENUM_SERVREST <FuncEnum>	Text box	This entry specifies the application enumeration: 0 = Other applications 1 = Distribution Power Flow 2 = Protection Validation 4 = Load and Volt/VAR Management 5 = Fault Isolation and Service Restoration 6 = Automatic Feeder Reconfiguration 7 = Planned Outage Study 8 = FISR Contingency Analysis 10 = Fault Location 11 = Short Circuit Calculation <i>i</i> Note This field is exposed only in the old DNAF Configurator, not in the new DNAF Configuration Editor.

6. Automated Feeder Reconfiguration

6.1. AFR Plan Generation

This application provides plans to optimally unload overloaded segments, including power transformers and feeder main conductors. It can provide a plan to reduce real power losses in a station and its neighboring stations. It can also provide a plan to return a station to its nominal configuration (Return to Normal).

6.1.1. Reduce Overloads and Violations Problem Formulation

For the Reduce Overloads and Violations problem formulation, AFR uses an approach similar to FISR. AFR first creates a parallel circuit, then breaks the circuit and restores it to a radial condition. This is to avoid dropping load in any step.

The AFR application with the Reduce Overloads and Violations problem formulation tries to transfer part of the load downstream of the overloaded feeder main devices not in an AFR disabled feeder or station. There are three levels of load transfer.

Level 0: Transfer to own feeder

If the downstream load is connected to the upstream part of the overloaded feeder through a normal open switch, close that switch and open a switch between this closed switch and the overloaded device. This is illustrated in figure below.

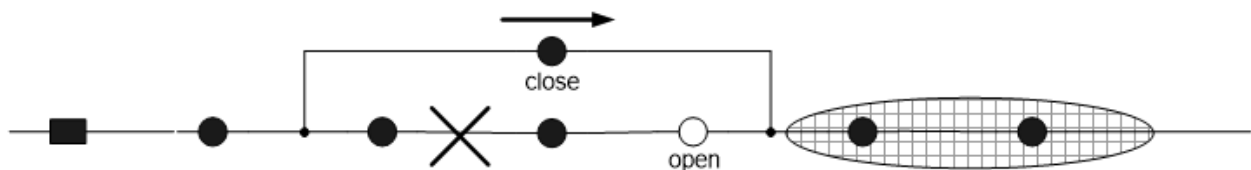


Figure 131. Level 0 AFR Reduce Overloads and Violations

Level 1: Transfer to one adjacent feeder

If the downstream load is connected to an adjacent feeder through an open tie switch, it is possible that the downstream load can be transferred to that feeder. This is illustrated in figure below.

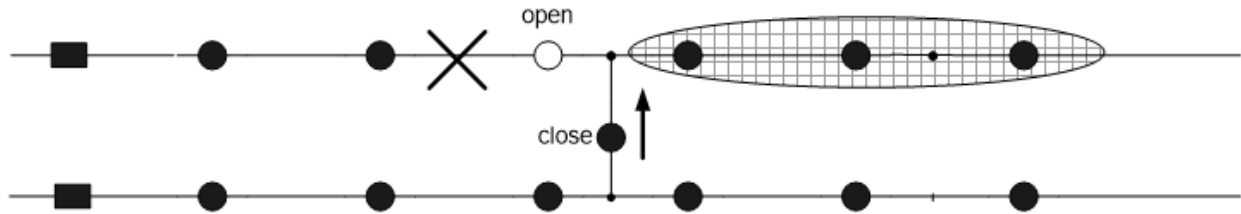


Figure 132. Level 1 AFR Reduce Overloads and Violations

Level 2: Transfer to two adjacent feeders or with load transfer

There are two cases at this level.

Case 1 involves two or more adjacent feeders. It transfers the load to two or more adjacent feeders by closing the related tie switches and opening one or more normally closed switches in the downstream area. Figure below illustrates the case of transferring load to two adjacent feeders.

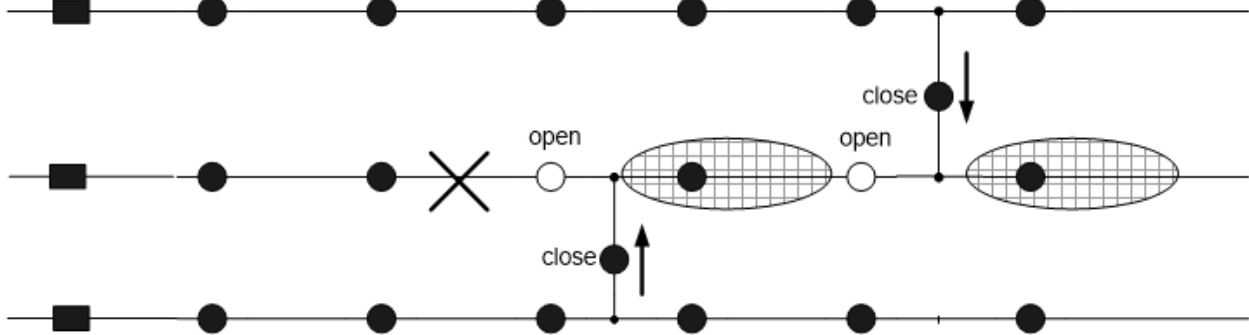


Figure 133. Level 2 AFR Reduce Overloads and Violations Case 1

Case 2 involves adjacent feeders and other feeders that are two levels away from the overloaded feeder. In this case, the adjacent feeders do not have enough transfer capacity to pick up all dropped load. The solution is to transfer some live load in to adjacent feeders, allowing enough capacity on the source feeder to pick-up the downstream load. This is illustrated in figure below.

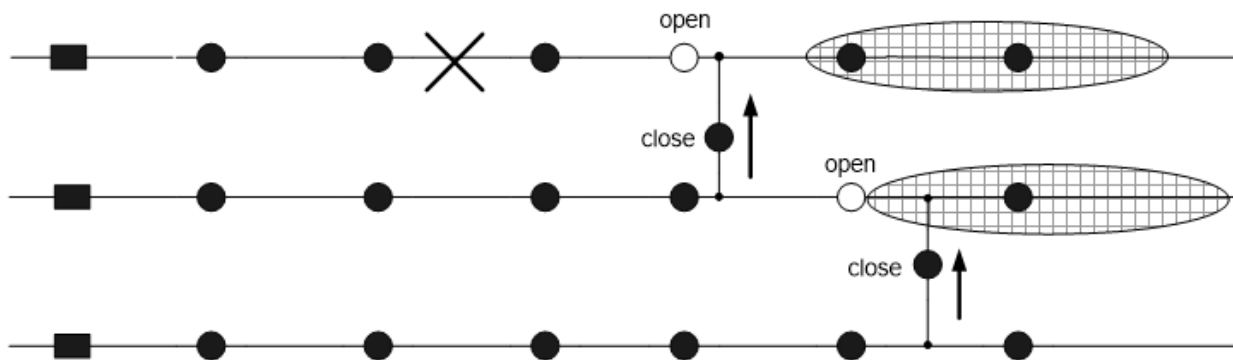


Figure 134. Level 2 AFR Reduce Overloads and Violations Case 2

These levels are the major approaches that AFR with Reduce Overloads and Violations uses to solve

a single overload violation. In the case of multiple violations in the same network selection that need to be solved, AFR internally solves them one by one starting with the worst violated devices; it then combines the steps to generate plans in one solution.

6.1.2. Reduce Real Power Losses Problem Formulation

For the Reduce Real Power Losses problem formulation, AFR uses a heuristic approach. This approach is enabled when the "Reduce Real Power Losses" option is checked in the selected AFR Problem Formulation.

The goal of the reduce power loss problem formulation is to identify the open/close status of three-phase feeder main switching devices indicated in the problem formulation which results in the greatest real power loss reduction in the system.

The reduce power loss problem formulation starts by ranking current open switching devices based on the sum of the voltage magnitude difference of the three individual phases across the device, as shown in the equation below:

$$|V_{1a} - V_{2a}| + |V_{1b} - V_{2b}| + |V_{1c} - V_{2c}|$$

Then each device is selected one by one from the ranked list. The selected switch is first closed to form a loop, and then a list containing all eligible closed switches on the loop is found by topology search. For each closed switch in the loop, a DPF is performed to calculate the real power loss if the switch is opened to break the loop (without dropping load). In addition to reducing the real power reduce loss, the selection of the close/open device pair must not create new flow or voltage violations in the system, nor can it make the existing flow or voltage violations worse. The best close/open device pair which satisfies the above condition is selected and added to the generated plan.

The plan generation will stop when the optimization engine cannot find additional switching pairs to further reduce the real power losses, the number of moves in the plan "Maximum Allowed Number of Open/Close Pairs" configuration parameter in the Switching Reconfiguration Problem Formulations section in the "DNAF_PARMS.xml" file has been reached, or the real power loss reduction amount in the "Loss Reduction Amount (%)" configuration parameter defined in the AFR section in the "DNAF_PARMS.xml" file has been reached.

6.1.3. Other Problem Formulations

The Load Shedding and Return to Normal problem formulations are implemented in an iterative method. Three parameters control how many total plans are studied and how many plans are studied in each iteration (depth). These parameters are stored in the optalg.dat file in the server directory; see Optalg.dat in the *Distribution Network Analysis Functions (DNAF) Standard Applications Analyst Guide* for a description of the file.

- **MaxSwitchAttempts:** Maximum number of switching plans to be generated.

- **MaxPlansPerDepth:** Maximum number of switching plans that are studied in each iteration.
- **MaxPlansPerDepth2:** This number multiplied by depth is added to MaxPlansPerDepth for the actual test; this allows solving more plans at greater depths.

The method starts with a zero-step plan and goes to the first iteration by adding one more switching devices to the plan. Some heuristic rules are used to reduce the number of candidate switches that can be added to the plan.

- All switching operations should not drop load.
- If an open switch is parallel to a closed switch, the open switch should not be added.
- If the plan generated new loops in the previous steps, only the switches that can break the loops are added.
- Switches that are downstream of the branch with the worst flow violation are added.
- Switches downstream of the branch with the worst low voltage violation are added.

In each iteration, there are six steps to follow:

1. Get the plan with the best performance index generated in the previous iteration.
2. Get all candidate switches suitable for this plan using the above heuristic rules.
3. Sort the list of candidate switches by switching cost from low to high (also consider available transfer capacity for the first iteration).
4. Get the first switch in the list in order, and add one to the best plan picked in Step 1 to generate a new plan. Delete the switch from the list.
5. If the number of total plans generated is larger than MaxSwitchAttempts, exit AFR. If the number of plans generated in this iteration (depth) is larger than $\text{MaxPlansPerDepth} + \text{depth} * \text{MaxPlansPerDepth2}$, exit this iteration and go to Step 1.
6. If all the candidate switches are used, get the next-best plan generated in the previous iteration and go to Step 2.

After this iterative method is finished, the top plans with the smallest total PI are displayed to the user. The maximum number of generated plans and the selected plans are specified by the Number of Selected Top Plans and the Maximum Number of Internally Generated Plans fields in the DNAF Configuration Editor. In addition, a results file is created for each application execution.

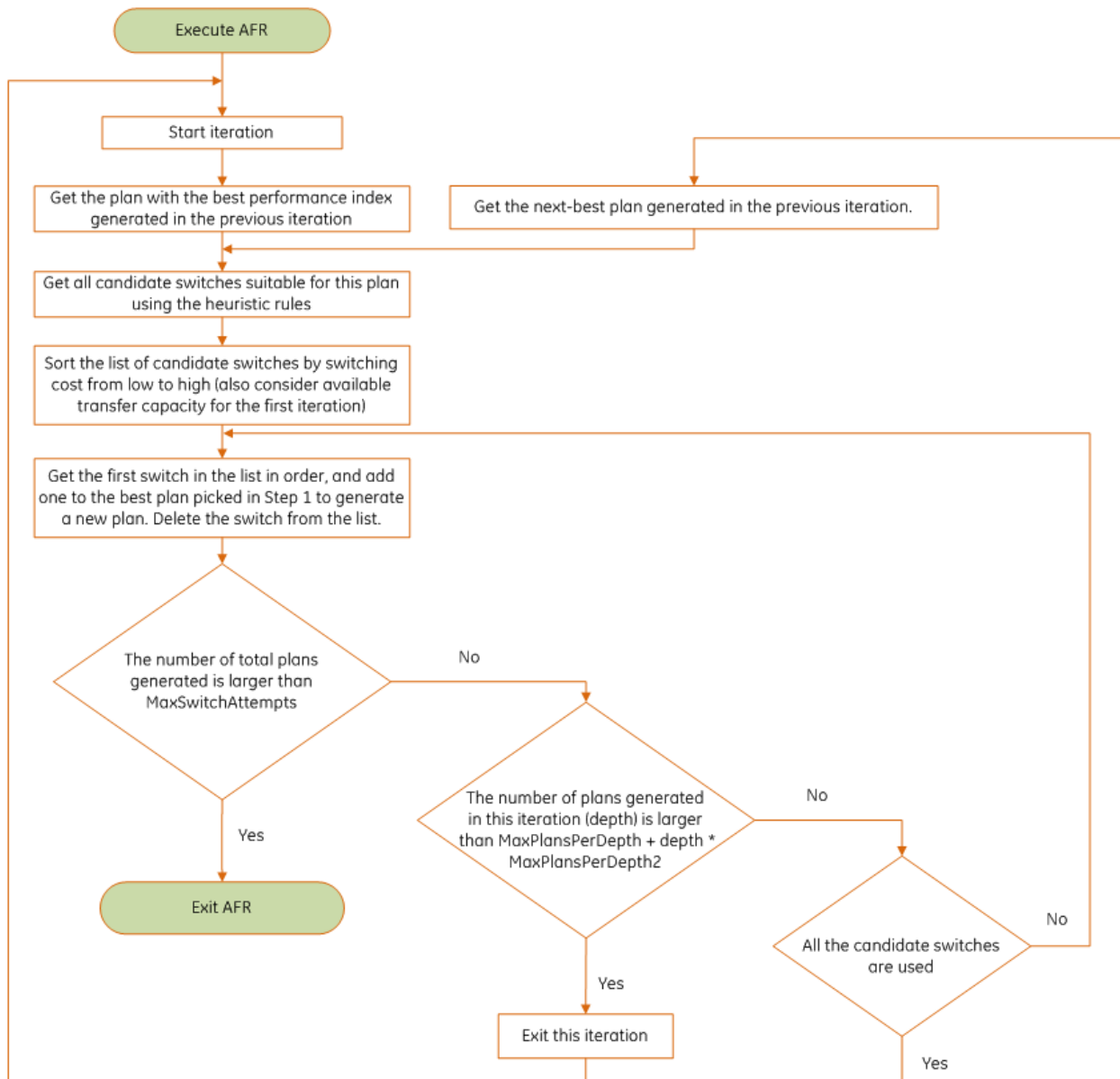


Figure 135. Iterative Method for Generating AFR Plans

6.2. Modes of Operations

This section describes the modes of operation for AFR. In both real-time and study mode, DPF must be solved before AFR is run.

The AFR application can be run in real-time mode or study mode, depending on the selected problem formulation. The automatic triggering for overloads is only supported in real-time mode.

6.2.1. Reduce Overloads and Violations Problem Formulation

AFR with the Reduce Overloads and Violations problem formulation can be run in both real-time mode and study mode.

The application can be configured to trigger automatically by event. In this case, after DPF is run for a station, the results are analyzed for any overloads on feeder main lines or power transformers not in an AFR disabled feeder or station. If any overloads are found, AFR is executed to eliminate these violations.

AFR can be invoked manually by selecting any component in the station or the station circle, and then clicking the Run Network Analysis Application button in the toolbar. This opens the DNAF pop-up display from which you can run AFR. The overload violations that can be solved are overloads of the branch components within the station fence or along the feeder main backbone. These components do not include lateral branch devices such as tap fuses or distribution transformers or any device in an AFR disabled feeder or station.

The overload violations are solved in the following priority order:

- If the station circle or any component inside the station fence (not belonging to any feeder) is selected, the following logic is applied:
 - If there are any overload violations for the station power transformers, only these violations are to be resolved.
 - If there is no power transformer overload violation, the worst overload violation in the station is to be resolved. This worst overload is decided by the percentage of the flow over the device's lowest flow limit.
 - If there is no overload violation in the station, AFR returns and displays a message stating that there is no overload violation in the station.
- If any component in the field (belonging to a feeder) is selected, the following logic is applied:
 - If there are any overload violations for the station power transformers, only these violations are to be resolved.
 - If there is no power transformer overload violation, the worst overload violation in the selected feeder is to be resolved. This worst overload is decided by the percentage of the flow over the device's lowest flow limit.
 - If there is no overload violation in the feeder, AFR returns and displays a message stating that there is no overload violation in the feeder.

6.2.2. Reduce Real Power Losses Problem Formulation

The Reduce Real Power Losses problem formulation can be executed in both real time and study mode.

The problem formulation can only be requested manually. The generated plan can be executed automatically via the Plan Manager subsystem if configured to do so.

6.2.3. Other Problem Formulations

The AFR application with other problem formulations can be executed in study mode only. When it is invoked, AFR tries to solve the selected problem formulation for the entire network selection area.

6.3. AFR Alarms

In real time, the AFR application can generate several alarms and send them to the alarm subsystem.

Table below lists the supported AFR alarms. Note that the alarms can be enabled or disabled at the station level using the DNAF System Configuration. More information on the Alarm naming format is detailed in "DNAF Alarm Format" in the *Distribution Network Analysis Functions (DNAF) Standard Applications Analyst Guide*.

Table 35. AFR Alarms

Alarm	Description
Enabled By SCADA	When AFR is enabled for a feeder by the StatusDAEnabled status measurement, this alarm is issued for the feeder.
Disabled By SCADA	When AFR is disabled for a feeder by the StatusDAEnabled status measurement, this alarm is issued for the feeder.
Disabled By DMS	When AFR is disabled for a feeder by the DMS server due to auto-disabling checks, this alarm is issued for the feeder.
Enabled Status Not Updated	When AFR is disabled for a feeder by the StatusDAEnabled status measurement, but this feeder is already disabled by the DMS server due to auto-disabling checks, this alarm is issued for the feeder.
AFR Started	When AFR starts to solve for a station, this alarm is issued for the station. There are several reasons for AFR to be triggered: • AFR is event triggered due to a power transformer and/or line overload detected using the DPF limit monitor (if LIM is enabled in DCE). • A user manually runs AFR from the DNAF pop-up display.
AFR Completed	When AFR solves for a station, this alarm is issued for the station. There are several potential outcomes: • AFR solves but no plan is generated (for example, due to non-converged BLA islands). • AFR solves and plans are generated, but all plans have voltage and/or flow violations. • AFR solves and plans are generated, and at least one plan does not have any voltage or flow violations. • AFR solves and plans are generated, but there are topology changes in the network selection when AFR is solving.

Alarm	Description
AFR Failed	When AFR fails to solve for a station, this alarm is issued for the station. There are several possible reasons: • There is no valid DPF solution for the station. • The AFR internal power flow does not converge. • There are too many feeders in the network selection. • AFR does not find an overloaded component in the network.
Plan Approved	If in closed loop mode or in semi-automatic mode and the AFR plan is approved to be implemented by the user, this alarm is issued. The approval comment is also included in the alarm message.
Plan Rejected	If in semi-automatic mode, the AFR plan is rejected by user, this alarm is issued. The rejection comment is also included in the alarm message.
Implementation Complete	When AFR plan implementation is completed successfully, this alarm is issued.
Implementation Failed	When AFR plan implementation fails to complete, this alarm is issued.
Make Before Break Low Flow	After a make-before-break switching step in an AFR plan is implemented, the measured flow through that switching device is checked against the flow threshold specified in the DNAF configuration editor. If the measured flow is smaller than the threshold, this alarm is issued.
LVM Support Mode	If LVM integration with AFR is enabled, and AFR cannot generate a plan without flow or voltage violations, LVM may be able to generate plans to correct the violations in the new system configuration (due to switching). In this scenario, when a correctable AFR plan is to be implemented, this alarm is issued.
LVM Support Mode Return To Normal	If LVM integration with AFR is enabled, and AFR cannot generate a plan without flow or voltage violations, LVM may be able to generate plans to correct the violations in the new system configuration (due to switching). In this scenario, the correctable AFR plan is implemented. Later when the return to normal plan is generated and implemented, and all the switching devices are back to their normal state, this alarm is issued.
Plan Canceled	When AFR plan implementation is canceled due to a restrictive execution mode is applied because of third-party owned equipment, this alarm is issued.

6.4. AFR Configuration

This section describes the Automated Feeder Reconfiguration configuration parameters.

Base Applications

- Bus Load Allocation (BLA)
- Distribution Power Flow (DPF)
- Limit Monitor (LIM)
- Fault Location (FL)
- Protection Validation (PRV)
- Short Circuit Calculation (SCC)

Advanced Applications

- Surgical Load Shed (SLS)
- Switching Reconfiguration**
 - Fault Isolation and Service Restoration (FISR)
 - Automatic Feeder Reconfiguration (AFR)**
 - FISR Contingency Analysis (FISCA)
 - Planned Outage Study (PLOS)
 - Faulted Section Determination (FSD)
 - Problem Formulations
 - Operation Modes
- Load and Volt/VAR Management**
 - Load and Volt/VAR Management (LVM)
 - Problem Formulations
 - Operation Modes
- Plugin Applications**
 - Configuration

Solution Formulation

Objectives

- Default Station Problem Formulation: Reduce Overloads and Violations
- Default Transformer Problem Formulation: None: No Transformer Objective
- Default Feeder Problem Formulation: None: No Feeder Objective

Convergence Criteria

- Maximum Real Power Mismatch (kW): 2
- Maximum Reactive Power Mismatch (kVAR): 2
- Maximum Voltage Mismatch (p.u.): 0.0001

Switching Plan Generation Parameters

- Number of Selected Top Plans: 3
- Maximum Number of Internally Generated Plans: 3
- Maximum Number of Feeders: 150
- Plan Life - Real Time (hr): 4
- Look Ahead Duration (hr): 4
- Switch Transfer Flow Margin (%): 20
- Maximum Allowed Number of Open/Close Pairs: 2
- Real Power Loss Reduction Amount (%): 4

Preprocessing Checks

- ☐ Automatic Disabling by Feeder
- Minimum Number of Unavailable Devices: 1
- Device Unavailability Conditions:
 - ☒ Respect Tags
 - ☒ Hot Line

Figure 136. Automated Feeder Reconfiguration Configuration

Table below lists the AFR configuration parameters. The <Params xsi:type = "AFRParams"> element in the configuration file DNAF_PARAMS.xml contains all listed parameters.

Note

All AFR problem formulations are shown on the Switching and Reconfiguration Problem Formulations page of the DNAF Configuration Editor.

Table 36. AFR Configuration Parameters

Configuration Editor UI Name	CSV File Field Name and XML File Attribute Name	Type	Description
Objectives			
Default Station Problem Formulation	DEFPROBFORM_FDRCONF <DefaultTransProbIndex>	Drop-down list	This list shows the default AFR problem formulation for stations. If an event triggers AFR, the setting specified for the station in the DNAF System Configuration (DSC) is checked. If the DSC indicates default problem formulation, the problem formulation specified by this field is used for the station.

Configuration Editor UI Name	CSV File Field Name and XML File Attribute Name	Type	Description
Default Transformer Problem Formulation	<DefaultTransProbIndex>	Drop-down list	This list shows the default AFR problem formulation for power transformers. To inherit the station's problem formulation, select "None". If AFR runs due to an event trigger, the setting specified for the station transformers in the DNAF System Configuration (DSC) is checked. If the DSC indicates default problem formulation, the problem formulation specified by this field is used for the station transformers.
Default Feeder Problem Formulation	<DefaultFeederProbIndex>	Drop-down list	This list shows the default AFR problem formulation for feeders. To inherit the station's or transformer's problem formulation, select "None". If AFR runs due to an event trigger, the setting specified for the station transformers in the DNAF System Configuration (DSC) is checked. If the DSC indicates default problem formulation, the problem formulation specified by this field is used for the station feeders.
Convergence Criteria			
Maximum Real Power Mismatch	PMISMATCH_FDRCONF <PMismatch>	Text box	This entry specifies the internal power flow's real power convergence criterion in kW for the AFR application. The power flow checks the calculated real power difference for every load between the current iteration and the previous iteration and the mismatch between the real power target and calculated output of every PQ and PV generator. The AFR solution status is set to Diverged if any load's real power difference between internal power flow iterations or any generator's real power mismatch between its target and output is larger than this specified value. The allowable range is 0-10.

Configuration Editor UI Name	CSV File Field Name and XML File Attribute Name	Type	Description
Maximum Reactive Power Mismatch	QMISMATCH_FDRCONF <QMismatch>	Text box	This entry specifies the internal power flow's reactive power convergence criterion in kVAR for the AFR application. The power flow checks the calculated reactive power difference for every load between the current iteration and the previous iteration and the mismatch between the reactive power target and calculated output of every PQ generator. The AFR solution status is set to Diverged if any load's reactive power difference between internal power flow iterations or any generator's reactive power mismatch between its target and output is larger than this specified value. The allowable range is 0-10.
Maximum Voltage Mismatch	VMISMATCH_FDRCONF <VMismatch>	Text box	This entry specifies the internal power flow's voltage convergence criterion in per unit for the AFR application. The power flow checks the differences between the voltage target and the calculated voltage at the regulated node for every PV generator and reference bus interface. The AFR solution status is set to Diverged if the voltage mismatch is larger than this specified value. The allowable range is 0-0.1.
Switching Plan Generation Parameters			
Number of Selected Top Plans	TOPPLANEVAL_FDRCONF <NoTopPlan>	Text box	This entry specifies the number of top-ranked fully evaluated AFR plans to be listed in the tabular output displays. Plans with smaller performance indexes are ranked higher. If there are both feasible and infeasible plans, the plan with the smallest performance index among the plans that are qualified to be executed in closed loop is ranked as the first. The other plans are ranked next based on their performance indexes. The allowable range is 1-50.
Maximum Number of Internally Generated Plans	<MaxNoGeneratedPlan>	Text box	This entry specifies the number of plans to be generated and fully evaluated by AFR and listed in the AFR results file. The allowable range is 1-50.

Configuration Editor UI Name	CSV File Field Name and XML File Attribute Name	Type	Description
Maximum Number of Feeders	<MaxNumFeeder>	Text box	This entry specifies the maximum number of feeders that are considered in the network selection for AFR. If the number of feeders in the network selection is larger than this entry, AFR will not be solved. As AFR includes complete stations in its network selection, this number should be large (the default value is 150 feeders). The allowable range is 1-200.
Plan Life - Real Time	PLANLIFE_FDRCONF <PlanLife>	Text box	This value specifies the number of hours an AFR plan is saved in the real time DMS. After this period expires, the plan is deleted. This parameter applies only in real time. In study mode, an AFR plan is never deleted. The allowable range is 0.5-168.
Look Ahead Duration	LOOKAHEAD_FDRCONF <LookAheadDuration>	Text box	This value specifies the look-ahead duration in hours to find the peak loading time used in the AFR analysis. AFR is solved at the peak loading time is to make sure the plan is sufficient from a loading perspective once the plan is implemented in the field, especially if the plan requires manual switching. If this is set to 0, AFR will use the current loading values. An entry of 0 typically means that the devices are all automated or loading is not a concern. The allowable range is 0-24.
Switch Transfer Flow Margin (%)	<SwitchTransferMargin>	Text box	This entry specifies the margin in percentage of transfer tie switch flow limit when determining if an AFR plan should be considered. The parameter will affect AFR plan generation under several conditions: • When a level 1 plan restores downstream load to an adjacent feeder, AFR will check if the downstream load is less than the tie switch flow limit plus the margin. • When AFR generates a level 2 split 2, 3, 4, or 5 section plan, the tie switch flow limit will be checked. • When AFR generates a level 3 restoration plan that transfers part of the load on an adjacent feeder to an additional adjacent feeder (a far distance for a tie switch). The allowable range is 0-200.

Configuration Editor UI Name	CSV File Field Name and XML File Attribute Name	Type	Description
Maximum Allowed Number of Open/Close Pairs	<MaximumAllowedOpenClosePairs>	Text box	This entry specifies the maximum allowable number of open/close switching device pairs. Plan generation will stop if the number of open/close pairs in the plan reaches this limit. This parameter only applies to the reduce real power losses problem formulation. The allowable range is 1-100. The default value is 2.
Loss Reduction Amount (%)	<LossReductionAmountPercent>	Text box	This entry specifies the threshold percentage at which the plan generation will stop if the loss reduction reaches this value. The threshold percentage is calculated as the estimated loss reduction divided by the original loss in the network selection. This parameter only applies to the reduce real power losses problem formulation. The allowable range is 0.1-10.0. The default value is 2.0.
Preprocessing Checks			
Automatic Disabling by Feeder	AUTODISABLEBYFEEDER_FDRCONF <AutoDisableByFeeder>	Check box	This check box specifies whether to allow automatically disabling AFR by feeder. If it is checked and the number of optimization ineligible devices in one feeder is greater than or equal to the specified Minimum Number of Unavailable Devices parameter, DMS will disable AFR automatically for this feeder.
Minimum Number of Unavailable Devices	MAXNUMUNAVAILABLEDEVICE_FDRCONF <MaxUnavailableDevice>	Text box	This entry specifies the minimum number of optimization ineligible devices. If automatically disabling AFR by feeder is allowed and the number of optimization ineligible devices in one feeder is greater than or equal to this parameter, DMS will disable AFR automatically for this feeder. The allowable range is 1-20.

Configuration Editor UI Name	CSV File Field Name and XML File Attribute Name	Type	Description
Respect Tags	DISABLESWITCHTAGGED_FDRCONF <DisableSwitchTagged>	Check box	This check box specifies whether switching devices with advanced application eligibility blocking tags are counted as optimization ineligible devices. The types of tags are specified by the TAGTYPE_CONTROLRESTRICTIONS_n parameters in the Main server and DNAF server configuration files. For a tag to block optimization operations, the seventh bit of this parameter must be set. If this box is checked, a switching device with a block advanced application eligibility tag is counted as an optimization ineligible device. If the total number of optimization ineligible devices for one feeder is greater than or equal to the specified Minimum Number of Unavailable Devices, AFR will be automatically disabled for this feeder.
Hot Line	DISABLESWITCHHOTLINETAGGED_FDRCONF <DisableSwitchHotLineTagged>	Check box	This check box specifies whether switching devices with an applied hot line are counted as optimization ineligible devices. If this box is checked, a switching device with an applied hot line is counted as an optimization ineligible device. If the total number of optimization ineligible devices in one feeder is greater than or equal to the specified Minimum Number of Unavailable Devices, AFR will be automatically disabled for this feeder.
In Alarm	DISABLESWITCHINALARM_FDRCONF <DisableSwitchInAlarm>	Check box	This check box specifies whether switching devices with an enabled alarm status measurement are counted as optimization ineligible devices. If this box is checked, a switching device with an enabled alarm status measurement is counted as an optimization ineligible device. If the total number of optimization ineligible devices in one feeder is greater than or equal to the specified Minimum Number of Unavailable Devices, AFR will not be automatically disabled for this feeder.

Configuration Editor UI Name	CSV File Field Name and XML File Attribute Name	Type	Description
Remote Control Disabled	DISABLESWITCHLOCALCTRL_FDRCONF <DisableSwitchLocalControl>	Check box	This check box specifies whether switching devices with a remote/local status measurement set to local are counted as optimization ineligible devices. If this box is checked, a switching device with remote/local status measurement set to local is counted as an optimization ineligible device. If the total number of optimization ineligible devices in one feeder is greater than or equal to the specified Minimum Number of Unavailable Devices, AFR will be automatically disabled for this feeder.
Reclosing Disabled	<DisableSwitchReclosing>	Check box	This check box specifies whether switching devices with a reclosing status measurement set to disabled are counted as optimization ineligible devices. If this box is checked, a switching device with disabled reclosing status measurement is counted as an optimization ineligible device. If the total number of optimization ineligible devices in one feeder is greater than or equal to the specified Minimum Number of Unavailable Devices, AFR will not be automatically disabled for this feeder.
Telemetered Switch Evaluated as a Manual Switch			
Suspect Quality - Status	<EligibleForManualDevice WithSuspectStatusMeas>	Check box	This check box specifies whether telemetered switching devices with a suspect quality switch status measurement are treated as manual switching devices.
Suspect Quality - Hot Line	<EligibleForManualDevice WithSuspectHotLineTagMeas>	Check box	This check box specifies whether telemetered switching devices with a suspect quality Hot Line status measurement are treated as manual switching devices.
Suspect Quality - In Alarm	<EligibleForManualDevice WithSuspectInAlarmMeas>	Check box	This check box specifies whether telemetered switching devices with a suspect quality enabled alarm status measurement are treated as manual switching devices.
Suspect Quality - Remove/Local	<EligibleForManualDevice WithSuspectRemoteLocalMeas>	Check box	This check box specifies whether telemetered switching devices with a suspect quality remote/local status measurement are treated as manual switching devices.
Suspect Quality - Control Available	<EligibleForManualDevice WithSuspectCtrlAvailMeas>	Check box	This check box specifies whether telemetered switching devices with a suspect quality reclosing disabled status measurement are treated as manual switching devices.

Configuration Editor UI Name	CSV File Field Name and XML File Attribute Name	Type	Description
Suspect Quality - Supervisory Cutout	<EligibleForManualDeviceWithSuspectSupervisoryCutoutMeas>	Check box	This check box specifies whether telemetered switching devices with a suspect quality supervisory cutout status measurement are treated as manual switching devices.
Suspect Quality - Auto/Manual	<EligibleForManualDeviceWithSuspectAutoManualMeas>	Check box	This check box specifies whether telemetered switching devices with a suspect quality auto/manual status measurement are treated as manual switching devices.
Good Quality with Abnormal Value - Hot Line	<EligibleForManualDeviceHotLineTagged>	Check box	This check box specifies whether telemetered switching devices with an abnormal Hot Line status measurement with good quality are treated as manual switching devices.
Good Quality with Abnormal Value - In Alarm	<EligibleForManualDeviceInAlarm>	Check box	This check box specifies whether telemetered switching devices with an abnormal alarm status measurement with good quality are treated as manual switching devices.
Good Quality with Abnormal Value - Control Available	<EligibleForManualDeviceCtrlNotAvail>	Check box	This check box specifies whether telemetered switching devices with an abnormal reclosing disabled status measurement with good quality are treated as manual switching devices.
Other			
Integrate With LVM	INTEGRATELVM_SERVREST <IntegrateLVM>	Check box	This check box specifies whether to run LVM inside the AFR application when AFR cannot generate a plan that reduces overload or returns the system to normal without flow or voltage violations. If this box is checked and AFR cannot generate a plan that can restore all de-energized loads without violation, LVM is automatically executed within AFR to check whether LVM can correct the violations after the best AFR plan has been implemented.
LVM Problem Formulation	<IntegrateLVMProbForm>	Drop-down list	This selection specifies the default LVM problem formulation when the Integrate with LVM option is enabled for AFR. This problem formulation automatically becomes the default LVM problem formulation until the station topology is returned to its nominal state.

Configuration Editor UI Name	CSV File Field Name and XML File Attribute Name	Type	Description
Check Flow After Make Before Break Switching	<MakeBeforeBreakCheck>	Check box	This check box specifies whether a load check is enabled as part of make-before-break switching operations. When a distribution dispatcher performs a make-before-break switching, after closing a recloser the dispatcher will check for loading. This is a good check that a recloser jumper did not burn off because of the sudden loading change. In some cases, the recloser may have either a light loading situation or a significant customer who is slower in energizing their equipment. If the threshold specified by the Flow Threshold option is not exceeded, a message is sent to the dispatcher.
Flow Threshold	<MakeBeforeBreakAmps>	Text box	This entry specifies the minimum single-phase Amp flow that should be exceeded when closing a device as part of a make-before-break switching sequence. The allowable range is 0.5-200.
Overload Trigger			

Configuration Editor UI Name	CSV File Field Name and XML File Attribute Name	Type	Description
Plan Execution	PLANEXEOVERLOAD_FDRCONF <OverloadPlanExecution>	Drop-down list	This selection specifies how the overload triggered AFR plan is executed in real time. There are four options: • Advisory: The plan cannot be executed automatically; it is displayed in the AFR plans tabular display. The operator can execute the plan manually (automatic and/or manual devices). • Semi-Automatic: The plan can be executed automatically after operator approval. The plan will appear in the AFR plans tabular display and can be implemented by clicking the "Implement the Plan" button if all devices in the plan are remotely controlled. • Closed Loop: The best plan without voltage and flow violations is automatically executed if all devices in the plan are remotely controlled. For this option, if no plan is without violations, the plans can be automatically executed after operator approval, similar to the Semi-Automatic option. • Closed Loop - Calculated Voltage Limit Violation Not Worse: Works similarly to the Closed Loop option, but also considers the plans with voltage violations. A plan can be executed automatically if the calculated voltage violations are not made worse by the plan. A voltage margin specified by the Violation Not Worse Margin (%) parameter is used to determine if a plan resulting in voltage violations is acceptable. The plan can be turned into a switch order for all four options.

Configuration Editor UI Name	CSV File Field Name and XML File Attribute Name	Type	Description
Settings Group Check	INCLUDESETTING GROUPOVERLOAD_FDRCONF <OverloadIncorpProtectCoord>	Drop-down list	This selection specifies how the protection settings group steps are added to an overload triggered AFR execution. The options are: • None: No settings group steps are added to the AFR plan. • By Feeder Source: Settings group steps are added to the AFR plan. • By Minimum Fault Current: If this option is set, Protective Validation (PRV) is forced to run after the AFR plan is generated. The active protection settings group is selected by finding the settings group with the highest phase trip current, which is below the minimum phase fault current on the protected protection area (multiplied by the appropriate safety factor) and also above the estimated load current at the peak load time and the continuous current limit of the protective device.

Configuration Editor UI Name	CSV File Field Name and XML File Attribute Name	Type	Description
Settings Group Execution	<OverloadSettingsGroupExecution>	Drop-down list	This selection specifies how the protection settings group steps are executed for an overload triggered AFR. The selection options depend on the Settings Group Selection option. The options are: • None: Execution options are disabled when no settings group steps are added to the AFR plan. • Always Before Switching: Settings group steps are executed before the switching step that results in the settings group change. • Always After Switching: Settings group steps are executed after the switching step that results in the settings group change. • Switch Before for Trip Raise, Switch After for Trip Lower: (This option is only valid when setting groups are checked by minimum fault current.) Settings group steps are executed before the switching step that results in the settings group change if the phase trip settings are being raised. Settings group steps are executed after the switching step that results in the settings group change if the phase trip settings are being lowered. Note Raise is the phase trip setting change to a higher value (for example, from 300 A to 500 A). Lower is the phase trip setting change to a lower value (for example, from 500 A to 300 A).
Power Transformer Trigger	OVERLOADTRIGGERPOWER TF_FDRCONF <OverloadTriggerByPowerTF>	Check box	This check box specifies whether the overload of a power transformer not in an AFR disabled feeder or station will trigger AFR with the Reduce Overloads and Violations problem formulation.

Configuration Editor UI Name	CSV File Field Name and XML File Attribute Name	Type	Description
Line or Switch Trigger	OVERLOADTRIGGERLINE_FDRCONF <OverloadTriggerByLine>	Check box	This check box specifies whether the overload of feeder main lines or switching devices will trigger AFR as long as they are not in an AFR disabled feeder. If both this option and the Power Transformer Trigger are enabled and there is an overload for a power transformer on an AFR enabled feeder, AFR will solve only for this power transformer and will not solve for overloaded feeder main lines or switches. When triggering due to this option, AFR will solve for the line or switching device with the largest loading percentage.
Use Triggering Feeder Plan Execution Mode	<OverloadTriggerUseFeederPlanExecutionMode>	Check box	This check box specifies whether only the plan execution mode specified for the triggering feeder on the DNAF System Configuration display is considered for the AFR overload trigger. If selected, the triggering feeder's plan execution mode is used. If cleared, the most restrictive execution mode among the feeders in the plan is used. For more details, see Plan Execution Mode.
Allow Closed Loop Execution if Protection Validation Violations Exist	<AllowClosedLoopOverloadTriggerAFRIfPRVViolation>	Check box	This check box specifies whether the closed loop execution is allowed for overload-triggered AFR plans with any type of protection validation violation in the post-plan Protection Validation (PRV) function (maximum interruption violation, minimum phase or ground pickup violation, protection peak load margin violation, or protection reach violation). If this option is selected and the plan execution mode is set to closed-loop, AFR automatically implements valid plans (without overloads or high/low voltage violations) regardless of PRV violations.

Manual Trigger

Configuration Editor UI Name	CSV File Field Name and XML File Attribute Name	Type	Description
Plan Execution	<ManualPlanExecution>	Drop-down list	This selection specifies how the plan for a manually triggered AFR is executed in real time. There are four options: • Advisory: The plan cannot be executed automatically; it appears in the AFR plans tabular display. The plan can be executed manually by the operator (automatic or manual devices) or possibly turned into a switch order. • Semi-Automatic: The plan can be executed automatically after operator approval. The plan will appear in the AFR plans tabular display and can be implemented by clicking the "Implement the Plan" button if all devices in the plan are remotely controlled. • Closed Loop: The best plan without voltage and flow violations is executed automatically if all devices in the plan are remotely controlled. • Closed Loop - Calculated Voltage Limit Violation Not Worse: Works similarly to the Closed Loop option, but also considers the plans with voltage violations. A plan can be executed automatically if the calculated voltage violations are not made worse by the plan. A voltage margin specified by the Violation Not Worse Margin (%) parameter is used to determine if a plan resulting in voltage violations is acceptable. The plan can be turned into a switch order for all four options.

Configuration Editor UI Name	CSV File Field Name and XML File Attribute Name	Type	Description
Settings Group Check	<ManualIncorpProtectCoord>	Drop-down list	This selection specifies how the protection settings group steps are added to a manually triggered AFR execution. The options are: • None: No settings group steps are added to the AFR plan. • By Feeder Source: Settings group steps are added to the AFR plan. • By Minimum Fault Current: If this option is set, Protective Validation (PRV) is forced to run after the AFR plan is generated. The active protection settings group is selected by finding the settings group with the highest phase trip current, which is below the minimum phase fault current on the protected protection area (multiplied by the appropriate safety factor) and also above the estimated load current at the peak load time and the continuous current limit of the protective device.

Configuration Editor UI Name	CSV File Field Name and XML File Attribute Name	Type	Description
Settings Group Execution	<ManualSettingsGroupExecution>	Drop-down list	This selection specifies how the protection settings group steps are executed for a manually triggered AFR. The selection options depend on the Settings Group Selection option. The options are: • None: Execution options are disabled when no settings group steps are added to the AFR plan. • Always Before Switching: Settings group steps are executed before the switching step that results in the settings group change. • Always After Switching: Settings group steps are executed after the switching step that results in the settings group change. • Switch Before for Trip Raise, Switch After for Trip Lower: (This option is only valid when setting groups are checked by minimum fault current.) Settings group steps are executed before the switching step that results in the settings group change if the phase trip settings are being raised. Settings group steps are executed after the switching step that results in the settings group change if the phase trip settings are being lowered.


Note

Raise is the phase trip setting change to a higher value (for example, from 300 A to 500 A. Lower is the phase trip setting change to a lower value (for example, from 500 A to 300 A).

Configuration Editor UI Name	CSV File Field Name and XML File Attribute Name	Type	Description
Allow Closed Loop Execution if Protection Validation Violations Exist	<AllowClosedLoopManualTriggerAFRIfPRVViolation>	Check box	This check box specifies whether the closed loop execution is allowed for manually-triggered AFR plans with any type of protection validation violation in the post-plan Protection Validation (PRV) function (maximum interruption violation, minimum phase or ground pickup violation, protection peak load margin violation, or protection reach violation). If this option is selected and the plan execution mode is set to closed-loop, AFR automatically implements valid plans (without overloads or high/low voltage violations) regardless of PRV violations.

Solution Results			
Create AFR Result File	PRINTOUT_FDRCONF <PrintResult>	Check box	This check box specifies whether to create an AFR results file. If this box is checked, a results file is generated. The results file is created in the folder specified by the parameter DNAF_OUTPUT_PATH in the appropriate server configuration file. The file name format depends on whether "Use Unique File Name" is enabled. Refer to the description for "Use Unique File Name" for more details.
Use Unique File Name	UNIQUELOG_FDRCONF <UniqueLog>	Check box	This check box specifies whether to create a unique AFR results file. If this box is checked, a unique results file is created for every AFR execution. In this case, the file name is in the format "AFR Results_<stationname>_<executiontimestamp>.dat". If this check box is cleared, the file name is in the format "AFR Results_<stationname>.dat". The results file is overwritten each time AFR is executed for that station.
Use Detailed Result Reporting	DETAILLOG_FDRCONF <DetailLog>	Check box	This check box specifies whether to create a detailed AFR results file. If it is checked, the AFR results file contains additional information, including the list of components with voltage and flow violations.
Top High Bus/Load Voltages Reported	<MaxNumHighVoltReported>	Text box	This entry specifies the maximum number of top ranking high bus or load voltages to be reported in the AFR results file.
Top Low Bus/Load Voltages Reported	<MaxNumLowVoltReported>	Text box	This entry specifies the maximum number of top ranking low bus or load voltages to be reported in the AFR results file.

Configuration Editor UI Name	CSV File Field Name and XML File Attribute Name	Type	Description
Top Branch Loading Levels Reported	<MaxNumBranchLoading LevelsReported>	Text box	This entry specifies the maximum number of top-ranking branch segment loading levels to be reported in the AFR results file.
Debug Files			
Maximum Number of Debug Files	MAXUPFOUTPUTFILE_FDRCONF <MaxNumUpfOutFile>	Text box	This entry specifies the maximum number of accumulated debug output files created when AFR does not solve. These debug files are output in the folder specified by the parameter DNAF_OUTPUT_PATH in the appropriate server configuration file and can be identified by the prefixes "upfin_export_", "optalg_export_" and "alg_export_" with station name and application name in the file name. Once the limit is reached, subsequent debug output files are not created. The allowable range is 0-50.
Manual Execution			
Allow in Real Time	REALTIME_FDRCONF <RealTime>	Check box	This check box indicates if the AFR application can be invoked through the RT DNAF Solution pop-up display (checked means the application is shown). Note AFR can still be executed by event triggers in RT by setting the appropriate triggering parameters in the server configuration file.
Allow in Study Mode	STUDYMODE_FDRCONF <Study>	Check box	This check box indicates if the AFR application can be invoked through the ST DNAF Solution pop-up display (checked means the application is shown).
Old and Deprecated Parameters			
Function Name	NAME_FDRCONF <Name>	Text box	This entry specifies the application name in the DNAF pop-up display. This field is exposed only in the old DNAF Configurator, not in the new DNAF Configuration Editor.

Configuration Editor UI Name	CSV File Field Name and XML File Attribute Name	Type	Description
Function Enumeration	FUNCENUM_FDRCONF <FuncEnum>	Text box	This entry specifies the application enumeration: 0 = Other applications 1 = Distribution Power Flow 2 = Protection Validation 4 = Load and Volt/VAR Management 5 = Fault Isolation and Service Restoration 6 = Automatic Feeder Reconfiguration 7 = Planned Outage Study 8 = FISIR Contingency Analysis 10 = Fault Location 11 = Short Circuit Calculation This field is exposed only in the old DNAF Configurator, not in the new DNAF Configuration Editor.

7. Planned Outage Study

7.1. PLOS Plan Generation

The Planned Outage Study (PLOS) applications assist the user in determining the best circuit configuration that isolates components in the distribution model (for example, for planned maintenance).

PLOS uses the same problem formulations as FISR. It also uses the same method as FISR described in [FISR Plan Generation](#); the only difference is that PLOS first transfers the affected load and then isolates the component to be isolated. This is to prevent the unnecessary dropping of load for reliability reasons.

The components to be isolated by the PLOS application can be defined by one or more of the following individual definitions. These definitions can be specified from either the ADMS client and/or the Hazard Management System (HMS) depending on the definition:

1. A complete station. PLOS will isolate the power transformers and all the components downstream of the power transformers. This definition can only be invoked from the ADMS geographic display by selecting the station circle in the geographic display and bringing up the DNAF popup to execute PLOS.
2. A complete feeder. PLOS will isolate all the components downstream of the feeder head breaker or feeder head node if there is no feeder head breaker defined for the feeder. This definition can only be invoked from the HMS.
3. A bus. PLOS will isolate the selected bus. This definition can only be invoked from the DMS geographic display by selecting the bus in the geographic display and bringing up the DNAF popup to execute PLOS.
4. A device defining the topmost point of a zone (zone will extend down to end of a feeder). PLOS will isolate all the components downstream of this device. This definition can only be invoked from the HMS.
5. A set of boundary devices within a feeder that defines the extent (edges) of the zone. PLOS will isolate all the components inside the zone bounded by the boundary devices. This definition can only be invoked from the HMS.
6. One component. PLOS will isolate this selected component. This definition can be invoked within the ADMS by selecting the component in the geographic display and bringing up the DNAF popup to execute PLOS.
7. A group of components. PLOS will isolate all of the selected components. This definition can only be invoked from the HMS.

7.2. Modes of Operations

This section describes the modes of operation for PLOS. In both real-time and study mode, DPF

must be solved first before executing PLOS.

When the PLOS application is invoked from the DMS geographic display or HMS, it generates isolation switching plans to take the selected substation/feeder/components out of service.

The current system topology and peak load time (within the pre-defined, look-ahead time window) are utilized in the analysis. If PLOS is invoked from the ADMS geographic display, the peak load time is determined using the evaluation time and the Look Ahead Duration defined in the Planned Outage Study (PLOS) tab in the DNAF Configuration Editor (DCE). If PLOS is invoked from HMS, the peak load time is determined based on the Hazard Event start time and end time taken from HMS.

The PLOS application only uses the energized portions of the network when looking for alternate energy sources during load transfers.

7.3. PLOS Alarms

The following alarms are currently supported for PLOS in the real-time environment:

Table 37. PLOS Alarms

Alarm	Description
Disabled by DMS	When PLOS is disabled by DMS operator, this alarm is issued.
PLOS Not Triggered	When PLOS is not triggered due to insufficient information, this alarm is issued.
PLOS Started	When PLOS execution is started, this alarm is issued.
PLOS Success	When PLOS execution completed without errors, this alarm is issued.
PLOS Failed	When PLOS execution fails due to internal errors, this alarm is issued.

7.4. PLOS Configuration

This section describes the Planned Outage Study configuration parameters.

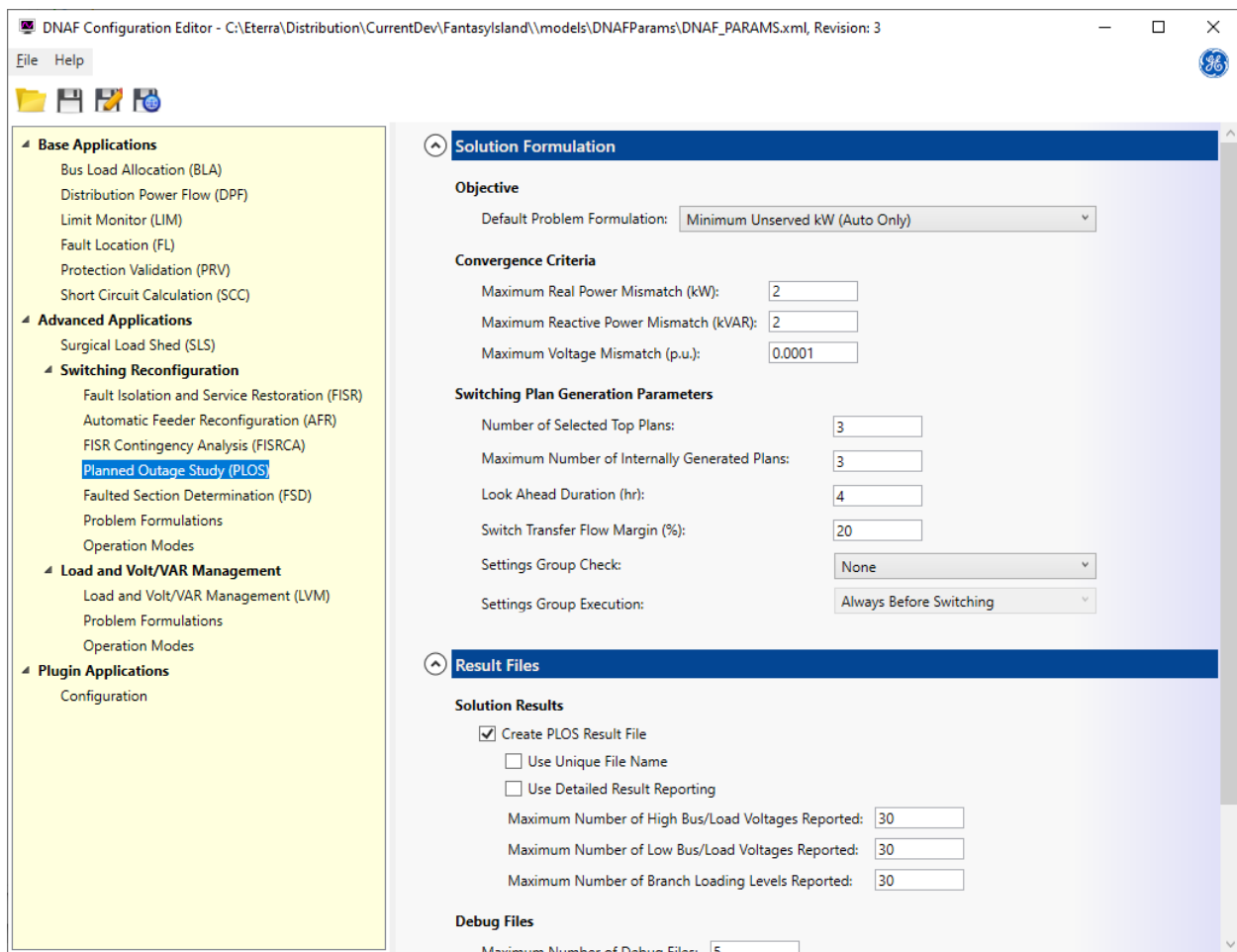


Figure 137. PLOS Configuration

Table below lists the PLOS configuration parameters. The <Params xsi:type = "POSPParams"> element in the configuration file DNAF_PARAMS.xml contains all listed parameters.

Table 38. PLOS Configuration Parameters

Configuration Editor UI Name	CSV File Field Name / XML File Element	Type	Description
Objective			
Default Problem Formulation	<DefaultProbIndex>	Drop-down list	This list specifies the default problem formulation to use for a PLOS analysis. This value will be overridden by any changes specified for individual stations in the Study Mode DNAF System Configuration display, and by a specific problem formulation requested in the DNAF pop-up display.
Convergence			

Configuration Editor UI Name	CSV File Field Name / XML File Element	Type	Description
Maximum Real Power Mismatch	PMISMATCH_PLANOUT <PMismatch>	Text box	This entry specifies the internal power flow's real power convergence criterion in kW for the PLOS application. The power flow checks the calculated real power difference for every load between the current iteration and the previous iteration and the mismatch between the real power target and calculated output of every PQ and PV generator. The PLOS solution status is set to Diverged if any load's real power difference between internal power flow iterations or any generator's real power mismatch between its target and output is larger than this specified value. The allowable range is 0-10.
Maximum Reactive Power Mismatch	QMISMATCH_PLANOUT <QMismatch>	Text box	This entry specifies the internal power flow's reactive power convergence criterion in kVAR for the PLOS application. The power flow checks the calculated reactive power difference for every load between the current iteration and the previous iteration and the mismatch between the reactive power target and calculated output of every PQ generator. The PLOS solution status is set to Diverged if any load's reactive power difference between internal power flow iterations or any generator's reactive power mismatch between its target and output is larger than this specified value. The allowable range is 0-10.
Maximum Voltage Mismatch	VMISMATCH_PLANOUT <VMismatch>	Text box	This entry specifies the internal power flow's voltage convergence criterion in per unit for the PLOS application. The power flow checks the differences between the voltage target and the calculated voltage at the regulated node for every PV generator and reference bus interface. The PLOS solution status is set to Diverged if the voltage mismatch is larger than this specified value. The allowable range is 0-0.1.
Parameters			

Configuration Editor UI Name	CSV File Field Name / XML File Element	Type	Description
Number of Selected Top Plans	TOPPLANEVAL_PLANOUT <NoTopPlan>	Text box	This entry specifies the number of top-ranked fully evaluated PLOS plans to be listed in the tabular output displays. Plans with smaller performance indexes are ranked higher. If there are both feasible and infeasible plans, the plan with the smallest performance index among the plans that are qualified to be executed in closed loop is ranked as the first. The other plans are ranked next based on their performance indexes. The allowable range is 1-50.
Maximum Number of Internally Generated Plans	<MaxNoGeneratedPlan>	Text box	This entry specifies the number of plans to be generated and fully evaluated by PLOS and listed in the PLOS results file. The allowable range is 1-50.
Look Ahead Duration	LOOKAHEAD_PLANOUT <LookAheadDuration>	Text box	This entry specifies the look-ahead duration in hours to find the peak loading time used in the PLOS analysis. PLOS is solved at the peak loading time to make sure the plan is sufficient from a loading perspective once the plan is implemented in the field (especially if the plan requires manual switching). If this value is set to 0, PLOS will use the current loading. A value of 0 typically means that the devices are all automated or loading is not a concern. The allowable range is 0-24.
Switch Transfer Flow Margin (%)	<SwitchTransferMargin>	Text box	This entry specifies the margin in percentage of transfer tie switch flow limit when determining if a PLOS plan should be considered. The parameter will affect PLOS plan generation under several conditions: • When a level 1 plan restores downstream load to an adjacent feeder, PLOS will check if the downstream load is less than the tie switch flow limit plus the margin. • When PLOS generates a level 2 split 2, 3, 4, or 5 section plan, the tie switch flow limit will be checked. • When PLOS generates a level 3 restoration plan that transfers part of the load on an adjacent feeder to an additional adjacent feeder (a far distance for a tie switch). The allowable range is 10-200.

Solution Results

Configuration Editor UI Name	CSV File Field Name / XML File Element	Type	Description
Create PLOS Result File	PRINTOUT_PLANOUT <PrintResult>	Check box	This check box specifies whether to create a PLOS results file. If this box is checked, a results file is generated. The results file is created in the folder specified by the DNAF_OUTPUT_PATH parameter in the appropriate server configuration file. The file name format depends on whether "Use Unique File Name" is enabled. Refer to the description for "Use Unique File Name" for more details.
Use Unique File Name	UNIQUELOG_PLANOUT <UniqueLog>	Check box	This check box specifies whether to create a unique PLOS results file. If this box is checked, a unique results file is created for every PLOS execution. In this case, the file name is in the format "PLOS Results_<stationname>_<executiontimestamp>.dat". If this check box is cleared, the file name is in the format "PLOS Results_<stationname>.dat". The results file is overwritten each time PLOS is executed for the station.
Use Detailed Result Reporting	DETAILLOG_PLANOUT <DetailLog>	Check box	This check box specifies whether to create a detailed PLOS results file. If it is checked, the PLOS results file includes additional information, such as the list of components with voltage and flow violations.
Debug Files			
Maximum Number of Debug Files	MAXUPFOUTPUTFILE_PLANOUT <MaxNumUpfOutFile>	Text box	This entry specifies the maximum number of accumulated debug output files created when PLOS does not solve. These debug files are output in the folder specified by the parameter DNAF_OUTPUT_PATH in the appropriate server configuration file and can be identified by the prefixes "upfin_export_", "optalg_export_" and "alg_export_" with station name and application name in the file name. Once the limit is reached, subsequent debug output files are not created. The allowable range is 0-50.
Manual Execution			
Allow in Study Mode	STUDYMODE_PLANOUT <Study>	Check box	This check box indicates if the PLOS application can be invoked through the ST DNAF Solution pop-up display (checked means the application is shown).
Old and Deprecated Parameters			

Configuration Editor UI Name	CSV File Field Name / XML File Element	Type	Description
N/A	NAME_PLANOUT <Name>	Text box	This entry specifies the application name in the DNAF pop-up display. This field is exposed only in the old DNAF Configurator, not in the new DNAF Configuration Editor.
N/A	FUNCENUM_PLANOUT <FuncEnum>	Integer	This value specifies the application enumeration: 0 = Other applications 1 = Distribution Power Flow 2 = Protection Validation 4 = Load and Volt/VAR Management 5 = Fault Isolation and Service Restoration 6 = Automatic Feeder Reconfiguration 7 = Planned Outage Study 8 = FISR Contingency Analysis 10 = Fault Location 11 = Short Circuit Calculation This field is exposed only in the old DNAF Configurator, not in the new DNAF Configuration Editor.

8. Fault Isolation and Service Restoration - Contingency Analysis

8.1. FISRCA Plan Generation

FISRCA simulates placing a fault in each non-looped feeder main switchable section in the feeder. FISRCA uses the same four-level restoration logic as FISR. The top restoration plan, if any, for each switchable section is shown in the results tabular displays.

8.2. Modes of Operations

This section describes the modes of operation for FISRCA. DPF must be solved before executing the FISRCA application.

The FISRCA application can be executed in study mode only and is manually triggered. It generates a solution summary for the studied substation along with switching plans for every evaluated switchable section.

8.3. Performance

The time FISRCA takes to solve a station is greatly impacted by the number of switchable sections solved in the station. For example, in the demo model (Fantasy Island), the number of switchable sections that are solved in one station ranges from 41 to 304, where the average is 132 switchable sections per station. Using a machine with 1 CPU @ 2.4 GHz with 8 cores and 64 GB of memory (RAM), the average solution time per switchable section is 16.9 seconds and the average solution time per station is 37.3 minutes utilizing the Minimum Unserved kW (Auto + Manual) problem formulation and all the default parameters in the DNAF Configuration Editor.

The following key configuration parameter changes may impact the FISRCA performance:

- Disabling the Network Reduction Algorithm option.
- Setting the Protection Validation performance index weight to a non-zero value, so that plan PRV violations are evaluated.
- Setting the Number of Adjacency Levels value to 1.
- Disabling the Settings Groups check.

8.4. FISRCA Configuration

This section describes the Fault Isolation and Service Restoration Contingency Analysis configuration parameters.

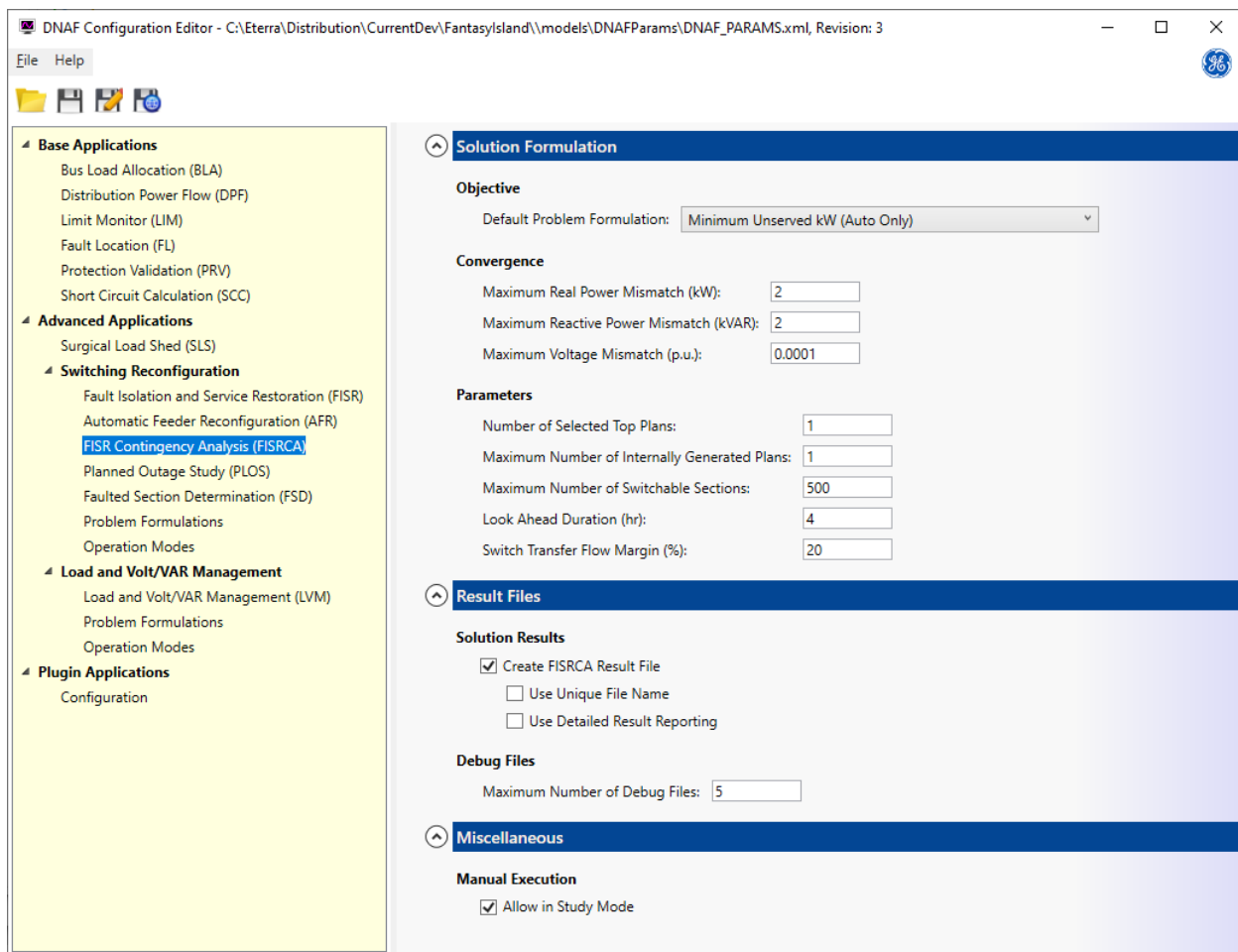


Figure 138. FISRCA Configuration

Table below lists the FISRCA configuration parameters. The <Params xsi:type = "FISRCAParams"> element in the DNAF_PARAMS.xml configuration file contains all listed parameters.

Table 39. FISRCA Configuration Parameters

Configuration Editor UI Name	CSV File Field Name / XML File Element	Type	Description
Objective			
Default Problem Formulation	<DefaultProbIndex>	Drop-down list	This selection specifies the default problem formulation to use for a FISRCA analysis. This value will be overridden by any changes specified for individual stations in the Study Mode DNAF System Configuration display, and by a specific problem formulation requested in the DNAF Solution pop-up display.
Convergence			

Configuration Editor UI Name	CSV File Field Name / XML File Element	Type	Description
Maximum Real Power Mismatch	PMISMATCH_FISRCA <PMismatch>	Text box	This entry specifies the internal power flow's real power convergence criterion in kW for the FISRCA application. The power flow checks the calculated real power difference for every load between the current iteration and the previous iteration and the mismatch between the real power target and calculated output of every PQ and PV generator. The FISRCA solution status is set to Diverged if any load's real power difference between internal power flow iterations or any generator's real power mismatch between its target and output is larger than this specified value. The allowable range is 0-10.
Maximum Reactive Power Mismatch	QMISMATCH_FISRCA <QMismatch>	Text box	This entry specifies the internal power flow's reactive power convergence criterion in kVAR for the FISRCA application. The power flow checks the calculated reactive power difference for every load between the current iteration and the previous iteration and the mismatch between the reactive power target and calculated output of every PQ generator. The FISRCA solution status is set to Diverged if any load's reactive power difference between internal power flow iterations or any generator's reactive power mismatch between its target and output is larger than this specified value. The allowable range is 0-10.
Maximum Voltage Mismatch	VMISMATCH_FISRCA <VMismatch>	Text box	This entry specifies the internal power flow's voltage convergence criterion in per unit for the FISRCA application. The power flow checks the differences between the voltage target and the calculated voltage at the regulated node for every PV generator and reference bus interface. The FISRCA solution status is set to Diverged if the voltage mismatch is larger than this specified value. The allowable range is 0-0.1.
Parameters			

Configuration Editor UI Name	CSV File Field Name / XML File Element	Type	Description
Number of Selected Top Plans	TOPPLANEVAL_FISRCA <NoTopPlan>	Text box	This entry specifies the number of top-ranked fully evaluated FISR plans for every internal FISR to be listed in the tabular output displays. Plans with smaller performance indexes are ranked higher. If there are both feasible and infeasible plans, the plan with the smallest performance index among the plans that are qualified to be executed in closed loop and is ranked as the first. The other plans are ranked next based on their performance indexes. The allowable range is 1-50.
Maximum Number of Internally Generated Plans	<MaxNoGeneratedPlan>	Text box	This entry specifies the number of plans to be generated and fully evaluated by FISRCA for every internal FISR and listed in the FISRCA results file. The allowable range is 1-50.
Maximum Number of Switchable Sections	MAXSWSECTION_FISRCA <MaxNoSwitchableSection>	Text box	This entry specifies the maximum number of switchable sections that are to be studied in one FISRCA execution. If this value is set to 1, only the selected switchable section is analyzed. The allowable range is 1-2000.
Look Ahead Duration	LOOKAHEAD_FISRCA <LookAheadDuration>	Text box	This entry specifies the look-ahead duration in hours to find the peak loading time used in the FISRCA analysis. FISRCA is solved at peak loading time to make sure the plans are sufficient from a loading perspective once a plan is implemented in the field, especially if the plan requires manual switching. If this is set to 0, FISRCA will use current loading values. An entry of 0 typically means that the devices are all automated or loading is not a concern. The allowable range is 0-24.

Configuration Editor UI Name	CSV File Field Name / XML File Element	Type	Description
Switch Transfer Flow Margin (%)	<SwitchTransferMargin>	Text box	This entry specifies margin in percentage of transfer tie switch flow limit when determining if a FISRCA plan should be considered. The parameter will affect FISRCA plan generation under several conditions: • When a level 1 plan restores downstream load to an adjacent feeder, FISRCA will check if the downstream load is less than the tie switch flow limit plus the margin. • When FISRCA generates a level 2 split 2, 3, 4, or 5 section plan, the tie switch flow limit will be checked. • When FISRCA generates a level 3 restoration plan that transfers part of the load on an adjacent feeder to an additional adjacent feeder (a far distance for a tie switch). The allowable range is 10-200.

Solution Results

Create FISRCA Result File	PRINTOUT_FISRCA <PrintResult>	Check box	This check box specifies whether to create a FISRCA results file. If this box is checked, a results file is generated. The results file is created in the folder specified by the DNAF_OUTPUT_PATH parameter in the appropriate server configuration file. The file name format depends on whether "Use Unique File Name" is enabled. Refer to the description for "Use Unique File Name" for more details.
Use Unique File Name	UNIQUELOG_FISRCA <UniqueLog>	Check box	This check box specifies whether to create a unique FISRCA results file. If this box is checked, a unique results file is created for every FISRCA execution. In this case, the file name is in the format "FISRCA Results_<stationname>_<executiontimestamp>.dat". If this check box is cleared, the file name is in the format "FISRCA Results_<stationname>.dat". The results file is overwritten each time FISRCA is executed for that station.
Use Detailed Result Reporting	DETAILLOG_FISRCA <DetailLog>	Check box	This check box specifies whether to create a detailed FISRCA results file. If it is checked, the FISRCA results file includes additional information such as the list of components with voltage and flow violations.

Debug Files

Configuration Editor UI Name	CSV File Field Name / XML File Element	Type	Description
Maximum Number of Debug Files	MAXUPFOUTPUTFILE_FISRCA <MaxNumUpfOutFile>	Text box	This entry specifies the maximum number of accumulated debug output files created when FISRCA does not solve. These debug files are output in the folder specified by the parameter DNAF_OUTPUT_PATH in the appropriate server configuration file and can be identified by the prefixes "upfin_export_", "optalg_export_" and "alg_export_" with station name and application name in the file name. Once the limit is reached, subsequent debug output files are not created. The allowable range is 0-50.
Manual Execution			
Allow in Study Mode	STUDYMODE_FISRCA <Study>	Check box	This check box indicates if the FISRCA application can be invoked through the ST DNAF Solution pop-up display (checked means the application is shown).
Old and Deprecated Parameters			
N/A	NAME_FISRCA <Name>	Text box	This entry specifies the application name in the DNAF pop-up display. This field is exposed only in the old DNAF Configurator, not in the new DNAF Configuration Editor.
N/A	FUNCENUM_FISRCA <FuncEnum>	Integer	This value specifies the application enumeration: 0 = Other applications 1 = Distribution Power Flow 2 = Protection Validation 4 = Load and Volt/VAR Management 5 = Fault Isolation and Service Restoration 6 = Automatic Feeder Reconfiguration 7 = Planned Outage Study 8 = FISR Contingency Analysis 10 = Fault Location 11 = Short Circuit Calculation This field is exposed only in the old DNAF Configurator, not in the new DNAF Configuration Editor.

9. Faulted Section Determination

The Faulted Section Determination (FSD) function provides switching plans to identify a fault and isolate faulted sections. The FSD function applies a switching-based approach with step-by-step feeder breaker opening and closing switching cycles. At Stage 1, a faulted feeder downstream of a faulted power transformer is identified and isolated. At Stage 2, the faulted section is identified and isolated, the fault is placed in the faulted section.

The application is designed to operate for network structures with extensive use of compensated networks. In compensated networks with single phase-to-ground faults, the short circuit currents may be small and hardly detected by conventional fault passage indicators. Instead, the fault detection mechanism considers voltage presence at the Arc Suppression Coil (ASC or sometimes called Petersen coil) fault indicators on the load side (bus) of the station power transformer. When the ASC fault status is set, a fault is present in the downstream network, even though no breaker or recloser trip may have occurred.

The FSD application uses the following factors to determine a faulted switchable section:

- Fault indicator state
- ASC voltage presence

The FSD application operates at two independent stages, which are run at different conditions and perform different tasks:

- **Stage 1 FSD:** Identifies the faulted feeder when a fault is identified downstream of a station power transformer.
- **Stage 2 FSD:** After a faulted feeder is identified at FSD Stage 1, FSD Stage 2 identifies the faulted section and places a fault.

The FSD application is implemented as a plug-in to the DNaF server. To enable this application, set the APP_FSD_ENABLED parameter to 1 in the NetServerDnafConfig.txt file.

9.1. FSD Methodology

Stage 1: Faulted Feeder Identification

At FSD Stage 1, the system determines feeders downstream of the power transformer with a set ASC fault indicator. Then the feeders are ordered for the opening and closing switching cycles to identify a faulted feeder. Weighting factors are used for ranking and ordering feeders in FSD Stage 1.

Feeder ranks are calculated using the following equation:

$$\text{Feeder Rank} = FI * WFI + NC * WNC + TL * WTL + FDRL * WFDRL - NPC * WCP \quad (\text{Equation 9.1})$$

Where:

- **FI:** Set fault indicators: 1 if there are set fault indicators on a feeder; 0 if there are no set fault indicators on a feeder.
- **WFI:** The weighting factor for the number of set fault indicators.
- **NC:** The number of customers on a feeder.
- **WNC:** The weighting factor for the number of customers.
- **TL:** Total load (in kW) connected to a feeder.
- **WTL:** The weighting factor for total load.
- **FDRL:** Total feeder overhead conductor length (in feet if using imperial system, in meters if using metric system).
- **WFDRL:** The weighting factor for feeder overhead conductor length.
- **NPC:** Total number of priority customers on a feeder. It means all the feeder head breaker downstream load priority customers. Each load priority customer number = load priority * load customer.
- **WCP:** The weighting factor for the number of priority customers.

Stage 2: Faulted Section Identification

At FSD Stage 2, the system determines automated switching devices downstream of the open feeder breaker that are normally closed. Then the switches are ordered from the top feeder head to the bottom for the opening and closing switching cycles to identify a faulted section.

FSD Example

Figure below demonstrates the functionality of the FSD application.

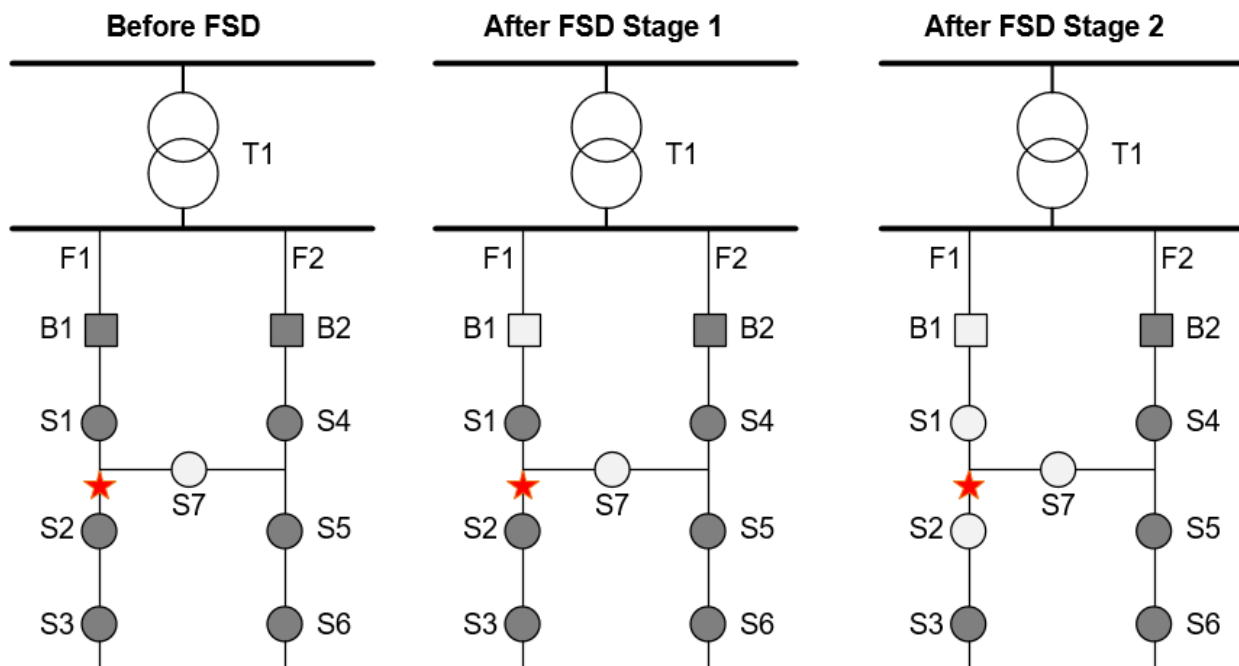


Figure 139. FSD Functionality Demonstration

A fault occurs between SCADA switches S1 and S2. ASC fault indicator on transformer T1 identifies a fault in downstream network.

First, FSD Stage 1 triggers, in which:

1. FSD determines two feeders, F1 and F2, under transformer T1.
2. FSD ranks and orders the feeders; for example, F2 is ranked higher.
3. FSD opens breaker B2 to isolate feeder F2.

The fault is still indicated on the transformer T1 ASC fault indicator.

4. FSD closes B2 and opens B1. The fault is no longer indicated.

The faulted feeder, F1, is identified and isolated. FSD Stage 1 is completed.

Then, FSD Stage 2 triggers, in which:

1. FSD determines three closed SCADA switches, S1, S2, and S3, under feeder breaker F1. FSD orders switches from the top to the bottom, S1 is the first, followed by S2, and then by S3.
2. FSD opens switch S1 and closes breaker B1 to test the section between B1 and S1 for a fault.

The breaker B1 is not tripped and the fault is not indicated on the transformer T1 ASC fault indicator. The fault is not between B1 and S1.

The faulted section is not yet determined.

3. FSD opens B1, closes S1, opens S2, and closes B1 to test the section between S1 and S2 for a fault. The breaker B1 is tripped.

The fault is identified to be between S1 and S2. FSD opens S1 to fully isolate the fault.

4. FSD places the fault in the faulted section.

FSD Stage 2 is completed.

FISR can now be triggered to restore de-energized sections.

9.2. Modes of Operations

The FSD application can be executed in real-time mode only.

The FSD application does not require DPF to be run; FSD is supported for stations that are not DNAF capable. However, DPF is used if FISR is triggered after FSD.

The application can be configured to trigger automatically by the following events.

- Power transformer ASC fault indicator status is updated from "Not Set" to "Set". (FSD Stage 1 is triggered)
- Feeder fault indicator status is updated from "Not Set" to "Set". (FSD Stage 2 is triggered)

FSD can also be run manually for sections downstream of set fault indicators. This can be done

from the DNAF Solution pop-up display.

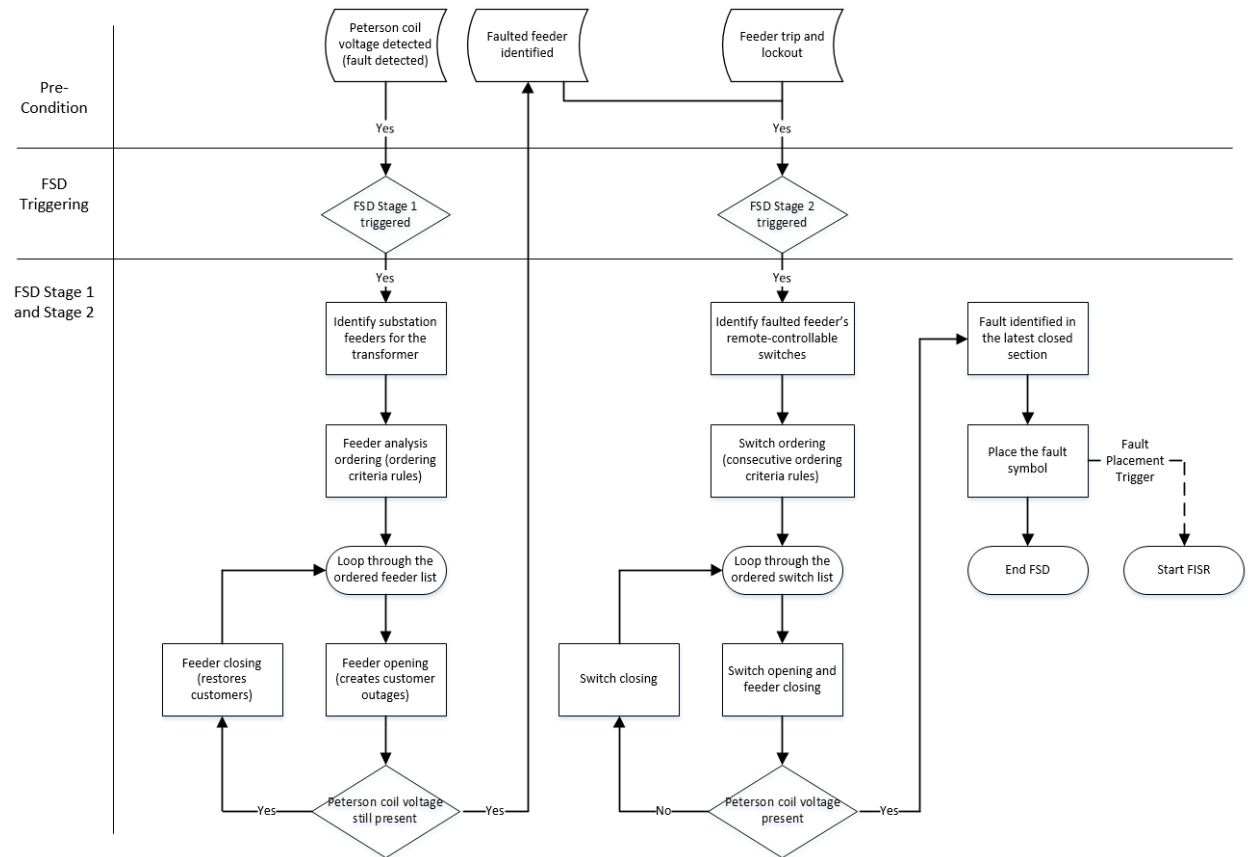


Figure 140. FSD Flowchart

Stage 1: Faulted Feeder Identification

FSD Stage 1 can run after a fault is detected by a station transformer's ASC fault indicator. No SCADA device trip in the downstream network is required at this stage.

FSD orders feeders connected to the station power transformer according to the feeder priority ranking. Feeders with higher rankings are considered the most probable fault location and are processed (tested) first. Feeder ranking weighting factors are configurable.

After a feeder order (ranked feeder list) is created, FSD cycles through the feeders until the faulted feeder is found. Each cycle includes opening a feeder breaker, checking whether the transformer ASC fault indicator status is not set, and closing the feeder breaker (for non-faulted feeders).

After the faulted feeder is identified, the FSD Stage 2 is activated for identifying the faulted section using SCADA-controlled switching devices on the feeder.

Stage 2: Faulted Section Identification

FSD Stage 2 can be run after a faulted feeder is identified (as a result of FSD Stage 1 or when a feeder breaker is tripped and locked out).

FSD recloses the feeder breaker and SCADA-controllable field switching devices on the faulted

feeder step-by-step from the top (feeder head) to the bottom (or from the section immediately upstream of the active fault indication to bottom) until the faulted sections are identified. The fault presence is checked by the breaker tripping and by the transformer's ASC fault indicator status.

During FSD Stage 2, to prevent creating temporary outages on other non-faulted feeders, no downstream restoration is performed through normally open switches. However, FISR may be automatically triggered for the faulted area after FSD is completed and a fault is placed.

9.3. FSD Alarms

In real time, the FSD application can generate several alarms and send them to the alarm subsystem.

Table below lists the supported FSD alarms.

Table 40. FSD Alarms

Alarm	Description
FSD Stage 1 starting	When FSD Stage 1 is triggered for a power transformer, this alarm is issued.
FSD Stage 1 successfully ended	When FSD Stage 1 is completed for a power transformer and a faulted feeder is identified, this alarm is issued.
FSD Stage 1 ended with no result	When FSD Stage 1 is completed for a power transformer and a faulted feeder is not identified, this alarm is issued.
SCADA control failure during FSD Stage 1	When a device control step failed during the FSD stage 1 execution, this alarm is issued.
FSD Stage 2 starting	When FSD Stage 1 is triggered for a feeder, this alarm is issued.
FSD Stage 2 successfully ended	When FSD Stage 1 is completed for a feeder and faulted sections are identified, this alarm is issued.
FSD Stage 2 ended with no result	When FSD Stage 1 is completed for a feeder and a faulted section is not identified, this alarm is issued.
SCADA control failure during FSD Stage 2	When a device control step failed during the FSD Stage 2 execution, this alarm is issued.

9.4. FSD Configuration

This section describes the Faulted Section Determination configuration parameters.

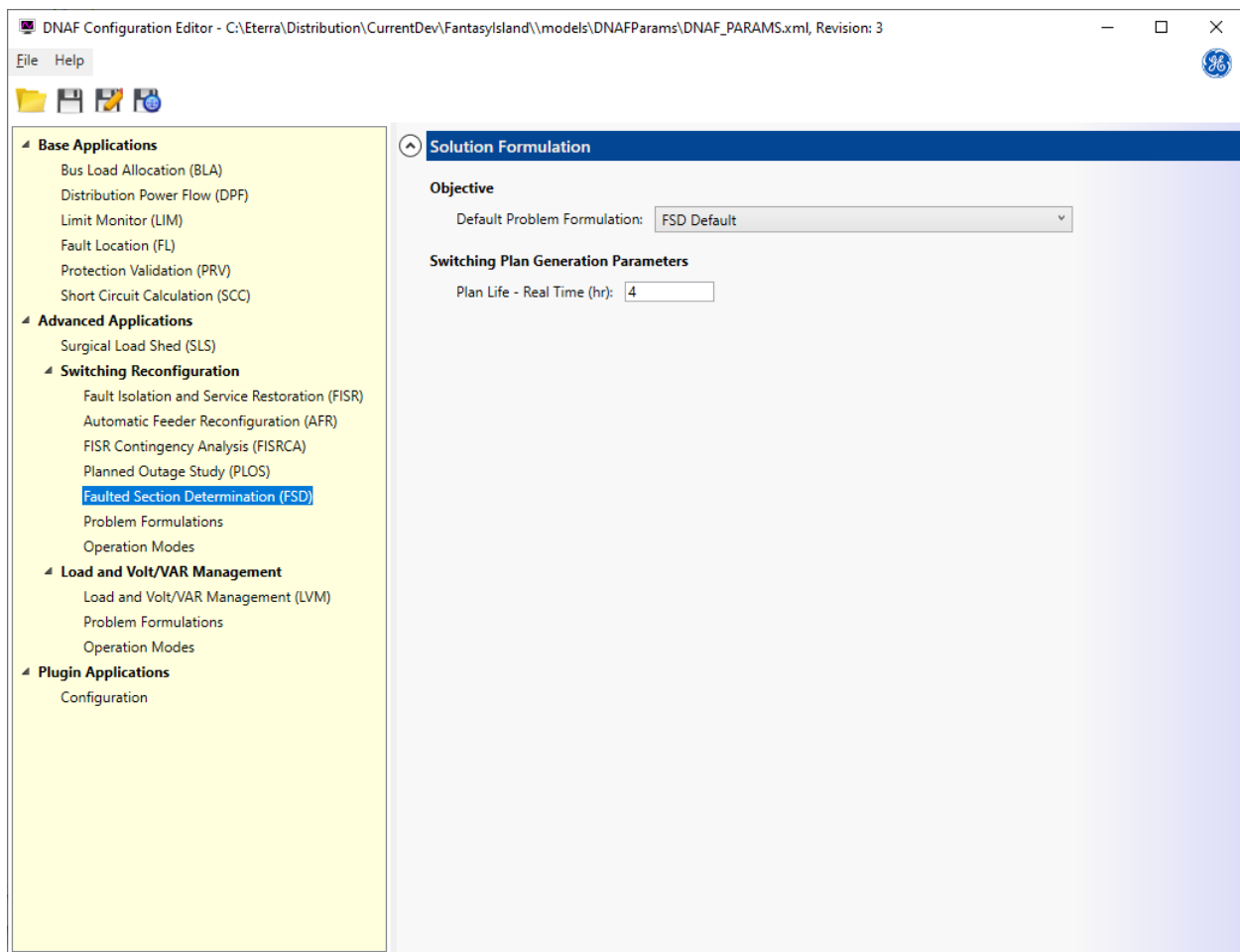


Figure 141. Faulted Section Determination Configuration

Table above lists the FSD configuration parameters. All listed parameters are contained by the <Params xsi:type = "FSDParams"> element in the configuration file DNAF_PARAMS.xml.

Note

All FSD problem formulations are shown on the Switching and Reconfiguration Problem Formulations page of the DNAF Configuration Editor.

Table 41. FSD Configuration Parameters

Configuration Editor UI Name	CSV File Field Name and XML File Attribute Name	Type	Description
Objectives			
Default Problem Formulation	DEFPROBFORM_FSD <DefaultProbIndex>	Drop-down list	This list shows the default FSD problem formulation. All FSD problem formulations are defined on the Switching and Reconfiguration Problem Formulations page.

10. Surgical Load Shed

The Surgical Load Shed (SLS) application is part of a combined DMS and SCADA application for prioritizing and shedding circuits to obtain a desired target reduction in system load.

- The Rotating Load Shed (RLS) SCADA application controls the actual load shedding operations, such as reclosing circuits and the shedding timing.
- The DMS Surgical Load Shed application controls configuration for enabling and disabling circuits for load shed and specifies the order in which the circuits are shed.

Unlike the other DNAF applications, the SLS application is not run to obtain a set of results. Instead, it is used to continually update SCADA with a set of enabling and ordering data, which can then be used by the SCADA RLS application.

10.1. Circuit Classification

One of the key concepts in load shed is Circuit Classification. It indicates the device priority for load shed based on the importance of the supplied load or equipment.

Circuit Classification is an integer value assigned to a load shed device (breaker or recloser). Typical values range from 1, indicating Do Not Shed, to 4, indicating the devices to be used for load shedding first. A special value of 0 indicates that the device is ignored and is not considered for load shed at all. The circuit classification can be configured to support up to 10 types (values 1-10).

There are four ways to set the circuit classification for a device:

- Using the static mode, by specifying the feeder's circuit classification field. This optional value represents the default value for all breakers or reclosers associated with the feeder.
- Using the static model, by specifying the device's circuit classification field. This optional value represents a base default value for the individual device.

This way is suitable for representing a special circuit that should not be shed, if the reason for keeping it supplied cannot be easily identified by the load shed rule set.

- Using the Real-Time DNAF System Configuration display, by selecting a rule set for the associated station.
- Using the DNAF Configuration Editor, by configuring the circuit classification for special conditions (for example, the number of critical customers) that may occur on the circuit.

The final classification assigned to a circuit is the most restrictive of the classifications in the static model data, the rule set classification, and the classification triggered by any of the conditional rules. When evaluating these rules, all downstream devices are checked, even if another load shed device is contained in that trace.

10.2. Circuit Classification and Topology

When calculating the circuit classification for a device, all downstream devices are checked, even if such devices are below another load shed device. Thus, the most restrictive load shed conditions propagate up to any upstream devices.

For example, consider a load shed-capable recloser supplying a hospital. The rule set associated with the device indicates that if a critical customer is present, the classification is set to 1 (Do Not Shed). This classification propagates up to the feeder head breaker - it is also assigned a classification of 1 (Do Not Shed), even though the circuit between the breaker and recloser may have no critical customers or special conditions.

As a counter example, consider a recloser supplying a purely industrial load. The rule set associated with the device indicates that the classification should be set to 2 (High) if more than 75% industrial load is present, and the amount of industrial load below the recloser triggers this condition. However, the entire feeder supplies a wider range of load types, so at the feeder head breaker the percentage of industrial load is much lower, and the rule condition is not triggered.

There are two main topological cases where two load shed devices (breakers or reclosers) may be in series: sub-fed stations and field reclosers.

If one distribution substation is fed from another one, it is necessary to specify the device to be targeted for shedding. For example, consider a higher-voltage feeder with 10 MVA of load, supplying 4 lower-voltage feeders, each with 8 MVA of load. If all devices are considered for shedding, there may be an erroneous assumption that 18MVA of load can be shed, when actually there is only 10MVA to shed. For this reason, the configuration requires choosing only one of the following shedding options:

- Higher-voltage feeders only.
- Lower-voltage feeders only.
- Consider the least restrictive option, which is to choose higher-voltage feeders when circuit classifications are the same or when at least one of the lower-voltage feeders is more restrictive. Lower-voltage feeders are used when they all have a less restrictive circuit classification.

The same type of situation occurs on a single feeder if both the feeder breaker and one or more field reclosers directly below the feeder breaker are capable of load shedding. The configuration gives the following options:

- Only consider the feeder breaker (and ignore the reclosers for load shed).
- Use the reclosers if they have a less restrictive classification.

10.3. Rule Sets

Rule sets are used to store a collection of rules that define how circuit classifications are assigned. Different rule sets can be assigned to different stations. The rule set associated with a station can

be configured at run time using the DNAF System Configuration display.

For example, two rule sets may be defined:

- The default rule set to be used under normal circumstances.
- A more permissive rule set to enable more devices for load shed. This rule set can be used when a large amount of load needs to be shed from the system and the default rule set prevents too many devices from being considered.

Rule sets contain two types of rules:

- Conditional rules that assign a new classification depending on different conditions on the circuit.
- Topology inclusion rules that determine which devices are selected for load shed if two or more devices are in series.

Note

In the case of sub-fed stations (one station supplies another one), the rule associated with the supplying station (i.e., the higher voltage station) is used to determine the topological rule to use.

10.3.1. Conditional Rules

Each of the five conditional rules is optional. They determine the circuit classification to be assigned if a particular condition is met.

The conditional rules are described below.

Critical Customers

If there are too many critical customers on a circuit, a more restrictive classification can be assigned. The number of critical customers needed to trigger the rule is configurable.

Presence of DG/DER

If there is too much distributed generation or distributed energy resource power on a circuit, a more restrictive classification can be assigned. The total nominal kVA of generators and DERs found on the circuit is used to trigger the rule.

Reserved Capacity

Reserved capacity objects represent utility customers who have a contractual requirement for a given power supply via one or more feeders. Typically, a reserved capacity object specifies a primary supply and one or more alternate supplies.

If a reserved capacity object is found on the circuit, a more restrictive classification can be assigned. The rule can be applied to circuits that contain a primary supply feed, an alternate supply feed, or either type of feed.

Load Category

If there is too much load of a particular type on a circuit, a different classification can be assigned. The rule is triggered if too high a percentage of load is found belonging to one or more specified load categories. A typical use for this rule would be to restrict a device for load shed if the circuit contains mostly industrial or commercial loads. Load magnitude is measured from load measurements if available, or power flow results, or nominal load if no other source is present.

Cold Load Pickup (CLPU)

The cold load pickup grade is a simple representation of how easy it would be to restore load on a circuit and avoid tripping protective devices. The CLPU Grade is defined as the ratio of the protection phase trip setting (maximum of available settings groups if more than one) divided by the current load on the circuit.

$$CLPU_{grade} = \frac{Ph Trip_{max}(A)}{Load(A)}$$

If the CLPU Grade is too low, it may be difficult to restore load on the circuit without extra work to open and restore it in sections. This would make it a poor candidate for use with rotating load shed.

This conditional rule allows a more restrictive classification to be assigned to a circuit if the CLPU Grade is smaller than a specified limit.

When there are other load shed devices (breakers or reclosers) downstream, the cold load pickup grade of all devices is considered. The lowest (i.e., worst) value is used.

10.3.2. Topology Rules

The Surgical Load Shed topology rules ensure that the best device is selected for load shed if two or more devices are in series.

If the devices are in series (that is, one is upstream of the other), they cannot be independently used for load shed. Shedding an upstream device would de-energize downstream devices, and shedding a downstream device would change the shedable load available on the upstream device (and also make restoration more complicated). To simplify management of devices, only one device in series can be considered for shedding. The series conditions typically occur for sub-fed stations and field reclosers.

Sub-Fed Stations

The first topology configuration option relates to sub-fed stations. In this case a distribution substation containing one or more load shed devices is fed from another distribution substation that also contains load shed devices.

For sub-fed stations, one of the following options can be selected:

- Use devices on the higher voltage feeding station
- Use devices on the lower voltage fed station(s)
- Use the feeding station (HV) device unless the fed stations (LV) device classifications are all less restrictive.

For example, consider the system illustrated below:

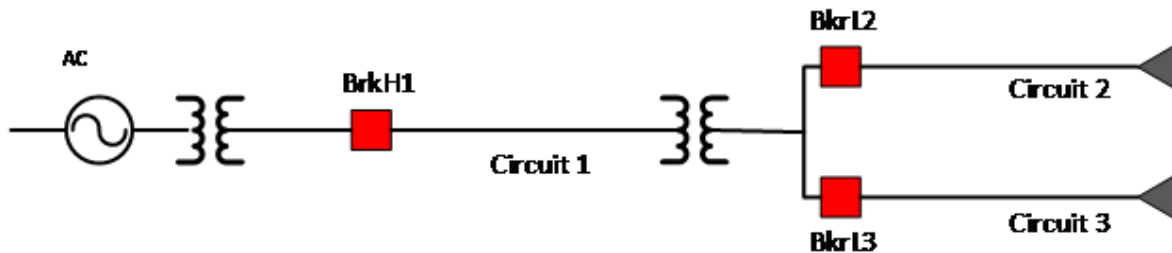


Figure 142. SLS Topology Rules for Sub-Fed Stations

For this system, the following table shows the behavior of the different options when classifications are different for the three circuits.

Breaker	Default Circuit Classification	1 - Only Consider Downstream	2 - Only Consider Upstream	3 - Prefer Less Restrictive Downstream
BkrH1	3	1*	3	3
BkrL2	2	2	0**	0***
BkrL3	4	4	0**	0***

*- Upstream circuit not considered. Set classification to 1 to disable for load shed.

** - Downstream circuit not considered. Set classification to 0 to ignore for load shed.

*** - Downstream circuits are not all less restrictive. Set to 0 to ignore for load shed.

The following table shows the behavior when classifications are the same on all three circuits.

Breaker	Default Circuit Classification	1 - Only Consider Downstream	2 - Only Consider Upstream	3 - Prefer Less Restrictive Downstream
BkrH1	2	1*	2	2
BkrL2	2	2	0**	0***
BkrL3	2	2	0**	0***

*- Upstream circuit not considered. Set classification to 1 to disable for load shed.

** - Downstream circuit not considered. Set classification to 0 to ignore for load shed.

*** - Ignored because downstream devices were not all less restrictive.

Field Reclosers

This topology configuration relates to individual feeders where both the feeder breaker and one or more reclosers are available for load shed. In this case, there are two options:

- Always use the breaker and ignore any reclosers.
- Use the first recloser found on the feeder if its circuit classification is less restrictive than the breaker's.

10.3.3. Selecting a Rule Set

Using the Real-Time DNAF System Configuration (DSC) display, you can enable/disable the SLS and select a rule set for individual stations, divisions, or for the entire system at run time. If no rule set is selected, the default rule set is used.

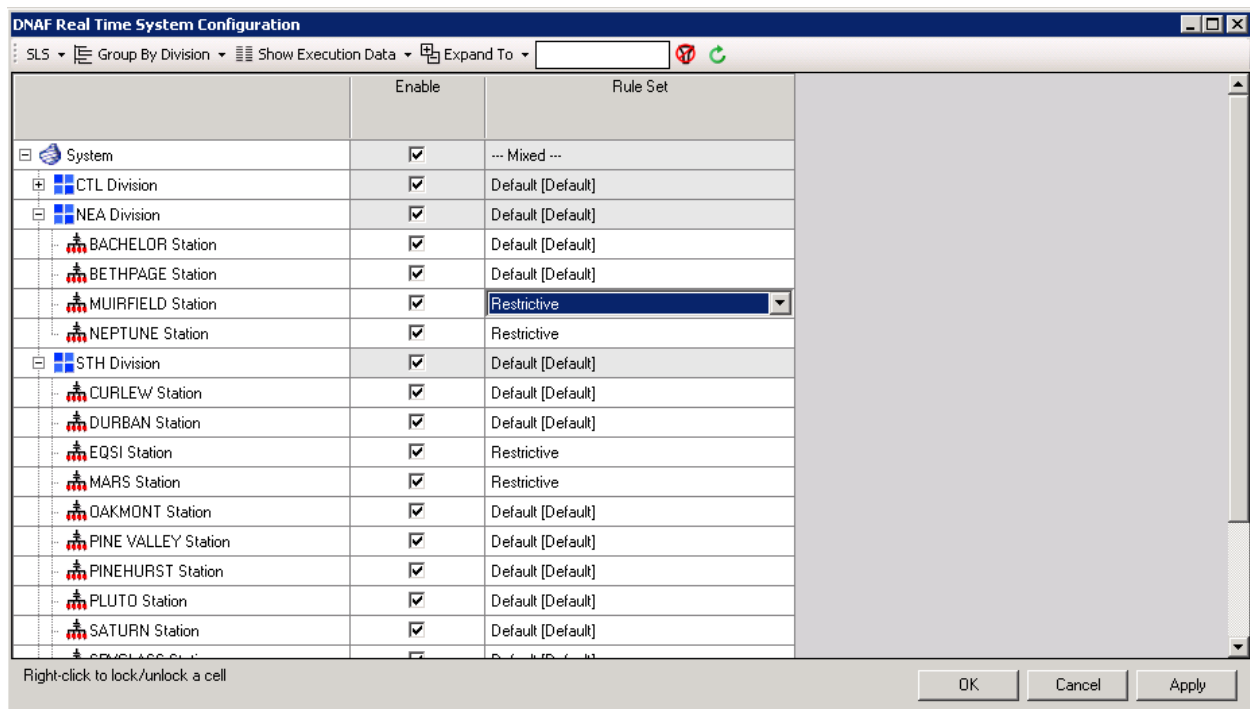


Figure 143. Configuring SLS at Run Time

Note that the DSC also allows you to disable load shed for entire stations or system areas. When a station is disabled, load shed is prevented for all devices in that station.

10.4. Do Not Shed Tag

A breaker or recloser is not considered for load shed if it or a downstream load shed device has a tag attached to it with a Do Not Shed control bit.

10.5. Update Triggers

Information about circuit classification and sort order is updated whenever relevant DMS data changes. Several different situations trigger the recalculation.

- When the system topology changes, SLS recalculates the load shed enabling and sort values for all devices in affected stations. This also applies to loading of new model files and model loading when starting the server.
- Whenever a load flow solution is computed for a station, all load shed devices in that station are updated.
- When the DNAF parameters configuration file changes, SLS recalculates the load shed enabling and sort values for all load shed devices in the entire model.

10.6. Sorting

The SLS configuration can specify how to sort the load shed devices to determine which should be shed first. Up to four levels of sorting are available, so that the exact order in which devices are considered for load shed can be fine-tuned. The available sort options are described below.

- **Order by System Division (1):** Sorts devices according the area (division) of the system they belong to.

A division is typically a geographic area of the system containing multiple stations. Using this sort option, all devices in one division are selected for load shed before selecting devices in the next division.

Note

You cannot use this option to control the sequence in which divisions are used for load shed. The order of the divisions is essentially random.

- **Order Across System Divisions (2):** Using this option, one device is selected from each division before selecting a second device from a division, etc.

This option can spread out the circuits used for load shed geographically across the system. This is the opposite of the previous option.

- **Order by Station (3):** Like option (1) except applying to stations, this option picks all items from one station before picking items from the next station, etc.

Note

The order in which stations are chosen is essentially random.

- **Order Across Stations (4):** Like option (2) except for stations, this option sorts load shed devices so that sequential devices in the sorted list come from different stations. One device from each station is selected before picking a second device from any station.

- **Order by Operations Center (5):** Much like (1) and (3), this option groups load shed devices by the operation center to which the load shed device belongs.

Note

Operations Center values for devices are defined in the static model.

- **Order Across Operations Center (6):** Much like (2) and (4), this option spreads load shed devices across operation centers.

This option can be used to avoid burdening operators at a single operation center during load shed operations.

- **Order by Decreasing Circuit Classification (7):** This is a key sort option. It uses the load shed device circuit classification, which is set using rule sets as described above.

This option orders devices from least critical classification (the largest classification number) to most critical classification (the lowest number applicable to shedding - 2). Devices with a classification of 1 (Do Not Shed) are never used for load shed.

- **Order by Decreasing Cold Load Pickup (8):** This sort option orders load shed devices by decreasing CLPU Grade, so devices that are less likely to trip protection during restoration are selected first.

This option uses sort buckets to group CLPU Grade into a handful of ranges (for example, 0-2, 3-4, 5-7, 7+) so that it can be used for multi-criteria sorting more easily.

- **Order by Decreasing Circuit Current (9):** This option sorts circuits by decreasing current flow, so that circuits with higher current flow are selected for load shed first.

This minimizes the number of devices that must be operated to drop load. As with (8), actual current (A) values are assigned to a small number of "buckets" for sort purposes.

- **Order by Increasing Circuit Current (10):** This option sorts circuits by increasing current (A) flow, so that smaller circuits are selected first.

This option can provide finer control over the amount of load shed, although it requires operating more devices.

As with (8), actual currents are assigned to a small number of "buckets" for sort purposes.

- **Order by Decreasing Circuit Power (kVA) (11):** Much like (9), this option sorts circuits by decreasing power flow to minimize the number of devices shed.

Shedding devices based on power instead of current may be more effective if the system contains load shed devices at many different voltage levels.

As with (8), the kVA values are assigned to a small number of "buckets" for sort purposes.

- **Order by Increasing Circuit Power (kVA) (12):** As with (10), this option sorts circuits by increasing power flow, which may require more devices to be shed to obtain a target load reduction.

10.7. SCADA Measurements

The Surgical Load Shed application passes data to the Rotating Load Shed (RLS) SCADA application using two measurement types: SetLoadShedEnabled and SetLoadShedSortOrder.

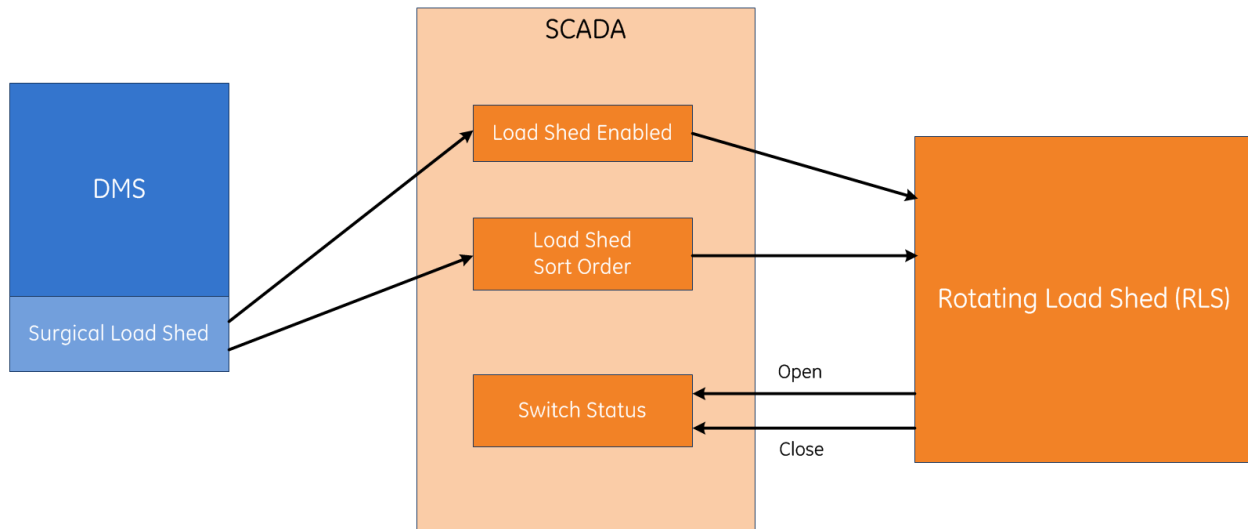


Figure 144. SLS and RLS Data Flow

RLS uses the values in these measurements to decide which circuits to shed. It sends the actual commands to open circuits to shed load and to close circuits to restore load.

Load Shed Enabled Measurement

The SetLoadShedEnabled status measurement determines whether load shed is enabled for the breaker or recloser. This value will be false if either load shed is disabled on the station containing the device (via the DNAF System Configuration display) or if the device has been assigned a load shed circuit classification of 1 (meaning Do Not Shed).

Load Shed Sort Order Measurement

The SetLoadShedSortOrder is an analog measurement used to store a value that the Rotating Load Shed SCADA application can use to sort the load shed devices, to determine which to shed first. The value set in this measurement is a combination of individual values determined from the different sort order criteria configured, where each digit in the value represents one of the sort criteria. The actual sort order value is not a meaningful value for human viewing; it is just a sortable value that RLS can use to order the circuits. However, to better understand where it comes from, consider the following example.

Consider a configuration with 3 sort criteria: 1) by circuit classification, 2) by station, 3) by CLPU grade, and consider that the CLPU grade has sort buckets defined as (1 = 0.0 to 2.0, 2 = 2.0 to 4.0, 3 = 4.0 - 6.0, 4 = 6.0+). If a specific breaker has the following characteristics (classification = 3, station = RAVEN, CLPU Grade = 3.26) and there are between 10 and 99 stations in the system, then the resulting sort order value might be 3092, where the first digit (3) represents the classification, 09 = station number (2-digit numbers are randomly assigned to each station), and the last digit (2)

represents the CLPU Grade (in the 2.0 to 4.0 bucket).

10.8. Surgical Load Shed Configuration

This section lists the SLS configuration parameters.

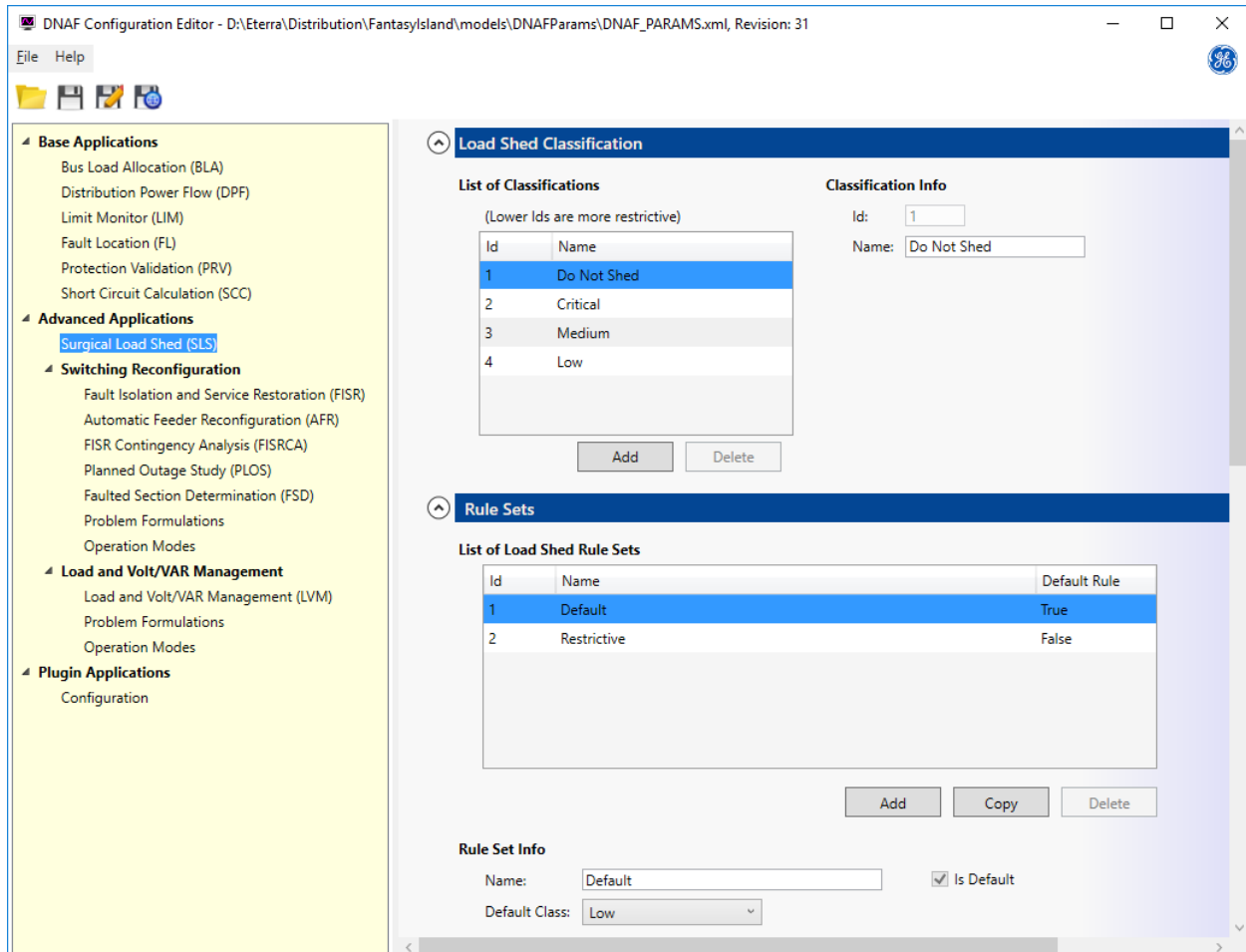


Figure 145. SLS Configuration

General parameters are stored in the `<Params xsi:type="">` element in the `DNAF_PARAMS.xml` configuration file. Rule set data is stored in the `<Params xsi:type="LoadShedRuleSetParams">` element.

Table 42. Load Shed Configuration Parameters

Configuration Editor UI Name	CSV File Field Name / XML File Element	Type	Description
Load Shed Classification			
Classification Id	<ClassificationIDs>	Text Box	This field contains the index for a load shed circuit classification. A value of 1 always means "Do Not Shed". Higher indexes represent less critical circuits where load shed is preferred.

Configuration Editor UI Name	CSV File Field Name / XML File Element	Type	Description
Classification Name	<ClassificationDescs>	Text Box	This field contains the name of the associated classification index.
Rule Sets			
Name	<Name>	Text Box	This field contains the name of the rule set.
Is Default	<IsDefault>	Check Box	This field identifies the current rule set as the default rule set. Only one rule set can be the default. The default rule set is assigned to all stations where no other rule set has been specifically selected.
Default Class	<DefaultClassificationId>	List Box	This field specifies the default classification for circuits using this rule set. Circuits are assigned the most restrictive classification from the following: • Static modeled classification • This default classification • The classification assigned based on the rule set rules
Critical Customers	<CriticalCustomerUse>	Check Box	This field determines whether to consider the number of critical customers on the circuit when determining the circuit classification.
Number of Critical Customers	<CriticalCustomerNumber>	Text Box	This field specifies how many critical customers are allowed on a circuit before making the classification more restrictive. This option is only considered if the Critical Customers check box is checked. The allowable range is 0-200.
Critical Customer Classification	<CriticalCustomerClass>	List Box	This field determines which classification to assign (if more restrictive than the static modeled or default classification) to a circuit if the number of critical customers exceeds the allowed number. This option is only considered if the Critical Customers check box is checked.
DG/DER	<DERUse>	Check Box	This field determines whether to consider the presence of distributed generation and distributed energy resources on the circuit when determining the circuit classification.
DG/DER kVA	<DERkVA>	Text Box	This field specifies how much rated kVA of generator/DER must be present to affect the classification of the circuit. This option is only considered if the DG/DER check box is checked. The allowable range is 0-10000.

Configuration Editor UI Name	CSV File Field Name / XML File Element	Type	Description
DG/DER Classification	<DERClass>	List Box	This field determines which classification to assign (if more restrictive than the static modeled or default classification) to a circuit if the size of attached generator/DER exceeds the specified kVA. This option is only considered if the DG/DER check box is checked.
Reserved Capacity	<ReserveCapUse>	Check Box	This field determines whether to consider the presence of reserved capacity loads on the circuit when determining the circuit classification.
Reserved Capacity Supply	<ReserveCapSupply>	List Box	This selection determines which types of reserved capacity supplies to consider. Options are: • Nominal (or primary) feed • Alternate (or secondary) feed • Both - any type of reserve capacity feed. This option is only considered if the Reserved Capacity check box is checked.
Reserved Capacity Classification	<ReserveCapClass>	List Box	This selection determines which classification to assign (if more restrictive than the static modeled or default classification) to a circuit if reserved capacity loads are present. This option is only considered if the Reserved Capacity check box is checked.
Load Category	<LoadTypeUse>	Check Box	This field determines whether to consider the proportion of industrial and commercial loads on the circuit when determining the circuit classification.
Load Category Percentage	<LoadTypePercent>	Text Box	This field specifies how much (in percent) of the specified type of loads must be present to affect the classification of the circuit. This option is only considered if the Load Category check box is checked. For more information, see Conditional Rules. The allowable range is 0-100.
Load Categories	<LoadTypeCategories>	Text Box	This field specifies the names of the load categories to be considered when determining the circuit classification. These names are the load category names specified in the globals.csv system definitions file. For example, RES, SCI, MCI, and LCI. This option is only considered if the Load Category check box is checked.

Configuration Editor UI Name	CSV File Field Name / XML File Element	Type	Description
Load Category Classification	<LoadTypeClass>	List Box	This field determines which classification to assign (if more restrictive than the static modeled or default classification) to a circuit if more than the specified percentage and types of loads are present. This option is only considered if the Load Category check box is checked.
Cold Load Pickup	<ColdLoadPickupUse>	Check Box	This field determines whether to consider the cold load pickup grade of the circuit when determining the circuit classification.
Cold Load Pickup Grade	<ColdLoadPickupGrade>	Text Box	This entry specifies the maximum CLPU grade for triggering a classification change. Any CLPU grade below this value causes the rule to apply a new classification (if more restrictive). This option is only considered if the Cold Load Pickup check box is selected. For more information, see Conditional Rules. The allowable range is 0-10.
Cold Load Pickup Classification	<ColdLoadPickupClass>	List Box	This field determines which classification to assign (if more restrictive than the static modeled or default classification) to a circuit if CLPU grade is lower than the specified value. This option is only considered if the Cold Load Pickup check box is selected.

Configuration Editor UI Name	CSV File Field Name / XML File Element	Type	Description
Downstream Substation-Fed Feeders	<SubFedFeederHandling>	Option Buttons	These options determine how to handle the situation where a feeder (and typically its entire substation) is fed from another distribution substation. This is required to avoid double-counting the load available to be shed on both the higher and lower voltage feeders. The options determine whether the higher or lower voltage feeders are available for load shed. 1 - Only Consider Downstream Feeders Only downstream feeders are allowed to shed load. All supplying (higher voltage) feeders are marked as disabled for load shed and customers on the higher voltage feeders will never be shed. 2 - Only Consider Supplying Feeders Only higher voltage supplying feeders are considered for load shed. Lower voltage feeders supplied from this feeder will never be enabled for independent load shedding. Note When the higher voltage feeder is shed, all customers on the lower voltage fed feeders are consequently de-energized. Also, any conditions on the lower voltage feeders that affect load classification affect the classification of the higher-voltage feeders. 3 - Prefer Load Shed on Downstream Feeders if All Downstream Feeders have Less Restrictive Classification In a sub-fed feeder case, this option allows either Higher Medium Voltage (HVM) or Lower Medium Voltage (LMV) circuits to be available for load shed, depending on the relative circuit classifications of each. If the load classification of the HVM circuit is the same or less restrictive than that of the LVM circuits, then the HVM circuit is available for shedding. If one or more of the LMV circuits have a less restrictive classification, they are available for shedding and the HVM circuit is disabled.

Configuration Editor UI Name	CSV File Field Name / XML File Element	Type	Description
Downstream Reclosers on a Feeders	<PreferFeederReclosers>	Check Box	This field provides the option to use feeder reclosers as well as circuit breakers. If selected, this option makes load-shed capable feeder reclosers available for load shed if the classification of the circuit below the recloser is less restrictive than the section above the recloser. If the classification of both sections of the feeder are equal, then the breaker is considered for load shed and the recloser is ignored. If the option is not selected, any reclosers are ignored and only the feeder head breaker is considered for load shed.

Load Drop Ordering

Configuration Editor UI Name	CSV File Field Name / XML File Element	Type	Description
First Sort Criteria	<SortOrder1>	List Boxes	These options contain choices for the sorting of circuits for load shed, as follows:
Second Sort Criteria	<SortOrder2>		[0 - No sort].
Third Sort Criteria	<SortOrder3>		[1 - By System Division] Sort by system division. Circuits in one division are selected for load shed before circuits in another division.
Fourth Sort Criteria	<SortOrder4>		[2 - Across System Divisions] Sort so that one circuit is selected from each division before selecting additional circuits.
			[3 - By Station] Sort so that circuits in one station are selected before circuits in another station.
			[4 - Across Station] Sort so that one circuit is selected from each station before choosing additional circuits.
			[5 - By Operations Center] Sort so that circuits are grouped by operation centers.
			[6 - Across Operations Center] Sort so that one circuit is taken from each operation center before picking additional circuits.
			[7 - Decreasing Circuit Classification] Order by circuit class (for example, 4's then 3's then 2's).
			[8 - Decreasing Cold Load Pickup Grade] Order by decreasing CLPU Grade.
			[9 - By Decreasing Circuit Current] Order circuits to be shed from largest to smallest current flow.
			[10 - By Increasing Circuit Current] Order circuits from smallest to largest current flow.
			[11 - By Decreasing Circuit kVA] Order circuits from largest to smallest kVA power.
			[12 - By Increasing Circuit kVA] Order circuits from smallest to largest kVA power.
Sort Buckets	<SortOrderH2LCurrentBuckets> <SortOrderL2HCurrentBuckets> <SortOrderCLPUBuckets> <SortOrderH2LkVABuckets> <SortOrderL2HkVABuckets>	Special List	These options determine how to group floating point values for sorting. Click the Edit button to change the number of sort "buckets" and the values associated with them. The Values column shows the range of values and the Index column shows the associated index used for actual sorting. Note that sorting of circuits by Index is from high (prefer to shed) to low (do not shed).

11. Reliability Index Calculations

To implement a quick reliability assessment of the switching optimization functions, an approximate technique is utilized. This technique makes several assumptions about the reliability of the protection system and the switching system, and about any post-fault restoration practices.

To explain the algorithm, consider the distribution system shown in figure below. This system consists of one substation and one feeder. There is one normally closed sectionalizing switch (S1). The feeder main is protected by breaker B1 (which has reclosing capability), and the laterals (lines L5-L8) are protected by fuses. There are eight customers (two customers per lateral).

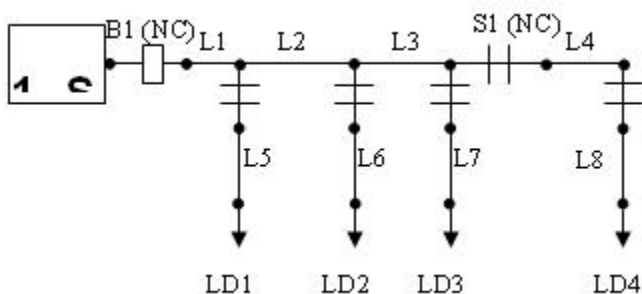


Figure 146. Small Distribution System

To calculate the reliability of the system (the system can be an individual feeder, a group of feeders, or a group of components), three reliability parameters are required for each line section. These parameters are as follows:

- Temporary fault rate (λ_T in failures/unit length-year): The number of temporary faults that cause a momentary outage in a given year.
- Permanent fault rate (λ_P in failures/unit length-year): The number of permanent faults in a given year resulting in a sustained outage (an outage greater than 5 minutes or the necessary time to manually close a protection device that opened).
- Mean time to repair (μ in hours/repair): The mean repair time for a sustained outage.

These three attributes are currently defined in both the conductor model and as default override values on the individual line sections.

The reliability calculation described in this appendix makes several assumptions, including:

- It is assumed that there is only one protection device at the feeder head. This device is perfect (the device responds correctly to faults 100.0% of the time, all temporary faults result in only momentary outages, and all permanent faults result in sustained outages), and it has reclosing capability for clearing temporary faults.
- There is no post-fault switching. Protection devices operate and are opened for the entire duration (for a sustained outage).

Post-fault switching scenarios are very difficult to generate. They require knowing other attributes such as switching reliabilities, the probability of back-feed (the percentage of time a back-feed path is available is based on the mean time to switch, the mean time to repair, and the historical loading

levels), and utility operating practices. This type of detailed analysis is better suited for the planning environment.

The output of the reliability assessment is three reliability performance indices: MAIFI, SAIFI, and SAIDI. These indices can be calculated by feeder, by station, or by a group of stations.

The following information about reliability performance indices was taken from the IEEE *Trial Use Guide for Electric Power Reliability Indices*, P-1366.

SAIFI - System Average Interruption Frequency Index

The system average interruption frequency index is designed to give information about the average frequency of sustained interruptions per customer over a predefined area. In words, the definition is as follows:

$$SAIFI = \frac{\text{Total Number of Customer Interruptions}}{\text{Total Number of Customers Served}} \quad (\text{Equation A.1})$$

SAIDI - System Average Interruption Duration Index

This index is commonly referred to as "Customer Minutes of Interruption" or "Customer Hours", and it is designed to provide information about the average time the customers are interrupted. In words, the definition is as follows:

$$SAIDI = \frac{\sum \text{Customer Interruption Durations}}{\text{Total Number of Customers Served}} \quad (\text{Equation A.2})$$

MAIFI - Momentary Average Interruption Frequency Index

This index is very similar to SAIFI, except that it tracks the average frequency of momentary interruptions. In words, the definition is as follows:

$$MAIFI = \frac{\text{Total Number of Customer Momentary Interruptions}}{\text{Total Number of Customers Served}} \quad (\text{Equation A.3})$$

The simplified reliability calculation requires the following steps:

1. Sum up the number of momentary interruptions (temporary fault rates) for each fused line section (a fused line section includes all lines that are downstream from the fuse). The equation is:

$$TMI_{\text{Lateral}} = \sum_{\substack{\text{Lateral} \\ \text{Line Sections}}} \lambda_T \quad (\text{Equation A.3})$$

2. Sum up the number of sustained interruptions (permanent fault rates) for each fused line section. The equation is:

$$TSI_{Lateral} = \sum_{\substack{\text{Lateral} \\ \text{Line Sections}}} \lambda_p \quad (\text{Equation A.4})$$

3. Sum up the weighted repair times (outage durations) for each fused line section. The equation is:

$$WSOD_{Lateral} = \sum_{\substack{\text{Lateral} \\ \text{Line Sections}}} \lambda_p \cdot \mu \quad (\text{Equation A.5})$$

4. Sum up the total number of customers for each fused line section. The equation is:

$$TC_{Lateral} = \sum_{\substack{\text{All Lateral} \\ \text{Transformers}}} \text{Customer Count} \quad (\text{Equation A.6})$$

5. Repeat calculations 1-4 for the feeder main (backbone section).

When performing the calculations on the backbone section, all other devices except for the feeder breaker should be ignored. The following values are determined:

TMI_{Main}

TSI_{Main}

$WSOD_{Main}$

TC_{Main} - This number is probably close to zero.

6. Calculate the reliability indices as follows:

$$MAIFI_{Feeder} = \frac{\left[\sum_{\text{Laterals}} (TMI_{Main} + TMI_{\text{All Laterals}} + TSI_{\text{Other Laterals}}) \cdot TC_{Lateral} + (TMI_{Main} + TMI_{\text{All Laterals}} + TSI_{\text{All Laterals}}) \cdot TC_{Main} \right]}{TC_{Main} + \sum_{\text{Laterals}} TC_{Lateral}} \quad (\text{Equation A.7})$$

$$SAIFI_{Feeder} = \frac{\sum_{\text{Laterals}} (TSI_{Lateral} + TSI_{Main}) \cdot TC_{Lateral} + (TSI_{Main} \cdot TC_{Main})}{TC_{Main} + \sum_{\text{Laterals}} TC_{Lateral}} \quad (\text{Equation A.8})$$

$$SAIDI_{Feeder} = \frac{\sum_{\text{Laterals}} (WSOD_{Lateral} + WSOD_{Main}) \cdot TC_{Lateral} + (WSOD_{Main} \cdot TC_{Main})}{TC_{Main} + \sum_{\text{Laterals}} TC_{Lateral}} \quad (\text{Equation A.9})$$

$$MAIFI_{Substation} = \frac{\sum_{\text{Feeders}} MAIFI_{Feeder}}{\sum_{\text{Feeders}} \text{Total_Customers}_{Feeder}} \quad (\text{Equation A.10})$$

$$SAIFI_{Substation} = \frac{\sum_{\text{Feeders}} SAIFI_{Feeder}}{\sum_{\text{Feeders}} \text{Total_Customers}_{Feeder}} \quad (\text{Equation A.11})$$

$$SAIDI_{Substation} = \frac{\sum_{\text{Feeders}} SAIDI_{Feeder}}{\sum_{\text{Feeders}} \text{Total_Customers}_{Feeder}} \quad (\text{Equation A.12})$$

As an example, consider the following data for the simple system shown in figure above.

Table 43. Reliability Input Data for the System

Line Section	Temporary Fault Rate (failures/year)	Permanent Fault Rate (failures/year)	Mean Time to Repair (hours/repair)
L1	0.15	0.20	2.0
L2	0.20	0.15	2.0
L3	0.25	0.15	2.0
L4	0.20	0.10	3.0
L5	0.15	0.10	2.0
L6	0.15	0.10	2.0
L7	0.15	0.10	2.0
L8	0.15	0.10	2.0

The reliability index calculations yield the numbers shown in Table below.

Table 44. Intermediate Reliability Calculations

Lateral/Main	TMI	TSI	WSOD	TC
L5	0.15	0.10	0.20	2
L6	0.15	0.10	0.20	2
L7	0.15	0.10	0.20	2
L8	0.15	0.10	0.20	2
Main	0.80	0.60	1.30	0

From the data given in Table above, the reliability indices are calculated and shown in Table below.

Table 45. Station Reliability Indices

Reliability Index	Value
MAIFI (customer interruptions/year)	1.70
SAIFI (customer interruptions/year)	0.70
SAIDI (customer hours/year)	1.50

12. Problem Formulations

There are several problem formulations for the advanced applications. Every problem formulation has its own goal or set of goals. The goals of every problem formulation are decided by the Performance Indices (PI). One or more plans are recommended by the reconfiguration applications; these plans are ranked based on total PI from small to large. For the LVM application, only one plan is recommended; however, the performance indices are also used during the plan generation process for plan selection.

The total PI is the sum of weighted individual PIs.

$$PI_{total} = \sum_{i=1}^n (w_i \cdot PI_i) \quad (\text{Equation B.1})$$

The weight of a PI can be any real number from 0 to infinity. If the weight is 0, it means that the optimization function does not take this PI into consideration. The larger the weight, the more important the PI is.

Table below describes all PIs used in the advanced applications.

Table 46. Description of Performance Indices

Performance Index	Description
Flow Violations – Final	The flow violation PI after the final step. It is calculated using the following equation: $PI_{flow} = \sum_1^n (R_{overload})^2$ Where: $R_{overload} = 0$ if the branch is not overloaded, and $R_{overload} = \text{flow} - \text{lim}$ if the branch is overloaded. The flow violations and limits are in per unit.
Flow Violations – Intermediate	The flow violation PI for the intermediate steps. It is the maximum of all the flow violation PIs for the intermediate steps. For every intermediate step, the equation is the same as the one for the final step.

Performance Index	Description
Bus Voltage Violations – Final	The bus voltage violation PI after the final step. It is calculated using the following equation: $PI_{busvolt} = \sum_1^n (R_{busvoltviol})^2$ Where: $R_{busvoltviol} = 0$ if the bus voltage is within limit, and $R_{busvoltviol} = \text{bus voltage} - \text{lim}$ if the bus voltage upper or lower limit is violated. The voltage violations and limits are in per unit.
Bus Voltage Violations – Intermediate	The bus voltage violation PI for intermediate steps. It is the maximum of all the voltage violation PIs for the intermediate steps. For every intermediate step, the equation is the same as the one for the final step.
Load Voltage Violations – Final	The load voltage violation PI after the final step. It is calculated using the following equation: $PI_{volt} = \sum_1^n (R_{loadvoltviol})^2$ Where: $R_{loadvoltviol} = 0$ if the load voltage is within limit, and $R_{loadvoltviol} = \text{load voltage} - \text{lim}$ if the load voltage upper or lower limit is violated. The voltage violations and limits are in per unit.
Load Voltage Violations – Intermediate	The load voltage violation PI for intermediate steps. It is the maximum of all the voltage violation PIs for the intermediate steps. For every intermediate step, the equation is the same as the one for the final step.
Total Cost – Final	The total cost of control moves. It is the sum of costs of all the control moves in the plan. The cost of the control is decided by the adjacency to the reconfigured area, whether it is an automatic control or a manual control and the operation cost defined in the model.
Lost Load – Final	The total real power of all de-energized loads after the final step, in kW.
Lost Load – Intermediate	The maximum total real power of all de-energized loads for intermediate steps, in kW.
Dropped Customer Count – Final	The number of customers de-energized after the final step.
Dropped Customer Count – Intermediate	The maximum number of customers de-energized for intermediate steps.
Dropped Customer Priority – Final	The dropped customer priority after the final step. The customer priorities are assigned based on the costs of uninterrupted service that are given for each load category. (Customer priority is modeled with the PCOSTINT and ECOSTINT fields in the load category object.)

Performance Index	Description
Dropped Customer Priority – Intermediate	The maximum dropped customer priority for intermediate steps.
SumFlow Final	The sum of active or reactive flow value for different flow check points after the final step. The check points are created according to the selected problem formulation on the station, transformer, and feeder levels.
SumFlow Intermediate	The maximum sum of active or reactive flow value for different flow check points in the intermediate step. The check points are created according to the selected problem formulation on the station, transformer, and feeder levels.
Radial Condition – Final	The total number of loops after the final step. It is calculated as follows: $PI_{R,F} = SF_{R,F} * n$ Where: $PI_{R,F}$ is the Final Radial PI value $SF_{R,F}$ is the Final Radial Scaling Factor n is the number of unique loops in the system after the final step
Radial Condition – Intermediate	The maximum number of loops for intermediate steps. It is calculated as follows: $PI_{R,I} = SF_{R,I} * n$ Where: $PI_{R,I}$ is the Final Radial PI value $SF_{R,I}$ is the Final Radial Scaling Factor n is the number of unique loops in the system after the final step
Maximum Margin	The maximum flow margin of all branches after the final step. The margin is the difference between flow value and limit. If the branch is not overloaded, the margin is positive; if the branch is overloaded, the margin is negative. The value is in percent. The Maximum margin PI the value is calculated by using the equation $PI_{MaxMargin} = \min(0, 100.0 - margin)$
Protection Margin	The protection margin performance index considers the distance from the current/load flow (multiplied by a safety factor) to the flow limit. The value is calculated by using the equation $PI_{ProtectionMargin} = \sum_{i=1}^n u(Load * LoadSafetyFactor - PhaseTrip) * (Load * LoadSafetyFactor - PhaseTrip)$ where $u(x)$ is the step function that equals to zero when $x \leq 0$, and equals to 1 when $x > 0$.
Number of Control Moves	The total number of steps in the plan.

Performance Index	Description
Third-Party Owned Equipment Number of Control Moves	The total number of steps involving third-party equipment in the plan.
Losses	The losses in kW after the final step.
Protection Validation	The result related to protection validation after the final step. If there is no violation found in protection validation, the value is 0. If a violation is found, the result is proportional to the violation value in per unit. When a maximum interruption, maximum phase pickup, or minimum ground pickup violation is found, the value is calculated by using the equation $PI_{PRV} = \sum_{i=1}^n R_{Max} + R_{MinPh} + R_{MinGnd}$ Where: $R_{Max} = \left(\frac{MaxFaultCurrent}{MaxInterruptSetting} - 1.0 \right) * 100.0$ $R_{MinPh} = \frac{MinPhasePickupSetting}{MinPhaseFaultCurrent} * 100.0$ $R_{MinGnd} = \left(\frac{MinGroundPickupSetting}{MinGroundFaultCurrent} - 1.0 \right) * 100.0$
Protection Reach	The protection reach performance index checks whether a protection settings group can satisfy the minimum fault current. The value is calculated by using the equations: $I_{diff} = Reach\ Safety\ Factor * Phase\ Trip - Min\ Phase\ Fault\ Current$ $PI_{Reach} = \sum_{Protection\ Zones} u(I_{diff}) * I_{diff}$ where $u(x)$ is the step function that equals to zero when $x \leq 0$, and equals to 1 when $x > 0$.
Voltage Equalization	The voltage difference at the end of feeders that are connected to a common regulation transformer. The voltages are in per unit.
CMI	The Customer Minutes Interruption for the plan. It is the sum of the interrupted minutes for all de-energized customers. The automatic and manual switch switching operation time can be specified in the problem formulation via the DNAF Configuration Editor.
LMI	The Load Minutes Interruption for the plan. The value is calculated by using the equation $PI_{LMI} = \sum_{i=1}^n (kW_i * t_i)$ in which kW is the load de-energized in kW, and t is the duration the load is de-energized in minutes.

Performance Index	Description
Priority CMI	The Customer Minutes Interruption with priority for the plan. The value is calculated by using the equation $PI_{P_LMI} = \sum_{i=1}^n (Priority_i * t_i)$ in which t is the duration the customer is de-energized, and $priority$ is the priority of that customer. The customer priorities are assigned based on the costs of uninterrupted service that are given for each load category.
MAIFI	The Momentary Average Interruption Frequency Index for the plan calculated for the entire network selection. For more information, see Reliability Index Calculations.
SAIFI	The System Average Interruption Frequency Index for the plan calculated for the entire network selection. For more information, see Reliability Index Calculations.
SAIDI	The System Average Interruption Duration Index for the plan calculated for the entire network selection. For more information, see Reliability Index Calculations.
Feeder Voltage Drop	The Feeder voltage drop for the plan. The feeder voltage drop is calculated for every feeder by subtracting the minimum voltage along the feeder from the voltage at feeder head node, and the PI is the largest voltage drop among all these feeders. It is in per unit.
Non-Converged BLA Islands	The number of non-converged BLA measurement islands involved in a switching plan. The value is calculated using the following equation: $PI_{NonConvergedBLA} = Count_{NonConvergedBLAIsland} * Weight_{NonConvergedBLA}$
Reverse Flow for Non-Reversible Recloser/Breaker	The number of non-reversible devices with reverse flow. The value is calculated using the following equation: $PI_{RevFlowNonRevDev} = Count_{RevFlowNonRevDev} * Weight_{RevFlowNonRevDev}$
Switching Validation Check - Ferroresonance	The number of switching steps with the "Ferroresonance switching conditions" switching check violation. The value is calculated using the following equation: $PI_{FerroResonanceCheck} = Count_{FerroResonanceCheckViolation} * Weight_{FerroResonanceCheckPI}$
Switching Validation Check - Grounded Load	The number of switching steps with the "Feeding single phase-ground loads from 3 or 4 wire feeder sections" switching check violation. The value is calculated using the following equation: $PI_{GroundedLoadCheckPI} = Count_{GroundedLoadCheckViolation} * Weight_{GroundedLoadCheckPI}$

Performance Index	Description
Switching Validation Check - Parallel Recloser	The number of switching steps with the "Paralleling circuits with reclosing capable protective devices" switching check violation. The value is calculated using the following equation:
$PI_{ParallelRecloserCheckPI} = Count_{ParallelRecloserCheckViolation} * Weight_{ParallelRecloserCheckPI}$	

The problem formulation scaling costs are used by the LVM. Table below describes all scaling costs.

Table 47. Scaling Factors

Scaling Cost	Description
Bus Low Voltage Violation	The scaling factor for the bus low voltage violation cost curve. For more information about violation penalties and costs, see Cost Model .
Bus High Voltage Violation	The scaling factor for the bus high voltage violation cost curve.
Load Low Voltage Violation	The scaling factor for the load low voltage violation cost curve.
Load High Voltage Violation	The scaling factor for the load high voltage violation cost curve.
Low Voltage Violation Count	The scaling factor for the number of customers associated with a load. This scaling factor is multiplied by the number of customers associated with a load, and then applied to the load low voltage violation cost curve.
High Voltage Violation Count	The scaling factor for the number of customers associated with a load. This scaling factor is multiplied by the number of customers associated with a load, and then applied to the load high voltage violation cost curve.
Critical Customer Low Voltage Violation	The scaling factor for critical customers associated with a load. This scaling factor is multiplied by the active power of a load with critical customers, and then applied to the load low voltage violation cost curve.
Critical Customer High Voltage Violation	The scaling factor for critical customers associated with a load. This scaling factor is multiplied by the active power of a load with critical customers, and then applied to the load high voltage violation cost curve.
Flow Violation	The scaling cost for branch flow violations for monitored branches (lines/cables, switching devices, regulators, power transformers, step up/down transformers, and auto transformers).
Voltage	The scaling factor for voltage control setting (DER/generators).
Active Flow	The scaling cost for real power loss (used in the minimize loss and minimize/maximize demand problem formulations).
Reactive Flow	The scaling factor for the station's reactive power limit violation.
Tap Changer	The scaling cost for LTC/regulator tap movement before it exceeds the maximum daily switching count. To obtain the effective LTC/regulator movement cost that is used in formulating the Linear Programming model in Optimization LVM, this scaling factor is multiplied by the LTC/regulator movement cost from the LTC/regulator model and by a factor equal to (1 + CurrentDailyOperations/MaximumDailyOperations).

Scaling Cost	Description
Tap Changer Movement (Exceed Daily Count)	The scaling cost for LTC/regulator tap movement after it exceeds the maximum daily switching count. To obtain the effective LTC/regulator movement cost that is used in formulating the Linear Programming model in Optimization LVM, this scaling factor is multiplied by the LTC/regulator movement cost from the LTC/regulator model and by a factor equal to $(1 + \text{CurrentDailyOperations}/\text{MaximumDailyOperations})$.
Capacitor Switching	The scaling cost for capacitor switching before it exceeds the maximum daily switching count. To obtain the effective capacitor switching cost that is used in formulating the Linear Programming model in Optimization LVM, this scaling factor is multiplied by the capacitor switching cost from the capacitor model and by a factor equal to $(1 + \text{CurrentDailyOperations}/\text{MaximumDailyOperations})$.
Capacitor Switching (Exceed Daily Count)	The scaling cost for capacitor switching after it exceeds the maximum daily switching count. To obtain the effective capacitor switching cost that is used in formulating the Linear Programming model in Optimization LVM, this scaling factor is multiplied by the capacitor switching cost from the capacitor model and by a factor equal to $(1 + \text{CurrentDailyOperations}/\text{MaximumDailyOperations})$.
Generator VAR Change	The scaling cost for a generator's VAR output change.
DER VAR Change	The scaling cost for a distributed energy resource's VAR output change.
Large Loads	The scaling cost for constraining the voltage limits of large loads.
Regulator Group Voltage Imbalance	The scaling cost for voltage imbalance in regulation groups.
Local Control Capacitors (Amps)	The scaling cost for constraining the switching of locally regulating capacitors of the Amp control type.
Local Control Capacitors (Volt)	The scaling cost for constraining the switching of locally regulating capacitors of the Volt control type.
Local Control Capacitors (VAR)	The scaling cost for constraining the switching of locally regulating capacitors of the VAR control type.
Local Control Capacitors (PF)	The scaling cost for constraining the switching of locally regulating capacitors of the PF control type.

13. LVM Plan Selection Truth Tables

This section provides "truth tables" for different LVM goals. These tables show an exhaustive list of different combinations of situations (pre-LVM and post-LVM voltage limit violations, magnitude of voltage change, magnitude of demand reduction), with the corresponding decision with regards to the acceptance of the LVM plan.

Table 48. Truth Table for Minimizing Violations

All pre-plan voltages within limits?	All post-plan voltages within limits?	Delta V > deadband	Pre-plan PF within limits?	Post-plan PF within limits?	Accept plan?	Comments
Yes	Yes	Yes	Yes	Yes	Yes	
Yes	Yes	Yes	Yes	No	No	The plan is discarded because the power factor changed from inside the target range to outside.
Yes	Yes	Yes	No	NA	Yes	The plan is accepted so that power factor violation is resolved over the course of multiple plans.
Yes	Yes	No	Yes	NA	No	If the power factor is within limits in the initial state, the plan is discarded if there is not enough voltage improvement.
Yes	Yes	No	No	Yes	Yes	If the power factor is not within limits in the initial state, the plan is accepted if the power factor improved despite insufficient voltage improvement.
Yes	Yes	No	No	No	No	The plan is discarded because of insufficient voltage improvement.
Yes	No	NA	NA	NA	No	The plan is discarded because it creates a voltage violation.
No	Yes	NA	NA	NA	Yes	The plan is accepted because it removes an initial voltage violation.
No	No	Yes	NA	NA	Yes	The plan is accepted because there is enough voltage improvement.
No	No	No	Yes/No	Yes/No	No	The plan is discarded because of insufficient voltage improvement.

Table 49. Truth Table for Minimizing Demand

All pre-plan voltages within limits?	All post-plan voltages within limits?	Delta V > deadband	Delta P > deadband	Pre-plan PF within limits?	Post-plan PF within limits?	Accept plan?	Comments
Yes	Yes	Yes	Yes	Yes	Yes	Yes	The plan is accepted because voltage and demand are improved and the power factor is within the target range.

All pre-plan voltages within limits?	All post-plan voltages within limits?	Delta V > deadband	Delta P > deadband	Pre-plan PF within limits?	Post-plan PF within limits?	Accept plan?	Comments
Yes	Yes	Yes	Yes	Yes	No	No	The plan is discarded because it changes the power factor to a value outside the acceptable range.
Yes	Yes	Yes	Yes	No	NA	Yes	The plan is accepted so that power factor violation is resolved over the course of multiple plans.
Yes	Yes	Yes	No	NA	NA	No	The plan is discarded due to insufficient demand reduction.
Yes	Yes	No	Yes	Yes	Yes	Yes	The plan is accepted due to sufficient demand reduction.
Yes	Yes	No	Yes	Yes	No	No	The plan is discarded due to insufficient voltage improvement and insufficient power factor changes.
Yes	Yes	No	Yes	No	NA	Yes	The plan is accepted due to sufficient demand reduction.
Yes	Yes	No	No	NA	NA	No	The plan is discarded due to insufficient voltage improvement and insufficient demand reduction.
Yes	No	NA	NA	NA	NA	No	The plan is discarded because it creates a new voltage violation.
No	Yes	NA	NA	NA	NA	Yes	The plan is accepted because it removes the voltage violation.
No	No	Yes	NA	NA	NA	Yes	The plan is accepted due to the voltage improvement.
No	No	No	NA	NA	NA	No	The plan is discarded due to insufficient voltage improvement.

14. DNAF Output Files

This section details the contents of the output files for the advanced DNAF applications.

The default location of the output files is %IDMS_ROOT%\log\dnaf\. This location can be changed by modifying the parameter DNAF_OUTPUT_PATH in the DNAF server configuration file.

14.1. LVM Output File

LVM results for stations are recorded in files named LVM Results_<StationName>_<TimeStamp>.dat. The data in this file is grouped in sections as follows:

- LVM EXECUTION SUMMARY

This section shows the substation ID, substation name, substation area (service center), substation permission area, LVM station group, and LVM station group schedule.

- LVM SOLUTION SUMMARY

This section contains triggering information (evaluation time, execution time, LVM solution type, problem formulation, LVM execution reason, and triggering device ID).

- DNAF INPUT PARAMETER FILE LOCATIONS

This section details the file locations for all the input parameters. The files included are DNAF_PARAMS.xml, alg.dat, msgdef.dat, optalg.dat, phid.dat, and tftype.dat. The time last modified and revision number for the DNAF_PARAMS.xml file is also included, as well as the DNAF Configuration Editor (DCE) version.

- PROBLEM FORMULATION PARAMETERS - LVM

This section details the values of the problem formulation parameters from the DNAF Configuration Editor: performance index weights, violation penalties and costs, eligible devices, number of infeasible constraints, respect voltage/flow violations, voltage limits, and branch flow limits.

- LVM FAILSAFE STATUS

This section provides failsafe status, problem formulation, reversion method, and reason for a substation and the substation's power transformers and feeders.

- BUS VOLTAGE MISMATCH

This section provides comparison information for the measured and calculated bus voltages to help the user verify whether the LVM failsafe operation is performed correctly. This section provides the following information for station internal and external buses: bus ID, name, nominal station and feeder, energizing station and feeder, indication whether this is a station internal bus, maximum voltage mismatch limit, indication whether the mismatch limit is exceeded, measurement base, and phase A/B/C voltage measurement ID, measured value, calculated value, and voltage mismatch.

- **ALARM CONFIGURATION**

The section lists the status (enabled or disabled) of the following alarms: Failed to Complete, Failed To Complete RTN (Return To Normal), No Plan, No Plan RTN, Maximum Consecutive Infeasible Plans, Maximum Daily Infeasible Plans, Maximum Consecutive Discarded Plans, Maximum Daily Discarded Plans, Maximum Capacitor Moves, Maximum Tap Changer Moves, LVM Optimization Group Throttling Started, LVM Optimization Group Throttling Ended, LVM Failsafe Ended, LVM Failsafe Started, LVM Plan Execution Failure Retrigger, Feeder Disabled, Feeder Disabled - Not Enough Controls, Tap Control Resend Failure, Capacitor Control Resend Failure, Regulator Group Voltage Imbalance, Regulator Delay Tap Move Canceled, Controller Delta Voltage Mismatch, Energized Sensor Zero Voltage, Energized Sensor Zero Voltage RTN, LVM Failsafe Started, LVM Failsafe Ended, Plan Execution Failure Retrigger, Plan State Check Failure, and Plan Canceled.

- **OPERATION MODE**

This section details the values of the problem formulation operation mode parameters, including the plan execution method, plan and step timeouts, maximum execution attempts for capacitors and LTC/regulators, active and reactive power throttling, monitoring upstream reactive power, and monitoring load side voltage for tap position confirmation.

- **EXECUTION MODE**

This section gives the plan execution mode, such as closed-loop mode.

- **STATION/POWER TRANSFORMER/FEEDER ELIGIBILITY SETTINGS**

This section details the enabled status, problem formulation, and power factor range for each power transformer and feeder in the substation.

- **SELECTED OPTIMIZATION ALGORITHM OPTIONS**

This section details the values of the parameters used for LVM from file optalg.dat.

- **DNAF PARAMETERS**

This section details the values of the DPF parameters from the DNAF Configuration Editor used in LVM including P, Q, and V mismatches.

- **LVM PARAMETERS**

This section details the values of the LVM parameters from the DNAF Configuration Editor.

- **ANALOG MEASUREMENTS**

This section includes a summary of analog measurements in the network selection, grouped by substation.

- **STATUS MEASUREMENTS**

This section includes a summary of status measurements in the network selection, grouped by substation.

- **INVALID ANALOG MEASUREMENTS**

This section includes a summary of analog measurements in the network selection with invalid status or quality, grouped by substation.

- **BUS VOLTAGE MEASUREMENTS**

This section includes a summary of analog measurements for all buses in the network selection.

- **LOAD VOLTAGE MEASUREMENTS**

This section includes a summary of analog measurements for all loads in the network selection.

- **SYSTEM MONITORING INFORMATION**

This section provides system monitoring information from the LVM solution, including Time Since Last Plan was Created, Consecutive Infeasible Solutions, Daily Infeasible Solutions, Consecutive Discarded Solutions, Daily Discarded Solutions, Capacitor Time Monitoring Duration, Capacitor Moves during Monitoring Duration, Regulator/LTC Time Monitoring Duration, and Regulator/LTC Moves during Monitoring Duration (Limit).

- **ELIGIBLE CONTROLS**

This section summarizes the eligible controllable devices included in the network selection. The controllable devices include capacitors, LTC/regulators, generators, and Distributed Energy Resources (DERs).

- **INELIGIBLE CONTROLS**

This section summarizes the ineligible controllable devices included in the network selection. The controllable devices include capacitors, LTC/regulators, generators, and DERs.

- **MAXIMUM CAPACITOR AND TAPCHANGER OPERATING TIMES CHECK**

This section details the checks performed on capacitors and LTC/regulators regarding limits on their operations that may impact the LVM solution.

- **CAPACITOR SWITCHING COST DETAIL**

This section details the optimization switching cost for each capacitor in the network selection.

- **TAP CHANGER MOVEMENT COST DETAIL**

This section details the optimization LTC/regulator movement cost for each LTC/regulator in the network selection.

- **REGULATION ZONES**

This section lists regulation zones and upstream and downstream regulating devices for each zone.

- **VOLTAGE LIMITS**

This section details the voltage limits along with the measured and calculated voltage values for each bus and load in the network selection.

- ITERATION INFORMATION

This section provides details on each LVM and BLA/DPF iteration. Within each iteration, P, Q, and V mismatch values are provided along with transformer tap and capacitor moves. Constraint information is also provided for each LVM iteration.

- PLAN INFORMATION

This section provides a summary of the proposed plan produced by LVM. The plan ID, status, number of control moves, device information, performance indices, statistics, and post plan bus voltages and limits are provided.

- LVM EVALUATION DATA

- VIOLATION SUMMARY BEFORE PLAN

This section details the limit violations (those included in the problem formulation) in the network selection before the plan is implemented, ordered from highest to lowest violation. Below the summary and for each limit violation, the sensitivity of each control device with respect to enforcement of the limit constraint is detailed.

- VIOLATION SUMMARY AFTER PLAN

This section details the limit violations in the network selection after the plan is implemented, ordered from highest to lowest violation.

- LVM PLAN ENHANCEMENT

This section details the results of the plan enhancement process (see [Plan Enhancement](#)). Information from each step of the plan enhancement process that was performed is included (it is not always necessary to run all the steps). The five steps are Direct Discretization, Fix Capacitors, Adjust Taps, Increase Margins, and Discrete Move Improvement. The control moves and violations from each step are included.

- LVM PLAN SELECTION

This section details the results of the plan selection process (see [Plan Selection](#)). For the proposed plan, a status is given (accepted or rejected) along with the control moves, information on the final objective function value, information on variables within the problem formulation, and an overall summary of the plan's effectiveness.

14.2. LVM System Monitor Implementation Result File

LVM System Monitor implementation results are recorded in files named LVMSystemMonitor_<TimeStamp>.dat. The data in this file is grouped in sections as follows:

- PARAMETERS

This section lists LVM System Monitor settings from the DNAF Configuration Editor.

- LOCKED STATIONS

This section lists stations whose problem formulations are locked.

- LOCKED TRANSFORMER BANKS

This section lists station power transformers whose problem formulations are locked.

- LOCKED FEEDERS

This section lists station feeders whose problem formulations are locked.

- OVERALL SYSTEM RESULTS

This section provides implementation results for the entire system: initial real power, final real power, and voltage and real and reactive power changes over the implementation period.

- AREA RESULTS

This section provides implementation results by optimization areas: initial real power, final real power, and voltage and real and reactive power changes over the implementation period.

- STATION RESULTS

This section provides implementation results for stations: original problem formulation, new problem formulation, time implemented, steps implemented, steps failed, initial real power, initial reactive power, initial voltage, final real power, final reactive power, final voltage, locked status, locked power transformers, locked feeders, and excluded devices.

14.3. FISR Output File

FISR results for stations are recorded in files named FISR Results_<StationName>_<TimeStamp>.dat.

The FISR results file contains the FISR solution information and FISR triggering information from the FISR Fault Event Triggering Details file or FISR Loss of Voltage Event Triggering Details file (depending on the FISR triggering event type).

When FISR is triggered by a fault event, the FISR results file contains the following sections from the FISR Fault Event Triggering Details file (for more details, see [FISR Fault Event Triggering Details](#)):

- FISR FAULT TRIGGERING PARAMETERS
- TRIGGERING EVENT
- LOCKOUT INFORMATION
- FAULT INDICATOR STATUS
- TRIPPED PROTECTIVE DEVICES
- FAULT LOCATION RESULTS
- FAULT PREDICTION
- FAULT PLACEMENT

When FISR is triggered by a loss of voltage or station outage event, the FISR results file contains the

following sections from the FISR Loss of Voltage Event Triggering Details file (for more details, see [FISR Loss of Voltage Event Triggering Details](#)):

- FISR LOSS OF VOLTAGE TRIGGERING PARAMETERS
- TRIGGERING EVENT
- ASSOCIATED PROTECTIVE DEVICE
- FEEDER CURRENT MEASUREMENT INFORMATION AFTER LOSS OF VOLTAGE DELAY
- SWITCH VOLTAGE/CURRENT MEASUREMENT INFORMATION AFTER LOSS OF VOLTAGE DELAY
- LOSS OF VOLTAGE PLACEMENT
- STATION OUTAGE EVENT

The FISR results file contains the following sections related to the FISR solution:

- FISR EXECUTION SUMMARY

This section shows the substation ID, substation name, substation area (service center), and substation permission area.
- FISR SOLUTION SUMMARY

This section contains triggering information: evaluation time, peak evaluation time, execution time, execution reason, and solution status.
- FISR TRIGGERING PARAMETERS

This section lists FISR triggering settings, including fault indicator, fault location, and loss of voltage event triggers.
- PROBLEM FORMULATION

This section details the problem formulation, including the default formulation, the active formulation, and the formulation that was used in this solution.
- DNAF INPUT PARAMETER FILE LOCATIONS

This section details the file locations for the input parameters. The files included are DNAF_PARAMS.xml, alg.dat, msgdef.dat, optalg.dat, phid.dat, and tftype.dat. The DNAF_PARAMS.xml file's time last modified and revision number are also included, as well as the DNAF Configuration Editor (DCE) version.
- DNAF INPUT PARAMETERS - FAULT ISOLATION AND SERVICE RESTORATION

This section details the FISR parameter values from the DNAF Configuration Editor.
- DNAF INPUT PARAMETERS - PROTECTION VALIDATION

This section details the values of the PRV parameters from the DNAF Configuration Editor used in FISR, including P, Q, and V mismatches, safety factor, and a flag to simulate minimum number of faults.
- PROBLEM FORMULATION PARAMETERS – FISR

This section details the values of the problem formulation parameters from the DNAF Configuration Editor: performance index weights, analysis options, switching costs, plan parameters, limit strategy, and limit adjustment.

- **BREAKER AND RECLOSER LIMITS**

This section details the values of the breaker and recloser Limit Monitor configuration parameters from the DNAF Configuration Editor.

- **ALARM CONFIGURATION**

The section lists the status (enabled or disabled) of the following alarms: Enabled by SCADA Event, Disabled by DMS Event, Disabled by SCADA Event, Enabled Status Not Updated Event, FISIR Started Event, FISIR Success Event, FISIR Failed Event, Plan Approved Event, Plan Rejected Event, Implementation Complete Event, Implementation Failed Event, Make Before Break Low Flow Event, LVM Support Mode Event, and LVM Support Mode RTN Event.

- **OPERATION MODE**

This section details the values of the problem formulation operation mode parameters, including the plan execution method, plan and step timeouts, and maximum execution attempts for switching devices and settings groups.

- **DA SUPPORT**

This section specifies whether the LVM is solved as part of the FISIR solution.

- **EXECUTION MODE**

This section gives the plan execution mode for permanent fault and loss of voltage event triggers and provides the LVM DA support mode.

- **STATION/POWER TRANSFORMER/FEEDER ELIGIBILITY, PROBLEM FORMULATION CONFIGURATION, AND PLAN EXECUTION SETTINGS**

This section provides the setting level, SCADA Enabled status, Application Enabled status, problem formulation, permanent fault plan execution type, loss of voltage plan execution type, and FISIR fault event trigger status for each substation, power transformer, and feeder in the network selection.

- **BUS VOLTAGE MEASUREMENTS**

This section provides bus voltage measurements.

- **FAULT INDICATOR MEASUREMENT STATE BEFORE/AFTER FISIR SOLUTION**

This section provides details for fault indicators attached to protective devices before and after the FISIR solution, including measurement ID, indication whether the state changed, and fault indicator ID, name, phasing, initial status, final status, initial quality, and final quality.

- **VALID FAULTED COMPONENT**

This section provides details for the faulted component, including station name, feeder name,

device name and ID, original line ID, and the restore load status.

- **NETWORK SELECTION**

This section provides details for elements that are included in the network selection: stations, feeders, measurement islands, and third-party equipment (when the detailed logging is enabled). It also details feeder adjacency levels and lists open wye delta distribution transformers that are not in the network selection.

- **SOURCE VOLTAGES**

This section provides phase voltages and voltage angles for reference bus interface objects.

- **FEEDER HEAD BRANCH FLOWS**

This section provides real power, reactive power, and current values for feeder head switching devices.

- **EQUIVALENT LOAD INFORMATION**

This section provides real power and reactive power load values and capacitor reactive power injection values for feeder nodes.

- **PEAK LOADING DETERMINATION**

This section provides the feeder look-ahead loading summary, including the station load category nominal schedule multipliers and data pairs of total load and forecasted load for station feeders. It also provides data pairs of forecasted load and scheduled load for all feeders and the total load for each time step. The Look Ahead Hour starts from the beginning of the hour when the event occurs (Real time) or the time of evaluation (Study environment), which is logged as Start Time just under the section header.

- **INELIGIBLE SWITCHING DEVICES**

This section provides details for ineligible switching devices included in the network selection. If a feeder is disabled by FISR, it will be not included in network selection.

- **ELIGIBLE SWITCHING DEVICES**

This section provides details for eligible switching devices included in the network selection, including station name, switch name, switch ID, switch type, reversible status, bypass switch ID, and the associated settings group template. If a feeder is disabled by FISR, it will be not included in network selection.

- **INELIGIBLE SETTINGS GROUPS**

This section provides details for ineligible settings groups that are associated to switching devices included in the network selection.

- **ELIGIBLE SETTINGS GROUPS**

This section provides details for eligible settings groups in reference to the associated switching devices included in the network selection, including station name, switch name, switch ID,

switch type, active settings group, and the associated settings group template.

- **SETTINGS GROUP TEMPLATES**

This section provides a summary for protection settings group templates, including template name, ID, and number of settings groups.

- **PROTECTION SETTINGS GROUP INFORMATION**

This section provides details for settings group templates. For each template, it lists settings groups and provides settings group name, template name, settings group value, phase trip setting, ground trip setting, mode, and available transitions between settings groups.

- **PLAN INFORMATION**

This section provides the number of plans in the current solution. For each plan, it provides the number of moves, plan feasibility status, execution type, control moves, performance indices, statistics, highest and lowest bus and load voltages, initial and final highest branch loadings after DER re-energization, re-energized generators and DERs, re-energized embedded generation in loads, protection validation results, non-converged BLA and scheduled loading measurement islands, and protection settings group changes. For each closed control move, it provides the list of restored customers and loads.

- **PLAN GENERATION INFORMATION**

This section provides plan generation details, including the isolation device and associated tie devices. It provides restoration details per device for each plan implementation level. The additional processing for composite switches is also included in this section, if applicable. Below the restoration information, the details for the best valid FISR plan are provided.

- **ANALOG MEASUREMENTS**

This section includes a summary of analog measurements in the network selection, grouped by substation.

- **STATUS MEASUREMENTS**

This section includes a summary of status measurements in the network selection, grouped by substation.

- **ABNORMAL ANALOG MEASUREMENTS**

This section includes a summary of analog measurements in the network selection with invalid status or quality, grouped by substation.

14.3.1. FISR Fault Event Triggering Details

When the fault event triggering conditions for FISR are met, the event information, such as measurements or conditions, is recorded in a unique results file. If the application is actually triggered, this trigger information file is then appended at the top of the existing FISR results file and the triggering detail file is deleted. If for some reason an application is not triggered (for

example, due to incomplete or inconsistent data or an invalid SCADA measurement quality), the event triggering information is available in a unique results file for later analysis.

FISR fault event triggering details are recorded in files named FISR Fault Triggering_<StationName>_FIND_<LineID><TimeStamp>.dat. The data in this file is grouped in sections as follows:

- **FISR FAULT TRIGGERING PARAMETERS**

This section lists FISR fault triggering parameters specified in the DNAF server configuration file and DNAF Configuration Editor. These are:

- DNAF_FISR_FAULTIND_EVENT_TRIGGER
- DNAF_FISR_FL_EVENT_TRIGGER
- DNAF_FL_EVENT_TRIGGER
- DNAF_EVENT_DELAY (sec)
- DNAF_FL_DELAY (sec)
- DNAF_FISR_MAX_LOCKOUT_WAITTIME (sec)
- DNAF_FISR_LOCKOUT_FAULTPLACEMENT
- DNAF_FISR_NO_LOCKOUT_WAITTIME (sec)
- DNAF_FISR_CHECK_LOCKOUT_PERIOD (sec)
- Handle Miscoordinations
- Time Difference Tolerance (sec)
- Time Tolerance for Old Events Not Considered (sec)
- Display System Generated Fault Symbol
- Teams and Schemes - Enable FISR Integration with Local Schemes
- Teams and Schemes - FISR Wait for Local Schemes Delay (sec)
- Teams and Schemes - FISR Can Use Team Members

- **TRIGGERING EVENT**

This section provides FISR fault indicator event triggering details. For each event, it provides the station name, set fault indicator name and Id, set phases, associated device name, associated device Id, associated device type, and NTP fault indicator event time stamp.

- **FEEDER MAIN SWITCHING DEVICE ENERGIZATION**

This section provides the energization status of devices on the main feeder. For each device, the section provides the station name, Id, type, phases, open phases, node 1 Id, node 1 energized phases, node 2 Id, and node 2 energized phases.

- **LOCKOUT INFORMATION**

This section provides information for protective device lockout status measurements that are

associated with the triggering event both within and outside the user-configured old events time tolerance. For each lockout measurement, the section provides the measurement Id, measurement value, measurement phasing, measurement quality, measurement time stamp, device name, device Id, and device open phases.

- SWITCH STATUS MEASUREMENTS

This section provides measurement details for switching devices that are associated with the triggering event both within and outside the user-configured old events time tolerance. For each measurement, the section provides the station name, status measurement ID, phase, measurement quality, measurement value, measurement time stamp, switch name, switch Id, switch type, and switch status.

- FAULT INDICATORS

This section provides the details for fault indicators that are 1) inside or at the boundary of the faulted area, 2) upstream of the faulted area, and 3) the valid set fault indicators in the substation. For each fault indicator, the section provides the station name, ID, phase, status, forward set phases, forward held fault indication phases, reverse set phases, reverse held fault indication phases, and disregard phasing (phase agnostic) flag.

- FAULT INDICATOR STATUS MEASUREMENTS

This section provides the measurement details for fault indicators that are associated with the triggering event. For each measurement, the section provides the station name, fault indicator status measurement ID, phase, measurement quality, measurement value, measurement time stamp, fault indicator Id, fault indicator phases, fault indicator status, fault indicator set phases, fault indicator disregard phasing (phase agnostic) flag, device name, device Id, and device type.

- TRIPPED PROTECTIVE DEVICES

This section lists tripped protective devices.

- FAULT LOCATION RESULTS

This section provides a summary of the Fault Location application results: station, branch ID, feeder main/lateral flag, location range, faulted phases, energized phases, fault indicator halo phases, measured fault current, calculated fault current, and fault current difference.

- FAULT PREDICTION

This section provides the fault prediction information including the fault object name, fault object phases, branch Id, branch type (a line or an inline/series transformer such as a step up/down transformer, auto transformer, or a regulator), upstream restoration is possible flag, tripped protective device, miscoordination occurred status, closest protective device, and fault object location (geographic x/y coordinates).

- FAULT PLACEMENT

This section provides the fault placement information: fault object name, faulted section has been found and a fault can be placed flag, and the trigger FISR flag.

14.3.2. FISR Loss of Voltage Event Triggering Details

When the loss of voltage event triggering conditions for FISR are met, the event information, such as measurements or conditions, is recorded in a unique results file. If the application is triggered, this trigger information file is then appended at the top of the existing FISR results file and the triggering detail file is deleted. If for some reason an application is not triggered (for example, due to incomplete or inconsistent data or an invalid SCADA measurement quality), the event triggering information is available in a unique results file for later analysis.

FISR loss of voltage event triggering details are recorded in files named FISR Loss of Voltage Triggering_<StationName>_<Voltage Measurement Name>_<TimeStamp>.dat. The data in this file is grouped in sections as follows:

- FISR LOSS OF VOLTAGE TRIGGERING PARAMETERS

This section lists FISR loss of voltage triggering parameters specified in the DNAF server configuration file and DNAF Configuration Editor. These are:

- DNAF_FISR_FAULTIND_EVENT_TRIGGER
- DNAF_FISR_FL_EVENT_TRIGGER
- DNAF_FL_EVENT_TRIGGER
- DNAF_FISR_LOV_EVENT_TRIGGER
- DNAF_LOV_DELAY (sec)
- DNAF_LOV_2ND_DELAY (sec)
- Display System Generated Cut Symbol
- Voltage Measurement Threshold (%)
- Use Current Measurement Threshold
- Current Measurement Threshold (A)

- TRIGGERING EVENT

This section provides FISR loss of voltage event triggering details. For each event, it provides the station name, voltage measurement name and ID, measurement phase, old and new measurement values, and loss of voltage event time stamp.

- STATION OUTAGE EVENT TRIGGER ERROR INFO (this section only appears in FISR Station Outage Event Triggering files)

This section describes the reason for stopping the FISR solution triggering for the station outage event.

- FEEDER FAULT WITH ONLY BREAKER STATUS MEASUREMENT (this section only appears in feeder fault triggered loss of voltage FISR triggering files)

This section provides placement details for the feeder fault triggered loss of voltage predicted cut. For each cut, it provides the station name, switch ID, switch name, upstream branch ID, and

upstream branch name.

- ASSOCIATED PROTECTIVE DEVICE

This section provides details for breakers and reclosers affected by the loss of voltage event. For each device, it provides the associated device name, ID, type, status, commanded operation and initiated by load shed flags, and last operation time.

- FEEDER CURRENT MEASUREMENT INFORMATION AFTER LOSS OF VOLTAGE DELAY

This section provides current measurement details for feeders affected by the loss of voltage event. For each feeder, it provides the station name, feeder name and ID, phase A, B, and C flow values, current measurement ID, current measurement value, current measurement quality, and current measurement time stamp.

- SWITCH VOLTAGE/CURRENT MEASUREMENT INFORMATION AFTER LOSS OF VOLTAGE DELAY

This section provides voltage and current details for components affected by the loss of voltage event. For each device, it provides the station name, device name, device ID, device type, voltage measurement ID, voltage measurement name, voltage measurement value, measurement phasing, voltage measurement quality, measurement time stamp, a flag for whether the voltage measurement was used to trigger the loss of voltage event for FISIR, indication if the measurement was found to be consistent (for measurements that were used for consistency check), current measurement ID, current measurement name, current measurement value, current measurement quality, current measurement time stamp, and a flag for whether the current measurement was used to trigger the loss of voltage event for FISIR.

- LOSS OF VOLTAGE PLACEMENT (this section only appears in FISIR Loss of Voltage Triggering files)

This section provides placement details for the loss of voltage predicted cut. For each cut, it provides the station name, switch name and ID, and predicted cut branch name and ID.

- STATION OUTAGE EVENT (this section only appears in FISIR Station Outage Event Triggering files)

This section provides summary for protection devices affected by the station outage event. For each cut, it provides the station name, switch name and ID, and upstream branch name and ID.

- INFO (this section only appears in FISIR Loss of Voltage Triggering files)

This section describes the reason for stopping the FISIR solution triggering for the loss of voltage event.

14.3.3. FISIR Station Outage Event Triggering Details

When the station outage event triggering conditions for FISIR are met, the event information, such as measurements or conditions, is recorded in a unique results file. If the application is actually triggered, this trigger information file is then appended at the top of the existing FISIR results file

and the triggering detail file is deleted. If for some reason an application is not triggered (for example, due to incomplete or inconsistent data or an invalid SCADA measurement quality), the event triggering information is available in a unique results file for later analysis.

FISR station outage event triggering details are recorded in files named FISR Station Outage Event Triggering_<StationName>_<Voltage Measurement Name>_<TimeStamp>.dat.

The FISR Station Outage Event Triggering Details file contains the same sections as the FISR Loss of Voltage Event Triggering Details file. For more details, see [FISR Loss of Voltage Event Triggering Details](#).

14.4. AFR Output File

AFR results for stations are recorded in files named AFR Results_<StationName>_<TimeStamp>.dat. The data in this file is grouped in sections as follows:

- AFR SOLUTION SUMMARY

This section contains triggering information (evaluation time, peak evaluation time, execution time, and execution reason).

- PROBLEM FORMULATION

This section details the problem formulation, including the default formulation, the active formulation, and the formulation that was used in this solution.

- DNAF INPUT PARAMETER FILE LOCATIONS

This section details the file locations for all the input parameters. The files included are DNAF_PARAMS.xml, alg.dat, msgdef.dat, optalg.dat, phid.dat, and tftype.dat. The time last modified and revision number for the DNAF_PARAMS.xml file is also included, as well as the DNAF Configuration Editor (DCE) version.

- DNAF INPUT PARAMETERS - AUTOMATIC FEEDER RECONFIGURATION

This section details the values of the AFR parameters from the DNAF Configuration Editor.

- DNAF INPUT PARAMETERS - PROTECTION VALIDATION

This section details the values of the PRV parameters from the DNAF Configuration Editor used in AFR, including P, Q, and V mismatches, safety factor, and a flag to simulate minimum number of faults.

- PROBLEM FORMULATION PARAMETERS – AFR

This section details the values of the problem formulation parameters from the DNAF Configuration Editor: performance index weights, analysis options, switching costs, plan parameters, limit strategy, and limit adjustment.

- ALARM CONFIGURATION

The section lists the status (enabled or disabled) of the following alarms: Enabled by SCADA Event, Disabled by DMS Event, Disabled by SCADA Event, Enabled Status Not Updated Event, AFR Started Event, AFR Success Event, AFR Failed Event, Plan Approved Event, Plan Rejected Event, Implementation Complete Event, Implementation Failed Event, Make Before Break Low Flow Event, LVM Support Mode Event, and LVM Support Mode RTN Event.

- OPERATION MODE

This section details the values of the problem formulation operation mode parameters, including the plan execution method, plan and step timeouts, and maximum execution attempts for switching devices and settings groups.

- EXECUTION MODE

This section gives the plan execution mode for permanent fault and loss of voltage event triggers and provides the LVM DA support mode.

- STATION/POWER TRANSFORMER/FEEDER ELIGIBILITY SETTINGS

This section details the enabled status, problem formulation, and power factor range for each power transformer and feeder in the substation.

- BUS VOLTAGE MEASUREMENTS

This section provides bus voltage measurements.

- OVERLOAD COMPONENT

This section provides details for the overload component.

- NETWORK SELECTION

This section lists elements that are included in the network selection: stations, feeders, and measurement islands, and third-party equipment (when the detailed logging is enabled). It also details feeder adjacency levels and lists partially removed components.

- SOURCE VOLTAGES

This section provides voltages and voltage angles for reference bus interface objects.

- FEEDER HEAD BRANCH FLOWS

This section provides real and reactive power and current values for feeder head devices.

- PEAK LOADING DETERMINATION

This section provides the feeder look-ahead loading summary, including the station load category nominal schedule multipliers and data pairs of total load and forecasted load for station feeders. It also provides data pairs of forecasted load and scheduled load for all feeders and the total load for each time step. The Look Ahead Hour starts from the beginning of the hour when the event occurs (Real time) or the time of evaluation (Study environment), which is logged as Start Time just under the section header.

- INELIGIBLE SWITCHING DEVICES

This section provides details for ineligible switching devices included in the network selection.

- **ELIGIBLE SWITCHING DEVICES**

This section provides details for eligible switching devices included in the network selection, including station name, switch name, switch ID, switch type, reversible status, bypass switch ID, and the associated settings group template.

- **INELIGIBLE SETTINGS GROUPS**

This section provides details for ineligible settings groups that are associated to switching devices included in the network selection.

- **ELIGIBLE SETTINGS GROUPS**

This section provides details for eligible settings groups in reference to the associated switching devices included in the network selection, including station name, switch name, switch ID, switch type, active settings group, and the associated settings group template.

- **SETTINGS GROUP TEMPLATES**

This section provides a summary for protection settings group templates, including template name, ID, and number of settings groups.

- **PROTECTION SETTINGS GROUP INFORMATION**

This section provides details for settings group templates. For each template, it lists settings groups and provides settings group name, template name, settings group value, phase trip setting, ground trip setting, mode, and available transitions between settings groups.

- **PLAN INFORMATION**

This section provides the number of plans in the current solution. For each plan, it provides the number of moves, plan status, control moves, performance indices, statistics, protection validation results, and non-converged BLA and scheduled loading measurement islands.

- **PLAN GENERATION INFORMATION**

This section provides load transfer details per device for each plan implementation level.

- **ANALOG MEASUREMENTS**

This section includes a summary of analog measurements in the network selection, grouped by substation.

- **STATUS MEASUREMENTS**

This section includes a summary of status measurements in the network selection, grouped by substation.

- **INVALID ANALOG MEASUREMENTS**

This section includes a summary of analog measurements in the network selection with invalid status or quality, grouped by substation.

14.5. PLOS Output File

PLOS results for stations are recorded in files named PLOS Results_<StationName>_<TimeStamp>.dat. The data in this file is grouped in sections as follows:

- PLOS SOLUTION SUMMARY

This section contains triggering information (evaluation time, peak evaluation time, execution time, and execution reason).

- PROBLEM FORMULATION

This section details the problem formulation, including the default formulation, the active formulation, and the formulation that was used in this solution.

- DNAF INPUT PARAMETER FILE LOCATIONS

This section details the file locations for all the input parameters. The files included are DNAF_PARAMS.xml, alg.dat, msgdef.dat, optalg.dat, phid.dat, and tftype.dat. The time last modified and revision number for the DNAF_PARAMS.xml file is also included, as well as the DNAF Configuration Editor (DCE) version.

- DNAF INPUT PARAMETERS - PLANNED OUTAGE STUDY

This section details the values of the PLOS parameters from the DNAF Configuration Editor.

- DNAF INPUT PARAMETERS - PROTECTION VALIDATION

This section details the values of the PRV parameters from the DNAF Configuration Editor used in PLOS, including P, Q, and V mismatches, safety factor, and a flag to simulate minimum number of faults.

- PROBLEM FORMULATION PARAMETERS – PLOS

This section details the values of the problem formulation parameters from the DNAF Configuration Editor: performance index weights, analysis options, switching costs, plan parameters, limit strategy, and limit adjustment.

- BUS VOLTAGE MEASUREMENTS

This section provides bus voltage measurements.

- VALID OUTAGED COMPONENT

This section provides details for the disconnected component.

- NETWORK SELECTION

This section lists elements that are included in the network selection: stations, feeders, and measurement islands, and third-party equipment (when the detailed logging is enabled). It also details feeder adjacency levels and lists open wye delta distribution transformers not in the network selection.

-

SOURCE VOLTAGES

This section provides voltages and voltage angles for reference bus interface objects.

- FEEDER HEAD BRANCH FLOWS

This section provides real and reactive power and current values for feeder head devices.

- EQUIVALENT LOAD INFORMATION

This section provides real and reactive power load and capacitor reactive power injection values for nodes.

- PEAK LOADING DETERMINATION

This section provides the feeder look-ahead loading summary, including the station load category nominal schedule multipliers and data pairs of total load and forecasted load for station feeders. It also provides data pairs of forecasted load and scheduled load for all feeders and the total load for each time step. The Look Ahead Hour starts from the beginning of the hour when the event occurs (Real time) or the time of evaluation (Study environment), which is logged as Start Time just under the section header.

- INELIGIBLE SWITCHING DEVICES

This section provides details for ineligible switching devices included in the network selection.

- ELIGIBLE SWITCHING DEVICES

This section provides details for eligible switching devices included in the network selection.

- SETTINGS GROUP TEMPLATES

This section provides a summary for protection settings group templates, including template name, ID, and number of settings groups.

- PROTECTION SETTINGS GROUP INFORMATION

This section provides details for settings group templates. For each template, it lists settings groups and provides the settings group name, template name, settings group value, phase trip setting, ground trip setting, mode, and available transitions between settings groups.

- PLAN INFORMATION

This section provides the number of plans in the current solution. For each plan, it provides the number of moves, plan status, control moves, performance indices, statistics, protection validation results, and non-converged BLA and scheduled loading measurement islands.

- PLAN GENERATION INFORMATION

This section provides plan generation details, including the isolation device and associated tie devices. It provides restoration details per device for each plan implementation level. The additional processing for composite switches is also included in this section, if applicable. Below the restoration information, the details for the best valid PLOS plan are provided.

- ANALOG MEASUREMENTS

This section includes a summary of analog measurements in the network selection, grouped by substation.

- STATUS MEASUREMENTS

This section includes a summary of status measurements in the network selection, grouped by substation.

- INVALID ANALOG MEASUREMENTS

This section includes a summary of analog measurements in the network selection with invalid status or quality, grouped by substation.

14.6. FISRCA Output File

FISRCA results for stations are recorded in files named FISRCA Results_<StationName>_<TimeStamp>.dat. The data in this file is grouped in sections as follows:

- FISRCA SOLUTION SUMMARY

This section contains triggering information (evaluation time, peak evaluation time, execution time, and execution reason).

- PROBLEM FORMULATION

This section details the problem formulation, including the default formulation, the active formulation, and the formulation that was used in this solution.

- DNAF INPUT PARAMETER FILE LOCATIONS

This section details the file locations for all the input parameters. The files included are DNAF_PARAMS.xml, alg.dat, msgdef.dat, optalg.dat, phid.dat, and tftype.dat. The time last modified and revision number for the DNAF_PARAMS.xml file is also included, as well as the DNAF Configuration Editor (DCE) version.

- DNAF INPUT PARAMETERS - FISRCA

This section details the values of the FISRCA parameters from the DNAF Configuration Editor.

- SWITCHABLE SECTIONS

This section details the switchable section information for the station.

This section provides the number of switchable sections in the substation. For each switchable section, it provides the name and ID of included bounding switches.

- BUS VOLTAGE MEASUREMENTS

This section provides bus voltage measurements.

- VALID FAULTED COMPONENT

This section provides details for the faulted component.

- NETWORK SELECTION

This section lists elements that are included in the network selection: stations, feeders, and measurement islands. It also details feeder adjacency levels and lists open wye delta distribution transformers not in the network selection.

- SOURCE VOLTAGES

This section provides voltages and voltage angles for reference bus interface objects.

- FEEDER HEAD BRANCH FLOWS

This section provides real and reactive power and current values for feeder head devices.

- EQUIVALENT LOAD INFORMATION

This section provides real and reactive power load and capacitor reactive power injection values for nodes.

- PEAK LOADING DETERMINATION

This section provides the feeder look-ahead loading summary, including the station load category nominal schedule multipliers and data pairs of total load and forecasted load for station feeders. It also provides data pairs of forecasted load and scheduled load for all feeders and the total load for each time step. The Look Ahead Hour starts from the beginning of the hour when the event occurs (Real time) or the time of evaluation (Study environment), which is logged as Start Time just under the section header.

- INELIGIBLE SWITCHING DEVICES

This section provides details for ineligible switching devices included in the network selection.

- ELIGIBLE SWITCHING DEVICES

This section provides details for eligible switching devices included in the network selection.

- SETTINGS GROUP TEMPLATES

This section provides a summary for protection settings group templates, including template name, ID, and number of settings groups.

- PROTECTION SETTINGS GROUP INFORMATION

This section provides details for settings group templates. For each template, it lists settings groups and provides the settings group name, template name, settings group value, phase trip setting, ground trip setting, mode, and available transitions between settings groups.

- PLAN INFORMATION

This section provides the number of plans in the current solution. For each plan, it provides the number of moves, plan status, control moves, performance indices, statistics, protection validation results, and non-converged BLA and scheduled loading measurement islands.

- PLAN GENERATION INFORMATION

This section provides plan generation details, including the isolation device and associated tie devices. It provides restoration details per device for each plan implementation level. Below the restoration information, the details for the best valid FISR plan are provided.

- ANALOG MEASUREMENTS

This section includes a summary of analog measurements in the network selection, grouped by substation.

- STATUS MEASUREMENTS

This section includes a summary of status measurements in the network selection, grouped by substation.

- INVALID ANALOG MEASUREMENTS

This section includes a summary of analog measurements in the network selection with invalid status or quality, grouped by substation.

14.7. FSD Stage 1 Output File

FSD Stage 1 results for station power transformers are recorded in files named FSD STAGE 1 Results_<StationName>_<TransformerName>_<TimeStamp>.dat. The data in this file is grouped in sections as follows:

- FSD STAGE 1 EXECUTION SUMMARY

This section shows the substation ID, substation name, substation area (service center), substation permission area, and execution time.

- PARAMETER

This section provides details for the values of the problem formulation parameters from the DNAF Configuration Editor. The parameters are fault indicator weight, customer count weight, total load weight, customer priority weight, and feeder length weight.

- RANKED BREAKERS

This section provides details for feeder breakers and feeders downstream of the transformer. The parameters are device ID and name, customer count, feeder customer priority, existence of set fault indicators, overhead feeder length, total load, ranking, eligibility status, and ineligibility reason.

- PLAN DETAILS

This section provides details for the FSD Stage 1 plan. These details are plan id, triggering component ID and name, solution ID, plan creation time, and plan objective function name.

This section also provides plan implementation details. These details are step ID, control type,

plan step number, associated switching step number, initial value, final value, device phases, and feeder ID.

14.8. FSD Stage 2 Output File

FSD Stage 2 results for station power transformers are recorded in files named FSD STAGE 2 Results_<StationName>_<FeederHeadBrakerID>_<TimeStamp>.dat. The data in this file is grouped in sections as follows:

- FSD STAGE 2 EXECUTION SUMMARY

This section shows the substation ID, substation name, substation area (service center), substation permission area, and execution time.

- PLAN DETAILS

This section provides details for the FSD Stage 2 plan. These details are plan id, triggering component ID and name, solution ID, plan creation time, and plan objective function name.

This section also provides plan implementation details. These details are step ID, control type, plan step number, associated switching step number, initial value, final value, device phases, and feeder ID.